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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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- x the first digit:
 - 1 presented to TSG for information;
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- Y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

The present document specifies the radio network layer signalling protocol for the NG interface. The NG Application Protocol (NGAP) supports the functions of the NG interface by signalling procedures defined in this document. NGAP is developed in accordance to the general principles stated in TS 38.401 [2] and TS 38.410 [3].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [2] 3GPP TS 38.401: "NG-RAN; Architecture description".
- [3] 3GPP TS 38.410: "NG-RAN; NG general aspects and principles".
- [4] ITU-T Recommendation X.691 (07/2002): "Information technology – ASN.1 encoding rules: Specification of Packed Encoding Rules (PER)".
- [5] ITU-T Recommendation X.680 (07/2002): "Information technology – Abstract Syntax Notation One (ASN.1): Specification of basic notation".
- [6] ITU-T Recommendation X.681 (07/2002): "Information technology – Abstract Syntax Notation One (ASN.1): Information object specification".
- [7] 3GPP TR 25.921 (version.7.0.0): "Guidelines and principles for protocol description and error handling".
- [8] 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description; Stage 2".
- [9] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".
- [10] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".
- [11] 3GPP TS 32.422: "Trace control and configuration management".
- [12] 3GPP TS 38.304: "NR; User Equipment (UE) procedures in idle mode and in RRC inactive state".
- [13] 3GPP TS 33.501: "Security architecture and procedures for 5G System".
- [14] 3GPP TS 38.414: "NG-RAN; NG data transport".
- [15] 3GPP TS 29.281: "General Packet Radio System (GPRS); Tunnelling Protocol User Plane (GTPv1-U)".
- [16] 3GPP TS 36.413: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP)".
- [17] 3GPP TS 36.300: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Overall description; Stage 2".
- [18] 3GPP TS 38.331: "NG-RAN; Radio Resource Control (RRC) Protocol Specification".
- [19] 3GPP TS 38.455: "NG-RAN; NR Positioning Protocol A (NRPPa)".

- [20] Void
- [21] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA) Radio Resource Control (RRC); Protocol specification".
- [22] 3GPP TS 23.041: "Technical realization of Cell Broadcast Service (CBS)".
- [23] 3GPP TS 23.003: "Numbering, addressing and identification".
- [24] 3GPP TS 38.423: "NG-RAN; Xn Application Protocol (XnAP)".
- [25] IETF RFC 5905 (2010-06): "Network Time Protocol Version 4: Protocol and Algorithms Specification".
- [26] 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".
- [27] 3GPP TS 33.401: "3GPP System Architecture Evolution (SAE); Security architecture".
- [28] 3GPP TS 25.413: "UTRAN Iu interface RANAP signalling".
- [29] 3GPP TS 36.304: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) procedures in idle mode".
- [30] 3GPP TS 29.531: "5G System; Network Slice Selection Services; Stage 3".
- [31] 3GPP TS 23.216: "Single Radio Voice Call Continuity (SRVCC); Stage 2".
- [32] 3GPP TS 37.340: "Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity; Stage 2".
- [33] 3GPP TS 23.287: "Architecture enhancements for 5G System (5GS) to support Vehicle-to-Everything (V2X) services".
- [34] 3GPP TS 23.316: "Wireless and wireline convergence access support for the 5G System (5GS)".
- [35] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".
- [36] 3GPP TS 29.510: "5G System; Network Function Repository Services; Stage 3".
- [37] CableLabs WR-TR-5WWC-ARCH: "5G Wireless Wireline Converged Core Architecture".
- [38] 3GPP TS 36.401: "E-UTRAN Architecture Description".
- [39] 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception".
- [40] 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 Application Protocol (X2AP)".
- [41] 3GPP TS 37.320: "Universal Terrestrial Radio Access (UTRA), Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Radio measurement collection for Minimization of Drive Tests (MDT); Overall description; Stage 2".
- [42] 3GPP TS 36.306: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio access capabilities".
- [43] 3GPP TS 29.244: "Interface between the Control Plane and the User Plane Nodes; Stage 3".
- [44] 3GPP TS 23.247: "Architectural enhancements for 5G multicast-broadcast services; Stage 2".
- [45] 3GPP TS 28.405: "Telecommunication management; Quality of Experience (QoE) measurement collection; Control and configuration".
- [46] 3GPP TS 26.247: "Transparent end-to-end Packet-switched Streaming Service (PSS); Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH)".
- [47] 3GPP TS 23.304: "Proximity based Services (ProSe) in the 5G System (5GS)".
- [48] 3GPP TS 38.314: "NR; Layer 2 Measurements".

- [49] 3GPP TS 36.314: "Evolved Universal Terrestrial Radio Access (E-UTRA); Layer 2 - Measurements".
- [50] 3GPP TS 23.203: "Policy and charging control architecture".
- [51] 3GPP TS 26.114: "IP Multimedia Subsystem (IMS); Multimedia Telephony; Media handling and interaction".
- [52] 3GPP TS 26.118: "Virtual Reality (VR) profiles for streaming applications".
- [53] IETF RFC 4122: "A Universally Unique IDentifier (UUID) URN Namespace".
- [54] 3GPP TS 23.256: "Support of Uncrewed Aerial Systems (UAS) connectivity, identification and tracking; Stage 2".
- [55] IEEE Std 1588: "IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems", Edition 2019.
- [56] 3GPP TS 29.585: "5G System (5GS); Session Management Function (SMF) / Centralized User Configuration (CUC) to Access Network Talker Listener (AN-TL) and Core Network Talker Listener (CN-TL) protocol aspects; Stage 3".
- [57] 3GPP TS 23.527: "5G System; Restoration procedures".
- [58] 3GPP TS 24.007: "Mobile radio interface signalling layer 3; General aspects".
- [59] 3GPP TS 37.355: "LTE Positioning Protocol (LPP)".
- [60] 3GPP TS 23.369: "Architecture support for Ambient power-enabled Internet of Things; Stage 2".
- [61] 3GPP TS 33.369: "Security aspects of Ambient Internet of Things (AIoT) services for isolated private networks".
- [62] 3GPP TS 33.545: "Security aspects of NR Femto".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

ACL functionality: as defined in TS 36.413 [16].

A-IoT CN node: as defined in TS 38.300 [8].

A-IoT NAS: as defined in TS 23.369 [60].

CAG cell: as defined in TS 38.300 [8].

DAPS Handover: as defined in TS 38.300 [8].

Elementary Procedure: NGAP consists of Elementary Procedures (EPs). An Elementary Procedure is a unit of interaction between the NG-RAN node and the AMF. These Elementary Procedures are defined separately and are intended to be used to build up complete sequences in a flexible manner. If the independence between some EPs is restricted, it is described under the relevant EP description. Unless otherwise stated by the restrictions, the EPs may be invoked independently of each other as standalone procedures, which can be active in parallel. The usage of several NGAP EPs together or together with EPs from other interfaces is specified in stage 2 specifications (e.g., TS 38.401 [2], TS 38.410 [3] and TS 38.300 [8]).

An EP consists of an initiating message and possibly a response message. Two kinds of EPs are used:

- **Class 1:** Elementary Procedures with response (success and/or failure).
- **Class 2:** Elementary Procedures without response.

For Class 1 EPs, the types of responses can be as follows:

Successful:

- A signalling message explicitly indicates that the elementary procedure successfully completed with the receipt of the response.

Unsuccessful:

- A signalling message explicitly indicates that the EP failed.
- On time supervision expiry (i.e., absence of expected response).

Successful and Unsuccessful:

- One signalling message reports both successful and unsuccessful outcome for the different included requests. The response message used is the one defined for successful outcome.

Class 2 EPs are considered always successful.

en-gNB: as defined in TS 37.340 [32].

gNB: as defined in TS 38.300 [8].

MBS session resource: as defined in TS 38.401 [2].

Mobile IAB-MT: as defined in TS 38.300 [8].

Mobile IAB-node: as defined in TS 38.300 [8].

NB-IoT: as defined in TS 36.300 [17].

ng-eNB: as defined in TS 38.300 [8].

NG-RAN node: as defined in TS 38.300 [8].

Non-CAG cell: as defined in TS 38.300 [8].

PDU session resource: as defined in TS 38.401 [2].

Public Network Integrated NPN: as defined in TS 23.501 [9].

Stand-alone Non-Public Network: as defined in TS 23.501 [9].

WAB-gNB: as defined in TS 38.401 [2].

WAB-node: as defined in TS 38.401 [2].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5GC	5G Core Network
5QI	5G QoS Identifier
ACL	Access Control List
A-IoT	Ambient IoT
AIOTF	Ambient IoT Function
AMF	Access and Mobility Management Function
AUN3	Authenticable Non-3GPP
CAG	Closed Access Group

CGI	Cell Global Identifier
CP	Control Plane
DAPS	Dual Active Protocol Stacks
DC	Dual Connectivity
DL	Downlink
EPC	Evolved Packet Core
FN-RG	Fixed Network Residential Gateway
GUAMI	Globally Unique AMF Identifier
HFC	Hybrid Fiber-Coax
IAB	Integrated Access and Backhaul
IoT	Internet of Things
IMEISV	International Mobile station Equipment Identity and Software Version number
LMF	Location Management Function
LP-WUSPS	Low Power Wake Up Signal with Paging Subgrouping
MBS	Multicast/Broadcast Service
MT	Mobile Terminated
N3IWF	Non 3GPP InterWorking Function
NB-IoT	Narrow Band Internet of Things
NGAP	NG Application Protocol
NID	Network Identifier
NPN	Non-Public Network
NRPPa	NR Positioning Protocol Annex
NSAG	Network Slice AS Group
NSCI	New Security Context Indicator
NSSAI	Network Slice Selection Assistance Information
OTDOA	Observed Time Difference of Arrival
PEIPS	Paging Early Indication with Paging Subgrouping
PNI-NPN	Public Network Integrated Non-Public Network
ProSe	Proximity Services
PSCell	Primary SCG Cell
PTM	Point to Multipoint
PTP	Point to Point
QMC	QoE Measurement Collection
QoE	Quality of Experience
RedCap	Reduced Capability
RIM	Remote Interference Management
RIM-RS	RIM Reference Signal
RSN	Redundancy Sequence Number
RSPP	Ranging/SL Positioning Protocol
SCG	Secondary Cell Group
SCTP	Stream Control Transmission Protocol
SgNB	Secondary gNB
SMF	Session Management Function
S-NG-RAN node	Secondary NG-RAN node
SNPN	Stand-alone Non-Public Network
S-NSSAI	Single Network Slice Selection Assistance Information
TAC	Tracking Area Code
TAI	Tracking Area Identity
TNAP	Trusted Non-3GPP Access Point
TNGF	Trusted Non-3GPP Gateway Function
TNLA	Transport Network Layer Association
TSS	Timing Synchronisation Status
TWAP	Trusted WLAN Access Point
TWIF	Trusted WLAN Interworking Function
UE	User Equipment
UL	Uplink
ULI	User Location Information
UP	User Plane
UPF	User Plane Function
V2X	Vehicle-to-Everything
W-AGF	Wireline Access Gateway Function
WUS	Wake Up Signal

4 General

4.1 Procedure Specification Principles

The principle for specifying the procedure logic is to specify the functional behaviour of the terminating node exactly and completely. Any rule that specifies the behaviour of the originating node shall be possible to be verified with information that is visible within the system.

The following specification principles have been applied for the procedure text in clause 8:

- The procedure text discriminates between:

- 1) Functionality which "shall" be executed

The procedure text indicates that the receiving node "shall" perform a certain function Y under a certain condition. If the receiving node supports procedure X but cannot perform functionality Y requested in the REQUEST message of a Class 1 EP, the receiving node shall respond with the message used to report unsuccessful outcome for this procedure, containing an appropriate cause value.

- 2) Functionality which "shall, if supported" be executed

The procedure text indicates that the receiving node "shall, if supported," perform a certain function Y under a certain condition. If the receiving node supports procedure X, but does not support functionality Y, the receiving node shall proceed with the execution of the EP, possibly informing the requesting node about the not supported functionality.

- Any required inclusion of an optional IE in a response message is explicitly indicated in the procedure text. If the procedure text does not explicitly indicate that an optional IE shall be included in a response message, the optional IE shall not be included. For requirements on including *Criticality Diagnostics* IE, see clause 10.

4.2 Forwards and Backwards Compatibility

The forwards and backwards compatibility of the protocol is assured by mechanism where all current and future messages, and IEs or groups of related IEs, include ID and criticality fields that are coded in a standard format that will not be changed in the future. These parts can always be decoded regardless of the standard version.

4.3 Specification Notations

For the purposes of the present document, the following notations apply:

Procedure	When referring to an elementary procedure in the specification the Procedure Name is written with the first letters in each word in upper case characters followed by the word "procedure", e.g., Procedure Name procedure.
Message	When referring to a message in the specification the MESSAGE NAME is written with all letters in upper case characters followed by the word "message", e.g., MESSAGE NAME message.
IE	When referring to an information element (IE) in the specification the <i>Information Element Name</i> is written with the first letters in each word in upper case characters and all letters in <i>Italic font</i> followed by the abbreviation "IE", e.g., <i>Information Element</i> IE.
Value of an IE	When referring to the value of an information element (IE) in the specification the "Value" is written as it is specified in subclause 9.2 enclosed by quotation marks, e.g., "Value".

5 NGAP Services

NGAP provides the signalling service between the NG-RAN node and the AMF or between the NG-RAN node and the AIOTF that is required to fulfil the NGAP functions described in TS 38.410 [3]. NGAP services are divided into two groups:

- Non UE-associated services: They are related to the whole NG interface instance between the NG-RAN node and AMF or between the NG-RAN node and the AIOTF utilising a non UE-associated signalling connection.
- UE-associated services: They are related to one UE. NGAP functions that provide these services are associated with a UE-associated signalling connection that is maintained for the UE in question.

6 Services Expected from Signalling Transport

The signalling connection shall provide in sequence delivery of NGAP messages. NGAP shall be notified if the signalling connection breaks.

7 Functions of NGAP

The functions of NGAP are described in TS 38.410 [3].

8 NGAP Procedures

8.1 List of NGAP Elementary Procedures

In the following tables, all EPs are divided into Class 1 and Class 2 EPs (see subclause 3.1 for explanation of the different classes):

Table 8.1-1: Class 1 procedures

Elementary Procedure	Initiating Message	Successful Outcome	Unsuccessful Outcome
		Response message	Response message
AMF Configuration Update	AMF CONFIGURATION UPDATE	AMF CONFIGURATION UPDATE ACKNOWLEDGE	AMF CONFIGURATION UPDATE FAILURE
RAN Configuration Update	RAN CONFIGURATION UPDATE	RAN CONFIGURATION UPDATE ACKNOWLEDGE	RAN CONFIGURATION UPDATE FAILURE
Handover Cancellation	HANDOVER CANCEL	HANDOVER CANCEL ACKNOWLEDGE	
Handover Preparation	HANDOVER REQUIRED	HANDOVER COMMAND	HANDOVER PREPARATION FAILURE
Handover Resource Allocation	HANDOVER REQUEST	HANDOVER REQUEST ACKNOWLEDGE	HANDOVER FAILURE
Initial Context Setup	INITIAL CONTEXT SETUP REQUEST	INITIAL CONTEXT SETUP RESPONSE	INITIAL CONTEXT SETUP FAILURE
NG Reset	NG RESET	NG RESET ACKNOWLEDGE	
NG Setup	NG SETUP REQUEST	NG SETUP RESPONSE	NG SETUP FAILURE
Path Switch Request	PATH SWITCH REQUEST	PATH SWITCH REQUEST ACKNOWLEDGE	PATH SWITCH REQUEST FAILURE

Elementary Procedure	Initiating Message	Successful Outcome	Unsuccessful Outcome
		Response message	Response message
PDU Session Resource Modify	PDU SESSION RESOURCE MODIFY REQUEST	PDU SESSION RESOURCE MODIFY RESPONSE	
PDU Session Resource Modify Indication	PDU SESSION RESOURCE MODIFY INDICATION	PDU SESSION RESOURCE MODIFY CONFIRM	
PDU Session Resource Release	PDU SESSION RESOURCE RELEASE COMMAND	PDU SESSION RESOURCE RELEASE RESPONSE	
PDU Session Resource Setup	PDU SESSION RESOURCE SETUP REQUEST	PDU SESSION RESOURCE SETUP RESPONSE	
UE Context Modification	UE CONTEXT MODIFICATION REQUEST	UE CONTEXT MODIFICATION RESPONSE	UE CONTEXT MODIFICATION FAILURE
UE Context Release	UE CONTEXT RELEASE COMMAND	UE CONTEXT RELEASE COMPLETE	
Write-Replace Warning	WRITE-REPLACE WARNING REQUEST	WRITE-REPLACE WARNING RESPONSE	
PWS Cancel	PWS CANCEL REQUEST	PWS CANCEL RESPONSE	
UE Radio Capability Check	UE RADIO CAPABILITY CHECK REQUEST	UE RADIO CAPABILITY CHECK RESPONSE	
UE Context Suspend	UE CONTEXT SUSPEND REQUEST	UE CONTEXT SUSPEND RESPONSE	UE CONTEXT SUSPEND FAILURE
UE Context Resume	UE CONTEXT RESUME REQUEST	UE CONTEXT RESUME RESPONSE	UE CONTEXT RESUME FAILURE
UE Radio Capability ID Mapping	UE RADIO CAPABILITY ID MAPPING REQUEST	UE RADIO CAPABILITY ID MAPPING RESPONSE	
Broadcast Session Setup	BROADCAST SESSION SETUP REQUEST	BROADCAST SESSION SETUP RESPONSE	BROADCAST SESSION SETUP FAILURE
Broadcast Session Modification	BROADCAST SESSION MODIFICATION REQUEST	BROADCAST SESSION MODIFICATION RESPONSE	BROADCAST SESSION MODIFICATION FAILURE
Broadcast Session Release	BROADCAST SESSION RELEASE REQUEST	BROADCAST SESSION RELEASE RESPONSE	
Distribution Setup	DISTRIBUTION SETUP REQUEST	DISTRIBUTION SETUP RESPONSE	DISTRIBUTION SETUP FAILURE
Distribution Release	DISTRIBUTION RELEASE REQUEST	DISTRIBUTION RELEASE RESPONSE	
Multicast Session Activation	MULTICAST SESSION ACTIVATION REQUEST	MULTICAST SESSION ACTIVATION RESPONSE	MULTICAST SESSION ACTIVATION FAILURE
Multicast Session Deactivation	MULTICAST SESSION DEACTIVATION REQUEST	MULTICAST SESSION DEACTIVATION RESPONSE	
Multicast Session Update	MULTICAST SESSION UPDATE REQUEST	MULTICAST SESSION UPDATE RESPONSE	MULTICAST SESSION UPDATE FAILURE
Timing Synchronisation Status	TIMING SYNCHRONISATION STATUS REQUEST	TIMING SYNCHRONISATION STATUS RESPONSE	TIMING SYNCHRONISATION STATUS FAILURE
MT Communication Handling	MT COMMUNICATION HANDLING REQUEST	MT COMMUNICATION HANDLING RESPONSE	MT COMMUNICATION HANDLING FAILURE
Broadcast Session Transport	BROADCAST SESSION TRANSPORT REQUEST	BROADCAST SESSION TRANSPORT RESPONSE	BROADCAST SESSION TRANSPORT FAILURE

Elementary Procedure	Initiating Message	Successful Outcome	Unsuccessful Outcome
		Response message	Response message
NG Removal	NG REMOVAL REQUEST	NG REMOVAL RESPONSE	NG REMOVAL FAILURE
Inventory Request	INVENTORY REQUEST	INVENTORY RESPONSE	INVENTORY FAILURE
Command Request	COMMAND REQUEST	COMMAND RESPONSE	COMMAND FAILURE
A-IoT Session Release	AIOT SESSION RELEASE COMMAND	AIOT SESSION RELEASE COMPLETE	

Table 8.1-2: Class 2 procedures

Elementary Procedure	Message
Downlink RAN Configuration Transfer	DOWNLINK RAN CONFIGURATION TRANSFER
Downlink RAN Status Transfer	DOWNLINK RAN STATUS TRANSFER
Downlink NAS Transport	DOWNLINK NAS TRANSPORT
Error Indication	ERROR INDICATION
Uplink RAN Configuration Transfer	UPLINK RAN CONFIGURATION TRANSFER
Uplink RAN Status Transfer	UPLINK RAN STATUS TRANSFER
Handover Notification	HANDOVER NOTIFY
Initial UE Message	INITIAL UE MESSAGE
NAS Non Delivery Indication	NAS NON DELIVERY INDICATION
Paging	PAGING
PDU Session Resource Notify	PDU SESSION RESOURCE NOTIFY
Reroute NAS Request	REROUTE NAS REQUEST
UE Context Release Request	UE CONTEXT RELEASE REQUEST
Uplink NAS Transport	UPLINK NAS TRANSPORT
AMF Status Indication	AMF STATUS INDICATION
PWS Restart Indication	PWS RESTART INDICATION
PWS Failure Indication	PWS FAILURE INDICATION
Downlink UE Associated NRPPa Transport	DOWNLINK UE ASSOCIATED NRPPa TRANSPORT
Uplink UE Associated NRPPa Transport	UPLINK UE ASSOCIATED NRPPa TRANSPORT
Downlink Non UE Associated NRPPa Transport	DOWNLINK NON UE ASSOCIATED NRPPa TRANSPORT
Uplink Non UE Associated NRPPa Transport	UPLINK NON UE ASSOCIATED NRPPa TRANSPORT
Trace Start	TRACE START
Trace Failure Indication	TRACE FAILURE INDICATION
Deactivate Trace	DEACTIVATE TRACE
Cell Traffic Trace	CELL TRAFFIC TRACE
Location Reporting Control	LOCATION REPORTING CONTROL
Location Reporting Failure Indication	LOCATION REPORTING FAILURE INDICATION
Location Report	LOCATION REPORT
UE TNLA Binding Release	UE TNLA BINDING RELEASE REQUEST
UE Radio Capability Info Indication	UE RADIO CAPABILITY INFO INDICATION
RRC Inactive Transition Report	RRC INACTIVE TRANSITION REPORT
Overload Start	OVERLOAD START
Overload Stop	OVERLOAD STOP
Secondary RAT Data Usage Report	SECONDARY RAT DATA USAGE REPORT
Uplink RIM Information Transfer	UPLINK RIM INFORMATION TRANSFER
Downlink RIM Information Transfer	DOWNLINK RIM INFORMATION TRANSFER
Retrieve UE Information	RETRIEVE UE INFORMATION
UE Information Transfer	UE INFORMATION TRANSFER
RAN CP Relocation Indication	RAN CP RELOCATION INDICATION
Connection Establishment Indication	CONNECTION ESTABLISHMENT INDICATION
AMF CP Relocation Indication	AMF CP RELOCATION INDICATION
Handover Success	HANDOVER SUCCESS
Uplink RAN Early Status Transfer	UPLINK RAN EARLY STATUS TRANSFER
Downlink RAN Early Status Transfer	DOWNLINK RAN EARLY STATUS TRANSFER
Multicast Group Paging	MULTICAST GROUP PAGING
Broadcast Session Release Required	BROADCAST SESSION RELEASE REQUIRED
Timing Synchronisation Status Report	TIMING SYNCHRONISATION STATUS REPORT
RAN Paging Request	RAN PAGING REQUEST
Inventory Report	INVENTORY REPORT
A-IoT Session Release Request	AIOT SESSION RELEASE REQUEST

8.2 PDU Session Management Procedures

8.2.1 PDU Session Resource Setup

8.2.1.1 General

The purpose of the PDU Session Resource Setup procedure is to assign resources on Uu and NG-U for one or several PDU sessions and the corresponding QoS flows, and to setup corresponding DRBs for a given UE. The procedure uses UE-associated signalling.

8.2.1.2 Successful Operation

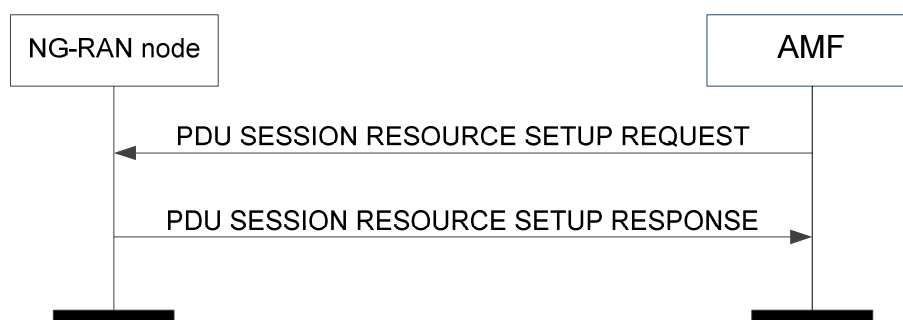


Figure 8.2.1.2-1: PDU session resource setup: successful operation

The AMF initiates the procedure by sending a PDU SESSION RESOURCE SETUP REQUEST message to the NG-RAN node.

The PDU SESSION RESOURCE SETUP REQUEST message shall contain the information required by the NG-RAN node to setup the PDU session related NG-RAN configuration consisting of at least one PDU session resource and include each PDU session resource to setup in the *PDU Session Resource Setup Request List* IE.

Upon reception of the PDU SESSION RESOURCE SETUP REQUEST message, if resources are available for the requested configuration, the NG-RAN node shall execute the requested NG-RAN configuration and allocate associated resources over NG and over Uu for each PDU session listed in the *PDU Session Resource Setup Request List* IE.

If the *RAN Paging Priority* IE is included in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node may use it to determine a priority for paging the UE in RRC_INACTIVE state.

For each requested PDU session, if resources are available for the requested configuration, the NG-RAN node shall establish at least one DRB and associate each accepted QoS flow of the PDU session which is not associated with an MBS QoS flow to a DRB established.

For each PDU session successfully established the NG-RAN node shall pass to the UE the *PDU Session NAS-PDU* IE, if included. The NG-RAN node shall not send to the UE the PDU Session NAS PDUs associated to the failed PDU sessions.

If the *NAS-PDU* IE is included in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall pass it to the UE.

For each PDU session the NG-RAN node shall store the *UL NG-U UP TNL Information* IE included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message and use it as the uplink termination point for the user plane data for this PDU session.

For each PDU session, if the *Additional UL NG-U UP TNL Information* IE is included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node may allocate for this split PDU session resources for an additional NG-U transport bearer for some or all of the QoS flows present in the *QoS Flow Setup Request List* IE and it shall indicate these QoS flows in the *Additional DL QoS Flow per TNL Information* IE in the *PDU Session Resource Setup Response Transfer* IE. In case the *Additional DL QoS Flow per TNL Information* IE is not included the SMF shall consider the proposed additional UL NG-U UP TNL information as available again.

For each PDU session, if the *Network Instance* IE is included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message and the *Common Network Instance* IE is not present, the NG-RAN node shall, if supported, use it when selecting transport network resource as specified in TS 23.501 [9].

For each PDU session, if the *Common Network Instance* IE is included in the *PDU Session Resource Setup Request Transfer* IE or in the *Additional UL NG-U UP TNL Information* IE, or in the *Additional Redundant UL NG-U UP TNL Information* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, use it when selecting transport network resource for the concerned NG-U transport bearer as specified in TS 23.501 [9].

For each PDU session, if the *Redundant UL NG-U UP TNL Information* IE is included in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, use it as the uplink termination point for the user plane data for this PDU session for the redundant transmission and it shall include the *Redundant QoS Flow per TNL Information* IE in the *PDU Session Resource Setup Response Transfer* IE as described in TS 23.501 [9].

For each PDU session, if the *Additional Redundant UL NG-U UP TNL Information* IE is included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node may allocate for this split PDU session resources for an additional redundant NG-U transport bearer for some or all of the QoS flows present in the *QoS Flow Setup Request List* IE and it shall indicate these QoS flows in the *Additional Redundant DL QoS Flow per TNL Information* IE in the *PDU Session Resource Setup Response Transfer* IE. In case the *Additional Redundant DL QoS Flow per TNL Information* IE is not included the SMF shall consider the proposed additional Redundant UL NG-U UP TNL information as available again.

For each PDU session, if the *Redundant Common Network Instance* IE is included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, use it when selecting transport network resource for the redundant transmission as specified in TS 23.501 [9].

For each PDU session, if the *TSC Traffic Characteristics* IE is included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, store it and use it as specified in TS 23.501 [9]. For each QoS flow which has been successfully established, if the *Burst Arrival Time* IE is included in the *TSC Assistance Information Uplink* IE of the *TSC Traffic Characteristics* IE, the NG-RAN node may include the *AN Packet Delay Budget Uplink* IE for the same QoS flow within the PDU SESSION RESOURCE SETUP RESPONSE message as specified in TS 23.501 [9]. If the *RAN Feedback Type* IE is included in the *TSC Traffic Characteristics* IE, the NG-RAN node shall, if supported, take this information into account when determining the feedback to provide in the *TSC Traffic Characteristics Feedback* IE in the *PDU Session Resource Setup Response Transfer* IE in the PDU SESSION RESOURCE SETUP RESPONSE message as specified in TS 23.501 [9].

For each PDU session, if the *PDU Session Type* IE included in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message is set to "ethernet", the NG-RAN node may perform appropriate header compression for the concerned PDU session, or if it is set to "unstructured", the NG-RAN node shall not perform header compression for the concerned PDU session.

For each PDU session for which the *Security Indication* IE is included in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, and the *Integrity Protection Indication* IE or *Confidentiality Protection Indication* IE is set to "required", then the NG-RAN node shall perform user plane integrity protection or ciphering, respectively, for the concerned PDU session. If the NG-RAN node cannot perform the user plane integrity protection or ciphering, it shall reject the setup of the PDU session resources with an appropriate cause value. If the NG-RAN node is an ng-eNB, it shall behave as specified in TS 33.501 [13].

For each PDU session for which the *Security Indication* IE is included in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, and the *Integrity Protection Indication* IE or *Confidentiality Protection Indication* IE is set to "preferred", then the NG-RAN node should, if supported, perform user plane integrity protection or ciphering, respectively, for the concerned PDU session and shall notify whether it performed the user plane integrity protection or ciphering by including the *Integrity Protection Result* IE or *Confidentiality Protection Result* IE, respectively, in the *PDU Session Resource Setup Response Transfer* IE of the PDU SESSION RESOURCE SETUP RESPONSE message.

For each PDU session for which the *Maximum Integrity Protected Data Rate Downlink* IE or the *Maximum Integrity Protected Data Rate Uplink* IE are included in the *Security Indication* IE in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall store the

respective information and, if integrity protection is to be performed for the PDU session, it shall enforce the traffic limits corresponding to the received values, for the concerned PDU session and concerned UE, as specified in TS 23.501 [9].

For each PDU session for which the *Security Indication* IE is included in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message:

- if the *Integrity Protection Indication* IE is set to "not needed", then the NG-RAN node shall not perform user plane integrity protection for the concerned PDU session;
- if the *Confidentiality Protection Indication* IE is set to "not needed", then the NG-RAN node shall not perform user plane ciphering for the concerned PDU session.

For each PDU session for which the *PDU Session Aggregate Maximum Bit Rate* IE is included in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall store the received value in the UE context and use it when enforcing traffic policing for Non-GBR QoS flows for the concerned UE as specified in TS 23.501 [9].

For each PDU session in the PDU SESSION RESOURCE SETUP REQUEST message, if the *Additional QoS Flow Information* IE is included in the *QoS Flow Level QoS Parameters* IE in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node may consider it for the DRB allocation process. It is up to NG-RAN node implementation to decide whether and how to use it.

For each PDU session in the PDU SESSION RESOURCE SETUP REQUEST message, if the *Alternative QoS Parameters Set List* IE is included in the *GBR QoS Flow Information* IE in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node may accept the setup of the QoS flow when notification control has been enabled if the requested QoS parameters or at least one of the alternative QoS parameters sets can be fulfilled at the time of setup. In case the NG-RAN node accepts the setup fulfilling one of the alternative QoS parameters it shall indicate the alternative QoS parameters set which it currently fulfils in the *Current QoS Parameters Set Index* IE within the *PDU Session Resource Setup Response Transfer* IE of the PDU SESSION RESOURCE SETUP RESPONSE message, as specified in TS 23.501 [9]. If the NG-RAN node receives a PDU SESSION RESOURCE SETUP REQUEST message containing the *Delay Critical* IE in the *Dynamic 5QI Descriptor* IE of the *QoS Flow Level QoS Parameters* IE of the *PDU Session Resource Setup Request Transfer* IE set to the value "delay critical" but the *Maximum Data Burst Volume* IE is not present in the *Alternative QoS Parameters Set List* IE, the NG-RAN node shall use the *Maximum Data Burst Volume* IE included in the *Dynamic 5QI Descriptor* IE.

For each QoS flow which has been successfully established, the NG-RAN node shall, if supported, store the *Redundant QoS Flow Indicator* IE if included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message and consider it for the redundant transmission as specified in TS 23.501 [9].

For each QoS flow which has been successfully established, if the *QoS Monitoring Request* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall store this information, and, if supported, perform delay measurement and QoS monitoring, as specified in TS 23.501 [9]. If the *QoS Monitoring Reporting Frequency* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall store this information and, if supported, use it for RAN part delay reporting.

For each GBR QoS flow which has been successfully established, if the *Monitoring Request on Available Bitrate* IE was included in the *GBR QoS Flow Information* IE in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, store this information and perform available bitrate monitoring, as specified in TS 23.501 [9].

For each QoS flow requested to be setup, if the *PDU Set QoS Parameters* IE is included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, store the received PDU Set QoS Parameters in the UE context and use it as specified in TS 23.501 [9]. If the *ECN Marking or Congestion Information Reporting Request* IE is included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, use it accordingly for the specific QoS flow. If the *ECN Marking or Congestion Information Reporting Status* IE is included in the *PDU Session Resource Setup Response Transfer* IE, the SMF shall, if supported, use it to deduce if ECN marking at NG-RAN or ECN marking at UPF or congestion information reporting is active or not active.

For each QoS flow requested to be setup the NG-RAN node shall take into account the received *QoS Flow Level QoS Parameters* IE. For each QoS flow the NG-RAN node shall establish or modify the resources according to the values of the *Allocation and Retention Priority* IE (priority level and pre-emption indicators) and the resource situation as follows:

- The NG-RAN node shall consider the priority level of the requested QoS flow, when deciding on the resource allocation.
- The priority levels and the pre-emption indicators may (individually or in combination) be used to determine whether the QoS flow setup has to be performed unconditionally and immediately. If the requested QoS flow is marked as "may trigger pre-emption" and the resource situation requires so, the NG-RAN node may trigger the pre-emption procedure which may then cause the forced release of a lower priority QoS flow which is marked as "pre-emptable". Whilst the process and the extent of the pre-emption procedure are operator-dependent, the pre-emption indicators shall be treated as follows:
 1. The values of the last received *Pre-emption Vulnerability* IE and *Priority Level* IE shall prevail.
 2. If the *Pre-emption Capability* IE is set to "may trigger pre-emption", then this allocation request may trigger the pre-emption procedure.
 3. If the *Pre-emption Capability* IE is set to "shall not trigger pre-emption", then this allocation request shall not trigger the pre-emption procedure.
 4. If the *Pre-emption Vulnerability* IE is set to "pre-emptable", then this QoS flow shall be included in the pre-emption process.
 5. If the *Pre-emption Vulnerability* IE is set to "not pre-emptable", then this QoS flow shall not be included in the pre-emption process.
- The NG-RAN node pre-emption process shall keep the following rules:
 1. The NG-RAN node shall only pre-empt QoS flows with lower priority, in ascending order of priority.
 2. The pre-emption may be done for QoS flows belonging to the same UE or to other UEs.

For each QoS flow which has been successfully established, the NG-RAN node shall store the mapped E-RAB ID if included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message and use it as specified in TS 38.300 [8].

For each PDU session, if the *Redundant PDU Session Information* IE is included in the *PDU Session Resource Setup Request Transfer* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, store the received information in the UE context and setup the redundant user plane for the redundant PDU session as specified in TS38.300 [8] and TS 23.501 [9]. If the *PDU Session Type* IE is set to "ethernet" and the redundancy requirement is fulfilled using a secondary NG-RAN node, the NG-RAN node shall, if supported, include the *Global RAN Node ID of Secondary NG-RAN Node* IE in the *PDU Session Resource Setup Response Transfer* IE of the PDU SESSION RESOURCE SETUP RESPONSE message. If the *PDU Session Pair ID* IE is included in the *Redundant PDU Session Information* IE, the NG-RAN node may use it to identify the paired PDU sessions.

The NG-RAN node shall report to the AMF in the PDU SESSION RESOURCE SETUP RESPONSE message the result for each PDU session resource requested to be setup:

- For each PDU session resource successfully setup, the *PDU Session Resource Setup Response Transfer* IE shall be included containing:
 1. The NG-U UP transport layer information to be used for the PDU session and associated list of QoS flows which have been successfully established, in the *QoS Flow per TNL Information* IE.
 2. The list of QoS flows which failed to be established, if any, in the *QoS Flow Failed to Setup List* IE. When the NG-RAN node reports unsuccessful establishment of a QoS flow, the cause value should be precise enough to enable the SMF to know the reason for the unsuccessful establishment.
- For each PDU session resource which failed to be setup, the *PDU Session Resource Setup Unsuccessful Transfer* IE shall be included containing a cause value that should be precise enough to enable the SMF to know the reason for the unsuccessful establishment.

Upon reception of the PDU SESSION RESOURCE SETUP RESPONSE message the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transfer transparently the *PDU Session Resource Setup Response Transfer* IE or *PDU Session Resource Setup Unsuccessful Transfer* IE to the SMF associated with the concerned PDU session.

Upon reception of the PDU SESSION RESOURCE SETUP REQUEST message to setup a QoS flow for IMS voice, if the NG-RAN node is not able to support IMS voice, the NG-RAN node shall initiate EPS fallback or RAT fallback for IMS voice procedure as specified in TS 23.501 [9] and report unsuccessful establishment of the QoS flow in the *PDU Session Resource Setup Response Transfer* IE or in the *PDU Session Resource Setup Unsuccessful Transfer* IE with cause value "IMS voice EPS fallback or RAT fallback triggered".

For each PDU session for which the *Global RAN Node ID of Secondary NG-RAN Node* IE is included in the *PDU Session Resource Setup Response Transfer* IE of the PDU SESSION RESOURCE SETUP RESPONSE message, the SMF shall, if supported, handle this information as specified in TS 23.501 [9].

The *UE Aggregate Maximum Bit Rate* IE should be sent to the NG-RAN node if the AMF has not sent it previously. If it is included in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall store the UE Aggregate Maximum Bit Rate in the UE context, and use the received UE Aggregate Maximum Bit Rate for all Non-GBR QoS flows for the concerned UE as specified in TS 23.501 [9].

For each PDU session, if the *PDU Session Expected UE Activity Behaviour* IE is included in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, handle this information as specified in TS 23.501 [9].

If the *UE Slice Maximum Bit Rate List* IE is included in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, store the UE Slice Maximum Bit Rate List in the UE context, and use it for each S-NSSAI for the concerned UE as specified in TS 23.501 [9].

For each PDU session, if the *Downlink TL Container* IE is included in the *PDU Session Resource Setup Request Transfer* IE of the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, handle it as specified in TS 29.585 [56] and shall include the *Uplink TL Container* IE in the *PDU Session Resource Setup Response Transfer* IE as specified in TS 29.585 [56].

If the *PDU Set QoS Parameters* IE or the *DL PDU Set Information Marking Support Indication* IE set to "true" is included in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, report in the PDU SESSION RESOURCE SETUP RESPONSE message the *PDU Set based Handling Indicator* IE in the *PDU Session Resource Setup Response Transfer* IE. If the *PDU Set based Handling Indicator* IE is included in the *PDU Session Resource Setup Response Transfer* IE in the PDU SESSION RESOURCE SETUP RESPONSE message, the SMF shall, if supported, handle this information as specified in TS 23.501 [9].

For each QoS flow requested to be setup, if the *MMSID* IE is included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, consider that the QoS flow is related to a multi-modal service, as described in TS 23.501 [9] and TS 38.300 [8].

For each QoS flow requested to be setup, if the *Indication of Bitrate Adaptation* IE is included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE SETUP REQUEST message, the NG-RAN node shall, if supported, store the received indication and use it for uplink rate control, as described in TS 23.501 [9] and TS 38.300 [8].

The NG-RAN node shall, if supported, report in the PDU SESSION RESOURCE SETUP RESPONSE message location information of the UE in the *User Location Information* IE.

If the *User Location Information* IE is included in the PDU SESSION RESOURCE SETUP RESPONSE message, the AMF shall handle this information as specified in TS 23.501 [9].

If the *MBS Support Indicator* IE is included in the *PDU Session Resource Setup Response Transfer* IE in the PDU SESSION RESOURCE SETUP RESPONSE message, the SMF shall, if supported, handle this information as specified in TS 23.247 [44].

Interactions with Handover Preparation procedure:

If a handover becomes necessary during the PDU Session Resource Setup procedure, the NG-RAN node may interrupt the ongoing PDU Session Resource Setup procedure and initiate the Handover Preparation procedure as follows:

1. The NG-RAN node shall send the PDU SESSION RESOURCE SETUP RESPONSE message in which the NG-RAN node shall indicate, if necessary, all the PDU session resources which failed to be setup with an appropriate

cause value, e.g. "NG intra-system handover triggered", "NG inter-system handover triggered" or "Xn handover triggered".

2. The NG-RAN node shall trigger the handover procedure.

8.2.1.3 Unsuccessful Operation

The unsuccessful operation is specified in the successful operation section.

8.2.1.4 Abnormal Conditions

If the NG-RAN node receives a PDU SESSION RESOURCE SETUP REQUEST message containing several *PDU Session ID* IEs (in the *PDU Session Resource Setup Request List* IE) set to the same value, the NG-RAN node shall report the establishment of the corresponding PDU sessions as failed in the PDU SESSION RESOURCE SETUP RESPONSE message with an appropriate cause value.

If the NG-RAN node receives a PDU SESSION RESOURCE SETUP REQUEST message containing a *PDU Session ID* IE (in the *PDU Session Resource Setup Request List* IE) set to a value that identifies an active PDU session (established before the PDU SESSION RESOURCE SETUP REQUEST message was received), the NG-RAN node shall report the establishment of the new PDU session as failed in the PDU SESSION RESOURCE SETUP RESPONSE message with an appropriate cause value.

If the NG-RAN node receives a PDU SESSION RESOURCE SETUP REQUEST message containing a *QoS Flow Setup Request List* IE in the *PDU Session Resource Setup Request Transfer* IE including at least one Non-GBR QoS flow but the *PDU Session Aggregate Maximum Bit Rate* IE is not present, the NG-RAN node shall report the establishment of the corresponding PDU session as failed in the PDU SESSION RESOURCE SETUP RESPONSE message with an appropriate cause value.

If the NG-RAN node receives a PDU SESSION RESOURCE SETUP REQUEST message containing a *QoS Flow Level QoS Parameters* IE in the *PDU Session Resource Setup Request Transfer* IE for a GBR QoS flow but the *GBR QoS Flow Information* IE is not present, the NG-RAN node shall report the establishment of the corresponding QoS flow as failed in the *PDU Session Resource Setup Response Transfer* IE of the PDU SESSION RESOURCE SETUP RESPONSE message with an appropriate cause value. If the NG-RAN node receives a PDU SESSION RESOURCE SETUP REQUEST message containing the *Delay Critical* IE in the *Dynamic 5QI Descriptor* IE of the *QoS Flow Level QoS Parameters* IE of the *PDU Session Resource Setup Request Transfer* IE set to the value "delay critical" but the *Maximum Data Burst Volume* IE is not present, the NG-RAN node shall report the establishment of the corresponding QoS flow as failed in the *PDU Session Resource Setup Response Transfer* IE of the PDU SESSION RESOURCE SETUP RESPONSE message with an appropriate cause value.

8.2.2 PDU Session Resource Release

8.2.2.1 General

The purpose of the PDU Session Resource Release procedure is to enable the release of already established PDU session resources for a given UE. The procedure uses UE-associated signalling.

8.2.2.2 Successful Operation

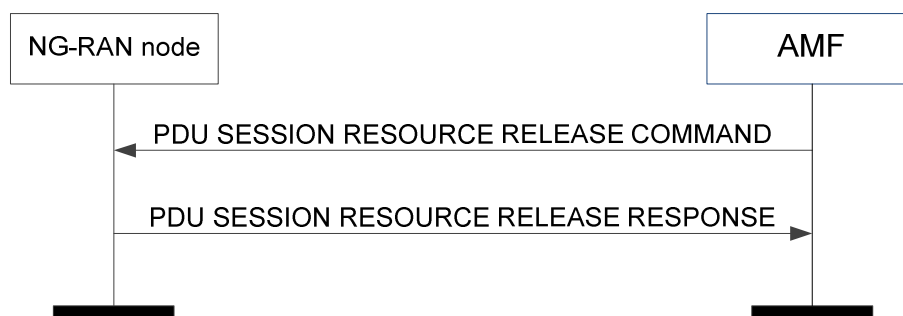


Figure 8.2.2.2-1: PDU session resource release: successful operation

The AMF initiates the procedure by sending a PDU SESSION RESOURCE RELEASE COMMAND message.

The PDU SESSION RESOURCE RELEASE COMMAND message shall contain the information required by the NG-RAN node to release at least one PDU session resource, and include each PDU session resource to release in the *PDU Session Resource to Release List* IE.

If a *NAS-PDU* IE is contained in the PDU SESSION RESOURCE RELEASE COMMAND message, the NG-RAN node shall pass it to the UE.

Upon reception of the PDU SESSION RESOURCE RELEASE COMMAND message the NG-RAN node shall execute the release of the requested PDU sessions. For each PDU session to be released the NG-RAN node shall release the corresponding resources over Uu and over NG, if any.

If the *RAN Paging Priority* IE is included in the PDU SESSION RESOURCE RELEASE COMMAND message, the NG-RAN node may use it to determine a priority for paging the UE in RRC_INACTIVE state.

The NG-RAN node shall, if supported, report in the PDU SESSION RESOURCE RELEASE RESPONSE message location information of the UE in the *User Location Information* IE.

After sending a PDU SESSION RESOURCE RELEASE RESPONSE message, the NG-RAN node shall be prepared to receive a PDU SESSION RESOURCE SETUP REQUEST message requesting establishment of a PDU session with a PDU Session ID corresponding to one of the PDU Session IDs that was present in the *PDU Session Resource to Release List* IE of the PDU SESSION RESOURCE RELEASE COMMAND message.

If the *User Location Information* IE is included in the PDU SESSION RESOURCE RELEASE RESPONSE message, the AMF shall handle this information as specified in TS 23.501 [9].

For each PDU session for which the *Secondary RAT Usage Information* IE is included in the *PDU Session Resource Release Response Transfer* IE, the SMF shall handle this information as specified in TS 23.502 [10].

For each PDU session, if the *Downlink TL Container* IE for a QoS flow is included in the *PDU Session Resource Release Command Transfer* IE of the PDU SESSION RESOURCE RELEASE COMMAND message, the NG-RAN node shall, if supported, handle it as specified in TS 29.585 [56] and shall include the *Uplink TL Container* IE for the indicated QoS flow in the *PDU Session Resource Release Response Transfer* IE as specified in TS 29.585 [56].

8.2.2.3 Unsuccessful Operation

The unsuccessful operation is specified in the successful operation section.

8.2.2.4 Abnormal Conditions

If the NG-RAN node receives a PDU SESSION RESOURCE RELEASE COMMAND message containing multiple *PDU Session ID* IEs (in the *PDU Session Resource to Release List* IE) set to the same value, the NG-RAN node shall initiate the release of one corresponding PDU session and ignore the duplication of the instances of the selected corresponding PDU sessions.

8.2.3 PDU Session Resource Modify

8.2.3.1 General

The purpose of the PDU Session Resource Modify procedure is to enable configuration modifications of already established PDU session(s) for a given UE. It is also to enable the setup, modification and release of the QoS flow for already established PDU session(s). The procedure uses UE-associated signalling.

8.2.3.2 Successful Operation

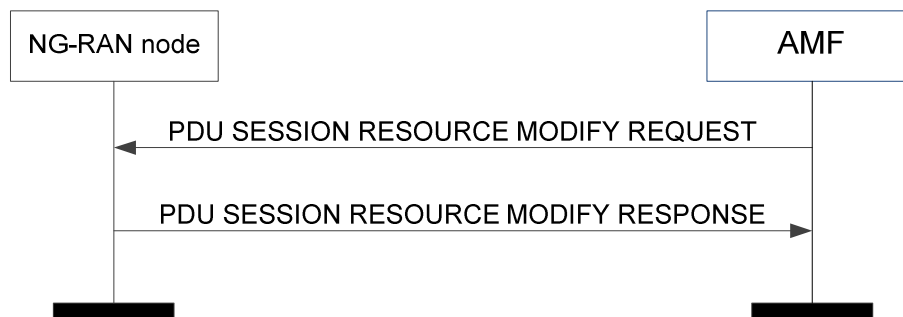


Figure 8.2.3.2-1: PDU session resource modify: successful operation

The AMF initiates the procedure by sending a PDU SESSION RESOURCE MODIFY REQUEST message to the NG-RAN node.

The PDU SESSION RESOURCE MODIFY REQUEST message shall contain the information required by the NG-RAN node, which may trigger the NG-RAN configuration modification for the existing PDU sessions listed in the *PDU Session Resource Modify Request List* IE.

Upon reception of the PDU SESSION RESOURCE MODIFY REQUEST message, if the NG-RAN configuration is triggered to be modified and if resources are available for the modified NG-RAN configuration, the NG-RAN node shall execute the configuration modification for the requested PDU session.

If the *RAN Paging Priority* IE is included in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node may use it to determine a priority for paging the UE in RRC_INACTIVE state.

For each PDU session, if the *S-NSSAI* IE is included in the *PDU Session Resource Modify Request Item* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall replace the previously provided S-NSSAI by the received S-NSSAI for the concerned PDU session and use it as specified in TS 23.502 [10].

For each PDU session, if the *Network Instance* IE is included in the *PDU Session Resource Modify Request Transfer* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message and the *Common Network Instance* IE is not present, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9].

For each PDU session, if the *Common Network Instance* IE is included in the *PDU Session Resource Modify Request Transfer* IE or in the *Additional UL NG-U UP TNL Information* IE, or in the *Additional Redundant UL NG-U UP TNL Information* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, use it when selecting transport network resource for the concerned NG-U transport bearer as specified in TS 23.501 [9].

For each PDU session, if the *Redundant Common Network Instance* IE is included in the *PDU Session Resource Modify Request Transfer* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, use it for the redundant transmission as specified in TS 23.501 [9].

For each PDU session, if the *TSC Traffic Characteristics* IE is included in the *PDU Session Resource Modify Request Transfer* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, store it and use it as specified in TS 23.501 [9]. If the *RAN Feedback Type* IE is included in the *TSC Traffic Characteristics* IE, the NG-RAN node shall, if supported, take this information into account when determining the feedback to provide in the *TSC Traffic Characteristics Feedback* IE in the *PDU Session Resource Modify Response Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message as specified in TS 23.501 [9].

For each PDU session, if the *Redundant QoS Flow Indicator* IE is included and set to “false” for all QoS flows, the NG-RAN node shall, if supported, stop the redundant transmission and release the redundant tunnel for the concerned PDU session as specified in TS 23.501 [9].

For each PDU session, if the *Downlink TL Container* IE for a QoS flow is included in the *PDU Session Resource Modify Request Transfer* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, handle it as specified in TS 29.585 [56] and shall include the *Uplink TL Container* IE for the indicated QoS flow in the *PDU Session Resource Modify Response Transfer* IE as specified in TS 29.585 [56].

For each PDU session in the PDU SESSION RESOURCE MODIFY REQUEST message, if the *Alternative QoS Parameters Set List* IE is included in the *GBR QoS Flow Information* IE in the *PDU Session Resource Modify Request Transfer* IE of the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node may accept the setup of the QoS flow when notification control has been enabled if the requested QoS parameters or at least one of the alternative QoS parameters sets can be fulfilled at the time of setup. In case the NG-RAN node accepts the setup fulfilling one of the alternative QoS parameters it shall indicate the alternative QoS parameters set which it currently fulfils in the *Current QoS Parameters Set Index* IE within the *PDU Session Resource Modify Response Transfer* IE of the PDU SESSION RESOURCE MODIFY RESPONSE message, as specified in TS 23.501 [9]. If the NG-RAN node receives a PDU SESSION RESOURCE MODIFY REQUEST message containing the *Delay Critical* IE in the *Dynamic 5QI Descriptor* IE of the *QoS Flow Level QoS Parameters* IE of the *PDU Session Resource Modify Request Transfer* IE set to the value "delay critical" but the *Maximum Data Burst Volume* IE is not present in the *Alternative QoS Parameters Set List* IE, the NG-RAN node shall use the *Maximum Data Burst Volume* IE included in the *Dynamic 5QI Descriptor* IE.

For each PDU session included in the *PDU Session Resource Modify Request List* IE:

- For each QoS flow included in the *QoS Flow Add or Modify Request List* IE, based on the *QoS Flow Level QoS Parameters* IE, the NG-RAN node may establish, modify or release the DRB configuration and may change allocation of resources on NG or Uu accordingly. The NG-RAN node shall associate each QoS flow accepted to setup or modify which is not associated with an MBS QoS flow with a DRB of the PDU session. The associated DRB for the QoS flow accepted to modify may not change. If the *PDU Set QoS Parameters* IE is received, the NG-RAN node shall, if supported, store the received PDU Set QoS Parameters in the UE context and use it as specified in TS 23.501 [9]. If the *ECN Marking or Congestion Information Reporting Request* IE is included in the *PDU Session Resource Modify Request Transfer* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, use it accordingly for the specific QoS flow. If the *ECN Marking or Congestion Information Reporting Status* IE is included in the *PDU Session Resource Modify Response Transfer* IE, the SMF shall, if supported, use it to deduce if ECN marking at NG-RAN or ECN marking at UPF or congestion information reporting is active or not active.
- For each QoS flow, if the *Redundant QoS Flow Indicator* IE is included, the NG-RAN node shall, if supported, store it and consider it for the redundant transmission as specified in TS 23.501 [9].
- For each QoS flow included in the *QoS Flow Add or Modify Request List* IE, if the *QoS Flow Add or Modify Request Item* IE is included for an existing *QoS Flow Identifier* IE, the NG-RAN node shall overwrite the content of the full *QoS Flow Add or Modify Request Item* IE.
- For each QoS flow included in the *QoS Flow to Release List* IE, the NG-RAN node shall de-associate the QoS flow with the previously associated DRB.
- For each QoS flow included in the *QoS Flow Add or Modify Request List* IE, if the *UL NG-U UP TNL Information* IE is included, the NG-RAN node shall, if supported, include the *DL NG-U UP TNL Information* IE in the *QoS Flow Add or Modify Response Item* IE for the QoS flow within the PDU SESSION RESOURCE MODIFY RESPONSE message as specified in TS 23.501 [9].
- For each QoS flow, if the *Burst Arrival Time* IE is included in the *TSC Assistance Information Uplink* IE of the *TSC Traffic Characteristics* IE, the NG-RAN node may include the *AN Packet Delay Budget Uplink* IE for the same QoS flow within the PDU SESSION RESOURCE MODIFY RESPONSE message as specified in TS 23.501 [9].
- If the *NAS-PDU* IE is received for the PDU session, the NG-RAN node shall pass it to the UE only if at least one of the requests included in the *PDU Session Resource Modify Request Transfer* IE is successful (i.e. the PDU session is included in the *PDU Session Resource Modify Response Item* IE of the PDU SESSION RESOURCE MODIFY RESPONSE message).
- The NG-RAN node may change allocation of resources on NG according to the requested target configuration.
- If the *PDU Session Aggregate Maximum Bit Rate* IE is included in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node shall store and use the received PDU Session Aggregate Maximum Bit Rate value when enforcing traffic policing for Non-GBR QoS flows for the concerned UE as specified in TS 23.501 [9].
- If the *UL NG-U UP TNL Information* IE in the *UL NG-U UP TNL Modify List* IE is included in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node shall update the transport layer information for the

uplink data accordingly for the concerned transport bearers identified by the *DL NG-U UP TNL Information* IE included in the *PDU Session Resource Modify Request Transfer* IE for the concerned PDU session.

- If the *Additional UL NG-U UP TNL Information* IE is included in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node may allocate resources for an additional NG-U transport bearer for some or all of the QoS flows present in the *QoS Flow Add or Modify Request List* IE and it shall indicate these QoS flows in the *Additional DL QoS Flow per TNL Information* IE in the *PDU Session Resource Modify Response Transfer* IE. In case the *Additional DL QoS Flow per TNL Information* IE is not included the SMF shall consider the proposed additional UL NG-U UP TNL information as available again.
- In case more than one NG-U transport bearers have been set up for the PDU session, if all the QoS flows associated to one existing NG-U transport bearer are included in the *QoS Flow to Release List* IE in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node and 5GC shall consider that the concerned NG-U transport bearer is removed for the PDU session, and both NG-RAN node and 5GC shall therefore consider the related NG-U UP TNL information as available again.
- If the *Redundant UL NG-U UP TNL Information* IE within the *UL NG-U UP TNL Modify List* IE is included in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node shall, if supported, update the transport layer information for the uplink data accordingly for the concerned transport bearer identified by the *Redundant DL NG-U UP TNL Information* IE included in the *PDU Session Resource Modify Request Transfer* IE for the concerned PDU session.
- If the *Additional Redundant UL NG-U UP TNL Information* IE is included in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node may allocate resources for an additional redundant NG-U transport bearer for some or all of the QoS flows present in the *QoS Flow Add or Modify Request List* IE and it shall, if supported, indicate these QoS flows in the *Additional Redundant DL QoS Flow per TNL Information* IE in the *PDU Session Resource Modify Response Transfer* IE. In case the *Additional Redundant DL QoS Flow per TNL Information* IE is not included the SMF shall consider the proposed additional Redundant UL NG-U UP TNL information as available again.
- If the *Redundant UL NG-U UP TNL Information* IE is included in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node may allocate resources for a redundant NG-U transport bearer for some or all of the QoS flows present in the *QoS Flow Add or Modify Request List* IE and it shall, if supported, indicate the corresponding NG-RAN endpoint of this NG-U transport bearer in the *Redundant DL NG-U UP TNL Information* IE in the *PDU Session Resource Modify Response Transfer* IE.
- If the *Security Indication* IE is included in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node shall, if supported, only update the maximum integrity protected data rate uplink and/or the maximum integrity protected data rate downlink, and take them into account as defined in the PDU Session Resource Setup procedure.
- If the *User Plane Failure Indication* IE is included in the *PDU Session Resource Modify Request Transfer* IE, the NG-RAN node shall, if supported, consider that the NG-U tunnel established for the PDU session experienced a user plane failure, and perform one of the below actions:
 - trigger the PDU Session Resource Notify procedure to release the tunnel and the associated QoS flows.
 - trigger the PDU Session Resource Modify Indication procedure to move the associated QoS flows to the other existing tunnel.
 - decide to establish a new NG-U tunnel to replace the failed one, in which case it shall include, as applicable, either the new *DL NG-U UP TNL Information* IE or the *Additional DL NG-U UP TNL Information* IE, and set the *User Plane Failure Indication Report* IE in the *PDU Session Resource Modify Response Transfer* IE to "new transport address allocated" and use the *UL NG-U UP TNL Information* IE in the *User Plane Failure Indication* IE for the uplink data.

For each QoS flow which has been successfully added or modified, if the *QoS Monitoring Request* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall store this information, and, if supported, perform delay measurement and QoS monitoring, as specified in TS 23.501 [9]. If the *QoS Monitoring Reporting Frequency* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall store this information and, if supported, use it for RAN part delay reporting.

For each GBR QoS flow which has been successfully added or modified, if the *Monitoring Request on Available Bitrate* IE was included in the *GBR QoS Flow Information* IE in the *PDU Session Resource Modify Request Transfer* IE of the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, store this information and perform available bitrate monitoring, as specified in TS 23.501 [9].

The NG-RAN node shall report to the AMF, in the PDU SESSION RESOURCE MODIFY RESPONSE message, the result for each PDU session requested to be modified listed in the PDU SESSION RESOURCE MODIFY REQUEST message:

- For each PDU session which is successfully modified, the *PDU Session Resource Modify Response Transfer* IE shall be included containing:
 1. The list of QoS flows which have been successfully setup or modified, if any, in the *QoS Flow Add or Modify Response List* IE in case the PDU Session Resource Modify procedure is triggered by QoS flow setup or modification.
 2. The list of QoS flows which have failed to be setup or modified, if any, in the *QoS Flow Failed to Add or Modify List* IE in case the PDU Session Resource Modify procedure is triggered by QoS flow setup or modification.
- For each PDU session which failed to be modified, the *PDU Session Resource Modify Unsuccessful Transfer* IE shall be included containing the failure cause.
- For each PDU session, if the *DL NG-U UP TNL Information* IE is included in the *PDU Session Resource Modify Response Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message, it shall be considered by the SMF as the new DL transport layer address for the PDU session. The NG-RAN also may indicate the mapping between each new DL transport layer address and the corresponding UL transport layer address assigned by the 5GC.
- For each PDU session, if the *Additional NG-U UP TNL Information* IE is included in the *PDU Session Resource Modify Response Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message, it shall, if supported, be considered by the SMF as the new DL transport layer address(es) for the PDU session. The NG-RAN also may indicate the mapping between each new DL transport layer address and the corresponding UL transport layer address assigned by the 5GC.
- For each PDU session, if the *Additional Redundant NG-U UP TNL Information* IE is included in the *PDU Session Resource Modify Response Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message, it shall, if supported, be considered by the SMF as the new DL transport layer address(es) for the PDU session for the redundant transmission. The NG-RAN also may indicate the mapping between each new redundant DL transport layer address and the corresponding redundant UL transport layer address assigned by the 5GC.
- For each PDU session, if the *User Plan Failure Indication Report* IE is included in the *PDU Session Resource Modify Response Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message, it shall, if supported, take it into account as specified in TS 23.527 [57].

Upon reception of the PDU SESSION RESOURCE MODIFY RESPONSE message the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transfer transparently the *PDU Session Resource Modify Response Transfer* IE or *PDU Session Resource Modify Unsuccessful Transfer* IE to each SMF associated with the concerned PDU session.

The NG-RAN node shall, if supported, report in the PDU SESSION RESOURCE MODIFY RESPONSE message location information of the UE in the *User Location Information* IE.

For a PDU session or a QoS flow which failed to be modified, the NG-RAN node shall fall back to the configuration of the PDU session or the QoS flow as it was configured prior to the reception of the PDU SESSION RESOURCE MODIFY REQUEST message.

Upon reception of the PDU SESSION RESOURCE MODIFY REQUEST message to setup a QoS flow for IMS voice, if the NG-RAN node is not able to support IMS voice, the NG-RAN node shall initiate EPS fallback or RAT fallback for IMS voice procedure as specified in TS 23.501 [9] and report unsuccessful establishment of the QoS flow in the *PDU Session Resource Modify Response Transfer* IE or in the *PDU Session Resource Modify Unsuccessful Transfer* IE with cause value "IMS voice EPS fallback or RAT fallback triggered".

For each PDU session, if the *PDU Session Expected UE Activity Behaviour* IE is included in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, handle this information as specified in TS 23.501 [9].

If the *PDU Set QoS Parameters* IE or the *DL PDU Set Information Marking Support Indication* IE set to "true" is included in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, report in the PDU SESSION RESOURCE MODIFY RESPONSE message the *PDU Set based Handling Indicator* IE in the *PDU Session Resource Modify Response Transfer* IE. If the *PDU Set based Handling Indicator* IE is included in the *PDU Session Resource Modify Response Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message, the SMF shall, if supported, handle this information as specified in TS 23.501 [9].

For each QoS flow to be added or modified, if the *MMSID* IE is included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, consider that the QoS flow is related to a multi-modal service, as described in TS 23.501 [9] and TS 38.300 [8].

For each QoS flow to be added or modified, if the *Indication of Bitrate Adaptation* IE is included in the *QoS Flow Level QoS Parameters* IE contained in the PDU SESSION RESOURCE MODIFY REQUEST message, the NG-RAN node shall, if supported, store the received indication and use it for uplink rate control, as described in TS 23.501 [9] and TS 38.300 [8].

If the *User Location Information* IE is included in the PDU SESSION RESOURCE MODIFY RESPONSE message, the AMF shall handle this information as specified in TS 23.501 [9].

For each PDU session for which the *Secondary RAT Usage Information* IE is included in the *PDU Session Resource Modify Response Transfer* IE, the SMF shall handle this information as specified in TS 23.502 [10].

If the *MBS Support Indicator* IE is included in the *PDU Session Resource Modify Response Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message, the SMF shall, if supported, handle this information as specified in TS 23.247 [44].

Interactions with Handover Preparation procedure:

If a handover becomes necessary during the PDU Session Resource Modify procedure, the NG-RAN node may interrupt the ongoing PDU Session Resource Modify procedure and initiate the Handover Preparation procedure as follows:

1. The NG-RAN node shall send the PDU SESSION RESOURCE MODIFY RESPONSE message in which the NG-RAN node shall indicate, if necessary, all the PDU sessions failed with an appropriate cause value, e.g. "NG intra-system handover triggered", "NG inter-system handover triggered" or "Xn handover triggered".
2. The NG-RAN node shall trigger the handover procedure.

Interactions with PDU Session Resource Notify procedure:

If the *User Plane Failure Indication* IE is included in the *PDU Session Resource Modify Request Transfer* IE and the NG-RAN node triggers the PDU Session Resource Notify procedure, the NG-RAN node shall set the *User Plane Failure Indication Report* IE in the *PDU Session Resource Modify Response Transfer* IE to "QoS flows and tunnel to be released".

Interactions with PDU Session Resource Modify Indication procedure:

If the *User Plane Failure Indication* IE is included in the *PDU Session Resource Modify Request Transfer* IE and the NG-RAN node triggers the PDU Session Resource Modify Indication procedure, the NG-RAN node shall set the *User Plane Failure Indication Report* IE in the *PDU Session Resource Modify Response Transfer* IE to "QoS flows to be moved".

8.2.3.3 Unsuccessful Operation

The unsuccessful operation is specified in the successful operation section.

8.2.3.4 Abnormal Conditions

If the NG-RAN node receives a PDU SESSION RESOURCE MODIFY REQUEST message containing several *PDU Session ID* IEs (in the *PDU Session Resource Modify Request List* IE) set to the same value, the NG-RAN node shall

report the modification of the corresponding PDU sessions as failed in the PDU SESSION RESOURCE MODIFY RESPONSE message with an appropriate cause value.

If the NG-RAN node receives a PDU SESSION RESOURCE MODIFY REQUEST message containing some *PDU Session ID* IEs (in the *PDU Session Resource Modify Request List* IE) that the NG-RAN node does not recognize, the NG-RAN node shall report the corresponding invalid PDU sessions as failed in the PDU SESSION RESOURCE MODIFY RESPONSE message with an appropriate cause value.

If the NG-RAN node receives a PDU SESSION RESOURCE MODIFY REQUEST message containing a *QoS Flow Level QoS Parameters* IE in the *PDU Session Resource Modify Request Transfer* IE for a GBR QoS flow but the *GBR QoS Flow Information* IE is not present, the NG-RAN node shall report the addition or modification of the corresponding QoS flow as failed in the *PDU Session Resource Modify Response Transfer* IE of the PDU SESSION RESOURCE MODIFY RESPONSE message with an appropriate cause value.

If the NG-RAN node receives a PDU SESSION RESOURCE MODIFY REQUEST message containing the *Delay Critical* IE in the *Dynamic 5QI Descriptor* IE of the *QoS Flow Level QoS Parameters* IE of the *PDU Session Resource Modify Request Transfer* IE set to the value “delay critical” but the *Maximum Data Burst Volume* IE is not present, the NG-RAN node shall report the addition or modification of the corresponding QoS flow as failed in the *PDU Session Resource Modify Response Transfer* IE of the PDU SESSION RESOURCE MODIFY RESPONSE message with an appropriate cause value.

If the NG-RAN node receives a PDU SESSION RESOURCE MODIFY REQUEST message containing a PDU session in the *PDU Session Resource Modify Request List* IE with the same QoS flow included both in the *QoS Flow Add or Modify Request List* IE and the *QoS Flow to Release List* IE, the NG-RAN node shall report the corresponding QoS flow as failed in the *QoS Flow Failed to Add or Modify List* IE in the *PDU Session Resource Modify Response Transfer* IE of the PDU SESSION RESOURCE MODIFY RESPONSE message with an appropriate cause value if the PDU session is modified successfully. The NG-RAN node shall not release the QoS flow when the corresponding QoS flow already exists.

8.2.4 PDU Session Resource Notify

8.2.4.1 General

The purpose of the PDU Session Resource Notify procedure is to notify that the already established QoS flow(s) or PDU session(s) for a given UE are released or not fulfilled anymore or fulfilled again by the NG-RAN node for which notification control is requested. It is also used to notify that the updated QoS parameters during the Path Switch Request procedure are not successfully accepted by the NG-RAN node. The procedure uses UE-associated signalling.

8.2.4.2 Successful Operation

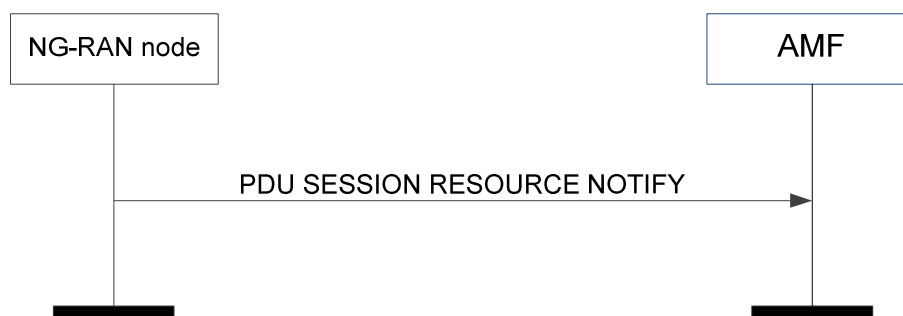


Figure 8.2.4.2-1: PDU session resource notify

The NG-RAN node initiates the procedure by sending a PDU SESSION RESOURCE NOTIFY message.

The PDU SESSION RESOURCE NOTIFY message shall contain the information of PDU session resources or QoS flows which are released or not fulfilled anymore or fulfilled again by the NG-RAN node.

- For each PDU session for which some QoS flows are released or not fulfilled anymore or fulfilled again by the NG-RAN node, the *PDU Session Resource Notify Transfer* IE shall be included containing:

1. The list of QoS flows which are released by the NG-RAN node, if any, in the *QoS Flow Released List* IE. If no more QoS flow is associated to an existing bearer after the release (e.g. split PDU session), the NG-RAN node and 5GC shall consider that the concerned NG-U transport bearer is removed and the related NG-U UP TNL information is available again.
 2. The list of GBR QoS flows which are not fulfilled anymore or fulfilled again by the NG-RAN node, if any, in the *QoS Flow Notify List* IE together with the *Notification Cause* IE. For a QoS flow indicated as not fulfilled anymore the NG-RAN node may also indicate an alternative QoS parameters set which it can currently fulfil in the *Current QoS Parameters Set Index* IE, as specified in TS 23.501 [9]. For a QoS flow indicated as not fulfilled anymore the NG-RAN node may also indicate the RAN feedback in the *TSC Traffic Characteristics Feedback* IE.
 3. The list of QoS flows for which the QoS parameters were updated but could not be successfully accepted by the NG-RAN node during the Path Switch Request procedure, if any, in the *QoS Flow Feedback List* IE which may be associated with a value it could offer.
- For each PDU session resource which is released by the NG-RAN node, the *PDU Session Resource Notify Released Transfer* IE shall be included containing the release cause in the *Cause* IE. If the *User Plane Error Indicator* IE is set to "GTP-U Error Indication Received", the SMF shall, if supported, consider that the PDU session is released due to GTP-U Error Indication received over the NG-U tunnel, as specified in 3GPP TS 23.527 [57].

The NG-RAN node shall, if supported, report in the PDU SESSION RESOURCE NOTIFY message location information of the UE in the *User Location Information* IE.

Upon reception of the PDU SESSION RESOURCE NOTIFY message, the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transfer transparently the *PDU Session Resource Notify Transfer* IE or *PDU Session Resource Notify Released Transfer* IE to the SMF associated with the concerned PDU session. Upon reception of *PDU Session Resource Notify Transfer* IE, the SMF normally initiate the appropriate release or modify procedure on the core network side for the PDU session(s) or QoS flow(s) identified as not fulfilled anymore.

For each PDU session for which the *Secondary RAT Usage Information* IE is included in the *PDU Session Resource Notify Transfer* IE or the *PDU Session Resource Notify Released Transfer* IE, the SMF shall handle this information as specified in TS 23.502 [10].

If the *User Location Information* IE is included in the PDU SESSION RESOURCE NOTIFY message, the AMF shall handle this information as specified in TS 23.501 [9].

8.2.4.3 Abnormal Conditions

Void.

8.2.5 PDU Session Resource Modify Indication

8.2.5.1 General

The purpose of the PDU Session Resource Modify Indication procedure is for the NG-RAN node to request modification of the established PDU session(s). The procedure uses UE-associated signalling.

8.2.5.2 Successful Operation

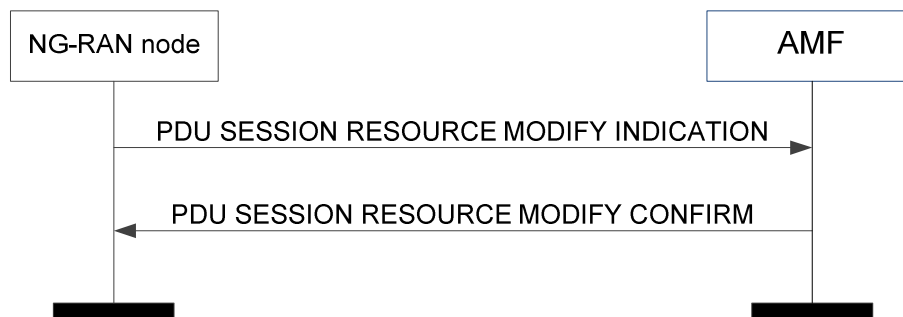


Figure 8.2.5.2-1: PDU session resource modify indication: successful operation

The NG-RAN node initiates the procedure by sending a PDU SESSION RESOURCE MODIFY INDICATION message. Upon reception of the PDU SESSION RESOURCE MODIFY INDICATION message, the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transparently transfer the *PDU Session Resource Modify Indication Transfer* IE to the SMF associated with the concerned PDU session.

For each PDU session for which the *DL QoS Flow per TNL Information* IE is included in the *PDU Session Resource Modify Indication Transfer* IE in the PDU SESSION RESOURCE MODIFY INDICATION message, the SMF shall, if the request is accepted, consider the included DL transport layer address as the DL transport layer address for the included associated QoS flows and provide the associated UL transport layer address in the *UL NG-U UP TNL Information* IE in the *PDU Session Resource Modify Confirm Transfer* IE in the PDU SESSION RESOURCE MODIFY CONFIRM message.

For each PDU session for which the *Additional DL QoS Flow per TNL Information* IE is included in the *PDU Session Resource Modify Indication Transfer* IE in the PDU SESSION RESOURCE MODIFY INDICATION message, the SMF shall, if supported, consider for this split PDU session each included DL transport layer address(es) as the DL transport layer address(s) for the included associated QoS flows and it may provide the associated UL transport layer address(s) in the *Additional NG-U UP TNL Information* IE in the *PDU Session Resource Modify Confirm Transfer* IE in the PDU SESSION RESOURCE MODIFY CONFIRM message.

In case more than one NG-U transport bearers have been set up for the PDU session, the *DL QoS Flow per TNL Information* IE and the *Additional DL QoS Flow per TNL Information* IE in the *PDU Session Resource Modify Indication Transfer* IE in the PDU SESSION RESOURCE MODIFY INDICATION message shall be included if at least one QoS flow is associated to their respective NG-U transport bearer; if no QoS flow is associated to one existing NG-U transport bearer after the modification, the NG-RAN node and 5GC shall consider that the concerned NG-U transport bearer is removed for the PDU session, and both NG-RAN node and 5GC shall therefore consider the related NG-U UP TNL information as available again.

For each PDU session for which the *Redundant DL QoS Flow per TNL Information* IE is included in the *PDU Session Resource Modify Indication Transfer* IE in the PDU SESSION RESOURCE MODIFY INDICATION message, the SMF shall, if supported, consider the included DL transport layer address as the new DL transport layer address for the included associated QoS flows for redundant transmission and it may provide the associated UL transport layer address in the *Redundant UL NG-U UP TNL Information* IE in the *PDU Session Resource Modify Confirm Transfer* IE in the PDU SESSION RESOURCE MODIFY CONFIRM message.

For each PDU session for which the *Additional Redundant DL QoS Flow per TNL Information* IE is included in the *PDU Session Resource Modify Indication Transfer* IE in the PDU SESSION RESOURCE MODIFY INDICATION message, the SMF shall, if supported, consider for this split PDU session each included DL transport layer address(es) as the new downlink termination point(s) for the included associated QoS flows and it may provide the associated UL transport layer address(s) in the *Additional Redundant NG-U UP TNL Information* IE in the *PDU Session Resource Modify Confirm Transfer* IE in the PDU SESSION RESOURCE MODIFY CONFIRM message for the redundant transmission.

For each PDU session for which the *Global RAN Node ID of Secondary NG-RAN Node* IE is included in the *PDU Session Resource Modify Indication Transfer* IE of the PDU SESSION RESOURCE MODIFY INDICATION message, the SMF shall, if supported, handle this information as specified in TS 23.501 [9].

If the *Security Result* IE is included in the *PDU Session Resource Modify Indication Transfer* IE in the PDU SESSION RESOURCE MODIFY INDICATION message, it shall be considered by the SMF as the new security status of the PDU session.

If the *ECN Marking or Congestion Information Reporting Status* IE is included in the *PDU Session Resource Modify Indication Transfer* IE in the PDU SESSION RESOURCE MODIFY INDICATION message, the SMF shall, if supported, use it to deduce if ECN marking at NG-RAN or ECN marking at UPF or congestion information reporting is active or not active as described in TS 23.501 [9].

For each PDU session for which the *Secondary RAT Usage Information* IE is included in the *PDU Session Resource Modify Indication Transfer* IE, the SMF shall handle this information as specified in TS 23.502 [10].

The AMF shall report to the NG-RAN node in the PDU SESSION MODIFY RESOURCE CONFIRM message the result for each PDU session listed in PDU SESSION RESOURCE MODIFY INDICATION message:

- For each PDU session which is successfully modified, the *PDU Session Resource Modify Confirm Transfer* IE shall be included containing:
 1. The list of QoS flows which have been successfully modified in the *QoS Flow Modify Confirm List* IE.
 2. The list of QoS flows which have failed to be modified, if any, in the *QoS Flow Failed to Modify List* IE.
- For each PDU session which failed to be modified, the *PDU Session Resource Modify Indication Unsuccessful Transfer* IE shall be included to report the failure cause.

Upon reception of the *PDU Session Resource Modify Confirm Transfer* IE for each PDU session listed in the PDU SESSION RESOURCE MODIFY CONFIRM message:

- If the *QoS Flow Failed To Modify List* IE is included, the NG-RAN node shall either
 1. de-associate the corresponding DRB for the concerned QoS flow, or
 2. keep the previous transport layer information before sending the PDU SESSION RESOURCE MODIFY INDICATION unchanged for the concerned QoS flow.

Upon reception of the *PDU Session Resource Modify Indication Unsuccessful Transfer* IE for each PDU session listed in the PDU SESSION RESOURCE MODIFY CONFIRM message, the NG-RAN node shall either:

1. release all corresponding NG-RAN configuration and resources for the concerned PDU session, or
2. keep the previous transport layer information before sending the PDU SESSION RESOURCE MODIFY INDICATION unchanged for the concerned PDU session.

The NG-RAN node shall, if supported, report in the PDU SESSION RESOURCE MODIFY INDICATION message location information of the UE in the *User Location Information* IE.

Interactions with PDU Session Resource Setup procedure or PDU Session Resource Modification procedure:

If the PDU SESSION RESOURCE MODIFY INDICATION message is received by the AMF during an ongoing PDU Session Resource Setup procedure or an ongoing PDU Session Resource Modification procedure, the AMF shall proceed with the PDU Session Resource Modify Indication procedure.

If the *PDU Session Resource Modify Indication Transfer* IE is received by the SMF during an ongoing PDU Session Resource Setup procedure or an ongoing PDU Session Resource Modification procedure, the SMF shall process the *PDU Session Resource Modify Indication Transfer* IE.

8.2.5.3 Unsuccessful Operation

The unsuccessful operation is specified in the successful operation section.

8.2.5.4 Abnormal Conditions

Void.

8.3 UE Context Management Procedures

8.3.1 Initial Context Setup

8.3.1.1 General

The purpose of the Initial Context Setup procedure is to establish the necessary overall initial UE context at the NG-RAN node, when required, including PDU session context, the Security Key, Mobility Restriction List, UE Radio Capability and UE Security Capabilities, etc. The AMF may initiate the Initial Context Setup procedure if a UE-associated logical NG-connection exists for the UE or if the AMF has received the *RAN UE NGAP ID* IE in an INITIAL UE MESSAGE message or if the NG-RAN node has already initiated a UE-associated logical NG-connection by sending an INITIAL UE MESSAGE message via another NG interface instance. The procedure uses UE-associated signalling.

For signalling only connections and if the *UE Context Request* IE is not received in the Initial UE Message, the AMF may be configured to trigger the procedure for all NAS procedures or on a per NAS procedure basis depending on operator's configuration.

8.3.1.2 Successful Operation

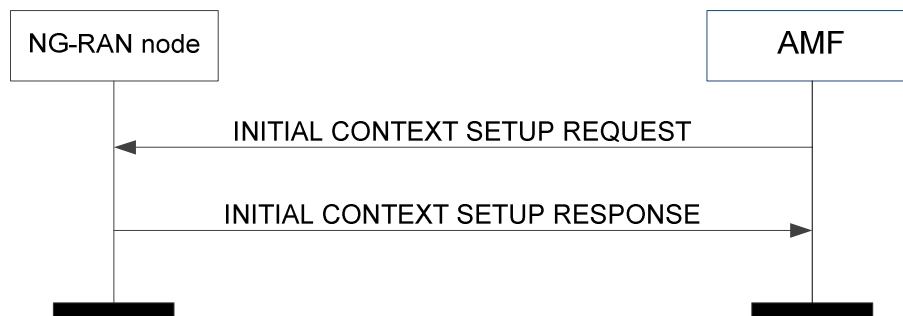


Figure 8.3.1.2-1: Initial context setup: successful operation

In case of the establishment of a PDU session the 5GC shall be prepared to receive user data before the INITIAL CONTEXT SETUP RESPONSE message has been received by the AMF. If no UE-associated logical NG-connection exists, the UE-associated logical NG-connection shall be established at reception of the INITIAL CONTEXT SETUP REQUEST message.

The INITIAL CONTEXT SETUP REQUEST message shall contain the *Index to RAT/Frequency Selection Priority* IE, if available in the AMF.

If the *NAS-PDU* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall pass it transparently towards the UE.

If the *Masked IMEISV* IE is contained in the INITIAL CONTEXT SETUP REQUEST message the target NG-RAN node shall, if supported, use it to determine the characteristics of the UE for subsequent handling.

Upon receipt of the INITIAL CONTEXT SETUP REQUEST message the NG-RAN node shall

- attempt to execute the requested PDU session configuration;
- store the received UE Aggregate Maximum Bit Rate in the UE context, and use the received UE Aggregate Maximum Bit Rate for Non-GBR QoS flows for the concerned UE as specified in TS 23.501 [9];
- store the received Mobility Restriction List in the UE context;
- store the received UE Radio Capability in the UE context;
- store the received Index to RAT/Frequency Selection Priority in the UE context and use it as defined in TS 23.501 [9];
- store the received UE Security Capabilities in the UE context;

- store the received Security Key in the UE context and, if the NG-RAN node is required to activate security for the UE, take this security key into use;
- if supported, store the received SRVCC Operation Possible in the UE context and use it as defined in TS 23.216 [31];
- store the received NR V2X Services Authorization information, if supported, in the UE context;
- store the received LTE V2X Services Authorization information, if supported, in the UE context;
- store the received NR A2X Services Authorization information, if supported, in the UE context;
- store the received LTE A2X Services Authorization information, if supported, in the UE context;
- store the received NR UE Sidelink Aggregate Maximum Bit Rate, if supported, in the UE context, and use it for the concerned UE's sidelink communication in network scheduled mode for NR V2X services;
- store the received LTE UE Sidelink Aggregate Maximum Bit Rate, if supported, in the UE context, and use it for the concerned UE's sidelink communication in network scheduled mode for LTE V2X services;
- store the received NR A2X UE PC5 Aggregate Maximum Bit Rate, if supported, in the UE context, and use it for the concerned UE's sidelink communication in network scheduled mode for NR A2X services;
- store the received LTE A2X UE PC5 Aggregate Maximum Bit Rate, if supported, in the UE context, and use it for the concerned UE's sidelink communication in network scheduled mode for LTE A2X services;
- store the received PC5 QoS Parameters, if supported, in the UE context and use it as defined in TS 23.287 [33];
- store the received A2X PC5 QoS Parameters, if supported, in the UE context and use it as defined in TS 23.256 [54].
- store the received Management Based MDT PLMN List information, if supported, in the UE context;
- if supported, store the received IAB Authorization information in the UE context, and use it accordingly for the IAB-MT as specified in TS 38.401 [2];
- store the received 5G ProSe Authorization information in the UE context, if supported, and use it for the concerned UE's sidelink communication in network scheduled mode for 5G ProSe services;
- store the received 5G ProSe UE PC5 Aggregate Maximum Bit Rate in the UE context, if supported, and use it for the concerned UE's sidelink communication in network scheduled mode for 5G ProSe services;
- store the received 5G ProSe PC5 QoS Parameters, if supported, in the UE context and use it as defined in TS 23.304 [47];
- store the received Ranging and Sidelink Positioning service information, if supported, in the UE context;
- store the received Network Controlled Repeater Authorization, if supported, in the UE context;
- if supported, store the received Mobile IAB Authorization information in the UE context, and use it accordingly for the mobile IAB-MT;
- store the received PDU Set QoS parameters, if supported, in the UE context and use it as specified in TS 23.501 [9].

For the Initial Context Setup an initial value for the Next Hop Chaining Count is stored in the UE context.

If the *PDU Session Resource Setup Request List* IE is contained in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall behave the same as defined in the PDU Session Resource Setup procedure. The NG-RAN node shall report to the AMF in the INITIAL CONTEXT SETUP RESPONSE message the result for each PDU session resource requested to be setup as defined in the PDU Session Resource Setup procedure.

Upon reception of the INITIAL CONTEXT SETUP RESPONSE message the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transfer transparently the *PDU Session Resource Setup Response Transfer* IE or *PDU Session Resource Setup Unsuccessful Transfer* IE to the SMF associated with the concerned PDU session. In case the splitting PDU session is not used by the NG-RAN node, the SMF should remove the Additional Transport Layer Information, if any.

The NG-RAN node shall use the information in the *Mobility Restriction List* IE if present in the INITIAL CONTEXT SETUP REQUEST message to

- determine a target for subsequent mobility action for which the NG-RAN node provides information about the target of the mobility action towards the UE;
- select a proper SCG during dual connectivity operation;
- assign proper RNA(s) for the UE when moving the UE to RRC_INACTIVE state.

If the *Mobility Restriction List* IE is not contained in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall consider that no roaming and no access restriction apply to the UE except for the PNI NPN mobility as described in TS 23.501 [9]. The NG-RAN node shall also consider that no roaming and no access restriction apply to the UE when:

- one of the QoS flows includes a particular ARP value (TS 23.501 [9]).

The NG-RAN node shall consider that roaming or access to CAG cells is only allowed if the *Allowed PNI-NPN List* IE is contained in the INITIAL CONTEXT SETUP REQUEST message, as described in TS 23.501 [9].

If the *Trace Activation* IE is included in the INITIAL CONTEXT SETUP REQUEST message the NG-RAN node shall, if supported, initiate the requested trace function as described in TS 32.422 [11]. In particular, the NG-RAN node shall, if supported:

- if the *Trace Activation* IE includes the *MDT Activation* IE set to "Immediate MDT and Trace", initiate the requested trace session and MDT session as described in TS 32.422 [11];
- if the *Trace Activation* IE includes the *MDT Activation* IE set to "Immediate MDT Only", "Logged MDT only", initiate the requested MDT session as described in TS 32.422 [11] and the NG-RAN node shall ignore the *Interfaces To Trace* IE and the *Trace Depth* IE;
- if the *Trace Activation* IE includes the *MDT Location Information* IE within the *MDT Configuration* IE, store this information and take it into account in the requested MDT session;
- if the *Trace Activation* IE includes the *Signalling Based MDT PLMN List* IE within the *MDT Configuration* IE, the NG-RAN node may use it to propagate the MDT Configuration as described in TS 37.320 [41].
- if the *Trace Activation* IE includes the *Bluetooth Measurement Configuration* IE within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [41].
- if the *Trace Activation* IE includes the *WLAN Measurement Configuration* IE within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [41].
- if the *Trace Activation* IE includes the *Sensor Measurement Configuration* IE within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [41].
- if the *Trace Activation* IE includes the *MDT Configuration* IE and if the NG-RAN node is a gNB at least the *MDT Configuration-NR* IE shall be present, while if the NG-RAN node is an ng-eNB at least the *MDT Configuration-EUTRA* IE shall be present.
- if the *Trace Activation* IE includes the *MN Only MDT Collection* IE and the *MN Only MDT Collection* IE is set to "MN only", consider that the *MDT Configuration-NR* IE or the *MDT Configuration-EUTRA* IE is only applicable for the MN if the UE is configured with MR-DC.

If the *UE Security Capabilities* IE included in the INITIAL CONTEXT SETUP REQUEST message only contains the EIA0 or NIA0 algorithm as defined in TS 33.501 [13] and if the EIA0 or NIA0 algorithm is defined in the configured list of allowed integrity protection algorithms in the NG-RAN node (TS 33.501 [13]), the NG-RAN node shall take it into use and ignore the keys received in the *Security Key* IE.

If the *QMC Configuration Information* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, use it for QoE management, as described in TS 38.300 [8]. If the *Assistance Information for QoE Measurement* IE is included in the *UE Application Layer Measurement Configuration Information* IE within the *QMC Configuration Information* IE, the NG-RAN node may take it into account for controlling the reporting of application layer measurement when the NG-RAN node is overloaded, as described in TS 38.300 [8].

If the *Core Network Assistance Information for RRC INACTIVE* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context and use it for the RRC_INACTIVE state decision and RNA configuration for the UE, and RAN paging, if any, for a UE in RRC_INACTIVE state, as specified in TS 38.300 [8]. In particular,

- if the *UE Specific DRX* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it for RAN paging.
- if the *MICO Mode Indication* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, consider that the UE is configured with MICO mode by the AMF as specified in TS 23.501 [9].
- if the *Expected UE Behaviour* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it to assist the UE RRC state transition decision.
- if the *E-UTRA Paging eDRX Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, take it into account to configure the RAN Paging when the UE is in RRC_INACTIVE as specified in TS 36.300 [17].
- if the *Extended UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it for RAN Paging.
- if the *UE Radio Capability for Paging* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, check and update it with all the UE radio capabilities for paging of all the RAT(s) the UE supports in the serving PLMN according to the received *UE Radio Capability* IE, and use it for subsequent RAN paging.
- if the *MICO All PLMN* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE the NG-RAN node shall, if supported, consider that the registration area for the UE is the full PLMN and ignore the *TAI List for RRC Inactive* IE.
- if the *NR Paging eDRX Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, take it into account to configure the RAN Paging when the UE is in RRC_INACTIVE as described in TS 38.300 [8].
- if the *Paging Cause Indication for Voice Service* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store and use it as specified in TS 38.300 [8].
- if the *PEIPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].
- if the *Hashed UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it to determine the start point of PTW.
- if the *CN MT Communication Handling* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and may subsequently request, based on implementation, the CN for MT communication handling as described in TS 23.502 [10].
- if the *LP-WUSPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for LP-WUS paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].
- if the *Further Extended UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it as specified in TS 38.304 [12].
- If the *LP-WUS Disable Indication* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it to disable LP-WUS as described in TS 38.300 [8].

If the *CN Assisted RAN Parameters Tuning* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node may use it as described in TS 23.501 [9].

If the *RRC Inactive Transition Report Request* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context.

If the *Emergency Fallback Indicator* IE is included in the INITIAL CONTEXT SETUP REQUEST message, it indicates that the UE context to be set up is subject to emergency service fallback as described in TS 23.501 [9] and the NG-RAN node may, if supported, take the appropriate mobility actions.

If the *Old AMF* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall consider that this UE-associated logical NG-connection was redirected to this AMF from another AMF identified by the *Old AMF* IE. If the *Extended Old AMF* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, consider that this UE-associated logical NG-connection was redirected to this AMF from another AMF identified by the *Extended Old AMF* IE.

If the *Redirection for Voice EPS Fallback* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store it and use it in a subsequent decision of EPS fallback for voice as specified in TS 23.502 [10].

If the *Location Reporting Request Type* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node should perform the requested location reporting functionality for the UE as described in subclause 8.12.

If the *Enhanced Coverage Restriction* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *Extended Connected Time* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, use it as described in TS 23.501 [9].

If the *UE Differentiation Information* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context for further use according to TS 23.501 [9].

If the *CE-mode-B Restricted* IE is included in the INITIAL CONTEXT SETUP REQUEST message and the *Enhanced Coverage Restriction* IE is not set to "restricted" and the Enhanced Coverage Restriction information stored in the UE context is not set to "restricted", the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *UE User Plane CIoT Support Indicator* IE is included in the INITIAL CONTEXT SETUP REQUEST message the NG-RAN node shall, if supported, store this information in the UE context and consider that User Plane CIoT 5GS Optimisation as specified in TS 23.501 [9] is supported for the UE.

If the *Management Based MDT PLMN List* IE is contained in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, use it to allow subsequent selection of the UE for management based MDT defined in TS 32.422 [11].

If the INITIAL CONTEXT SETUP REQUEST message contains the *UE Radio Capability ID* IE, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9] and TS 23.502 [10].

For each PDU session, if the *PDU Session Expected UE Activity Behaviour* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, handle this information as specified in TS 23.501 [9].

If the *Time Synchronisation Assistance Information* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store the information in the UE context and use it as defined in TS 23.501 [9].

If the *Target NSSAI Information* IE is contained in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node may use this information as specified in TS 23.501 [9].

If the *Partially Allowed NSSAI* IE is contained in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, deduce from it the partially allowed network slices for the UE, store and replace any previously received Partially Allowed NSSAI and use it as specified in TS 23.501 [9].

If the *UE Slice Maximum Bit Rate List* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store the received UE Slice Maximum Bit Rate List in the UE context, and use it for each S-NSSAI for the concerned UE as specified in TS 23.501 [9].

If the *Aerial UE Subscription Information* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 38.300 [8].

If the *PNI-NPN Area Scope of MDT* IE is included in the *MDT Configuration-NR* IE included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, use it to derive the MDT area scope for MDT measurement collection in PNI-NPN areas. Upon reception of the *PNI-NPN Area Scope of MDT* IE, the NG-RAN node shall consider that the area scope for MDT measurement collection in PNI-NPN areas is defined only by the areas included in the *PNI-NPN Area Scope of MDT* IE.

If the *UE Radio Capability for Paging* IE is included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9].

If the *Network Slice Area Scope of MDT* IE is included in the *MDT Configuration-NR* IE included in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, use it to derive the MDT area scope for MDT measurement collection. Upon reception of the *Network Slice Area Scope of MDT* IE, the NG-RAN node shall consider that the area scope for MDT measurement collection is defined only by the *Network Slice Area Scope of MDT* IE and the *Area Scope of MDT* IE.

If the *AMF UE NGAP ID 2* IE is contained in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context and use it for subsequent Xn handovers.

Interactions with Initial UE Message procedure:

The NG-RAN node shall use the *AMF UE NGAP ID* IE and *RAN UE NGAP ID* IE received in the INITIAL CONTEXT SETUP REQUEST message as identification of the logical connection even if the *RAN UE NGAP ID* IE had been allocated in an INITIAL UE MESSAGE message sent over a different NG interface instance.

Interactions with RRC Inactive Transition Report procedure:

If the *RRC Inactive Transition Report Request* IE is included in the INITIAL CONTEXT SETUP REQUEST message and set to "subsequent state transition report", the NG-RAN node shall, if supported, send the RRC INACTIVE TRANSITION REPORT message to the AMF to report the RRC state of the UE when the UE enters or leaves RRC_INACTIVE state.

Interactions with UE Radio Capability Info Indication procedure:

If the *UE Radio Capability for Paging* IE is included in the INITIAL CONTEXT SETUP REQUEST message, which does not include all the UE radio capability for paging of all the NG-RAN RAT(s) that the UE supports according to the received *UE Radio Capability* IE, the NG-RAN node shall, if supported, send the UE RADIO CAPABILITY INFO INDICATION message to the AMF containing all the UE radio capability for paging of all the NG-RAN RAT(s) the UE supports in the serving PLMN.

If the INITIAL CONTEXT SETUP REQUEST message is received with neither the *UE Radio Capability* IE nor the *UE Radio Capability ID* IE, the NG-RAN node shall, if supported, request the UE Radio Capability information from the UE and send it to the AMF via the UE RADIO CAPABILITY INFO INDICATION message, as specified in TS 23.501 [9].

8.3.1.3 Unsuccessful Operation

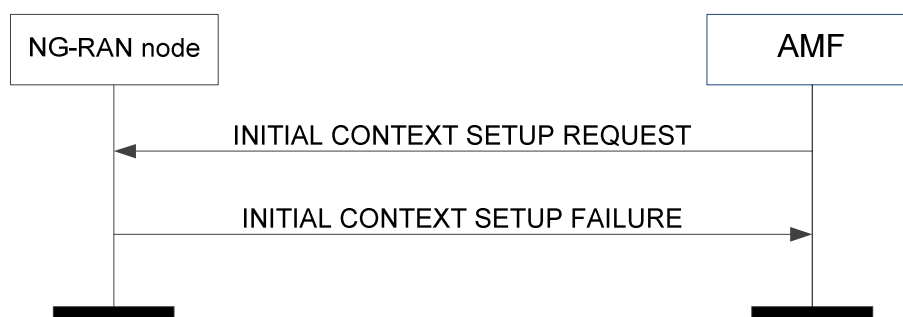


Figure 8.3.1.3-1: Initial context setup: unsuccessful operation

If the NG-RAN node is not able to establish an NG UE context, it shall consider the procedure as failed and reply with the INITIAL CONTEXT SETUP FAILURE message.

If the *PDU Session Resource Setup Request List* IE is contained in the INITIAL CONTEXT SETUP REQUEST message, the NG-RAN node shall report to the AMF, in the INITIAL CONTEXT SETUP FAILURE message, the unsuccessful establishment result for each PDU session resource requested to be setup as defined in the PDU Session Resource Setup procedure.

Upon reception of the INITIAL CONTEXT SETUP FAILURE message the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transfer transparently the *PDU Session Resource Setup Unsuccessful Transfer* IE to the SMF associated with the concerned PDU session and may consider that the NAS PDU included in the INITIAL CONTEXT SETUP REQUEST message was not delivered.

8.3.1.4 Abnormal Conditions

If the supported algorithms for encryption defined in the *Encryption Algorithms* IE in the *UE Security Capabilities* IE, plus the mandated support of EEA0 and NEA0 in all UEs (TS 33.501 [13]), do not match any allowed algorithms defined in the configured list of allowed encryption algorithms in the NG-RAN node (TS 33.501 [13]), the NG-RAN node shall reject the procedure using the INITIAL CONTEXT SETUP FAILURE message.

If the *Partially Allowed NSSAI* IE is received in the INITIAL CONTEXT SETUP REQUEST message and the total number of S-NSSAIs included in the Allowed NSSAI and Partially Allowed NSSAI exceeds eight, the NG-RAN node shall consider the procedure as failed.

If any of the S-NSSAI which is present in the *Partially Allowed NSSAI* IE is also present in the *Allowed NSSAI* IE, the NG-RAN node shall consider the procedure as failed.

If the supported algorithms for integrity defined in the *Integrity Protection Algorithms* IE in the *UE Security Capabilities* IE, plus the mandated support of the EIA0 and NIA0 algorithm in all UEs (TS 33.501 [13]), do not match any allowed algorithms defined in the configured list of allowed integrity protection algorithms in the NG-RAN node (TS 33.501 [13]), the NG-RAN node shall reject the procedure using the INITIAL CONTEXT SETUP FAILURE message.

If the *PNI-NPN Area Scope of MDT* IE is included in the *MDT Configuration-NR* IE in the INITIAL CONTEXT SETUP REQUEST message, and the *Area Scope of MDT* IE is set to "PNI-NPN Based MDT", the NG-RAN node shall, if supported, use the *Area Scope of MDT* IE to derive the MDT area scope for MDT measurement collection in PNI-NPN areas, and ignore the *PNI-NPN Area Scope of MDT*.

If the *Network Slice Area Scope of MDT* IE is included in the *MDT Configuration-NR* IE in the INITIAL CONTEXT SETUP REQUEST message, and the *MDT Activation* IE is set to "Logged MDT only", the NG-RAN node shall ignore the *Network Slice Area Scope of MDT* IE.

8.3.2 UE Context Release Request (NG-RAN node initiated)

8.3.2.1 General

The purpose of the UE Context Release Request procedure is to enable the NG-RAN node to request the AMF to release the UE-associated logical NG-connection due to NG-RAN node generated reasons. The procedure uses UE-associated signalling.

8.3.2.2 Successful Operation

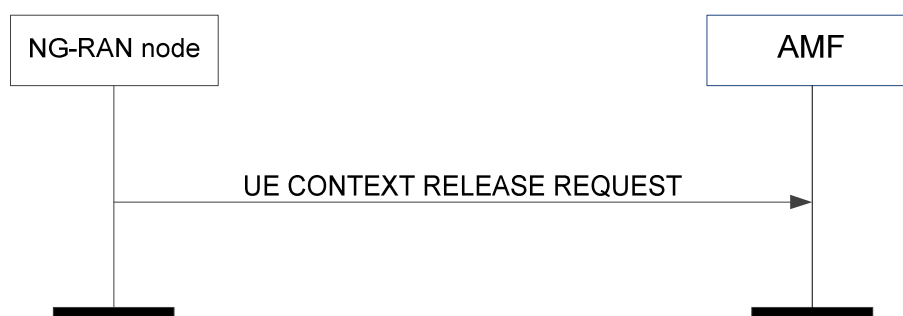


Figure 8.3.2.2-1: UE context release request

The NG-RAN node controlling a UE-associated logical NG-connection initiates the procedure by sending a UE CONTEXT RELEASE REQUEST message towards the affected AMF.

The UE CONTEXT RELEASE REQUEST message shall indicate the appropriate cause value, e.g., "TXnRELOCOverall Expiry", "Redirection", for the requested UE-associated logical NG-connection release.

If the *PDU Session Resource List* IE is included in the UE CONTEXT RELEASE REQUEST message, the AMF shall handle this information as specified in TS 23.502 [10].

If the *GW Context Release Indication* IE is included in the UE CONTEXT RELEASE REQUEST message, the AMF shall, if supported, consider that the UE context may be released as specified in TS 38.300 [8].

Interactions with UE Context Release procedure:

The UE Context Release procedure should be initiated upon reception of a UE CONTEXT RELEASE REQUEST message with the *Cause* IE set to a value different than "User inactivity". The UE Context Release procedure should be initiated upon reception of a UE CONTEXT RELEASE REQUEST message with the *Cause* IE set to "User inactivity" and there is no downlink signaling, as specified in TS 23.502 [10]. If the UE was configured with DC radio resources at the time UE Context Release Request procedure was triggered, and the PSCell information was available, the NG-RAN node shall store the PSCell information in the UE context.

8.3.2.3 Abnormal Conditions

Void.

8.3.3 UE Context Release (AMF initiated)

8.3.3.1 General

The purpose of the UE Context Release procedure is to enable the AMF to order the release of the UE-associated logical NG-connection due to various reasons, e.g., completion of a transaction between the UE and the 5GC, or release of the old UE-associated logical NG-connection when the UE has initiated the establishment of a new UE-associated logical NG-connection, etc. The procedure uses UE-associated signalling.

8.3.3.2 Successful Operation

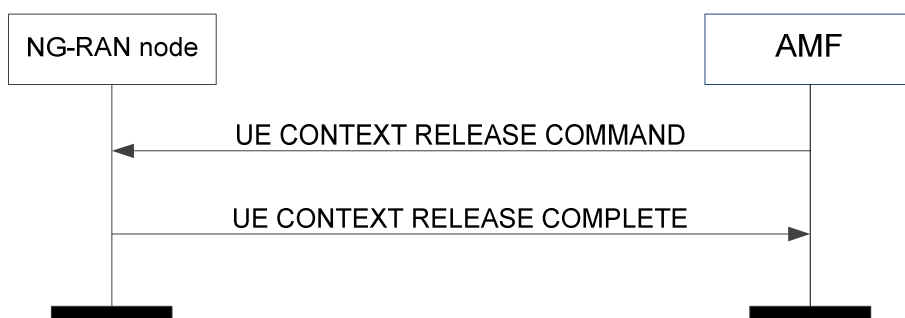


Figure 8.3.3.2-1: UE context release: successful operation

The AMF initiates the procedure by sending the UE CONTEXT RELEASE COMMAND message to the NG-RAN node.

The UE CONTEXT RELEASE COMMAND message shall contain both the AMF UE NGAP ID IE and the *RAN UE NGAP ID* IE if available, otherwise the message shall contain the *AMF UE NGAP ID* IE.

Upon reception of the UE CONTEXT RELEASE COMMAND message, the NG-RAN node shall release all related signalling and user data transport resources and reply with the UE CONTEXT RELEASE COMPLETE message.

If the *PDU Session Resource List* IE is included in the UE CONTEXT RELEASE COMPLETE message, the AMF shall handle this information as specified in TS 23.502 [10].

If the *User Location Information* IE is included in the UE CONTEXT RELEASE COMPLETE message, the AMF shall handle this information as specified in TS 23.502 [10].

If the *Information on Recommended Cells and RAN Nodes for Paging* IE is included in the UE CONTEXT RELEASE COMPLETE message, the AMF shall, if supported, store it and may use it for subsequent paging.

For each PDU session for which the *Secondary RAT Usage Information* IE is included in the *PDU Session Resource Release Response Transfer* IE, the SMF shall handle this information as specified in TS 23.502 [10].

If the *Paging Assistance Data for CE Capable UE* IE is included in the UE CONTEXT RELEASE COMPLETE message, the AMF shall, if supported, store it and use it for subsequent paging, as specified in TS 23.502 [10].

8.3.3.3 Unsuccessful Operation

Not applicable.

8.3.3.4 Abnormal Conditions

If the UE Context Release procedure is not initiated towards the NG-RAN node before the expiry of the timer $T_{NG_RELOC_Overall}$, the NG-RAN node shall request the AMF to release the UE context.

If the UE returns to the NG-RAN node before the reception of the UE CONTEXT RELEASE COMMAND message or the expiry of the timer $T_{NG_RELOC_Overall}$, the NG-RAN node shall stop the timer $T_{NG_RELOC_Overall}$ and continue to serve the UE.

8.3.4 UE Context Modification

8.3.4.1 General

The purpose of the UE Context Modification procedure is to partly modify the established UE context. The procedure uses UE-associated signalling.

8.3.4.2 Successful Operation

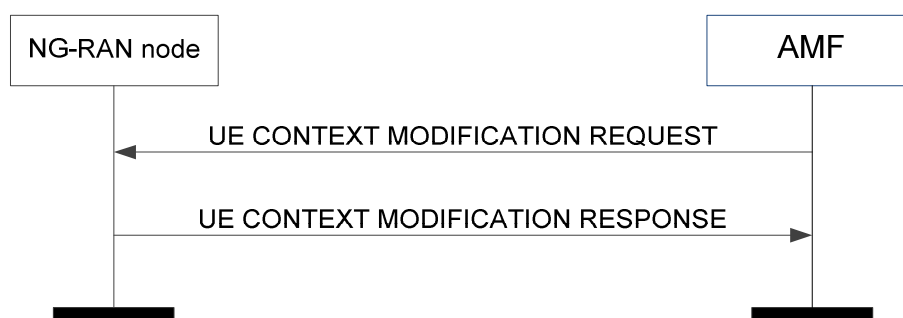


Figure 8.3.4.2-1: UE context modification: successful operation

Upon receipt of the UE CONTEXT MODIFICATION REQUEST message the NG-RAN node shall

- if supported, store the received IAB Authorization information in the UE context and use it as specified in TS 38.401 [2].
- if supported, store the received Mobile IAB Authorization information in the UE context. If the *Mobile IAB Authorized* IE is set to "not authorized" for a mobile IAB-MT, the NG-RAN node shall, if supported, initiate actions to ensure that the mobile IAB-node will not serve any UE(s).

If the *Security Key* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall store it and perform AS key re-keying according to TS 33.501 [13].

If the *UE Security Capabilities* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall store them and take them into use together with the received keys according to TS 33.501 [13].

If the *Index to RAT/Frequency Selection Priority* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, use it as defined in TS 23.501 [9].

If the *RAN Paging Priority* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node may use it to determine a priority for paging the UE in RRC_INACTIVE state.

If the *UE Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall

- replace the previously provided UE Aggregate Maximum Bit Rate by the received UE Aggregate Maximum Bit Rate in the UE context;
- use the received UE Aggregate Maximum Bit Rate for all Non-GBR QoS flows for the concerned UE as specified in TS 23.501 [9].

If the *Core Network Assistance Information for RRC INACTIVE* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, replace the previously provided Core Network Assistance Information for RRC INACTIVE and use it for the RRC_INACTIVE state decision and RNA configuration for the UE, and RAN paging, if any, for a UE in RRC_INACTIVE state, as specified in TS 38.300 [8]. In particular,

- if the *UE Specific DRX* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it for RAN paging.
- if the *MICO Mode Indication* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, consider that the UE is configured with MICO mode by the AMF as specified in TS 23.501 [9].
- if the *Expected UE Behaviour* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it to assist the UE RRC state transition decision.
- if the *E-UTRA Paging eDRX Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, take it into account to configure the RAN Paging when the UE is in RRC_INACTIVE as specified in TS 36.300 [17].
- if the *Extended UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it for RAN Paging.
- if the *UE Radio Capability for Paging* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it for subsequent RAN paging.
- if the *MICO All PLMN* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE the NG-RAN node shall, if supported, consider that the registration area for the UE is the full PLMN and ignore the *TAI List for RRC Inactive* IE.
- if the *NR Paging eDRX Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, take it into account to configure the RAN Paging when the UE is in RRC_INACTIVE as described in TS 38.300 [8].
- if the *Paging Cause Indication for Voice Service* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store and use it as specified in TS 38.300 [8].
- if the *PEIPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].
- if the *Hashed UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it to determine the start point of PTW.
- if the *CN MT Communication Handling* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and may subsequently request, based on implementation, the CN for MT communication handling as described in TS 23.502 [10].
- if the *LP-WUSPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for LP-WUS paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].

- if the *Further Extended UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it as specified in TS 38.304 [12].
- If the *LP-WUS Disable Indication* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it to disable LP-WUS as described in TS 38.300 [8].

If the *CN Assisted RAN Parameters Tuning* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node may use it as described in TS 23.501 [9].

If the *RRC Inactive Transition Report Request* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context and report to the AMF the *User Location Information* IE and the *RRC State* IE in the UE CONTEXT MODIFICATION RESPONSE message.

If the *RRC Inactive Transition Report Request* IE is included in the UE CONTEXT MODIFICATION REQUEST message and set to "cancel report", the NG-RAN node shall, if supported, stop reporting to the AMF the RRC state of the UE.

The NG-RAN node shall report, in the UE CONTEXT MODIFICATION RESPONSE message to the AMF, the successful update of the UE context.

If the *Emergency Fallback Indicator* IE is included in the UE CONTEXT MODIFICATION REQUEST message, it indicates that the concerned UE context is subject to emergency service fallback as described in TS 23.501 [9] and the NG-RAN node may, if supported, take the appropriate mobility actions taking into account the *Emergency Service Target CN* IE if provided.

If the *New AMF UE NGAP ID* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall use the received value for future signalling with the AMF.

If the *New GUAMI* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall replace the previously stored GUAMI as specified in TS 23.501 [9].

If the *SRVCC Operation Possible* IE is included in UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, store the content of the received *SRVCC Operation Possible* IE in the UE context and use it as defined in TS 23.216 [31].

If the *NR V2X Services Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, update its V2X services authorization information for the UE accordingly. If the *NR V2X Services Authorized* IE includes one or more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *LTE V2X Services Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, update its V2X services authorization information for the UE accordingly. If the *LTE V2X Services Authorized* IE includes one or more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *NR A2X Services Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, update its A2X services authorization information for the UE accordingly. If the *NR A2X Services Authorized* IE includes one or more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *LTE A2X Services Authorized* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, update its A2X services authorization information for the UE accordingly. If the *LTE A2X Services Authorized* IE includes one or more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *NR UE Sidelink Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported:

- replace the previously provided NR UE Sidelink Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for NR V2X services.

If the *LTE UE Sidelink Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported:

- replace the previously provided LTE UE Sidelink Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for LTE V2X services.

If the *NR A2X UE PC5 Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported:

- replace the previously provided NR A2X UE PC5 Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for NR A2X services.

If the *LTE A2X UE PC5 Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported:

- replace the previously provided LTE A2X UE PC5 Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for LTE A2X services.

If the *PC5 QoS Parameters* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, use it as defined in TS 23.287 [33].

If the *A2X PC5 QoS Parameters* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, use it as defined in TS 23.256 [54].

If the UE CONTEXT MODIFICATION REQUEST message contains the *UE Radio Capability ID* IE, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9] and TS 23.502 [10].

If the *Time Synchronisation Assistance Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, store the information in the UE context and use it as defined in TS 23.501 [9].

If the *QMC Configuration Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, use it for QoE management, as described in TS 38.300 [8]. If the *Assistance Information for QoE Measurement* IE is included in the *UE Application Layer Measurement Configuration Information* IE within the *QMC Configuration Information* IE, the NG-RAN node may take it into account for controlling the reporting of application layer measurement when the NG-RAN node is overloaded, as described in TS 38.300 [8].

If the *QMC Deactivation* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, deactivate the QMC configurations therein.

If the *UE Slice Maximum Bit Rate List* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported:

- store and replace the previously provided UE Slice Maximum Bit Rate List, if any, by the received UE Slice Maximum Bit Rate List in the UE context;
- use the received UE Slice Maximum Bit Rate List for each S-NSSAI for the concerned UE as specified in TS 23.501 [9].

If the *Management Based MDT PLMN Modification List* IE is contained in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, overwrite any previously stored Management Based MDT PLMN List information in the UE context and use the received information to determine subsequent selection of the UE for management based MDT defined in TS 32.422 [11].

If the *5G ProSe Authorized* IE is included in UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, update the 5G ProSe authorization information for the UE accordingly. If the *5G ProSe*

Authorized IE includes one or more IEs set to “not authorized”, the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant 5G ProSe service(s).

If the *5G ProSe UE PC5 Aggregate Maximum Bit Rate* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported:

- replace the previously provided 5G ProSe UE PC5 Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE’s sidelink communication in network scheduled mode for 5G ProSe services.

If the *5G ProSe PC5 QoS Parameters* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, use it as defined in TS 23.304 [47].

If the *Network Controlled Repeater Authorized* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, update its Network Controlled Repeater Authorization information for the UE accordingly and take it into account when configuring UE information.

If the *Aerial UE Subscription Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, store the information or overwrite any previously stored information in the UE context and use it as defined in TS 38.300 [8].

If the *Ranging and Sidelink Positioning Service Information* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node shall, if supported, update the Ranging and Sidelink Positioning service information for the UE accordingly. If the *Ranging and Sidelink Positioning Authorized* IE within the *Ranging and Sidelink Positioning Service Information* IE is set to “not authorized”, the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the Ranging and Sidelink Positioning service.

Interactions with RRC Inactive Transition Report procedure:

If the *RRC Inactive Transition Report Request* IE is included in the UE CONTEXT MODIFICATION REQUEST message and set to “single RRC connected state report”, the NG-RAN node shall, if supported and if the UE is in RRC_INACTIVE state, send one subsequent RRC INACTIVE TRANSITION REPORT message to the AMF when the RRC state transitions to RRC_CONNECTED state.

If the *RRC Inactive Transition Report Request* IE is included in the UE CONTEXT MODIFICATION REQUEST message and set to “subsequent state transition report”, the NG-RAN node shall, if supported, send the RRC INACTIVE TRANSITION REPORT message to the AMF to report the RRC state of the UE when the UE enters or leaves RRC_INACTIVE state.

8.3.4.3 Unsuccessful Operation

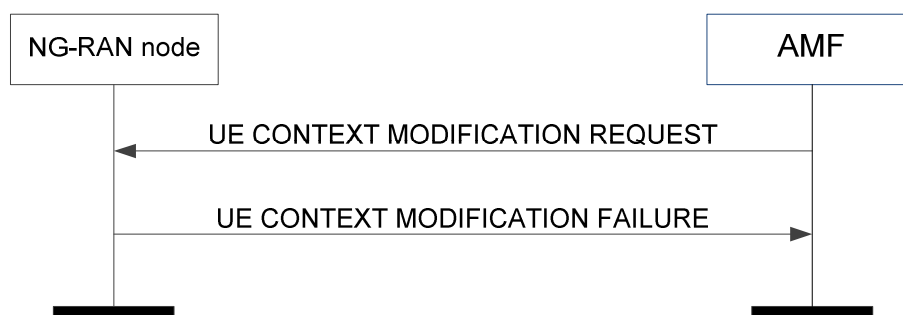


Figure 8.3.4.3-1: UE context modification: unsuccessful operation

In case the UE context update cannot be performed successfully, the NG-RAN node shall respond with the UE CONTEXT MODIFICATION FAILURE message to the AMF with an appropriate cause value in the *Cause* IE.

If the *New AMF UE NGAP ID* IE is included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node may use the received *New AMF UE NGAP ID* IE or *Old AMF UE NGAP ID* IE in the UE CONTEXT MODIFICATION FAILURE message.

8.3.4.4 Abnormal Conditions

If the UE CONTEXT MODIFICATION REQUEST message including the *New AMF UE NGAP ID* IE is received after the NG-RAN node has initiated another class 1 NGAP EP, the NG-RAN node shall be prepared to receive the response message containing an AMF UE NGAP ID with the value received in the *New AMF UE NGAP ID* IE.

NOTE: If the *Emergency Fallback Indicator* IE and the *Security Key* IE are both included in the UE CONTEXT MODIFICATION REQUEST message, the NG-RAN node may handle only the *Emergency Fallback Indicator* IE.

8.3.5 RRC Inactive Transition Report

8.3.5.1 General

The purpose of the RRC Inactive Transition Report procedure is to notify the AMF when the UE enters or leaves RRC_INACTIVE state. The procedure uses UE-associated signalling.

8.3.5.2 Successful Operation

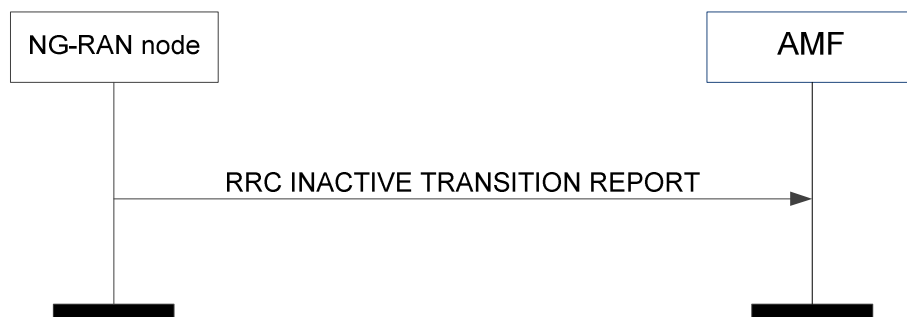


Figure 8.3.5.2-1: RRC Inactive transition report

The NG-RAN node initiates the procedure by sending an RRC INACTIVE TRANSITION REPORT message to the AMF. Upon reception of the RRC INACTIVE TRANSITION REPORT message, the AMF shall take appropriate actions based on the information indicated by the *RRC State* IE.

8.3.5.3 Abnormal Conditions

Void.

8.3.6 Connection Establishment Indication

8.3.6.1 General

The purpose of the Connection Establishment Indication procedure is to enable the AMF to complete the establishment of the UE-associated logical NG-connection. The procedure uses UE-associated signalling. This procedure applies only if the NG-RAN node is an ng-eNB.

8.3.6.2 Successful Operation

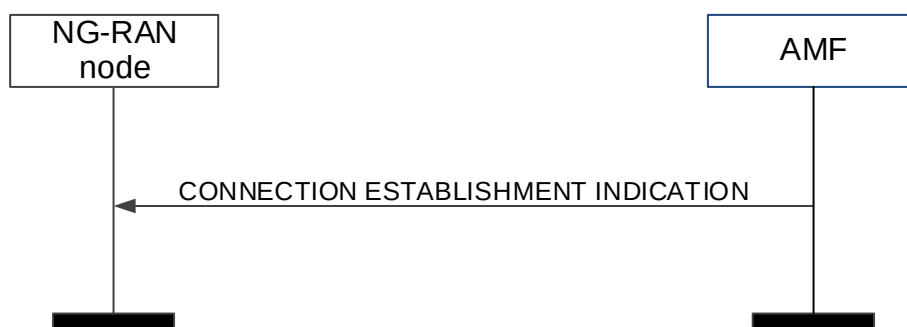


Figure 8.3.6.2-1: Connection Establishment Indication procedure. Successful operation.

The AMF initiates the procedure by sending a CONNECTION ESTABLISHMENT INDICATION message to the NG-RAN node.

If the UE-associated logical NG-connection is not established, the AMF shall allocate a unique AMF UE NGAP ID to be used for the UE and include it in the CONNECTION ESTABLISHMENT INDICATION message.

If the *UE Radio Capability* IE is included in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall store this information in the UE context, and use it as defined in TS 38.300 [8].

If the *End Indication* IE is included in the CONNECTION ESTABLISHMENT INDICATION message and set to "no further data", the NG-RAN node shall consider that there are no further NAS PDUs to be transmitted for this UE.

If the *S-NSSAI* IE is contained in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall store this information in the UE context, and use it as specified in TS 23.501 [9].

If the *Allowed NSSAI* IE is contained in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall store this information in the UE context, and use it as specified in TS 23.501 [9].

If the *Partially Allowed NSSAI* IE is contained in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall, if supported, deduce from it the partially allowed network slices for the UE, store and replace any previously received Partially Allowed NSSAI and use it as specified in TS 23.501 [9].

If the *Ranging and Sidelink Positioning Service Information* IE is contained in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, take it into account for the Ranging and Sidelink Positioning service.

If the *UE Differentiation Information* IE is included in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall, if supported, store this information in the UE context for further use according to TS 23.501 [9].

If the *DL CP Security Information* IE is included in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall forward this information to the UE as described in TS 36.300 [17].

If the *NB-IoT UE Priority* IE is contained in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall, if supported, store this information in the UE context, and use it as specified in TS 23.501 [9].

If the *Enhanced Coverage Restriction* IE is included in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *CE-mode-B Restricted* IE is included in the CONNECTION ESTABLISHMENT INDICATION message and the *Enhanced Coverage Restriction* IE is not set to "restricted" and the Enhanced Coverage Restricted information stored in the UE context is not set to "restricted", the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *UE Radio Capability ID* IE is contained in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9] and TS 23.502 [10].

If the *Masked IMEISV* IE is contained in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall, if supported, use it to determine the characteristics of the UE for subsequent handling.

If the *Old AMF* IE is included in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall consider that this UE-associated logical NG-connection was redirected to this AMF from another AMF identified by the *Old AMF* IE. If the *Extended Old AMF* IE is included in the CONNECTION ESTABLISHMENT INDICATION message, the NG-RAN node shall, if supported, consider that this UE-associated logical NG-connection was redirected to this AMF from another AMF identified by the *Extended Old AMF* IE.

Interactions with UE Radio Capability Info Indication procedure:

If the CONNECTION ESTABLISHMENT INDICATION message is received with neither the *UE Radio Capability* IE nor the *UE Radio Capability ID* IE, the NG-RAN node shall, if supported, request the UE Radio Capability information from the UE, and send it to the AMF via the UE RADIO CAPABILITY INFO INDICATION message, as specified in TS 23.501 [9].

8.3.6.3 Abnormal Conditions

If the *Partially Allowed NSSAI* IE is received in the CONNECTION ESTABLISHMENT INDICATION message and the total number of S-NSSAIs included in the Allowed NSSAI and Partially Allowed NSSAI exceeds eight, the NG-RAN node shall consider the procedure as failed.

If any of the S-NSSAI which is present in the *Partially Allowed NSSAI* IE is also present in the *Allowed NSSAI* IE, the NG-RAN node shall consider the procedure as failed.

8.3.7 AMF CP Relocation Indication

8.3.7.1 General

The purpose of the AMF CP Relocation Indication procedure is to inform the NG-RAN node that the UE's connection is to be relocated to another NG-RAN node as described in TS 38.300 [8], for a UE using Control Plane CIoT 5GS Optimisation. This procedure applies only if the NG-RAN node is an ng-eNB.

The procedure uses UE-associated signalling.

8.3.7.2 Successful Operation

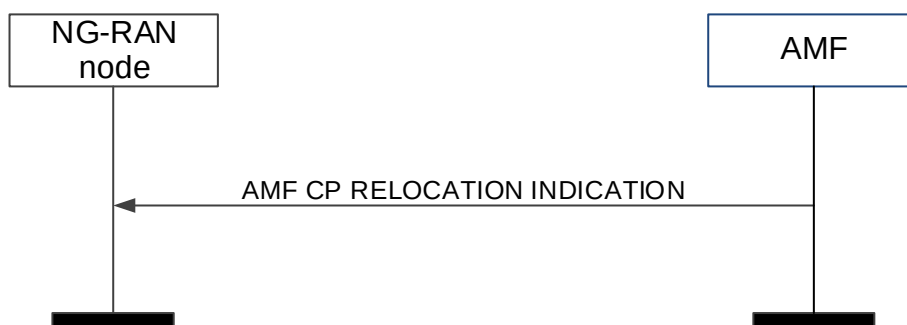


Figure 8.3.7.2-1: AMF CP Relocation Indication. Successful operation.

The AMF initiates the procedure by sending an AMF CP RELOCATION INDICATION message to the NG-RAN node.

Upon reception of the AMF CP RELOCATION INDICATION message, the NG-RAN node shall terminate the delivery of NAS messages that have been received from the AMF.

Interactions with NAS Non Delivery Indication procedure:

On reception of the AMF CP RELOCATION INDICATION message, the NG-RAN node may initiate NAS Non Delivery Indication procedure(s) to report the non-delivery of any NAS PDUs previously received from the AMF.

8.3.7.3 Abnormal Conditions

Void.

8.3.8 RAN CP Relocation Indication

8.3.8.1 General

The purpose of the RAN CP Relocation Indication procedure is to request the AMF to authenticate the UE's re-establishment request, and trigger the establishment of the respective UE-associated logical NG-connection, for a NB-IoT UE using Control Plane CIoT 5GS Optimisation. This procedure applies only if the NG-RAN node is an ng-eNB.

The procedure uses UE-associated signalling.

8.3.8.2 Successful Operation

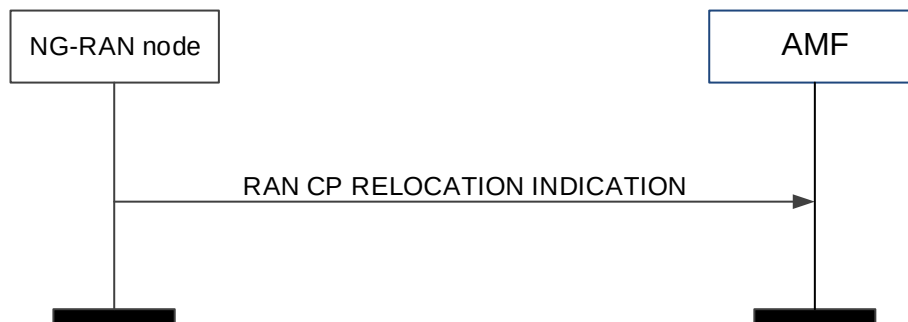


Figure 8.3.8.2-1: RAN CP Relocation Indication.

The NG-RAN node initiates the procedure by sending a RAN CP RELOCATION INDICATION message to the AMF.

The NG-RAN node shall allocate a unique RAN UE NGAP ID to be used for the UE and the NG-RAN node shall include this identity in the RAN CP RELOCATION INDICATION message.

Upon receiving the RAN CP RELOCATION INDICATION message, the AMF shall authenticate the request using the NAS-level security information received in the *UL CP Security Information* IE and if the authentication is successful initiate the Connection Establishment Indication procedure including NAS-level security information in the *DL CP Security Information* IE.

In case the AMF cannot authenticate the UE's request, the CONNECTION ESTABLISHMENT INDICATION message does not contain security information, and the NG-RAN node shall fail the RRC Re-establishment.

In case of authentication failure, the NG-RAN node and the AMF should locally release the allocated NG resources, if any.

Interactions with the AMF CP Relocation and UE Context Release procedures:

In case of successful UE authentication, the AMF initiates the UE Context Release procedure to release the UE's NG-connection in the old NG-RAN node. The AMF may initiate the AMF CP Relocation procedure before the release procedure in order to trigger the old NG-RAN node to return non-delivered NAS PDUs to the AMF.

8.3.8.3 Abnormal Conditions

Void.

8.3.9 Retrieve UE Information

8.3.9.1 General

The purpose of the Retrieve UE Information procedure is for the NG-RAN node to request the UE information including NB-IoT UE Priority and UE Radio Capability from the AMF, for a NB-IoT UE using Control Plane CIoT 5GS Optimisation. The procedure uses non UE-associated signalling. This procedure applies only if the NG-RAN node is an ng-eNB.

8.3.9.2 Successful Operation

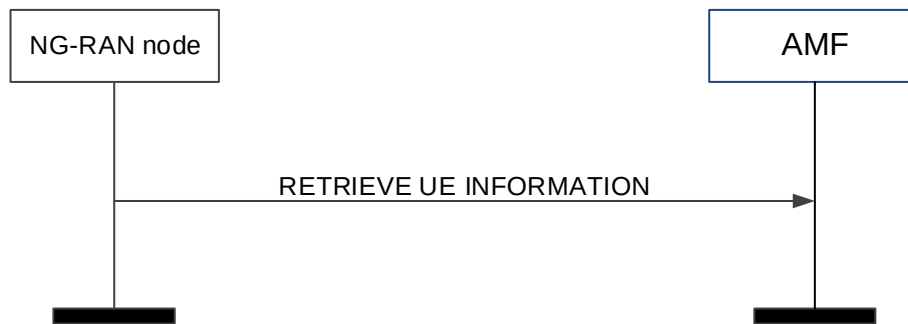


Figure 8.3.9.2-1: Retrieve UE Information

The NG-RAN node initiates the procedure by sending the RETRIEVE UE INFORMATION message to the AMF.

8.3.9.3 Abnormal Conditions

Void.

8.3.10 UE Information Transfer

8.3.10.1 General

The purpose of the UE Information Transfer procedure is for the AMF to send the UE information including NB-IoT UE Priority and UE Radio Capability to the NG-RAN node, for a NB-IoT UE using Control Plane CIoT 5GS Optimisation. The procedure uses non UE-associated signalling. This procedure applies only if the NG-RAN node is an ng-eNB.

8.3.10.2 Successful Operation

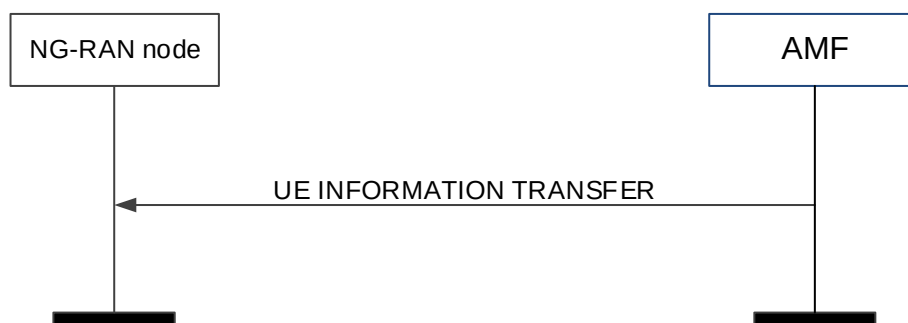


Figure 8.3.10.2-1: UE Information Transfer

The AMF initiates the procedure by sending the UE INFORMATION TRANSFER message to the NG-RAN node.

If the *NB-IoT UE Priority* IE is contained in the UE INFORMATION TRANSFER message, the NG-RAN node shall store this information in the UE context, and use it as specified in TS 23.501 [9].

If the *UE Radio Capability* IE is contained in the UE INFORMATION TRANSFER message, the NG-RAN node shall store this information in the UE context, and use it as specified in TS 23.501 [9].

If the *S-NSSAI* IE is contained in the UE INFORMATION TRANSFER message, the NG-RAN node shall store this information in the UE context, and use it as specified in TS 23.501 [9].

If the *Allowed NSSAI* IE is contained in the UE INFORMATION TRANSFER message, the NG-RAN node shall store this information in the UE context, and use it as specified in TS 23.501 [9].

If the *Partially Allowed NSSAI* IE is contained in the UE INFORMATION TRANSFER message, the NG-RAN node shall, if supported, deduce from it the partially allowed network slices for the UE, store and replace any previously received Partially Allowed NSSAI and use it as specified in TS 23.501 [9].

If the *UE Differentiation Information* IE is included in the UE INFORMATION TRANSFER message, the NG-RAN node shall, if supported, store this information in the UE context for further use according to TS 23.501 [9].

If the *Masked IMEISV* IE is contained in the UE INFORMATION TRANSFER message, the NG-RAN node shall, if supported, use it to determine the characteristics of the UE for subsequent handling.

8.3.10.3 Abnormal Conditions

If the *Partially Allowed NSSAI* IE is received in the UE INFORMATION TRANSFER message and the total number of S-NSSAIs included in the Allowed NSSAI and Partially Allowed NSSAI exceeds eight, the NG-RAN node shall consider the procedure as failed.

If any of the S-NSSAI which is present in the *Partially Allowed NSSAI* IE is also present in the *Allowed NSSAI* IE, the NG-RAN node shall consider the procedure as failed.

8.3.11 UE Context Suspend

8.3.11.1 General

The purpose of the UE Context Suspend procedure is to suspend the UE-associated logical NG-connection and the NG-U transport bearer with the 5GC while keeping the UE context in the NG-RAN node. The procedure uses UE-associated signalling.

In this version of the specification, this procedure applies only if the NG-RAN node is an ng-eNB.

8.3.11.2 Successful Operation

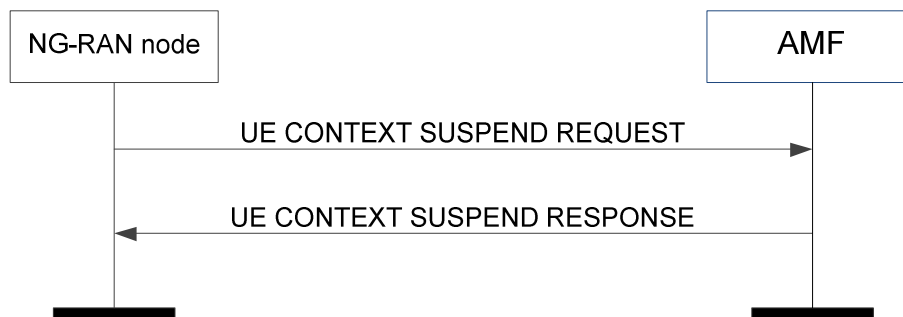


Figure 8.3.11.2-1: UE Context Suspend: Successful operation.

The NG-RAN node initiates the procedure by sending the UE CONTEXT SUSPEND REQUEST message to the AMF.

Upon receipt of the UE CONTEXT SUSPEND REQUEST message the AMF shall act as defined in TS 23.502 [10].

Upon receipt of the UE CONTEXT SUSPEND RESPONSE message the NG-RAN node shall suspend the UE context, the UE-associated logical NG-connection and the related PDU session contexts and send the UE to RRC_IDLE.

If the *Information on Recommended Cells and RAN Nodes for Paging* IE is included in the UE CONTEXT SUSPEND REQUEST message, the AMF shall, if supported, store it and may use it for subsequent paging.

If the *Paging Assistance Data for CE Capable UE* IE is included in the UE CONTEXT SUSPEND REQUEST message, the AMF shall, if supported, store it and use it for subsequent paging, as specified in TS 23.502 [10].

If the *Security Context* IE is included in the UE CONTEXT SUSPEND RESPONSE message, the NG-RAN node shall store the received *Security Context* IE in the UE context and remove any existing unused stored {NH, NCC} as specified in TS 33.501 [13].

If the *Suspend Indicator* IE is included in the UE CONTEXT SUSPEND REQUEST message, the SMF shall, if supported, consider the associated PDU session as suspended.

If the *User Location Information* IE is included in the UE CONTEXT SUSPEND REQUEST message, the AMF shall, if supported, handle this information as specified in TS 23.501 [9].

8.3.11.3 Unsuccessful Operation

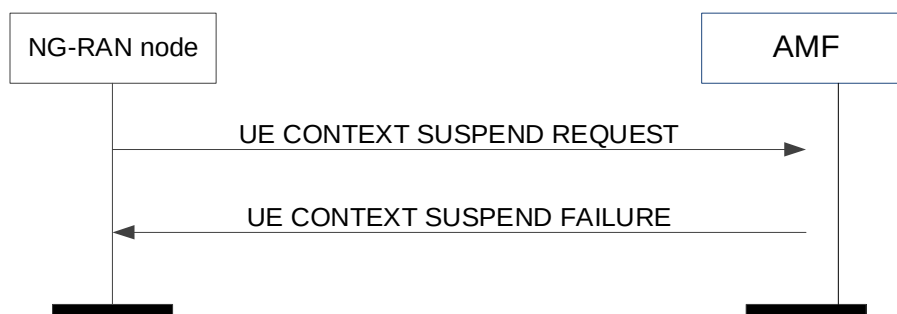


Figure 8.3.11.3-1: UE Context Suspend: unsuccessful operation.

If the AMF decides to not suspend the connection e.g. due to pending downlink data to be sent, it shall send the UE CONTEXT SUSPEND FAILURE message to the NG-RAN node.

8.3.11.4 Abnormal Conditions

Void.

8.3.12 UE Context Resume

8.3.12.1 General

The purpose of the UE Context Resume procedure is to resume the UE context, the suspended UE-associated logical NG-connection and the related NG-U transport bearer in the 5GC for this UE. The procedure uses UE-associated signalling.

In this version of the specification, this procedure applies only if the NG-RAN node is an ng-eNB.

8.3.12.2 Successful Operation

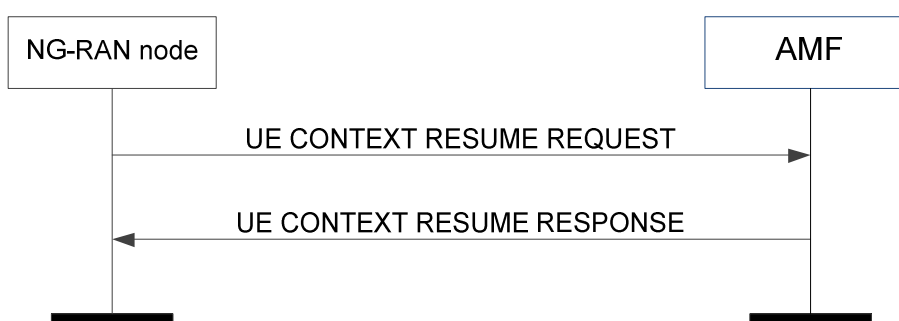


Figure 8.3.12.2-1: UE Context Resume procedure. Successful operation.

The NG-RAN node initiates the procedure by sending the UE CONTEXT RESUME REQUEST message to the AMF. If the NG-RAN node is not able to admit any suspended PDU sessions, the NG-RAN node shall indicate this in the *PDU Session Resource Failed to Resume List* IE. If the NG-RAN node is not able to admit certain QoS flows for a PDU session, the NG-RAN node shall indicate this in the *QoS Flow Failed to Resume List* IE included in the *UE Context Resume Request Transfer* IE for that PDU session.

Upon receipt of the UE CONTEXT RESUME REQUEST message the AMF shall act as defined in TS 23.502 [10] and respond with the UE CONTEXT RESUME RESPONSE message. If the AMF is not able to admit any suspended PDU sessions, the AMF shall indicate this in the *PDU Session Resource Failed to Resume List* IE. If the SMF is not able to admit certain QoS flows for a PDU session, the SMF shall indicate this in the *QoS Flow Failed to Resume List* IE included in the *UE Context Resume Response Transfer* IE for that PDU session.

The NG-RAN node shall release resources for each PDU session or QoS flow failed to resume and shall assume that the 5GC has released respective resources as well.

If the *Security Context* IE is included in the UE CONTEXT RESUME RESPONSE message, the NG-RAN node shall store the received *Security Context* IE in the UE context and the NG-RAN node shall use it for the next suspend/resume or Xn handover or Intra NG-RAN node handovers as specified in TS 33.501 [13].

If the *Suspend Request Indication* IE is included in the UE CONTEXT RESUME REQUEST message, the AMF shall, if supported, consider that the NG-RAN node is requesting immediate transition to RRC IDLE with Suspend as specified in TS 23.502 [10]. If the *Suspend Response Indication* IE is included in the UE CONTEXT RESUME RESPONSE message, the NG-RAN node shall suspend the UE context, the UE-associated logical NG-connection and the related PDU session contexts and send the UE to RRC_IDLE.

If the *Information on Recommended Cells and RAN Nodes for Paging* IE is included in the UE CONTEXT RESUME REQUEST message, the AMF shall, if supported, store it and may use it for subsequent paging.

If the *Paging Assistance Data for CE Capable UE* IE is included in the UE CONTEXT RESUME REQUEST message, the AMF shall, if supported, store it and use it for subsequent paging, as specified in TS 23.502 [10].

If the *Extended Connected Time* IE is included in the UE CONTEXT RESUME RESPONSE message, the NG-RAN node shall, if supported, use it as described in TS 23.501 [9].

If the *User Location Information* IE is included in the UE CONTEXT RESUME REQUEST message, the AMF shall, if supported, handle this information as specified in TS 23.501 [9].

8.3.12.3 Unsuccessful Operation

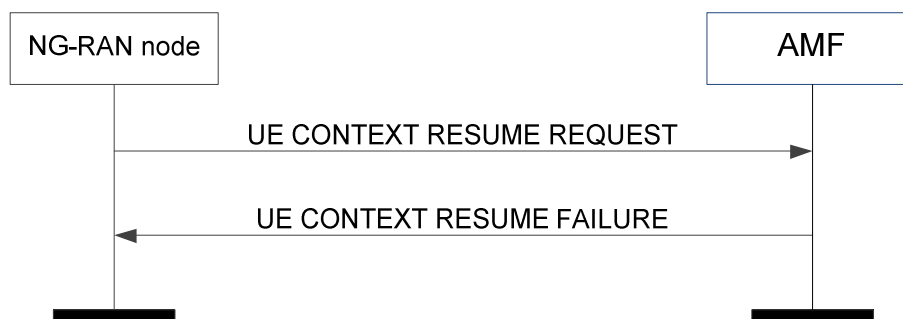


Figure 8.3.12.3-1: UE Context resume: unsuccessful operation.

If the AMF is not able to resume a single PDU session, it releases the UE-associated logical NG-connection by sending the UE CONTEXT RESUME FAILURE message to the NG-RAN node. Upon reception of the UE CONTEXT RESUME FAILURE message the NG-RAN node shall release the RRC connection as specified in TS 36.331 [21] and release all related signalling and user data transport resources.

8.3.12.4 Abnormal Conditions

Void.

8.3.13 MT Communication Handling

8.3.13.1 General

The purpose of the MT Communication Handling procedure is to request the AMF to activate or deactivate the CN based MT communication handling for a UE in RRC_INACTIVE state with eDRX beyond 10.24 seconds as specified in TS 23.501 [9]. The procedure uses UE-associated signalling.

8.3.13.2 Successful Operation

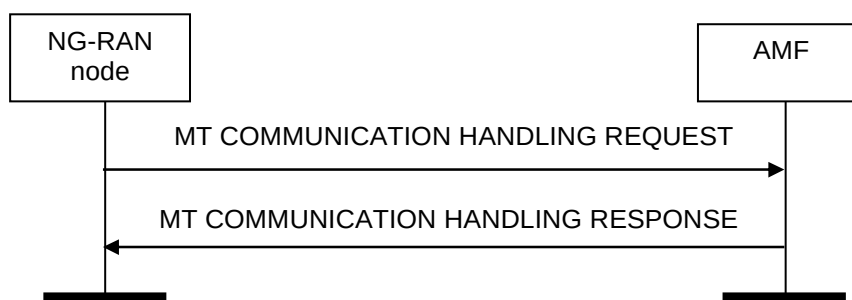


Figure 8.3.13.2-1: MT Communication Handling procedure. Successful operation.

The NG-RAN node initiates the procedure by sending the MT COMMUNICATION HANDLING REQUEST message to the AMF.

If the *5GC Action* IE included in the MT COMMUNICATION HANDLING REQUEST message is set to "HLCOM Activate", the AMF shall activate MT communication handling for the PDU session(s) indicated in the *PDU Session Resource List* IE as specified in TS 23.501 [9] and take into account the *NR Paging Long eDRX Information for RRC INACTIVE* IE when applying MT communication handling as specified in TS 38.304 [12] and TS 23.502 [10].

If the *5GC Action* IE included in the MT COMMUNICATION HANDLING REQUEST message is set to "HLCOM Deactivate", the AMF shall deactivate MT communication handling for the PDU session(s) indicated in the *PDU Session Resource List* IE as specified in TS 23.501 [9].

8.3.13.3 Unsuccessful Operation

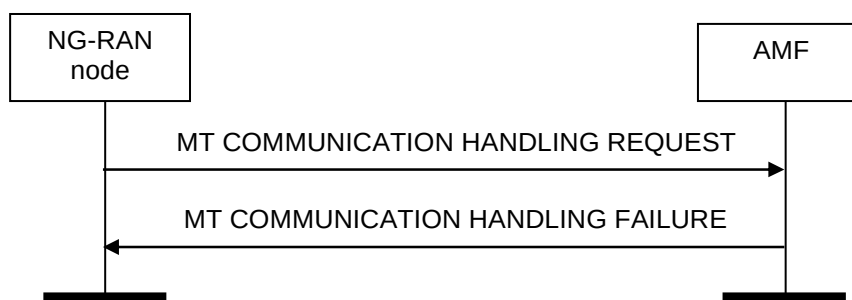


Figure 8.3.13.3-1: MT Communication Handling: unsuccessful operation.

If the AMF is not able to activate CN based MT communication handling for the UE configured with eDRX cycle value longer than 10.24 seconds in RRC_INACTIVE state, it shall send a MT COMMUNICATION HANDLING FAILURE message to the NG-RAN node.

8.3.13.4 Abnormal Conditions

Void.

8.3.14 RAN Paging Request

8.3.14.1 General

This procedure is initiated by the AMF to indicate that there is DL data buffered in 5GC or DL signalling arrives for the UE. The procedure uses UE-associated signalling.

8.3.14.2 Successful Operation

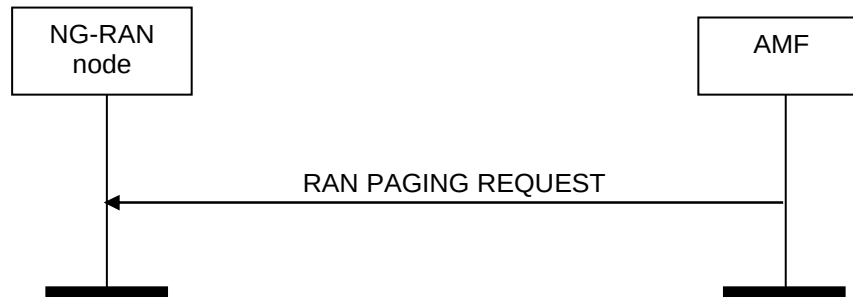


Figure 8.3.14.2-1: RAN Paging Request

The AMF initiates the RAN Paging Request procedure by sending the RAN PAGING REQUEST message to the NG-RAN node.

Upon reception of the RAN PAGING REQUEST message, the NG-RAN node shall perform RAN Paging for the UE in RRC_INACTIVE state.

If the *Paging Policy Differentiation* IE is included in the RAN PAGING REQUEST message, the NG-RAN node shall, if supported, take it into account when performing RAN Paging for the UE in RRC_INACTIVE state.

If the *DL Signalling* IE is included in the RAN PAGING REQUEST message, the NG-RAN node shall, if supported, take it into account when performing RAN Paging for the UE in RRC_INACTIVE state.

8.3.14.3 Abnormal Conditions

Void.

8.4 UE Mobility Management Procedures

8.4.1 Handover Preparation

8.4.1.1 General

The purpose of the Handover Preparation procedure is to request the preparation of resources at the target side via the 5GC. There is only one Handover Preparation procedure ongoing at the same time for a certain UE. The procedure uses UE-associated signalling.

8.4.1.2 Successful Operation



Figure 8.4.1.2-1: Handover preparation: successful operation

The source NG-RAN node initiates the handover preparation by sending the **HANDOVER REQUIRED** message to the serving AMF. When the source NG-RAN node sends the **HANDOVER REQUIRED** message, it shall start the timer $T_{\text{NGRELOCprep}}$. The source NG-RAN node shall indicate the appropriate cause value for the handover in the *Cause* IE.

Upon reception of the **HANDOVER REQUIRED** message the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transparently transfer the *Handover Required Transfer* IE to the SMF associated with the concerned PDU session. If the UE is a mobile IAB-MT and the *PDU Session ID* IE contained in the **HANDOVER REQUIRED** message indicates no PDU session identity assigned as defined in TS 24.007 [58], the AMF shall, if supported, consider the mobile IAB-MT has no PDU session, ignore the *PDU Session Resource List* IE, and behave as specified in TS 23.501 [9].

In case of intra-system handover, the information in the *Source to Target Transparent Container* IE shall be encoded according to the definition of the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE.

If the *DL Forwarding* IE is included for a given QoS flow in the *PDU Session Resource Information Item* IE within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE of the **HANDOVER REQUIRED** message and it is set to "DL forwarding proposed", it indicates that the source NG-RAN node proposes forwarding of downlink data for that QoS flow.

If the *UL Forwarding* IE is included for a given QoS flow in the *PDU Session Resource Information Item* IE within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE of the **HANDOVER REQUIRED** message and it is set to "UL forwarding proposed", it indicates that the source NG-RAN node proposes forwarding of uplink data for that QoS flow.

If the *DRBs to QoS Flows Mapping List* IE is included in the *PDU Session Resource Information Item* IE within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE of the **HANDOVER REQUIRED** message, it implicitly indicates that the source NG-RAN node proposes forwarding of downlink data for those DRBs.

If the *QoS Flow Mapping Indication* IE for a QoS flow is included in the *Associated QoS Flow List* IE within the *DRBs to QoS Flows Mapping List* IE within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE of the **HANDOVER REQUIRED** message, it indicates that the source NG-RAN node has mapped only the uplink or downlink of the QoS flow to the DRB.

The source NG-RAN node shall, for each MRB of each MBS session contained in the *MBS Session Information Target to Source List* IE, start data forwarding to the TNL address contained in the *DL Forwarding UP TNL Information* IE. If the *MRB Progress Information* IE is contained for an MRB in the *Data Forwarding Response MRB List* IE in the *MBS Session Information Target to Source List* IE, the source NG-RAN node may use this information to determine when to stop data forwarding.

In case of intra-system handover, if the **HANDOVER COMMAND** message contains the *DL Forwarding UP TNL Information* IE for a given DRB within the *Data Forwarding Response DRB List* IE in the *Handover Command Transfer* IE, the source NG-RAN node shall consider that the forwarding of downlink data for this DRB is accepted by the target NG-RAN node. If the **HANDOVER COMMAND** message contains the *UL Forwarding UP TNL Information* IE for a given DRB in the *Data Forwarding Response DRB List* IE within the *Handover Command Transfer* IE, it means the target NG-RAN node has requested the forwarding of uplink data for this DRB.

In case direct data forwarding is applied for inter-system handover, if the *Data Forwarding Response E-RAB List* IE in the *Handover Command Transfer* IE is included in the HANDOVER COMMAND message, the source NG-RAN node shall consider that forwarding of downlink data for this E-RAB is accepted by the target eNB.

If the HANDOVER COMMAND message contains the *UL Forwarding UP TNL Information* IE for a given PDU session within the *Handover Command Transfer* IE, the source NG-RAN node shall consider that the forwarding of uplink data of the QoS flows is accepted by the target NG-RAN node.

In case of inter-system handover to LTE, the information in the *Source to Target Transparent Container* IE shall be encoded according to the *Source eNB to Target eNB Transparent Container* IE definition as specified in TS 36.413 [16].

If the *Direct Forwarding Path Availability* IE is included in the HANDOVER REQUIRED message the AMF shall handle it as specified in TS 23.502 [10].

If the *Direct Forwarding Path Availability* IE is included within the *Handover Required Transfer* IE of the HANDOVER REQUIRED message the SMF shall handle it as specified in TS 23.502 [10].

When the preparation, including the reservation of resources at the target side is ready, the AMF responds with the HANDOVER COMMAND message to the source NG-RAN node. In case of intra-system handover, the AMF shall include the *PDU Session Resource Handover List* IE in the HANDOVER COMMAND message.

Upon reception of the HANDOVER COMMAND message the source NG-RAN node shall stop the timer TNG_{RELOCprep} and start the timer TNG_{RELOCoverall}.

If there are any PDU sessions that could not be admitted in the target, they shall be indicated in the *PDU Session Resource to Release List* IE.

NOTE: As an exception in case of inter-system handover to LTE, the AMF generates the *Handover Preparation Unsuccessful Transfer* IE in the *PDU Session Resource to Release List* IE.

If the HANDOVER COMMAND message contains the *QoS Flow to be Forwarded List* IE and/or *Data Forwarding Response DRB List* IE within the *Handover Command Transfer* IE for a given PDU session, then the source NG-RAN node should initiate data forwarding for the QoS flows as specified in TS 38.300 [8].

If the HANDOVER COMMAND message contains the *Additional DL Forwarding UP TNL Information* IE within the *Handover Command Transfer* IE, the source NG-RAN node should initiate data forwarding of the PDU session split in different tunnel and shall use the received UP transport layer information for the forwarding QoS flows associated to it.

If the HANDOVER COMMAND message contains the *Additional UL Forwarding UP TNL Information* IE within the *Handover Command Transfer* IE, the source NG-RAN node should initiate data forwarding of the PDU session split in different tunnels using the received UP transport layer information.

If the *NAS Security Parameters from NG-RAN* IE is included in the HANDOVER COMMAND message the NG-RAN node shall use it as specified in TS 33.501 [13].

If the *Target to Source Transparent Container* IE has been received by the AMF from the handover target then the transparent container shall be included in the HANDOVER COMMAND message.

If the HANDOVER COMMAND message contains the *QoS Flow Failed to Setup List* IE within the *Handover Command Transfer* IE, the source NG-RAN node shall consider that the listed QoS flows are failed to be handed over.

In case of inter-system handover to LTE, the information in the *Target to Source Transparent Container* IE shall be encoded according to the definition of the *Target eNB to Source eNB Transparent Container* IE as specified in TS 36.413 [16].

If the *Index to RAT/Frequency Selection Priority* IE is contained in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE, the target NG-RAN node shall store the content of the received *Index to RAT/Frequency Selection Priority* IE in the UE context and use it as defined in TS 23.501 [9].

If the *DAPS Request Information* IE is included for a DRB in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUIRED message, it indicates that the request concerns a DAPS Handover for that DRB, as described in TS 38.300 [8].

Interactions with other NGAP procedures:

If, after a HANDOVER REQUIRED message is sent and before the Handover Preparation procedure is terminated, the source NG-RAN node receives an AMF initiated PDU Session Management procedure on the same UE-associated signalling connection, the source NG-RAN node shall either:

1. Cancel the Handover Preparation procedure by executing the Handover Cancellation procedure with an appropriate cause value. After successful completion of the Handover Cancellation procedure, the source NG-RAN node shall continue the AMF initiated PDU Session Management procedure.

or

2. Terminate the AMF initiated PDU Session Management procedure by sending the appropriate response message with an appropriate cause value, e.g. "NG intra-system handover triggered" or "NG inter-system handover triggered" to the AMF and then the source NG-RAN node shall continue with the handover procedure.

8.4.1.3 Unsuccessful Operation

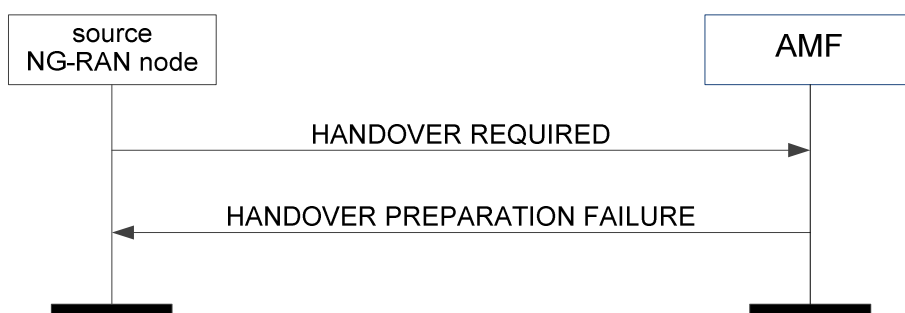


Figure 8.4.1.3-1: Handover preparation: unsuccessful operation

If the 5GC or the target side is not able to accept any of the PDU session resources or a failure occurs during the Handover Preparation, the AMF sends the HANDOVER PREPARATION FAILURE message with an appropriate cause value to the source NG-RAN node.

If the *Target to Source Failure Transparent Container* IE has been received by the AMF from the handover target then the transparent container shall be included in the HANDOVER PREPARATION FAILURE message.

If the *Target to Source Failure Transparent Container* IE is received in the HANDOVER PREPARATION FAILURE message including the *Cell CAG Information* IE, the source NG-RAN node shall, if supported, store and replace the PNI-NPN information associated with the indicated cell.

Interaction with Handover Cancel procedure:

If there is no response from the AMF to the HANDOVER REQUIRED message before timer $T_{NG_{RELOC}_{prep}}$ expires in the source NG-RAN node, the source NG-RAN node should cancel the Handover Preparation procedure by initiating the Handover Cancel procedure with the appropriate value for the *Cause* IE. The source NG-RAN node shall ignore any HANDOVER COMMAND message or HANDOVER PREPARATION FAILURE message received after the initiation of the Handover Cancel procedure.

8.4.1.4 Abnormal Conditions

In case of inter-system handover, if the NG-RAN node receives at least one PDU Session ID included in the *PDU Session Resource Handover List* IE without at least one valid associated GTP tunnel address pair (in either UL or DL), then the NG-RAN node shall consider it as a logical error and act as described in subclause 10.4. A GTP tunnel address pair is considered valid if both the *GTP-TEID* IE and the *Endpoint IP Address* IE are present.

8.4.2 Handover Resource Allocation

8.4.2.1 General

The purpose of the Handover Resource Allocation procedure is to reserve resources at the target NG-RAN node for the handover of a UE. The procedure uses UE-associated signalling.

8.4.2.2 Successful Operation



Figure 8.4.2.2-1: Handover resource allocation: successful operation

The AMF initiates the procedure by sending the HANDOVER REQUEST message to the target NG-RAN node.

If the *Masked IMEISV* IE is contained in the HANDOVER REQUEST message the target NG-RAN node shall, if supported, use it to determine the characteristics of the UE for subsequent handling.

Upon receipt of the HANDOVER REQUEST message the target NG-RAN node shall

- attempt to execute the requested PDU session configuration and associated security;
- store the received UE Aggregate Maximum Bit Rate in the UE context, and use the received UE Aggregate Maximum Bit Rate for all Non-GBR QoS flows for the concerned UE as specified in TS 23.501 [9];
- store the received Mobility Restriction List in the UE context;
- store the received UE Security Capabilities in the UE context;
- store the received Security Context in the UE context and take it into use as defined in TS 33.501 [13];
- if supported, store the received UE Slice Maximum Bit Rate List in the UE context and use the received UE Slice Maximum Bit Rate List for each S-NSSAI for the concerned UE as specified in TS 23.501 [9].
- if supported, store the received PDU Set QoS parameters in the UE context and use it as specified in TS 23.501 [9].

Upon reception of the *UE History Information* IE, which is included within the *Source to Target Transparent Container* IE of the HANDOVER REQUEST message, the target NG-RAN node shall collect the information defined as mandatory in the *UE History Information* IE and shall, if supported, collect the information defined as optional in the *UE History Information* IE, for as long as the UE stays in one of its cells, and store the collected information to be used for future handover preparations.

Upon receiving the *PDU Session Resource Setup List* IE contained in the HANDOVER REQUEST message and the HANDOVER REQUEST message does not contain the *No PDU Session Indication* IE, the target NG-RAN node shall behave the same as defined in the PDU Session Resource Setup procedure. The target NG-RAN node shall report to the AMF in the HANDOVER REQUEST ACKNOWLEDGE message the result for each PDU session resource requested to be setup. In particular, for each PDU session resource successfully setup, it shall include the *Handover Request Acknowledge Transfer* IE containing the following information:

- The list of QoS flows which have been successfully established in the *QoS Flow Setup Response List* IE.
- The *Data Forwarding Accepted* IE if the data forwarding for the QoS flow is accepted.
- The list of QoS flows which have failed to be established, if any, in the *QoS Flow Failed to Setup List* IE.
- The UP transport layer information to be used for the PDU session.
- The security result associated to the PDU session.
- The redundant UP transport layer information to be used for the redundant transmission for the PDU session.

- The PDU Set based Handling Indicator if the HANDOVER REQUEST message includes the *PDU Set QoS Parameters* IE or the *DL PDU Set Information Marking Support Indication* IE set to "true".
- The ECN Marking or Congestion Information Reporting Status if the HANDOVER REQUEST message includes the *ECN Marking or Congestion Information Reporting Request* IE.

For each PDU session resource which failed to be setup, the *Handover Resource Allocation Unsuccessful Transfer* IE shall be included in the HANDOVER REQUEST ACKNOWLEDGE message containing a cause value that should be precise enough to enable the SMF to know the reason for the unsuccessful establishment.

For each PDU session included in the HANDOVER REQUEST ACKNOWLEDGE message, if the *Current QoS Parameters Set Index* IE is included for a QoS flow in the *QoS Flow Setup Response List* IE within the *Handover Request Acknowledge Transfer* IE the SMF shall consider it as the currently fulfilled QoS parameters set among the alternative QoS parameters for the involved QoS flow.

Upon reception of the HANDOVER REQUEST ACKNOWLEDGE message the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transfer transparently the *Handover Request Acknowledge Transfer* IE or *Handover Resource Allocation Unsuccessful Transfer* IE to the SMF associated with the concerned PDU session.

If the HANDOVER REQUEST message contains the *Data Forwarding Not Possible* IE associated with a given PDU session within the *Handover Request Transfer* IE set to "data forwarding not possible", the target NG-RAN node may not include the *DL Forwarding UP TNL Information* IE and for intra-system handover the *Data Forwarding Response DRB List* IE within the *Handover Request Acknowledge Transfer* IE in the HANDOVER REQUEST ACKNOWLEDGE message for that PDU session.

If the HANDOVER REQUEST message contains the *Redundant PDU Session Information* IE associated with a given PDU session within the *Handover Request Transfer* IE, the target NG-RAN node shall, if supported, store the received information in the UE context and use it for redundant PDU session setup as specified in TS38.300 [8] and TS 23.501 [9]. If the *PDU Session Type* IE is set to "ethernet" and the redundancy requirement is fulfilled using a secondary NG-RAN node, the NG-RAN node shall, if supported, include the *Global RAN Node ID of Secondary NG-RAN Node* IE in the *Handover Request Acknowledge Transfer* IE of the HANDOVER REQUEST ACKNOWLEDGE message. If the *PDU Session Pair ID* IE is included in the *Redundant PDU Session Information* IE, the NG-RAN node may use it to identify the paired PDU sessions.

For each PDU session for which the *Global RAN Node ID of Secondary NG-RAN Node* IE is included in the *Handover Request Acknowledge Transfer* IE of the HANDOVER REQUEST ACKNOWLEDGE message, the SMF shall, if supported, handle this information as specified in TS 23.501 [9].

In case of intra-system handover, if the target NG-RAN node accepts the downlink data forwarding for at least one QoS flow for which the *DL Forwarding* IE is set to "DL forwarding proposed", it may include the *DL Forwarding UP TNL Information* IE in the *Handover Request Acknowledge Transfer* IE as forwarding tunnel for the QoS flows listed in the *QoS Flow Setup Response List* IE of the HANDOVER REQUEST ACKNOWLEDGE message.

In case of intra-system handover, if the target NG-RAN node accepts the uplink data forwarding for at least one QoS flow for which the *UL Forwarding* IE is set to "UL forwarding proposed", it may include the *UL Forwarding UP TNL Information* IE in the *Handover Request Acknowledge Transfer* IE for the PDU session within the *PDU Session Resource Admitted List* IE of the HANDOVER REQUEST ACKNOWLEDGE message.

In case of intra-system handover, for each PDU session for which the *Additional DL UP TNL Information for HO List* IE is included in the *Handover Request Acknowledge Transfer* IE of the HANDOVER REQUEST ACKNOWLEDGE message, the SMF shall consider the included *Additional DL NG-U UP TNL Information* IE as the downlink termination point for the associated flows indicated in the *Additional QoS Flow Setup Response List* IE for this PDU session split in different tunnels and shall consider the *Additional DL Forwarding UP TNL Information* IE, if included, as the forwarding tunnel associated to these QoS flows.

In case of intra-system handover, for each PDU session for which the *Additional UL Forwarding UP TNL Information* IE is included in the *Handover Request Acknowledge Transfer* IE of the HANDOVER REQUEST ACKNOWLEDGE message, the SMF shall consider it as the termination points for the uplink forwarding tunnels for this PDU session split in different tunnels.

In case of intra-system handover, if the target NG-RAN node accepts the data forwarding for a successfully configured DRB, the target NG-RAN node may include the *DL Forwarding UP TNL Information* IE for the DRB within the *Data Forwarding Response DRB List* IE within *Handover Request Acknowledge Transfer* IE of the HANDOVER REQUEST ACKNOWLEDGE message.

In case of intra-system handover, if the target NG-RAN node receives the *Direct Forwarding Path Availability* IE set to "direct path available" within the *PDU Session Resource Setup Request Transfer* IE, the target NG-RAN node shall, if supported, assign the UP transport layer information for intra-system direct data forwarding from the appropriate address space, if applicable.

If the HANDOVER REQUEST ACKNOWLEDGE message contains the *UL Forwarding UP TNL Information* IE for a given DRB in the *Data Forwarding Response DRB List* IE within the *Handover Request Acknowledge Transfer* IE, it indicates the target NG-RAN node has requested the forwarding of uplink data for the DRB.

In case of inter-system handover from E-UTRAN, if the *PDU Session Resource Setup Request Transfer* IE contains the *Direct Forwarding Path Availability* IE set to "direct path available", the target NG-RAN node shall, if supported, and if it accepts downlink data forwarding for the QoS flows mapped to an E-RAB of an admitted PDU session, include the *DL Forwarding UP TNL Information* IE in the *Data Forwarding Response E-RAB List* IE in the *Handover Request Acknowledge Transfer* IE in the HANDOVER REQUEST ACKNOWLEDGE message for that mapped E-RAB.

In case of inter-system handover from E-UTRAN, the target NG-RAN node includes the *Data Forwarding Accepted* IE for each QoS flow that the *DL Forwarding* IE is set to "DL forwarding proposed" for the corresponding E-RAB in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE and that the target NG-RAN node has admitted the proposed forwarding of downlink data for the QoS flow. If indirect data forwarding is applied for inter-system handover, if the target NG-RAN node accepts the downlink data forwarding for at least one QoS flow of an admitted PDU session it shall include the *DL Forwarding UP TNL Information* IE in the *PDU Session Resource Setup Response Transfer* IE for that PDU session within the *PDU Session Resources Admitted List* IE of the HANDOVER REQUEST ACKNOWLEDGE message.

In case of inter-system handover from E-UTRAN with direct forwarding, if the target NG-RAN node receives the *SgNB UE X2AP ID* IE in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE, it may use it for internal forwarding as described in TS 37.340 [32].

In case of inter-system handover from E-UTRAN, if the target cell is a CAG cell, the target NG-RAN node shall include the *NPN Access Information* IE in the HANDOVER REQUEST ACKNOWLEDGE message, and the AMF shall consider that the included information is associated to the target cell and to the UE's serving PLMN Identity, and use it as specified in TS 23.501 [9].

The target NG-RAN node shall use the information in the *Mobility Restriction List* IE if present in the HANDOVER REQUEST message to

- determine a target for subsequent mobility action for which the target NG-RAN node provides information about the target of the mobility action towards the UE;
- select a proper SCG during dual connectivity operation;
- assign proper RNA(s) for the UE when moving the UE to RRC_INACTIVE state.

If the *Mobility Restriction List* IE is not contained in the HANDOVER REQUEST message, the target NG-RAN node shall consider that no roaming and no access restriction apply to the UE except for the PNI NPN mobility as described in TS 23.501 [9]. The target NG-RAN node shall also consider that no roaming and no access restriction apply to the UE when:

- one of the QoS flows includes a particular ARP value (TS 23.501 [9]).

The NG-RAN node shall consider that roaming or access to CAG cells is only allowed if the *Allowed PNI-NPN List* IE is contained in the HANDOVER REQUEST message, as described in TS 23.501 [9].

If the *Trace Activation* IE is included in the HANDOVER REQUEST message the target NG-RAN node shall, if supported, initiate the requested trace function as described in TS 32.422 [11]. In particular, the NG-RAN node shall, if supported:

- if the *Trace Activation* IE includes the *MDT Activation* IE set to "Immediate MDT and Trace", initiate the requested trace session and MDT session as described in TS 32.422 [11];
- if the *Trace Activation* IE includes the *MDT Activation* IE set to "Immediate MDT Only", "Logged MDT only", initiate the requested MDT session as described in TS 32.422 [11] and the target NG-RAN node shall ignore the *Interfaces To Trace* IE and the *Trace Depth* IE;

- if the *Trace Activation* IE includes the *MDT Location Information* IE within the *MDT Configuration* IE, store this information and take it into account in the requested MDT session;
- if the *Trace Activation* IE includes the *Signalling Based MDT PLMN List* IE within the *MDT Configuration* IE, the NG-RAN node may use it to propagate the MDT Configuration as described in TS 37.320 [41].
- if the *Trace Activation* IE includes the *Bluetooth Measurement Configuration* IE within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [41].
- if the *Trace Activation* IE includes the *WLAN Measurement Configuration* IE within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [41].
- if the *Trace Activation* IE includes the *Sensor Measurement Configuration* IE within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [41].
- if the *Trace Activation* IE includes the *MDT Configuration* IE and if the NG-RAN node is a gNB at least the *MDT Configuration-NR* IE shall be present, while if the NG-RAN node is an ng-eNB at least the *MDT Configuration-EUTRA* IE shall be present.
- if the *Trace Activation* IE includes the *MN Only MDT Collection* IE and the *MN Only MDT Collection* IE is set to "MN only", consider that the *MDT Configuration-NR* IE or the *MDT Configuration-EUTRA* IE is only applicable for the MN if the UE is configured with MR-DC.

If the *Location Reporting Request Type* IE is included in the HANDOVER REQUEST message, the target NG-RAN node should perform the requested location reporting functionality for the UE as described in subclause 8.12.

If the *Core Network Assistance Information for RRC INACTIVE* IE is included in the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, store this information in the UE context and use it for the RRC_INACTIVE state decision and RNA configuration for the UE, and RAN paging, if any, for a UE in RRC_INACTIVE state, as specified in TS 38.300 [8]. In particular,

- if the *UE Specific DRX* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it for RAN paging.
- if the *MICO Mode Indication* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, consider that the UE is configured with MICO mode by the AMF as specified in TS 23.501 [9].
- if the *Expected UE Behaviour* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it to assist the UE RRC state transition decision.
- if the *E-UTRA Paging eDRX Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, take it into account to configure the RAN Paging when the UE is in RRC_INACTIVE as specified in TS 36.300 [17].
- if the *Extended UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it for RAN Paging.
- if the *UE Radio Capability for Paging* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it for subsequent RAN paging.
- if the *MICO All PLMN* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE the NG-RAN node shall, if supported, consider that the registration area for the UE is the full PLMN and ignore the *TAI List for RRC Inactive* IE.
- if the *NR Paging eDRX Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, take it into account to configure the RAN Paging when the UE is in RRC_INACTIVE as described in TS 38.300 [8].
- if the *Paging Cause Indication for Voice Service* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store and use it as specified in TS 38.300 [8].
- if the *PEIPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].

- if the *Hashed UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it to determine the start point of PTW.
- if the *CN MT Communication Handling* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and may subsequently request, based on implementation, the CN for MT communication handling as described in TS 23.502 [10].
- if the *LP-WUSPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for LP-WUS paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].
- if the *Further Extended UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it as specified in TS 38.304 [12].
- If the *LP-WUS Disable Indication* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it to disable LP-WUS as described in TS 38.300 [8].

If the *CN Assisted RAN Parameters Tuning* IE is included in the HANDOVER REQUEST message, the NG-RAN node may use it as described in TS 23.501 [9].

If the *New Security Context Indicator* IE is included in the HANDOVER REQUEST message, the target NG-RAN node shall use the information as specified in TS 33.501 [13].

If the *NASC* IE is included in the HANDOVER REQUEST message, the target NG-RAN node shall use it towards the UE as specified in TS 33.501 [13].

If the *RRC Inactive Transition Report Request* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context.

If the *Redirection for Voice EPS Fallback* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, store it and use it in a subsequent decision of EPS fallback for voice as specified in TS 23.502 [10].

If the *SRVCC Operation Possible* IE is included in the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, store the content of the received *SRVCC Operation Possible* IE in the UE context and use it as defined in TS 23.216 [31].

If the *IAB Authorized* IE is contained in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, consider that the handover is for an IAB node and use it as specified in TS 38.401 [2].

If the *Mobile IAB Authorized* IE is contained in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, consider that the handover is for a mobile IAB-node. In addition, if the *No PDU Session Indication* IE is contained in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, consider the mobile IAB-MT does not have any PDU sessions, ignore the *PDU Session Resource Setup List* IE, and it shall not take any action with respect to PDU session setup. Subsequently, the AMF shall, if supported, ignore the *PDU Session Resources Admitted List* IE in the HANDOVER REQUEST ACKNOWLEDGE message.

If the *Enhanced Coverage Restriction* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *UE Differentiation Information* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context for further use according to TS 23.501 [9].

If the *UE User Plane CIoT Support Indicator* IE is included in the HANDOVER REQUEST message the NG-RAN node shall, if supported, store this information in the UE context and consider that User Plane CIoT 5GS Optimisation as specified in TS 23.501 [9] is supported for the UE.

Upon reception of the *UE History Information from UE* IE, which is included within the *Source to Target Transparent Container* IE of the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, store the collected information and use it for future handover preparations.

After all necessary resources for the admitted PDU session resources have been allocated, the target NG-RAN node shall generate the HANDOVER REQUEST ACKNOWLEDGE message.

If the *RedCap Indication* IE or the *eRedCap Indication* IE is included in the HANDOVER REQUEST ACKNOWLEDGE message, the AMF shall, if supported, consider the UE respectively as a RedCap UE or an eRedCap UE that was previously served by a E-UTRA cell, and use the IE according to TS 23.501 [9].

For each QoS flow which has been established in the target NG-RAN node, if the *QoS Monitoring Request* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the HANDOVER REQUEST message, the target NG-RAN node shall store this information, and, if supported, perform delay measurement and QoS monitoring, as specified in TS 23.501 [9]. If the *QoS Monitoring Reporting Frequency* IE was included in the *QoS Flow Level QoS Parameters* IE contained in the HANDOVER REQUEST message, the target NG-RAN node shall store this information and, if supported, use it for RAN part delay reporting.

If the *NR V2X Services Authorized* IE is contained in the HANDOVER REQUEST message and it contains one or more IEs set to "authorized", the NG-RAN node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *LTE V2X Services Authorized* IE is contained in the HANDOVER REQUEST message and it contains one or more IEs set to "authorized", the NG-RAN node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *NR A2X Services Authorized* IE is contained in the HANDOVER REQUEST message and it contains one or more IEs set to "authorized", the NG-RAN node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *LTE A2X Services Authorized* IE is contained in the HANDOVER REQUEST message and it contains one or more IEs set to "authorized", the NG-RAN node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *NR UE Sidelink Aggregate Maximum Bit Rate* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use the received value for the concerned UE's sidelink communication in network scheduled mode for NR V2X services.

If the *LTE UE Sidelink Aggregate Maximum Bit Rate* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use the received value for the concerned UE's sidelink communication in network scheduled mode for LTE V2X services.

If the *NR A2X UE PC5 Aggregate Maximum Bit Rate* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use the received value for the concerned UE's sidelink communication in network scheduled mode for NR A2X services.

If the *LTE A2X UE PC5 Aggregate Maximum Bit Rate* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use the received value for the concerned UE's sidelink communication in network scheduled mode for LTE A2X services.

If the *PC5 QoS Parameters* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use it as defined in TS 23.287 [33].

If the *A2X PC5 QoS Parameters* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use it as defined in TS 23.256 [54].

If the *CE-mode-B Restricted* IE is included in the HANDOVER REQUEST message and the *Enhanced Coverage Restriction* IE is not set to "restricted" and the Enhanced Coverage Restriction information stored in the UE context is not set to "restricted", the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *Management Based MDT PLMN List* IE is contained in the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, store the received information in the UE context, and use this information to allow subsequent selections of the UE for management based MDT defined in TS 32.422 [11].

If the HANDOVER REQUEST message contains the *UE Radio Capability ID* IE, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9] and TS 23.502 [10].

If the *DAPS Request Information* IE is included for a DRB in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, the target NG-RAN node shall consider that the request concerns a DAPS Handover for that DRB, as described in TS 38.300 [8]. The target NG-RAN node shall include the *DAPS Response information List* IE in the *Target NG-RAN Node to Source NG-RAN Node Transparent*

Container IE within the HANDOVER REQUEST ACKNOWLEDGE message, containing the *DAPS Response Information* IE for each DRB requested to be configured with DAPS Handover.

If the *Extended Connected Time* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use it as described in TS 23.501 [9].

If the target NG-RAN node receives the *UE Context Reference at Source* IE in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, it may use it to identify an existing UE.

If the *Source Node ID* IE is included in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, use it to decide whether direct forwarding path is available between the target NG-RAN node and this source RAN node. If the direct forwarding path is available, the target NG-RAN node shall include the *Direct Forwarding Path Availability* IE in the *Target NG-RAN Node to Source NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST ACKNOWLEDGE message.

In case there are MBS sessions the UE has joined, for all the MBS sessions the UE has joined, the SMF shall, if supported, include the *MBS Session Setup Request List* IE within the *PDU Session Resource Setup Request Transfer* IE in the HANDOVER REQUEST message.

If the HANDOVER REQUEST message contains the *MBS Session Setup Request List* IE in a *PDU Session Resource Setup Request Transfer* IE the NG-RAN node shall, if supported, use it as specified in TS 23.247 [44] and TS 38.300 [8].

If the *MBS Active Session Information Source to Target List* IE is contained in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, assume the indicated MBS sessions to be active and establish MBS session resources as specified in TS 23.247 [44] and TS 38.300 [8], if applicable. The target NG-RAN node shall, if supported, consider that the MBS sessions the UE has joined which are not included in the *MBS Active Session Information Source to Target List* IE are inactive.

If the *MBS Area Session ID* IE is included in the *MBS Active Session Information Source to Target List* IE in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, the target NG-RAN shall use this information as indication from which MBS Area Session ID the UE is handed over.

If the *MBS Service Area* IE is included in the *MBS Active Session Information Source to Target List* IE in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, the target NG-RAN shall use this information to setup respective MBS session resources, if applicable.

If the target NG-RAN node decides to allocate resource for data forwarding for an active MBS session, respective information is provided for that MBS session within the *Data Forwarding Response MRB List* IE in the *MBS Active Session Information Target to Source List* IE in the *Target NG-RAN Node to Source NG-RAN Node Transparent Container* IE.

If the *Time Synchronisation Assistance Information* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, store the information in the UE context and use it as defined in TS 23.501 [9].

If the *5G ProSe Authorized* IE is contained in the HANDOVER REQUEST message and it contains one or more IEs set to "authorized", the NG-RAN node shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *5G ProSe UE PC5 Aggregate Maximum Bit Rate* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use the received value for the concerned UE's sidelink communication in network scheduled mode for 5G ProSe services.

If the *5G ProSe PC5 QoS Parameters* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use it as defined in TS 23.304 [47].

If for a given QoS flow the *Source Transport Layer Address* IE is included within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE of the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, store this information and use it as part of its ACL functionality configuration actions for direct data forwarding, if such ACL functionality is deployed and if direct forwarding path is available between the target NG-RAN node and this source RAN node.

If for a given QoS flow the *Source Node Transport Layer Address* IE is included within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE of the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, store this information and use it as part of its ACL functionality configuration actions for direct data forwarding, if such ACL functionality is deployed and if direct forwarding path is available between the target NG-RAN node and this source RAN node.

If for a given E-RAB the *Source Transport Layer Address* IE is included within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE of the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, store this information and use it as part of its ACL functionality configuration actions for direct data forwarding, if such ACL functionality is deployed and if direct forwarding path is available between the target NG-RAN node and this source RAN node.

If for a given E-RAB the *Source Node Transport Layer Address* IE is included within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE of the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, store this information and use it as part of its ACL functionality configuration actions for direct data forwarding, if such ACL functionality is deployed and if direct forwarding path is available between the target NG-RAN node and this source RAN node.

If the HANDOVER REQUEST message contains within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE the *NGAP IE Support Information Request List* IE, the target NG-RAN node shall, if supported and the target NG-RAN node accepts the request for handover, for each included NGAP Protocol IE-Id provided within the *Target NG-RAN Node to Source NG-RAN Node Transparent Container* IE in the HANDOVER REQUEST ACKNOWLEDGE message

- set the *NGAP Protocol IE Support Information* IE to "supported" if the target NG-RAN node has information that the functionality associated with the indicated IE is supported
- set the *NGAP Protocol IE Support Information* IE to "not-supported" if the target NG-RAN node has information that the functionality associated with the indicated IE is not supported

on the interface instance via which the HANDOVER REQUEST message has been received, and

- set the *NGAP Protocol IE Presence Information* IE to "present" if the target NG-RAN node has received the respective NGAP Protocol IE-Id in the HANDOVER REQUEST message, and "not-present" otherwise.

If the HANDOVER REQUEST message contains within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE the *Time Based Handover Information* IE, the target NG-RAN node may use this information to allocate necessary resources for the incoming handover.

If the *Candidate Relay UE Information List* IE is included in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, use it to configure the path switch to indirect path as specified in TS 38.300 [8].

If the *QMC Configuration Information* IE is included in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, take it into account for QoE management handling, as described in TS 38.300 [8].

If the *Source SN to Target SN QMC Information* IE is included in the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE within the HANDOVER REQUEST message, the target NG-RAN node shall, if supported, take it into account for QoE management handling, as described in TS 37.340 [32].

If the *Aerial UE Subscription Information* IE is included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 38.300 [8].

If the *PNI-NPN Area Scope of MDT* IE is included in the *MDT Configuration-NR* IE included in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, use it to derive the MDT area scope for MDT measurement collection in PNI-NPN areas. Upon reception of the *PNI-NPN Area Scope of MDT* IE, the NG-RAN node shall consider that the area scope for MDT measurement collection in PNI-NPN areas is defined only by the areas included in the *PNI-NPN Area Scope of MDT* IE.

If the *Partially Allowed NSSAI* IE is contained in the HANDOVER REQUEST message, the NG-RAN node shall, if supported, deduce from it the partially allowed network slices for the UE, store and replace any previously received Partially Allowed NSSAI and use it as specified in TS 23.501 [9].

If the *AMF UE NGAP ID 2* IE is included in the *HANDOVER REQUEST* message, the target NG-RAN node shall, if supported, store this information in the UE context and use it for subsequent Xn handovers.

If the *Target RAN Node ID* IE is included in the *HANDOVER REQUEST* message, the target NG-RAN node shall, if supported, use it for routing the *HANDOVER REQUEST* message, as specified in TS 38.300 [8].

If the *Network Slice Area Scope of MDT* IE is included in the *MDT Configuration-NR* IE included in the *HANDOVER REQUEST* message, the NG-RAN node shall, if supported, use it to derive the MDT area scope for MDT measurement collection. Upon reception of the *Network Slice Area Scope of MDT* IE, the NG-RAN node shall consider that the area scope for MDT measurement collection is defined only by the *Network Slice Area Scope of MDT* IE and the *Area Scope of MDT* IE.

If the *MBS Support Indicator* IE is included in the *Handover Request Acknowledge Transfer* IE in the *HANDOVER REQUEST ACKNOWLEDGE* message, the SMF shall, if supported, handle this information as specified in TS 23.247 [44].

If the *ECN Marking or Congestion Information Reporting Status* IE is included in the *Handover Request Acknowledge Transfer* IE, the SMF shall, if supported, use it to deduce if ECN marking at NG-RAN or ECN marking at UPF or congestion information reporting is active or not active as described in TS 23.501 [9].

If the S-NSSAI dedicated to WAB-MT's backhaul PDU session(s) is included in the *PDU Session Resource Setup List* IE in the *HANDOVER REQUEST* message, the target NG-RAN node shall act as defined in TS 38.401 [2].

Interactions with RRC Inactive Transition Report procedure:

If the *RRC Inactive Transition Report Request* IE is included in the *HANDOVER REQUEST* message and set to "subsequent state transition report", the NG-RAN node shall, if supported, send the *RRC INACTIVE TRANSITION REPORT* message to the AMF to report the RRC state of the UE when the UE enters or leaves *RRC_INACTIVE* state.

8.4.2.3 Unsuccessful Operation



Figure 8.4.2.3-1: Handover resource allocation: unsuccessful operation

If the target NG-RAN node does not admit any of the PDU session resources, or a failure occurs during the Handover Preparation, it shall send the *HANDOVER FAILURE* message to the AMF with an appropriate cause value.

If the *HANDOVER REQUEST* message contains within the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE the *NGAP IE Support Information Request List* IE, the target NG-RAN node shall, if supported and the target NG-RAN node does not accept the request for handover, for each included NGAP Protocol IE-Id provided within the *Target NG-RAN Node to Source NG-RAN Node Failure Transparent Container* IE in the *HANDOVER FAILURE* message

- set the *NGAP Protocol IE Support Information* IE to "supported" if the target NG-RAN node has information that the functionality associated with the indicated IE is supported
- set the *NGAP Protocol IE Support Information* IE to "not-supported" if the target NG-RAN node has information that the functionality associated with the indicated IE is not supported

on the interface instance via which the *HANDOVER REQUEST* message has been received, and

- set the *NGAP Protocol IE Presence Information* IE to "present" if the target NG-RAN node has received the respective NGAP Protocol IE-Id in the *HANDOVER REQUEST* message, and "not-present" otherwise.

8.4.2.4 Abnormal Conditions

If the supported algorithms for encryption defined in the *Encryption Algorithms* IE in the *UE Security Capabilities* IE, plus the mandated support of EEA0 and NEA0 in all UEs (TS 33.501 [13]), do not match any allowed algorithms defined in the configured list of allowed encryption algorithms in the NG-RAN node (TS 33.501 [13]), the target NG-RAN node shall reject the procedure using the HANDOVER FAILURE message.

If the supported algorithms for integrity defined in the *Integrity Protection Algorithms* IE in the *UE Security Capabilities* IE, plus the mandated support of the EIA0 and NIA0 algorithm in all UEs (TS 33.501 [13]), do not match any allowed algorithms defined in the configured list of allowed integrity protection algorithms in the NG-RAN node (TS 33.501 [13]), the target NG-RAN node shall reject the procedure using the HANDOVER FAILURE message.

If the target NG-RAN node receives a HANDOVER REQUEST message which does not contain the *Mobility Restriction List* IE, and the serving PLMN cannot be determined otherwise by the NG-RAN node, the target NG-RAN node shall reject the procedure using the HANDOVER FAILURE message.

If the target NG-RAN node receives a HANDOVER REQUEST message containing the *Mobility Restriction List* IE, and the serving PLMN indicated is not supported by the target cell, the target NG-RAN node shall reject the procedure using the HANDOVER FAILURE message.

If the target NG-RAN node receives a HANDOVER REQUEST message containing an *Allowed PN1-NPN List* IE in the *Mobility Restriction List* IE which does not allow access to the cell indicated in the *Target Cell ID* IE, the target NG-RAN node shall reject the procedure using the HANDOVER FAILURE message with an appropriate cause value and may include the *Cell CAG Information* IE corresponding to this cell and the selected PLMN.

If the target NG-RAN node receives a HANDOVER REQUEST message containing a *Serving PLMN* IE and *Serving NID* IE in the *Mobility Restriction List* IE which does not allow access to the cell indicated in the *Target Cell ID* IE, the target NG-RAN node shall reject the procedure using the HANDOVER FAILURE message with an appropriate cause value.

If the *PN1-NPN Area Scope of MDT* IE is included in the *MDT Configuration-NR* IE in the HANDOVER REQUEST message, and the *Area Scope of MDT* IE is set to "PN1-NPN Based MDT", the NG-RAN node shall, if supported, use the *Area Scope of MDT* IE to derive the MDT area scope for MDT measurement collection in PN1-NPN areas, and ignore the *PN1-NPN Area Scope of MDT* IE.

If the *Partially Allowed NSSAI* IE is received in the HANDOVER REQUEST message and the total number of S-NSSAIs included in the Allowed NSSAI and Partially Allowed NSSAI exceeds eight, the NG-RAN node shall consider the procedure as failed.

If any of the S-NSSAI which is present in the *Partially Allowed NSSAI* IE is also present in the *Allowed NSSAI* IE, the NG-RAN node shall consider the procedure as failed.

If the *Network Slice Area Scope of MDT* IE is included in the *MDT Configuration-NR* IE in the HANDOVER REQUEST message, and the *MDT Activation* IE is set to "Logged MDT only", the NG-RAN node shall ignore the *Network Slice Area Scope of MDT* IE.

8.4.3 Handover Notification

8.4.3.1 General

The purpose of the Handover Notification procedure is to indicate to the AMF that the UE has arrived to the target cell and the NG-based handover has been successfully completed. The procedure uses UE-associated signalling.

8.4.3.2 Successful Operation



Figure 8.4.3.2-1: Handover notification

The target NG-RAN node shall send the HANDOVER NOTIFY message to the AMF when the UE has been identified in the target cell and the NG-based handover has been successfully completed.

Interactions with Handover Success procedure:

If the *Notify Source NG-RAN Node* IE is included in the HANDOVER NOTIFY message, the AMF shall, if supported, notify the source NG-RAN node that the UE has successfully accessed the target NG-RAN node by sending the HANDOVER SUCCESS message.

8.4.3.3 Abnormal Conditions

Void.

8.4.4 Path Switch Request

8.4.4.1 General

The purpose of the Path Switch Request procedure is to establish a UE associated signalling connection to the 5GC and, if applicable, to request the switch of the downlink termination point of the NG-U transport bearer towards a new termination point. The procedure uses UE-associated signalling.

8.4.4.2 Successful Operation

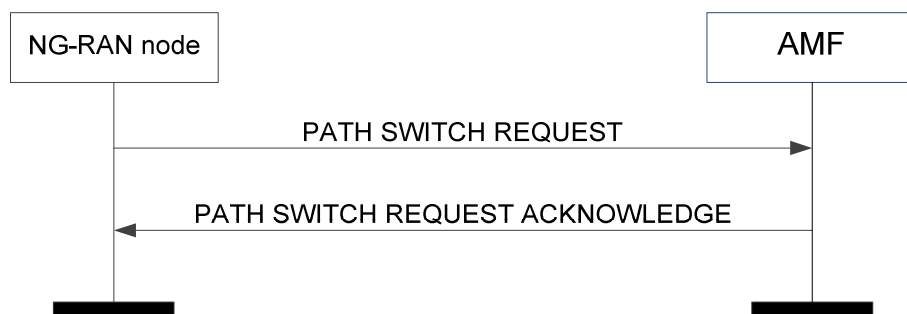


Figure 8.4.4.2-1: Path switch request: successful operation

The NG-RAN node initiates the procedure by sending the PATH SWITCH REQUEST message to the AMF. Upon reception of the PATH SWITCH REQUEST message the AMF shall, for each PDU session indicated in the *PDU Session ID* IE, transparently transfer the *Path Switch Request Transfer* IE to the SMF associated with the concerned PDU session.

Upon reception of the PATH SWITCH REQUEST message the AMF shall deactivate MT communication handling if already activated, as specified in TS 23.502 [10].

If the *RRC Resume Cause* IE is included in the PATH SWITCH REQUEST message, the AMF shall, if supported, use it as described in TS 23.502 [10] for User Plane ClIoT 5GS Optimisation when the NG-RAN node is an ng-eNB.

If the *RedCap Indication* IE or the *eRedCap Indication* IE is included in the PATH SWITCH REQUEST message, the AMF shall, if supported, consider the UE respectively as a RedCap UE or an eRedCap UE that was previously served by a E-UTRA cell, and use the IE according to TS 23.501 [9].

After all necessary updates including the UP path switch have been successfully completed in the 5GC for at least one of the PDU session resources included in the PATH SWITCH REQUEST, the AMF shall send the PATH SWITCH REQUEST ACKNOWLEDGE message to the NG-RAN node and the procedure ends.

The list of accepted QoS flows shall be included in the PATH SWITCH REQUEST message within the *Path Switch Request Transfer* IE. The SMF shall handle this information as specified in TS 23.502 [10].

If the GUAMI of the AMF currently serving the UE is available at the gNB (see TS 38.300 [8]), the NG-RAN node shall, if supported, include the *Source AMF GUAMI* IE within the PATH SWITCH REQUEST message.

For an IAB-MT or a mobile IAB-MT for which the *PDU Session ID* IE contained in the PATH SWITCH REQUEST message indicates no PDU session identity assigned as defined in TS 24.007 [58], the AMF shall, if supported, consider that the IAB-MT or the mobile IAB-MT, respectively, has no PDU session, and behave as specified in TS 23.501 [9]. Subsequently, the NG-RAN node shall, if supported, ignore the *PDU Session Resource Switched List* IE in the PATH SWITCH REQUEST ACKNOWLEDGE message.

For each PDU session for which the *Additional DL QoS Flow per TNL Information* IE is included in the *Path Switch Request Transfer* IE of the PATH SWITCH REQUEST message, the SMF may use each included UP transport layer information as the downlink termination point for the included associated QoS flows for this PDU session split in different tunnels.

The list of PDU sessions which failed to be setup, if any, shall be included in the PATH SWITCH REQUEST message within the *Path Switch Request Setup Failed Transfer* IE. The AMF shall handle this information as specified in TS 23.502 [10].

For each PDU session for which the *User Plane Security Information* IE is included in the *Path Switch Request Transfer* IE of the PATH SWITCH REQUEST message, the SMF shall behave as specified in TS 33.501 [13] and may send back the *Security Indication* IE within the *Path Switch Request Acknowledge Transfer* IE of the PATH SWITCH REQUEST ACKNOWLEDGE message.

For each PDU session for which the *DL NG-U TNL Information Reused* IE set to "true" is included in the *Path Switch Request Transfer* IE of the PATH SWITCH REQUEST message, the SMF shall, if supported, consider that the DL TNL information contained in the *DL NG-U UP TNL Information* IE has been reused.

For each PDU session for which the *Additional Redundant DL QoS Flow per TNL Information* IE is included in the *Path Switch Request Transfer* IE of the PATH SWITCH REQUEST message, the SMF may use each included UP transport layer information as the downlink termination point for the included associated QoS flows for this PDU session split in different tunnels for the redundant transmission.

For each PDU session for which the *Redundant DL NG-U TNL Information Reused* IE is included in the *Path Switch Request Transfer* IE of the PATH SWITCH REQUEST message, the SMF shall, if supported, consider the included DL transport layer address as the DL transport layer address for the redundant transmission as specified in TS 23.501 [9].

For each PDU session for which the *Global RAN Node ID of Secondary NG-RAN Node* IE is included in the *Path Switch Request Transfer* IE of the PATH SWITCH REQUEST message, the SMF shall, if supported, handle this information as specified in TS 23.501 [9].

For each PDU session included in the PATH SWITCH REQUEST message, if the *Current QoS Parameters Set Index* IE is included in the *Path Switch Request Transfer* IE the SMF shall consider it as the currently fulfilled QoS parameters set among the alternative QoS parameters for the involved QoS flow.

The NG-RAN node shall, if supported, report in the PATH SWITCH REQUEST message the *PDU Set based Handling Indicator* IE in the *Path Switch Request Transfer* IE. If the *PDU Set based Handling Indicator* IE is included in the *Path Switch Request Transfer* IE in the PATH SWITCH REQUEST message, the SMF shall, if supported, handle this information as specified in TS 23.501 [9].

If the *MBS Support Indicator* IE is included in the *Path Switch Request Transfer* IE in the PATH SWITCH REQUEST message, the SMF shall, if supported, handle this information as specified in TS 23.247 [44].

The NG-RAN node shall, if supported, report in the PATH SWITCH REQUEST message the *ECN Marking or Congestion Information Reporting Status IE* in the *Path Switch Request Transfer IE*. If the *ECN Marking or Congestion Information Reporting Status IE* is included in the *Path Switch Request Transfer IE* in the PATH SWITCH REQUEST message, the SMF shall, if supported, use it to deduce if ECN marking at NG-RAN or ECN marking at UPF or congestion information reporting is active or not active as described in TS 23.501 [9].

If the *Security Indication IE* is included within the *Path Switch Request Acknowledge Transfer IE* of the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall behave as specified in TS 33.501 [13].

If the *UL NG-U UP TNL Information IE* is included within the *Path Switch Request Acknowledge Transfer IE* of the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall store this information and use it as the uplink termination point for the user plane data for this PDU session.

If the *Additional NG-U UP TNL Information IE* is included within the *Path Switch Request Acknowledge Transfer IE* of the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall store this information and use the included *UL NG-U UP TNL Information IE(s)* as the uplink termination point(s) of the user plane data for this PDU session split in different tunnel.

If the *Redundant UL NG-U UP TNL Information IE* is included within the *Path Switch Request Acknowledge Transfer IE* of the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store this information and use it as the uplink termination point for the user plane data for the redundant transmission for this PDU session as specified in TS 23.501 [9].

If the *Additional Redundant NG-U UP TNL Information IE* is included within the *Path Switch Request Acknowledge Transfer IE* of the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store this information and use the included *UL NG-U UP TNL Information IE(s)* as the uplink termination point(s) of the user plane data for this PDU session split in different tunnel.

If the *CN Packet Delay Budget Downlink IE* is included within the *Path Switch Request Acknowledge Transfer IE* of the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, replace the previously provided CN Packet Delay Budget Downlink if any and use it as specified in TS 23.502 [10].

If the *CN Packet Delay Budget Uplink IE* is included within the *Path Switch Request Acknowledge Transfer IE* of the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, replace the previously provided CN Packet Delay Budget Uplink if any and use it as specified in TS 23.502 [10].

If the *Burst Arrival Time Downlink IE* is included within the *Path Switch Request Acknowledge Transfer IE* of the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, replace the previously provided value if any and use it as specified in TS 23.502 [10].

If the *Core Network Assistance Information for RRC INACTIVE IE* is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store this information in the UE context and use it for the RRC_INACTIVE state decision and RNA configuration for the UE, and RAN paging, if any, for a UE in RRC_INACTIVE state, as specified in TS 38.300 [8]. In particular,

- if the *UE Specific DRX IE* is included in the *Core Network Assistance Information for RRC INACTIVE IE*, the NG-RAN node shall, if supported, use it for RAN paging.
- if the *MICO Mode Indication IE* is included in the *Core Network Assistance Information for RRC INACTIVE IE*, the NG-RAN node shall, if supported, consider that the UE is configured with MICO mode by the AMF as specified in TS 23.501 [9].
- if the *Expected UE Behaviour IE* is included in the *Core Network Assistance Information for RRC INACTIVE IE*, the NG-RAN node shall, if supported, use it to assist the UE RRC state transition decision.
- if the *E-UTRA Paging eDRX Information IE* is included in the *Core Network Assistance Information for RRC INACTIVE IE*, the NG-RAN node shall, if supported, take it into account to configure the RAN Paging when the UE is in RRC_INACTIVE as specified in TS 36.300 [17].
- if the *Extended UE Identity Index Value IE* is included in the *Core Network Assistance Information for RRC INACTIVE IE*, the NG-RAN node shall, if supported, use it for RAN Paging.
- if the *UE Radio Capability for Paging IE* is included in the *Core Network Assistance Information for RRC INACTIVE IE*, the NG-RAN node shall, if supported, use it for subsequent RAN paging.

- if the *MICO All PLMN* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE the NG-RAN node shall, if supported, consider that the registration area for the UE is the full PLMN and ignore the *TAI List for RRC Inactive* IE.
- if the *NR Paging eDRX Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, take it into account to configure the RAN Paging when the UE is in RRC_INACTIVE as described in TS 38.300 [8].
- if the *Paging Cause Indication for Voice Service* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store and use it as specified in TS 38.300 [8]. If the *PEIPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].
- if the *PEIPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].
- if the *Hashed UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, use it to determine the start point of PTW.
- if the *CN MT Communication Handling* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and may subsequently request, based on implementation, the CN for MT communication handling as described in TS 23.502 [10].
- if the *LP-WUSPS Assistance Information* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it for LP-WUS paging subgrouping the UE in RRC_INACTIVE state, as specified in TS 38.300 [8].
- if the *Further Extended UE Identity Index Value* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it as specified in TS 38.304 [12].
- If the *LP-WUS Disable Indication* IE is included in the *Core Network Assistance Information for RRC INACTIVE* IE, the NG-RAN node shall, if supported, store it and use it to disable LP-WUS as described in TS 38.300 [8].

If the *CN Assisted RAN Parameters Tuning* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node may use it as described in TS 23.501 [9].

If the *RRC Inactive Transition Report Request* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store this information in the UE context.

If the *New Security Context Indicator* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall use the information as specified in TS 33.501 [13].

Upon reception of the PATH SWITCH REQUEST ACKNOWLEDGE message the NG-RAN node shall store the received *Security Context* IE in the UE context and the NG-RAN node shall use it as specified in TS 33.501 [13].

If the *UE Security Capabilities* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall handle it accordingly (TS 33.501 [13]).

If the *Redirection for Voice EPS Fallback* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store it and use it in a subsequent decision of EPS fallback for voice as specified in TS 23.502 [10].

If the *PDU Session Resource Released List* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall release the corresponding QoS flows and regard the PDU session(s) indicated in the *PDU Session Resource Released List* IE as being released. The appropriate cause value for each PDU session released is included in the *Path Switch Request Unsuccessful Transfer* IE contained in the PATH SWITCH REQUEST ACKNOWLEDGE message.

If the *SRVCC Operation Possible* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store the content of the received *SRVCC Operation Possible* IE in the UE context and use it as defined in TS 23.216 [31].

If the *Enhanced Coverage Restriction* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *Extended Connected Time* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, use it as described in TS 23.501 [9].

If the *UE Differentiation Information* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store this information in the UE context for further use according to TS 23.501 [9].

If the *NR V2X Services Authorized* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, update its NR V2X services authorization information for the UE accordingly. If the *NR V2X Services Authorized* IE includes one or more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *LTE V2X Services Authorized* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, update its LTE V2X services authorization information for the UE accordingly. If the *LTE V2X Services Authorized* IE includes one or more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *NR A2X Services Authorized* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, update its NR A2X services authorization information for the UE accordingly. If the *NR A2X Services Authorized* IE includes one or more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *LTE A2X Services Authorized* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, update its LTE A2X services authorization information for the UE accordingly. If the *LTE A2X Services Authorized* IE includes one or more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant service(s).

If the *NR UE Sidelink Aggregate Maximum Bit Rate* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported:

- replace the previously provided UE Sidelink Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for NR V2X services.

If the *LTE UE Sidelink Aggregate Maximum Bit Rate* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported:

- replace the previously provided UE Sidelink Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for LTE V2X services.

If the *NR A2X UE PC5 Aggregate Maximum Bit Rate* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported:

- replace the previously provided NR A2X UE PC5 Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for NR A2X services.

If the *LTE A2X UE PC5 Aggregate Maximum Bit Rate* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported:

- replace the previously provided LTE A2X UE PC5 Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for LTE A2X services.

If the *PC5 QoS Parameters* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, use it as defined in TS 23.287 [33].

If the *A2X PC5 QoS Parameters* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, use it as defined in TS 23.256 [54].

If the *CE-mode-B Restricted* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message and the *Enhanced Coverage Restriction* IE is not set to "restricted" and the Enhanced Coverage Restriction information stored in the UE context is not set to "restricted", the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *UE User Plane CIoT Support Indicator* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message the NG-RAN node shall, if supported, store this information in the UE context and consider that User Plane CIoT 5GS Optimisation as specified in TS 23.501 [9] is supported for the UE.

If the PATH SWITCH REQUEST ACKNOWLEDGE message contains the *UE Radio Capability ID* IE, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9] and TS 23.502 [10].

If the PATH SWITCH REQUEST ACKNOWLEDGE message contains the *Alternative QoS Parameters Set List* IE, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9].

For each PDU session, if the *PDU Session Expected UE Activity Behaviour* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, handle this information as specified in TS 23.501 [9].

If the PATH SWITCH REQUEST ACKNOWLEDGE message contains the *Management Based MDT PLMN List* IE, the NG-RAN node shall store it in the UE context, and if supported, use it to allow subsequent selection of the UE for management based MDT defined in TS 32.422 [11].

If the PATH SWITCH REQUEST ACKNOWLEDGE message contains the *Management Based MDT PLMN Modification List* IE, the NG-RAN node shall, if supported, use it to overwrite any previously stored management based MDT PLMN list information in the UE context and use the received information to allow subsequent selection of the UE for management based MDT defined in TS 32.422 [11].

If the *Time Synchronisation Assistance Information* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store the information in the UE context and use it as defined in TS 23.501 [9].

If the *5G ProSe Authorized* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, update its ProSe authorization information for the UE accordingly. If the *5G ProSe Authorized* IE includes one and more IEs set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the relevant 5G ProSe service(s).

If the *5G ProSe UE PC5 Aggregate Maximum Bit Rate* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported:

- replace the previously provided 5G ProSe UE PC5 Aggregate Maximum Bit Rate, if available in the UE context, with the received value;
- use the received value for the concerned UE's sidelink communication in network scheduled mode for 5G ProSe services.

If the *5G ProSe PC5 QoS Parameters* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, use it as defined in TS 23.304 [47].

If the *Aerial UE Subscription Information* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store the information or overwrite any previously stored information in the UE context and use it as defined in TS 38.300 [8].

If the *IAB Authorized* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store the received IAB Authorization information in the UE context and use it as specified in TS 38.401 [2].

If the *Mobile IAB Authorized* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store the received mobile IAB Authorization status in the UE context of the mobile IAB-

MT. If the *Mobile IAB Authorized* IE for a mobile IAB-MT is set to "not authorized", the NG-RAN node shall, if supported, ensure that the mobile IAB-node will not serve any UE(s).

If the *Partially Allowed NSSAI* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, deduce from it the partially allowed network slices for the UE, store and replace any previously received Partially Allowed NSSAI and use it as specified in TS 23.501 [9].

If the *Ranging and Sidelink Positioning Service Information* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, update the Ranging and Sidelink Positioning service information for the UE accordingly. If the *Ranging and Sidelink Positioning Authorized* IE within the *Ranging and Sidelink Positioning Service Information* IE is set to "not authorized", the NG-RAN node shall, if supported, initiate actions to ensure that the UE is no longer accessing the Ranging and Sidelink Positioning service.

If the *AMF UE NGAP ID 2* IE is contained in the PATH SWITCH REQUEST ACKNOWLEDGE message, the NG-RAN node shall, if supported, store this information in the UE context and use it for subsequent Xn handovers.

Interactions with RRC Inactive Transition Report procedure:

If the *RRC Inactive Transition Report Request* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message and set to "single RRC connected state report" and the UE is in RRC_CONNECTED state, the NG-RAN node shall, if supported, send one RRC INACTIVE TRANSITION REPORT message to the AMF to report the RRC state of the UE.

If the *RRC Inactive Transition Report Request* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message and set to "single RRC connected state report" and the UE is in RRC_INACTIVE state, the NG-RAN node shall, if supported, send to the AMF one RRC INACTIVE TRANSITION REPORT message plus one subsequent RRC INACTIVE TRANSITION REPORT message when the RRC state transitions to RRC_CONNECTED state.

If the *RRC Inactive Transition Report Request* IE is included in the PATH SWITCH REQUEST ACKNOWLEDGE message and set to "subsequent state transition report", the NG-RAN node shall, if supported, send one RRC INACTIVE TRANSITION REPORT message to the AMF to report the RRC state of the UE and subsequent RRC INACTIVE TRANSITION REPORT messages to report the RRC state of the UE when the UE enters or leaves RRC_INACTIVE state.

Interactions with PDU Session Resource Notify procedure:

If the QoS related parameters (e.g. the *CN Packet Delay Budget Downlink* IE or the *CN Packet Delay Budget Uplink* IE) are included in the *Path Switch Request Acknowledge Transfer* IE of the PATH SWITCH REQUEST ACKNOWLEDGE message, but cannot be successfully accepted by the NG-RAN node, the NG-RAN node should continue to use the old values received from the source NG-RAN node, if any. The NG-RAN node shall, if supported, send the PDU SESSION RESOURCE NOTIFY message to notify the AMF.

8.4.4.3 Unsuccessful Operation

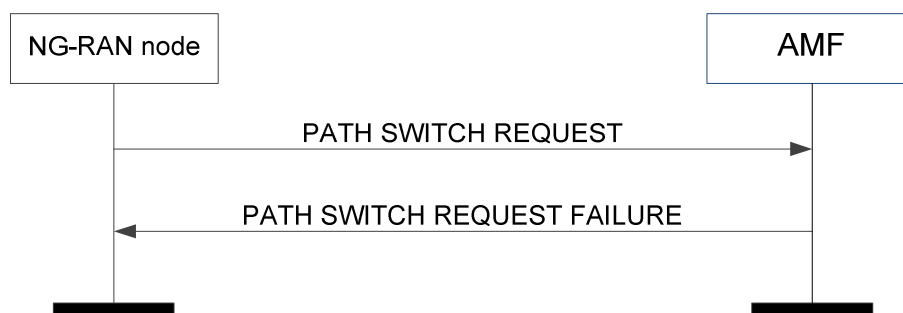


Figure 8.4.4.3-1: Path switch request: unsuccessful operation

If the 5GC fails to switch the downlink termination point of the NG-U transport bearer towards a new termination point for all PDU session resources, the AMF shall send the PATH SWITCH REQUEST FAILURE message to the NG-RAN node.

The NG-RAN node shall release the corresponding QoS flows and regard the PDU session(s) indicated in the *PDU Session Resource Released List* IE included in the PATH SWITCH REQUEST FAILURE message as being released.

The appropriate cause value for each PDU session released is included in the *Path Switch Request Unsuccessful Transfer* IE contained in the PATH SWITCH REQUEST FAILURE message.

8.4.4.4 Abnormal Conditions

If the AMF receives a PATH SWITCH REQUEST message containing several *PDU Session ID* IEs (in the *PDU Session Resource to be Switched in Downlink List* IE) set to the same value, the AMF shall send the PATH SWITCH REQUEST FAILURE message to the NG-RAN node.

NOTE: As an exception, the AMF generates the *Path Switch Request Unsuccessful Transfer* IE.

If the *Partially Allowed NSSAI* IE is received in the PATH SWITCH REQUEST ACKNOWLEDGE message and the total number of S-NSSAIs included in the Allowed NSSAI and Partially Allowed NSSAI exceeds eight, the NG-RAN node shall consider the procedure as failed.

If any of the S-NSSAI which is present in the *Partially Allowed NSSAI* IE is also present in the *Allowed NSSAI* IE, the NG-RAN node shall consider the procedure as failed.

8.4.5 Handover Cancellation

8.4.5.1 General

The purpose of the Handover Cancellation procedure is to enable a source NG-RAN node to cancel an ongoing handover preparation or an already prepared handover. The procedure uses UE-associated signalling.

8.4.5.2 Successful Operation

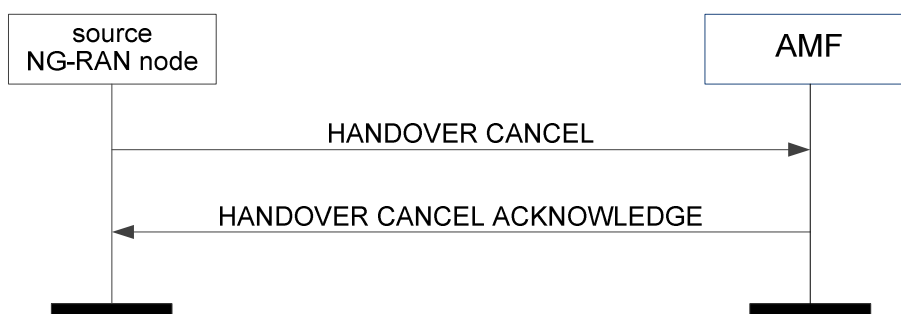


Figure 8.4.5.2-1: Handover cancel: successful operation

The source NG-RAN node initiates the procedure by sending a HANDOVER CANCEL message to the AMF.

8.4.5.3 Unsuccessful Operation

Not applicable.

8.4.5.4 Abnormal Conditions

If the source NG-RAN node becomes aware of the fact that an expected HANDOVER CANCEL ACKNOWLEDGE message is missing, the source NG-RAN node shall consider the Handover Cancellation procedure as successfully terminated.

8.4.6 Uplink RAN Status Transfer

8.4.6.1 General

The purpose of the Uplink RAN Status Transfer procedure is to enable lossless NG-based handover. The procedure uses UE-associated signalling.

8.4.6.2 Successful Operation

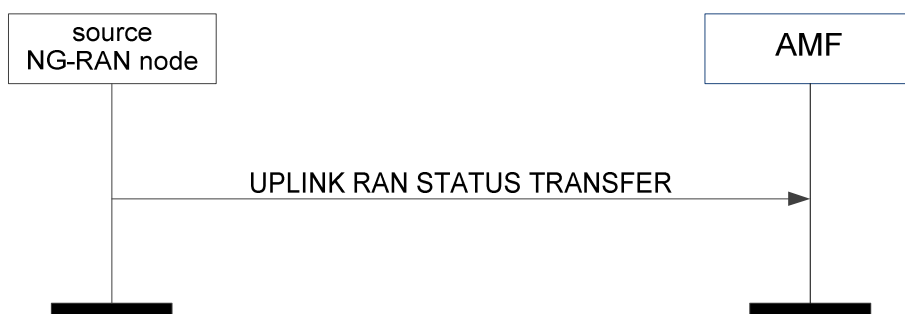


Figure 8.4.6.2-1: Uplink RAN status transfer

The source NG-RAN node initiates the procedure by stopping the assigning of PDCP-SNs to downlink SDUs and sending the UPLINK RAN STATUS TRANSFER message to the AMF at the point in time when it considers the transmitter/receiver status to be frozen.

For each DRB for which PDCP-SN and HFN status preservation applies, the source NG-RAN node shall include the *DRB ID* IE, the *UL COUNT Value* IE and the *DL COUNT Value* IE within the *DRBs Subject to Status Transfer List* IE in the *RAN Status Transfer Transparent Container* IE of the UPLINK RAN STATUS TRANSFER message.

The source NG-RAN node may also include in the UPLINK RAN STATUS TRANSFER message the missing and the received uplink SDUs in the *Receive Status of UL PDCP SDUs* IE for each DRB for which the source NG-RAN node has accepted the request from the target NG-RAN node for uplink forwarding.

8.4.6.3 Abnormal Conditions

Void.

8.4.7 Downlink RAN Status Transfer

8.4.7.1 General

The purpose of the Downlink RAN Status Transfer procedure is to enable lossless NG-based handover. The procedure uses UE-associated signalling.

8.4.7.2 Successful Operation

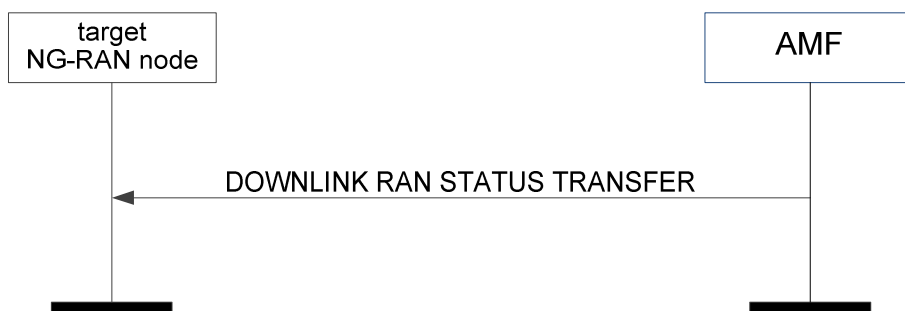


Figure 8.4.7.2-1: Downlink RAN status transfer

The AMF initiates the procedure by sending the DOWNLINK RAN STATUS TRANSFER message to the target NG-RAN node. The target NG-RAN node using Full Configuration for this handover as per TS 38.300 [8] shall ignore the information received in this message.

For each DRB in the *DRBs Subject to Status Transfer List* IE within the *RAN Status Transfer Transparent Container* IE, the target NG-RAN node shall not deliver any uplink packet which has a PDCP-SN lower than the value of the *UL Count Value* IE.

For each DRB in the *DRBs Subject to Status Transfer List* IE within the *RAN Status Transfer Transparent Container* IE, the target NG-RAN node shall use the value of the *DL COUNT Value* IE for the first downlink packet for which there is no PDCP-SN yet assigned.

If the *Receive Status of UL PDCP SDUs* IE is included for at least one DRB in the *RAN Status Transfer Transparent Container* IE of the DOWNLINK RAN STATUS TRANSFER message, the target NG-RAN node may use it in a Status Report message sent to the UE over the radio interface.

8.4.7.3 Abnormal Conditions

If the target NG-RAN node receives this message for a UE for which no prepared handover exists at the target NG-RAN node, the target NG-RAN node shall ignore the message.

8.4.8 Handover Success

8.4.8.1 General

The Handover Success procedure is used during a DAPS Handover, to inform the source NG-RAN node that the UE has successfully accessed the target NG-RAN node. The procedure uses UE-associated signalling.

8.4.8.2 Successful Operation



Figure 8.4.8.2-1: Handover Success

The AMF initiates the procedure by sending the HANDOVER SUCCESS message to the source NG-RAN node.

8.4.8.3 Abnormal Conditions

If the HANDOVER SUCCESS message refers to a context that does not exist, the source NG-RAN node shall ignore the message.

8.4.9 Uplink RAN Early Status Transfer

8.4.9.1 General

The purpose of the Uplink RAN Early Status Transfer procedure is to transfer the COUNT of the first downlink SDU that the source NG-RAN node forwards to the target NG-RAN node, from the source NG-RAN node to the target NG-RAN node via the AMF during NG DAPS Handover, or the COUNT for discarding of already forwarded downlink SDUs for respective DRB(s) during NG-based handover with time-based trigger condition. The procedure uses UE-associated signalling.

8.4.9.2 Successful Operation



Figure 8.4.9.2-1: Uplink RAN Early Status Transfer

The source NG-RAN node initiates the procedure by sending the UPLINK RAN EARLY STATUS TRANSFER message to the AMF when it considers at least a DRB to be simultaneously served by the source and the target NG-RAN nodes during NG DAPS Handover or sending the UPLINK RAN EARLY STATUS TRANSFER message to the AMF after it initiated a handover preparation with time-based trigger condition.

For each DRB for which DAPS Handover applies, the source NG-RAN node shall include the *DRB ID IE* and the *FIRST DL COUNT Value IE* within the *DRBs Subject To Early Status Transfer Item IE* in the *Early Status Transfer Transparent Container IE* of the UPLINK RAN EARLY STATUS TRANSFER message.

8.4.9.3 Abnormal Conditions

Void.

8.4.10 Downlink RAN Early Status Transfer

8.4.10.1 General

The purpose of the Downlink RAN Early Status Transfer procedure is to transfer the COUNT of the first downlink SDU that the source NG-RAN node forwards to the target NG-RAN node, from the source NG-RAN node to the target NG-RAN node via the AMF during NG DAPS Handover, or the COUNT for discarding of already forwarded downlink SDUs for respective DRB during NG-based handover with time-based trigger condition. The procedure uses UE-associated signalling.

8.4.10.2 Successful Operation

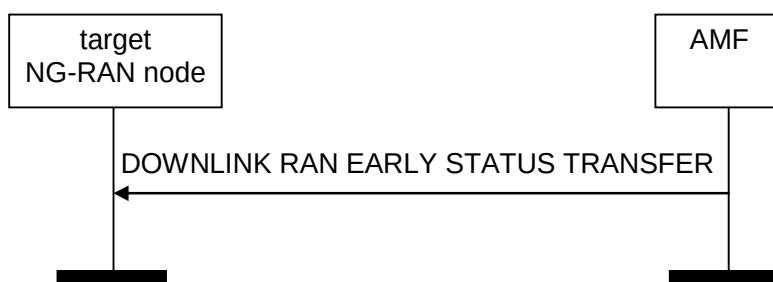


Figure 8.4.10.2-1: Downlink RAN Early Status Transfer

The AMF initiates the procedure by sending the DOWNLINK RAN EARLY STATUS TRANSFER message to the target NG-RAN node.

For each DRB for which the *FIRST DL COUNT Value IE* is received in the DOWNLINK RAN EARLY STATUS TRANSFER message, the target NG-RAN node shall use it as the COUNT of the first downlink SDU that the source NG-RAN node forwards to the target NG-RAN node.

For each DRB for which the *DISCARD DL COUNT Value IE* is received in the DOWNLINK RAN EARLY STATUS TRANSFER message, the target NG-RAN node does not transmit forwarded downlink SDUs to the UE whose COUNT is less than the provided *DISCARD DL COUNT Value IE* and discards them if transmission has not been attempted.

8.4.10.3 Abnormal Conditions

If the target NG-RAN node receives this message for a UE for which no prepared handover exists at the target NG-RAN node, the target NG-RAN node shall ignore the message.

8.5 Paging Procedures

8.5.1 Paging

8.5.1.1 General

The purpose of the Paging procedure is to enable the AMF to page a UE in the specific NG-RAN node.

8.5.1.2 Successful Operation

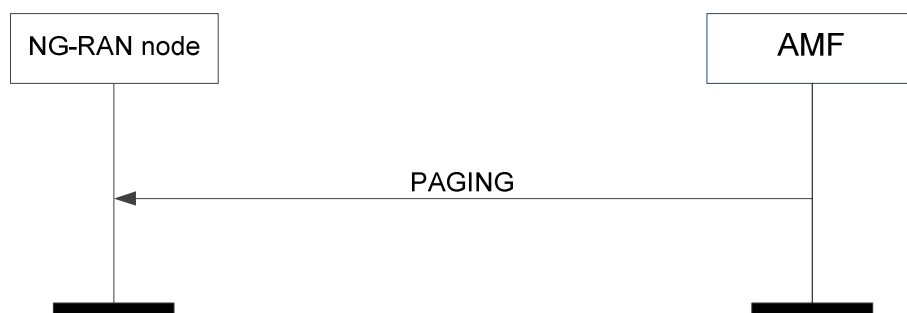


Figure 8.5.1.2-1: Paging

The AMF initiates the Paging procedure by sending the PAGING message to the NG-RAN node.

At the reception of the PAGING message, the NG-RAN node shall perform paging of the UE in cells which belong to tracking areas as indicated in the *TAI List for Paging* IE.

If the *Paging DRX* IE is included in the PAGING message, the NG-RAN node shall use it according to TS 38.304 [12] and TS 36.304 [29].

For each cell that belongs to any of the tracking areas indicated in the *TAI List for Paging* IE, the NG-RAN node shall generate one page on the radio interface.

If the *Paging Priority* IE is included in the PAGING message, the NG-RAN node may use it according to TS 23.501 [9].

If the *UE Radio Capability for Paging* IE is included in the PAGING message, the NG-RAN node may use it to apply specific paging schemes.

If the *Assistance Data for Recommended Cells* IE is included in the *Assistance Data for Paging* IE it may be used, together with the *Paging Attempt Information* IE if also present, according to TS 38.300 [8].

If the *Next Paging Area Scope* IE is included in the *Paging Attempt Information* IE it may be used for paging the UE according to TS 38.300 [8].

If the *Paging Origin* IE is included in the PAGING message, the NG-RAN node shall transfer it to the UE according to TS 38.331 [18] and TS 36.331 [21].

If the *NB-IoT Paging eDRX Information* IE is included in the PAGING message, the NG-RAN node shall, if supported, use it according to TS 36.304 [29]. If the *NB-IoT Paging Time Window* IE is included in the *NB-IoT Paging eDRX Information* IE, the NG-RAN node shall take this information into account to determine the UE's paging occasion according to TS 36.304 [29]. The NG-RAN node should take into account the reception time of the PAGING message on the NG interface to determine when to page the UE.

If the *NB-IoT Paging DRX* IE is included in the PAGING message, the NG-RAN node shall use it according to TS 36.304 [29].

If the *Enhanced Coverage Restriction* IE is included in the PAGING message, the NG-RAN node shall, if supported, use it as defined in TS 23.501 [9].

If the *Paging Assistance Data for CE Capable UE* IE is included in the *Assistance Data for Paging* IE in the PAGING message, it may be used for paging the indicated CE capable UE, according to TS 36.300 [17].

If the *WUS Assistance Information* IE is included in the PAGING message, the NG-RAN node shall, if supported, use it to determine the WUS group for the UE, as specified in TS 36.304 [29].

If the *E-UTRA Paging eDRX Information* IE is included in the PAGING message, the NG-RAN node shall, if supported, use it according to TS 36.304 [29]. If the *E-UTRA Paging Time Window* IE is included in the *E-UTRA Paging eDRX Information* IE, the NG-RAN node shall take this information into account to determine the UE's paging occasion according to TS 36.304 [29]. The NG-RAN node should take into account the reception time of the PAGING message on the NGAP interface to determine when to page the UE.

If the *CE-mode-B Restricted* IE is included in the PAGING message and the *Enhanced Coverage Restriction* IE is not set to "restricted", the NG-RAN node shall, if supported, use it as defined in TS 23.501 [9].

If the *NPN Paging Assistance Information* IE is included in the *Assistance Data for Paging* IE, the NG-RAN node may take it into account when determining the cells where paging will be performed.

If the *NR Paging eDRX Information* IE is included in the PAGING message, the NG-RAN node shall, if supported, use it according to TS 38.304 [12] and TS 23.501 [9]. If the *NR Paging Time Window* IE is included in the *NR Paging eDRX Information* IE, the NG-RAN node shall take this information into account to determine the UE's paging occasion according to TS 38.304 [12].

If the *Paging Cause* IE is included in the PAGING message, the NG-RAN node shall, if supported, transfer it to the UE according to TS 38.331 [18] and TS 36.331 [21].

If the *PEIPS Assistance Information* IE is included in the PAGING message, the NG-RAN node shall, if supported, use it for paging subgrouping of the UE, as specified in TS 38.300 [8].

If the *LP-WUSPS Assistance Information* IE is included in the PAGING message, the NG-RAN node shall, if supported, use it for LP-WUS paging subgrouping of the UE, as specified in TS 38.300 [8].

If the *LP-WUS Disable Indication* IE is included in the PAGING message, the NG-RAN node shall, if supported, store it and use it to disable LP-WUS as described in TS 38.300 [8].

8.5.1.3 Abnormal Conditions

Void.

8.5.2 Multicast Group Paging

8.5.2.1 General

The purpose of the Multicast Group Paging procedure is to enable the AMF to notify CM-IDLE UEs which have joined a multicast MBS session about its activation. The procedure uses non-UE associated signalling.

8.5.2.2 Successful Operation



Figure 8.5.2.2-1: Multicast Group Paging

The AMF initiates the Multicast Group Paging procedure by sending the MULTICAST GROUP PAGING message to the NG-RAN node.

At the reception of the MULTICAST GROUP PAGING message, the NG-RAN node shall perform multicast group paging of the MBS session identified by the *MBS Session ID* IE utilising information provided by the AMF.

If the *Paging DRX* IE is included in the MULTICAST GROUP PAGING message, the NG-RAN node shall use it according to TS 38.304 [12].

If the *MBS Service Area* IE is included in the MULTICAST GROUP PAGING message, the NG-RAN node shall take it into account during multicast group paging, as specified in TS 23.247 [44].

If the *UE Paging List* IE is included in the MULTICAST GROUP PAGING message, the NG-RAN node shall, if supported, use it according to TS 38.304 [12]. If absent, the NG-RAN node shall perform multicast group paging of the MBS session in all paging occasions within at least one default paging cycle, as specified in TS 38.304 [12].

8.5.2.3 Abnormal Conditions

Void.

8.6 Transport of NAS Messages Procedures

8.6.1 Initial UE Message

8.6.1.1 General

The Initial UE Message procedure is used when the NG-RAN node has received from the radio interface the first uplink NAS message to be forwarded to an AMF.

8.6.1.2 Successful Operation



Figure 8.6.1.2-1: Initial UE message

The NG-RAN node initiates the procedure by sending an INITIAL UE MESSAGE message to the AMF. The NG-RAN node shall allocate a unique RAN UE NGAP ID to be used for the UE and the NG-RAN node shall include this identity in the INITIAL UE MESSAGE message.

The *NAS-PDU* IE contains a UE – AMF message that is transferred without interpretation in the NG-RAN node.

In case of network sharing, the selected PLMN is indicated by the *PLMN Identity* IE within the *TAI* IE included in the INITIAL UE MESSAGE message, or by the *Serving PLMN* IE within the *NR NTN TAI Information* IE included in the same message for NTN.

When the NG-RAN node has received from the radio interface the *5G-S-TMSI* IE, it shall include it in the INITIAL UE MESSAGE message.

If the *AMF Set ID* IE is included in the INITIAL UE MESSAGE message this indicates that the message is a rerouted message and the AMF shall, if supported, use the IE as described in TS 23.502 [10].

If the NG-RAN node does not support the NAS Node Selection Function:

- If the NG-RAN node has received from the radio interface the *GUAMI* IE, the NG-RAN node shall include it in the INITIAL UE MESSAGE message.
- If the NG-RAN node has received from the radio interface the *GUAMI Type* IE, the NG-RAN node shall include it in the INITIAL UE MESSAGE message.
- If the NG-RAN node has received from the radio interface the *Requested NSSAI* IE, the NG-RAN node shall include it in the INITIAL UE MESSAGE message.

If the *UE Context Request* IE is included in the INITIAL UE MESSAGE message the AMF shall trigger an Initial Context Setup procedure towards the NG-RAN node.

If the *Allowed NSSAI* IE is included in the INITIAL UE MESSAGE message the AMF shall use the IE as defined in TS 23.502 [10].

If the *Partially Allowed NSSAI* IE is contained in the INITIAL UE MESSAGE message, the AMF shall, if supported, use the IE as defined in TS 23.502 [10].

If the *Source to Target AMF Information Reroute* IE is included in the INITIAL UE MESSAGE message the AMF shall use the IE as defined in TS 23.502 [10].

If the *IAB Node Indication* IE is included in the INITIAL UE MESSAGE message, the AMF shall consider that the message is related to an IAB node.

If the *CE-mode-B Support Indicator* IE is included in the INITIAL UE MESSAGE message and set to "supported", the AMF shall, if supported, use the extended NAS timer settings for the UE as specified in TS 23.501 [9].

If the *LTE-M indication* IE is included in the INITIAL UE MESSAGE message the AMF shall, if supported, use it according to TS 23.501 [9].

If the *EDT Session* IE set to "true" is included in the INITIAL UE MESSAGE message and the NG-RAN node is an ng-eNB, the AMF shall, if supported, consider that the message has been received as a result of an EDT session initiated by the UE.

If PNI-NPN related information within the *NPN Access Information* IE is received in the INITIAL UE MESSAGE message, the AMF shall, if supported, consider that the included information is associated to the cell via which the UE has sent the first NAS message, and to the PLMN Identity which is indicated within the *TAI* IE, and use the included information as specified in TS 23.501 [9].

In case of network sharing for SNPNs, the selected SNPN is indicated within the *User Location Information* IE included in the INITIAL UE MESSAGE message by the *PLMN Identity* IE within the *TAI* IE and the *NID* IE.

If the *RedCap Indication* IE or the *eRedCap Indication* IE is included in the INITIAL UE MESSAGE message, the AMF shall, if supported, use it according to TS 23.501 [9].

If the *Mobile IAB Node Indication* IE is included in the INITIAL UE MESSAGE message, the AMF shall consider that the message is related to a mobile IAB-node.

8.6.1.3 Abnormal Conditions

If the 5G-S-TMSI is not received by the AMF in the INITIAL UE MESSAGE message whereas expected, the AMF shall consider the procedure as failed.

If the *Partially Allowed NSSAI* IE is received in the INITIAL UE MESSAGE message and the total number of S-NSSAIs included in the Allowed NSSAI and Partially Allowed NSSAI exceeds eight, the AMF shall consider the procedure as failed.

If any of the S-NSSAI which is present in the *Partially Allowed NSSAI* IE is also present in the *Allowed NSSAI* IE, the AMF shall consider the procedure as failed.

8.6.2 Downlink NAS Transport

8.6.2.1 General

The Downlink NAS Transport procedure is used when the AMF only needs to send a NAS message transparently via the NG-RAN node to the UE, and a UE-associated logical NG-connection exists for the UE or the AMF has received the *RAN UE NGAP ID* IE in an INITIAL UE MESSAGE message or if the NG-RAN node has already initiated a UE-associated logical NG-connection by sending an INITIAL UE MESSAGE message via another NG interface instance.

8.6.2.2 Successful Operation

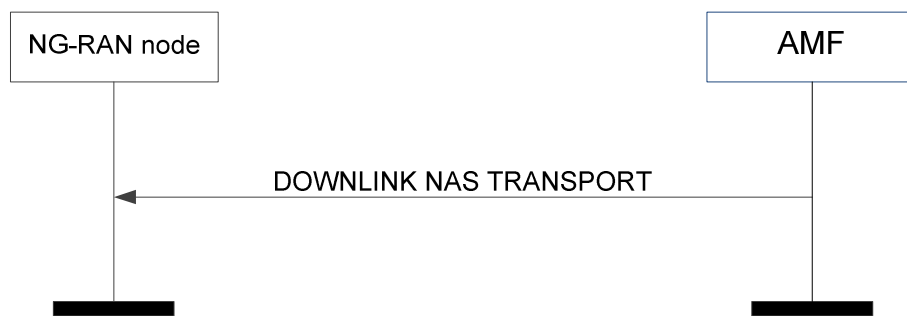


Figure 8.6.2.2-1: Downlink NAS transport

The AMF initiates the procedure by sending a DOWNLINK NAS TRANSPORT message to the NG-RAN node. If the UE-associated logical NG-connection is not established, the AMF shall allocate a unique AMF UE NGAP ID to be used for the UE and include that in the DOWNLINK NAS TRANSPORT message; by receiving the *AMF UE NGAP ID* IE in the DOWNLINK NAS TRANSPORT message, the NG-RAN node establishes the UE-associated logical NG-connection.

If the *RAN Paging Priority* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node may use it to determine a priority for paging the UE in RRC_INACTIVE state.

The *NAS-PDU* IE contains an AMF – UE message that is transferred without interpretation in the NG-RAN node.

If the *Mobility Restriction List* IE is contained in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall overwrite any previously stored mobility restriction information in the UE context. The NG-RAN node shall use the information in the *Mobility Restriction List* IE if present in the DOWNLINK NAS TRANSPORT message to:

- determine a target for subsequent mobility action for which the NG-RAN node provides information about the target of the mobility action towards the UE;
- select a proper SCG during dual connectivity operation;
- assign proper RNA(s) for the UE when moving the UE to RRC_INACTIVE state.

If the *Mobility Restriction List* IE is not contained in the DOWNLINK NAS TRANSPORT message and there is no previously stored mobility restriction information, the NG-RAN node shall consider that no roaming and no access restriction apply to the UE except for the PNI NPN mobility as described in TS 23.501 [9].

The NG-RAN node shall consider that roaming or access to CAG cells is only allowed if the *Allowed PNI-NPN List* IE is contained in the DOWNLINK NAS TRANSPORT message, as described in TS 23.501 [9].

If the *Index to RAT/Frequency Selection Priority* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, use it as defined in TS 23.501 [9].

The *UE Aggregate Maximum Bit Rate* IE should be sent to the NG-RAN node if the AMF has not sent it previously. If it is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall store the UE Aggregate Maximum Bit Rate in the UE context, and use the received UE Aggregate Maximum Bit Rate for all Non-GBR QoS flows for the concerned UE as specified in TS 23.501 [9].

If the *Old AMF* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall consider that this UE-associated logical NG-connection was redirected to this AMF from another AMF identified by the *Old AMF* IE. If the *Extended Old AMF* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, consider that this UE-associated logical NG-connection was redirected to this AMF from another AMF identified by the *Extended Old AMF* IE.

If the *SRVCC Operation Possible* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, store the content of the received *SRVCC Operation Possible* IE in the UE context and use it as defined in TS 23.216 [31].

If the *Extended Connected Time* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, use it as described in TS 23.501 [9].

If the *Enhanced Coverage Restriction* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *UE Differentiation Information* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, store this information in the UE context for further use according to TS 23.501 [9].

If the *CE-mode-B Restricted* IE is included in the DOWNLINK NAS TRANSPORT message and the *Enhanced Coverage Restriction* IE is not set to "restricted" and the Enhanced Coverage Restricted information stored in the UE context is not set to "restricted", the NG-RAN node shall, if supported, store this information in the UE context and use it as defined in TS 23.501 [9].

If the *UE Radio Capability* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall store this information in the UE context, and use it as defined in TS 38.300 [8].

If the *End Indication* IE is included in the DOWNLINK NAS TRANSPORT message and set to "no further data", the NG-RAN node shall consider that besides the included NAS PDU in this message, there are no further NAS PDUs to be transmitted for this UE.

If the DOWNLINK NAS TRANSPORT message contains the *UE Radio Capability ID* IE, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9] and TS 23.502 [10].

If the *Target NSSAI Information* IE is contained in the DOWNLINK NAS TRANSPORT message, the NG-RAN node may use this information as specified in TS 23.501 [9].

If the *Partially Allowed NSSAI* IE is contained in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, deduce from it the partially allowed network slices for the UE, store and replace any previously received Partially Allowed NSSAI and use it as specified in TS 23.501 [9].

If the *Masked IMEISV* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, use it to determine the characteristics of the UE for subsequent handling.

If the *Mobile IAB Authorized* IE is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall, if supported, store the received Mobile IAB Authorization status in the UE context. If the *Mobile IAB Authorized* IE is set to "not authorized" for a mobile IAB-MT, the NG-RAN node shall, if supported, ensure that the mobile IAB node will not serve any UE(s).

Interactions with Initial UE Message procedure:

The NG-RAN node shall use the *AMF UE NGAP ID* IE and *RAN UE NGAP ID* IE received in the DOWNLINK NAS TRANSPORT message as identification of the logical connection even if the *RAN UE NGAP ID* IE had been allocated in an INITIAL UE MESSAGE message sent over a different NG interface instance.

Interaction with the UE Radio Capability Info Indication procedure:

If the *UE Capability Info Request* IE set to "requested" is included in the DOWNLINK NAS TRANSPORT message, the NG-RAN node shall trigger the UE Radio Capability Info Indication procedure if UE capability related information was successfully retrieved from the UE.

8.6.2.3 Abnormal Conditions

If the *Partially Allowed NSSAI* IE is received in the DOWNLINK NAS TRANSPORT message and the total number of S-NSSAIs included in the Allowed NSSAI and Partially Allowed NSSAI exceeds eight, the NG-RAN node shall consider the procedure as failed.

If any of the S-NSSAI which is present in the *Partially Allowed NSSAI* IE is also present in the *Allowed NSSAI* IE, the NG-RAN node shall consider the procedure as failed.

8.6.3 Uplink NAS Transport**8.6.3.1 General**

The Uplink NAS Transport procedure is used when the NG-RAN node has received from the radio interface a NAS message to be forwarded to the AMF to which a UE-associated logical NG-connection for the UE exists.

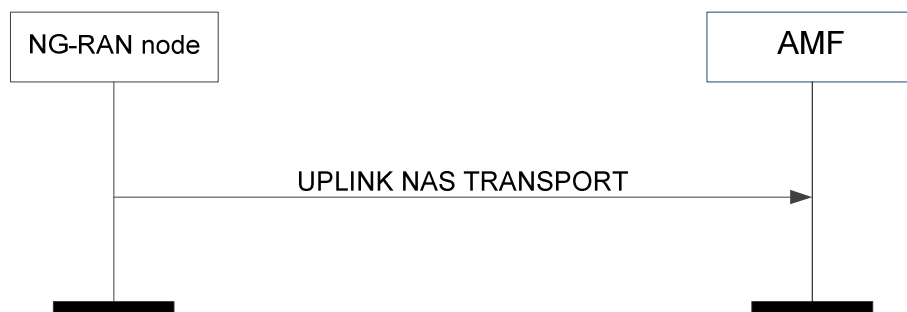
8.6.3.2 Successful Operation

Figure 8.6.3.2-1: Uplink NAS transport

The NG-RAN node initiates the procedure by sending an UPLINK NAS TRANSPORT message to the AMF.

The *NAS-PDU* IE contains a UE – AMF message that is transferred without interpretation in the NG-RAN node.

8.6.3.3 Abnormal Conditions

Void.

8.6.4 NAS Non Delivery Indication**8.6.4.1 General**

The NAS Non Delivery Indication procedure is used when the NG-RAN node decides not to start the delivery of a NAS message that has been received over a UE-associated logical NG-connection or the NG-RAN node is unable to ensure that the message has been received by the UE.

8.6.4.2 Successful Operation



Figure 8.6.4.2-1: NAS non delivery indication

The NG-RAN node initiates the procedure by sending a NAS NON DELIVERY INDICATION message to the AMF. The NG-RAN node shall report the non-delivery of a NAS message by including the non-delivered NAS message within the *NAS-PDU* IE and an appropriate cause value within the *Cause* IE, e.g., "NG intra system handover triggered", "NG inter system handover triggered" or "Xn handover triggered".

8.6.4.3 Abnormal Conditions

Void.

8.6.5 Reroute NAS Request

8.6.5.1 General

The purpose of the Reroute NAS Request procedure is to enable the AMF to request for a rerouting of the INITIAL UE MESSAGE message to another AMF.

8.6.5.2 Successful Operation

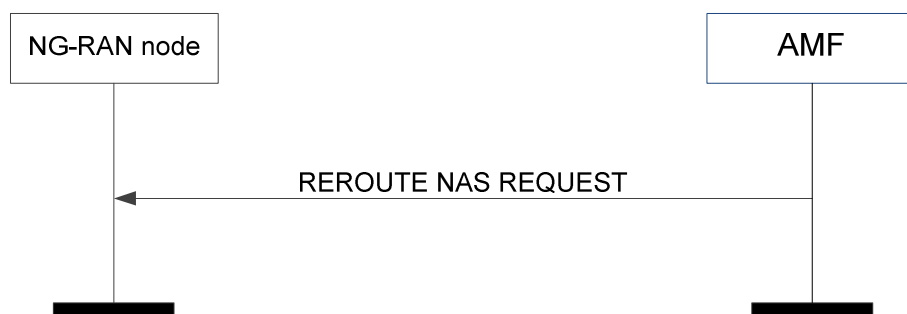


Figure 8.6.5.2-1: Reroute NAS request

The AMF initiates the procedure by sending a REROUTE NAS REQUEST message to the NG-RAN node. The NG-RAN node shall, if supported, reroute the INITIAL UE MESSAGE message to an AMF indicated by the *AMF Set ID* IE as described in TS 23.501 [9].

If the *Allowed NSSAI* IE is included in the REROUTE NAS REQUEST message, then the NG-RAN node shall propagate it in the rerouted INITIAL UE MESSAGE message as defined in TS 23.502 [10].

If the *Partially Allowed NSSAI* IE is contained in the REROUTE NAS REQUEST message, the NG-RAN node shall, if supported, propagate it in the rerouted INITIAL UE MESSAGE message as defined in TS 23.502 [10].

If the *Source to Target AMF Information Reroute* IE is included in the REROUTE NAS REQUEST message, then the NG-RAN node shall propagate it in the rerouted INITIAL UE MESSAGE message as defined in TS 23.502 [10].

8.6.5.3 Abnormal Conditions

If the *Partially Allowed NSSAI* IE is received in the REROUTE NAS REQUEST message and the total number of S-NSSAIs included in the Allowed NSSAI and Partially Allowed NSSAI exceeds eight, the NG-RAN node shall consider the procedure as failed.

If any of the S-NSSAI which is present in the *Partially Allowed NSSAI* IE is also present in the *Allowed NSSAI* IE, the NG-RAN node shall consider the procedure as failed.

8.7 Interface Management Procedures

8.7.1 NG Setup

8.7.1.1 General

The purpose of the NG Setup procedure is to exchange application level data needed for the NG-RAN node and the AMF to correctly interoperate on the NG-C interface. This procedure shall be the first NGAP procedure triggered after the TNL association has become operational. The procedure uses non-UE associated signalling.

This procedure erases any existing application level configuration data in the two nodes, replaces it by the one received and clears AMF overload state information at the NG-RAN node. If the NG-RAN node and AMF do not agree on retaining the UE contexts this procedure also re-initialises the NGAP UE-related contexts (if any) and erases all related signalling connections in the two nodes like an NG Reset procedure would do.

If the NG-RAN node supports A-IoT and is communicating directly with the AIOTF, as specified in TS 23.369 [60], the NG Setup procedure, as depicted in Figures 8.7.1.2-2 and 8.7.1.3-2 and specified in the respective sections, is executed between the NG-RAN node and the AIOTF.

8.7.1.2 Successful Operation

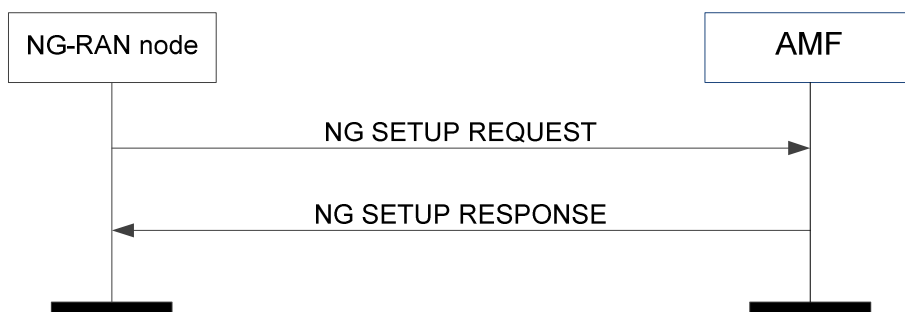


Figure 8.7.1.2-1: NG setup: successful operation with the AMF

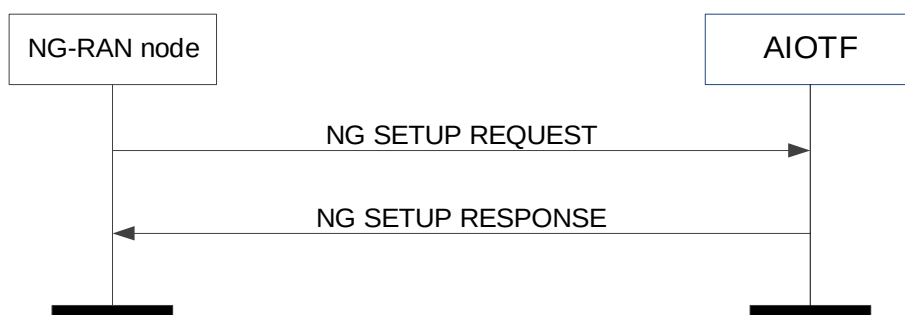


Figure 8.7.1.2-2: NG setup: successful operation with the AIOTF

If the NG Setup procedure is executed between the NG-RAN node and the AMF:

- The NG-RAN node initiates the procedure by sending an NG SETUP REQUEST message including the appropriate data to the AMF. The AMF responds with an NG SETUP RESPONSE message including the appropriate data.
- If the *Configured TAC Indication* IE set to "true" is included for a Tracking Area contained in the *Supported TA List* IE in the NG SETUP REQUEST message, the AMF may take it into account to optimise NG-C signalling towards this NG-RAN node.
- If the *UE Retention Information* IE set to "ues-retained" is included in the NG SETUP REQUEST message, the AMF may accept the proposal to retain the existing UE related contexts and signalling connections by including the *UE Retention Information* IE set to "ues-retained" in the NG SETUP RESPONSE message.
- If the AMF supports IAB, the AMF shall include the *IAB Supported* IE in the NG SETUP RESPONSE message. If the *IAB Supported* IE is included in the NG SETUP RESPONSE message, the NG-RAN node shall, if supported, store this information and use it for further AMF selection for the IAB-MT.
- The AMF shall include the *Backup AMF Name* IE, if available, in the *Served GUAMI List* IE in the NG SETUP RESPONSE message. The NG-RAN node shall, if supported, consider the AMF as indicated by the *Backup AMF Name* IE when performing AMF reselection, as specified in TS 23.501 [9]. If the *Extended Backup AMF Name* IE is included in the *Served GUAMI List* IE in the NG SETUP RESPONSE message, the NG-RAN node shall, if supported, consider the AMF as indicated by the *Extended Backup AMF Name* IE when performing AMF reselection, as specified in TS 23.501 [9].
- If the *GUAMI Type* IE is included in the NG SETUP RESPONSE message, the NG-RAN node shall store the received value and use it for further AMF selection as defined in TS 23.501 [9].
- If the *RAN Node Name* IE is included in the NG SETUP REQUEST message, the AMF may use this IE as a human readable name of the NG-RAN node. If the *Extended RAN Node Name* IE is included in the NG SETUP REQUEST message, the AMF may use this IE as a human readable name of the NG-RAN node and shall ignore the *RAN Node Name* IE if also included.
- If the *Extended AMF Name* IE is included in the NG SETUP RESPONSE message, the NG-RAN node may use this IE as a human readable name of the AMF and shall ignore the *AMF Name* IE.
- If the *NB-IoT Default Paging DRX* IE is included in the NG SETUP REQUEST message, the AMF shall take it into account for paging.
- If the *RAT Information* IE is included in the NG SETUP REQUEST message, the AMF shall handle this information as specified in TS 23.502 [10].
- If the *NID* IE within the *NPN Support* IE is included within a *Broadcast PLMN Item* IE in the NG SETUP REQUEST message, the AMF shall consider that the NG-RAN node supports the indicated S-NSSAI(s) for the corresponding tracking area code for the SNPN identified by the *PLMN Identity* IE and the *NID* IE.
- If the *NID* IE within the *NPN Support* IE is included within a *PLMN Support Item* IE in the NG SETUP RESPONSE message, the NG-RAN node shall consider that the AMF supports the SNPN identified by the *PLMN Identity* IE and the *NID* IE.
- If the *Onboarding Support* IE is also included within the same *PLMN Support Item* IE, the NG-RAN node shall, if supported, consider that the AMF supports UE onboarding for the identified SNPN, as specified in TS 23.501 [9].
- If the *TAI NSAG Support List* IE is included in the *Broadcast PLMN Item* IE in the NG SETUP REQUEST message, the AMF shall, if supported, use this information as specified in TS 23.501 [9].
- If the AMF supports mobile IAB, the AMF shall include the *Mobile IAB Supported* IE in the NG SETUP RESPONSE message. If the *Mobile IAB Supported* IE is included in the NG SETUP RESPONSE message, the NG-RAN node shall, if supported, store this information and further use it for AMF selection for the mobile IAB-MT.
- If the *Additional ULI* IE is included in the NG SETUP REQUEST message, the AMF shall, if supported, store this information, consider that this transmitting NG-RAN node supports the WAB-gNB functionality, and take this information into account for determining the location of the UEs served by the NG-RAN node, as specified in TS 23.501 [9].

If the NG Setup procedure is executed between the NG-RAN node and the AIOTF:

- The NG-RAN node initiates the procedure by sending an NG SETUP REQUEST message including the appropriate data to the AIOTF. The AIOTF responds with an NG SETUP RESPONSE message including the *AIOTF Name* IE and the *AIOTF Identifier* IE. The NG-RAN node shall ignore the not applicable IEs in the NG SETUP RESPONSE message as specified in clause 9.2.6.2.

If the NG Setup procedure is triggered by an NG-RAN node supporting A-IoT:

- If the *A-IoT Support* IE is included in the NG SETUP REQUEST message and set to "A-IoT only", the receiving node shall, if supported, consider that the NG-RAN node only supports A-IoT radio, and ignores the not applicable IEs as specified in clause 9.2.6.1.
- If the *A-IoT Support* IE is included in the NG SETUP REQUEST message and set to "A-IoT and NR Uu", the receiving node shall, if supported, consider that the NG-RAN node supports both A-IoT radio and NR Uu radio.

8.7.1.3 Unsuccessful Operation

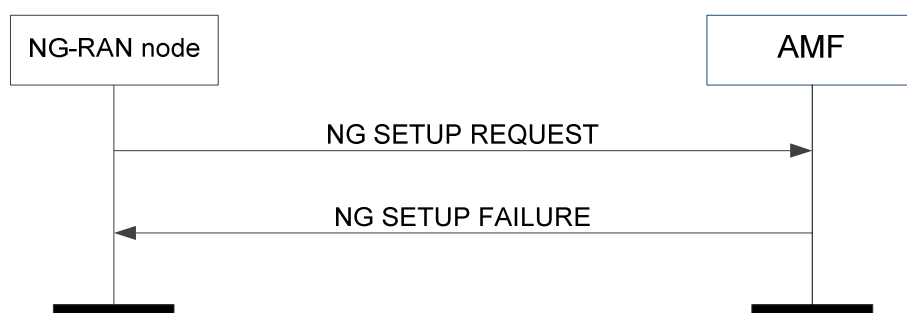


Figure 8.7.1.3-1: NG setup: unsuccessful operation with the AMF

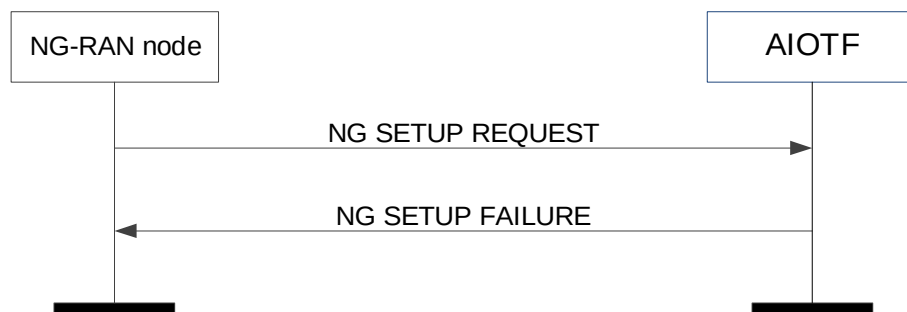


Figure 8.7.1.3-2: NG setup: unsuccessful operation with the AIOTF

If the AMF cannot accept the setup, it should respond with an NG SETUP FAILURE message and appropriate cause value.

If the NG SETUP FAILURE message includes the *Time to Wait* IE, the NG-RAN node shall wait at least for the indicated time before reinitiating the NG Setup procedure towards the same AMF.

If the NG Setup procedure is executed between the NG-RAN node and the AIOTF, the specification text above concerning the AMF applies for the AIOTF.

8.7.1.4 Abnormal Conditions

If the AMF does not identify any of the PLMNs/SNPNs indicated in the NG SETUP REQUEST message, it shall reject the NG Setup procedure with an appropriate cause value.

If none of the RATs indicated by the NG-RAN node in the NG SETUP REQUEST message is supported by the AMF, then the AMF shall fail the NG Setup procedure with an appropriate cause value.

If the NG Setup procedure is executed between the NG-RAN node and the AIOTF, the specification text above concerning the AMF applies for the AIOTF.

8.7.2 RAN Configuration Update

8.7.2.1 General

The purpose of the RAN Configuration Update procedure is to update application level configuration data needed for the NG-RAN node and the AMF to interoperate correctly on the NG-C interface. This procedure does not affect existing UE-related contexts, if any. The procedure uses non UE-associated signalling.

If the NG-RAN node supports A-IoT and is communicating directly with an AIOTF, as specified in TS 23.369 [60], the RAN Configuration Update procedure, as depicted in Figures 8.7.2.2-2 and 8.7.2.3-2 and specified in the respective sections, is executed between the NG-RAN node and the AIOTF.

8.7.2.2 Successful Operation

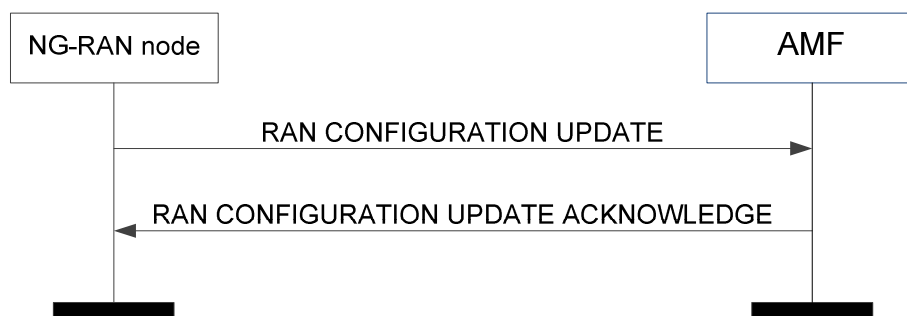


Figure 8.7.2.2-1: RAN configuration update: successful operation with the AMF

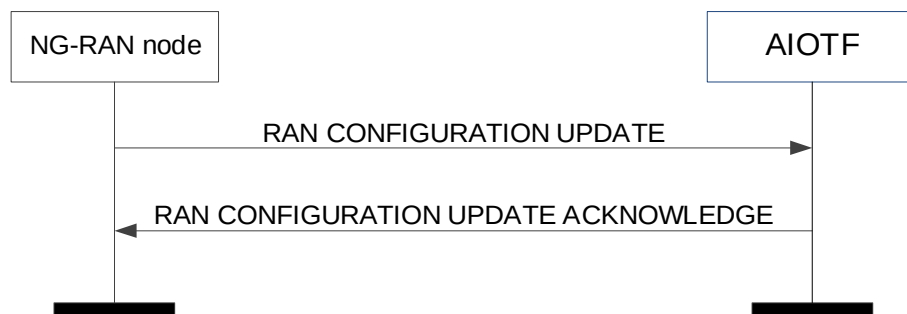


Figure 8.7.2.2-2: RAN configuration update: successful operation with the AIOTF

If the RAN Configuration Update procedure is executed between the NG-RAN node and the AMF:

- The NG-RAN node initiates the procedure by sending a RAN CONFIGURATION UPDATE message to the AMF including an appropriate set of updated configuration data that it has just taken into operational use. The AMF responds with a RAN CONFIGURATION UPDATE ACKNOWLEDGE message to acknowledge that it successfully updated the configuration data. If an information element is not included in the RAN CONFIGURATION UPDATE message, the AMF shall interpret that the corresponding configuration data is not changed and shall continue to operate the NG-C interface with the existing related configuration data.
- If the *Supported TA List* IE is included in the RAN CONFIGURATION UPDATE message, the AMF shall overwrite the whole list of supported TAs and the corresponding list of supported slices for each TA, and use them for subsequent registration area management of the UE.
- If the *Configured TAC Indication* IE set to "true" is included for a Tracking Area contained in the *Supported TA List* IE in the RAN CONFIGURATION UPDATE message, the AMF may take it into account to optimise NG-C signalling towards this NG-RAN node.

- If the *Global RAN Node ID* IE is included in the RAN CONFIGURATION UPDATE message, the AMF shall associate the TNLA to the NG-C interface instance using the Global RAN Node ID.
- If the RAN CONFIGURATION UPDATE message includes the *NG-RAN TNL Association to Remove List* IE, the AMF shall, if supported, initiate removal of the TNL association(s) indicated by NG-RAN TNL endpoint(s) and AMF TNL endpoint(s) if the *TNL Association Transport Layer Address at AMF* IE is present, or the TNL association(s) indicated by NG-RAN TNL endpoint(s) if the *TNL Association Transport Layer Address at AMF* IE is absent:
 - if the received *TNL Association Transport Layer Address* IE includes the *Port Number* IE, the NG-RAN TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the NG-RAN TNL endpoints correspond to all NG-RAN TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).
 - if the received *TNL Association Transport Layer Address at AMF* IE includes the *Port Number* IE, the AMF TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the AMF TNL endpoints correspond to all AMF TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).
- If the RAN CONFIGURATION UPDATE message includes the *RAN Node Name* IE, the AMF may store it or update this IE value if already stored, and use it as a human readable name of the NG-RAN node. If the RAN CONFIGURATION UPDATE message includes the *Extended RAN Node Name* IE, the AMF may store it or update this IE value if already stored, and use it as a human readable name of the NG-RAN node and shall ignore the *RAN Node Name* IE if also included.
- If the *NB-IoT Default Paging DRX* IE is included in the RAN CONFIGURATION UPDATE message, the AMF shall overwrite any previously stored NB-IoT default paging DRX value for the NG-RAN node.
- If the *RAT Information* IE is included in the RAN CONFIGURATION UPDATE message, the AMF shall handle this information as specified in TS 23.502 [10].
- If the *NID* IE within the *NPN Support* IE is included within a *Broadcast PLMN Item* IE in the RAN CONFIGURATION UPDATE message, the AMF shall consider that the NG-RAN node supports the indicated S-NSSAI(s) for the corresponding tracking area code for the SNPN identified by the *PLMN Identity* IE and the *NID* IE.
- If the *TAI NSAG Support List* IE is included in the *Broadcast PLMN Item* IE in the RAN CONFIGURATION UPDATE message, the AMF shall, if supported, use this information as specified in TS 23.501 [9].

If the RAN Configuration Update procedure is executed between the NG-RAN node and the AIOTF:

- The NG-RAN node initiates the procedure by sending a RAN CONFIGURATION UPDATE message to the AIOTF including an appropriate set of updated configuration data that it has just taken into operational use. The AIOTF responds with a RAN CONFIGURATION UPDATE ACKNOWLEDGE message to acknowledge that it successfully updated the configuration data. If an information element is not included in the RAN CONFIGURATION UPDATE message, the AIOTF shall interpret that the corresponding configuration data is not changed and shall continue to operate the NG-C interface with the existing related configuration data.

If the RAN Configuration Update procedure is triggered by an NG-RAN node supporting A-IoT:

- If the *A-IoT Support* IE is included in the RAN CONFIGURATION UPDATE message and set to "A-IoT only", the receiving node shall, if supported, consider that the NG-RAN node only supports A-IoT radio.
- If the *A-IoT Support* IE is included in the RAN CONFIGURATION UPDATE message and set to "A-IoT and NR Uu", the receiving node shall, if supported, consider that the NG-RAN node supports both A-IoT radio and NR Uu radio.

If the *Additional ULI* IE is included in the RAN CONFIGURATION UPDATE message, the AMF shall, if supported, store this information, and take it into account for determining the location of UEs served by the NG-RAN node, as specified in TS 23.501 [9].

8.7.2.3 Unsuccessful Operation

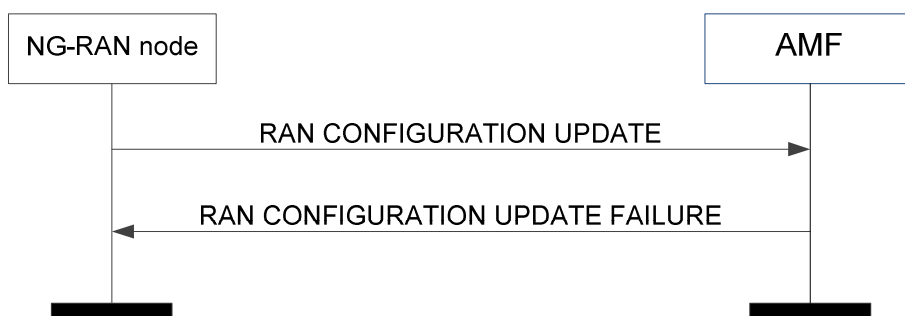


Figure 8.7.2.3-1: RAN configuration update: unsuccessful operation with the AMF

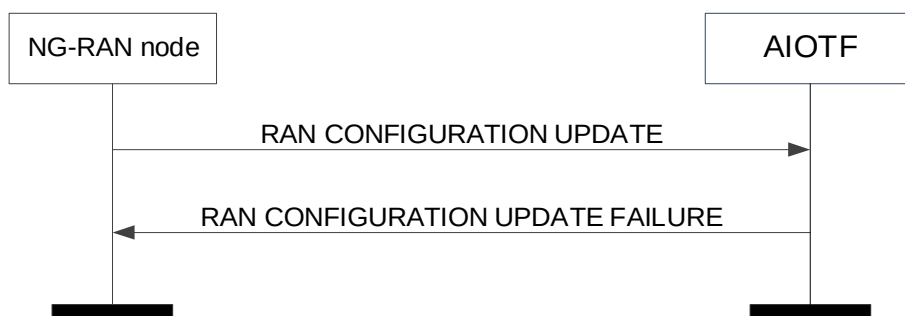


Figure 8.7.2.3-2: RAN configuration update: unsuccessful operation with the AIOTF

If the AMF cannot accept the update, it shall respond with a RAN CONFIGURATION UPDATE FAILURE message and appropriate cause value.

If the *Time to Wait* IE is included in the RAN CONFIGURATION UPDATE FAILURE message, the NG-RAN node shall wait at least for the indicated time before reinitiating the RAN Configuration Update procedure towards the same AMF.

If the RAN Configuration Update procedure is executed between the NG-RAN node and the AIOTF, the specification text above concerning the AMF applies for the AIOTF.

8.7.2.4 Abnormal Conditions

If the NG-RAN node, after initiating the RAN Configuration Update procedure, receives neither a RAN CONFIGURATION UPDATE ACKNOWLEDGE nor a RAN CONFIGURATION UPDATE FAILURE message, the NG-RAN node may reinitiate a further RAN Configuration Update procedure towards the same AMF, provided that the content of the new RAN CONFIGURATION UPDATE message is identical to the content of the previously unacknowledged RAN CONFIGURATION UPDATE message.

If the RAN Configuration Update procedure is executed between the NG-RAN node and the AIOTF, the specification text above concerning the AMF applies for the AIOTF.

8.7.3 AMF Configuration Update

8.7.3.1 General

The purpose of the AMF Configuration Update procedure is to update application level configuration data needed for the NG-RAN node and AMF to interoperate correctly on the NG-C interface. This procedure does not affect existing UE-related contexts, if any. The procedure uses non UE-associated signalling.

8.7.3.2 Successful Operation

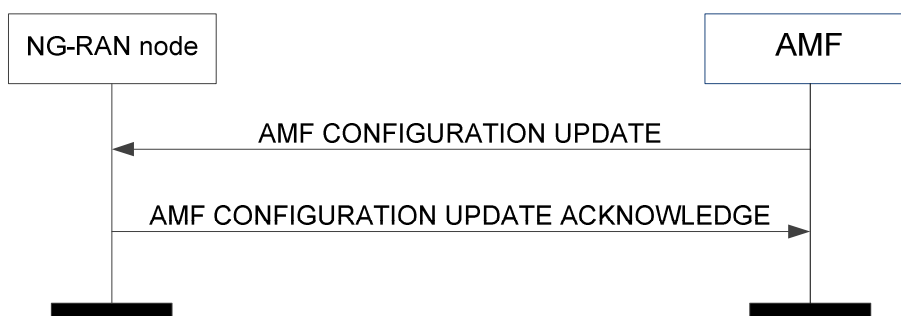


Figure 8.7.3.2-1: AMF configuration update: successful operation

The AMF initiates the procedure by sending an AMF CONFIGURATION UPDATE message including the appropriate updated configuration data to the NG-RAN node. The NG-RAN node responds with an AMF CONFIGURATION UPDATE ACKNOWLEDGE message to acknowledge that it successfully updated the configuration data. Unless stated otherwise, if an information element is not included in the AMF CONFIGURATION UPDATE message, the NG-RAN node shall interpret that the corresponding configuration data is not changed and shall continue to operate the NG-C interface with the existing related configuration data.

If the *PLMN Support List* IE is included in the AMF CONFIGURATION UPDATE message, the NG-RAN node shall overwrite the whole list of supported PLMN/SNPN Identities and the corresponding list of AMF slices and, if present, other associated information for each PLMN/SNPN Identity and use the received values for further network slice selection and AMF selection.

If the *AMF TNL Association to Add List* IE is included in the AMF CONFIGURATION UPDATE message, the NG-RAN node shall, if supported, use it to establish the TNL association(s) with the AMF. If the *AMF TNL Association to Add List* IE is included in the AMF CONFIGURATION UPDATE message, and if the *AMF TNL Association Address* IE does not include the *Port Number* IE, the NG-RAN node shall assume that port number value 38412 is used for the endpoint. The NG-RAN node shall report to the AMF, in the AMF CONFIGURATION UPDATE ACKNOWLEDGE message, the successful establishment of the TNL association(s) with the AMF as follows:

- A list of successfully established TNL associations shall be included in the *AMF TNL Association Setup List* IE;
- A list of TNL associations that failed to be established shall be included in the *AMF TNL Association Failed to Setup List* IE.

If the AMF CONFIGURATION UPDATE message includes the *AMF TNL Association to Remove List* IE, the NG-RAN node shall, if supported, initiate removal of the TNL association(s) indicated by AMF TNL endpoint(s) and NG-RAN node TNL endpoint(s) if the *TNL Association Transport Layer Address NG-RAN* IE is present, or the TNL association(s) indicated by AMF TNL endpoint(s) if the *TNL Association Transport Layer Address NG-RAN* IE is absent:

- if the received *AMF TNL Association Address* IE includes the *Port Number* IE, the AMF TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the AMF TNL endpoints correspond to all AMF TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).
- if the received *TNL Association Transport Layer Address NG-RAN* IE includes the *Port Number* IE, the NG-RAN node TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the NG-RAN node TNL endpoints correspond to all NG-RAN node TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).

If the *AMF Name* IE is included in the AMF CONFIGURATION UPDATE message, the NG-RAN node shall overwrite the previously stored AMF name and use it to identify the AMF.

If the AMF CONFIGURATION UPDATE message includes the *AMF Name* IE, the NG-RAN node may store it or update this IE value if already stored, and use it as a human readable name of the AMF. If the AMF CONFIGURATION UPDATE message includes the *Extended AMF Name* IE, the NG-RAN node may store it or update this IE value if already stored, and use it as a human readable name of the AMF and shall ignore the *AMF Name* IE if also included.

If the *Served GUAMI List* IE is included in the AMF CONFIGURATION UPDATE message, the NG-RAN node shall overwrite the whole list of GUAMIs served by the AMF by the new list and use the received values for further AMF management and AMF selection as defined in TS 23.501 [9].

If the *Relative AMF Capacity* IE is included in the AMF CONFIGURATION UPDATE message, the NG-RAN node may use it as defined in TS 23.501 [9].

If the *AMF TNL Association to Update List* IE is included in the AMF CONFIGURATION UPDATE message the NG-RAN node shall, if supported, update the TNL association(s) indicated by the received AMF Transport Layer information towards the AMF:

- if the received *AMF TNL Association Address* IE includes the *Port Number* IE, the AMF TNL endpoint is identified by the *Endpoint IP Address* IE and the *Port Number* IE. Otherwise, the AMF TNL endpoints correspond to all AMF TNL endpoints identified by the *Endpoint IP Address* IE and any port number(s).

If the *TNL Association Usage* IE or the *TNL Address Weight Factor* IE is included in the *AMF TNL Association to Add List* IE or the *AMF TNL Association to Update List* IE, the NG-RAN node shall, if supported, consider it as defined in TS 23.502 [10].

If the *NID* IE within the *NPN Support* IE is included within a *PLMN Support Item* IE in the AMF CONFIGURATION UPDATE message, the NG-RAN node shall consider that the AMF supports the SNPN identified by the *PLMN Identity* IE and the *NID* IE.

If the *Onboarding Support* IE is also included within the same *PLMN Support Item* IE, the NG-RAN node shall, if supported, consider that the AMF supports UE onboarding for the identified SNPN, as specified in TS 23.501 [9].

8.7.3.3 Unsuccessful Operation

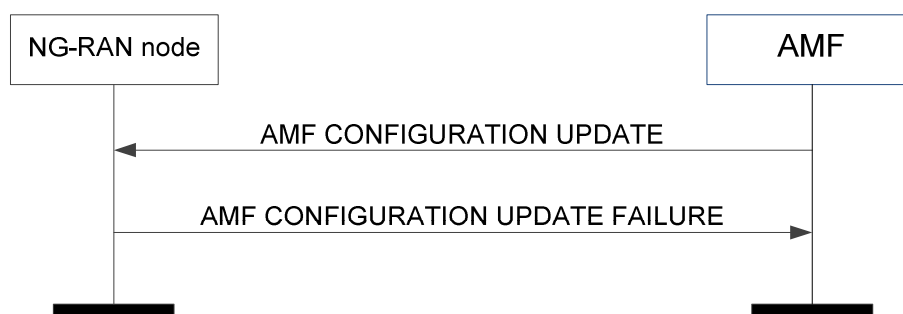


Figure 8.7.3.3-1: AMF configuration update: unsuccessful operation

If the NG-RAN node cannot accept the update, it shall respond with an AMF CONFIGURATION UPDATE FAILURE message and appropriate cause value.

If the *Time to Wait* IE is included in the AMF CONFIGURATION UPDATE FAILURE message, the AMF shall wait at least for the indicated time before reinitiating the AMF Configuration Update procedure towards the same NG-RAN node.

8.7.3.4 Abnormal Conditions

If the AMF receives neither an AMF CONFIGURATION UPDATE ACKNOWLEDGE nor an AMF CONFIGURATION UPDATE FAILURE message, the AMF may reinitiate the AMF Configuration Update procedure towards the same NG-RAN node provided that the content of the new AMF CONFIGURATION UPDATE message is identical to the content of the previously unacknowledged AMF CONFIGURATION UPDATE message.

8.7.4 NG Reset

8.7.4.1 General

The purpose of the NG Reset procedure is to initialise or re-initialise the RAN, or part of RAN NGAP UE-related contexts, in the event of a failure in the 5GC or vice versa. This procedure does not affect the application level configuration data exchanged during, e.g., the NG Setup procedure. The procedure uses non-UE associated signalling.

If the NG-RAN node supports A-IoT and is communicating directly with an AIOTF, as specified in TS 23.369 [60], the NG Reset procedure, as depicted in Figures 8.7.4.2.1-2 and 8.7.4.2.2-2 and specified in the respective sections, is executed between the NG-RAN node and the AIOTF.

8.7.4.2 Successful Operation

8.7.4.2.1 NG Reset initiated by the AMF

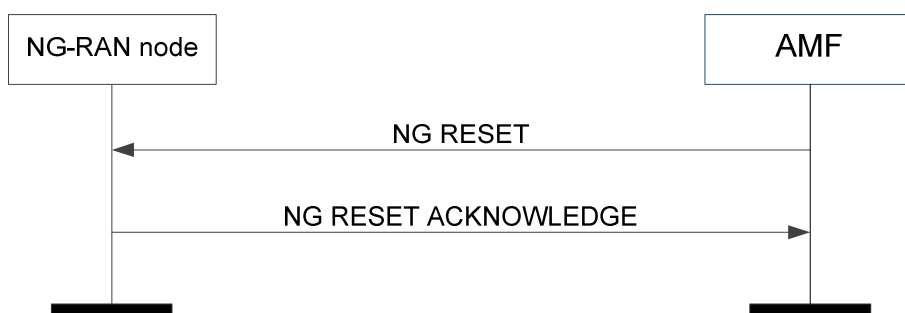


Figure 8.7.4.2.1-1: NG reset initiated by the AMF: successful operation

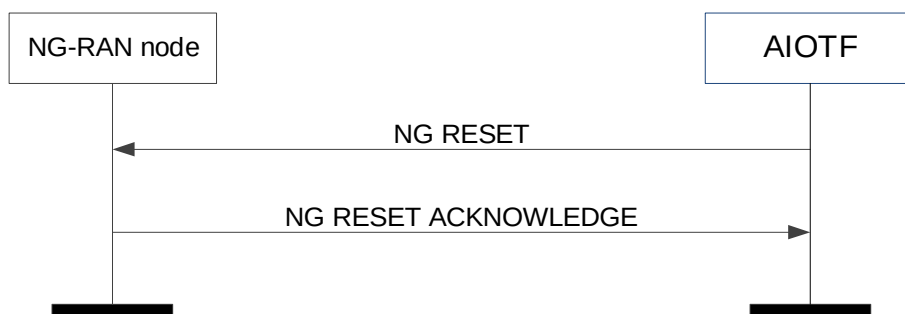


Figure 8.7.4.2.1-2: NG reset initiated by the AIOTF: successful operation

If the NG Reset procedure is executed between the NG-RAN node and the AMF:

- In the event of a failure at the AMF which has resulted in the loss of some or all transaction reference information, an NG RESET message shall be sent to the NG-RAN node.

If the NG Reset procedure is executed between the NG-RAN node and the AIOTF:

- In the event of a failure at the AIOTF which has resulted in the loss of some or all transaction reference information, an NG RESET message shall be sent to the NG-RAN node.

At reception of the NG RESET message the NG-RAN node shall release all allocated resources on NG and Uu related to the UE association(s) indicated explicitly or implicitly in the NG RESET message and remove the indicated UE contexts including NGAP ID, and if applicable, release all the related allocated A-IoT radio resources.

After the NG-RAN node has released all assigned NG resources and the UE NGAP IDs for all indicated UE associations which can be used for new UE-associated logical NG-connections over the NG interface, the NG-RAN node shall respond with the NG RESET ACKNOWLEDGE message. The NG-RAN node does not need to wait for the release of radio resources to be completed before returning the NG RESET ACKNOWLEDGE message.

If the NG RESET message contains the *UE-associated Logical NG-connection List* IE, then:

- The NG-RAN node shall use the *AMF UE NGAP ID* IE and/or the *RAN UE NGAP ID* IE to explicitly identify the UE association(s) to be reset.
- The NG-RAN node shall include in the NG RESET ACKNOWLEDGE message, for each UE association to be reset, the *UE-associated Logical NG-connection Item* IE in the *UE-associated Logical NG-connection List* IE. The *UE-associated Logical NG-connection Item* IEs shall be in the same order as received in the NG RESET message and shall include also unknown UE-associated logical NG-connections. Empty *UE-associated Logical NG-connection Item* IEs, received in the NG RESET message, may be omitted in the NG RESET ACKNOWLEDGE message.
- If the *AMF UE NGAP ID* IE is included in the *UE-associated Logical NG-connection Item* IE for a UE association, the NG-RAN node shall include the *AMF UE NGAP ID* IE in the corresponding *UE-associated Logical NG-connection Item* IE in the NG RESET ACKNOWLEDGE message.
- If the *RAN UE NGAP ID* IE is included in the *UE-associated Logical NG-connection Item* IE for a UE association, the NG-RAN node shall include the *RAN UE NGAP ID* IE in the corresponding *UE-associated Logical NG-connection Item* IE in the NG RESET ACKNOWLEDGE message.

Interactions with other procedures:

If the NG RESET message is received, any other ongoing procedure (except for another NG Reset procedure) on the same NG interface related to a UE association, indicated explicitly or implicitly in the NG RESET message, shall be aborted.

8.7.4.2.2 NG Reset initiated by the NG-RAN node

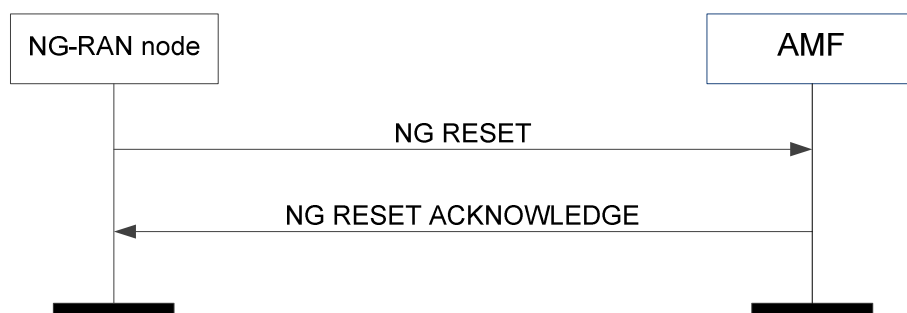


Figure 8.7.4.2.2-1: NG reset initiated by the NG-RAN node: successful operation with the AMF

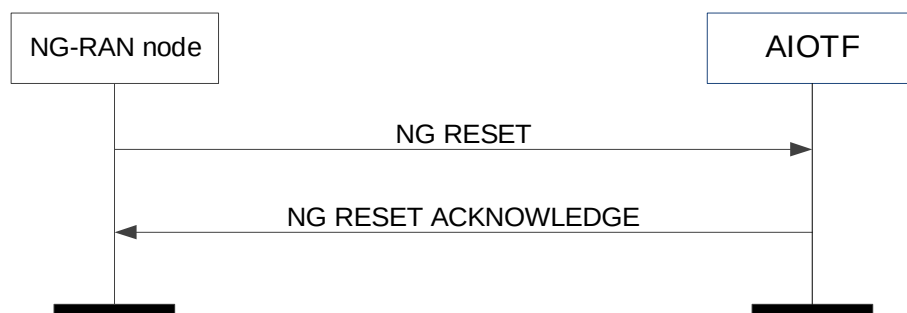


Figure 8.7.4.2.2-2: NG reset initiated by the NG-RAN node: successful operation with the AIOTF

If the NG Reset procedure is executed between the NG-RAN node and the AMF:

- In the event of a failure at the NG-RAN node which has resulted in the loss of some or all transaction reference information, an NG RESET message shall be sent to the AMF.
- At reception of the NG RESET message the AMF shall release all allocated resources on NG related to the UE association(s) indicated explicitly or implicitly in the NG RESET message and remove the NGAP ID for the indicated UE associations.

If the NG Reset procedure is executed between the NG-RAN node and the AIOTF:

- At reception of the NG RESET message the AIOTF shall release all allocated NG resources.

After the AMF has released all assigned NG resources and the UE NGAP IDs for all indicated UE associations which can be used for new UE-associated logical NG-connections over the NG interface, the AMF shall respond with the NG RESET ACKNOWLEDGE message.

After the AIOTF has released all allocated NG resources, the AIOTF shall respond with the NG RESET ACKNOWLEDGE message.

If the NG RESET message contains the *UE-associated Logical NG-connection List* IE, then:

- The AMF shall use the *AMF UE NGAP ID* IE and/or the *RAN UE NGAP ID* IE to explicitly identify the UE association(s) to be reset.
- The AMF shall include in the NG RESET ACKNOWLEDGE message, for each UE association to be reset, the *UE-associated Logical NG-connection Item* IE in the *UE-associated Logical NG-connection List* IE. The *UE-associated Logical NG-connection Item* IEs shall be in the same order as received in the NG RESET message and shall include also unknown UE-associated logical NG-connections. Empty *UE-associated Logical NG-connection Item* IEs, received in the NG RESET message, may be omitted in the NG RESET ACKNOWLEDGE message.
- If the *AMF UE NGAP ID* IE is included in the *UE-associated Logical NG-connection Item* IE for a UE association, the AMF shall include the *AMF UE NGAP ID* IE in the corresponding *UE-associated Logical NG-connection Item* IE in the NG RESET ACKNOWLEDGE message.
- If the *RAN UE NGAP ID* IE is included in a *UE-associated Logical NG-connection Item* IE for a UE association, the AMF shall include the *RAN UE NGAP ID* IE in the corresponding *UE-associated Logical NG-connection Item* IE in the NG RESET ACKNOWLEDGE message.

Interactions with other procedures:

If the NG RESET message is received, any other ongoing procedure (except for another NG Reset procedure) on the same NG interface related to a UE association, indicated explicitly or implicitly in the NG RESET message, shall be aborted.

8.7.4.3 Unsuccessful Operation

Not applicable.

8.7.4.4 Abnormal Conditions

8.7.4.4.1 Abnormal Condition at the 5GC

If the NG RESET message includes the *UE-associated Logical NG-connection List* IE, but neither the *AMF UE NGAP ID* IE nor the *RAN UE NGAP ID* IE is present for a *UE-associated Logical NG-connection Item* IE, then the AMF shall ignore the *UE-associated Logical NG-connection Item* IE. The AMF may return the empty *UE-associated Logical NG-connection Item* IE in the *UE-associated Logical NG-connection List* IE in the NG RESET ACKNOWLEDGE message.

8.7.4.4.2 Abnormal Condition at the NG-RAN

If the NG RESET message includes the *UE-associated Logical NG-connection List* IE, but neither the *AMF UE NGAP ID* IE nor the *RAN UE NGAP ID* IE is present for a *UE-associated Logical NG-connection Item* IE, then the NG-RAN node shall ignore the *UE-associated Logical NG-connection Item* IE. The NG-RAN node may return the empty *UE-associated Logical NG-connection Item* IE in the *UE-associated Logical NG-connection List* IE in the NG RESET ACKNOWLEDGE message.

8.7.4.4.3 Crossing of NG RESET Messages

If an NG Reset procedure is ongoing in the NG-RAN node and the NG-RAN node receives an NG RESET message from the peer entity on the same NG interface related to one or several UE associations previously requested to be reset, indicated explicitly or implicitly in the received NG RESET message, the NG-RAN node shall respond with the NG RESET ACKNOWLEDGE message as described in 8.7.4.2.1.

If an NG Reset procedure is ongoing in the AMF and the AMF receives an NG RESET message from the peer entity on the same NG interface related to one or several UE associations previously requested to be reset, indicated explicitly or implicitly in the received NG RESET message, the AMF shall respond with the NG RESET ACKNOWLEDGE message as described in 8.7.4.2.2.

If the NG-RAN node supports A-IoT and is communicating directly with an AIOTF and if an NG Reset procedure is ongoing in the NG-RAN node and the NG-RAN node receives an NG RESET message from the peer entity on the same NG interface, the NG-RAN node shall respond with the NG RESET ACKNOWLEDGE message as described in 8.7.4.2.1.

If the NG-RAN node supports A-IoT and is communicating directly with an AIOTF and if an NG Reset procedure is ongoing in the AIOTF and the AIOTF receives an NG RESET message from the NG-RAN node on the same NG interface, the AIOTF shall respond with the NG RESET ACKNOWLEDGE message as described in 8.7.4.2.2.

8.7.5 Error Indication

8.7.5.1 General

The Error Indication procedure is initiated by a node in order to report detected errors in one incoming message, provided they cannot be reported by an appropriate failure message.

If the error situation arises due to reception of a message utilising UE-associated signalling, then the Error Indication procedure uses UE-associated signalling. Otherwise the procedure uses non-UE associated signalling.

If the NG-RAN node supports A-IoT and is communicating directly with an AIOTF, as specified in TS 23.369 [60], the Error Indication procedure, as depicted in Figures 8.7.5.2-3 and 8.7.5.2-4 is executed between the NG-RAN node and the AIOTF.

8.7.5.2 Successful Operation

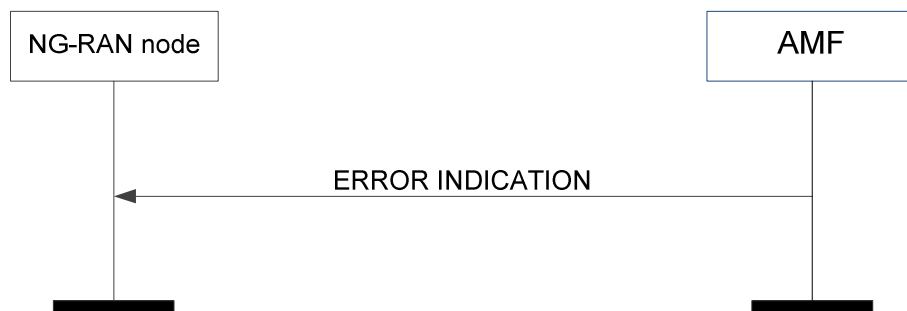


Figure 8.7.5.2-1: Error indication initiated by the AMF

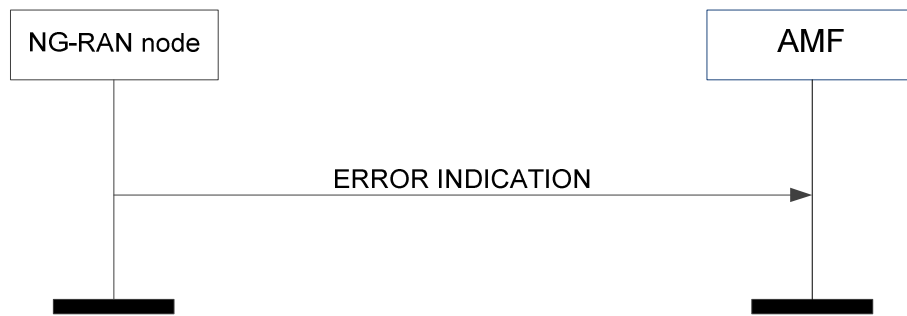


Figure 8.7.5.2-2: Error indication initiated by the NG-RAN node to the AMF

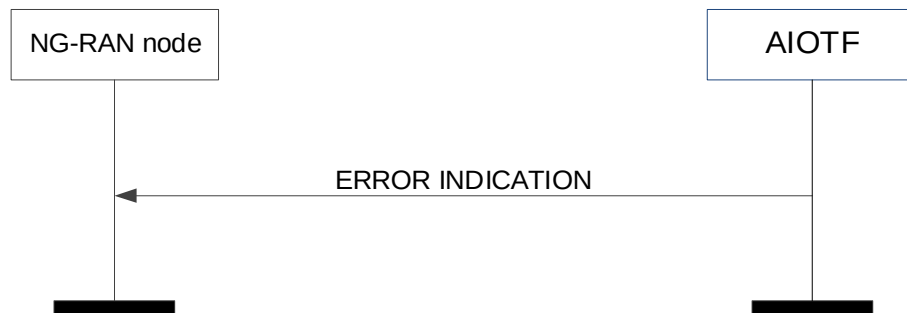


Figure 8.7.5.2-3: Error indication initiated by the AIOTF

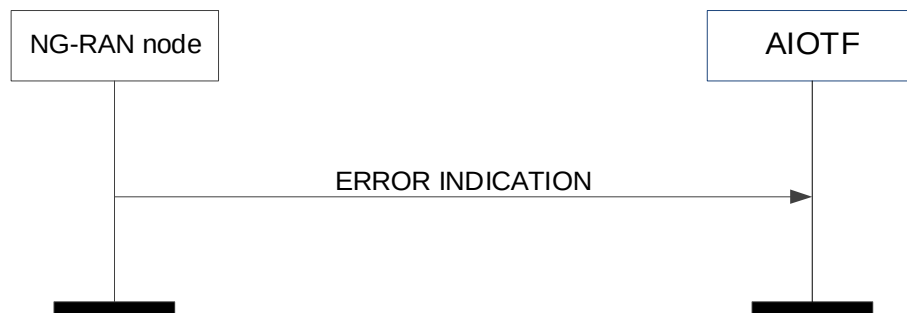


Figure 8.7.5.2-4: Error indication initiated by the NG-RAN node to the AIOTF

When the conditions defined in clause 10 are fulfilled, the Error Indication procedure is initiated by an ERROR INDICATION message sent from the receiving node.

The ERROR INDICATION message shall contain at least either the *Cause* IE or the *Criticality Diagnostics* IE. In case the Error Indication procedure is triggered by utilising UE-associated signalling the *AMF UE NGAP ID* IE and the *RAN UE NGAP ID* IE shall be included in the ERROR INDICATION message. In the case of NB-IoT UE using Control Plane CIoT 5GS Optimisation, upon receiving the *5G-S-TMSI* IE in the ERROR INDICATION message, the receiving node shall use the 5G-S-TMSI to identify the UE. If one or both of the *AMF UE NGAP ID* IE and the *RAN UE NGAP ID* IE are not correct, the cause shall be set to an appropriate value, e.g., "Unknown local UE NGAP ID" or "Inconsistent remote UE NGAP ID".

8.7.5.3 Abnormal Conditions

Void.

8.7.6 AMF Status Indication

8.7.6.1 General

The purpose of the AMF Status Indication procedure is to support AMF management functions. The procedure uses non UE-associated signalling.

8.7.6.2 Successful Operation

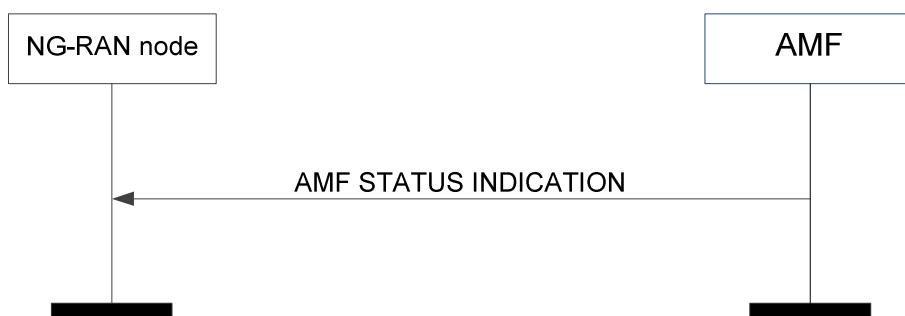


Figure 8.7.6.2-1: AMF status indication

The AMF initiates the procedure by sending an AMF STATUS INDICATION message to the NG-RAN node.

Upon receipt of the AMF STATUS INDICATION message, the NG-RAN node shall consider the indicated GUAMI(s) will be unavailable and perform AMF reselection as defined in TS 23.501 [9].

The NG-RAN node shall, if supported, act accordingly as specified in TS 23.501 [9], based on the presence or absence of the *Timer Approach for GUAMI Removal* IE.

If the *Backup AMF Name* IE is included in the AMF STATUS INDICATION message, the NG-RAN node shall, if supported, perform AMF reselection considering the AMF as indicated by the *Backup AMF Name* IE as specified in TS 23.501 [9]. If the *Extended Backup AMF Name* IE is included in the AMF STATUS INDICATION message, the NG-RAN node shall, if supported, perform AMF reselection considering the AMF as indicated by the *Extended Backup AMF Name* IE as specified in TS 23.501 [9].

8.7.6.3 Abnormal Conditions

Void.

8.7.7 Overload Start

8.7.7.1 General

The purpose of the Overload Start procedure is to inform an NG-RAN node to reduce the signalling load towards the concerned AMF. The procedure uses non-UE associated signalling.

8.7.7.2 Successful Operation



Figure 8.7.7.2-1: Overload start

The NG-RAN node receiving the OVERLOAD START message shall assume the AMF from which it receives the message as being in an overloaded state.

If the *Overload Action* IE is included the *AMF Overload Response* IE within the OVERLOAD START message, the NG-RAN node shall use it to identify the related signalling traffic. When the *Overload Action* IE is set to

- "reject RRC connection establishments for non-emergency mobile originated data transfer" (i.e., reject traffic corresponding to RRC cause "mo-data", "mo-SMS", "mo-VideoCall" and "mo-VoiceCall" in TS 38.331 [18] or "mo-data" and "mo-VoiceCall" in TS 36.331 [21]), or
- "reject RRC connection establishments for signalling" (i.e., reject traffic corresponding to RRC cause "mo-data", "mo-SMS", "mo-signalling", "mo-VideoCall" and "mo-VoiceCall" in TS 38.331 [18] or "mo-data", "mo-signalling" and "mo-VoiceCall" in TS 36.331 [21]), or
- "only permit RRC connection establishments for emergency sessions and mobile terminated services" (i.e., only permit traffic corresponding to RRC cause "emergency" and "mt-Access" in TS 38.331 [18] or in TS 36.331 [21]), or
- "only permit RRC connection establishments for high priority sessions and mobile terminated services" (i.e., only permit traffic corresponding to RRC cause "highPriorityAccess", "mps-PriorityAccess", "mcs-PriorityAccess" and "mt-Access" in TS 38.331 [18] or "highPriorityAccess", "mo-ExceptionData" and "mt-Access" in TS 36.331 [21]),

the NG-RAN node shall:

- if the *AMF Traffic Load Reduction Indication* IE is included in the OVERLOAD START message, reduce the signalling traffic by the indicated percentage,
- otherwise ensure that only the signalling traffic not indicated as to be rejected is sent to the AMF.

If the *Overload Start NSSAI List* IE is included in the OVERLOAD START message, the NG-RAN node shall:

- if the *Slice Traffic Load Reduction Indication* IE is present, reduce the signalling traffic by the indicated percentage for the UE(s) whose requested NSSAI only include S-NSSAI(s) contained in the *Overload Start NSSAI List* IE, and the signalling traffic indicated as to be reduced by the *Overload Action* IE in the *Slice Overload Response* IE if the IE is present,
- otherwise ensure that only the signalling traffic from UE(s) whose requested NSSAI includes S-NSSAI(s) other than the ones contained in the *Overload Start NSSAI List* IE, or the signalling traffic not indicated as to be reduced by the *Overload Action* IE in the *Slice Overload Response* IE for the UE(s) if the requested NSSAI matched, is sent to the AMF.

If the *GUAMI List* IE is present, the NG-RAN node shall, if supported, use this information to identify to which traffic the above defined rejections shall be applied.

If an overload action is ongoing and the NG-RAN node receives a further OVERLOAD START message, the NG-RAN node shall replace the ongoing overload action with the newly requested one. If the *GUAMI List* IE is present, the NG-RAN node replaces applicable ongoing actions according to TS 38.300 [8], clause 4.10.2.

If an overload control is ongoing and the NG-RAN node receives a further OVERLOAD START message, the NG-RAN node shall replace the contents of the previously received information with the new one.

8.7.7.3 Abnormal Conditions

Void.

8.7.8 Overload Stop

8.7.8.1 General

The purpose of the Overload Stop procedure is to signal to an NG-RAN node the AMF is connected to that the overload situation at the AMF has ended and normal operation shall resume. The procedure uses non-UE associated signalling.

8.7.8.2 Successful Operation



Figure 8.7.8.2-1: Overload stop

The NG-RAN node receiving the OVERLOAD STOP message shall assume that the overload situation at the AMF from which it receives the message has ended and shall resume normal operation for the applicable traffic towards this AMF.

If the *GUAMI List* IE is present, the NG-RAN node shall, if supported, use this information to identify which traffic to cease rejecting, and proceed according to TS 38.300 [8], clause 4.10.2. If no overload action is ongoing for a specific GUAMI value, the NG-RAN node shall ignore this value.

8.7.8.3 Abnormal Conditions

Void.

8.7.9 NG Removal

8.7.9.1 General

The purpose of the NG Removal procedure is to remove the interface between the NG-RAN node and the AMF in a controlled manner. If successful, this procedure erases any existing application level configuration data in the two nodes.

The procedure uses non-UE associated signaling.

8.7.9.2 Successful Operation

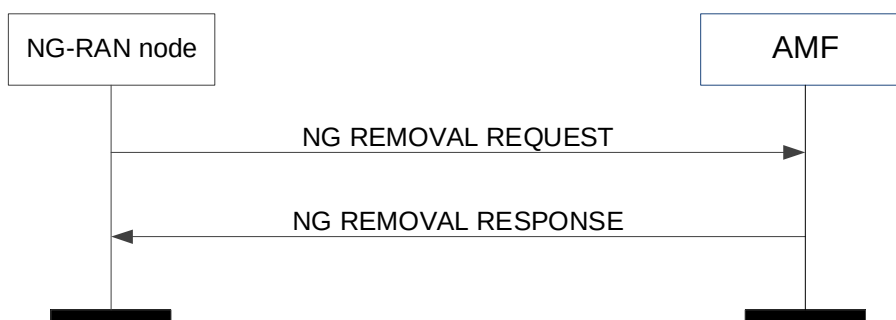


Figure 8.7.9.2-1: NG Removal, successful operation

The NG-RAN node initiates the procedure by sending the NG REMOVAL REQUEST message to the AMF. Upon reception of the NG REMOVAL REQUEST message, the AMF shall reply with the NG REMOVAL RESPONSE message. After receiving the NG REMOVAL RESPONSE message, the NG-RAN node shall initiate removal of the TNL association towards the AMF and may remove all resources associated with that interface. The AMF may then remove all resources associated with the NG-RAN node.

If the *Extended AMF Name* IE is included in the NG REMOVAL RESPONSE message, the NG-RAN node may use this IE as a human readable name of the AMF and shall ignore the *AMF Name* IE.

8.7.9.3 Unsuccessful Operation

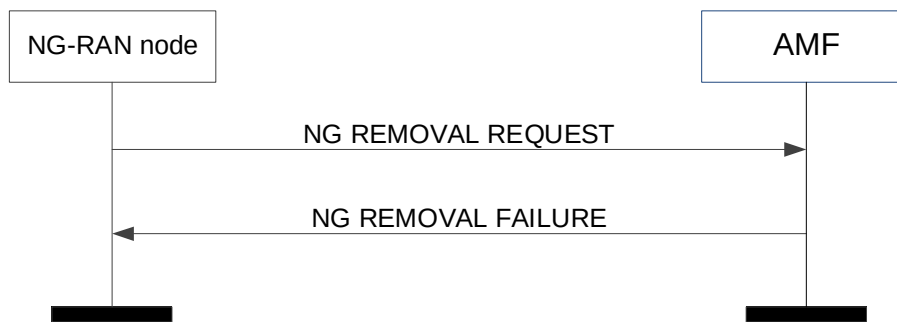


Figure 8.7.9.3-1: NG Removal, unsuccessful operation

If the AMF cannot accept to remove the interface with NG-RAN node, it shall respond with an NG REMOVAL FAILURE message with an appropriate cause value.

8.7.9.4 Abnormal Conditions

Void.

8.8 Configuration Transfer Procedures

8.8.1 Uplink RAN Configuration Transfer

8.8.1.1 General

The purpose of the Uplink RAN Configuration Transfer procedure is to transfer RAN configuration information from the NG-RAN node to the AMF. The AMF does not interpret the transferred RAN configuration information. This procedure uses non-UE associated signalling.

8.8.1.2 Successful Operation

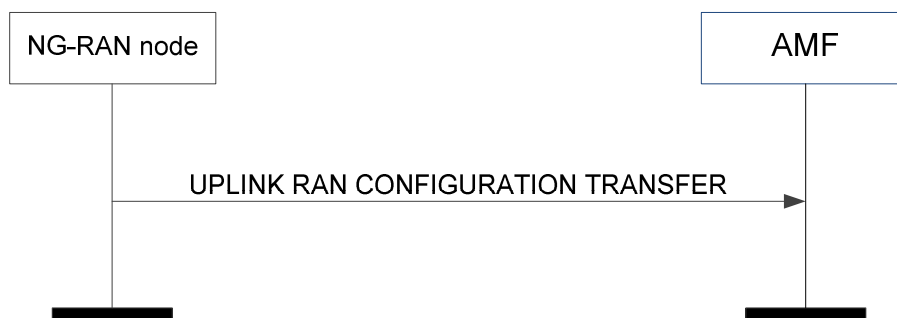


Figure 8.8.1.2-1: Uplink RAN configuration transfer

The NG-RAN node initiates the procedure by sending the UPLINK RAN CONFIGURATION TRANSFER message to the AMF.

If the AMF receives the *SON Configuration Transfer* IE, it shall transparently transfer the *SON Configuration Transfer* IE towards the NG-RAN node indicated in the *Target RAN Node ID* IE which is included in the *SON Configuration Transfer* IE. If the *NR CGI* IE is included within the *Target RAN Node ID* IE, the AMF shall, if supported, ignore the *Global RAN Node ID* IE within the *Target RAN Node ID* IE, and use it to identify the target gNB as described in TS 38.300 [8].

If the AMF receives the *EN-DC SON Configuration Transfer* IE, it shall transparently transfer the *EN-DC SON Configuration Transfer* IE towards an MME serving the eNB indicated in the *Target eNB-ID* IE which is included in the *EN-DC SON Configuration Transfer* IE.

If the AMF receives the *Inter-system SON Configuration Transfer* IE, it shall transparently transfer the *Inter-system SON Configuration Transfer* IE towards an MME serving the eNB indicated in the *Target eNB-ID* IE which is included in the *Inter-system SON Configuration Transfer* IE.

8.8.1.3 Abnormal Conditions

Void.

8.8.2 Downlink RAN Configuration Transfer

8.8.2.1 General

The purpose of the Downlink RAN Configuration Transfer procedure is to transfer RAN configuration information from the AMF to the NG-RAN node. This procedure uses non-UE associated signalling.

8.8.2.2 Successful Operation

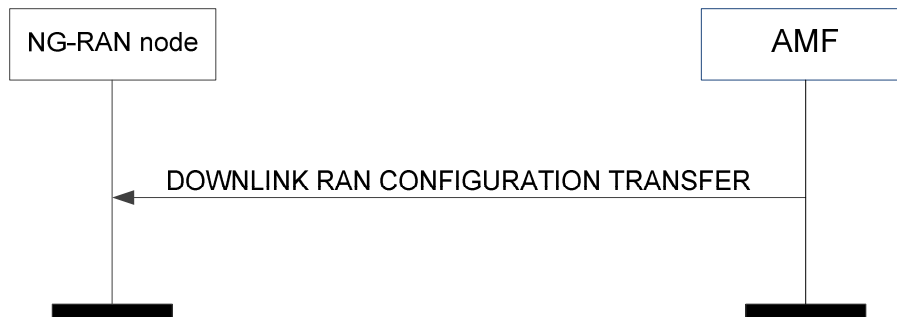


Figure 8.8.2.2-1: Downlink RAN configuration transfer

The procedure is initiated with an *DOWNLINK RAN CONFIGURATION TRANSFER* message sent from the AMF to the NG-RAN node.

If the NG-RAN node receives, in the *SON Configuration Transfer* IE or in the *EN-DC SON Configuration Transfer* IE, the *SON Information* IE containing the *SON Information Request* IE, it may transfer back the requested information either towards the NG-RAN node indicated in the *Source RAN Node ID* IE of the *SON Configuration Transfer* IE or towards an eNB indicated in the *Source eNB-ID* IE of the *EN-DC SON Configuration Transfer* IE by initiating the Uplink RAN Configuration Transfer procedure.

If the NG-RAN node receives, in the *SON Configuration Transfer* IE, the *Xn TNL Configuration Info* IE containing the *Xn Extended Transport Layer Addresses* IE, it may use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If the NG-RAN node receives, in the *SON Configuration Transfer* IE, the *SON Information* IE containing the *SON Information Reply* IE including the *Xn TNL Configuration Info* IE as an answer to a former request, it may use it to initiate the Xn TNL establishment.

In case the *IP-Sec Transport Layer Address* IE is present and the *GTP Transport Layer Addresses* IE within the *Xn Extended Transport Layer Addresses* IE is not empty, GTP traffic is conveyed within an IP-Sec tunnel terminated at the IP-Sec tunnel endpoint given in the *IP-Sec Transport Layer Address* IE.

In case the *IP-Sec Transport Layer Address* IE is not present, GTP traffic is terminated at the endpoints given by the list of addresses in the *Xn GTP Transport Layer Addresses* IE within the *Xn Extended Transport Layer Addresses* IE.

In case the *Xn GTP Transport Layer Addresses* IE is empty and the *IP-Sec Transport Layer Address* IE is present, SCTP traffic is conveyed within an IP-Sec tunnel terminated at the IP-Sec tunnel endpoint given in the *IP-Sec Transport Layer Address* IE, within the *Xn Extended Transport Layer Addresses* IE.

In case the *Xn SCTP Transport Layer Addresses* IE is present and the *IP-Sec Transport Layer Address* IE is also present, the concerned SCTP traffic is conveyed within an IP-Sec tunnel terminated at the IP-Sec tunnel endpoint given in this *IP-Sec Transport Layer Address* IE, within the *Xn Extended Transport Layer Addresses* IE.

If the NG-RAN node receives the *SON Information* IE containing the *SON Information Report* IE it may use it as specified in TS 38.300 [8].

If the NG-RAN node receives the *Inter-system SON Information* IE containing the *Inter-system SON Information Report* IE it may use it as specified in TS 38.300 [8].

If the NG-RAN node receives the *Inter-system SON Information* IE containing the *Inter-system SON Information Request* IE or the *Inter-system SON Information Reply* IE, it may use it as specified in TS 38.300 [8]. If the *Reporting System* IE in the *Inter-system SON Information Request* IE is set to "No Reporting", the DOWNLINK RAN CONFIGURATION TRANSFER message shall be ignored.

If the NG-RAN node is configured to use one IPsec tunnel for all NG and Xn traffic (IPsec star topology) then the traffic to the peer NG-RAN node shall be routed through this IPsec tunnel and the *IP-Sec Transport Layer Address* IE shall be ignored.

Interaction with the Uplink RAN Configuration Transfer procedure:

If the NG-RAN node receives the *SON Configuration Transfer* IE containing the *Requested TNL Information* IE and the *SON Information Request* IE set to "Xn TNL Configuration Info", the NG-RAN node shall, if supported, take the PLMN ID contained in the *Requested TNL Information* IE into account and include Xn TNL configuration information within the UPLINK RAN CONFIGURATION TRANSFER message associated with this PLMN.

8.8.2.3 Abnormal Conditions

If the NG-RAN node receives the *SON Configuration Transfer* IE containing the *Requested TNL Information* IE and the *SON Information Request* IE set to "Xn TNL Configuration Info", and If the PLMN ID within the *Requested TNL Information* IE neither matches the PLMN ID included in the Selected TAI in the *Target RAN Node ID SON* IE in the *SON Configuration Transfer* IE nor the PLMN ID included in the Selected TAI in the *Source RAN Node-ID* IE in the *SON Configuration Transfer* IE, the behaviour of the receiving NG-RAN node is not specified.

8.9 Warning Message Transmission Procedures

8.9.1 Write-Replace Warning

8.9.1.1 General

The purpose of Write-Replace Warning procedure is to start or overwrite the broadcasting of warning messages. The procedure uses non UE-associated signalling.

8.9.1.2 Successful Operation

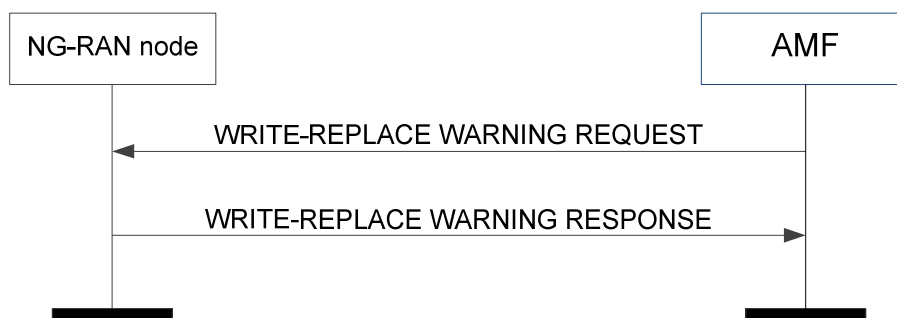


Figure 8.9.1.2-1: Write-Replace Warning procedure: successful operation

The AMF initiates the procedure by sending a WRITE-REPLACE WARNING REQUEST message to the NG-RAN node.

Upon receipt of the WRITE-REPLACE WARNING REQUEST message, the NG-RAN node shall prioritise its resources to process the warning message.

If, in a certain area, broadcast of a warning message is already ongoing and the NG-RAN node receives a WRITE-REPLACE WARNING REQUEST message with *Message Identifier* IE and/or *Serial Number* IE which are different from those in the warning message being broadcast, and if the *Concurrent Warning Message Indicator* IE is not present, the NG-RAN node shall replace the warning message being broadcast with the newly received one for that area.

If the NG-RAN node receives a WRITE-REPLACE WARNING REQUEST message with a warning message identified by the *Message Identifier* IE and *Serial Number* IE and if there are no prior warning messages being broadcast in any of the warning areas indicated in the *Warning Area List* IE, the NG-RAN node shall broadcast the received warning message for those area(s).

If, in a certain area, broadcast of one or more warning messages are already ongoing and the NG-RAN node receives a WRITE-REPLACE WARNING REQUEST message with a *Message Identifier* IE and/or *Serial Number* IE which are different from those in any of the warning messages being broadcast, and if the *Concurrent Warning Message Indicator* IE is present, the NG-RAN node shall schedule the received warning message for broadcast, for that area.

If the *Concurrent Warning Message Indicator* IE is present and if a value "0" is received in the *Number of Broadcasts Requested* IE, the NG-RAN node shall broadcast the received warning message indefinitely until requested otherwise to stop broadcasting, except if the *Repetition Period* IE is set to "0".

If, in a certain area, broadcast of one or more warning messages are already ongoing and the NG-RAN node receives a WRITE-REPLACE WARNING REQUEST message with *Message Identifier* IE and *Serial Number* IE which correspond to one of the warning messages already being broadcast in that area, the NG-RAN node shall not start a new broadcast or replace an existing one but it shall still reply by sending a WRITE-REPLACE WARNING RESPONSE message which includes the *Broadcast Completed Area List* IE set according to the ongoing broadcast.

If the *Warning Area List* IE is not included in the WRITE-REPLACE WARNING REQUEST message, the NG-RAN node shall broadcast the indicated message in all of the cells within the NG-RAN node.

If the *Warning Type* IE is included in the WRITE-REPLACE WARNING REQUEST message, the NG-RAN node shall broadcast the Primary Notification irrespective of the setting of the *Repetition Period* IE and the *Number of Broadcasts Requested* IE, and process the Primary Notification according to TS 36.331 [21] and TS 38.331 [18].

If the *Data Coding Scheme* IE and the *Warning Message Contents* IE are both included in the WRITE-REPLACE WARNING REQUEST message, the NG-RAN node shall schedule a broadcast of the warning message according to the value of the *Repetition Period* IE and the *Number of Broadcasts Requested* IE and process the warning message according to TS 36.331 [21] and TS 38.331 [18].

If the *Warning Area Coordinates* IE is included in the WRITE-REPLACE WARNING REQUEST message, the NG-RAN node shall include this information together with the warning message being broadcast according to TS 36.331 [21] and TS 38.331 [18].

The NG-RAN node acknowledges the WRITE-REPLACE WARNING REQUEST message by sending a WRITE-REPLACE WARNING RESPONSE message to the AMF.

If the *Broadcast Completed Area List* IE is not included in the WRITE-REPLACE WARNING RESPONSE message, the AMF shall consider that the broadcast is unsuccessful in all the cells within the NG-RAN node.

8.9.1.3 Unsuccessful Operation

Not applicable.

8.9.1.4 Abnormal Conditions

If the *Concurrent Warning Message Indicator* IE is not present and if a value "0" is received in the *Number of Broadcasts Requested* IE, the NG-RAN node shall not broadcast the received secondary notification.

If the *Concurrent Warning Message Indicator* IE is included and if a value "0" is received in the *Repetition Period* IE, the NG-RAN node shall not broadcast the received warning message except if the *Number of Broadcasts Requested* IE is set to "1".

If the *Concurrent Warning Message Indicator* IE is not included and if a value "0" is received in the *Repetition Period* IE, the NG-RAN node shall not broadcast the received secondary notification except if the *Number of Broadcasts Requested* IE is set to "1".

8.9.2 PWS Cancel

8.9.2.1 General

The purpose of the PWS Cancel procedure is to cancel an already ongoing broadcast of a warning message. The procedure uses non UE-associated signalling.

8.9.2.2 Successful Operation

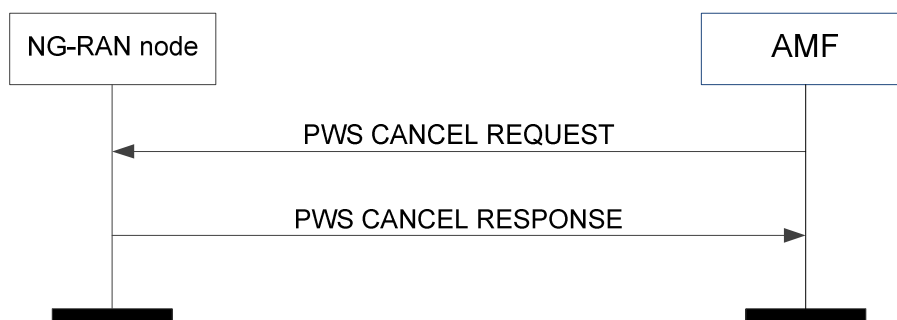


Figure 8.9.2.2-1: PWS Cancel procedure: successful operation

The AMF initiates the procedure by sending a PWS CANCEL REQUEST message to the NG-RAN node.

If the NG-RAN node receives a PWS CANCEL REQUEST message and broadcast of the warning message identified by the *Message Identifier* and *Serial Number* IE is ongoing in an area indicated within the *Warning Area List* IE, the NG-RAN node shall stop broadcasting the warning message within that area and discard the warning message for that area.

If the *Warning Area List* IE is not included in the PWS CANCEL REQUEST message, the NG-RAN node shall stop broadcasting and discard the warning message identified by the *Message Identifier* IE and the *Serial Number* IE in all of the cells in the NG-RAN node.

The NG-RAN node shall acknowledge the PWS CANCEL REQUEST message by sending the PWS CANCEL RESPONSE message, with the *Message Identifier* IE and the *Serial Number* IE copied from the PWS CANCEL REQUEST message and shall, if there is an area to report where an ongoing broadcast was stopped successfully, include the *Broadcast Cancelled Area List* IE.

If an area included in the *Warning Area List* IE in the PWS CANCEL REQUEST message does not appear in the *Broadcast Cancelled Area List* IE, the AMF shall consider that the NG-RAN node had no ongoing broadcast to stop for the same *Message Identifier* and *Serial Number* in that area.

If the *Broadcast Cancelled Area List* IE is not included in the PWS CANCEL RESPONSE message, the AMF shall consider that the NG-RAN node had no ongoing broadcast to stop for the same *Message Identifier* and *Serial Number*.

If the *Cancel-All Warning Messages Indicator* IE is present in the PWS CANCEL REQUEST message, then the NG-RAN node shall stop broadcasting and discard all warning messages for the area as indicated in the *Warning Area List* IE or in all the cells of the NG-RAN node if the *Warning Area List* IE is not included. The NG-RAN node shall acknowledge the PWS CANCEL REQUEST message by sending the PWS CANCEL RESPONSE message, with the *Message Identifier* IE and the *Serial Number* IE copied from the PWS CANCEL REQUEST message and shall, if there is area to report where an ongoing broadcast was stopped successfully, include the *Broadcast Cancelled Area List* IE with the *Number of Broadcasts* IE set to 0.

8.9.2.3 Unsuccessful Operation

Not applicable.

8.9.2.4 Abnormal Conditions

Void.

8.9.3 PWS Restart Indication

8.9.3.1 General

The purpose of the PWS Restart Indication procedure is to inform the AMF that PWS information for some or all cells of the NG-RAN node may be reloaded from the CBC if needed. The procedure uses non UE-associated signalling.

8.9.3.2 Successful Operation

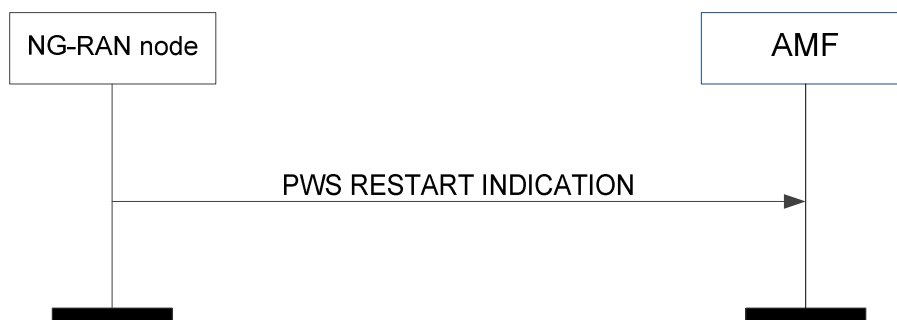


Figure 8.9.3.2-1: PWS restart indication

The NG-RAN node initiates the procedure by sending a PWS RESTART INDICATION message to the AMF. On receipt of a PWS RESTART INDICATION message, the AMF shall act as defined in TS 23.527 [57].

If the Emergency Area ID is available, the NG-RAN node shall also include it in the *Emergency Area ID List for Restart* IE.

8.9.3.3 Abnormal Conditions

Void.

8.9.4 PWS Failure Indication

8.9.4.1 General

The purpose of the PWS Failure Indication procedure is to inform the AMF that ongoing PWS operation for one or more cells of the NG-RAN node has failed. The procedure uses non UE-associated signalling.

8.9.4.2 Successful Operation

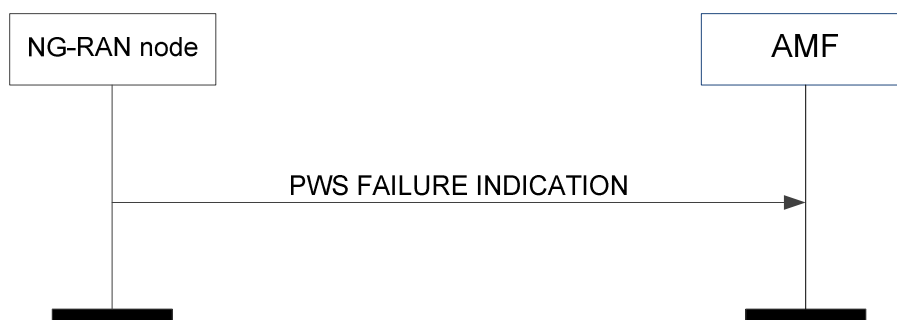


Figure 8.9.4.2-1: PWS failure indication

The NG-RAN node initiates the procedure by sending a PWS FAILURE INDICATION message to the AMF. On receipt of a PWS FAILURE INDICATION message, the AMF shall act as defined in TS 23.041 [22].

8.9.4.3 Abnormal Conditions

Void.

8.10 NRPPa Transport Procedures

8.10.1 General

The purpose of the NRPPa Transport procedures is to carry NRPPa signalling (defined in TS 38.455 [19]) between the NG-RAN node and the LMF over the NG interface.

The Downlink UE Associated NRPPa Transport procedure and the Uplink UE Associated NRPPa Transport procedure use UE-associated signalling. The UE-associated signalling is used to support E-CID Location Information Transfer, Positioning Information Transfer, Measurement Preconfiguration Information Transfer, and Reporting of General Error Situations due to reception of an NRPPa message that utilized UE-associated signalling.

The Downlink Non UE Associated NRPPa Transport procedure and the Uplink Non UE Associated NRPPa Transport procedure use non-UE associated signalling. The non-UE associated signalling is used to support OTDOA Information Transfer, Assistance Information Transfer, TRP Information Transfer, Measurement Information Transfer, PRS Information Transfer, Area-specific Information Transfer, Positioning Data Collection Information Transfer and Reporting of General Error Situations due to reception of an NRPPa message that utilized non-UE associated signalling.

8.10.2 Successful Operations

8.10.2.1 DOWNLINK UE ASSOCIATED NRPPa TRANSPORT

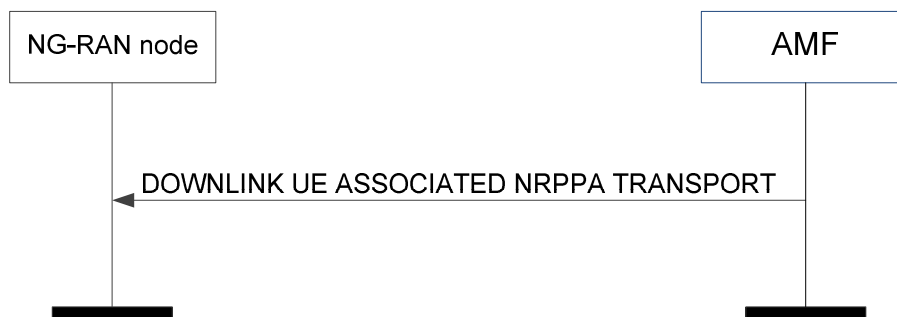


Figure 8.10.2.1-1: Downlink UE-associated NRPPa transport

The AMF initiates the procedure by sending the DOWNLINK UE ASSOCIATED NRPPa TRANSPORT message to the NG-RAN node.

8.10.2.2 UPLINK UE ASSOCIATED NRPPA TRANSPORT

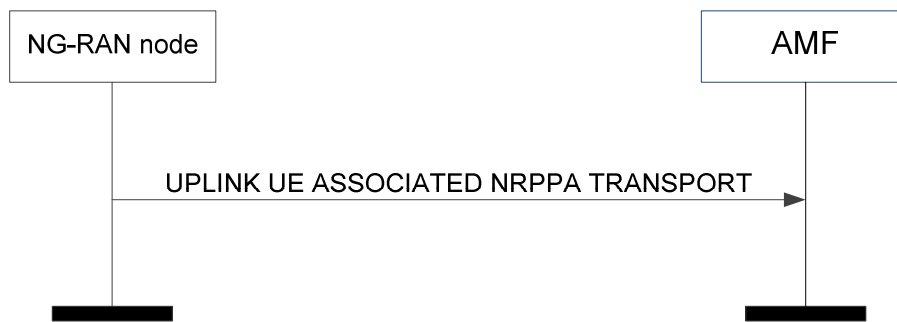


Figure 8.10.2.2-1: Uplink UE-associated NRPPa transport

The NG-RAN node initiates the procedure by sending the UPLINK UE ASSOCIATED NRPPA TRANSPORT message to the AMF.

8.10.2.3 DOWNLINK NON UE ASSOCIATED NRPPA TRANSPORT



Figure 8.10.2.3-1: Downlink non UE-associated NRPPa transport

The AMF initiates the procedure by sending the DOWNLINK NON UE ASSOCIATED NRPPA TRANSPORT message to the NG-RAN node.

8.10.2.4 UPLINK NON UE ASSOCIATED NRPPA TRANSPORT

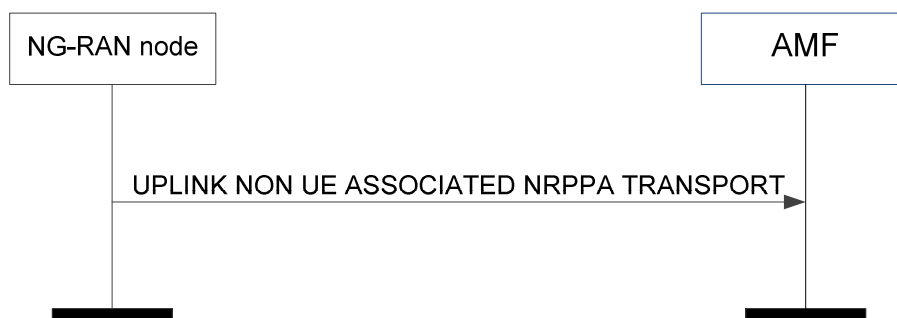


Figure 8.10.2.4-1: Uplink non UE-associated NRPPa transport

The NG-RAN node initiates the procedure by sending the UPLINK NON UE ASSOCIATED NRPPA TRANSPORT message to the AMF.

8.10.3 Unsuccessful Operations

Not applicable.

8.10.4 Abnormal Conditions

If an AMF receives an UPLINK UE ASSOCIATED NRPPA TRANSPORT message with an unknown Routing ID for the UE, the AMF shall ignore the message.

If an AMF receives an UPLINK NON UE ASSOCIATED NRPPA TRANSPORT message indicating an unknown or unreachable Routing ID, the AMF shall ignore the message.

8.11 Trace Procedures

8.11.1 Trace Start

8.11.1.1 General

The purpose of the Trace Start procedure is to allow the AMF to request the NG-RAN node to initiate a trace session for a UE. The procedure uses UE-associated signalling. If no UE-associated logical NG-connection exists, the UE-associated logical NG-connection shall be established as part of the procedure.

8.11.1.2 Successful Operation



Figure 8.11.1.2-1: Trace start

The AMF initiates the procedure by sending a TRACE START message. Upon reception of the TRACE START message, the NG-RAN node shall initiate the requested trace session as described in TS 32.422 [11].

If the *Trace Activation* IE is included in the TRACE START message which includes the *MDT Activation* IE set to "Immediate MDT and Trace", the NG-RAN node shall, if supported, initiate the requested trace session and MDT session as described in TS 32.422 [11].

If the *Trace Activation* IE is included in the TRACE START message which includes the *MDT Activation* IE set to "Immediate MDT Only", "Logged MDT only", the NG-RAN node shall, if supported, initiate the requested MDT session as described in TS 32.422 [11] and the NG-RAN node shall ignore the *Interfaces To Trace* IE and the *Trace Depth* IE.

If the *Trace Activation* IE includes the *MDT Location Information* IE within the *MDT Configuration* IE, the NG-RAN node shall, if supported, store this information and take it into account in the requested MDT session.

If the *Trace Activation* IE is included in the TRACE START message which includes the *MDT Activation* IE set to "Immediate MDT Only", "Logged MDT only" and if the *Signalling Based MDT PLMN List* IE is included in the *MDT Configuration* IE, the NG-RAN node may use it to propagate the MDT Configuration as described in TS 37.320 [41].

If the *Trace Activation* IE includes the *Bluetooth Measurement Configuration* IE within the *MDT Configuration* IE, the NG-RAN node shall, if supported, take it into account for MDT Configuration as described in TS 37.320 [41].

If the *Trace Activation* IE includes the *WLAN Measurement Configuration* IE within the *MDT Configuration* IE, the NG-RAN node shall, if supported, take it into account for MDT Configuration as described in TS 37.320 [41].

If the *Trace Activation* IE includes the *Sensor Measurement Configuration* IE within the *MDT Configuration* IE, the NG-RAN node shall, if supported, take it into account for MDT Configuration as described in TS 37.320 [41].

If the *Trace Activation IE* includes the *MDT Configuration IE* and if the NG-RAN node is a gNB at least the *MDT Configuration-NR IE* shall be present, while if the NG-RAN node is an ng-eNB at least the *MDT Configuration-EUTRA IE* shall be present.

If the *PNI-NPN Area Scope of MDT IE* is included in the *MDT Configuration-NR IE* included in the TRACE START message, the NG-RAN node shall, if supported, use it to derive the MDT area scope for MDT measurement collection in PNI-NPN areas. Upon reception of the *PNI-NPN Area Scope of MDT IE*, the NG-RAN node shall consider that the area scope for MDT measurement collection in PNI-NPN areas is defined only by the areas included in the *PNI-NPN Area Scope of MDT IE*.

If the *Trace Activation IE* includes the *MN Only MDT Collection IE* and the *MN Only MDT Collection IE* is set to "MN only", the NG-RAN node shall, if supported, consider that the *MDT Configuration-NR IE* or the *MDT Configuration-EUTRA IE* is only applicable for the MN if the UE is configured with MR-DC.

If the *Network Slice Area Scope of MDT IE* is included in the *MDT Configuration-NR IE* included in the TRACE START message, the NG-RAN node shall, if supported, use it to derive the MDT area scope for MDT measurement collection. Upon reception of the *Network Slice Area Scope of MDT IE*, the NG-RAN node shall consider that the area scope for MDT measurement collection is defined only by the *Network Slice Area Scope of MDT IE* and the *Area Scope of MDT IE*.

Interactions with other procedures:

If the NG-RAN node is not able to initiate the trace session due to ongoing handover of the UE to another NG-RAN node, the NG-RAN node shall initiate a Trace Failure Indication procedure with the appropriate cause value.

8.11.1.3 Abnormal Conditions

If the *PNI-NPN Area Scope of MDT IE* is included in the *MDT Configuration-NR IE* in the TRACE START message, and the *Area Scope of MDT IE* is set to "PNI-NPN Based MDT", the NG-RAN node shall, if supported, use the *Area Scope of MDT IE* to derive the MDT area scope for MDT measurement collection in PNI-NPN areas, and ignore the *PNI-NPN Area Scope of MDT IE*.

If the *Network Slice Area Scope of MDT IE* is included in the *MDT Configuration-NR IE* in the TRACE START message, and the *MDT Activation IE* is set to "Logged MDT only", the NG-RAN node shall ignore the *Network Slice Area Scope of MDT IE*.

8.11.2 Trace Failure Indication

8.11.2.1 General

The purpose of the Trace Failure Indication procedure is to allow the NG-RAN node to inform the AMF that a Trace Start procedure or a Deactivate Trace procedure or an ongoing trace has failed due to an interaction with a handover procedure or, while the UE is in RRC_INACTIVE state, due to reception of multiple trace activations. The procedure uses UE-associated signalling.

8.11.2.2 Successful Operation

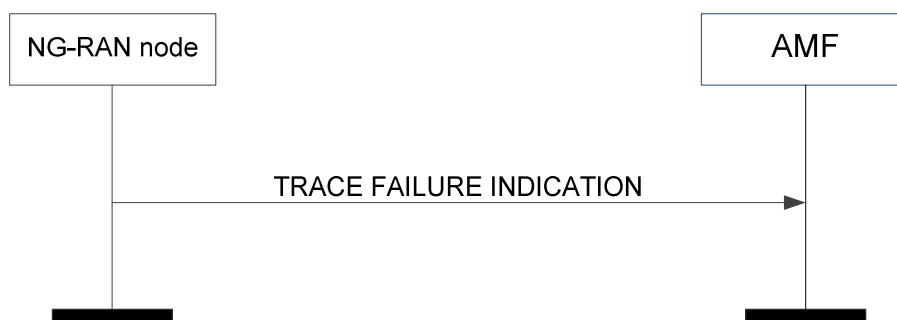


Figure 8.11.2.2-1: Trace failure indication

The NG-RAN node initiates the procedure by sending a TRACE FAILURE INDICATION message. Upon reception of the TRACE FAILURE INDICATION message, the AMF shall take appropriate actions based on the failure reason indicated by the *Cause* IE.

8.11.2.3 Abnormal Conditions

Void.

8.11.3 Deactivate Trace

8.11.3.1 General

The purpose of the Deactivate Trace procedure is to allow the AMF to request the NG-RAN node to stop the trace session for the indicated trace reference. The procedure uses UE-associated signalling.

8.11.3.2 Successful Operation

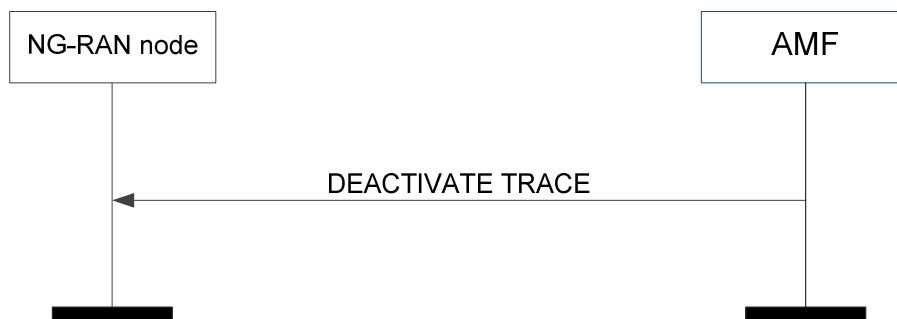


Figure 8.11.3.2-1: Deactivate trace

The AMF initiates the procedure by sending a DEACTIVATE TRACE message to the NG-RAN node as described in TS 32.422 [11]. Upon reception of the DEACTIVATE TRACE message, the NG-RAN node shall stop the trace session for the indicated trace reference in the *NG-RAN Trace ID* IE.

Interactions with other procedures:

If the NG-RAN node is not able to stop the trace session due to ongoing handover of the UE to another NG-RAN node, the NG-RAN node shall initiate a Trace Failure Indication procedure with the appropriate cause value.

8.11.3.3 Abnormal Conditions

Void.

8.11.4 Cell Traffic Trace

8.11.4.1 General

The purpose of the Cell Traffic Trace procedure is to send the allocated Trace Recording Session Reference and the Trace Reference to the AMF. The procedure uses UE-associated signalling.

8.11.4.2 Successful Operation

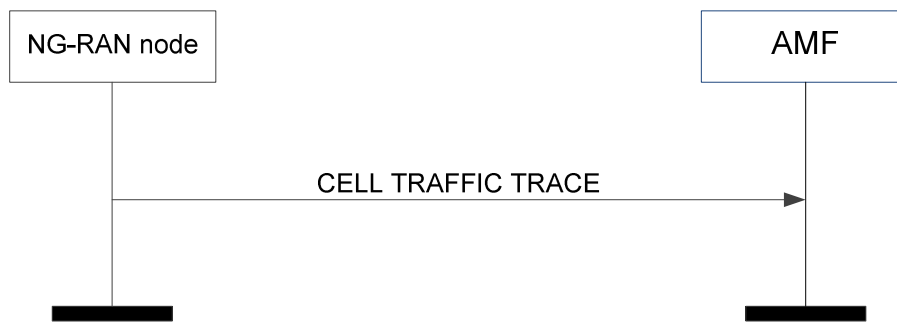


Figure 8.11.4.2-1: Cell traffic trace

The NG-RAN node initiates the procedure by sending a CELL TRAFFIC TRACE message.

If the *Privacy Indicator* IE is included in the message, the AMF shall take the information into account for anonymization of MDT data as described in TS 32.422 [11].

8.11.4.3 Abnormal Conditions

Void.

8.12 Location Reporting Procedures

8.12.1 Location Reporting Control

8.12.1.1 General

The purpose of the Location Reporting Control procedure is to allow the AMF to request the NG-RAN node to report the UE's current location, or the UE's last known location with time stamp, or the UE's presence in the area of interest while in CM-CONNECTED state, or the Aerial UE's flight information as specified in TS 23.501 [9] and TS 23.502 [10]. The procedure uses UE-associated signalling.

8.12.1.2 Successful Operation

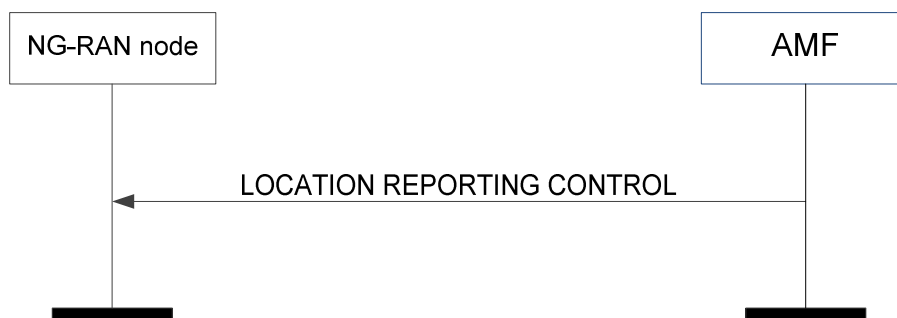


Figure 8.12.1.2-1: Location reporting control

The AMF initiates the procedure by sending a LOCATION REPORTING CONTROL message to the NG-RAN node. On receipt of the LOCATION REPORTING CONTROL message the NG-RAN node shall perform the requested location reporting control action for the UE.

The *Location Reporting Request Type* IE indicates to the NG-RAN node whether:

- to report directly;
- to report upon change of serving cell;

- to report UE presence in the area of interest;
- to stop reporting at change of serving cell;
- to stop reporting UE presence in the area of interest;
- to cancel location reporting for the UE;
- to report upon change of serving cell and to report UE presence in the area of interest;
- to report flight information for the Aerial UE;
- to cancel flight information reporting for the Aerial UE.

If the *Area Of Interest List* IE is included in the *Location Reporting Request Type* IE in the LOCATION REPORTING CONTROL message, the NG-RAN node shall store this information and use it to track the UE's presence in the area of interest as defined in TS 23.502 [10].

NOTE: The NG-RAN reports the UE presence for all set of Location Reporting Reference IDs for inter-NG-RAN node handover.

If the *Additional Location Information* IE is included in the LOCATION REPORTING CONTROL message and set to "Include PSCell" then, if Dual Connectivity is activated, the NG-RAN node shall include the current PSCell in the report. If a report upon change of serving cell is requested, the NG-RAN node shall provide the report also whenever the UE changes the PSCell, and when Dual Connectivity is activated.

If reporting upon change of serving cell is requested, the NG-RAN node shall send a report immediately and shall send a report whenever the UE's location changes.

If the *Event Type* IE is set to "stop UE presence in the area of interest", and if the *Additional Cancelled Location Reporting Reference ID List* IE is included in the *Location Reporting Request Type* IE within the LOCATION REPORTING CONTROL message, the NG-RAN node shall, if supported, stop reporting UE presence for all received Location Reporting Reference IDs.

If the *Event Type* IE is present as "report aerial UE flight information", and the *Aerial UE Flight Information report control* IE is included in the *Location Reporting Request Type* IE in the LOCATION REPORTING CONTROL message, the NG-RAN node shall, if supported, store this information and use it to report the aerial UE flight information as defined in TS 23.502 [10].

8.12.1.3 Abnormal Conditions

Interactions with Location Reporting Failure Indication procedure:

If the NG-RAN node receives a LOCATION REPORTING CONTROL message containing several *Location Reporting Reference ID* IE set to the same value, the NG-RAN node shall send the LOCATION REPORTING FAILURE INDICATION message with an appropriate cause value.

If the *Location Reporting Request Type* IE in the received LOCATION REPORTING CONTROL message contains the *Event Type* IE set to neither "UE presence in the area of interest" nor "change of serving cell and UE presence in the area of interest", but the *Area of Interest List* IE is present, the NG-RAN node shall ignore the *Area of Interest List* IE. and proceed with the Location Reporting Procedure.

8.12.2 Location Reporting Failure Indication

8.12.2.1 General

The purpose of the Location Reporting Failure Indication procedure is to allow the NG-RAN node to inform the AMF that the location reporting request contained in the Location Reporting Control procedure, the Handover Resource Allocation procedure or the Initial Context Setup procedure has failed. The procedure is also used to inform the AMF in case an ongoing aerial UE flight information reporting is failed. The procedure uses UE-associated signalling.

8.12.2.2 Successful Operation

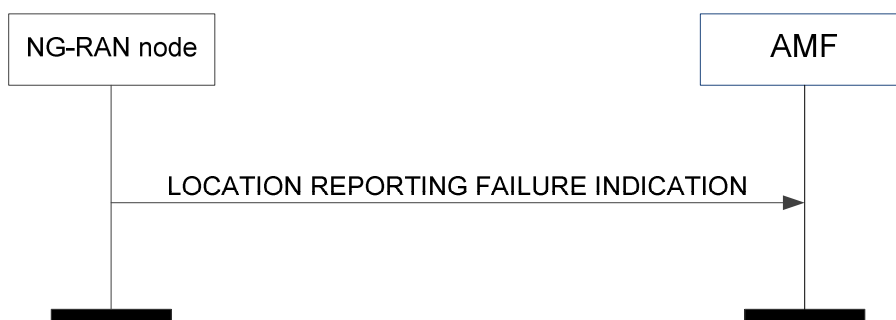


Figure 8.12.2.2-1: Location reporting failure indication

The NG-RAN node initiates the procedure by sending a LOCATION REPORTING FAILURE INDICATION message to the AMF. Upon reception of the LOCATION REPORTING FAILURE INDICATION message the AMF shall, based on the failure reason indicated by the *Cause IE*, take appropriate action.

If an ongoing aerial UE flight information reporting is failed, the NG-RAN node initiates the procedure by sending a LOCATION REPORTING FAILURE INDICATION message to the AMF. Upon reception of the LOCATION REPORTING FAILURE INDICATION message, the AMF shall, based on the *Aerial UE Flight Information Ongoing Reporting Failed IE*, take appropriate action according to TS 23.501 [9].

8.12.2.3 Abnormal Conditions

Void.

8.12.3 Location Report

8.12.3.1 General

The purpose of the Location Report procedure is to provide the UE's current location, the UE's last known location with time stamp, or the UE's presence in the area of interest to the AMF. The procedure uses UE-associated signalling. For Aerial UE, the Location Reporting procedure is to provide the UE's flight formation.

8.12.3.2 Successful Operation



Figure 8.12.3.2-1: Location report

The NG-RAN node initiates the procedure by sending a LOCATION REPORT message to the AMF. The LOCATION REPORT message may be used as a response to the LOCATION REPORTING CONTROL message.

8.12.3.3 Abnormal Conditions

Void.

8.13 UE TNLA Binding Procedures

8.13.1 UE TNLA Binding Release

8.13.1.1 General

The purpose of the UE TNLA Binding Release procedure is to request the NG-RAN node to release the NGAP UE TNLA binding, while requesting the NG-RAN node to maintain NG-U (user plane connectivity) and UE context information as specified in TS 23.502 [10]. The procedure uses UE-associated signalling.

8.13.1.2 Successful Operation



Figure 8.13.1.2-1: UE TNLA binding release request

At reception of the UE TNLA BINDING RELEASE REQUEST message, the NG-RAN node shall release the UE TNLA binding for the UE indicated in the UE TNLA BINDING RELEASE REQUEST message. The NG-RAN node shall keep the NG-U (user plane connectivity) and UE context information for the UE, and behave according to TS 23.502 [10].

8.13.1.3 Abnormal Conditions

Void.

8.14 UE Radio Capability Management Procedures

8.14.1 UE Radio Capability Info Indication

8.14.1.1 General

The purpose of the UE Radio Capability Info Indication procedure is to enable the NG-RAN node to provide to the AMF UE radio capability-related information. The procedure uses UE-associated signalling.

8.14.1.2 Successful Operation



Figure 8.14.1.2-1: UE radio capability info indication

The NG-RAN node controlling a UE-associated logical NG connection initiates the procedure by sending a UE RADIO CAPABILITY INFO INDICATION message to the AMF including the UE radio capability information.

The UE RADIO CAPABILITY INFO INDICATION message may also include paging specific UE radio capability information within the *UE Radio Capability for Paging* IE. If the *UE Radio Capability for Paging* IE includes the *UE Radio Capability for Paging of NR* IE and the *UE Radio Capability for Paging of E-UTRA* IE, the AMF shall, if supported, use it as specified in TS 23.501 [9].

The UE radio capability information received by the AMF shall replace previously stored corresponding UE radio capability information in the AMF for the UE, as described in TS 23.501 [9].

If the UE RADIO CAPABILITY INFO INDICATION message includes the *UE Radio Capability – E-UTRA Format* IE, the AMF shall, if supported, use it as specified in TS 23.501 [9].

If the UE RADIO CAPABILITY INFO INDICATION message includes the *XR Device with 2Rx* IE, the AMF shall, if supported, store this information and use it accordingly.

8.14.1.3 Abnormal Conditions

Void.

8.14.2 UE Radio Capability Check

8.14.2.1 General

The purpose of the UE Radio Capability Check procedure is for the AMF to request the NG-RAN node to derive and provide an indication to the AMF on whether the UE radio capabilities are compatible with the network configuration for IMS voice. The procedure uses UE-associated signalling.

8.14.2.2 Successful Operation

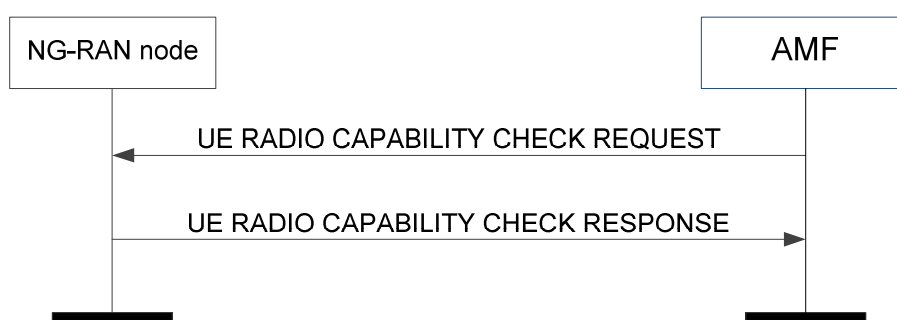


Figure 8.14.2.2-1: UE radio capability check procedure: successful operation

The AMF initiates the procedure by sending a UE RADIO CAPABILITY CHECK REQUEST message. If the UE-associated logical NG-connection is not established, the AMF shall allocate a unique AMF UE NGAP ID to be used for the UE and include the *AMF UE NGAP ID* IE in the UE RADIO CAPABILITY CHECK REQUEST message; by receiving the *AMF UE NGAP ID* IE in the UE RADIO CAPABILITY CHECK REQUEST message, the NG-RAN node establishes the UE-associated logical NG-connection.

Upon receipt of the UE RADIO CAPABILITY CHECK REQUEST message, the NG-RAN node checks whether the UE radio capabilities are compatible with the network configuration for IMS voice, and responds with a UE RADIO CAPABILITY CHECK RESPONSE message, as defined in TS 23.502 [10].

If the *UE Radio Capability* IE is contained in the UE RADIO CAPABILITY CHECK REQUEST message, the NG-RAN node shall use it to determine the value of the *IMS Voice Support Indicator* IE to be included in the UE RADIO CAPABILITY CHECK RESPONSE message.

If the UE RADIO CAPABILITY CHECK REQUEST message contains the *UE Radio Capability ID* IE, the NG-RAN node shall, if supported, use it as specified in TS 23.501 [9] and TS 23.502 [10].

Interactions with UE Radio Capability Info Indication procedure:

If the UE RADIO CAPABILITY CHECK REQUEST message is received with neither the *UE Radio Capability* IE nor the *UE Radio Capability ID* IE, and there is no UE Radio Capability information available in the NG-RAN node, the NG-RAN node shall, if supported, request the UE Radio Capability information from the UE, and send it to the AMF via the UE RADIO CAPABILITY INFO INDICATION message, as specified in TS 23.501 [9].

8.14.2.3 Unsuccessful Operation

Not applicable.

8.14.2.4 Abnormal Conditions

Void.

8.14.3 UE Radio Capability ID Mapping

8.14.3.1 General

The purpose of the UE Radio Capability ID Mapping procedure is for the NG-RAN node to request from the AMF UE Radio Capability information mapped to the UE Radio Capability ID.

The procedure uses non UE-associated signalling.

8.14.3.2 Successful Operation

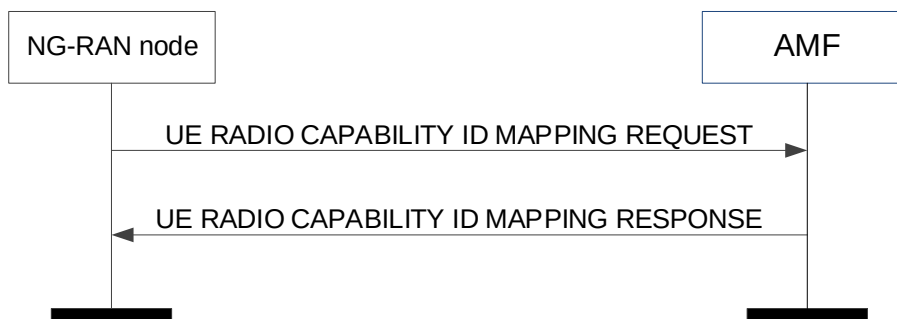


Figure 8.14.3.2-1: UE Radio Capability ID Mapping procedure: successful operation

The NG-RAN node initiates the procedure by sending a UE RADIO CAPABILITY ID MAPPING REQUEST message.

Upon receipt of the UE RADIO CAPABILITY ID MAPPING REQUEST message, the AMF shall provide within the UE RADIO CAPABILITY ID MAPPING RESPONSE message the UE Radio Capability information mapped to the UE Capability ID indicated in the UE RADIO CAPABILITY ID MAPPING REQUEST message.

8.14.3.3 Unsuccessful Operation

Not applicable.

8.14.3.4 Abnormal Conditions

Void.

8.15 Data Usage Reporting Procedures

8.15.1 Secondary RAT Data Usage Report

8.15.1.1 General

The purpose of the Secondary RAT Data Usage Report procedure is to provide information on the used resources of the secondary RAT (e.g. NR resources during MR-DC operation) as specified in TS 23.501 [9]. The procedure uses UE-associated signalling.

8.15.1.2 Successful Operation

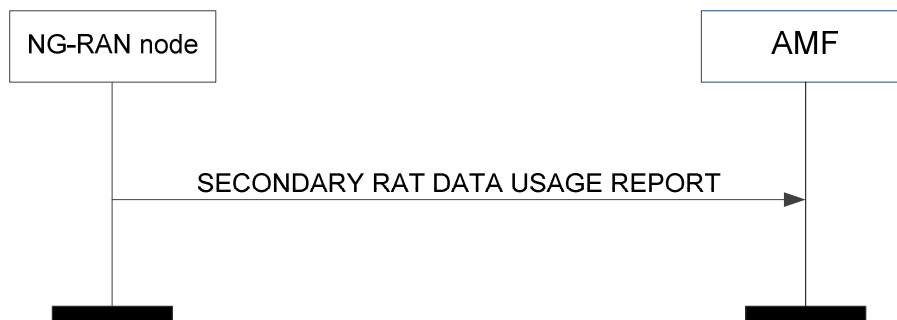


Figure 8.15.1.2-1: Secondary RAT data usage report

The NG-RAN node initiates the procedure by sending the SECONDARY RAT DATA USAGE REPORT message to the AMF.

If the *Handover Flag* IE is included in the SECONDARY RAT DATA USAGE REPORT message, it indicates that for each PDU session the AMF should buffer the *Secondary RAT Data Usage Report Transfer* IE since the secondary RAT data usage report is sent due to handover as defined in TS 23.502 [10].

For each PDU session for which the *Secondary RAT Usage Information List* IE is included in the *Secondary RAT Data Usage Transfer* IE, the SMF shall handle this information as specified in TS 23.502 [10].

The NG-RAN node shall, if supported, report in the SECONDARY RAT DATA USAGE REPORT message location information of the UE in the *User Location Information* IE.

8.15.1.3 Abnormal Conditions

Void.

8.16 RIM Information Transfer Procedures

8.16.1 Uplink RIM Information Transfer

8.16.1.1 General

The purpose of the Uplink RIM Information Transfer procedure is to transfer RIM information from the NG-RAN node to the AMF. The AMF does not interpret the transferred RIM information. This procedure uses non-UE associated signalling.

8.16.1.2 Successful Operation

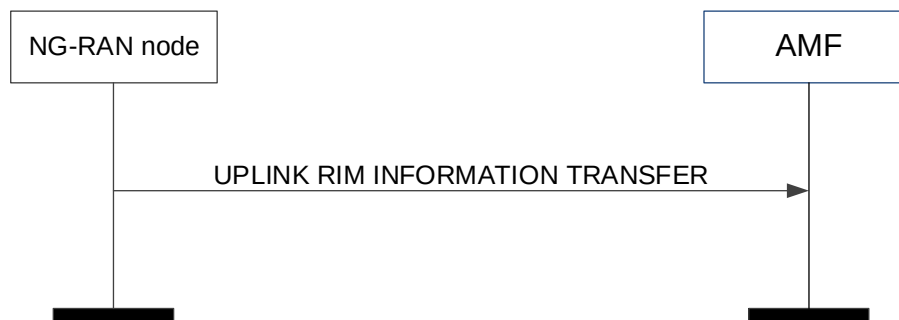


Figure 8.16.1.2-1: Uplink RIM Information Transfer

The NG-RAN node initiates the procedure by sending an UPLINK RIM INFORMATION TRANSFER message to the AMF.

Upon reception of the UPLINK RIM INFORMATION TRANSFER message, the AMF shall transparently transfer it towards the NG-RAN node indicated in the *Target RAN Node ID* IE.

8.16.1.3 Abnormal Conditions

Void.

8.16.2 Downlink RIM Information Transfer

8.16.2.1 General

The purpose of the Downlink RIM Information Transfer procedure is to transfer RIM information from the AMF to the NG-RAN node. This procedure uses non-UE associated signalling.

8.16.2.2 Successful Operation

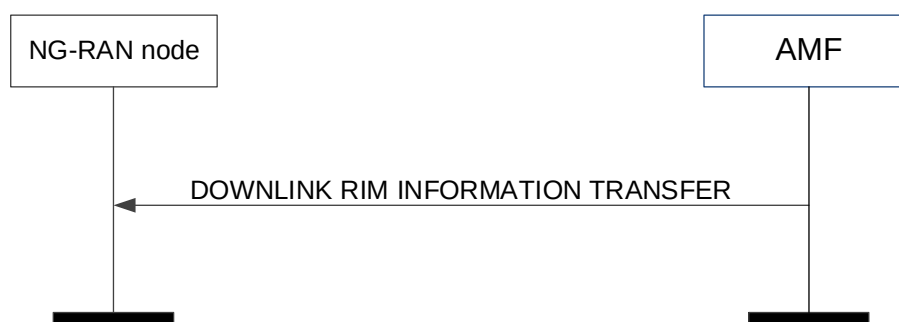


Figure 8.16.2.2-1: Downlink RIM Information Transfer

The AMF initiates the procedure by sending a DOWNLINK RIM INFORMATION TRANSFER message to the NG-RAN node. The NG-RAN node may use the RIM information in the received DOWNLINK RIM INFORMATION TRANSFER message for executing the RIM functionality, as specified in TS 38.300 [8].

8.16.2.3 Abnormal Conditions

Void.

8.17 Broadcast Session Management Procedures

8.17.1 Broadcast Session Setup

8.17.1.1 General

The purpose of the Broadcast Session Setup procedure is to request the NG-RAN node to setup MBS session resources for a broadcast MBS session. The procedure uses non-UE associated signalling.

8.17.1.2 Successful Operation

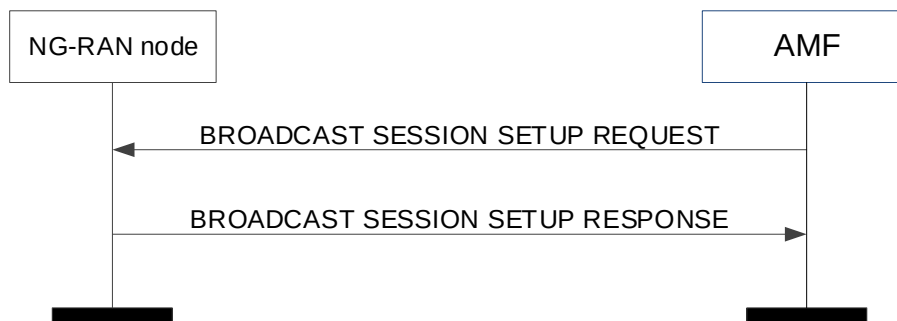


Figure 8.17.1.2-1: Broadcast Session Setup, successful operation.

The AMF initiates the procedure by sending a BROADCAST SESSION SETUP REQUEST message to the NG-RAN node. If the NG-RAN node accepts all the MBS QoS flows in the MBS session at least in one of its cells, the NG-RAN node responds with the BROADCAST SESSION SETUP RESPONSE message.

If the *MBS Service Area* IE is included in the BROADCAST SESSION SETUP REQUEST message, the NG-RAN node shall take it into account as specified in TS 23.247 [44], and TS 38.300 [8] for NTN operation.

If the *MBS Session FSA ID List* IE is included in the BROADCAST SESSION SETUP REQUEST message, the NG-RAN node shall take it into account to determine cells/frequencies within the MBS service area to broadcast MBS session data as specified in TS 23.247 [44].

If the *Supported UE Type List* IE is included in the BROADCAST SESSION SETUP REQUEST message, the NG-RAN node shall, if supported, take it into account for configuring MBS session resources.

If the *Associated Session ID* IE is included in the BROADCAST SESSION SETUP REQUEST message the NG-RAN node shall, if supported, take this information into account to determine whether MBS session resource sharing is possible, as specified in TS 38.300 [8].

8.17.1.3 Unsuccessful Operation

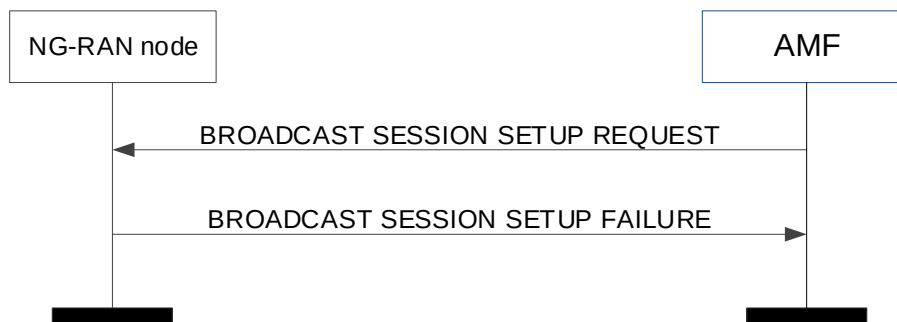


Figure 8.17.1.3-1: Broadcast Session Setup, unsuccessful operation.

If the NG-RAN node is not able to provide the requested MBS session resources for all the MBS QoS flows in the MBS session in any of its cells, it shall send the BROADCAST SESSION SETUP FAILURE message.

8.17.1.4 Abnormal Conditions

Void.

8.17.2 Broadcast Session Modification

8.17.2.1 General

The purpose of the Broadcast Session Modification procedure is to request the NG-RAN node to update the MBS session resources or the area related to a previously established broadcast MBS session. The procedure uses non-UE associated signalling.

8.17.2.2 Successful Operation



Figure 8.17.2.2-1: Broadcast Session Modification, successful operation.

The AMF initiates the procedure by sending a BROADCAST SESSION MODIFICATION REQUEST message to the NG-RAN node.

If the *MBS Service Area* IE is included in the BROADCAST SESSION MODIFICATION REQUEST message, the NG-RAN node shall update the MBS service area as needed and send the BROADCAST SESSION MODIFICATION RESPONSE message. For NTN operation, the NG-RAN node shall, if supported, update the intended service area information as specified in TS 38.300 [8].

If the *MBS Session Modification Request Transfer* IE is included in the BROADCAST SESSION MODIFICATION REQUEST message, the NG-RAN node shall replace the previously provided information by the newly received one and update the MBS session resources and area as requested and send the BROADCAST SESSION MODIFICATION RESPONSE message.

If the *Supported UE Type List* IE is included in the BROADCAST SESSION MODIFICATION REQUEST message, the NG-RAN node shall, if supported, take it into account for configuring MBS session resources.

If the *MBS NG-U Failure Indication* IE is included in the *BROADCAST SESSION MODIFICATION REQUEST* message within the *MBS Session Setup or Modification Request Transfer* IE and set to "NG-U path failure detected", the NG-RAN node may provide new NG-U transport layer information to replace the failed one(s) or switch the data delivery to another 5GC as per the Broadcast MBS session restoration procedure for N3mb Path Failure specified in TS 23.527 [57].

8.17.2.3 Unsuccessful Operation

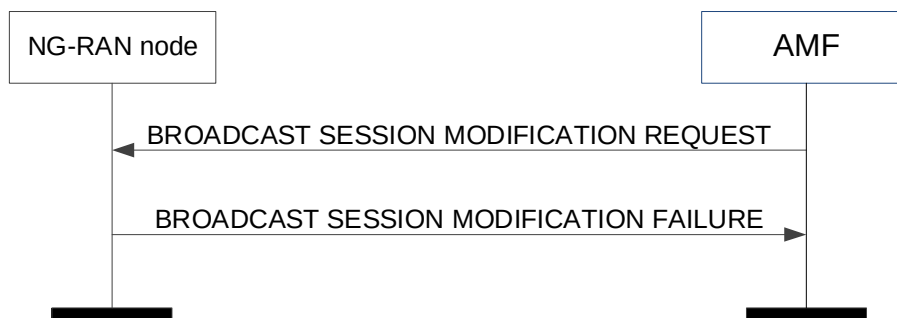


Figure 8.17.2.3-1: Broadcast Session Modification, unsuccessful operation.

If the NG-RAN node fails to update any requested modification, the NG-RAN node shall send the *BROADCAST SESSION MODIFICATION FAILURE* message.

8.17.2.4 Abnormal Conditions

Void.

8.17.3 Broadcast Session Release

8.17.3.1 General

The purpose of the Broadcast Session Release procedure is to release the MBS session resources related to a previously established broadcast MBS session. The procedure uses non-UE associated signalling.

8.17.3.2 Successful Operation

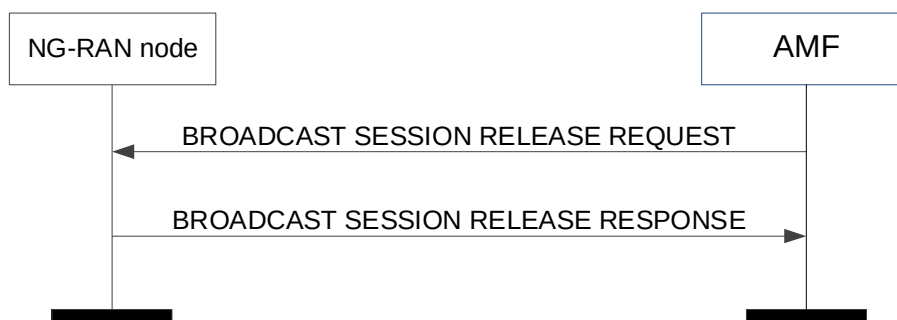


Figure 8.17.3.2-1: Broadcast Session Release, successful operation.

The AMF initiates the procedure by sending a *BROADCAST SESSION RELEASE REQUEST* message to the NG-RAN node.

Upon reception of the *BROADCAST SESSION RELEASE REQUEST* message, the NG-RAN node shall respond with the *BROADCAST SESSION RELEASE RESPONSE* message. The NG-RAN node shall stop broadcasting and release all MBS session resources associated with the broadcast session.

Upon reception of the *BROADCAST SESSION RELEASE RESPONSE* message, the AMF shall transfer transparently the *Broadcast Session Release Response Transfer* IE, if available, to the MB-SMF.

8.17.3.3 Unsuccessful Operation

Not applicable.

8.17.3.4 Abnormal Conditions

Void.

8.17.4 Broadcast Session Release Required

8.17.4.1 General

The purpose of the Broadcast Session Release Required procedure is to trigger the AMF to release the MBS session resources related to a previously established broadcast MBS session. The procedure uses non-UE associated signalling.

8.17.4.2 Successful Operation



Figure 8.17.4.2-1: Broadcast Session Release Required, successful operation.

The NG-RAN node initiates the procedure by sending a BROADCAST SESSION RELEASE REQUIRED message to the AMF.

Upon reception of the BROADCAST SESSION RELEASE REQUIRED message, the AMF shall realize that the NG-RAN node is lacking adequate MBS session resources for a previously established broadcast MBS session and initiate the release of the MBS session resources.

8.17.4.3 Abnormal Conditions

Void.

8.17.5 Broadcast Session Transport

8.17.5.1 General

The purpose of the Broadcast Session Transport procedure is to assign NG-U resources for a broadcast MBS session. The procedure uses non-UE associated signalling.

8.17.5.2 Successful Operation

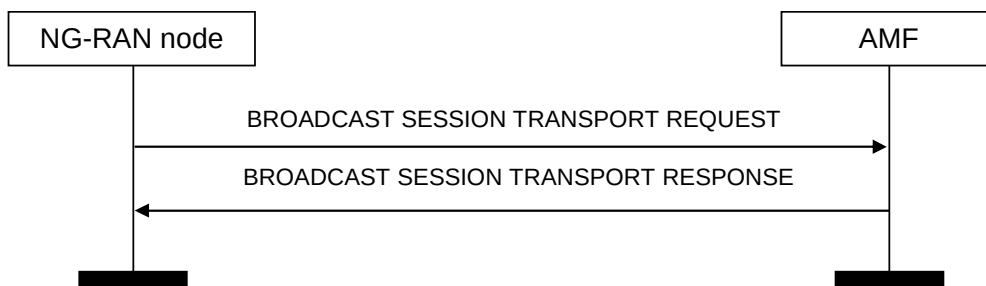


Figure 8.17.5.2-1: Broadcast Session Transport, successful operation.

The NG-RAN node initiates the procedure by sending a BROADCAST SESSION TRANSPORT REQUEST message to the AMF. The AMF responds with a BROADCAST SESSION TRANSPORT RESPONSE message.

If the *Shared NG-U Unicast TNL Information* IE is included in the *MBS Session TNL Information NG-RAN* IE within the *MBS Session Transport Request Transfer* IE in the BROADCAST SESSION TRANSPORT REQUEST message, the MB-SMF shall use the included information as the downlink termination point for the shared NG-U transport.

8.17.5.3 Unsuccessful Operation

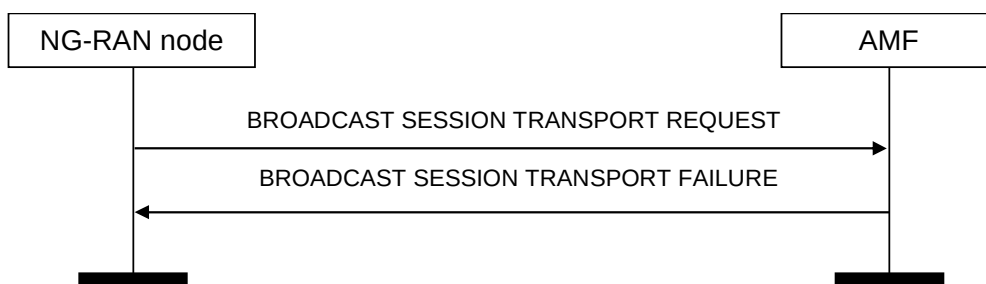


Figure 8.17.5.3-1: Broadcast Session Transport, unsuccessful operation.

In case the shared NG-U transport cannot be setup successfully, the AMF shall respond with the BROADCAST SESSION TRANSPORT FAILURE message to the NG-RAN node with an appropriate cause value.

8.17.5.4 Abnormal Conditions

Void.

8.18 Multicast Session Management Procedures

8.18.1 Distribution Setup

8.18.1.1 General

The purpose of the Distribution Setup procedure is to assign NG-U resources for a multicast MBS session. The procedure uses non-UE-associated signalling.

8.18.1.2 Successful Operation

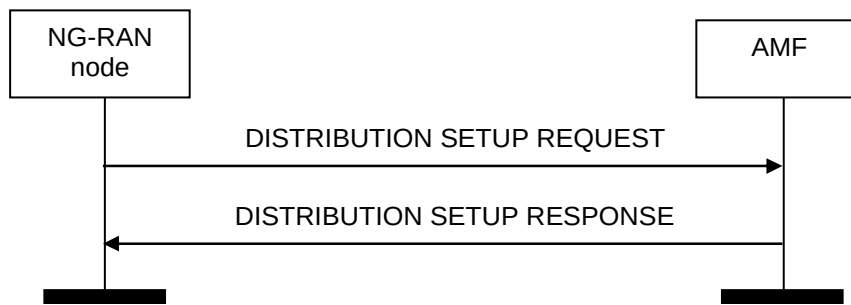


Figure 8.18.1.2-1: Distribution Setup, successful operation.

The NG-RAN node initiates the procedure by sending a DISTRIBUTION SETUP REQUEST message to the AMF. The AMF responds with a DISTRIBUTION SETUP RESPONSE message.

For location dependent multicast sessions, the NG-RAN node shall include the *MBS Area Session ID* IE in the DISTRIBUTION SETUP REQUEST message, and the AMF shall provide the same value of the *MBS Area Session ID* IE in the DISTRIBUTION SETUP RESPONSE message.

If the *Shared NG-U Unicast TNL Information* IE is included in the *MBS Distribution Setup Request Transfer* IE in the DISTRIBUTION SETUP REQUEST message, the MB-SMF shall use the included information as the downlink termination point for the shared NG-U transport.

If the *Shared NG-U Unicast TNL Information* IE is not included in the *MBS Distribution Setup Request Transfer* IE in the DISTRIBUTION SETUP REQUEST message, the MB-SMF shall interpret that the IP multicast is used for this shared NG-U transport, and include the *Shared NG-U Multicast TNL Information* IE in the *MBS Distribution Setup Response Transfer* IE in the DISTRIBUTION SETUP RESPONSE message.

If the *TAI MBS Support List* IE is included in the *MBS Distribution Setup Request Transfer* IE in the DISTRIBUTION SETUP REQUEST message, the MB-SMF shall, if supported, use the included information as specified in TS 23.247 [44].

8.18.1.3 Unsuccessful Operation

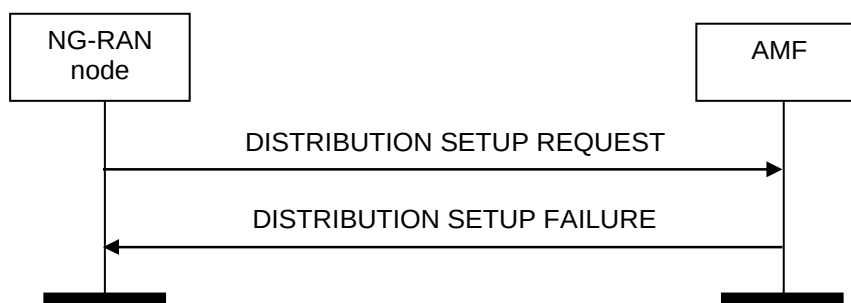


Figure 8.18.1.3-1: Distribution Setup, unsuccessful operation.

In case the shared NG-U transport cannot be setup successfully, the AMF shall respond with the DISTRIBUTION SETUP FAILURE message to the NG-RAN node with an appropriate cause value.

8.18.1.4 Abnormal Conditions

Void.

8.18.2 Distribution Release

8.18.2.1 General

The purpose of the Distribution Release procedure is to enable the release of already established NG-U resources for a given multicast MBS session, or for a given area session of the multicast MBS session. The procedure uses non-UE-associated signalling.

8.18.2.2 Successful Operation

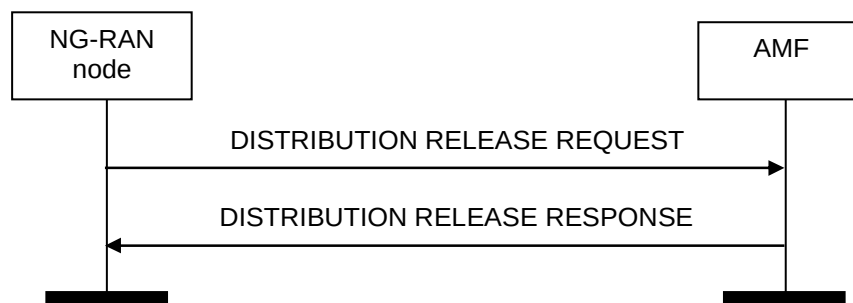


Figure 8.18.2.2-1: Distribution Release, successful operation.

The NG-RAN node initiates the procedure by sending a DISTRIBUTION RELEASE REQUEST message.

Upon receipt of the DISTRIBUTION RELEASE REQUEST message, the AMF shall send the DISTRIBUTION RELEASE RESPONSE message after successfully removing the corresponding NG-U resource for the MBS session.

For location dependent multicast session, the NG-RAN node shall include the *MBS Area Session ID* IE in the DISTRIBUTION RELEASE REQUEST message, and the AMF shall provide the same value of the *MBS Area Session ID* IE in the DISTRIBUTION RELEASE RESPONSE message.

If unicast shared NG-U transport is used, the NG-RAN node shall include the *Shared NG-U TNL Information* IE in the *MBS Distribution Release Request Transfer* IE in the DISTRIBUTION RELEASE REQUEST message, and the MB-SMF shall release the corresponding shared NG-U transport as specified in TS 23.247 [44].

8.18.2.3 Unsuccessful Operation

Not applicable.

8.18.2.4 Abnormal Conditions

Void.

8.18.3 Multicast Session Activation

8.18.3.1 General

The purpose of the Multicast Session Activation procedure is to request a NG-RAN node to activate the MBS session resources of a multicast MBS session. The procedure uses non-UE-associated signalling.

8.18.3.2 Successful Operation



Figure 8.18.3.2-1: Multicast Session Activation, successful operation.

The AMF initiates the procedure by sending a MULTICAST SESSION ACTIVATION REQUEST message to the NG-RAN node.

Upon receipt of the MULTICAST SESSION ACTIVATION REQUEST, the NG-RAN node activates the previously requested MBS session resources corresponding to the MBS session indicated in the MULTICAST SESSION ACTIVATION REQUEST message and indicates in the MULTICAST SESSION ACTIVATION RESPONSE message for which MBS session the request was fulfilled.

8.18.3.3 Unsuccessful Operation



Figure 8.18.3.3-1: Multicast Session Activation, unsuccessful operation.

If the NG-RAN node cannot activate the previously requested MBS session resources indicated by the MULTICAST SESSION ACTIVATION REQUEST message, it shall respond with a MULTICAST SESSION ACTIVATION FAILURE message with an appropriate cause value.

8.18.3.4 Abnormal Conditions

Void.

8.18.4 Multicast Session Deactivation

8.18.4.1 General

The purpose of the Multicast Session Deactivation procedure is to request a NG-RAN node to deactivate the multicast MBS session resources of one MBS session. The procedure uses non-UE-associated signalling.

8.18.4.2 Successful Operation



Figure 8.18.4.2-1: Multicast Session Deactivation, successful operation.

The AMF initiates the procedure by sending a MULTICAST SESSION DEACTIVATION REQUEST message to the NG-RAN node.

Upon receipt of this message, the NG-RAN node shall deactivate the previously requested MBS session resources corresponding to the MBS session indicated in the MULTICAST SESSION DEACTIVATION REQUEST message if the MBS resources are active and shall indicate in the MULTICAST SESSION DEACTIVATION RESPONSE message for which MBS session the request was fulfilled.

8.18.4.3 Unsuccessful Operation

Not applicable.

8.18.4.4 Abnormal Conditions

Void.

8.18.5 Multicast Session Update

8.18.5.1 General

The purpose of the Multicast Session Update procedure is to request the NG-RAN node to update the NG-RAN MBS session resources or area related to a multicast MBS session. The procedure uses non-UE associated signalling.

8.18.5.2 Successful Operation



Figure 8.18.5.2-1: Multicast Session Update, successful operation.

The AMF initiates the procedure by sending a MULTICAST SESSION UPDATE REQUEST message to the NG-RAN node.

Upon receipt of the MULTICAST SESSION UPDATE REQUEST message, the NG-RAN node shall, if requested, update the MBS QoS profile and/or MBS Service Area for the multicast service and send the MULTICAST SESSION UPDATE RESPONSE message to the AMF.

For location dependent multicast session, the AMF shall include the *MBS Area Session ID* IE in the MULTICAST SESSION UPDATE REQUEST message, and the NG-RAN node shall provide the same value of the *MBS Area Session ID* IE in the MULTICAST SESSION UPDATE RESPONSE message.

In case the *MBS Service Area Information* IE is included in the *Multicast Session Update Request Transfer* IE in the MULTICAST SESSION UPDATE REQUEST message, the NG-RAN node shall update the stored MBS Service Area Information for that service, as specified in TS 23.247 [44].

In case the *MBS QoS Flows To Be Setup or Modify List* IE is included in the *Multicast Session Update Request Transfer* IE in the MULTICAST SESSION UPDATE REQUEST message, the NG-RAN node shall setup or modify the MBS QoS profile accordingly.

In case the *MBS QoS Flows To Be Release List* IE is included in the *Multicast Session Update Request Transfer* IE in the MULTICAST SESSION UPDATE REQUEST message, the NG-RAN node shall release the indicated MBS QoS flows.

In case the *MBS Session TNL Information 5GC* IE is included in the *Multicast Session Update Request Transfer* IE in the MULTICAST SESSION UPDATE REQUEST message, the NG-RAN node shall take the information into account and update related NG-U resources.

Interactions with other procedures:

If the *MBS User Plane Failure Indication* IE is included in the *Multicast Session Update Request Transfer* IE in the MULTICAST SESSION UPDATE REQUEST message and set to "NG-U path failure detected", the NG-RAN node may provide new NG-U transport layer information to replace the failed one via the Distribution Setup procedure, or release the failed one via the Distribution Release procedure, as per the Multicast MBS session restoration procedure for N3mb Path Failure specified in TS 23.527 [57].

8.18.5.3 Unsuccessful Operation

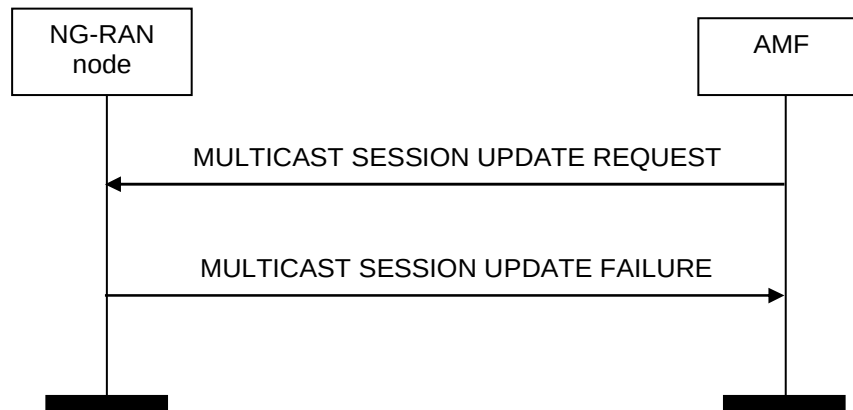


Figure 8.18.5.3-1: Multicast Session Update, unsuccessful operation.

If the NG-RAN node is not able to perform any of the requested update, it shall respond with a MULTICAST SESSION UPDATE FAILURE message with an appropriate cause value.

8.18.5.4 Abnormal Conditions

Void.

8.19 Timing Synchronisation Status Reporting Procedures

8.19.1 Timing Synchronisation Status

8.19.1.1 General

The purpose of the Timing Synchronisation Status procedure is to enable the AMF to request the NG-RAN node to start or stop reporting of RAN timing synchronisation status information as specified in TS 23.501 [9] and TS 23.502 [10]. The procedure uses non-UE associated signalling.

8.19.1.2 Successful Operation

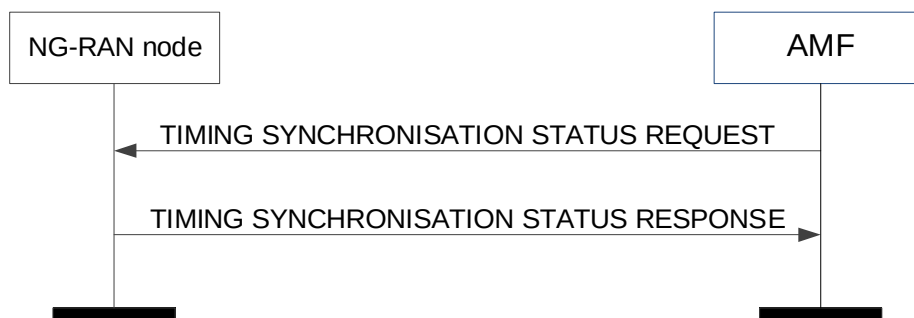


Figure 8.19.1.2-1: Timing synchronisation status procedure: successful operation

The AMF initiates the procedure by sending a TIMING SYNCHRONISATION STATUS REQUEST message to the NG-RAN node.

If the *RAN TSS Request Type* IE included in the TIMING SYNCHRONISATION STATUS REQUEST message is set to "start", the NG-RAN node shall start the RAN TSS reporting for the TSCTSF indicated by the *Routing ID* IE. If the *RAN TSS Request Type* IE is set to "stop", the NG-RAN node shall stop the reporting for the TSCTSF indicated by the *Routing ID* IE.

8.19.1.3 Unsuccessful Operation

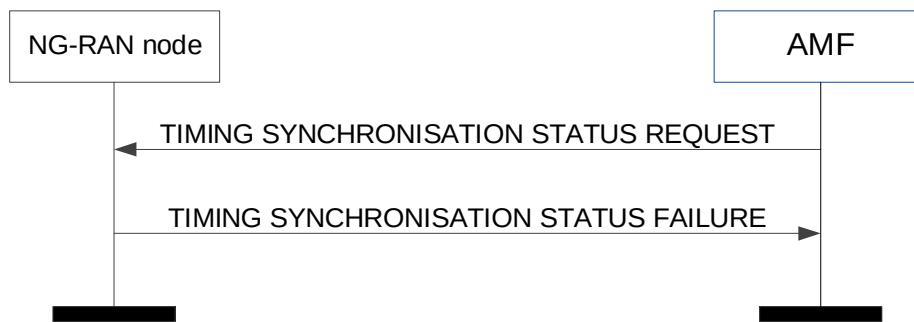


Figure 8.19.1.3-1: Timing synchronisation status procedure: unsuccessful operation

If the NG-RAN node is not able to report timing synchronisation status, it shall consider the procedure as failed and reply with the TIMING SYNCHRONISATION STATUS FAILURE message.

8.19.1.4 Abnormal Conditions

Void.

8.19.2 Timing Synchronisation Status Report

8.19.2.1 General

The purpose of the Timing Synchronisation Status Report procedure is to enable the NG-RAN node to provide RAN timing synchronisation status information to the AMF as specified in TS 23.501 [9] and TS 23.502 [10]. The procedure uses non-UE associated signalling.

8.19.2.2 Successful Operation



Figure 8.19.2.2-1: Timing synchronisation status report

The NG-RAN node initiates the procedure by sending a TIMING SYNCHRONISATION STATUS REPORT message to the AMF for the TSCTSF indicated by the *Routing ID* IE.

8.19.2.3 Abnormal Conditions

Void.

8.20 Ambient IoT Procedures

8.20.1 Inventory Request

8.20.1.1 General

The purpose of the Inventory Request procedure is to request the NG-RAN node to perform A-IoT paging over the A-IoT radio. This procedure applies only if the NG-RAN node is a gNB. The procedure uses non-UE associated signalling.

8.20.1.2 Successful Operation

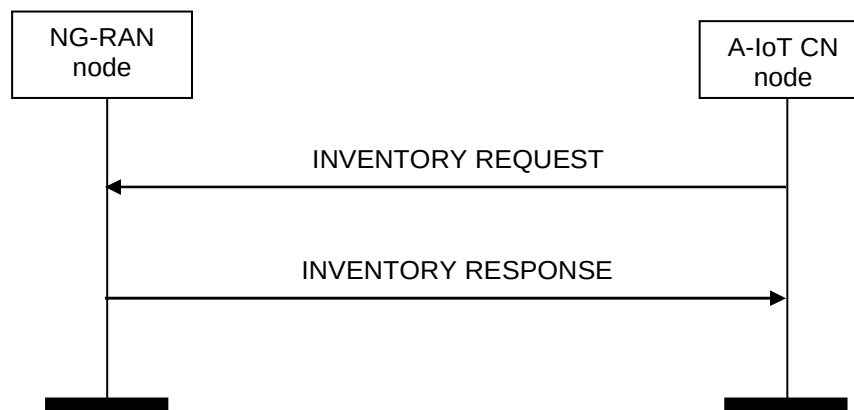


Figure 8.20.1.2-1: Inventory Request, successful operation.

The A-IoT CN node triggers the procedure by sending an INVENTORY REQUEST message to the NG-RAN node. If the NG-RAN node accepts the request, the NG-RAN node responds with the INVENTORY RESPONSE message.

The NG-RAN node shall, include the same values of the *AIOTF Identifier* IE and *A-IoT Correlation Identifier* IE in the INVENTORY RESPONSE message as received from the INVENTORY REQUEST message.

Upon receiving the *Inventory Assistance Information* IE contained in the *Inventory Request Transfer* IE in the INVENTORY REQUEST message, the NG-RAN node may take the information into account as defined in TS 38.300 [8].

Interaction with Inventory Report procedure

When starting the requested inventory, the NG-RAN node shall, include the same values of the *AIOTF Identifier* IE and *A-IoT Correlation Identifier* IE in the INVENTORY REPORT message(s) as received from the INVENTORY REQUEST message

If the *Follow on Command Indication* IE set to "true" is included in the *Inventory Request Transfer* IE in the INVENTORY REQUEST message, the NG-RAN node shall consider that there will be command to be transmitted to the device(s), and shall allocate and include a *RAN A-IoT Device NGAP ID* IE in each *Device Report Item* IE within the *Inventory Report Transfer* IE in the INVENTORY REPORT message(s).

8.20.1.3 Unsuccessful Operation

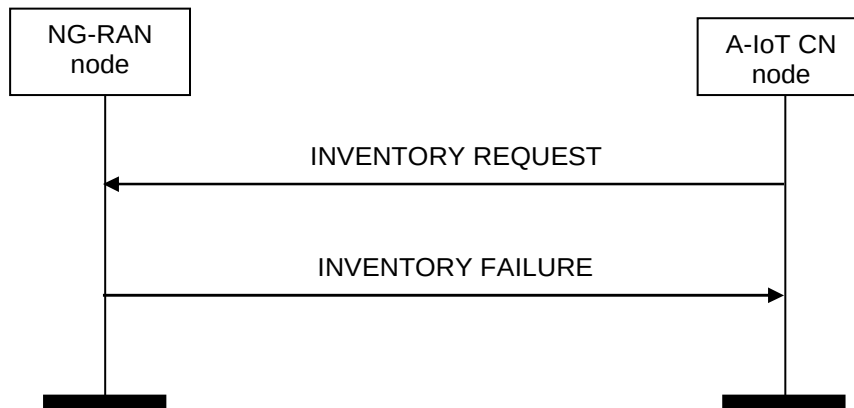


Figure 8.20.1.3-1: Inventory Request, unsuccessful operation.

If the NG-RAN node is not able to perform the inventory operation, it shall send the INVENTORY FAILURE message with an appropriate cause value.

8.20.1.4 Abnormal Conditions

Void.

8.20.2 Inventory Report

8.20.2.1 General

The purpose of the Inventory Report procedure is to enable the NG-RAN node to provide inventory report related information to the A-IoT CN node, following a successful Inventory Request procedure. This procedure applies only if the NG-RAN node is a gNB. The procedure uses non-UE associated signalling.

8.20.2.2 Successful Operation



Figure 8.20.2.2-1: Inventory Report

The NG-RAN node initiates the procedure by sending an INVENTORY REPORT message for the Inventory service.

If the *Reader Report List* IE is included in the *Inventory Report Transfer* IE contained in the INVENTORY REPORT message, the A-IoT CN node shall take it into account as specified in TS 23.369 [60].

If the *Inventory Complete Indication* IE set to “true” is included in the *Inventory Report Transfer* IE contained in the INVENTORY REPORT message, the A-IoT CN node shall consider that the inventory operation for the A-IoT session is completed in the NG-RAN node.

Interactions with the A-IoT Session Release procedure:

In case the *Inventory Complete Indication* IE is included in the *Inventory Report Transfer* IE contained in the INVENTORY REPORT message and set to “true”, if there is no follow-on command to be transmitted, or all the follow-on command transmissions have been completed, the A-IoT CN node should initiate the A-IoT Session Release procedure for the A-IoT session denoted by the *AIOTF Identifier* IE and the *A-IoT Correlation Identifier* IE contained in the INVENTORY REPORT message.

8.20.2.3 Abnormal Conditions

Void.

8.20.3 Command Request

8.20.3.1 General

The purpose of the Command Request procedure is to request the NG-RAN node to transfer the command (A-IoT NAS PDU) to the A-IoT device over the A-IoT radio. This procedure applies only if the NG-RAN node is a gNB. The procedure uses non-UE associated signalling.

8.20.3.2 Successful Operation

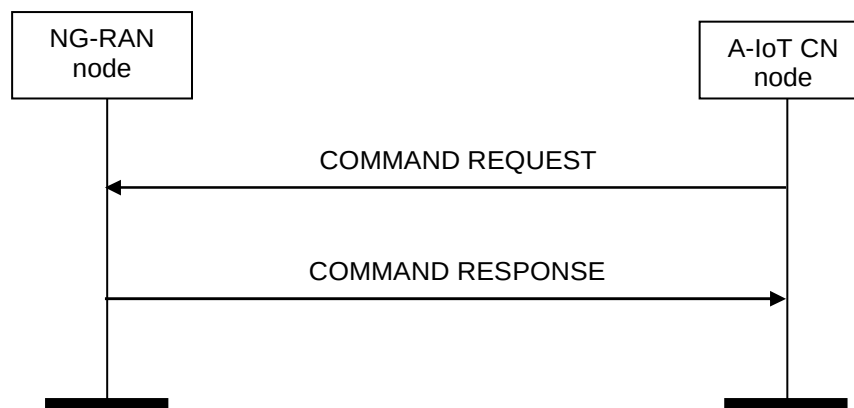


Figure 8.20.3.2-1: Command Request, successful operation.

The A-IoT CN node triggers the procedure by sending a COMMAND REQUEST message to the NG-RAN node. In case of successful operation, the NG-RAN node responds with the COMMAND RESPONSE message.

The NG-RAN node shall, include the same values of the *AIOTF Identifier* IE, the *A-IoT Correlation Identifier* IE and the *RAN A-IoT Device NGAP ID* IE in the COMMAND RESPONSE message as received from the COMMAND REQUEST message.

8.20.3.3 Unsuccessful Operation

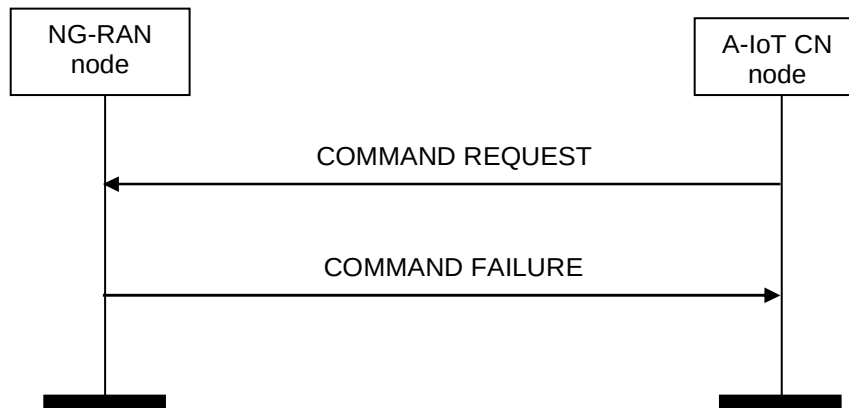


Figure 8.20.3.3-1: Command Request, unsuccessful operation.

If the NG-RAN node is not able to transmit the received *A-IoT NAS PDU* in the *Command Request Transfer* IE in the *COMMAND REQUEST* message towards the A-IoT device or if it does not receive the A-IoT NAS response from the A-IoT device, it shall send the *COMMAND FAILURE* message with an appropriate cause value.

8.20.3.4 Abnormal Conditions

Void.

8.20.4 A-IoT Session Release

8.20.4.1 General

The purpose of the A-IoT Session Release procedure is to enable the A-IoT CN node to terminate all activities related to the indicated A-IoT session. This procedure applies only if the NG-RAN node is a gNB. The procedure uses non-UE associated signalling.

8.20.4.2 Successful Operation



Figure 8.20.4.2-1: A-IoT Session Release, successful operation.

The A-IoT CN node triggers the procedure by sending an *AIOT SESSION RELEASE COMMAND* message to the NG-RAN node. Upon reception of an *AIOT SESSION RELEASE COMMAND* message the NG-RAN node shall terminate all activities and release the context and resources related to the A-IoT session denoted by the *AIOTF Identifier* IE and the *A-IoT Correlation Identifier* IE and respond with an *AIOT SESSION RELEASE COMPLETE* message.

8.20.4.3 Unsuccessful Operation

Not applicable.

8.20.4.4 Abnormal Conditions

Void.

8.20.5 A-IoT Session Release Request

8.20.5.1 General

The purpose of the A-IoT Session Release Request procedure is to enable the NG-RAN node to request the A-IoT CN node to release the A-IoT session. This procedure applies only if the NG-RAN node is a gNB. The procedure uses non-UE associated signalling.

8.20.5.2 Successful Operation



Figure 8.20.5.2-1: A-IoT Session Release Request

The NG-RAN node initiates the procedure by sending an AIOT SESSION RELEASE REQUEST message to the A-IoT CN node.

Interactions with the A-IoT Session Release procedure:

Upon reception of the AIOT SESSION RELEASE REQUEST message, the A-IoT CN node should initiate the A-IoT Session Release procedure for the A-IoT session denoted by the *AIOTF Identifier* IE and the *A-IoT Correlation Identifier* contained in the AIOT SESSION RELEASE REQUEST message.

8.20.5.3 Abnormal Conditions

Void.

9 Elements for NGAP Communication

9.0 General

Subclauses 9.2 and 9.3 present the NGAP message and IE definitions in tabular format. The corresponding ASN.1 definition is presented in subclause 9.4. In case there is contradiction between the tabular format and the ASN.1 definition, the ASN.1 shall take precedence, except for the definition of conditions for the presence of conditional IEs, where the tabular format shall take precedence.

The messages have been defined in accordance to the guidelines specified in TR 25.921 [7].

When specifying IEs which are to be represented by bitstrings, if not otherwise specifically stated in the semantics description of the concerned IE or elsewhere, the following principle applies with regards to the ordering of bits:

- The first bit (leftmost bit) contains the most significant bit (MSB);
- The last bit (rightmost bit) contains the least significant bit (LSB);
- When importing bitstrings from other specifications, the first bit of the bitstring contains the first bit of the concerned information;

9.1 Tabular Format Contents

9.1.1 Presence

All IEs are marked mandatory, optional or conditional according to table 9.1.1-1.

Table 9.1.1-1: Meaning of content within "Presence" column

Abbreviation	Meaning
M	IEs marked as Mandatory (M) shall always be included in the message.
O	IEs marked as Optional (O) may or may not be included in the message.
C	IEs marked as Conditional (C) shall be included in a message only if the condition is satisfied. Otherwise the IE shall not be included.

9.1.2 Criticality

Each IE or group of IEs may have criticality information applied to it according to table 9.1.2-1.

Table 9.1.2-1: Meaning of content within "Criticality" column

Abbreviation	Meaning
–	No criticality information is applied explicitly.
YES	Criticality information is applied. This is usable only for non-repeatable IEs
GLOBAL	The IE and all its repetitions together have one common criticality information. This is usable only for repeatable IEs.
EACH	Each repetition of the IE has its own criticality information. It is not allowed to assign different criticality values to the repetitions. This is usable only for repeatable IEs.

9.1.3 Range

The Range column indicates the allowed number of copies of repetitive IEs/IE groups.

9.1.4 Assigned Criticality

The Assigned Criticality column provides the actual criticality information as defined in subclause 10.3.2, if applicable.

9.2 Message Functional Definition and Content

9.2.1 PDU Session Management Messages

9.2.1.1 PDU SESSION RESOURCE SETUP REQUEST

This message is sent by the AMF and is used to request the NG-RAN node to assign resources on Uu and NG-U for one or several PDU session resources.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
NAS-PDU	O		9.3.3.4		YES	reject
PDU Session Resource Setup Request List		1			YES	reject
>PDU Session Resource Setup Request Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session NAS-PDU	O		NAS-PDU 9.3.3.4		-	
>>S-NSSAI	M		9.3.1.24		-	
>>PDU Session Resource Setup Request Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Setup Request Transfer</i> IE specified in subclause 9.3.4.1.	-	
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore
UE Aggregate Maximum Bit Rate	O		9.3.1.58		YES	ignore
UE Slice Maximum Bit Rate List	O		9.3.1.231		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.2 PDU SESSION RESOURCE SETUP RESPONSE

This message is sent by the NG-RAN node as a response to the request to assign resources on Uu and NG-U for one or several PDU session resources.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Setup Response List		<i>0..1</i>			YES	ignore
>PDU Session Resource Setup Response Item		<i>1..<maxno of PDUSessions></i>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Response Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Setup Response Transfer</i> IE specified in subclause 9.3.4.2.	-	
PDU Session Resource Failed to Setup List		<i>0..1</i>			YES	ignore
>PDU Session Resource Failed to Setup Item		<i>1..<maxno of PDUSessions></i>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Unsuccessful Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Setup Unsuccessful Transfer</i> IE specified in subclause 9.3.4.16.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.3 PDU SESSION RESOURCE RELEASE COMMAND

This message is sent by the AMF and is used to request the NG-RAN node to release already established PDU session resources for a given UE.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
NAS-PDU	O		9.3.3.4		YES	ignore
PDU Session Resource to Release List		1			YES	reject
>PDU Session Resource to Release Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Release Command Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Release Command Transfer</i> IE specified in subclause 9.3.4.12.	-	

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.4 PDU SESSION RESOURCE RELEASE RESPONSE

This message is sent by the NG-RAN node as a response to the request to release already established PDU session resources for a given UE.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Released List		1			YES	ignore
>PDU Session Resource Released Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Release Response Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Release Response Transfer</i> IE specified in subclause 9.3.4.21.	-	
User Location Information	O		9.3.1.16		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.5 PDU SESSION RESOURCE MODIFY REQUEST

This message is sent by the AMF and is used to request the NG-RAN node to enable modifications of already established PDU session resources for a given UE.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
PDU Session Resource Modify Request List		1			YES	reject
>PDU Session Resource Modify Request Item		<i>1..<maxno of PDUSessions></i>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>NAS-PDU	O		9.3.3.4		-	
>>PDU Session Resource Modify Request Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Modify Request Transfer</i> IE specified in subclause 9.3.4.3.	-	
>>S-NSSAI	O		9.3.1.24		YES	reject
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.6 PDU SESSION RESOURCE MODIFY RESPONSE

This message is sent by the NG-RAN node and is used to report the outcome of the request from the PDU SESSION RESOURCE MODIFY REQUEST message.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Modify Response List		0..1			YES	ignore
>PDU Session Resource Modify Response Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Response Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Modify Response Transfer</i> IE specified in subclause 9.3.4.4.	-	
PDU Session Resource Failed to Modify List		0..1			YES	ignore
>PDU Session Resource Failed to Modify Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Unsuccessful Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Modify Unsuccessful Transfer</i> IE specified in subclause 9.3.4.17.	-	
User Location Information	O		9.3.1.16		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.7 PDU SESSION RESOURCE NOTIFY

This message is sent by the NG-RAN node to notify that the QoS requirements of already established GBR QoS flow(s) for which notification control has been requested are either not fulfilled anymore or fulfilled again by the NG-RAN node. This message can also be sent by the NG-RAN node to notify that PDU session resource(s) for a given UE are released.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
PDU Session Resource Notify List		0..1			YES	reject
>PDU Session Resource Notify Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Notify Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Notify Transfer</i> IE specified in subclause 9.3.4.5.	-	
PDU Session Resource Released List		0..1			YES	ignore
>PDU Session Resource Released Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Notify Released Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Notify Released Transfer</i> IE specified in subclause 9.3.4.13.	-	
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.8 PDU SESSION RESOURCE MODIFY INDICATION

This message is sent by the NG-RAN node and is used to request the AMF to enable modifications of already established PDU session resources for a given UE.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
PDU Session Resource Modify Indication List		1			YES	reject
>PDU Session Resource Modify Indication Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Indication Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Modify Indication Transfer</i> IE specified in subclause 9.3.4.6.	-	
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.9 PDU SESSION RESOURCE MODIFY CONFIRM

This message is sent by the AMF and is used to confirm the outcome of the request from the PDU SESSION RESOURCE MODIFY INDICATION message.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Modify Confirm List		0..1			YES	ignore
>PDU Session Resource Modify Confirm Item		1..<maxno ofPDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Confirm Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Modify Confirm Transfer</i> IE specified in subclause 9.3.4.7.	-	
PDU Session Resource Failed to Modify List		0..1			YES	ignore
>PDU Session Resource Failed to Modify Item		1..<maxno ofPDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Indication Unsuccessful Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Modify Indication Unsuccessful Transfer</i> IE specified in subclause 9.3.4.22.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2 UE Context Management Messages

9.2.2.1 INITIAL CONTEXT SETUP REQUEST

This message is sent by the AMF to request the setup of a UE context.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Old AMF	O		AMF Name 9.3.3.21	This IE is ignored if the <i>Extended Old AMF</i> IE is present.	YES	reject
UE Aggregate Maximum Bit Rate	C- ifPDUses sionReso urceSetu p		9.3.1.58		YES	reject
Core Network Assistance Information for RRC INACTIVE	O		9.3.1.15		YES	ignore
GUAMI	M		9.3.3.3		YES	reject
PDU Session Resource Setup Request List		0..1			YES	reject
>>PDU Session Resource Setup Request Item		1..<maxno ofPDUSes sions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session NAS-PDU	O		NAS-PDU 9.3.3.4		-	
>>S-NSSAI	M		9.3.1.24		-	
>>PDU Session Resource Setup Request Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Setup Request Transfer</i> IE specified in subclause 9.3.4.1.	-	
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore
Allowed NSSAI	M		9.3.1.31	Indicates the S-NSSAIs permitted by the network	YES	reject
UE Security Capabilities	M		9.3.1.86		YES	reject
Security Key	M		9.3.1.87		YES	reject
Trace Activation	O		9.3.1.14		YES	ignore
Mobility Restriction List	O		9.3.1.85		YES	ignore
UE Radio Capability	O		9.3.1.74		YES	ignore
Index to RAT/Frequency Selection Priority	O		9.3.1.61		YES	ignore
Masked IMEISV	O		9.3.1.54		YES	ignore
NAS-PDU	O		9.3.3.4		YES	ignore
Emergency Fallback Indicator	O		9.3.1.26		YES	reject
RRC Inactive Transition Report Request	O		9.3.1.91		YES	ignore
UE Radio Capability for Paging	O		9.3.1.68		YES	ignore
Redirection for Voice EPS Fallback	O		9.3.1.116		YES	ignore
Location Reporting Request Type	O		9.3.1.65		YES	ignore
CN Assisted RAN Parameters Tuning	O		9.3.1.119		YES	ignore
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
IAB Authorized	O		9.3.1.129		YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
Extended Connected Time	O		9.3.3.31		YES	ignore

UE Differentiation Information	O		9.3.1.144		YES	ignore
NR V2X Services Authorized	O		9.3.1.146		YES	ignore
LTE V2X Services Authorized	O		9.3.1.147		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.148	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.149	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 QoS Parameters	O		9.3.1.150	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE User Plane CIoT Support Indicator	O		9.3.1.160		YES	ignore
RG Level Wireline Access Characteristics	O		OCTET STRING	Specified in TS 23.316 [34]. Indicates the wireline access technology specific QoS information corresponding to a specific wireline access subscription.	YES	ignore
Management Based MDT PLMN List	O		MDT PLMN List 9.3.1.168		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Time Synchronisation Assistance Information	O		9.3.1.220		YES	ignore
QMC Configuration Information	O		9.3.1.223		YES	ignore
Target NSSAI Information	O		9.3.1.229		YES	ignore
UE Slice Maximum Bit Rate List	O		9.3.1.231		YES	ignore
5G ProSe Authorized	O		9.3.1.233		YES	ignore
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 QoS Parameters	O		9.3.1.234	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
Network Controlled Repeater Authorized	O		9.3.1.245		YES	ignore
Aerial UE Subscription Information	O		9.3.1.246		YES	ignore
NR A2X Services Authorized	O		9.3.1.247		YES	ignore
LTE A2X Services Authorized	O		9.3.1.248		YES	ignore
NR A2X UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for NR A2X services.	YES	ignore

LTE A2X UE PC5 Aggregate Maximum Bit Rate	O		LTE UE Sidelink Aggregate Maximum Bit Rate 9.3.1.149	This IE applies only if the UE is authorized for LTE A2X services.	YES	ignore
A2X PC5 QoS Parameters	O		9.3.1.249	This IE applies only if the UE is authorized for A2X services.	YES	ignore
Mobile IAB Authorized	O		9.3.1.259		YES	ignore
Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network.	YES	ignore
Ranging and Sidelink Positioning Service Information	O		9.3.1.269	This IE applies only if the UE is authorized for NR V2X services and/or 5G ProSe services.	YES	ignore
Extended Old AMF	O		Extended AMF Name 9.3.3.51		YES	ignore
AMF UE NGAP ID 2	O		AMF UE NGAP ID 9.3.3.1	This IE indicates the AMF UE NGAP ID assigned by the AMF.	YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

Condition	Explanation
ifPDUSessionResourceSetup	This IE shall be present if the <i>PDU Session Resource Setup List</i> IE is present.

9.2.2.2 INITIAL CONTEXT SETUP RESPONSE

This message is sent by the NG-RAN node to confirm the setup of a UE context.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Setup Response List		0..1			YES	ignore
>PDU Session Resource Setup Response Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Response Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Setup Response Transfer</i> IE specified in subclause 9.3.4.2.	-	
PDU Session Resource Failed to Setup List		0..1			YES	ignore
>PDU Session Resource Failed to Setup Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Unsuccessful Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Setup Unsuccessful Transfer</i> IE specified in subclause 9.3.4.16.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.3 INITIAL CONTEXT SETUP FAILURE

This message is sent by the NG-RAN node to indicate that the setup of the UE context was unsuccessful.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Failed to Setup List		0..1			YES	ignore
>PDU Session Resource Failed to Setup Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Unsuccessful Transfer	M		OCTET STRING	Containing the <i>PDU Session Resource Setup Unsuccessful Transfer</i> IE specified in subclause 9.3.4.16.	-	
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.4 UE CONTEXT RELEASE REQUEST

This message is sent by the NG-RAN node to request the release of the UE-associated logical NG-connection over the NG interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
PDU Session Resource List		0..1			YES	reject
>PDU Session Resource Item		1..<maxnoofPDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
Cause	M		9.3.1.2		YES	ignore
GW Context Release Indication	O		9.3.1.274		YES	reject

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.5 UE CONTEXT RELEASE COMMAND

This message is sent by the AMF to request the release of the UE-associated logical NG-connection over the NG interface.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
CHOICE UE NGAP IDs	M				YES	reject
>UE NGAP ID pair						
>>AMF UE NGAP ID	M		9.3.3.1		-	
>>RAN UE NGAP ID	M		9.3.3.2		-	
>AMF UE NGAP ID						
>>AMF UE NGAP ID	M		9.3.3.1		-	
Cause	M		9.3.1.2		YES	ignore

9.2.2.6 UE CONTEXT RELEASE COMPLETE

This message is sent by the NG-RAN node to confirm the release of the UE-associated logical NG-connection over the NG interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore
Information on Recommended Cells and RAN Nodes for Paging	O		9.3.1.100		YES	ignore
PDU Session Resource List		<i>0..1</i>			YES	reject
>PDU Session Resource Item		<i>1..<maxno of PDUSessions></i>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Release Response Transfer	O		OCTET STRING	Containing the <i>PDU Session Resource Release Response Transfer</i> IE specified in subclause 9.3.4.21.	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Paging Assistance Data for CE Capable UE	O		9.3.1.141		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.7 UE CONTEXT MODIFICATION REQUEST

This message is sent by the AMF to provide UE Context information changes to the NG-RAN node.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
Security Key	O		9.3.1.87		YES	reject
Index to RAT/Frequency Selection Priority	O		9.3.1.61		YES	ignore
UE Aggregate Maximum Bit Rate	O		9.3.1.58		YES	ignore
UE Security Capabilities	O		9.3.1.86		YES	reject
Core Network Assistance Information for RRC INACTIVE	O		9.3.1.15		YES	ignore
Emergency Fallback Indicator	O		9.3.1.26		YES	reject
New AMF UE NGAP ID	O		AMF UE NGAP ID 9.3.3.1		YES	reject
RRC Inactive Transition Report Request	O		9.3.1.91		YES	ignore
New GUAMI	O		GUAMI 9.3.3.3		YES	reject
CN Assisted RAN Parameters Tuning	O		9.3.1.119		YES	ignore
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
IAB Authorized	O		9.3.1.129		YES	ignore
NR V2X Services Authorized	O		9.3.1.146		YES	ignore
LTE V2X Services Authorized	O		9.3.1.147		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.148	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.149	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 QoS Parameters	O		9.3.1.150	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
RG Level Wireline Access Characteristics	O		OCTET STRING	Specified in TS 23.316 [34]. Indicates the wireline access technology specific QoS information corresponding to a specific wireline access subscription.	YES	ignore
Time Synchronisation Assistance Information	O		9.3.1.220		YES	ignore
QMC Configuration Information	O		9.3.1.223		YES	ignore
QMC Deactivation	O		9.3.1.222		YES	ignore
UE Slice Maximum Bit Rate List	O		9.3.1.231		YES	ignore
Management Based MDT PLMN Modification List	O		MDT PLMN Modification List 9.3.1.243		YES	ignore
5G ProSe Authorized	O		9.3.1.233		YES	ignore

5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 QoS Parameters	O		9.3.1.234	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
Network Controlled Repeater Authorized	O		9.3.1.245		YES	ignore
Aerial UE Subscription Information	O		9.3.1.246		YES	ignore
NR A2X Services Authorized	O		9.3.1.247		YES	ignore
LTE A2X Services Authorized	O		9.3.1.248		YES	ignore
NR A2X UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for NR A2X services.	YES	ignore
LTE A2X UE PC5 Aggregate Maximum Bit Rate	O		LTE UE Sidelink Aggregate Maximum Bit Rate 9.3.1.149	This IE applies only if the UE is authorized for LTE A2X services.	YES	ignore
A2X PC5 QoS Parameters	O		9.3.1.249	This IE applies only if the UE is authorized for A2X services.	YES	ignore
Mobile IAB Authorized	O		9.3.1.259		YES	ignore
Ranging and Sidelink Positioning Service Information	O		9.3.1.269	This IE applies only if the UE is authorized for NR V2X services and/or 5G ProSe services.	YES	ignore

9.2.2.8 UE CONTEXT MODIFICATION RESPONSE

This message is sent by the NG-RAN node to confirm the performed UE context updates.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
RRC State	O		9.3.1.92		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.9 UE CONTEXT MODIFICATION FAILURE

This message is sent by the NG-RAN node in case the performed UE context update is not successful.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.10 RRC INACTIVE TRANSITION REPORT

This message is sent by the NG-RAN node to notify the 5GC the UE enters or leaves RRC_INACTIVE state.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RRC State	M		9.3.1.92		YES	ignore
User Location Information	M		9.3.1.16		YES	ignore

9.2.2.11 CONNECTION ESTABLISHMENT INDICATION

This message is sent by the AMF to complete the establishment of the UE-associated logical NG-connection.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
UE Radio Capability	O		9.3.1.74		YES	ignore
End Indication	O		9.3.3.32		YES	ignore
S-NSSAI	O		9.3.1.24		YES	ignore
Allowed NSSAI	O		9.3.1.31	Indicates the S-NSSAIs permitted by the network	YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
DL CP Security Information	O		9.3.3.49		YES	ignore
NB-IoT UE Priority	O		9.3.1.145		YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Masked IMEISV	O		9.3.1.54		YES	ignore
Old AMF	O		AMF Name 9.3.3.21	This IE is ignored if the <i>Extended Old AMF</i> IE is present.	YES	reject
Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network.	YES	ignore
Extended Old AMF	O		Extended AMF Name 9.3.3.51		YES	ignore

9.2.2.12 AMF CP RELOCATION INDICATION

This message is sent by the AMF to inform the NG-RAN node that the UE is to be relocated as described in TS 38.300 [8].

Direction: AMF → NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
S-NSSAI	O		9.3.1.24	In this version of the specification, this IE is not used, and shall be ignored if received.	YES	ignore
Allowed NSSAI	O		9.3.1.31	Indicates the S-NSSAIs permitted by the network. In this version of the specification, this IE is not used, and shall be ignored if received.	YES	ignore
Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network. In this version of the specification, this IE is not used, and shall be ignored if received.	YES	ignore

9.2.2.13 RAN CP RELOCATION INDICATION

This message is sent by the NG-RAN node to initiate the establishment of a UE-associated logical NG-connection, following the reception of re-establishment request.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
5G-S-TMSI	M		9.3.3.20		YES	reject
E-UTRA CGI	M		9.3.1.9		YES	ignore
TAI	M		9.3.3.11		YES	ignore
UL CP Security Information	M		9.3.3.48		YES	reject

9.2.2.14 RETRIEVE UE INFORMATION

The message is sent by the NG-RAN node to request UE information over the NG interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
5G-S-TMSI	M		9.3.3.20		YES	reject

9.2.2.15 UE INFORMATION TRANSFER

The message is sent by the AMF to transfer UE information over the NG interface.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
5G-S-TMSI	M		9.3.3.20		YES	reject
NB-IoT UE Priority	O		9.3.1.145		YES	ignore
UE Radio Capability	O		9.3.1.74		YES	ignore
S-NSSAI	O		9.3.1.24		YES	ignore
Allowed NSSAI	O		9.3.1.31	Indicates the S-NSSAIs permitted by the network	YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
Masked IMEISV	O		9.3.1.54		YES	ignore
Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network.	YES	ignore

9.2.2.16 UE CONTEXT SUSPEND REQUEST

This message is sent by the NG-RAN node to request the AMF to suspend the UE context and the related PDU session contexts.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Information on Recommended Cells and RAN Nodes for Paging	O		9.3.1.100		YES	ignore
Paging Assistance Data for CE Capable UE	O		9.3.1.141		YES	ignore
PDU Session Resource Suspend List		0..1			YES	reject
>PDU Session Resource Suspend Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>UE Context Suspend Request Transfer	M		Containing the <i>UE Context Suspend Request Transfer</i> IE specified in subclause 9.3.4.26.		-	
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.17 UE CONTEXT SUSPEND RESPONSE

This message is sent by the AMF to indicate to the NG-RAN node the UE context and the related PDU session contexts have been suspended.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Security Context	O		9.3.1.88		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.18 UE CONTEXT SUSPEND FAILURE

This message is sent by the AMF to indicate to the NG-RAN node that suspension of the UE context has failed in the 5GC.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.19 UE CONTEXT RESUME REQUEST

This message is sent by the NG-RAN node to request the AMF to resume the UE-associated logical NG-connection and UE context.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RRC Resume Cause	M		RRC Establishment Cause 9.3.1.111		YES	ignore
PDU Session Resource Resume List		0..1			YES	reject
>PDU Session Resource Resume Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>UE Context Resume Request Transfer	M		OCTET STRING	Containing the <i>UE Context Resume Request Transfer</i> IE specified in subclause 9.3.4.24	-	
PDU Session Resource Failed to Resume List		0..1			YES	reject
>PDU Session Resource Failed to Resume Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Cause	M		9.3.1.2		-	
Suspend Request Indication	O		9.3.1.158		YES	ignore
Information on Recommended Cells and RAN Nodes for Paging	O		9.3.1.100		YES	ignore
Paging Assistance Data for CE Capable UE	O		9.3.1.141		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.20 UE CONTEXT RESUME RESPONSE

This message is sent by the AMF to indicate to the NG-RAN node that the UE context and the related PDU session contexts have been resumed in the 5GC.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Resume List		0..1			YES	reject
>PDU Session Resource Resume Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>UE Context Resume Response Transfer	M		OCTET STRING	Containing the <i>UE Context Resume Response Transfer</i> IE specified in subclause 9.3.4.25	-	
PDU Session Resource Failed to Resume List		0..1			YES	reject
>PDU Session Resource Failed to Resume Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Cause	M		9.3.1.2		-	
Security Context	O		9.3.1.88		YES	reject
Suspend Response Indication	O		9.3.1.159		YES	ignore
Extended Connected Time	O		9.3.3.31		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.21 UE CONTEXT RESUME FAILURE

This message is sent by the AMF to indicate to the NG-RAN node that resumption of the UE context and the related PDU session contexts has failed in the 5GC.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.22 MT COMMUNICATION HANDLING REQUEST

This message is sent by the NG-RAN node to the AMF to request activating or deactivating CN based MT communication handling for UEs in RRC_INACTIVE state with long eDRX beyond 10.24 seconds as specified in TS 23.501 [9].

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
CHOICE 5GC Action	M				YES	reject
>HLCOM Activate						
>>NR Paging Long eDRX Information for RRC INACTIVE	M		9.3.3.64		-	
>HLCOM Deactivate						
>>UE Reachability Indication	M		ENUMERATED (true, ...)		-	
PDU Session Resource List		1			YES	reject
>PDU Session Resource Item		1..<maxno ofPDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.23 MT COMMUNICATION HANDLING RESPONSE

This message is sent by the AMF to indicate that CN based MT Communication handling was successfully activated or deactivated, as specified in TS 23.501 [9].

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.24 MT COMMUNICATION HANDLING FAILURE

This message is sent by the AMF to indicate that CN based MT Communication handling was unsuccessful.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.25 RAN PAGING REQUEST

This message is sent by the AMF to request RAN Paging due to MT data buffered in 5GC or the arrival of DL signalling for the UE.

Direction: AMF → NG-RAN

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Paging Policy Differentiation	O		9.3.3.63		YES	ignore
DL Signalling	O		ENUMERATED (true, ...)		YES	ignore

9.2.3 UE Mobility Management Messages

9.2.3.1 HANDOVER REQUIRED

This message is sent by the source NG-RAN node to the AMF to request the preparation of resources at the target.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Handover Type	M		9.3.1.22		YES	reject
Cause	M		9.3.1.2		YES	ignore
Target ID	M		9.3.1.25		YES	reject
Direct Forwarding Path Availability	O		9.3.1.64		YES	ignore
PDU Session Resource List		1			YES	reject
>PDU Session Resource Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Required Transfer	M		OCTET STRING	Containing the <i>Handover Required Transfer</i> IE specified in subclause 9.3.4.14.	-	
Source to Target Transparent Container	M		9.3.1.20		YES	reject

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.2 HANDOVER COMMAND

This message is sent by the AMF to inform the source NG-RAN node that resources for the handover have been prepared at the target side.

Direction: AMF → NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Handover Type	M		9.3.1.22		YES	reject
NAS Security Parameters from NG-RAN	C- iftoEPSU TRA		9.3.3.26		YES	reject
PDU Session Resource Handover List		0..1			YES	ignore
>PDU Session Resource Handover Item		1..<maxno ofPDUSes sions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Command Transfer	M		OCTET STRING	Containing the <i>Handover Command Transfer</i> IE specified in subclause 9.3.4.10.	-	
PDU Session Resource to Release List		0..1			YES	ignore
>PDU Session Resource to Release Item		1..<maxno ofPDUSes sions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Preparation Unsuccessful Transfer	M		OCTET STRING	Containing the <i>Handover Preparation Unsuccessful Transfer</i> IE specified in subclause 9.3.4.18.	-	
Target to Source Transparent Container	M		9.3.1.21		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

Condition	Explanation
iftoEPSUTRA	This IE shall be present if the <i>Handover Type</i> IE is set to the value "5GStoEPS" or "5GtoUTRA".

9.2.3.3 HANDOVER PREPARATION FAILURE

This message is sent by the AMF to inform the source NG-RAN node that the Handover Preparation has failed.

Direction: AMF → NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Target to Source Failure Transparent Container	O		9.3.1.186		YES	ignore

9.2.3.4 HANDOVER REQUEST

This message is sent by the AMF to the target NG-RAN node to request the preparation of resources.

Direction: AMF → NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
Handover Type	M		9.3.1.22		YES	reject
Cause	M		9.3.1.2		YES	ignore
UE Aggregate Maximum Bit Rate	M		9.3.1.58		YES	reject
Core Network Assistance Information for RRC INACTIVE	O		9.3.1.15		YES	ignore
UE Security Capabilities	M		9.3.1.86		YES	reject
Security Context	M		9.3.1.88		YES	reject
New Security Context Indicator	O		9.3.1.55		YES	reject
NASC	O		NAS-PDU 9.3.3.4	Refers to either the "Intra N1 mode NAS transparent container" or the "S1 mode to N1 mode NAS transparent container", the details of the IE definition and the encoding are specified in TS 24.501 [26].	YES	reject
PDU Session Resource Setup List		1			YES	reject
>PDU Session Resource Setup Item		1..<maxno of PDU Sessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>S-NSSAI	M		9.3.1.24		-	
>>Handover Request Transfer	M		OCTET STRING	Containing the PDU Session Resource Setup Request Transfer IE specified in subclause 9.3.4.1.	-	
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore
Allowed NSSAI	M		9.3.1.31	Indicates the S-NSSAIs permitted by the network.	YES	reject
Trace Activation	O		9.3.1.14		YES	ignore
Masked IMEISV	O		9.3.1.54		YES	ignore
Source to Target Transparent Container	M		9.3.1.20		YES	reject
Mobility Restriction List	O		9.3.1.85		YES	ignore
Location Reporting Request Type	O		9.3.1.65		YES	ignore
RRC Inactive Transition Report Request	O		9.3.1.91		YES	ignore
GUAMI	M		9.3.3.3		YES	reject
Redirection for Voice EPS Fallback	O		9.3.1.116		YES	ignore
CN Assisted RAN Parameters Tuning	O		9.3.1.119		YES	ignore
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
IAB Authorized	O		9.3.1.129		YES	reject
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore

UE Differentiation Information	O		9.3.1.144		YES	ignore
NR V2X Services Authorized	O		9.3.1.146		YES	ignore
LTE V2X Services Authorized	O		9.3.1.147		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.148	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.149	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 QoS Parameters	O		9.3.1.150	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE User Plane CIoT Support Indicator	O		9.3.1.160		YES	ignore
Management Based MDT PLMN List	O		MDT PLMN List 9.3.1.168		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Extended Connected Time	O		9.3.3.31		YES	ignore
Time Synchronisation Assistance Information	O		9.3.1.220		YES	ignore
UE Slice Maximum Bit Rate List	O		9.3.1.231		YES	ignore
5G ProSe Authorized	O		9.3.1.233		YES	ignore
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 QoS Parameters	O		9.3.1.234	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
Aerial UE Subscription Information	O		9.3.1.246		YES	ignore
NR A2X Services Authorized	O		9.3.1.247		YES	ignore
LTE A2X Services Authorized	O		9.3.1.248		YES	ignore
NR A2X UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for NR A2X services.	YES	ignore
LTE A2X UE PC5 Aggregate Maximum Bit Rate	O		LTE UE Sidelink Aggregate Maximum Bit Rate 9.3.1.149	This IE applies only if the UE is authorized for LTE A2X services.	YES	ignore
A2X PC5 QoS Parameters	O		9.3.1.249	This IE applies only if the UE is authorized for A2X services.	YES	ignore
Mobile IAB Authorized	O		9.3.1.259		YES	ignore
No PDU Session Indication	O		ENUMERATED (true, ...)	This IE applies only if the UE is a mobile IAB-MT.	YES	ignore

Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network.	YES	ignore
Ranging and Sidelink Positioning Service Information	O		9.3.1.269	This IE applies only if the UE is authorized for NR V2X services and/or 5G ProSe services.	YES	ignore
AMF UE NGAP ID 2	O		AMF UE NGAP ID 9.3.3.1	This IE indicates the AMF UE NGAP ID assigned by the AMF.	YES	ignore
Target RAN Node ID	O		Global RAN Node ID 9.3.1.5		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.5 HANDOVER REQUEST ACKNOWLEDGE

This message is sent by the target NG-RAN node to inform the AMF about the prepared resources at the target.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2	Allocated at the target NG-RAN node.	YES	ignore
PDU Session Resource Admitted List		1			YES	ignore
>PDU Session Resource Admitted Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Request Acknowledge Transfer	M		OCTET STRING	Containing the <i>Handover Request Acknowledge Transfer</i> IE specified in subclause 9.3.4.11.	-	
PDU Session Resource Failed to Setup List		0..1			YES	ignore
>PDU Session Resource Failed to Setup Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Resource Allocation Unsuccessful Transfer	M		OCTET STRING	Containing the <i>Handover Resource Allocation Unsuccessful Transfer</i> IE specified in subclause 9.3.4.19.	-	
Target to Source Transparent Container	M		9.3.1.21		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
NPN Access Information	O		9.3.3.46		YES	reject
RedCap Indication	O		9.3.1.228		YES	ignore
eRedCap Indication	O		ENUMERATED (true, ...)		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.6 HANDOVER FAILURE

This message is sent by the target NG-RAN node to inform the AMF that the preparation of resources has failed.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Target to Source Failure Transparent Container	O		9.3.1.186		YES	ignore

9.2.3.7 HANDOVER NOTIFY

This message is sent by the target NG-RAN node to inform the AMF that the UE has been identified in the target cell and the handover has been completed.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
Notify Source NG-RAN Node	O		ENUMERATED (NotifySource, ...)		YES	ignore

9.2.3.8 PATH SWITCH REQUEST

This message is sent by the NG-RAN node to inform the AMF of the new serving NG-RAN node and to transfer some NG-U DL tunnel termination point(s) to the SMF via the AMF for one or multiple PDU session resources.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Source AMF UE NGAP ID	M		AMF UE NGAP ID 9.3.3.1		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
UE Security Capabilities	M		9.3.1.86		YES	ignore
PDU Session Resource to be Switched in Downlink List		1			YES	reject
>PDU Session Resource to be Switched in Downlink Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Transfer	M		OCTET STRING	Containing the <i>Path Switch Request Transfer</i> IE specified in subclause 9.3.4.8.	-	
PDU Session Resource Failed to Setup List		0..1			YES	ignore
>PDU Session Resource Failed to Setup Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Setup Failed Transfer	M		OCTET STRING	Containing the <i>Path Switch Request Setup Failed Transfer</i> IE specified in subclause 9.3.4.15.	-	
RRC Resume Cause	O		RRC Establishment Cause 9.3.1.111		YES	ignore
RedCap Indication	O		9.3.1.228		YES	ignore
eRedCap Indication	O		ENUMERATED (true, ...)		YES	ignore
Source AMF GUAMI	O		GUAMI 9.3.3.3	This IE indicates the source AMF.	YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.9 PATH SWITCH REQUEST ACKNOWLEDGE

This message is sent by the AMF to inform the NG-RAN node that the path switch has been successfully completed in the 5GC.

Direction: AMF → NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
UE Security Capabilities	O		9.3.1.86		YES	reject
Security Context	M		9.3.1.88		YES	reject
New Security Context Indicator	O		9.3.1.55		YES	reject
PDU Session Resource Switched List		1			YES	ignore
>PDU Session Resource Switched Item		1..<maxno of PDUSes sions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Acknowledge Transfer	M		OCTET STRING	Containing the <i>Path Switch Request Acknowledge Transfer</i> IE specified in subclause 9.3.4.9.	-	
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore
PDU Session Resource Released List		0..1			YES	ignore
>PDU Session Resource Released Item		1..<maxno of PDUSes sions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Unsuccessful Transfer	M		OCTET STRING	Containing the <i>Path Switch Request Unsuccessful Transfer</i> IE specified in subclause 9.3.4.20.	-	
Allowed NSSAI	M		9.3.1.31	Indicates the S-NSSAIs permitted by the network.	YES	reject
Core Network Assistance Information for RRC INACTIVE	O		9.3.1.15		YES	ignore
RRC Inactive Transition Report Request	O		9.3.1.91		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Redirection for Voice EPS Fallback	O		9.3.1.116		YES	ignore
CN Assisted RAN Parameters Tuning	O		9.3.1.119		YES	ignore
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
Extended Connected Time	O		9.3.3.31		YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
NR V2X Services Authorized	O		9.3.1.146		YES	ignore
LTE V2X Services Authorized	O		9.3.1.147		YES	ignore

NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.148	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.149	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 QoS Parameters	O		9.3.1.150	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE User Plane CIoT Support Indicator	O		9.3.1.160		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Management Based MDT PLMN List	O		MDT PLMN List 9.3.1.168	This IE is ignored if the <i>Management Based MDT PLMN Modification List</i> IE is present.	YES	ignore
Time Synchronisation Assistance Information	O		9.3.1.220		YES	ignore
5G ProSe Authorized	O		9.3.1.233		YES	ignore
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 QoS Parameters	O		9.3.1.234	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
Management Based MDT PLMN Modification List	O		MDT PLMN Modification List 9.3.1.243		YES	ignore
IAB Authorized	O		9.3.1.129		YES	ignore
Aerial UE Subscription Information	O		9.3.1.246		YES	ignore
NR A2X Services Authorized	O		9.3.1.247		YES	ignore
LTE A2X Services Authorized	O		9.3.1.248		YES	ignore
NR A2X UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for NR A2X services.	YES	ignore
LTE A2X UE PC5 Aggregate Maximum Bit Rate	O		LTE UE Sidelink Aggregate Maximum Bit Rate 9.3.1.149	This IE applies only if the UE is authorized for LTE A2X services.	YES	ignore
A2X PC5 QoS Parameters	O		9.3.1.249	This IE applies only if the UE is authorized for A2X services.	YES	ignore
Mobile IAB Authorized	O		9.3.1.259		YES	ignore
Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network.	YES	ignore

Ranging and Sidelink Positioning Service Information	O		9.3.1.269	This IE applies only if the UE is authorized for NR V2X services and/or 5G ProSe services.	YES	ignore
AMF UE NGAP ID 2	O		AMF UE NGAP ID 9.3.3.1	This IE indicates the AMF UE NGAP ID assigned by the AMF.	YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.10 PATH SWITCH REQUEST FAILURE

This message is sent by the AMF to inform the NG-RAN node that a failure has occurred in the 5GC during the Path Switch Request procedure.

Direction: AMF → NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Released List		1			YES	ignore
>PDU Session Resource Released Item		1..<maxno ofPDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Unsuccessful Transfer	M		OCTET STRING	Containing the PDU session <i>Path Switch Request Unsuccessful Transfer</i> IE specified in subclause 9.3.4.20.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.11 HANDOVER CANCEL

This message is sent by the source NG-RAN node to the AMF to request the cancellation of an ongoing handover.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.3.12 HANDOVER CANCEL ACKNOWLEDGE

This message is sent by the AMF to the source NG-RAN node to confirm that the ongoing handover was cancelled.

Direction: AMF → NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.3.13 UPLINK RAN STATUS TRANSFER

This message is sent by the source NG-RAN node to transfer the uplink PDCP-SN and HFN receiver status and the downlink PDCP SN and HFN transmitter status during intra 5GC NG-based handover.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Status Transfer Transparent Container	M		9.3.1.108		YES	reject

9.2.3.14 DOWNLINK RAN STATUS TRANSFER

This message is sent by the AMF to the target NG-RAN node to transfer the uplink PDCP-SN and HFN receiver status and the downlink PDCP SN and HFN transmitter status during intra 5GC NG-based handover.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Status Transfer Transparent Container	M		9.3.1.108		YES	reject

9.2.3.15 HANDOVER SUCCESS

This message is sent by the AMF to the source NG-RAN node to indicate the successful access of the UE toward the target NG-RAN node.

Direction: AMF → NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject

9.2.3.16 UPLINK RAN EARLY STATUS TRANSFER

This message is sent by the source NG-RAN node to transfer the COUNT value(s) of the first forwarded downlink SDU(s) during NG DAPS Handover, and during NG-based handover with time-based trigger condition.

Direction: NG-RAN node → AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Early Status Transfer Transparent Container	M		9.3.1.190		YES	reject

9.2.3.17 DOWNLINK RAN EARLY STATUS TRANSFER

This message is sent by the AMF to transfer the COUNT value(s) of the first forwarded downlink SDU(s) during NG DAPS Handover, and during NG-based handover with time-based trigger condition.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Early Status Transfer Transparent Container	M		9.3.1.190		YES	reject

9.2.4 Paging Messages

9.2.4.1 PAGING

This message is sent by the AMF and is used to page a UE in one or several tracking areas.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
UE Paging Identity	M		9.3.3.18		YES	ignore
Paging DRX	O		9.3.1.90		YES	ignore
TAI List for Paging		1			YES	ignore
>TAI List for Paging Item		1..<maxno ofTAIforPaging>			-	
>>TAI	M		9.3.3.11		-	
Paging Priority	O		9.3.1.78		YES	ignore
UE Radio Capability for Paging	O		9.3.1.68		YES	ignore
Paging Origin	O		9.3.3.22		YES	ignore
Assistance Data for Paging	O		9.3.1.69		YES	ignore
NB-IoT Paging eDRX Information	O		9.3.1.138		YES	ignore
NB-IoT Paging DRX	O		9.3.1.139	If this IE is present, the <i>Paging DRX</i> IE is ignored.	YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
WUS Assistance Information	O		9.3.1.143		YES	ignore
E-UTRA Paging eDRX Information	O		9.3.1.154		YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
NR Paging eDRX Information	O		9.3.1.227		YES	ignore
Paging Cause	O		ENUMERATED (voice, ...)		YES	ignore
PEIPS Assistance Information	O		9.3.1.232		YES	ignore
LP-WUSPS Assistance Information	O		9.3.1.279		YES	ignore
LP-WUS Disable Indication	O		ENUMERATED (true, ...)		YES	ignore

Range bound	Explanation
maxnoofTAIforPaging	Maximum no. of TAIs for paging. Value is 16.

9.2.4.2 MULTICAST GROUP PAGING

This message is sent by the AMF and is used to notify involved UEs about the activation of a multicast MBS session.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
MBS Session ID	M		9.3.1.206		YES	ignore
MBS Service Area	O		9.3.1.208		YES	ignore
Multicast Group Paging Area List		1			YES	ignore
>Multicast Group Paging Area Item		1..<maxno ofPagingAreas>			-	
>>Multicast Group Paging Area	M		9.3.1.216		-	
>>UE Paging List		0..1			-	
>>>UE Paging Item		1..<maxno ofUEsforPaging>			-	
>>>>UE Identity Index Value	M		9.3.3.23		-	
>>>>Paging DRX	O		9.3.1.90		-	

Range bound	Explanation
maxnoofPagingAreas	Maximum no. of paging areas for multicast group paging. Value is 64.
maxnoofUEsforPaging	Maximum no. of UEs allowed within one paging area for multicast group paging. Value is 4096.

9.2.5 NAS Transport Messages

9.2.5.1 INITIAL UE MESSAGE

This message is sent by the NG-RAN node to transfer the initial layer 3 message to the AMF over the NG interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NAS-PDU	M		9.3.3.4		YES	reject
User Location Information	M		9.3.1.16		YES	reject
RRC Establishment Cause	M		9.3.1.111		YES	ignore
5G-S-TMSI	O		9.3.3.20		YES	reject
AMF Set ID	O		9.3.3.12		YES	ignore
UE Context Request	O		ENUMERATED (requested, ...)		YES	ignore
Allowed NSSAI	O		9.3.1.31		YES	reject
Source to Target AMF Information Reroute	O		9.3.3.27		YES	ignore
Selected PLMN Identity	O		PLMN Identity 9.3.3.5	Indicates the selected PLMN id for the non-3GPP access.	YES	ignore
IAB Node Indication	O		ENUMERATED (true, ...)	Indication of an IAB node	YES	reject
CE-mode-B Support Indicator	O		9.3.1.156		YES	reject
LTE-M Indication	O		9.3.1.157		YES	ignore
EDT Session	O		ENUMERATED (true, ...)		YES	ignore
Authenticated Indication	O		ENUMERATED (true, ...)	Indicates the FN-RG has been authenticated by the access network.	YES	ignore
NPN Access Information	O		9.3.3.46		YES	reject
RedCap Indication	O		9.3.1.228		YES	ignore
Selected NID	O		NID 9.3.3.42	Indicates together with the <i>Selected PLMN Identity</i> IE, the selected SNPN Identity for the non-3GPP access.	YES	ignore
Mobile IAB Node Indication	O		ENUMERATED (true, ...)	Indicates that the UE is a mobile IAB-node	YES	reject
Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network.	YES	ignore
eRedCap Indication	O		ENUMERATED (true, ...)		YES	ignore
AUN3 Device Access Information	O		9.3.3.66		YES	ignore
GUAMI	O		9.3.3.3		YES	ignore
GUAMI Type	O		ENUMERATED (native, mapped, ...)		YES	ignore
Requested NSSAI	O		9.3.1.275	Indicates the S-NSSAIs requested by the UE.	YES	ignore

9.2.5.2 DOWNLINK NAS TRANSPORT

This message is sent by the AMF and is used for carrying NAS information over the NG interface.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Old AMF	O		AMF Name 9.3.3.21	This IE is ignored if the <i>Extended Old AMF</i> IE is present.	YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
NAS-PDU	M		9.3.3.4		YES	reject
Mobility Restriction List	O		9.3.1.85		YES	ignore
Index to RAT/Frequency Selection Priority	O		9.3.1.61		YES	ignore
UE Aggregate Maximum Bit Rate	O		9.3.1.58		YES	ignore
Allowed NSSAI	O		9.3.1.31	Indicates the S-NSSAIs permitted by the network.	YES	reject
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
Extended Connected Time	O		9.3.3.31		YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE Radio Capability	O		9.3.1.74		YES	ignore
UE Capability Info Request	O		9.3.1.192		YES	ignore
End Indication	O		9.3.3.32		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Target NSSAI Information	O		9.3.1.229		YES	ignore
Masked IMEISV	O		9.3.1.54		YES	ignore
Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network.	YES	ignore
Mobile IAB Authorized	O		9.3.1.259		YES	ignore
Extended Old AMF	O		Extended AMF Name 9.3.3.51		YES	ignore

9.2.5.3 UPLINK NAS TRANSPORT

This message is sent by the NG-RAN node and is used for carrying NAS information over the NG interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NAS-PDU	M		9.3.3.4		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
W-AGF Identity Information	O		OCTET STRING	Containing the <i>WAgfInfo</i> IE specified in TS 29.510 [36].	YES	reject
TNGF Identity Information	O		OCTET STRING	Containing the <i>TngfInfo</i> IE specified in TS 29.510 [36].	YES	reject
TWIF Identity Information	O		OCTET STRING	Containing the <i>TwifInfo</i> IE specified in TS 29.510 [36].	YES	reject

9.2.5.4 NAS NON DELIVERY INDICATION

This message is sent by the NG-RAN node and is used for reporting the non-delivery of a NAS PDU previously received within a DOWNLINK NAS TRANSPORT message or the *NAS-PDU* IE previously received within the PDU SESSION RESOURCE SETUP REQUEST message over the NG interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NAS-PDU	M		9.3.3.4		YES	ignore
Cause	M		9.3.1.2		YES	ignore

9.2.5.5 REROUTE NAS REQUEST

This message is sent by the AMF in order to request for a rerouting of the INITIAL UE MESSAGE to another AMF.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
AMF UE NGAP ID	O		9.3.3.1		YES	ignore
NGAP Message	M		OCTET STRING	Contains the INITIAL UE MESSAGE	YES	reject
AMF Set ID	M		9.3.3.12		YES	reject
Allowed NSSAI	O		9.3.1.31		YES	reject
Source to Target AMF Information Reroute	O		9.3.3.27		YES	ignore
Partially Allowed NSSAI	O		9.3.1.261	Indicates the S-NSSAIs partially permitted by the network.	YES	ignore

9.2.6 Interface Management Messages

9.2.6.1 NG SETUP REQUEST

This message is sent by the NG-RAN node to transfer application layer information for an NG-C interface instance.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Global RAN Node ID	M		9.3.1.5	The AMF may use this information to verify the identity of the gNB, as specified in TS 33.545 [62].	YES	reject
RAN Node Name	O		PrintableString (SIZE(1..150, ...))		YES	ignore
Supported TA List		1		Supported TAs in the NG-RAN node. This IE is ignored if the <i>A-IoT Support</i> IE is present and set to "A-IoT only".	YES	reject
>Supported TA Item		1..<maxno ofTACs>			-	
>>TAC	M		9.3.3.10	Broadcast TAC	-	
>>Broadcast PLMN List		1			-	
>>>Broadcast PLMN Item		1..<maxno ofBPLMNs>			-	
>>>>PLMN Identity	M		9.3.3.5	Broadcast PLMN	-	
>>>>TAI Slice Support List	M		Slice Support List 9.3.1.17	Supported S-NSSAIs per TAC, per PLMN or per SNPN.	-	
>>>>NPN Support	O		9.3.3.44	If the <i>NID</i> IE is included, it identifies a SNPN together with the <i>PLMN Identity</i> IE.	YES	reject
>>>>Extended TAI Slice Support List	O		Extended Slice Support List 9.3.1.191	Additional Supported S-NSSAIs per TAC, per PLMN or per SNPN.	YES	reject
>>>>TAI NSAG Support List	O		9.3.1.238	NSAG information associated with the slices per TAC, per PLMN or per SNPN.	YES	ignore
>>Configured TAC Indication	O		9.3.3.50		YES	ignore
>>RAT Information	O		9.3.1.125	RAT information associated with the TAC of the indicated PLMN(s).	YES	reject
Default Paging DRX	M		Paging DRX 9.3.1.90	This IE is ignored if the <i>A-IoT Support</i> IE is present and set to "A-IoT only".	YES	ignore
UE Retention Information	O		9.3.1.117		YES	ignore
NB-IoT Default Paging DRX	O		9.3.1.137		YES	ignore
Extended RAN Node Name	O		9.3.1.193		YES	ignore
A-IoT Support	O		9.3.1.278		YES	reject
Additional ULI	O		9.3.1.280		YES	ignore

Range bound	Explanation
maxnoofTACs	Maximum no. of TACs. Value is 256.
maxnoofBPLMNs	Maximum no. of Broadcast PLMNs. Value is 12.

9.2.6.2 NG SETUP RESPONSE

This message is sent by the AMF to transfer application layer information for an NG-C interface instance.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF Name	M		9.3.3.21	This IE is ignored if the <i>Extended AMF Name</i> IE is present. This IE is ignored if the NG Setup procedure is executed between the NG-RAN node and the AIOTF. This IE is ignored if the NG-RAN node only supports A-IoT.	YES	reject
Served GUAMI List		1		This IE is ignored if the NG Setup procedure is executed between the NG-RAN node and the AIOTF. This IE is ignored if the NG-RAN node only supports A-IoT.	YES	reject
>Served GUAMI Item		1..<maxno of Served GUAMIs>			-	
>>GUAMI	M		9.3.3.3		-	
>>Backup AMF Name	O		AMF Name 9.3.3.21	This IE is ignored if the <i>Extended Backup AMF Name</i> IE is present.	-	
>>GUAMI Type	O		ENUMERATED (native, mapped, ...)		YES	ignore
>>Extended Backup AMF Name	O		Extended AMF Name 9.3.3.51		YES	ignore
Relative AMF Capacity	M		9.3.1.32	This IE is ignored if the NG Setup procedure is executed between the NG-RAN node and the AIOTF. This IE is ignored if the NG-RAN node only supports A-IoT.	YES	ignore
PLMN Support List		1		This IE is ignored if the NG Setup procedure is executed between the NG-RAN node and the AIOTF. This IE is ignored if the NG-RAN node only supports A-IoT.	YES	reject
>PLMN Support Item		1..<maxno of PLMNs>			-	
>>PLMN Identity	M		9.3.3.5		-	
>>Slice Support List	M		9.3.1.17	Supported S-NSSAIs per PLMN or per SNPN.	-	

>>NPN Support	O		9.3.3.44	If <i>NID</i> IE is included, it identifies a SNPN together with the <i>PLMN Identity</i> IE.	YES	reject
>>Extended Slice Support List	O		9.3.1.191	Additional Supported S-NSSAIs per PLMN or per SNPN.	YES	reject
>>Onboarding Support	O		ENUMERATED (true, ...)	Indication of onboarding support.	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
UE Retention Information	O		9.3.1.117		YES	ignore
IAB Supported	O		ENUMERATED (true, ...)	Indication of support for IAB.	YES	ignore
Extended AMF Name	O		9.3.3.51		YES	ignore
Mobile IAB Supported	O		ENUMERATED (true, ...)	Indication of support for mobile IAB.	YES	ignore
AIOTF Identifier	O		9.3.3.70		YES	reject
AIOTF Name	O		9.3.3.76		YES	reject

Range bound	Explanation
maxnoofServedGUAMIs	Maximum no. of GUAMIs served by an AMF. Value is 256.
maxnoofPLMNs	Maximum no. of PLMNs per message. Value is 12.

9.2.6.3 NG SETUP FAILURE

This message is sent by the AMF to indicate NG setup failure.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to Wait	O		9.3.1.56		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.4 RAN CONFIGURATION UPDATE

This message is sent by the NG-RAN node to transfer updated application layer information for an NG-C interface instance.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
RAN Node Name	O		PrintableString (SIZE(1..150, ...))		YES	ignore
Supported TA List		0..1		Supported TAs in the NG-RAN node.	YES	reject
>Supported TA Item		1..<maxno ofTACs>			-	
>>TAC	M		9.3.3.10	Broadcast TAC	-	
>>Broadcast PLMN List		1			-	
>>>Broadcast PLMN Item		1..<maxno ofBPLMNs >			-	
>>>>PLMN Identity	M		9.3.3.5	Broadcast PLMN	-	
>>>>TAI Slice Support List	M		Slice Support List 9.3.1.17	Supported S-NSSAIs per TAC, per PLMN or per SNPN.	-	
>>>>NPN Support	O		9.3.3.44	If the <i>NID</i> IE is included, it identifies a SNPN together with the <i>PLMN Identity</i> IE.	YES	reject
>>>>Extended TAI Slice Support List	O		Extended Slice Support List 9.3.1.191	Additional Supported S-NSSAIs per TAC, per PLMN or per SNPN.	YES	reject
>>>>TAI NSAG Support List	O		9.3.1.238	NSAG information associated with the slices per TAC, per PLMN or per SNPN.	YES	ignore
>>Configured TAC Indication	O		9.3.3.50		YES	ignore
>>RAT Information	O		9.3.1.125	RAT information associated with the TAC of the indicated PLMN(s).	YES	reject
Default Paging DRX	O		Paging DRX 9.3.1.90		YES	ignore
Global RAN Node ID	O		9.3.1.5		YES	ignore
NG-RAN TNL Association to Remove List		0..1			YES	reject
>NG-RAN TNL Association to Remove Item		1..<maxno ofTNLAssociations>			-	
>>TNL Association Transport Layer Address	M		CP Transport Layer Information 9.3.2.6	Transport layer address of the NG-RAN node.	-	
>>TNL Association Transport Layer Address at AMF	O		CP Transport Layer Information 9.3.2.6	Transport layer address of the AMF.	-	
NB-IoT Default Paging DRX	O		9.3.1.137		YES	ignore
Extended RAN Node Name	O		9.3.1.193		YES	ignore
Additional ULI	O		9.3.1.280		YES	ignore
A-IoT Support	O		9.3.1.278		YES	reject

Range bound	Explanation
maxnoofTACs	Maximum no. of TACs. Value is 256.
maxnoofBPLMNs	Maximum no. of Broadcast PLMNs. Value is 12.
maxnoofTNLAassociations	Maximum no. of TNL Associations between the NG-RAN node and the AMF. Value is 32.

9.2.6.5 RAN CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by the AMF to acknowledge the NG-RAN node transfer of updated information for an NG-C interface instance.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.6 RAN CONFIGURATION UPDATE FAILURE

This message is sent by the AMF to indicate RAN configuration update failure.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to Wait	O		9.3.1.56		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.7 AMF CONFIGURATION UPDATE

This message is sent by the AMF to transfer updated information for an NG-C interface instance.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF Name	O		9.3.3.21		YES	reject
Served GUAMI List		0..1			YES	reject
>Served GUAMI Item		1..<maxno of Served GUAMIs>			-	
>>GUAMI	M		9.3.3.3		-	
>>Backup AMF Name	O		AMF Name 9.3.3.21	This IE is ignored if the <i>Extended Backup AMF Name</i> IE is present.	-	
>>GUAMI Type	O		ENUMERATED (native, mapped, ...)		YES	ignore
>>Extended Backup AMF Name	O		Extended AMF Name 9.3.3.51		YES	ignore
Relative AMF Capacity	O		9.3.1.32		YES	ignore
PLMN Support List		0..1			YES	reject
>PLMN Support Item		1..<maxno of PLMNs>			-	
>>PLMN Identity	M		9.3.3.5		-	
>>Slice Support List	M		9.3.1.17	Supported S-NSSAIs per PLMN or per SNPN.	-	
>>NPN Support	O		9.3.3.44	If the <i>NID</i> IE is included, it identifies a SNPN together with the <i>PLMN Identity</i> IE.	YES	reject
>>Extended Slice Support List	O		9.3.1.191	Additional Supported S-NSSAIs per PLMN or per SNPN.	YES	reject
>>Onboarding Support	O		ENUMERATED (true, ...)	Indication of onboarding support.	YES	ignore
AMF TNL Association to Add List		0..1			YES	ignore
>AMF TNL Association to Add Item		1..<maxno of TNL Associations>			-	
>>AMF TNL Association Address	M		CP Transport Layer Information 9.3.2.6	AMF Transport Layer information used to set up the new TNL association.	-	
>>TNL Association Usage	O		9.3.2.9		-	
>>TNL Address Weight Factor	M		9.3.2.10		-	
AMF TNL Association to Remove List		0..1			YES	ignore
>AMF TNL Association to Remove Item		1..<maxno of TNL Associations>			-	
>>AMF TNL Association Address	M		CP Transport Layer Information 9.3.2.6	Transport Layer Address of the AMF.	-	
>>TNL Association Transport Layer Address NG-RAN	O		CP Transport Layer Information 9.3.2.6	Transport Layer Address of the NG-RAN node.	YES	reject

AMF TNL Association to Update List		0..1			YES	ignore
>AMF TNL Association to Update Item		1..<maxno ofTNLAssociations>			-	
>>AMF TNL Association Address	M		CP Transport Layer Information 9.3.2.6	AMF Transport Layer information used to identify the TNL association to be updated.	-	
>>TNL Association Usage	O		9.3.2.9		-	
>>TNL Address Weight Factor	O		9.3.2.10		-	
Extended AMF Name	O		9.3.3.51		YES	ignore

Range bound	Explanation
maxnoofServedGUAMIs	Maximum no. of GUAMIs served by an AMF. Value is 256.
maxnoofPLMNs	Maximum no. of PLMNs per message. Value is 12.
maxnoofTNLAssociations	Maximum no. of TNL Associations between the NG-RAN node and the AMF. Value is 32.

9.2.6.8 AMF CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by the NG-RAN node to acknowledge the AMF transfer of updated information for an NG-C interface instance.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF TNL Association Setup List		0..1			YES	ignore
>AMF TNL Association Setup Item		1..<maxno ofTNLAssociations>			-	
>>AMF TNL Association Address	M		CP Transport Layer Information 9.3.2.6	Previously received AMF Transport Layer information for the TNL association.	-	
AMF TNL Association Failed to Setup List	O		TNL Association List 9.3.2.7		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofTNLAssociations	Maximum no. of TNL Associations between the NG-RAN node and the AMF. Value is 32.

9.2.6.9 AMF CONFIGURATION UPDATE FAILURE

This message is sent by the NG-RAN node to indicate AMF configuration update failure.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to Wait	O		9.3.1.56		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.10 AMF STATUS INDICATION

This message is sent by the AMF to support AMF management functions.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Unavailable GUAMI List		1		Indicates the GUAMIs configured to be unavailable at the AMF	YES	reject
>Unavailable GUAMI Item		1..<maxno ofServedGUAMIs>			-	
>>GUAMI	M		9.3.3.3		-	
>>Timer Approach for GUAMI Removal	O		ENUMERATED (apply timer, ...)		-	
>>Backup AMF Name	O		AMF Name 9.3.3.21	This IE is ignored if the <i>Extended Backup AMF Name</i> IE is present.	-	
>>Extended Backup AMF Name	O		Extended AMF Name 9.3.3.51		YES	ignore

Range bound	Explanation
maxnoofServedGUAMIs	Maximum no. of GUAMIs served by an AMF. Value is 256.

9.2.6.11 NG RESET

This message is sent by both the NG-RAN node and the AMF to request that the NG interface, or parts of the NG interface, be reset.

Direction: NG-RAN node → AMF and AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
CHOICE Reset Type	M				YES	reject
>NG interface						
>>Reset All	M		ENUMERATED (Reset all, ...)		-	
>Part of NG interface						
>>UE-associated Logical NG-connection List	M		9.3.3.25		-	

9.2.6.12 NG RESET ACKNOWLEDGE

This message is sent by both the NG-RAN node and the AMF as a response to an NG RESET message.

Direction: NG-RAN node → AMF and AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
UE-associated Logical NG-connection List	O		9.3.3.25		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.13 ERROR INDICATION

This message is sent by both the NG-RAN node and the AMF to indicate that some error has been detected in the node.

Direction: NG-RAN node → AMF and AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	O		9.3.3.1		YES	ignore
RAN UE NGAP ID	O		9.3.3.2		YES	ignore
Cause	O		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
5G-S-TMSI	O		9.3.3.20	In the case of NB-IoT UE using Control Plane CIoT 5GS Optimisation, when only 5G-S-TMSI is used to identify UE, the 5G-S-TMSI shall be included.	YES	ignore

9.2.6.14 OVERLOAD START

This message is sent by the AMF and is used to indicate to the NG-RAN node that the AMF is overloaded.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF Overload Response	O		Overload Response 9.3.1.104		YES	reject
AMF Traffic Load Reduction Indication	O		Traffic Load Reduction Indication 9.3.1.106		YES	ignore
Overload Start NSSAI List		<i>0..1</i>			YES	ignore
>Overload Start NSSAI Item		<i>1..<maxno ofSliceltems></i>			-	
>>Slice Overload List	M		9.3.1.107		-	
>>Slice Overload Response	O		Overload Response 9.3.1.104		-	
>>Slice Traffic Load Reduction Indication	O		Traffic Load Reduction Indication 9.3.1.106		-	
GUAMI List		<i>0..1</i>			YES	ignore
>GUAMI List Item		<i>1..<maxno ofGUAMIs></i>			-	
>>GUAMI	M		9.3.3.3		-	

Range bound	Explanation
maxnoofSliceltems	Maximum no. of signalled slice support items. Value is 1024.
maxnoofGUAMIs	Maximum no. of GUAMIs. Value is 1024.

9.2.6.15 OVERLOAD STOP

This message is sent by the AMF and is used to indicate that the AMF is no longer overloaded.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
GUAMI List		<i>0..1</i>			YES	ignore
>GUAMI List Item		<i>1..<maxno ofGUAMIs></i>			-	
>>GUAMI	M		9.3.3.3		-	

Range bound	Explanation
maxnoofGUAMIs	Maximum no. of GUAMIs. Value is 1024.

9.2.6.16 NG REMOVAL REQUEST

This message is sent by the NG-RAN node to initiate the removal of the interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Global RAN Node ID	M		9.3.1.5		YES	reject

9.2.6.17 NG REMOVAL RESPONSE

This message is sent by the AMF to acknowledge the initiation of removal of the interface.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF Name	M		9.3.3.21	This IE is ignored if the <i>Extended AMF Name</i> IE is present.	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Extended AMF Name	O		9.3.3.51		YES	ignore

9.2.6.18 NG REMOVAL FAILURE

This message is sent by the AMF to indicate that removing the interface cannot be accepted.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.7 Configuration Transfer Messages

9.2.7.1 UPLINK RAN CONFIGURATION TRANSFER

This message is sent by the NG-RAN node in order to transfer RAN configuration information.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
SON Configuration Transfer	O		9.3.3.6		YES	ignore
EN-DC SON Configuration Transfer	O		OCTET STRING	Contains the <i>EN-DC SON Configuration Transfer</i> IE as defined in TS 36.413 [16].	YES	ignore
Inter-system SON Configuration Transfer	O		9.3.3.33		YES	ignore

9.2.7.2 DOWNLINK RAN CONFIGURATION TRANSFER

This message is sent by the AMF in order to transfer RAN configuration information.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
SON Configuration Transfer	O		9.3.3.6		YES	ignore
EN-DC SON Configuration Transfer	O		OCTET STRING	Contains the <i>EN-DC SON Configuration Transfer</i> IE as defined in TS 36.413 [16].	YES	ignore
Inter-system SON Configuration Transfer	O		9.3.3.33		YES	ignore

9.2.8 Warning Message Transmission Messages

9.2.8.1 WRITE-REPLACE WARNING REQUEST

This message is sent by the AMF to request the start or overwrite of the broadcast of a warning message.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Message Identifier	M		9.3.1.35		YES	reject
Serial Number	M		9.3.1.36		YES	reject
Warning Area List	O		9.3.1.37		YES	ignore
Repetition Period	M		9.3.1.49		YES	reject
Number of Broadcasts Requested	M		9.3.1.38		YES	reject
Warning Type	O		9.3.1.39		YES	ignore
Warning Security Information	O		OCTET STRING (SIZE(50))	This IE is not used in the specification. If received, the IE is ignored.	YES	ignore
Data Coding Scheme	O		9.3.1.41		YES	ignore
Warning Message Contents	O		9.3.1.42		YES	ignore
Concurrent Warning Message Indicator	O		9.3.1.46		YES	reject
Warning Area Coordinates	O		9.3.1.112		YES	ignore

9.2.8.2 WRITE-REPLACE WARNING RESPONSE

This message is sent by the NG-RAN node to acknowledge the AMF on the start or overwrite request of a warning message.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Message Identifier	M		9.3.1.35		YES	reject
Serial Number	M		9.3.1.36		YES	reject
Broadcast Completed Area List	O		9.3.1.43		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.8.3 PWS CANCEL REQUEST

This message is forwarded by the AMF to the NG-RAN node to cancel an already ongoing broadcast of a warning message.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Message Identifier	M		9.3.1.35		YES	reject
Serial Number	M		9.3.1.36		YES	reject
Warning Area List	O		9.3.1.37		YES	ignore
Cancel-All Warning Messages Indicator	O		9.3.1.47		YES	reject

9.2.8.4 PWS CANCEL RESPONSE

This message is sent by the NG-RAN node to indicate the list of warning areas where cancellation of the broadcast of the identified message was successful and unsuccessful.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Message Identifier	M		9.3.1.35		YES	reject
Serial Number	M		9.3.1.36		YES	reject
Broadcast Cancelled Area List	O		9.3.1.44		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.8.5 PWS RESTART INDICATION

This message is sent by the NG-RAN node to inform the AMF that PWS information for some or all cells of the NG-RAN node are available for reloading from the CBC if needed.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
CHOICE <i>Cell List for Restart</i>	M				YES	reject
>E-UTRA						
>>E-UTRA Cell List for Restart		1..<maxno ofCellsinngeNB>			-	
>>>E-UTRA CGI	M		9.3.1.9		-	
>NR						
>>NR Cell List for Restart		1..<maxno ofCellsingNB>			-	
>>>NR CGI	M		9.3.1.7		-	
Global RAN Node ID	M		9.3.1.5		YES	reject
TAI List for Restart		1..<maxno ofTAIforRestart>			YES	reject
>TAI	M		9.3.3.11		-	
Emergency Area ID List for Restart		0..<maxno ofEAIforRestart>			YES	reject
>Emergency Area ID	M		9.3.1.48		-	

Range bound	Explanation
maxnoofCellsinngeNB	Maximum no. of cells that can be served by an ng-eNB. Value is 256.
maxnoofCellsingNB	Maximum no. of cells that can be served by a gNB. Value is 16384.
maxnoofTAIforRestart	Maximum no. of TAIs subject for reloading warning message broadcast. Value is 2048.
maxnoofEAIforRestart	Maximum no. of Emergency Area IDs subject for reloading warning message broadcast. Value is 256.

9.2.8.6 PWS FAILURE INDICATION

This message is sent by the NG-RAN node to inform the AMF that ongoing PWS operation for one or more cells of the NG-RAN node has failed.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
CHOICE <i>PWS Failed Cell List</i>	M				YES	reject
>E-UTRA						
>>PWS Failed E-UTRA Cell List		1..<maxno ofCellsinngeNB>			-	
>>>E-UTRA CGI	M		9.3.1.9		-	
>NR						
>>PWS Failed NR Cell List		1..<maxno ofCellsingNB>			-	
>>>NR CGI	M		9.3.1.7		-	
Global RAN Node ID	M		9.3.1.5		YES	reject

Range bound	Explanation
maxnoofCellsinngeNB	Maximum no. of cells that can be served by an ng-eNB. Value is 256.
maxnoofCellsingNB	Maximum no. of cells that can be served by a gNB. Value is 16384.

9.2.9 NRPPa Transport Messages

9.2.9.1 DOWNLINK UE ASSOCIATED NRPPa TRANSPORT

This message is sent by the AMF and is used for carrying NRPPa message over the NG interface.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Routing ID	M		9.3.3.13		YES	reject
NRPPa-PDU	M		9.3.3.14		YES	reject

9.2.9.2 UPLINK UE ASSOCIATED NRPPa TRANSPORT

This message is sent by the NG-RAN node and is used for carrying NRPPa message over the NG interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Routing ID	M		9.3.3.13		YES	reject
NRPPa-PDU	M		9.3.3.14		YES	reject

9.2.9.3 DOWNLINK NON UE ASSOCIATED NRPPa TRANSPORT

This message is sent by the AMF and is used for carrying NRPPa message over the NG interface.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Routing ID	M		9.3.3.13		YES	reject
NRPPa-PDU	M		9.3.3.14		YES	reject

9.2.9.4 UPLINK NON UE ASSOCIATED NRPPa TRANSPORT

This message is sent by the NG-RAN node and is used for carrying NRPPa message over the NG interface.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Routing ID	M		9.3.3.13		YES	reject
NRPPa-PDU	M		9.3.3.14		YES	reject

9.2.10 Trace Messages

9.2.10.1 TRACE START

This message is sent by the AMF to initiate a trace session for a UE.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Trace Activation	M		9.3.1.14		YES	ignore

9.2.10.2 TRACE FAILURE INDICATION

This message is sent by the NG-RAN node to indicate that a Trace Start procedure or a Deactivate Trace procedure has failed for a UE.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	As per NG-RAN Trace ID in <i>Trace Activation</i> IE	YES	ignore
Cause	M		9.3.1.2		YES	ignore

9.2.10.3 DEACTIVATE TRACE

This message is sent by the AMF to deactivate a trace session.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	As per NG-RAN Trace ID in <i>Trace Activation</i> IE	YES	ignore

9.2.10.4 CELL TRAFFIC TRACE

This message is sent by the NG-RAN node to transfer trace specific information.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	This IE is composed of the following: Trace Reference defined in TS 32.422 [11] (leftmost 6 octets, with PLMN information encoded as in 9.3.3.5), and Trace Recording Session Reference defined in TS 32.422 [11] (last 2 octets).	YES	ignore
NG-RAN CGI	M		9.3.1.73		YES	ignore
Trace Collection Entity IP Address	M		Transport Layer Address 9.3.2.4	For File based Reporting. Defined in TS 32.422 [11]. This IE is ignored if the <i>Trace Collection Entity URI</i> IE is present	YES	ignore
Privacy Indicator	O		ENUMERATED (Immediate MDT, Logged MDT, ...)		YES	ignore
Trace Collection Entity URI	O		URI 9.3.2.14	For Streaming based Reporting. Defined in TS 32.422 [11].	YES	ignore

9.2.11 Location Reporting Messages

9.2.11.1 LOCATION REPORTING CONTROL

This message is used by the AMF to request the NG-RAN node to report the location of the UE.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Location Reporting Request Type	M		9.3.1.65		YES	ignore

9.2.11.2 LOCATION REPORTING FAILURE INDICATION

This message is sent by the NG-RAN node and is used to indicate the failure of location reporting.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Cause	M		9.3.1.2		YES	ignore
Aerial UE Flight Information Ongoing Reporting Failed	O		ENUMERATED (true, ...)		YES	ignore

9.2.11.3 LOCATION REPORT

This message is used to provide the UE's location.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
UE Presence in Area of Interest List	O		9.3.1.67		YES	ignore
Location Reporting Request Type	M		9.3.1.65	Contains the Location Reporting Request Type to which the Location Report refers.	YES	ignore

9.2.12 UE TNLA Binding Messages

9.2.12.1 UE TNLA BINDING RELEASE REQUEST

This message is sent by the AMF to request the NG-RAN node to release the TNLA binding for the respective UE.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject

9.2.13 UE Radio Capability Management Messages

9.2.13.1 UE RADIO CAPABILITY INFO INDICATION

This message is sent by the NG-RAN node to provide UE radio capability related information to the AMF.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
UE Radio Capability	M		9.3.1.74		YES	ignore
UE Radio Capability for Paging	O		9.3.1.68		YES	ignore
UE Radio Capability – E-UTRA Format	O		9.3.1.74a		YES	ignore
XR Device with 2Rx	O		ENUMERATED (true, ...)	Indicates the UE is a 2Rx XR UE as defined in TS 38.300 [8].	YES	ignore

9.2.13.2 UE RADIO CAPABILITY CHECK REQUEST

This message is sent by the AMF to request the NG-RAN node to check the compatibility between the UE radio capabilities and network configuration on IMS voice.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
UE Radio Capability	O		9.3.1.74		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject

9.2.13.3 UE RADIO CAPABILITY CHECK RESPONSE

This message is sent by the NG-RAN node to report IMS voice compatibility between the UE radio capabilities and network configuration.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
IMS Voice Support Indicator	M		9.3.1.89		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.13.4 UE RADIO CAPABILITY ID MAPPING REQUEST

This message is sent by the NG-RAN node to request the AMF to provide mapping information for the indicated UE Radio Capability ID.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
UE Radio Capability ID	M		9.3.1.142		YES	reject

9.2.13.5 UE RADIO CAPABILITY ID MAPPING RESPONSE

This message is sent by the AMF to provide UE Radio Capability information which is mapped to the UE Radio Capability ID indicated by the NG-RAN node in the UE RADIO CAPABILITY ID MAPPING REQUEST message.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
UE Radio Capability ID	M		9.3.1.142		YES	reject
UE Radio Capability	M		9.3.1.74		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14 Data Usage Reporting Messages

9.2.14.1 SECONDARY RAT DATA USAGE REPORT

This message is sent by the NG-RAN node to report Secondary RAT data usage.

Direction: NG-RAN → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Secondary RAT Usage List		1			YES	ignore
>PDU Session Resource Secondary RAT Usage Item		1..<maxno of PDUSessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Secondary RAT Data Usage Report Transfer	M		OCTET STRING	Containing the <i>Secondary RAT Data Usage Report Transfer</i> IE specified in subclause 9.3.4.23	-	
Handover Flag	O		ENUMERATED (handover_preparation, ...)		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.15 RIM Information Transfer Messages

9.2.15.1 UPLINK RIM INFORMATION TRANSFER

This message is sent by the NG-RAN node to the AMF to transfer the RIM Information.

Direction: NG-RAN → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
RIM Information Transfer	O		9.3.3.28		YES	ignore

9.2.15.2 DOWNLINK RIM INFORMATION TRANSFER

This message is sent by the AMF to the NG-RAN node to transfer the RIM Information.

Direction: AMF → NG-RAN

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
RIM Information Transfer	O		9.3.3.28		YES	ignore

9.2.16 Broadcast Session Management Messages

9.2.16.1 BROADCAST SESSION SETUP REQUEST

This message is sent by the AMF to establish MBS session resources for a broadcast MBS session.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
S-NSSAI	M		9.3.1.24		YES	reject
MBS Service Area	M		9.3.1.208		YES	reject
MBS Session Setup Request Transfer	M		OCTET STRING	Containing the <i>MBS Session Setup or Modification Request Transfer</i> IE specified in subclause 9.3.5.3	YES	reject
Associated Session ID	O		9.3.1.262		YES	ignore

9.2.16.2 BROADCAST SESSION SETUP RESPONSE

This message is sent by the NG-RAN node to report the successful outcome of the request from the BROADCAST SESSION SETUP REQUEST message.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Setup Response Transfer	O		OCTET STRING	Containing the <i>MBS Session Setup or Modification Response Transfer</i> IE specified in subclause 9.3.5.5	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.3 BROADCAST SESSION SETUP FAILURE

This message is sent by the NG-RAN node to report the unsuccessful outcome of the request from the BROADCAST SESSION SETUP REQUEST message.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Setup Failure Transfer	O		OCTET STRING	Containing the <i>MBS Session Setup or Modification Failure Transfer</i> IE specified in subclause 9.3.5.6	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.4 BROADCAST SESSION MODIFICATION REQUEST

This message is sent by the AMF to modify MBS session resources or the MBS session broadcast area of a previously established broadcast MBS session.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Service Area	O		9.3.1.208		YES	reject
MBS Session Modification Request Transfer	O		OCTET STRING	Containing the <i>MBS Session Setup or Modification Request Transfer</i> IE specified in subclause 9.3.5.3	YES	reject

9.2.16.5 BROADCAST SESSION MODIFICATION RESPONSE

This message is sent by the NG-RAN node to report the successful outcome of the request from the BROADCAST SESSION MODIFICATION REQUEST message.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Modification Response Transfer	O		OCTET STRING	Containing the <i>MBS Session Setup or Modification Response Transfer</i> IE specified in subclause 9.3.5.5	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.6 BROADCAST SESSION MODIFICATION FAILURE

This message is sent by the NG-RAN node to report the unsuccessful outcome of the request from the BROADCAST SESSION MODIFICATION REQUEST message.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Modification Failure Transfer	O		OCTET STRING	Containing the <i>MBS Session Setup or Modification Failure Transfer</i> IE specified in subclause 9.3.5.6	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.7 BROADCAST SESSION RELEASE REQUEST

This message is sent by the AMF to release the MBS session resources of a previously established broadcast MBS session.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.16.8 BROADCAST SESSION RELEASE RESPONSE

This message is sent by the NG-RAN node to acknowledge the BROADCAST SESSION RELEASE REQUEST message.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Release Response Transfer	O		OCTET STRING	Containing the <i>MBS Session Release Response Transfer</i> IE specified in subclause 9.3.5.14	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.9 BROADCAST SESSION RELEASE REQUIRED

This message is sent by the NG-RAN node to trigger the AMF to initiate the release of corresponding MBS session resources.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.16.10 BROADCAST SESSION TRANSPORT REQUEST

This message is sent by the NG-RAN node to request the setup of the NG-U transport for a broadcast MBS session.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Broadcast Transport Request Transfer	M		OCTET STRING	Containing the <i>Broadcast Transport Request Transfer</i> IE specified in subclause 9.3.5.15	YES	reject

9.2.16.11 BROADCAST SESSION TRANSPORT RESPONSE

This message is sent by the AMF to confirm the setup of the NG-U transport for the broadcast MBS session.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Broadcast Transport Response Transfer	M		OCTET STRING	Containing the <i>Broadcast Transport Response Transfer</i> IE specified in subclause 9.3.5.16	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.12 BROADCAST SESSION TRANSPORT FAILURE

This message is sent by the AMF to indicate that the setup of the NG-U transport was unsuccessful.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Broadcast Transport Failure Transfer	M		OCTET STRING	Containing the <i>Broadcast Transport Failure Transfer</i> IE specified in subclause 9.3.5.17	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17 Multicast Session Management Messages

9.2.17.1 DISTRIBUTION SETUP REQUEST

This message is sent by the NG-RAN node to request the setup of the NG-U transport for a multicast MBS session, or for one area session of a location dependent multicast MBS session.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
MBS Distribution Setup Request Transfer	M		OCTET STRING	Containing the <i>MBS Distribution Setup Request Transfer</i> IE specified in subclause 9.3.5.7.	YES	reject

9.2.17.2 DISTRIBUTION SETUP RESPONSE

This message is sent by the AMF to confirm the setup of the NG-U transport.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
MBS Distribution Setup Response Transfer	M		OCTET STRING	Containing the <i>MBS Distribution Setup Response Transfer</i> IE specified in subclause 9.3.5.8.	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.3 DISTRIBUTION SETUP FAILURE

This message is sent by the AMF to indicate that the setup of the NG-U transport was unsuccessful.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
MBS Distribution Setup Unsuccessful Transfer	M		OCTET STRING	Containing the <i>MBS Distribution Setup Unsuccessful Transfer</i> IE specified in subclause 9.3.5.9.	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.4 DISTRIBUTION RELEASE REQUEST

This message is sent by the NG-RAN node to request the release of the NG-U transport.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
MBS Distribution Release Request Transfer	M		OCTET STRING	Containing the <i>MBS Distribution Release Request Transfer</i> IE specified in subclause 9.3.5.10.	YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.17.5 DISTRIBUTION RELEASE RESPONSE

This message is sent by the AMF to confirm the release of the NG-U transport.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.6 MULTICAST SESSION ACTIVATION REQUEST

This message is sent by the AMF to a NG-RAN node to request for activating MBS session resources for a multicast MBS session.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Multicast Session Activation Request Transfer	M		OCTET STRING	Containing the <i>Multicast Session Activation Request Transfer</i> IE specified in subclause 9.3.5.11.	YES	reject

9.2.17.7 MULTICAST SESSION ACTIVATION RESPONSE

This message is sent by the NG-RAN node to the AMF to indicate that the MBS session resources have been activated.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.8 MULTICAST SESSION ACTIVATION FAILURE

This message is sent by the NG-RAN node to the AMF to indicate that the requested activation of the MBS session resources has failed.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.9 MULTICAST SESSION DEACTIVATION REQUEST

This message is sent by the AMF to a NG-RAN node to request to deactivate the MBS session resources of a multicast MBS session.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Multicast Session Deactivation Request Transfer	M		OCTET STRING	Containing the <i>Multicast Session Deactivation Request Transfer</i> IE specified in subclause 9.3.5.12.	YES	reject

9.2.17.10 MULTICAST SESSION DEACTIVATION RESPONSE

This message is sent by the NG-RAN node to the AMF to indicate that the MBS session resources have been deactivated.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.11 MULTICAST SESSION UPDATE REQUEST

This message is sent by the AMF to a NG-RAN node to update the MBS session resources information.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
Multicast Session Update Request Transfer	M		OCTET STRING	Containing the <i>Multicast Session Update Request Transfer</i> IE specified in subclause 9.3.5.13.	YES	reject

9.2.17.12 MULTICAST SESSION UPDATE RESPONSE

This message is sent by the NG-RAN node to the AMF to confirm the update of MBS information.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.13 MULTICAST SESSION UPDATE FAILURE

This message is sent by the NG-RAN node to the AMF to indicate multicast session update failure.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.18 Timing Synchronisation Status Reporting Messages

9.2.18.1 TIMING SYNCHRONISATION STATUS REQUEST

This message is sent by the AMF to request the NG-RAN node to start or stop reporting of RAN timing synchronization status information.

Direction: AMF → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Routing ID	M		9.3.3.13	Indicating the TSCTSF NF ID, refer to TS 23.502 [10]	YES	reject
RAN TSS Request Type	M		ENUMERATED (start, stop, ...)		YES	reject

9.2.18.2 TIMING SYNCHRONISATION STATUS RESPONSE

This message is sent by the NG-RAN node to confirm the request to start or stop reporting of RAN timing synchronization status information.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Routing ID	M		9.3.3.13	Indicating the TSCTSF NF ID, refer to TS 23.502 [10]	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.18.3 TIMING SYNCHRONISATION STATUS FAILURE

This message is sent by the NG-RAN node to indicate that reporting of RAN timing synchronisation status information cannot be initiated.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Routing ID	M		9.3.3.13	Indicating the TSCTSF NF ID, refer to TS 23.502 [10]	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.18.4 TIMING SYNCHRONISATION STATUS REPORT

This message is sent by the NG-RAN node to report previously requested RAN timing synchronisation status information.

Direction: NG-RAN node → AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Routing ID	M		9.3.3.13	Indicating the TSCTSF NF ID, refer to TS 23.502 [10]	YES	ignore
RAN Timing Synchronisation Status Information	M		9.3.1.252		YES	ignore
RAN TSS Scope	M		9.3.1.254		YES	ignore

9.2.19 A-IoT Messages

9.2.19.1 INVENTORY REQUEST

This message is sent by the A-IoT CN node to trigger the NG-RAN node to perform A-IoT paging over the A-IoT radio.

Direction: A-IoT CN node → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	reject
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
Inventory Request Transfer	M		OCTET STRING	Containing the <i>Inventory Request Transfer</i> IE specified in subclause 9.3.6.1	YES	reject

9.2.19.2 INVENTORY RESPONSE

This message is sent by the NG-RAN node to report the successful outcome of the request from the INVENTORY REQUEST message.

Direction: NG-RAN node → A-IoT CN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	reject
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
Inventory Response Transfer	M		OCTET STRING	Containing the <i>Inventory Response Transfer</i> IE specified in subclause 9.3.6.2	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.19.3 INVENTORY FAILURE

This message is sent by the NG-RAN node to report the unsuccessful outcome of the request from the INVENTORY REQUEST message.

Direction: NG-RAN node → A-IoT CN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	ignore
A-IoT Correlation Identifier	M		9.3.3.69		YES	ignore
Inventory Failure Transfer	M		OCTET STRING	Containing the <i>Inventory Failure Transfer</i> IE specified in subclause 9.3.6.3	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.19.4 INVENTORY REPORT

This message is sent by the NG-RAN node to provide the inventory report to the A-IoT CN node.

Direction: NG-RAN node → A-IoT CN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	reject
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
Inventory Report Transfer	M		OCTET STRING	Containing the <i>Inventory Report Transfer</i> IE specified in subclause 9.3.6.4	YES	reject

9.2.19.5 COMMAND REQUEST

This message is sent by the A-IoT CN node to the gNB to trigger the command transmission over the A-IoT radio.

Direction: A-IoT CN node → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	reject
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
RAN A-IoT Device NGAP ID	M		9.3.3.74		YES	reject
Command Request Transfer	M		OCTET STRING	Containing the <i>Command Request Transfer</i> IE specified in subclause 9.3.6.5.	YES	reject

9.2.19.6 COMMAND RESPONSE

This message is sent by the NG-RAN node to provide the command response to the A-IoT CN node.

Direction: NG-RAN node → A-IoT CN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	reject
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
RAN A-IoT Device NGAP ID	M		9.3.3.74		YES	reject
Command Response Transfer	M		OCTET STRING	Containing the <i>Command Response Transfer</i> IE specified in subclause 9.3.6.6.	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.19.7 COMMAND FAILURE

This message is sent by the NG-RAN node to report the unsuccessful transmission of the command to the A-IoT CN node.

Direction: NG-RAN node → A-IoT CN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	ignore
A-IoT Correlation Identifier	M		9.3.3.69		YES	ignore
RAN A-IoT Device NGAP ID	M		9.3.3.74		YES	ignore
Command Failure Transfer	M		OCTET STRING	Containing the <i>Command Failure Transfer</i> IE specified in subclause 9.3.6.7.	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.19.8 AIOT SESSION RELEASE COMMAND

This message is sent by the A-IoT CN node to the gNB to trigger the release of the A-IoT session.

Direction: A-IoT CN node → NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	reject
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
AIOT Session Release Command Transfer	M		OCTET STRING	Containing the <i>AIOT Session Release Command Transfer</i> IE specified in subclause 9.3.6.8.	YES	reject

9.2.19.9 AIOT SESSION RELEASE COMPLETE

This message is sent by the NG-RAN node to respond to the A-IoT session release command to the A-IoT CN node.

Direction: NG-RAN node → A-IoT CN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	ignore
A-IoT Correlation Identifier	M		9.3.3.69		YES	ignore
AIOT Session Release Complete Transfer	M		OCTET STRING	Containing the <i>AIOT Session Release Complete Transfer</i> IE specified in subclause 9.3.6.9.	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.19.10 AIOT SESSION RELEASE REQUEST

This message is sent by the gNB to request the A-IoT CN node to trigger the release of the A-IoT session.

Direction: NG-RAN node → A-IoT CN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AIOTF Identifier	M		9.3.3.70		YES	reject
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
AIOT Session Release Request Transfer	M		OCTET STRING	Containing the <i>AIOT Session Release Request Transfer</i> IE specified in subclause 9.3.6.10.	YES	reject

9.3 Information Element Definitions

9.3.1 Radio Network Layer Related IEs

9.3.1.1 Message Type

The *Message Type* IE uniquely identifies the message being sent. It is mandatory for all messages.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Procedure Code	M		INTEGER (0..255)	
Type of Message	M		CHOICE (Initiating Message, Successful Outcome, Unsuccessful Outcome, ...)	

9.3.1.2 Cause

The purpose of the *Cause* IE is to indicate the reason for a particular event for the NGAP protocol.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE Cause Group	M			
>Radio Network Layer				

>>Radio Network Layer Cause	M		ENUMERATED (Unspecified, TXnRELOCoverall expiry, Successful handover, Release due to NG-RAN generated reason, Release due to 5GC generated reason, Handover cancelled, Partial handover, Handover failure in target 5GC/NG-RAN node or target system, Handover target not allowed, TNGRELOCoverall expiry, TNGRELOCprep expiry, Cell not available, Unknown target ID, No radio resources available in target cell, Unknown local UE NGAP ID, Inconsistent remote UE NGAP ID, Handover desirable for radio reasons, Time critical handover, Resource optimisation handover, Reduce load in serving cell, User inactivity, Radio connection with UE lost, Radio resources not available, Invalid QoS combination, Failure in the radio interface procedure, Interaction with other procedure, Unknown PDU Session ID, Unknown QoS Flow ID, Multiple PDU Session ID Instances, Multiple QoS Flow ID Instances, Encryption and/or integrity protection algorithms not supported, NG intra-system handover triggered, NG inter-system handover triggered, Xn handover triggered, Not supported 5QI value, UE context transfer, IMS voice EPS fallback or RAT fallback triggered, UP integrity protection not possible, UP confidentiality protection not possible, Slice(s) not supported, UE in RRC_INACTIVE state not reachable, Redirection, Resources not available for the slice(s), UE maximum integrity protected data rate reason, Release due to CN-detected mobility, ..., N26 interface not available, Release due to pre-emption, Multiple Location Reporting Reference ID Instances, RSN not available for the UP, NPN access denied,	
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			CAG only access denied, Insufficient UE Capabilities, RedCap UE not supported, Unknown MBS Session ID, Indicated MBS Session Area Information not served by the gNB, Inconsistent slice info for the session, Misaligned association for the multicast and unicast sessions or flows, eRedCap UE not supported, 2Rx XR UE not supported, Aerial UE flight information reporting initiation failure, Unknown RAN A- IoT Device NGAP ID, Requested A- IoT service area information not served by the gNB, Unknown A-IoT Session, A-IoT device not reachable, Multiple A-IoT Session Instances)	
>Transport Layer				
>>Transport Layer Cause	M		ENUMERATED (Transport resource unavailable, Unspecified, ...)	
>NAS				
>>NAS Cause	M		ENUMERATED (Normal release, Authentication failure, Deregister, Unspecified, ..., UE not in PLMN serving area, Mobile IAB not authorized, IAB not authorized, No A-IoT NAS response)	
>Protocol				
>>Protocol Cause	M		ENUMERATED (Transfer syntax error, Abstract syntax error (reject), Abstract syntax error (ignore and notify), Message not compatible with receiver state, Semantic error, Abstract syntax error (falsely constructed message), Unspecified, ...)	
>Miscellaneous				
>>Miscellaneous Cause	M		ENUMERATED (Control processing overload, Not enough user plane processing resources, Hardware failure, O&M intervention, Unknown PLMN or SNPN, Unspecified, ...)	

The meaning of the different cause values is described in the following tables. In general, "not supported" cause values indicate that the related capability is missing. On the other hand, "not available" cause values indicate that the related capability is present, but insufficient resources were available to perform the requested action.

Radio Network Layer cause	Meaning
Unspecified	Sent for radio network layer cause when none of the specified cause values applies.
TXnRELOCOverall expiry	The timer guarding the handover that takes place over Xn has abnormally expired.
Successful handover	Successful handover.
Release due to NG-RAN generated reason	Release is initiated due to NG-RAN generated reason.
Release due to 5GC generated reason	Release is initiated due to 5GC generated reason.
Handover cancelled	The reason for the action is cancellation of Handover.
Partial handover	Provides a reason for the handover cancellation. The HANDOVER COMMAND message from AMF contained <i>PDU Session Resource to Release List</i> IE or <i>QoS flow to Release List</i> and the source NG-RAN node estimated service continuity for the UE would be better by not proceeding with handover towards this particular target NG-RAN node.
Handover failure in target 5GC/ NG-RAN node or target system	The handover failed due to a failure in target 5GC/NG-RAN node or target system.
Handover target not allowed	Handover to the indicated target cell is not allowed for the UE in question.
TNGRELOCOverall expiry	The reason for the action is expiry of timer TNGRELOCOverall.
TNGRELOCprep expiry	Handover Preparation procedure is cancelled when timer TNGRELOCprep expires.
Cell not available	The concerned cell is not available.
Unknown target ID	Handover rejected because the target ID is not known to the AMF.
No radio resources available in target cell	Load on target cell is too high.
Unknown local UE NGAP ID	The action failed because the receiving node does not recognise the local UE NGAP ID.
Inconsistent remote UE NGAP ID	The action failed because the receiving node considers that the received remote UE NGAP ID is inconsistent.
Handover desirable for radio reasons	The reason for requesting handover is radio related.
Time critical handover	Handover is requested for time critical reason i.e., this cause value is reserved to represent all critical cases where the connection is likely to be dropped if handover is not performed.
Resource optimisation handover	The reason for requesting handover is to improve the load distribution with the neighbour cells.
Reduce load in serving cell	Load on serving cell needs to be reduced. When applied to handover preparation, it indicates the handover is triggered due to load balancing.
User inactivity	The action is requested due to inactivity on all user data radio bearers (i.e., DRBs and, if applicable, MRBs as per section 16.10.5.2 in TS 38.300 [8]), e.g., NG is requested to be released in order to optimise the radio resources. For L2 U2N Relay UE, this action is requested due to user inactivity on all PDU sessions of L2 U2N Relay UE and its served remote UE(s).
Radio connection with UE lost	The action is requested due to losing the radio connection to the UE.
Radio resources not available	No requested radio resources are available.
Invalid QoS combination	The action was failed because of invalid QoS combination.
Failure in the radio interface procedure	Radio interface procedure has failed.
Interaction with other procedure	The action is due to an ongoing interaction with another procedure.
Unknown PDU Session ID	The action failed because the PDU Session ID is unknown in the NG-RAN node.
Unknown QoS Flow ID	The action failed because the QoS Flow ID is unknown in the NG-RAN node.
Multiple PDU Session ID instances	The action failed because multiple instances of the same PDU Session had been provided to/from the NG-RAN node.
Multiple QoS Flow ID instances	The action failed because multiple instances of the same QoS flow had been provided to the NG-RAN node.
Encryption and/or integrity protection algorithms not supported	The NG-RAN node is unable to support any of the encryption and/or integrity protection algorithms supported by the UE.
NG intra-system handover triggered	The action is due to a NG intra-system handover that has been triggered.
NG inter-system handover triggered	The action is due to a NG inter-system handover that has been triggered.
Xn handover triggered	The action is due to an Xn handover that has been triggered.
Not supported 5QI value	The QoS flow setup failed because the requested 5QI is not supported.
UE context transfer	The action is due to a UE resumes from the NG-RAN node different from the one which sent the UE into RRC INACTIVE state.
IMS voice EPS fallback or RAT fallback triggered	The setup of QoS flow is failed due to EPS fallback or RAT fallback for IMS voice using handover or redirection.
UP integrity protection not possible	The PDU session cannot be accepted according to the required user plane integrity protection policy.

Radio Network Layer cause	Meaning
UP confidentiality protection not possible	The PDU session cannot be accepted according to the required user plane confidentiality protection policy.
Slice(s) not supported	Slice(s) not supported.
UE in RRC_INACTIVE state not reachable	The action is requested due to RAN paging failure.
Redirection	The release is requested due to inter-system redirection or intra-system redirection.
Resources not available for the slice(s)	The requested resources are not available for the slice(s).
UE maximum integrity protected data rate reason	The request is not accepted in order to comply with the maximum data rate for integrity protection supported by the UE.
Release due to CN-detected mobility	The context release is requested by the AMF because the UE is already served by another CN node (same or different system), or another NG interface of the same CN node.
N26 interface not available	The action failed due to a temporary failure of the N26 interface.
Release due to pre-emption	Release is initiated due to pre-emption.
Multiple Location Reporting Reference ID Instances	The action failed because multiple areas of interest are set with the same Location Reporting Reference ID.
RSN not available for the UP	The redundant user plane resources indicated by RSN are not available.
NPN access denied	Access was denied, or release is requested, for NPN reasons.
CAG only access denied	Access was denied because the cell is a non-CAG cell and UE is only allowed to access CAG cells.
Insufficient UE Capabilities	The procedure can't proceed due to insufficient UE capabilities.
RedCap UE not supported	The action failed because target NG-RAN node does not support RedCap UE.
Unknown MBS Session ID	The action failed because the MBS Session ID is unknown.
Indicated MBS Service Area Information not served by the gNB	The action failed because none of the cells in the indicated MBS Service Area Information are served by the NG-RAN node.
Inconsistent slice info for the session	The action failed because the slice info of the multicast session is inconsistent.
Misaligned association for the multicast and unicast sessions or flows	The action failed because the Associated Unicast QoS Flow ID has already been used, or the Associated Unicast QoS Flow ID is not defined, or the Associated Unicast QoS Flow ID is not released, or multiple MBS QoS flows associated to the same unicast QoS flow, or same multicast session associated to multiple PDU Sessions.
eRedCap UE not supported	The action failed because target NG-RAN node does not support eRedCap UE.
2Rx XR UE not supported	The action failed because target NG-RAN node does not support 2Rx XR UE.
Aerial UE flight information reporting initiation failure	The action failed because gNB cannot initiate the aerial UE flight information reporting.
Unknown RAN A-IoT Device NGAP ID	The action failed because the receiving node does not recognise the RAN A-IoT Device NGAP ID for the involved A-IoT session.
Requested A-IoT service area information not served by the gNB	The action failed because neither the A-IoT Area(s) nor the reader(s) indicated in the Requested Service Area Information are served by the NG-RAN node.
Unknown A-IoT Session	The action failed because the A-IoT Session (identified by the combination of AIOTF Identifier and A-IoT Correlation Identifier) is unknown.
A-IoT device not reachable	The action failed because the A-IoT device is not reachable.
Multiple A-IoT Session instances	The action failed because multiple concurrent instances of the same A-IoT Session (identified by identical combinations of an AIOTF identifier and an A-IoT Correlation Identifier) have been provided to the NG-RAN node.

Transport Layer cause	Meaning
Transport resource unavailable	The required transport resources are not available.
Unspecified	Sent when none of the specified cause values applies but still the cause is Transport Network Layer related.

NAS cause	Meaning
Normal release	The release is normal.
Authentication failure	The action is due to authentication failure.
Deregister	The action is due to deregister.
Unspecified	Sent when none of the specified cause values applies but still the cause is NAS related.
UE not in PLMN serving area	The release is due to the UE not being within the serving area of its current PLMN (for NTN).
Mobile IAB not authorized	The release is due to the NG-RAN node having completed the operation for a non-authorized mobile IAB-node.
IAB not authorized	The action is requested due to the NG-RAN node having completed the operation for a non-authorized IAB-node.
No A-IoT NAS response	The action is due to no A-IoT NAS response received from the A-IoT device.

Protocol cause	Meaning
Transfer syntax error	The received message included a transfer syntax error.
Abstract syntax error (reject)	The received message included an abstract syntax error and the concerning criticality indicated "reject".
Abstract syntax error (ignore and notify)	The received message included an abstract syntax error and the concerning criticality indicated "ignore and notify".
Message not compatible with receiver state	The received message was not compatible with the receiver state.
Semantic error	The received message included a semantic error.
Abstract syntax error (falsely constructed message)	The received message contained IEs or IE groups in wrong order or with too many occurrences.
Unspecified	Sent when none of the specified cause values applies but still the cause is Protocol related.

Miscellaneous cause	Meaning
Control processing overload	Control processing overload.
Not enough user plane processing resources	Not enough resources are available related to user plane processing.
Hardware failure	Action related to hardware failure.
O&M intervention	The action is due to O&M intervention.
Unknown PLMN or SNPN	The AMF does not identify any PLMN or SNPN provided by the NG-RAN node.
Unspecified failure	Sent when none of the specified cause values applies and the cause is not related to any of the categories Radio Network Layer, Transport Network Layer, NAS or Protocol.

9.3.1.3 Criticality Diagnostics

The *Criticality Diagnostics* IE is sent by the NG-RAN node or the AMF when parts of a received message have not been comprehended or were missing, or if the message contained logical errors. When applicable, it contains information about which IEs were not comprehended or were missing.

For further details on how to use the *Criticality Diagnostics* IE, see clause 10.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Procedure Code	O		INTEGER (0..255)	Used if Criticality Diagnostics is part of Error Indication procedure, and not within the response message of the same procedure that caused the error.
Triggering Message	O		ENUMERATED (initiating message, successful outcome, unsuccessful outcome)	Used only if the Criticality Diagnostics is part of Error Indication procedure.
Procedure Criticality	O		ENUMERATED (reject, ignore, notify)	Used for reporting the Criticality of the Triggering message (Procedure).
Information Element Criticality Diagnostics		<i>0..<maxnoofErrors></i>		
>IE Criticality	M		ENUMERATED (reject, ignore, notify)	Used for reporting the criticality of the triggering IE. The value 'ignore' is not applicable.
>IE ID	M		INTEGER (0..65535)	The IE ID of the not understood or missing IE.
>Type of Error	M		ENUMERATED (not understood, missing, ...)	

Range bound	Explanation
maxnoofErrors	Maximum no. of IE errors allowed to be reported with a single message. Value is 256.

9.3.1.4 Bit Rate

This IE indicates the number of bits delivered by NG-RAN in UL or to NG-RAN in DL or by UE in sidelink within a period of time, divided by the duration of the period. It is used, for example, to indicate the maximum or guaranteed bit rate for a GBR QoS flow, or an aggregate maximum bit rate.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Bit Rate	M		INTEGER (0..4,000,000,000,000, ...)	The unit is: bit/s

9.3.1.5 Global RAN Node ID

This IE is used to globally identify an NG-RAN node (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>NG-RAN node</i>	M				-	
> <i>gNB</i>						
>>Global <i>gNB</i> ID	M		9.3.1.6		-	
> <i>ng-eNB</i>						
>>Global <i>ng-eNB</i> ID	M		9.3.1.8		-	
> <i>N3IWF</i>						
>>Global <i>N3IWF</i> ID	M		9.3.1.57		-	
> <i>TNGF</i>						
>>Global <i>TNGF</i> ID	M		9.3.1.161		YES	reject
> <i>TWIF</i>						
>>Global <i>TWIF</i> ID	M		9.3.1.163		YES	reject
> <i>W-AGF</i>						
>>Global <i>W-AGF</i> ID	M		9.3.1.162		YES	reject

9.3.1.6 Global gNB ID

This IE is used to globally identify a gNB (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE <i>gNB ID</i>	M			
> <i>gNB ID</i>				
>>gNB ID	M		BIT STRING (SIZE(22..32))	Equal to the leftmost bits of the <i>NR Cell Identity</i> IE contained in the <i>NR CGI</i> IE of each cell served by the gNB.

9.3.1.7 NR CGI

This IE is used to globally identify an NR cell (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
NR Cell Identity	M		BIT STRING (SIZE(36))	The leftmost bits of the <i>NR Cell Identity</i> IE correspond to the gNB ID (defined in subclause 9.3.1.6).

9.3.1.8 Global ng-eNB ID

This IE is used to globally identify an ng-eNB (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE <i>ng-eNB ID</i>	M			
> <i>Macro ng-eNB ID</i>				
>>Macro ng-eNB ID	M		BIT STRING (SIZE(20))	Equal to the 20 leftmost bits of the <i>E-UTRA Cell Identity</i> IE contained in the <i>E-UTRA CGI</i> IE of each cell served by the ng-eNB.
> <i>Short Macro ng-eNB ID</i>				
>>Short Macro ng-eNB ID	M		BIT STRING (SIZE(18))	Equal to the 18 leftmost bits of the <i>E-UTRA Cell Identity</i> IE contained in the <i>E-UTRA CGI</i> IE of each cell served by the ng-eNB.
> <i>Long Macro ng-eNB ID</i>				
>>Long Macro ng-eNB ID	M		BIT STRING (SIZE(21))	Equal to the 21 leftmost bits of the <i>E-UTRA Cell Identity</i> IE contained in the <i>E-UTRA CGI</i> IE of each cell served by the ng-eNB.

9.3.1.9 E-UTRA CGI

This IE is used to globally identify an E-UTRA cell (see TS 36.300 [17]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
E-UTRA Cell Identity	M		BIT STRING (SIZE(28))	The leftmost bits of the <i>E-UTRA Cell Identity</i> IE correspond to the ng-eNB ID (defined in subclause 9.3.1.8) and to the eNB ID (defined in subclause 9.3.1.165).

9.3.1.10 GBR QoS Flow Information

This IE indicates QoS parameters for a GBR QoS flow for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Maximum Flow Bit Rate Downlink	M		Bit Rate 9.3.1.4	Maximum Bit Rate in DL. Details in TS 23.501 [9].	-	
Maximum Flow Bit Rate Uplink	M		Bit Rate 9.3.1.4	Maximum Bit Rate in UL. Details in TS 23.501 [9].	-	
Guaranteed Flow Bit Rate Downlink	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate (provided there is data to deliver) in DL. Details in TS 23.501 [9].	-	
Guaranteed Flow Bit Rate Uplink	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate (provided there is data to deliver) in UL. Details in TS 23.501 [9].	-	
Notification Control	O		ENUMERATED (notification requested, ...)	Details in TS 23.501 [9].	-	
Maximum Packet Loss Rate Downlink	O		Packet Loss Rate 9.3.1.79	Indicates the maximum rate for lost packets that can be tolerated in the downlink direction. Details in TS 23.501 [9].	-	
Maximum Packet Loss Rate Uplink	O		Packet Loss Rate 9.3.1.79	Indicates the maximum rate for lost packets that can be tolerated in the uplink direction. Details in TS 23.501 [9].	-	
Alternative QoS Parameters Set List	O		9.3.1.151	Indicates alternative sets of QoS parameters for the QoS flow.	YES	ignore
Monitoring Request on Available Bitrate		0..1			YES	ignore
>Monitoring Request	M		ENUMERATED (ul, dl, both, stop, ...)	Indicates to monitor and report UL, or DL, or both UL/DL available bit rate for the associated QoS flow as specified in TS 23.501 [9], or stop the corresponding QoS monitoring.	-	
>DL Available Bitrate Report Thresholds	C- ifReportD L		Available Bitrate Report Threshold List 9.3.1.281		-	
>UL Available Bitrate Report Thresholds	C- ifReportU L		Available Bitrate Report Threshold List 9.3.1.281		-	

Condition	Explanation
ifReportDL	This IE shall be present if the <i>Monitoring Request</i> IE is set to the value “dl” or “both”.
ifReportUL	This IE shall be present if the <i>Monitoring Request</i> IE is set to the value “ul” or “both”.

9.3.1.11 Void

9.3.1.12 QoS Flow Level QoS Parameters

This IE defines the QoS parameters to be applied to a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE QoS Characteristics	M				-	
>Non-dynamic 5QI						
>>Non Dynamic 5QI Descriptor	M		9.3.1.28		-	
>Dynamic 5QI						
>>Dynamic 5QI Descriptor	M		9.3.1.18		-	
Allocation and Retention Priority	M		9.3.1.19		-	
GBR QoS Flow Information	O		9.3.1.10	This IE shall be present for GBR QoS flows and is ignored otherwise.	-	
Reflective QoS Attribute	O		ENUMERATED (subject to, ...)	Details in TS 23.501 [9]. This IE may be present in case of Non-GBR QoS flows and is ignored otherwise.	-	
Additional QoS Flow Information	O		ENUMERATED (more likely, ...)	This IE indicates that traffic for this QoS flow is likely to appear more often than traffic for other flows established for the PDU session. This IE may be present in case of Non-GBR QoS flows and is ignored otherwise.	-	
QoS Monitoring Request	O		ENUMERATED (UL, DL, Both, ..., stop)	Indicates to measure UL, or DL, or both UL/DL delays for the associated QoS flow or stop the corresponding QoS monitoring.	YES	ignore
QoS Monitoring Reporting Frequency	O		INTEGER (1..1800, ...)	Indicates the reporting frequency for RAN part delay for QoS monitoring. Units: second	YES	ignore
PDU Set QoS Parameters		0..1			YES	ignore
>UL PDU Set QoS Information	O		PDU Set QoS Information 9.3.1.264		-	
>DL PDU Set QoS Information	O		PDU Set QoS Information 9.3.1.264		-	
DL PDU Set Information Marking Support Indication	O		ENUMERATED (true, ...)		YES	ignore

MMSID	O		OCTET STRING (SIZE(1))	Multi-modal service ID from the application, used to indicate QoS flows are related to a multi-modal service, as specified in TS 23.501 [9] and TS 38.300 [8].	YES	ignore
Indication of Bitrate Adaptation	O		ENUMERATED (uplink, ...)	Indicates that the QoS Flow allows rate adaptation in the indicated direction.	YES	ignore

9.3.1.13 QoS Flow List with Cause

This IE contains a list of QoS flows with a cause value. It is used for example to indicate failed QoS flow(s) or QoS flow(s) to be released.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Item		<i>1..<maxnoofQoSFlows></i>		
>QoS Flow Identifier	M		9.3.1.51	
>Cause	M		9.3.1.2	

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.1.14 Trace Activation

This IE defines parameters related to a trace session activation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	This IE is composed of the following: Trace Reference defined in TS 32.422 [11] (leftmost 6 octets, with PLMN information encoded as in 9.3.3.5), and Trace Recording Session Reference defined in TS 32.422 [11] (last 2 octets).	-	
Interfaces to Trace	M		BIT STRING (SIZE(8))	Each position in the bitmap represents an NG-RAN node interface: first bit = NG-C, second bit = Xn-C, third bit = Uu, fourth bit = F1-C, fifth bit = E1: other bits reserved for future use. Value '1' indicates 'should be traced'. Value '0' indicates 'should not be traced'.	-	
Trace Depth	M		ENUMERATED (minimum, medium, maximum, minimumWithoutVendorSpecificExtension, mediumWithoutVendorSpecificExtension, maximumWithoutVendorSpecificExtension, ..., minimumOnlyVendorSpecificTraceRecord, mediumOnlyVendorSpecificTraceRecord, maximumOnlyVendorSpecificTraceRecord)	Defined in TS 32.422 [11].	-	
Trace Collection Entity IP Address	M		Transport Layer Address 9.3.2.4	For File based Reporting. Defined in TS 32.422 [11]. This IE is ignored if the <i>Trace Collection Entity URI</i> IE is present.	-	
MDT Configuration	O		9.3.1.167		YES	ignore

Trace Collection Entity URI	O		URI 9.3.2.14	For Streaming based Reporting. Defined in TS 32.422 [11].	YES	ignore
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9.3.1.15 Core Network Assistance Information for RRC INACTIVE

This IE provides assistance information for RRC_INACTIVE configuration.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UE Identity Index Value	M		9.3.3.23		-	
UE Specific DRX	O		Paging DRX 9.3.1.90		-	
Periodic Registration Update Timer	M		9.3.3.24		-	
MICO Mode Indication	O		9.3.1.23		-	
TAI List for RRC Inactive		1			-	
>TAI List for RRC Inactive Item		1..<maxno ofTAIforInactive>			-	
>>TAI	M		9.3.3.11		-	
Expected UE Behaviour	O		9.3.1.93		-	
E-UTRA Paging eDRX Information	O		9.3.1.154		YES	ignore
Extended UE Identity Index Value	O		9.3.3.52	This IE is ignored if the <i>Further Extended UE Identity Index Value</i> IE is present.	YES	ignore
UE Radio Capability for Paging	O		9.3.1.68		YES	ignore
MICO All PLMN	O		9.3.1.194		YES	ignore
NR Paging eDRX Information	O		9.3.1.227		YES	ignore
Paging Cause Indication for Voice Service	O		ENUMERATED (supported, ...)	This IE indicates that the UE supports the feature of indication of paging cause for voice service.	YES	ignore
PEIPS Assistance Information	O		9.3.1.232		YES	ignore
Hashed UE Identity Index Value	O		9.3.3.62		YES	ignore
CN MT Communication Handling	O		ENUMERATED (supported, ...)	This IE indicates the CN support of CN based MT communication handling.	YES	ignore
LP-WUSPS Assistance Information	O		9.3.1.279		YES	ignore
Further Extended UE Identity Index Value	O		9.3.3.77		YES	ignore
LP-WUS Disable Indication	O		ENUMERATED (true, ...)		YES	ignore

Range bound	Explanation
maxnoofTAIforInactive	Maximum no. of TAIs for RRC Inactive. Value is 16.

9.3.1.16 User Location Information

This IE is used to provide location information of the UE and the flight information of the Aerial UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>User Location Information</i>	M				-	
>E-UTRA <i>user location information</i>						
>>E-UTRA CGI	M		9.3.1.9		-	
>>TAI	M		9.3.3.11		-	
>>Age of Location	O		Time Stamp 9.3.1.75	Indicates the UTC time when the location information was generated.	-	
>>PSCell Information	O		NG-RAN CGI 9.3.1.73		YES	ignore
>NR <i>user location information</i>						
>>NR CGI	M		9.3.1.7		-	
>>TAI	M		9.3.3.11	This IE is ignored if the NR NTN TAI Information IE is present.	-	
>>Age of Location	O		Time Stamp 9.3.1.75	Indicates the UTC time when the location information was generated.	-	
>>PSCell Information	O		NG-RAN CGI 9.3.1.73		YES	ignore
>>NID	O		9.3.3.42		YES	reject
>>NR NTN TAI Information	O		9.3.3.53		YES	ignore
>>Mobile IAB-MT User Location Information	O		9.3.1.260	Indicates the user location information of a mobile IAB-MT, which is co-located with the mobile IAB-DU which serves the UE.	YES	ignore
>>Additional ULI	O		9.3.1.280	Indicates the Additional User Location Information for a UE served by a WAB-gNB.	YES	ignore
>>Aerial UE Flight information Reporting	O		9.3.1.282		YES	ignore
>N3IWF <i>user location information with port number</i>						
>>IP Address	M		Transport Layer Address 9.3.2.4	UE's local IP address used to reach the N3IWF	-	
>>Port Number	M		OCTET STRING (SIZE(2))	UDP source port number if NAT is detected.	-	
>>TAI	O		9.3.3.11		YES	ignore
>TNGF <i>user location information</i>					YES	ignore
>>TNAP ID	M		OCTET STRING	TNAP Identifier used to identify the TNAP. Details in TS 29.571 [35].	-	
>>IP Address	M		Transport Layer Address 9.3.2.4	UE's local IP address used to reach the TNGF.	-	

>>Port Number	O		OCTET STRING (SIZE(2))	UDP source port number if NAT is detected.	-	
>>TAI	O		9.3.3.11		YES	ignore
>TWIF user location information					YES	ignore
>>TWAP ID	M		OCTET STRING	TWAP Identifier used to identify the TWAP. Details in TS 29.571 [35].	-	
>>IP Address	M		Transport Layer Address 9.3.2.4	Non-5G-Capable over WLAN device's local IP address used to reach the TWIF.	-	
>>Port Number	O		OCTET STRING (SIZE(2))	UDP source port number if NAT is detected.	-	
>>TAI	O		9.3.3.11		YES	ignore
>W-AGF user location information				Indicates the location information via wireline access as specified in TS 23.316 [34].	YES	ignore
>>W-AGF user location information	M		9.3.1.164		-	
>N3IWF user location information without port number					YES	ignore
>>IP Address	M		Transport Layer Address 9.3.2.4	UE's local IP address used to reach the N3IWF.	-	
>>TAI	O		9.3.3.11		-	

9.3.1.17 Slice Support List

This IE indicates the list of supported slices.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Support Item		<i>1..<maxnoofSliceltems></i>		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofSliceltems	Maximum no. of signalled slice support items. Value is 1024.

9.3.1.18 Dynamic 5QI Descriptor

This IE indicates the QoS Characteristics for a Non-standardised or not pre-configured 5QI for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Priority Level	M		9.3.1.84	Priority Level is specified in TS 23.501 [9].	-	
Packet Delay Budget	M		9.3.1.80	Packet Delay Budget is specified in TS 23.501 [9]. This IE is ignored if the <i>Extended Packet Delay Budget</i> IE is present.	-	
Packet Error Rate	M		9.3.1.81	Packet Error Rate is specified in TS 23.501 [9].	-	
5QI	O		INTEGER (0..255, ...)	Indicates the dynamically assigned 5QI as specified in TS 23.501 [9].	-	
Delay Critical	C-ifGBRflow		ENUMERATED (delay critical, non-delay critical, ...)	Indicates whether the GBR QoS flow is delay critical as specified in TS 23.501 [9].	-	
Averaging Window	C-ifGBRflow		9.3.1.82	Averaging Window is specified in TS 23.501 [9].	-	
Maximum Data Burst Volume	O		9.3.1.83	Maximum Data Burst Volume is specified in TS 23.501 [9]. This IE shall be included if the <i>Delay Critical</i> IE is set to "delay critical" and is ignored otherwise.	-	
Extended Packet Delay Budget	O		9.3.1.135	Packet Delay Budget is specified in TS 23.501 [9].	YES	ignore
CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore

Condition	Explanation
ifGBRflow	This IE shall be present if the <i>GBR QoS Flow Information</i> IE is present in the <i>QoS Flow Level QoS Parameters</i> IE.

9.3.1.19 Allocation and Retention Priority

This IE specifies the relative importance of a QoS flow compared to other QoS flows for allocation and retention of NG-RAN resources.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Priority Level	M		INTEGER (1..15)	Desc.: This IE defines the relative importance of a resource request (see TS 23.501 [9]). Usage: Values are ordered in decreasing order of priority, i.e., with 1 as the highest priority and 15 as the lowest priority.
Pre-emption Capability	M		ENUMERATED (shall not trigger pre-emption, may trigger pre-emption, ...)	Desc.: This IE indicates the pre-emption capability of the request on other QoS flows (see TS 23.501 [9]). Usage: The QoS flow shall not pre-empt other QoS flows or, the QoS flow may pre-empt other QoS flows. Note: The Pre-emption Capability indicator applies to the allocation of resources for a QoS flow and as such it provides the trigger to the pre-emption procedures/processes of the NG-RAN node.
Pre-emption Vulnerability	M		ENUMERATED (not pre-emptable, pre-emptable, ...)	Desc.: This IE indicates the vulnerability of the QoS flow to pre-emption of other QoS flows (see TS 23.501 [9]). Usage: The QoS flow shall not be pre-empted by other QoS flows or the QoS flow may be pre-empted by other QoS flows. Note: The Pre-emption Vulnerability indicator applies for the entire duration of the QoS flow, unless modified and as such indicates whether the QoS flow is a target of the pre-emption procedures/processes of the NG-RAN node.

9.3.1.20 Source to Target Transparent Container

This IE is used to transparently pass radio related information from the handover source to the handover target through the core network; it is produced by the source RAN node and is transmitted to the target RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Source to Target Transparent Container	M		OCTET STRING	This IE includes a transparent container from the source RAN node to the target RAN node. The octets of the OCTET STRING are encoded according to the specifications of the target system. Note: In the current version of the specification, this IE may carry either the <i>Source NG-RAN Node to Target NG-RAN Node Transparent Container</i> IE or the <i>Source eNB to Target eNB Transparent Container</i> IE as defined in TS 36.413 [16] or the <i>Source RNC to Target RNC Transparent Container</i> IE as defined in TS 25.413 [28].

9.3.1.21 Target to Source Transparent Container

This IE is used to transparently pass radio related information from the handover target to the handover source through the core network; it is produced by the target RAN node and is transmitted to the source RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target to Source Transparent Container	M		OCTET STRING	This IE includes a transparent container from the target RAN node to the source RAN node. The octets of the OCTET STRING are encoded according to the specifications of the target system. Note: In the current version of the specification, this IE may carry either the <i>Target NG-RAN Node to Source NG-RAN Node Transparent Container</i> IE or the <i>Target eNB to Source eNB Transparent Container</i> IE as defined in TS 36.413 [16], or the <i>Target RNC to Source RNC Transparent Container</i> IE as defined in TS 25.413 [28].

9.3.1.22 Handover Type

This IE indicates which kind of handover was triggered in the source side.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Handover Type	M		ENUMERATED (Intra5GS, 5GStoEPS, EPSto5GS, ..., 5GStoUTRA)	Intra5GS: NG-RAN node to NG-RAN node 5GStoEPS: NG-RAN node to eNB EPSto5GS: eNB to NG-RAN node 5GStoUTRA: NG-RAN node to UTRA

9.3.1.23 MICO Mode Indication

This IE indicates that the UE is configured with MICO mode by the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MICO Mode Indication	M		ENUMERATED (true, ...)	

9.3.1.24 S-NSSAI

This IE indicates the S-NSSAI as defined in TS 23.003 [23].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SST	M		OCTET STRING (SIZE(1))	
SD	O		OCTET STRING (SIZE(3))	

9.3.1.25 Target ID

This IE identifies the target for the handover.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>Target ID</i>	M				-	
> <i>NG-RAN</i>						
>>Global RAN Node ID	M		9.3.1.5		-	
>>Selected TAI	M		TAI 9.3.3.11		-	
>>Selected Target SNPN Identity	O		NID 9.3.3.42	This IE together with the <i>PLMN Identity</i> IE included in the <i>Selected TAI</i> IE indicates the target SNPN.	YES	reject
> <i>E-UTRAN</i>						
>>Global eNB ID	M		Global ng-eNB ID 9.3.1.8		-	
>>Selected EPS TAI	M		EPS TAI 9.3.3.17		-	
> <i>Target RNC-ID</i>					YES	reject
>>LAI	M		9.3.3.30		-	
>>RNC-ID	M		9.3.1.123	This IE is ignored if the <i>Extended RNC-ID</i> IE is included in the <i>Target ID</i> IE.	-	
>>Extended RNC-ID	O		9.3.1.124	The <i>Extended RNC-ID</i> IE is used if the RNC identity has a value larger than 4095.	-	
> <i>Target Home eNB ID</i>					YES	reject
>>PLMN Identity	M		9.3.3.5		-	
>>Home eNB ID	M		BIT STRING (SIZE(28))	Equal to the <i>E-UTRA Cell Identity</i> IE contained in the <i>E-UTRA CGI</i> IE of the cell served by the eNB.	-	
>>Selected EPS TAI	M		EPS TAI 9.3.3.17		-	

9.3.1.26 Emergency Fallback Indicator

The IE indicates emergency service fallback.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Emergency Fallback Request Indicator	M		ENUMERATED (emergency fallback requested, ...)	
Emergency Service Target CN	O		ENUMERATED (5GC, EPC, ...)	

9.3.1.27 Security Indication

This IE contains the user plane integrity protection indication and confidentiality protection indication which indicates the requirements on UP integrity protection and ciphering for corresponding PDU sessions, respectively. Additionally, this IE contains the maximum integrity protected data rate per UE for integrity protection for DRBs.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Integrity Protection Indication	M		ENUMERATED (required, preferred, not needed, ...)	Indicates whether UP integrity protection shall apply, should apply or shall not apply for the concerned PDU session.	-	
Confidentiality Protection Indication	M		ENUMERATED (required, preferred, not needed, ...)	Indicates whether UP ciphering shall apply, should apply or shall not apply for the concerned PDU session.	-	
Maximum Integrity Protected Data Rate Uplink	C- if Integrity Protection Required or Preferred		Maximum Integrity Protected Data Rate 9.3.1.103	Indicates the maximum aggregate rate for integrity protected DRBs supported by the UE in UL. If the <i>Maximum Integrity Protected Data Rate Downlink</i> IE is absent, this IE applies to both UL and DL.	-	
Maximum Integrity Protected Data Rate Downlink	O		Maximum Integrity Protected Data Rate 9.3.1.103	Indicates the maximum aggregate rate for integrity protected DRBs supported by the UE in the DL.	YES	ignore

Condition	Explanation
if Integrity Protection Required or Preferred	This IE shall be present if the <i>Integrity Protection Indication</i> IE is set to the value "required" or "preferred".

9.3.1.28 Non Dynamic 5QI Descriptor

This IE indicates the QoS Characteristics for a standardized or pre-configured 5QI for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
5QI	M		INTEGER (0..255, ...)	Indicates the standardized or pre-configured 5QI as specified in TS 23.501 [9].	-	
Priority Level	O		9.3.1.84	Priority Level is specified in TS 23.501 [9]. When included, it overrides standardized or pre-configured value.	-	
Averaging Window	O		9.3.1.82	Averaging Window is specified in TS 23.501 [9]. When included, it overrides standardized or pre-configured value.	-	
Maximum Data Burst Volume	O		9.3.1.83	Maximum Data Burst Volume is specified in TS 23.501 [9]. When included, it overrides standardized or pre-configured value.	-	
CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore

9.3.1.29 Source NG-RAN Node to Target NG-RAN Node Transparent Container

This IE is produced by the source NG-RAN node and is transmitted to the target NG-RAN node. For inter-system handovers to 5G, the IE is transmitted from the external handover source to the target NG-RAN node.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
RRC Container	M		OCTET STRING	Includes the <i>HandoverPreparationInformation</i> message as defined in TS 38.331 [18] if the target is a gNB. Includes the <i>HandoverPreparationInformation</i> message as defined in TS 36.331 [21] if the target is an ng-eNB.	-	
PDU Session Resource Information List		0..1		For intra-system handovers in NG-RAN.	-	
>PDU Session Resource Information Item		1..<maxno of PDU Sessions>			-	
>>PDU Session ID	M		9.3.1.50		-	
>>QoS Flow Information List		1			-	
>>>QoS Flow Information Item		1..<maxno of QoS Flows>			-	
>>>>QoS Flow Identifier	M		9.3.1.51		-	
>>>>DL Forwarding	O		9.3.1.33		-	
>>>>UL Forwarding	O		9.3.1.118		YES	ignore
>>>>Source Transport Layer Address	O		Transport Layer Address 9.3.2.4	Identifies the TNL address used by the sending node for direct data forwarding towards the target NG-RAN node	YES	ignore
>>>>Source Node Transport Layer Address	O		Transport Layer Address 9.3.2.4	Identifies the TNL address used by the source SN node for direct data forwarding towards the target NG-RAN node	YES	ignore
>>DRBs to QoS Flows Mapping List	O		9.3.1.34		-	
E-RAB Information List		0..1		For inter-system handovers to 5G.	-	
>E-RAB Information Item		1..<maxno of E-RABs>			-	
>>E-RAB ID	M		9.3.2.3		-	
>>DL Forwarding	O		9.3.1.33		-	
>>Source Transport Layer Address	O		Transport Layer Address 9.3.2.4	Identifies the TNL address used by the sending node for direct data forwarding towards the target NG-RAN node	YES	ignore

>>Source Node Transport Layer Address	O		Transport Layer Address 9.3.2.4	Identifies the TNL address used by the source SN node for direct data forwarding towards the target NG-RAN node	YES	ignore
Target Cell ID	M		NG-RAN CGI 9.3.1.73		-	
Index to RAT/Frequency Selection Priority	O		9.3.1.61		-	
UE History Information	M		9.3.1.95		-	
SgNB UE X2AP ID	O		9.3.1.127	Allocated at the Source en-gNB	YES	ignore
UE History Information from UE	O		9.3.1.166		YES	ignore
Source Node ID	O		9.3.1.195	Source SN ID	YES	ignore
UE Context Reference at Source	O		RAN UE NGAP ID 9.3.3.2		YES	ignore
MBS Active Session Information Source to Target List		0..1			YES	ignore
>MBS Active Session Information Source to Target Item		1..<maxno ofMBSSessions ofUE>			-	
>>MBS Session ID	M		9.3.1.206		-	
>>MBS Area Session ID	O		9.3.1.207	If included, this IE indicates the MBS Area Session ID of the UE at the NG-RAN node from which the UE context is transferred	-	
>>MBS Service Area	O		9.3.1.208	Included if available in source NG-RAN node.	-	
>>MBS QoS Flows To Be Setup List	M		9.3.1.236		-	
>>MBS Mapping and Data Forwarding Request List		0..1			-	
>>>MBS Mapping and Data Forwarding Request Item		1..<maxno ofMRBs>			-	
>>>>MRB ID	M		9.3.1.218	Contains the MRB ID value allocated at the source NG-RAN node.	-	
>>>>MBS QoS Flow List		1..<maxno ofMBSQoSflows>			-	
>>>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51		-	
>>>>>MRB Progress Information	O		9.3.1.219	The SN information of the last packet which has already been delivered for the MRB.	-	

QMC Configuration Information	O		9.3.1.223	Used for passing the QoE measurement information from the source NG-RAN node to the target NG-RAN node.	YES	ignore
NGAP IE Support Information Request List		0..1			YES	ignore
>NGAP IE Support Information Request Item		1..<maxno of IESupportInfo>			-	
>>NGAP Protocol IE-Id	M		9.3.1.239		-	
Candidate Relay UE Information List		0..1			YES	reject
>Candidate Relay UE Information Item		1..<maxno of CandidateRelayUEs>			-	
>>Candidate Relay UE ID	M		BIT STRING (SIZE(24))	Includes the <i>SL-SourceIdentity</i> for the candidate relay UE as defined in TS 38.331 [18].	-	
Time Based Handover Information		0..1		This IE only applies to NTN.	YES	ignore
>Handover Window Start	M		INTEGER (0..549755813887)	Corresponds to information provided in <i>t1-Threshold</i> contained in the <i>ReportConfigNR</i> IE as defined in TS 38.331 [18]	-	
>Handover Window Duration	M		INTEGER (1..6000)	Corresponds to information provided in the <i>duration</i> contained in the <i>condEventT1</i> contained in the <i>ReportConfigNR</i> IE as defined in TS 38.331 [18]	-	
Source SN to Target SN QMC Information	O		QMC Configuration Information 9.3.1.223	This IE contains SN-related QMC configuration information to be forwarded to the target S-NG-RAN node upon inter-MN handover.	YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.
maxnoofE-RABs	Maximum no. of E-RABs allowed towards one UE. Value is 256.
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.
maxnoofMBSSessionsofUE	Maximum no. of MBS sessions allowed towards one UE. Value is 256.
maxnoofMBSQoSflows	Maximum no. of MBS QoS flows allowed within one MBS session. Value is 64.
maxnoofMRBs	Maximum no. of MRBs. Value is 32.
maxnoofIESupportInfo	Maximum no. of IE Support Information. Value is 32.
maxnoofCandidateRelayUEs	Maximum no. of Candidate Relay UEs. Value is 32.

9.3.1.30 Target NG-RAN Node to Source NG-RAN Node Transparent Container

This IE is produced by the target NG-RAN node and is transmitted to the source NG-RAN node. For inter-system handovers to 5G, the IE is transmitted from the target NG-RAN node to the external relocation source.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
RRC Container	M		OCTET STRING	Includes the <i>HandoverCommand</i> message as defined in TS 38.331 [18] if the target is a gNB. Includes the <i>HandoverCommand</i> message as defined in TS 36.331 [21] if the target is an ng-eNB.	-	
DAPS Response Information List		0..1			YES	ignore
>DAPS Response Information Item		1..<maxno of DRBs>			-	
>>DRB ID	M		9.3.1.53		-	
>>DAPS Response Information	M		9.3.1.189	Indicates the response to a requested DAPS Handover	-	
Direct Forwarding Path Availability	O		9.3.1.64	Indicates whether a direct forwarding path between the source SN and the target NG-RAN node is available for inter-system handover	YES	ignore
MBS Active Session Information Target to Source List		0..1			YES	ignore
>MBS Active Session Information Target to Source Item		1..<maxno of MBS Sessions of UE>			-	
>>MBS Session ID	M		9.3.1.206		-	
>>Data Forwarding Response MRB List		0..1			-	
>>>Data Forwarding Response MRB Item		1..<maxno of MRBs>			-	
>>>>MRB ID	M		9.3.1.218	Contains the MRB ID value allocated at the source NG-RAN node.	-	
>>>>DL Forwarding UP TNL Information	M		UP Transport Layer Information 9.3.2.2		-	
>>>>MRB Progress Information	O		9.3.1.219	This IE includes the information of the oldest packet available at the target NG-RAN node for the MRB.	-	
NGAP IE Support Information Response List	O		9.3.1.242		YES	ignore

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.
maxnoofMBSSessionsofUE	Maximum no. of MBS sessions allowed towards one UE. Value is 256.
maxnoofMRBs	Maximum no. of MRBs. Value is 32.

9.3.1.31 Allowed NSSAI

This IE contains the allowed NSSAI.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Allowed S-NSSAI Item		<i>1..<maxnoofAllowedS-NSSAIs></i>		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofAllowedS-NSSAIs	Maximum no. of allowed S-NSSAI. Value is 8.

9.3.1.32 Relative AMF Capacity

This IE indicates the relative processing capacity of an AMF with respect to the other AMFs in the AMF Set in order to load-balance AMFs within an AMF Set defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Relative AMF Capacity	M		INTEGER (0..255)	

9.3.1.33 DL Forwarding

This IE indicates that the QoS flow or E-RAB is proposed for forwarding of downlink packets.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL Forwarding	M		ENUMERATED (DL forwarding proposed, ...)	

9.3.1.34 DRBs to QoS Flows Mapping List

This IE contains a list of DRBs containing information about the mapped QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DRBs to QoS Flows Mapping Item		<i>1..<maxnoofDRBs></i>			-	
>DRB ID	M		9.3.1.53		-	
>Associated QoS Flow List	M		9.3.1.99	Contains information of the QoS flows mapped to the DRB	-	
>DAPS Request Information	O		9.3.1.188		YES	ignore

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.

9.3.1.35 Message Identifier

This IE identifies the warning message. It is set by the AMF and transferred to the UE by the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Message Identifier	M		BIT STRING (SIZE(16))	This IE is set by the 5GC, transferred to the UE by the NG-RAN node.

9.3.1.36 Serial Number

This IE identifies a particular message from the source and type indicated by the Message Identifier and is altered every time the message with a given Message Identifier is changed.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Serial Number	M		BIT STRING (SIZE(16))	

9.3.1.37 Warning Area List

This IE indicates the areas where the warning message needs to be broadcast or cancelled.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Warning Area</i>	M			
> <i>E-UTRA Cell IDs</i>				
>> <i>EUTRA CGI List for Warning</i>		1.. <i>maxnoofCellIDforWarning</i>		
>>> <i>E-UTRA CGI</i>	M		9.3.1.9	
> <i>NR Cell IDs</i>				
>> <i>NR CGI List for Warning</i>		1.. <i>maxnoofCellIDforWarning</i>		
>>> <i>NR CGI</i>	M		9.3.1.7	
> <i>TAIs for Warning</i>				
>> <i>TAI List for Warning</i>		1.. <i>maxnoofTAIforWarning</i>		
>>> <i>TAI</i>	M		9.3.3.11	
> <i>Emergency Area IDs</i>				
>> <i>Emergency Area ID List</i>		1.. <i>maxnoofEmergencyAreaID</i>		
>>> <i>Emergency Area ID</i>	M		9.3.1.48	

Range bound	Explanation
maxnoofCellIDforWarning	Maximum no. of Cell ID subject for warning message broadcast. Value is 65535.
maxnoofTAIforWarning	Maximum no. of TAI subject for warning message broadcast. Value is 65535.
maxnoofEmergencyAreaID	Maximum no. of Emergency Area ID subject for warning message broadcast. Value is 65535.

9.3.1.38 Number of Broadcasts Requested

This IE indicates the number of times a message is to be broadcast.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Number of Broadcasts Requested	M		INTEGER (0..65535)	

9.3.1.39 Warning Type

This IE indicates types of the disaster. This IE also indicates that a Primary Notification is included. This IE can be used by the UE to differentiate the type of alert according to the type of disaster.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Warning Type	M		OCTET STRING (SIZE(2))	

9.3.1.40 Void

9.3.1.41 Data Coding Scheme

This IE identifies the alphabet or coding employed for the message characters and message handling at the UE (it is passed transparently from the 5GC to the UE).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Coding Scheme	M		BIT STRING (SIZE(8))	

9.3.1.42 Warning Message Contents

This IE contains user information, e.g., the message with warning contents, and will be broadcast over the radio interface.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Warning Message Contents	M		OCTET STRING (SIZE(1..9600))	

9.3.1.43 Broadcast Completed Area List

This IE indicates the areas where either resources are available to perform the broadcast or where broadcast is performed successfully.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE Broadcast Completed Area	M			
>Cell ID Broadcast E-UTRA				
>>Completed Cell List		1..<maxnoofCellIDforWarning>		
>>>E-UTRA CGI	M		9.3.1.9	
>TAI Broadcast E-UTRA				
>>TAI Broadcast		1..<maxnoofTAIforWarning>		
>>>TAI	M		9.3.3.11	
>>>Completed Cell in TAI List		1..<maxnoofCellinTAI>		
>>>>E-UTRA CGI	M		9.3.1.9	
>Emergency Area ID Broadcast E-UTRA				
>>Emergency Area ID Broadcast		1..<maxnoofEmergencyAreaID>		
>>>Emergency Area ID	M		9.3.1.48	
>>>Completed Cell in Emergency Area ID List		1..<maxnoofCellinEAI>		
>>>>E-UTRA CGI	M		9.3.1.9	
>Cell ID Broadcast NR				
>>Completed Cell List		1..<maxnoofCellIDforWarning>		
>>>NR-CGI	M		9.3.1.7	
>TAI Broadcast NR				
>>TAI Broadcast		1..<maxnoofTAIforWarning>		
>>>TAI	M		9.3.3.11	
>>>Completed Cell in TAI List		1..<maxnoofCellinTAI>		
>>>>NR-CGI	M		9.3.1.7	
>Emergency Area ID Broadcast NR				
>>Emergency Area ID Broadcast		1..<maxnoofEmergencyAreaID>		
>>>Emergency Area ID	M		9.3.1.48	
>>>Completed Cell in Emergency Area ID List		1..<maxnoofCellinEAI>		
>>>>NR-CGI	M		9.3.1.7	

Range bound	Explanation
maxnoofCellIDforWarning	Maximum no. of Cell ID subject for warning message broadcast. Value is 65535.
maxnoofTAIforWarning	Maximum no. of TAI subject for warning message broadcast. Value is 65535.
maxnoofEmergencyAreaID	Maximum no. of Emergency Area ID subject for warning message broadcast. Value is 65535.
maxnoofCellinTAI	Maximum no. of Cell ID within a TAI. Value is 65535.
maxnoofCellinEAI	Maximum no. of Cell ID within an Emergency Area. Value is 65535.

9.3.1.44 Broadcast Cancelled Area List

This IE indicates the areas where broadcast was stopped successfully.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE Broadcast Cancelled Area	M			
>Cell ID Cancelled E-UTRA				
>>Cancelled Cell List		1..<maxnoofCellIDforWarning>		
>>>E-UTRA CGI	M		9.3.1.9	
>>>Number of Broadcasts	M		9.3.1.45	
>TAI Cancelled E-UTRA				
>>TAI Cancelled		1..<maxnoofTAIforWarning>		
>>>TAI	M		9.3.3.11	
>>>Cancelled Cell in TAI List		1..<maxnoofCellinTAI>		
>>>>E-UTRA CGI	M		9.3.1.9	
>>>>Number of Broadcasts	M		9.3.1.45	
>Emergency Area ID Cancelled E-UTRA				
>>Emergency Area ID Cancelled		1..<maxnoofEmergencyAreaID>		
>>>Emergency Area ID	M		9.3.1.48	
>>>Cancelled Cell in Emergency Area ID List		1..<maxnoofCellinEAI>		
>>>>E-UTRA CGI	M		9.3.1.9	
>>>>Number of Broadcasts	M		9.3.1.45	
>Cell ID Cancelled NR				
>>Cancelled Cell List		1..<maxnoofCellIDforWarning>		
>>>NR-CGI	M		9.3.1.7	
>>>Number of Broadcasts	M		9.3.1.45	
>TAI Cancelled NR				
>>TAI Cancelled		1..<maxnoofTAIforWarning>		
>>>TAI	M		9.3.3.11	
>>>Cancelled Cell in TAI List		1..<maxnoofCellinTAI>		
>>>>NR-CGI	M		9.3.1.7	
>>>>Number of Broadcasts	M		9.3.1.45	
>Emergency Area ID Cancelled NR				
>>Emergency Area ID Cancelled		1..<maxnoofEmergencyAreaID>		
>>>Emergency Area ID	M		9.3.1.48	
>>>Cancelled Cell in Emergency Area ID List		1..<maxnoofCellinEAI>		
>>>>NR-CGI	M		9.3.1.7	
>>>>Number of Broadcasts	M		9.3.1.45	

Range bound	Explanation
maxnoofCellIDforWarning	Maximum no. of Cell ID subject for warning message broadcast. Value is 65535.
maxnoofTAIforWarning	Maximum no. of TAI subject for warning message broadcast. Value is 65535.
maxnoofEmergencyAreaID	Maximum no. of Emergency Area ID subject for warning message broadcast. Value is 65535.
maxnoofCellinTAI	Maximum no. of Cell ID within a TAI. Value is 65535.
maxnoofCellinEAI	Maximum no. of Cell ID within an Emergency Area. Value is 65535.

9.3.1.45 Number of Broadcasts

This IE indicates the number of times that a particular message has been broadcast in a given warning area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Number of Broadcasts	M		INTEGER (0..65535)	This IE is set to '0' if valid results are not known or not available. It is set to 65535 if the counter results have overflowed.

9.3.1.46 Concurrent Warning Message Indicator

This IE indicates to the NG-RAN node that the received warning message is a new message to be scheduled for concurrent broadcast with any other ongoing broadcast of warning messages.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Concurrent Warning Message Indicator	M		ENUMERATED (true, ...)	This IE is used to identify a PWS type warning system which allows the broadcast of multiple concurrent warning messages over the radio.

9.3.1.47 Cancel-All Warning Messages Indicator

This IE indicates to the NG-RAN node to stop all already ongoing broadcast of warning messages in the NG-RAN node or in an area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cancel-All Warning Messages Indicator	M		ENUMERATED (true, ...)	

9.3.1.48 Emergency Area ID

This IE is used to indicate the area which has the emergency impact.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Emergency Area ID	M		OCTET STRING (SIZE(3))	Emergency Area ID may consist of several cells. Emergency Area ID is defined by the operator.

9.3.1.49 Repetition Period

This IE indicates the periodicity of the warning message to be broadcast.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Repetition Period	M		INTEGER (0..2 ¹⁷ -1)	The unit of value 1 to 2 ¹⁷ -1 is [second].

9.3.1.50 PDU Session ID

This IE identifies a PDU Session for a UE. The definition and use of the PDU Session ID is specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session ID	M		INTEGER (0..255)	

9.3.1.51 QoS Flow Identifier

This IE identifies a QoS flow within a PDU Session, or a MBS QoS flow within a MBS session. The definition and use of the QoS Flow Identifier is specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Identifier	M		INTEGER (0..63, ...)	

9.3.1.52 PDU Session Type

This IE indicates the PDU Session Type as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session Type	M		ENUMERATED (Ipv4, Ipv6, Ipv4v6, ethernet, unstructured, ...)	

9.3.1.53 DRB ID

This IE contains the DRB ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DRB ID	M		INTEGER (1..32, ...)	

9.3.1.54 Masked IMEISV

This IE contains the IMEISV value with a mask, to identify a terminal model without identifying an individual Mobile Equipment.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Masked IMEISV	M		BIT STRING (SIZE(64))	Coded as the International Mobile station Equipment Identity and Software Version Number (IMEISV) defined in TS 23.003 [23] with the last 4 digits of the SNR masked by setting the corresponding bits to 1. The first to fourth bits correspond to the first digit of the IMEISV, the fifth to eighth bits correspond to the second digit of the IMEISV, and so on.

9.3.1.55 New Security Context Indicator

This IE indicates that the AMF has activated a new 5G NAS security context as described in TS 33.501 [13].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
New Security Context Indicator	M		ENUMERATED (true, ...)	The NSCI as defined in TS 33.501 [13].

9.3.1.56 Time to Wait

This IE defines the minimum allowed waiting time.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Time to Wait	M		ENUMERATED (1s, 2s, 5s, 10s, 20s, 60s, ...)	

9.3.1.57 Global N3IWF ID

This IE is used to globally identify an N3IWF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE <i>N3IWF ID</i>	M			
> <i>N3IWF ID</i>				
>> <i>N3IWF ID</i>	M		BIT STRING (SIZE(16))	

9.3.1.58 UE Aggregate Maximum Bit Rate

This IE is applicable for all Non-GBR QoS flows per UE which is defined for the downlink and the uplink direction and a subscription parameter provided by the AMF to the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Aggregate Maximum Bit Rate		1		Applicable for Non-GBR QoS flows.
>UE Aggregate Maximum Bit Rate Downlink	M		Bit Rate 9.3.1.4	This IE indicates the UE Aggregate Maximum Bit Rate as specified in TS 23.501 [9] in the downlink direction.
>UE Aggregate Maximum Bit Rate Uplink	M		Bit Rate 9.3.1.4	This IE indicates the UE Aggregate Maximum Bit Rate as specified in TS 23.501 [9] in the uplink direction.

9.3.1.59 Security Result

This IE indicates whether the security policy indicated as "preferred" in the *Security Indication* IE is performed or not.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Integrity Protection Result	M		ENUMERATED (performed, not performed, ...)	Indicates whether UP integrity protection is performed or not for the concerned PDU session.
Confidentiality Protection Result	M		ENUMERATED (performed, not performed, ...)	Indicates whether UP ciphering is performed or not for the concerned PDU session.

9.3.1.60 User Plane Security Information

This IE indicates user plane security information related to security policy.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Security Result	M		9.3.1.59	
Security Indication	M		9.3.1.27	

9.3.1.61 Index to RAT/Frequency Selection Priority

This IE is used to define local configuration for RRM strategies such as camp priorities in Idle mode and control of inter-RAT/inter-frequency handover in Active mode (see TS 23.501 [9]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Index to RAT/Frequency Selection Priority	M		INTEGER (1..256, ...)	

9.3.1.62 Data Forwarding Accepted

This IE indicates that the NG-RAN node accepts the proposed DL data forwarding for the QoS flow which is subject to data forwarding.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Forwarding Accepted	M		ENUMERATED (data forwarding accepted, ...)	

9.3.1.63 Data Forwarding Not Possible

This IE indicates that the 5GC decided that the corresponding PDU session will not be subject to data forwarding.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Forwarding Not Possible	M		ENUMERATED (data forwarding not possible, ...)	

9.3.1.64 Direct Forwarding Path Availability

This IE indicates whether a direct forwarding path is available.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Direct Forwarding Path Availability	M		ENUMERATED (direct path available, ...)	

9.3.1.65 Location Reporting Request Type

This IE indicates the type of location request to be handled by the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Event Type	M		ENUMERATED (direct, change of serving cell, UE presence in the area of interest, stop change of serving cell, stop UE presence in the area of interest, cancel location reporting for the UE, ..., change of serving cell and UE presence in the area of interest, report the Aerial UE flight information, cancel the Aerial UE flight information reporting)		-	
Report Area	M		ENUMERATED (cell, ...)	Corresponds to the "Location Reporting Level" in TS 23.502 [10]	-	
Area of Interest List		<i>0..1</i>			-	
>Area of Interest Item		<i>1..<maxno ofAol></i>			-	
>>Area of Interest	M		9.3.1.66		-	
>>Location Reporting Reference ID	M		9.3.1.76		-	
Location Reporting Reference ID to be Cancelled	C-ifEventTypesIsStopUEPresenceInAol		Location Reporting Reference ID 9.3.1.76		-	
Additional Location Information	O		ENUMERATED (Include PSCell, ...)		YES	ignore
Additional Cancelled Location Reporting Reference ID List		<i>0..1</i>			YES	reject
>Additional Cancelled Location Reporting Reference ID Item		<i>1..<maxno ofAolMinusOne></i>			-	
>>Location Reporting Reference ID to be Cancelled	M		Location Reporting Reference ID 9.3.1.76		-	
Aerial UE Flight Information Reporting Control List		<i>0..1</i>			YES	ignore
>Aerial UE Flight Information Reporting Control Item		<i>1..<maxno ofFlightInfoReportControl></i>			-	
>>Aerial UE Flight Information Reporting Control	M		9.3.1.283		-	

Range bound	Explanation
maxnoofAol	Maximum no. of areas of interest. Value is 64.
maxnoofAolMinusOne	Maximum no. of areas of interest minus one. Value is 63.
maxnoofFlightInfoReportControl	Maximum no. of Aerial UE Flight Information Reporting Control. Value is 64.

Condition	Explanation
ifEventTypeisStopUEPresinAol	This IE shall be present if the <i>Event Type</i> IE is set to the value "stop UE presence in the area of interest".

9.3.1.66 Area of Interest

This IE indicates the area of interest.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Area of Interest TAI List		0..1		
>Area of Interest TAI Item		1..<maxnoofTAIinAol>		
>>TAI	M		9.3.3.11	
Area of Interest Cell List		0..1		
>Area of Interest Cell Item		1..<maxnoofCellinAol>		
>>NG-RAN CGI	M		9.3.1.73	
Area of Interest RAN Node List		0..1		
>Area of Interest RAN Node Item		1..<maxnoofRANNodeinAol>		
>>Global RAN Node ID	M		9.3.1.5	

Range bound	Explanation
maxnoofTAIinAol	Maximum no. of tracking areas in an area of interest. Value is 16.
maxnoofCellinAol	Maximum no. of cells in an area of interest. Value is 256.
maxnoofRANNodeinAol	Maximum no. of NG-RAN nodes in an area of interest. Value is 64.

9.3.1.67 UE Presence in Area of Interest List

This IE indicates the UE presence in the area of interest.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Presence in Area of Interest Item		1..<maxnoofAol>		
>Location Reporting Reference ID	M		9.3.1.76	
>UE Presence	M		ENUMERATED (in, out, unknown, ...)	

Range bound	Explanation
maxnoofAol	Maximum no. of areas of interest. Value is 64.

9.3.1.68 UE Radio Capability for Paging

This IE contains paging specific UE Radio Capability information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Radio Capability for Paging of NR	O		OCTET STRING	Includes the <i>UERadioPagingInformation</i> message as defined in TS 38.331 [18].
UE Radio Capability for Paging of E-UTRA	O		OCTET STRING	Includes the <i>UERadioPagingInformation</i> message as defined in TS 36.331 [21].
UE Radio Capability for Paging of NB-IoT	O		OCTET STRING	Includes the <i>UERadioPagingInformation-NB</i> message as defined in TS 36.331 [21].

9.3.1.69 Assistance Data for Paging

This IE provides assistance information for paging optimisation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Assistance Data for Recommended Cells	O		9.3.1.70		-	
Paging Attempt Information	O		9.3.1.72		-	
NPN Paging Assistance Information	O		9.3.1.183		YES	ignore
Paging Assistance Data for CE Capable UE	O		9.3.1.141		YES	ignore

9.3.1.70 Assistance Data for Recommended Cells

This IE provides assistance information for paging in recommended cells.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended Cells for Paging	M		9.3.1.71	

9.3.1.71 Recommended Cells for Paging

This IE contains the recommended cells for paging.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended Cell List		<i>1</i>		
>Recommended Cell Item		<i>1..<maxnoofRecommendedCells></i>		Includes visited and non-visited cells, where visited cells are listed in the order the UE visited them with the most recent cell being the first in the list. Non-visited cells are included immediately after the visited cell they are associated with.
>>NG-RAN CGI	M		9.3.1.73	
>>Time Stayed in Cell	O		INTEGER (0..4095)	This is included for visited cells and indicates the time a UE stayed in a cell in seconds. If the UE stays in a cell more than 4095 seconds, this IE is set to 4095.

Range bound	Explanation
maxnoofRecommendedCells	Maximum no. of recommended Cells. Value is 16.

9.3.1.72 Paging Attempt Information

This IE includes information related to the paging count over NG.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Attempt Count	M		INTEGER (1..16, ...)	Paging attempt count (see TS 38.300 [8]).
Intended Number of Paging Attempts	M		INTEGER (1..16, ...)	Intended number of paging attempts (see TS 38.300 [8]).
Next Paging Area Scope	O		ENUMERATED (same, changed, ...)	Indicates whether the paging area scope will change or not at next paging attempt. Usage specified in TS 38.300 [8].

9.3.1.73 NG-RAN CGI

This IE is used to globally identify a cell in NG-RAN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE NG-RAN CGI	M			
>NR				
>>NR CGI	M		9.3.1.7	
>E-UTRA				
>>E-UTRA CGI	M		9.3.1.9	

9.3.1.74 UE Radio Capability

This IE contains UE Radio Capability information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Radio Capability	M		OCTET STRING	Includes either the <i>UERadioAccessCapabilityInformation</i> message as defined in TS 38.331 [18], or the <i>UERadioAccessCapabilityInformation-NB</i> message as defined in TS 36.331 [21].

9.3.1.74a UE Radio Capability – E-UTRA Format

This IE contains UE Radio Capability information to support Mode of operation A as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Radio Capability – E-UTRA Format	M		OCTET STRING	Includes the <i>UERadioAccessCapabilityInformation</i> message as defined in TS 36.331 [21].

9.3.1.75 Time Stamp

This IE contains UTC time information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Time Stamp	M		OCTET STRING (SIZE(4))	Encoded in the same format as the first four octets of the 64-bit timestamp format as defined in section 6 of IETF RFC 5905 [25].

9.3.1.76 Location Reporting Reference ID

This IE contains the Location Reporting Reference ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Location Reporting Reference ID	M		INTEGER (1..64, ...)	

9.3.1.77 Data Forwarding Response DRB List

This IE indicates data forwarding related information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Forwarding Response DRB Item		<i>1..<maxnoofDRBs></i>		
>DRB ID	M		9.3.1.53	
>DL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	
>UL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.

9.3.1.78 Paging Priority

This element indicates the paging priority for paging a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Priority	M		ENUMERATED (PrioLevel1, PrioLevel2, PrioLevel3, PrioLevel4, PrioLevel5, PrioLevel6, PrioLevel7, PrioLevel8, ...)	Lower value codepoint indicates higher priority.

9.3.1.79 Packet Loss Rate

This IE indicates the Packet Loss Rate for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Packet Loss Rate	M		INTEGER (0..1000, ...)	Ratio of lost packets per number of packets sent, expressed in tenth of percent.

9.3.1.80 Packet Delay Budget

This IE indicates the Packet Delay Budget for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Packet Delay Budget	M		INTEGER (0..1023, ...)	Upper bound value for the delay that a packet may experience expressed in unit of 0.5ms.

9.3.1.81 Packet Error Rate

This IE indicates the Packet Error Rate for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Scalar	M		INTEGER (0..9, ...)	The packet error rate is expressed as <i>Scalar</i> x 10-k where k is the <i>Exponent</i> .
Exponent	M		INTEGER (0..9, ...)	

9.3.1.82 Averaging Window

This IE indicates the Averaging Window for a QoS flow, and applies to GBR QoS flows only.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Averaging Window	M		INTEGER (0..4095, ...)	Unit: ms. The default value of the IE is 2000ms.

9.3.1.83 Maximum Data Burst Volume

This IE indicates the Maximum Data Burst Volume for a QoS flow, and applies to delay critical GBR QoS flows only.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Maximum Data Burst Volume	M		INTEGER (0..4095, ..., 4096.. 2000000)	Unit: byte.

9.3.1.84 Priority Level

This IE indicates the Priority Level for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Priority Level	M		INTEGER (1..127, ...)	Values ordered in decreasing order of priority, i.e. with 1 as the highest priority and 127 as the lowest priority.

9.3.1.85 Mobility Restriction List

This IE defines roaming or access restrictions for subsequent mobility action for which the NG-RAN provides information about the target of the mobility action towards the UE, e.g., handover, or for SCG selection during dual connectivity operation or for assigning proper RNAs. NG-RAN behaviour upon receiving this IE is specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Serving PLMN	M		PLMN Identity 9.3.3.5		-	
Equivalent PLMNs		<i>0..<maxno ofEPLMNs></i>		Allowed PLMNs in addition to Serving PLMN. This list corresponds to the list of "equivalent PLMNs" as defined in TS 24.501 [26]. This list is part of the roaming restriction information. Roaming restrictions apply to PLMNs other than the Serving PLMN and Equivalent PLMNs.	-	
>PLMN Identity	M		9.3.3.5		-	
RAT Restrictions		<i>0..<maxno ofEPLMNs PlusOne></i>		This IE contains RAT restriction related information as specified in TS 23.501 [9].	-	
>PLMN Identity	M		9.3.3.5		-	
>RAT Restriction Information	M		BIT STRING { e-UTRA (0), nR (1), nR-unlicensed (2), nR-LEO (3), nR-MEO (4), nR-GEO (5), nR-OTHERSAT (6)} (SIZE(8, ...))	Each position in the bitmap represents a RAT. If a bit is set to "1", the respective RAT is restricted for the UE. If a bit is set to "0", the respective RAT is not restricted for the UE. Bit 7 reserved for future use.	-	
>Extended RAT Restriction Information	O		9.3.1.126	If this IE is included, the <i>RAT Restriction Information</i> IE is ignored.	YES	ignore
Forbidden Area Information		<i>0..<maxno ofEPLMNs PlusOne></i>		This IE contains Forbidden Area information as specified in TS 23.501 [9].	-	
>PLMN Identity	M		9.3.3.5		-	
> Forbidden TACs		<i>1..<maxno ofForbTACs></i>			-	
>>TAC	M		9.3.3.10	The TAC of the forbidden TAI.	-	
Service Area Information		<i>0..<maxno ofEPLMNs PlusOne></i>		This IE contains Service Area Restriction information as specified in TS 23.501 [9].	-	
>PLMN Identity	M		9.3.3.5		-	
> Allowed TACs		<i>0..<maxno ofAllowed Areas></i>			-	

>>TAC	M		9.3.3.10	The TAC of the allowed TAI.	-	
>Not Allowed TACs		0..<maxno of Allowed Areas>			-	
>>TAC	M		9.3.3.10	The TAC of the not-allowed TAI.	-	
Last E-UTRAN PLMN Identity	O		PLMN Identity 9.3.3.5	Indicates the E-UTRAN PLMN ID from where the UE formerly handed over to 5GS and which is preferred in case of subsequent mobility to EPS.	YES	ignore
Core Network Type Restriction for Serving PLMN	O		ENUMERATED (EPCForbidden, ...)	Indicates whether the UE is restricted to connect to EPC for the Serving PLMN as specified in TS 23.501 [9].	YES	ignore
Core Network Type Restriction for Equivalent PLMNs		0..<maxno of EPLMNs >			YES	ignore
>PLMN Identity	M		9.3.3.5	Includes any of the Equivalent PLMNs listed in the <i>Mobility Restriction List</i> IE for which CN Type restriction applies as specified in TS 23.501 [9].	-	
>Core Network Type Restriction	M		ENUMERATED (EPCForbidden, 5GCForbidden, ...)	Indicates whether the UE is restricted to connect to EPC or to 5GC for this PLMN.	-	
NPN Mobility Information	O		9.3.1.184		YES	reject

Range bound	Explanation
maxnoofEPLMNs	Maximum no. of equivalent PLMNs. Value is 15.
maxnoofEPLMNsPlusOne	Maximum no. of allowed PLMNs. Value is 16.
maxnoofForbTACs	Maximum no. of forbidden Tracking Area Codes. Value is 4096.
maxnoofAllowedAreas	Maximum no. of allowed or not allowed Tracking Areas. Value is 16.

9.3.1.86 UE Security Capabilities

This IE defines the supported algorithms for encryption and integrity protection in the UE. The Security Capabilities received from NAS signaling shall not be modified or truncated when forwarded to NG-RAN nodes and the NG-RAN nodes store and send the complete bitmaps without modification or truncation as specified in TS 38.300 [8].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR Encryption Algorithms	M		BIT STRING (SIZE(16, ...))	Each position in the bitmap represents an encryption algorithm: "all bits equal to 0" – UE supports no other algorithm than NEA0, "first bit" – 128-NEA1, "second bit" – 128-NEA2, "third bit" – 128-NEA3, "fourth to seventh bit" are mapped from bit 4 to bit 1 of octet 3 in the <i>UE Security Capability</i> IE defined in TS 24.501 [26], other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm. Algorithms are defined in TS 33.501 [13].
NR Integrity Protection Algorithms	M		BIT STRING (SIZE(16, ...))	Each position in the bitmap represents an integrity protection algorithm: "all bits equal to 0" – UE supports no other algorithm than NIA0, "first bit" – 128-NIA1, "second bit" – 128-NIA2, "third bit" – 128-NIA3, "fourth to seventh bit" are mapped from bit 4 to bit 1 of octet 4 in the <i>UE Security Capability</i> IE defined in TS 24.501 [26], other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm. Algorithms are defined in TS 33.501 [13].
E-UTRA Encryption Algorithms	M		BIT STRING (SIZE(16, ...))	Each position in the bitmap represents an encryption algorithm: "all bits equal to 0" – UE supports no other algorithm than EEA0, "first bit" – 128-EEA1, "second bit" – 128-EEA2, "third bit" – 128-EEA3, "fourth to seventh bit" are mapped from bit 4 to bit 1 of octet 5 in the <i>UE Security Capability</i> IE defined in TS 24.501 [26], other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm. Algorithms are defined in TS 33.401 [27].

E-UTRA Integrity Protection Algorithms	M		BIT STRING (SIZE(16, ...))	Each position in the bitmap represents an encryption algorithm: "all bits equal to 0" – UE supports no other algorithm than EIA0, "first bit" – 128-EIA1, "second bit" – 128-EIA2, "third bit" – 128-EIA3, "fourth to seventh bit" are mapped from bit 4 to bit 1 of octet 6 in the <i>UE Security Capability</i> IE defined in TS 24.501 [26], other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm. Algorithms are defined in TS 33.401 [27].
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9.3.1.87 Security Key

This IE is used to apply security in the NG-RAN for different scenarios as defined in TS 33.501 [13].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Security Key	M		BIT STRING (SIZE(256))	Key material for NG-RAN node or Next Hop Key as defined in TS 33.501 [13]

9.3.1.88 Security Context

This IE provides security related parameters to the NG-RAN node which are used to derive security keys for user plane traffic and RRC signalling messages and for security parameter generation for subsequent mobility, see TS 33.501 [13].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Next Hop Chaining Count	M		INTEGER (0..7)	Next Hop Chaining Counter (NCC) defined in TS 33.501 [13].
Next-Hop NH	M		Security Key 9.3.1.87	The NH together with the NCC is used to derive the security configuration as defined in TS 33.501 [13].

9.3.1.89 IMS Voice Support Indicator

This IE is set by the NG-RAN node to indicate whether the UE radio capabilities are compatible with the network configuration for IMS voice.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IMS Voice Support Indicator	M		ENUMERATED (Supported, Not Supported, ...)	

9.3.1.90 Paging DRX

This IE indicates the Paging DRX as defined in TS 38.304 [12] and TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging DRX	M		ENUMERATED (32, 64, 128, 256, ...)	Unit: [number of radioframes]

9.3.1.91 RRC Inactive Transition Report Request

This IE is used to request the NG-RAN node to report or stop reporting to the 5GC when the UE enters or leaves RRC_INACTIVE state.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC Inactive Transition Report Request	M		ENUMERATED (Subsequent state transition report, Single RRC connected state report, Cancel report, ...)	

9.3.1.92 RRC State

This IE indicates the RRC state of the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC State	M		ENUMERATED (Inactive, Connected, ...)	Indicates the current RRC state of the UE.

9.3.1.93 Expected UE Behaviour

This IE indicates the behaviour of a UE with predictable activity and/or mobility behaviour, to assist the NG-RAN node in e.g. determining the optimum RRC connection time or helping with the RRC_INACTIVE state transition and RNA configuration (e.g. size and shape of the RNA).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Expected UE Activity Behaviour	O		9.3.1.94	
Expected HO Interval	O		ENUMERATED (sec15, sec30, sec60, sec90, sec120, sec180, long-time, ...)	Indicates the expected time interval between inter NG-RAN node handovers. If "long-time" is included, the interval between inter NG-RAN node handovers is expected to be longer than 180 seconds.
Expected UE Mobility	O		ENUMERATED (stationary, mobile, ...)	Indicates whether the UE is expected to be stationary or mobile.
Expected UE Moving Trajectory		0..1		Indicates the UE's expected geographical movement.
>Expected UE Moving Trajectory Item		1..<maxnoofCellsUEMovingTrajectory>		Includes list of visited and non-visited cells, where visited cells are listed in the order the UE visited them with the most recent cell being the first in the list. Non-visited cells are included immediately after the visited cell they are associated with.
>>NG-RAN CGI	M		9.3.1.73	
>>Time Stayed in Cell	O		INTEGER (0..4095)	Included for visited cells and indicates the time a UE stayed in a cell in seconds. If the UE stays in a cell more than 4095 seconds, this IE is set to 4095.

Range bound	Explanation
maxnoofCellsUEMovingTrajectory	Maximum no. of cells of UE moving trajectory. Value is 16.

9.3.1.94 Expected UE Activity Behaviour

This IE indicates information about the expected "UE activity behaviour" of the UE or the PDU session as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Expected Activity Period	O		INTEGER (1..30 40 50 60 80 100 120 150 180 181, ...)	If set to "181" the expected activity time is longer than 180 seconds. The remaining values indicate the expected activity time in [seconds].
Expected Idle Period	O		INTEGER (1..30 40 50 60 80 100 120 150 180 181, ...)	If set to "181" the expected idle time is longer than 180 seconds. The remaining values indicate the expected idle time in [seconds].
Source of UE Activity Behaviour Information	O		ENUMERATED (subscription information, statistics, ...)	If "subscription information" is indicated, the information contained in the <i>Expected Activity Period</i> IE and the <i>Expected Idle Period</i> IE, if present, is derived from subscription information. If "statistics" is indicated, the information contained in the <i>Expected Activity Period</i> IE and the <i>Expected Idle Period</i> IE, if present, is derived from statistical information.

9.3.1.95 UE History Information

This IE contains information about cells that a UE has been served by in RRC_CONNECTED state prior to the target cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Last Visited Cell Item		1..<maxnoofCellsInUEHistoryInfo>		Most recent information is added to the top of this list.
>Last Visited Cell Information	M		9.3.1.96	

Range bound	Explanation
maxnoofCellsInUEHistoryInfo	Maximum no. of cells in the UE history information. Value is 16.

9.3.1.96 Last Visited Cell Information

This IE may contain cell specific information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Last Visited Cell Information</i>	M			
> <i>NG-RAN Cell</i>				
>>Last Visited NG-RAN Cell Information	M		9.3.1.97	
> <i>E-UTRAN Cell</i>				
>>Last Visited E-UTRAN Cell Information	M		OCTET STRING	Defined in TS 36.413 [16].
> <i>UTRAN Cell</i>				
>>Last Visited UTRAN Cell Information	M		OCTET STRING	Defined in TS 25.413 [28].
> <i>GERAN Cell</i>				
>>Last Visited GERAN Cell Information	M		OCTET STRING	Defined in TS 36.413 [16].

9.3.1.97 Last Visited NG-RAN Cell Information

This IE contains information about a cell. In case of NR cell, this IE contains information about a set of NR cells with the same NR ARFCN for reference point A, and the *Global Cell ID* IE identifies one of the NR cells in the set. The information is to be used for RRM purposes.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Global Cell ID	M		NG-RAN CGI 9.3.1.73		-	
Cell Type	M		9.3.1.98		-	
Time UE Stayed in Cell	M		INTEGER (0..4095)	The duration of time the UE stayed in the cell, or set of NR cells with the same NR ARFCN for reference point A, in seconds. If the duration is more than 4095s, this IE is set to 4095.	-	
Time UE Stayed in Cell Enhanced Granularity	O		INTEGER (0..40950)	The duration of time the UE stayed in the cell, or set of NR cells with the same NR ARFCN for reference point A, in 1/10 seconds. If the duration is more than 4095s, this IE is set to 40950.	-	
HO Cause Value	O		Cause 9.3.1.2	The cause for the handover.	-	
Last Visited PSCell List		0..<maxno ofPSCells PerPrimaryCellinUE HistoryInfo>		List of cells configured as PSCells. Most recent PSCell related information is added to the top of the list.	YES	ignore
>Last Visited PSCell Information	M		9.3.1.235	The PSCell related information.	-	

Range bound	Explanation
maxnoofPSCellsPerPrimaryCellinUEHistoryInfo	Maximum number of last visited PSCell information records that can be reported in the IE. Value is 8.

9.3.1.98 Cell Type

This IE provides the cell coverage area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell Size	M		ENUMERATED (verysmall, small, medium, large, ...)	

9.3.1.99 Associated QoS Flow List

This IE indicates the list of QoS flows associated with e.g. a DRB or UP TNL endpoint.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Associated QoS Flow Item		1..<maxno ofQoSFlows>			-	
>QoS Flow Identifier	M		9.3.1.51		-	
>QoS Flow Mapping Indication	O		ENUMERATED (ul, dl, ...)		-	
>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.152	Index to the currently fulfilled alternative QoS parameters set	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.1.100 Information on Recommended Cells and RAN Nodes for Paging

This IE provides information on recommended cells and NG-RAN nodes for paging.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended Cells for Paging	M		9.3.1.71	
Recommended RAN Nodes for Paging	M		9.3.1.101	

9.3.1.101 Recommended RAN Nodes for Paging

This IE contains recommended NG-RAN nodes for paging.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended RAN Node List		1		
>Recommended RAN Node Item		1..<maxnoofRecommendedRANNodes>		Includes visited and non-visited NG-RAN nodes, where visited NG-RAN nodes are listed in the order the UE visited them with the most recent NG-RAN node being the first in the list. Non-visited NG-RAN nodes are included after the visited NG-RAN node they are associated with.
>>CHOICE AMF Paging Target				The AMF paging target is either an NG-RAN node identity or a TAI as specified in TS 38.300 [8].
>>>RAN Node				
>>>>Global RAN Node ID	M		9.3.1.5	
>>>TAI				
>>>>TAI	M		9.3.3.11	

Range bound	Explanation
maxnoofRecommendedRANNodes	Maximum no. of recommended NG-RAN nodes. Value is 16.

9.3.1.102 PDU Session Aggregate Maximum Bit Rate

This IE is applicable for all Non-GBR QoS flows per PDU session which is defined for the downlink and the uplink direction and is provided by the SMF to the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session Aggregate Maximum Bit Rate		1		Applicable for Non-GBR QoS flows.
>PDU Session Aggregate Maximum Bit Rate Downlink	M		Bit Rate 9.3.1.4	Indicates the PDU session Aggregate Maximum Bit Rate as specified in TS 23.501 [9] in the downlink direction.
>PDU Session Aggregate Maximum Bit Rate Uplink	M		Bit Rate 9.3.1.4	Indicates the PDU session Aggregate Maximum Bit Rate as specified in TS 23.501 [9] in the uplink direction.

9.3.1.103 Maximum Integrity Protected Data Rate

This IE indicates the maximum aggregate data rate for integrity protected DRBs for a UE as defined in TS 38.300 [8].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Maximum Integrity Protected Data Rate	M		ENUMERATED (64kbps, max UE rate, ...)	Defines the upper bound of the aggregate data rate of user plane integrity protected data for either UL or DL.

9.3.1.104 Overload Response

This IE indicates the required behaviour of the NG-RAN node in an overload situation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Overload Response</i>	M			
> <i>Overload Action</i>				
>>Overload Action	M		9.3.1.105	

9.3.1.105 Overload Action

This IE indicates which signalling traffic is subject to rejection by the NG-RAN node in an AMF overload situation as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Overload Action	M		ENUMERATED (Reject RRC connection establishments for non-emergency MO DT, Reject RRC connection establishments for Signalling, Permit Emergency Sessions and mobile terminated services only, Permit High Priority Sessions and mobile terminated services only, ...)	

9.3.1.106 Traffic Load Reduction Indication

This IE indicates the percentage of the type of traffic relative to the instantaneous incoming rate at the NG-RAN node, as indicated in the *Overload Action* IE, to be rejected.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Traffic Load Reduction Indication	M		INTEGER (1..99)	

9.3.1.107 Slice Overload List

This IE indicates the list of overloaded slices.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Overload Item		<i>1..<maxnoofSliceItems></i>		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofSliceItems	Maximum no. of signalled slice support items. Value is 1024.

9.3.1.108 RAN Status Transfer Transparent Container

This IE is produced by the source NG-RAN node and is transmitted to the target NG-RAN node. It is used for intra 5GC NG-based handover.

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DRBs Subject to Status Transfer List		1			-	
>DRBs Subject to Status Transfer Item		<i>1..<maximum of DRBs></i>			-	
>>DRB ID	M		9.3.1.53		-	
>>CHOICE <i>UL DRB Status</i>	M				-	
>>>12 bits						
>>>>UL COUNT Value	M		COUNT Value for PDCP SN Length 12 9.3.1.109	PDCP-SN and HFN of the first missing UL PDCP SDU in case of 12 bit long PDCP-SN.	-	
>>>>Receive Status of UL PDCP SDUs	O		BIT STRING (SIZE(1..2048))	The IE is used in case of 12 bit long PDCP-SN. The first bit indicates the status of the SDU after the First Missing UL PDCP SDU. The N^{th} bit indicates the status of the UL PDCP SDU in position (N + First Missing SDU Number) modulo (1 + the maximum value of the PDCP-SN). 0: PDCP SDU has not been received. 1: PDCP SDU has been received correctly.	-	
>>>>18 bits						
>>>>UL COUNT Value	M		COUNT Value for PDCP SN Length 18 9.3.1.110	PDCP-SN and HFN of the first missing UL PDCP SDU in case of 18 bit long PDCP-SN.	-	
>>>>Receive Status of UL PDCP SDUs	O		BIT STRING (SIZE(1..131072))	The IE is used in case of 18 bit long PDCP-SN. The first bit indicates the status of the SDU after the First Missing UL PDCP SDU. The N^{th} bit indicates the status of the UL PDCP SDU in position (N + First Missing SDU Number) modulo (1 + the maximum value of the PDCP-SN). 0: PDCP SDU has not been received. 1: PDCP SDU has been received correctly.	-	

>>CHOICE <i>DL DRB Status</i>	M				-	
>>>12 bits						
>>>>DL COUNT Value	M		COUNT Value for PDCP SN Length 12 9.3.1.109	PDCP-SN and HFN that the target NG-RAN node should assign for the next DL PDCP SDU not having an SN yet in case of 12 bit long PDCP-SN.	-	
>>>>18 bits						
>>>>>DL COUNT Value	M		COUNT Value for PDCP SN Length 18 9.3.1.110	PDCP-SN and HFN that the target NG-RAN node should assign for the next DL PDCP SDU not having an SN yet in case of 18 bit long PDCP-SN.	-	
>>Old Associated QoS Flow List - UL End Marker Expected	O		Associated QoS Flow List 9.3.1.99	Indicates that the source NG-RAN node has initiated QoS flow re-mapping and has not yet received SDAP end markers, as described in TS 38.300 [8].	YES	ignore

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.

9.3.1.109 COUNT Value for PDCP SN Length 12

This IE contains a PDCP sequence number and a hyper frame number in case of 12 bit long PDCP-SN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDCP SN Length 12	M		INTEGER (0..4095)	
HFN for PDCP SN Length 12	M		INTEGER (0..1048575)	

9.3.1.110 COUNT Value for PDCP SN Length 18

This IE contains a PDCP sequence number and a hyper frame number in case of 18 bit long PDCP-SN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDCP SN Length 18	M		INTEGER (0..262143)	
HFN for PDCP SN Length 18	M		INTEGER (0..16383)	

9.3.1.111 RRC Establishment Cause

This IE indicates the reason for RRC Connection Establishment as received from the UE in the *EstablishmentCause* IE defined in TS 38.331 [18] or the *EstablishmentCause-5GC* IE defined in TS 36.331 [21], or the reason for RRC

Connection Resume as received from the UE in the *ResumeCause* IE defined in TS 38.331 [18] or the *ResumeCause-r15* IE defined in TS 36.331 [21] or the reason for Early Data Request as received from the UE in the *establishmentCause-r16* contained in the *RRCEarlyDataRequest-5GC-r16-IEs* IE defined in TS 36.331 [21].

In case of NB-IoT, this IE indicates the reason for RRC Connection Establishment as received from the UE in the *establishmentCause-r16* contained in the *RRCConnectionRequest-5GC-NB-r16-IEs* IE defined in TS 36.331 [21], or the reason for RRC Connection Resume as received from the UE in the *EstablishmentCause-NB-r13* IE defined in TS 36.331 [21], or the reason for Early Data Request as received from the UE in the *establishmentCause-r16* contained in the *RRCEarlyDataRequest-5GC-NB-r16-IEs* IE defined in TS 36.331 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC Establishment Cause	M		ENUMERATED (emergency, highPriorityAccess, mt-Access, mo-Signalling, mo-Data, mo-VoiceCall, mo-VideoCall, mo-SMS, mps-PriorityAccess, mcs-PriorityAccess, ..., notAvailable, mo-ExceptionData)	The <i>notAvailable</i> value is used in case the UE is re-establishing an RRC connection but there is fallback to RRC connection establishment as described in TS 38.331 [18], or the UE is resuming an RRC connection and the cause value received from the UE does not map to any other value of the <i>RRC Establishment Cause</i> IE.

9.3.1.112 Warning Area Coordinates

This IE contains the affected alert area coordinates of a warning message, and will be broadcast over the radio interface.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Warning Area Coordinates	M		OCTET STRING (SIZE(1..1024))	

9.3.1.113 Network Instance

This IE provides the network instance to be used by the NG-RAN node when selecting a particular transport network resource as described in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Network Instance	M		INTEGER (1..256, ...)	

9.3.1.114 Secondary RAT Usage Information

This IE provides information on the secondary resources used with MR-DC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session Usage Report		<i>0..1</i>		
>RAT Type	M		ENUMERATED (nR, e-UTRA, ..., nR-unlicensed, eUTRA-unlicensed)	
>PDU Session Timed Report List	M		Volume Timed Report List 9.3.1.115	
QoS Flows Usage Report List		<i>0..1</i>		
>QoS Flow Usage Report Item		<i>1..<maxnoofQoSFlows></i>		
>>QoS Flow Indicator	M		9.3.1.51	
>>RAT Type	M		ENUMERATED (nR, e-UTRA, ..., nR-unlicensed, eUTRA-unlicensed)	
>>QoS Flows Timed Report List	M		Volume Timed Report List 9.3.1.115	

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.1.115 Volume Timed Report List

This IE provides information on the data usage.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Volume Timed Report Item		<i>1..<maxnoofTimePeriods></i>		
>Start Timestamp	M		OCTET STRING (SIZE(4))	UTC time encoded in the same format as the first four octets of the 64-bit timestamp format as defined in section 6 of IETF RFC 5905 [25]. It indicates the start time of the collecting period of the included Usage Count UL IE and Usage Count DL IE.
>End Timestamp	M		OCTET STRING (SIZE(4))	UTC time encoded in the same format as the first four octets of the 64-bit timestamp format as defined in section 6 of IETF RFC 5905 [25]. It indicates the end time of the collecting period of the included Usage Count UL IE and Usage Count DL IE.
>Usage Count UL	M		INTEGER (0..2 ⁶⁴ -1)	The unit is: octets.
>Usage Count DL	M		INTEGER (0..2 ⁶⁴ -1)	The unit is: octets.

Range bound	Explanation
maxnoofTimePeriods	Maximum no. of time reporting periods. Value is 2.

9.3.1.116 Redirection for Voice EPS Fallback

This IE is used to indicate that the AMF and the UE support the redirection for voice for EPS Fallback.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Redirection for Voice EPS Fallback	M		ENUMERATED (possible, not-possible, ...)	

9.3.1.117 UE Retention Information

This IE allows the NG-RAN node and the AMF to indicate whether prior UE related contexts and related UE-associated logical NG-connections and RRC connections are intended to be retained.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Retention Information	M		ENUMERATED (ues-retained, ...)	

9.3.1.118 UL Forwarding

This IE indicates that the QoS flow is proposed for forwarding of uplink packets.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UL Forwarding	M		ENUMERATED (UL forwarding proposed, ...)	

9.3.1.119 CN Assisted RAN Parameters Tuning

This IE provides information for assisting in parameters tuning of the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Expected UE Behaviour	O		9.3.1.93	This IE may be present in case the <i>Core Network Assistance Information for RRC INACTIVE</i> IE is not included and is ignored otherwise.

9.3.1.120 Common Network Instance

This IE provides the common network instance to be used by the NG-RAN node when selecting a particular transport network resource as described in TS 23.501 [9] in a format common with 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Common Network Instance	M		OCTET STRING	The octets of OCTET STRING are encoded as the Network Instance field of the <i>Network Instance</i> IE specified in TS 29.244 [43]

9.3.1.121 Data Forwarding Response E-RAB List

This IE is used at inter-system HO to provide DL data forwarding address information, if direct data forwarding is applied.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Forwarding Response E-RAB List		<i>1..<maxnoofE-RABs></i>		The list may include the same DL Forwarding UP TNL Information for multiple E-RABs.
>E-RAB ID	M		9.3.2.3	
>DL Forwarding UP TNL Information	M		UP Transport Layer Information 9.3.2.2	

Range bound	Explanation
maxnoofE-RABs	Maximum no. of E-RABs. Value is 256.

9.3.1.122 gNB Set ID

The gNB Set ID IE is used to identify a group of gNBs which transmit the same RIM-RS.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB Set ID	M		BIT STRING (SIZE(22))	

9.3.1.123 RNC-ID

The RNC-ID is used to identify an RNC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RNC-ID	M		INTEGER (0..4095)	

9.3.1.124 Extended RNC-ID

This IE is used to identify an RNC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended RNC-ID	M		INTEGER (4096..65535)	This IE is used if the RNC identity has a value larger than 4095.

9.3.1.125 RAT Information

This IE provides RAT related information associated with a TAC, used as described in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAT Information	M		ENUMERATED (unlicensed, nb- IoT, ..., nR-LEO, nR-MEO, nR-GEO, nR-OTHERSAT)	

9.3.1.126 Extended RAT Restriction Information

This IE provides RAT restrictions as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Primary RAT Restriction	M		BIT STRING { e-UTRA (0), nR (1), nR-unlicensed (2), nR-LEO (3), nR-MEO (4), nR-GEO (5), nR-OTHERSAT (6), e-UTRA-LEO (7), e-UTRA-MEO (8), e-UTRA-GEO (9), e-UTRA-OTHERSAT (10)} (SIZE(8, ..., 16))	Each position in the bitmap represents a Primary RAT. If a bit is set to "1", the respective RAT is restricted for the UE. If a bit is set to "0", the respective RAT is not restricted for the UE. Bits 11-15 reserved for future use. The Primary RAT is the RAT used in the access cell, or target cell.
Secondary RAT Restriction	M		BIT STRING { e-UTRA (0), nR (1), e-UTRA-unlicensed (2), nR-unlicensed (3)} (SIZE(8, ...))	Each position in the bitmap represents a Secondary RAT. If a bit is set to "1", the respective RAT is restricted for the UE. If a bit is set to "0", the respective RAT is not restricted for the UE. Bits 4-7 reserved for future use. A Secondary RAT is a RAT, distinct from the UE's primary RAT, used in any cell serving the UE excluding the PCell.

9.3.1.127 SgNB UE X2AP ID

This IE uniquely identifies an UE over the X2 interface within an en-gNB. The usage of this IE is defined in TS 37.340 [32].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SgNB UE X2AP ID	M		INTEGER (0.. $2^{32}-1$)	

9.3.1.128 SRVCC Operation Possible

This IE indicates that both UE and AMF are SRVCC-capable. NG-RAN behaviour on receipt of this IE is specified in TS 23.216 [31].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SRVCC Operation Possible	M		ENUMERATED (Possible, not Possible, ...)	The value "Possible" indicates that UE and AMF are SRVCC capable.

9.3.1.129 IAB Authorized

This IE provides information about the authorization status of the IAB node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IAB Authorized	M		ENUMERATED (authorized, not authorized, ...)	Indicates the IAB node authorization status.

9.3.1.130 TSC Traffic Characteristics

This IE provides the traffic characteristics of TSC QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TSC Assistance Information Downlink	O		TSC Assistance Information 9.3.1.131	
TSC Assistance Information Uplink	O		TSC Assistance Information 9.3.1.131	

9.3.1.131 TSC Assistance Information

This IE provides the TSC assistance information for a TSC QoS flow in the uplink or downlink (see TS 23.501 [9]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Periodicity	M		9.3.1.132		-	
Burst Arrival Time	O		9.3.1.133		-	
Survival Time	O		9.3.1.221		YES	ignore
CHOICE <i>RAN Feedback Type</i>	O				YES	ignore
> <i>proactive</i>						
>>Burst Arrival Time Window	M		9.3.1.255		-	
>>Periodicity Range	O		9.3.1.256		-	
> <i>reactive</i>						
>>Capability for BAT Adaptation	M		ENUMERATED (true, ...)		-	
N6 Jitter Information	O		9.3.1.265		YES	ignore

9.3.1.132 Periodicity

This IE indicates the Periodicity of the TSC QoS flow as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodicity	M		INTEGER (0..640000, ...)	Periodicity expressed in units of 1 us.

9.3.1.133 Burst Arrival Time

This IE indicates the Burst Arrival Time of the TSC QoS flow as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Burst Arrival Time	M		OCTET STRING	Encoded in the same format as the <i>ReferenceTime</i> IE as defined in TS 38.331 [18]. The value is provided with 1 us accuracy.

9.3.1.134 Redundant QoS Flow Indicator

This IE provides the redundant QoS flow indicator for a QoS flow as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Redundant QoS Flow Indicator	M		ENUMERATED (true, false)	This IE indicates whether this QoS flow is requested for the redundant transmission. Value "true" indicates that redundant transmission is requested for this QoS flow. Value "false" indicates that redundant transmission is requested to be stopped if started.

9.3.1.135 Extended Packet Delay Budget

This IE indicates the Packet Delay Budget for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended Packet Delay Budget	M		INTEGER (1..65535, ..., 65536..109999)	Upper bound value for the delay that a packet may experience expressed in unit of 0.01ms.

9.3.1.136 Redundant PDU Session Information

This IE defines Redundancy information to be applied to a PDU session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
RSN	M		ENUMERATED (v1, v2, ...)		-	
PDU Session Pair ID	O		INTEGER (0..255, ...)	as defined in TS 23.501 [9]. This IE may be present in the request message and is ignored otherwise.	YES	ignore

9.3.1.137 NB-IoT Default Paging DRX

This IE indicates the NB-IoT Default Paging DRX as defined in TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NB-IoT Default Paging DRX	M		ENUMERATED (128, 256, 512, 1024, ...)	Unit: [number of radioframes]

9.3.1.138 NB-IoT Paging eDRX Information

This IE indicates the NB-IoT Paging eDRX parameters as defined in TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NB-IoT Paging eDRX Cycle	M		ENUMERATED (hf2, hf4, hf6, hf8, hf10, hf12, hf14, hf16, hf32, hf64, hf128, hf256, hf512, hf1024, ...)	T_{eDRX} defined in TS 36.304 [29]. Unit: [number of hyperframes].
NB-IoT Paging Time Window	O		ENUMERATED (s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, ...)	Unit: [2.56 seconds]

9.3.1.139 NB-IoT Paging DRX

This IE indicates the NB-IoT UE specific Paging DRX as defined in TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NB-IoT Paging DRX	M		ENUMERATED (32, 64, 128, 256, 512, 1024, ...)	Unit: [number of radioframes]

9.3.1.140 Enhanced Coverage Restriction

This IE provides information on the restriction information of using Coverage Enhancement.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Enhanced Coverage Restriction	M		ENUMERATED (restricted, ...)	Indicates whether the UE is restricted to use coverage enhancement. Value "restricted" indicates that the UE is not allowed to use coverage enhancement.

9.3.1.141 Paging Assistance Data for CE Capable UE

This IE provides Assistance Data for paging CE capable UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Global Cell ID	M		E-UTRA CGI 9.3.1.9	
Coverage Enhancement Level	M		OCTET STRING	Includes either the <i>UEPagingCoverageInformation</i> message as defined in TS 36.331 [21], or the <i>UEPagingCoverageInformation-NB</i> message as defined in TS 36.331 [21].

9.3.1.142 UE Radio Capability ID

This IE contains the UE Radio Capability ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Radio Capability ID	M		OCTET STRING	Defined in TS 23.003 [23].

9.3.1.143 WUS Assistance Information

This IE provides WUS Assistance Information to be used by the NG-RAN node for determining the WUS group for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Probability Information	M		ENUMERATED (p00, p05, p10, p15, p20, p25, p30, p35, p40, p45, p50, p55, p60, p65, p70, p75, p80, p85, p90, p95, p100, ...)	Unit: percentage

9.3.1.144 UE Differentiation Information

This IE is generated by the AMF based on the UE subscription information, it provides the Expected UE Behavior Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodic Communication Indicator	O		ENUMERATED (periodically, on demand, ...)	This IE indicates whether the UE communicates periodically or not, e.g. only on demand.
Periodic Time	O		INTEGER (1..3600, ...)	This IE indicates the interval time of periodic communication, the unit is: second
Scheduled Communication Time		0..1		This IE indicates the time zone and day of the week when the UE is available for communication.
>Day of Week	O		BIT STRING (SIZE(7))	Each position in the bitmap represents a day of the week: first bit = Mon, second bit =Tue, third bit =Wed, and so on. Value '1' indicates 'scheduled. Value '0' indicates 'not scheduled'. If Day-Of-Week is not provided, this is interpreted as every day of the week.
>Time of Day Start	O		INTEGER (0..86399, ...)	This IE indicates the time to start of the day, each value represents the corresponding second since 00:00 of the day. If Time-Of-Day-Start is not provided, starting time is start of the day(s) indicated by Day-Of-Week-Mask.
>Time of Day End	O		INTEGER (0..86399, ...)	This IE indicates the time to start of the day, each value represents the corresponding second since 00:00 of the day. The value of this IE should be bigger than the value of Time of Day Start IE. If Time-Of-Day-End is not provided, ending time is end of the day(s) indicated by Day-Of-Week-Mask.
Stationary Indication	O		ENUMERATED (stationary, mobile, ...)	
Traffic Profile	O		ENUMERATED (single packet, dual packets, multiple packets, ...)	"single packet" indicates single packet transmission (UL or DL), "dual packets" indicates dual packet transmission (UL with subsequent DL, or DL with subsequent UL), "multiple packets" indicates multiple packets transmission.
Battery Indication	O		ENUMERATED (battery powered, battery powered not rechargeable or replaceable, not battery powered, ...)	"battery powered" indicates that the UE is battery powered and the battery is rechargeable/replaceable, "battery powered not rechargeable or replaceable" indicates that the UE is battery powered but the battery is not rechargeable/replaceable, "not battery powered" indicates that the UE is not battery powered.

9.3.1.145 NB-IoT UE Priority

This IE provides the NB-IoT UE Priority and to be used by the NG-RAN to prioritise between UEs accessing via NB-IoT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NB-IoT UE Priority	M		INTEGER (0..255, ...)	Lower value indicates higher priority.

9.3.1.146 NR V2X Services Authorized

This IE provides information on the authorization status of the UE to use the NR sidelink for V2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Vehicle UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Vehicle UE
Pedestrian UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Pedestrian UE

9.3.1.147 LTE V2X Services Authorized

This IE provides information on the authorization status of the UE to use the LTE sidelink for V2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Vehicle UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Vehicle UE
Pedestrian UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Pedestrian UE

9.3.1.148 NR UE Sidelink Aggregate Maximum Bit Rate

This IE provides information on the Aggregate Maximum Bitrate of the UE's sidelink communication.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR UE Sidelink Aggregate Maximum Bit Rate	M		Bit Rate 9.3.1.4	Value 0 is not valid, and considered as a logical error by the receiving NG-RAN node.

9.3.1.149 LTE UE Sidelink Aggregate Maximum Bit Rate

This IE provides information on the Aggregate Maximum Bitrate of the UE's sidelink communication for LTE V2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LTE UE Sidelink Aggregate Maximum Bit Rate	M		Bit Rate 9.3.1.4	Value 0 is not valid, and considered as a logical error by the receiving NG-RAN node.

9.3.1.150 PC5 QoS Parameters

This IE provides information on the PC5 QoS parameters of the UE's sidelink communication for NR PC5.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PC5 QoS Flow List		<i>1</i>		
>PC5 QoS Flow Item		<i>1..<maxnoofPC5QoSFlows></i>		
>>PQI	M		INTEGER (0..255, ...)	PQI is a special 5QI as specified in TS 23.501 [9].
>>PC5 Flow Bit Rates		<i>0..1</i>		Only applies for GBR QoS Flows.
>>>Guaranteed Flow Bit Rate	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>>Maximum Flow Bit Rate	M		Bit Rate 9.3.1.4	Maximum Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>Range	O		ENUMERATED (m50, m80, m180, m200, m350, m400, m500, m700, m1000, ...)	Only applies for groupcast.
PC5 Link Aggregate Bit Rates	O		Bit Rate 9.3.1.4	Only applies for non-GBR QoS Flows.

Range bound	Explanation
maxnoofPC5QoSFlows	Maximum no. of PC5 QoS flows allowed towards one UE. Value is 2048.

9.3.1.151 Alternative QoS Parameters Set List

This IE contains alternative sets of QoS parameters which the NG-RAN node can indicate to be fulfilled when notification control is enabled and it cannot fulfil the requested list of QoS parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Alternative QoS Parameters Set Item		<i>1..<maxno ofQoSparaSets></i>			-	
>Alternative QoS Parameters Set Index	M		9.3.1.152		-	
>Guaranteed Flow Bit Rate Downlink	O		Bit Rate 9.3.1.4		-	
>Guaranteed Flow Bit Rate Uplink	O		Bit Rate 9.3.1.4		-	
>Packet Delay Budget	O		9.3.1.80		-	
>Packet Error Rate	O		9.3.1.81		-	
>Maximum Data Burst Volume	O		9.3.1.83	Maximum Data Burst Volume is specified in TS 23.501 [9]. This IE may be included if the <i>Delay Critical</i> IE is set to "delay critical" and is ignored otherwise.	YES	ignore
>PDU Set Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	This IE is included if the <i>PDU Set Packet Delay Budget</i> IE in the <i>DL PDU Set QoS Information</i> IE is present and is ignored otherwise.	YES	ignore
>PDU Set Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	This IE is included if the <i>PDU Set Packet Delay Budget</i> IE in the <i>UL PDU Set QoS Information</i> IE is present and is ignored otherwise.	YES	ignore
>PDU Set Error Rate Downlink	O		Packet Error Rate 9.3.1.81	This IE is included if the <i>PDU Set Error Rate</i> IE in the <i>DL PDU Set QoS Information</i> IE is present and is ignored otherwise.	YES	ignore
>PDU Set Error Rate Uplink	O		Packet Error Rate 9.3.1.81	This IE is included if the <i>PDU Set Error Rate</i> IE in the <i>UL PDU Set QoS Information</i> IE is present and is ignored otherwise.	YES	ignore

Range bound	Explanation
maxnoofQoSparaSets	Maximum no. of alternative sets of QoS Parameters allowed for the QoS profile. Value is 8.

9.3.1.152 Alternative QoS Parameters Set Index

This IE indicates the index of alternative QoS parameters set.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Alternative QoS Parameters Set Index	M		INTEGER (1..8, ...)	Values are ordered in decreasing order of priority, i.e., with 1 as the highest priority and 8 as the lowest priority.

9.3.1.153 Alternative QoS Parameters Set Notify Index

This IE indicates the QoS parameters set which can currently be fulfilled.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Alternative QoS Parameters Set Notify Index	M		INTEGER (0..8, ...)	Indicates the index of the item within the <i>Alternative QoS Parameters Set List</i> IE corresponding to the currently fulfilled alternative QoS parameters set. Value 0 indicates that NG-RAN cannot even fulfil the lowest priority alternative parameters set.

9.3.1.154 E-UTRA Paging eDRX Information

This IE indicates the E-UTRA Paging eDRX parameters as defined in TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-UTRA Paging eDRX Cycle	M		ENUMERATED (hfhalf, hf1, hf2, hf4, hf6, hf8, hf10, hf12, hf14, hf16, hf32, hf64, hf128, hf256, ...)	TeDRX defined in TS 36.304 [29]. Unit: [number of hyperframes].
E-UTRA Paging Time Window	O		ENUMERATED (s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, ...)	Unit: [1.28 second].

9.3.1.155 CE-mode-B Restricted

This IE provides information on the restriction information of using Coverage Enhancement Mode B.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CE-mode-B Restricted	M		ENUMERATED (restricted, not-restricted, ...)	Indicates whether the UE is restricted to use coverage enhancement. Value "restricted" indicates that the UE is not allowed to use coverage enhancement mode B. Value "not-restricted" indicates that the UE is allowed to use coverage enhancement mode B.

9.3.1.156 CE-mode-B Support Indicator

This IE indicates whether CE-mode-B as specified in TS 36.306[42] is supported for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CE-mode-B Support Indicator	M		ENUMERATED (supported, ...)	

9.3.1.157 LTE-M Indication

This IE is provided by the NG-RAN node to inform that the UE indicates category M1 or M2 in its UE Radio Capability.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LTE-M Indication	M		ENUMERATED (LTE-M, ...)	

9.3.1.158 Suspend Request Indication

This IE indicates that the NG-RAN node requests immediate transition to RRC idle with suspend, as specified in TS 23.502 [10].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Suspend Request Indication	M		ENUMERATED (suspend-requested, ...)	

9.3.1.159 Suspend Response Indication

This IE is used by the AMF to inform the NG-RAN node to suspend the UE and send it to RRC_IDLE, as specified in TS 23.502 [10].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Suspend Response Indication	M		ENUMERATED (suspend-indicated, ...)	

9.3.1.160 UE User Plane CIoT Support Indicator

This IE indicates whether User Plane CIoT 5GS Optimisation as specified in TS 23.501 [9] is supported for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE User Plane Clot Support Indicator	M		ENUMERATED (supported, ...)	

9.3.1.161 Global TNGF ID

This IE is used to globally identify a TNGF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE <i>TNGF ID</i>	M			
> <i>TNGF ID</i>				
>>TNGF ID	M		BIT STRING (SIZE(32, ...))	

9.3.1.162 Global W-AGF ID

This IE is used to globally identify a W-AGF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE <i>W-AGF ID</i>	M			
> <i>W-AGF ID</i>				
>>W-AGF ID	M		BIT STRING (SIZE(16, ...))	

9.3.1.163 Global TWIF ID

This IE is used to globally identify a TWIF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE <i>TWIF ID</i>	M			
> <i>TWIF ID</i>				
>>TWIF ID	M		BIT STRING (SIZE(32, ...))	

9.3.1.164 W-AGF User Location Information

This IE indicates the location information via wireline access as specified in TS 23.316 [34].

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>W-AGF User Location Information</i>	M				-	
> <i>Global Line ID</i>					-	
>>Global Line ID	M		OCTET STRING	Encoded as defined in TS 23.003 [23].	-	
>>Line Type	O		ENUMERATED (DSL, PON, ...)		-	
>>TAI	O		9.3.3.11		YES	ignore
> <i>HFC Node ID</i>					-	
>>HFC Node ID	M		OCTET STRING	Indicates the identifier of the HFC node as specified in [37]. Encoded as defined in TS 23.003 [23].	-	
> <i>Global Cable ID</i>					YES	ignore
>>Global Cable ID	M		OCTET STRING	Encoded as defined in TS 23.003 [23].	-	
> <i>HFC Node ID New</i>					YES	ignore
>>HFC Node ID	M		OCTET STRING	Indicates the identifier of the HFC node as specified in [37]. Encoded as defined in TS 23.003 [23].	-	
>>TAI	M		9.3.3.11		-	
>>Global Cable ID	O		OCTET STRING	Encoded as defined in TS 23.003 [23].	YES	ignore
> <i>Global Cable ID New</i>					YES	ignore
>>Global Cable ID	M		OCTET STRING	Encoded as defined in TS 23.003 [23].	-	
>>TAI	M		9.3.3.11		-	

9.3.1.165 Global eNB ID

This IE is used to globally identify an eNB (see TS 36.401 [38]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE <i>eNB ID</i>	M			
> <i>Macro eNB ID</i>				
>>Macro eNB ID	M		BIT STRING (SIZE(20))	Equal to the 20 leftmost bits of the <i>E-UTRA Cell Identity</i> IE contained in the <i>E-UTRA CGI</i> IE of each cell served by the eNB.
> <i>Home eNB ID</i>				
>>Home eNB ID	M		BIT STRING (SIZE(28))	Equal to the <i>E-UTRA Cell Identity</i> IE contained in the <i>E-UTRA CGI</i> IE of the cell served by the eNB.
> <i>Short Macro eNB ID</i>				
>>Short Macro eNB ID	M		BIT STRING (SIZE(18))	Equal to the 18 leftmost bits of the <i>E-UTRA Cell Identity</i> IE contained in the <i>E-UTRA CGI</i> IE of each cell served by the eNB.
> <i>Long Macro eNB ID</i>				
>>Long Macro eNB ID	M		BIT STRING (SIZE(21))	Equal to the 21 leftmost bits of the <i>E-UTRA Cell Identity</i> IE contained in the <i>E-UTRA CGI</i> IE of each cell served by the eNB.

9.3.1.166 UE History Information from UE

This IE contains information about mobility history report for a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>UE History Information from UE</i>	M			
> <i>NR</i>				
>>NR Mobility History Report	M		OCTET STRING	Includes the <i>mobilityHistoryReport</i> contained in the <i>UEInformationResponse</i> message as defined in TS 38.331 [18].

9.3.1.167 MDT Configuration

This IE defines the MDT configuration parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MDT Configuration-NR	O		9.3.1.169		-	
MDT Configuration-EUTRA	O		9.3.1.170		-	
MN Only MDT Collection	O		ENUMERATED (MN only, ...)	For MR-DC configurations, indicates that the <i>MDT Configuration-NR</i> IE or the <i>MDT Configuration-EUTRA</i> IE only applies to the MN.	YES	ignore

9.3.1.168 MDT PLMN List

The purpose of the *MDT PLMN List* IE is to provide the list of PLMN allowed for MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT PLMN List		1..<maxnoofMDTPLMNs>		
>PLMN Identity	M		9.3.3.5	

Range bound	Explanation
maxnoofMDTPLMNs	Maximum no. of PLMNs in the MDT PLMN list. Value is 16.

9.3.1.169 MDT Configuration-NR

This IE defines the MDT configuration parameters of NR.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MDT Activation	M		ENUMERATED (Immediate MDT only, Logged MDT only, Immediate MDT and Trace, ...)		-	
CHOICE Area Scope of MDT	M				-	
>Cell based				If <i>PNI-NPN Area Scope of MDT</i> IE is present, this IE covers non-CAG cells only, where non-CAG cells refer to cells that only provide public access.		
>>Cell ID List for MDT		1..<maxno ofCellIDfor MDT>			-	
>>>NR CGI	M		9.3.1.7		-	
>TA based				If <i>PNI-NPN Area Scope of MDT</i> IE is present, this IE covers non-CAG cells only, where non-CAG cells refer to cells that only provide public access.		
>>TA List for MDT		1..<maxno ofTAforMDT>			-	
>>>TAC	M		9.3.3.10	The TAI is derived using the current serving PLMN.	-	
>PLMN wide			NULL			
>TAI based				If <i>PNI-NPN Area Scope of MDT</i> IE is present, this IE covers non-CAG cells only, where non-CAG cells refer to cells that only provide public access.		
>>TAI List for MDT		1..<maxno ofTAforMDT>			-	
>>>TAI	M		9.3.3.11		-	
>PNI-NPN Based MDT					YES	ignore
>>CAG List for MDT			9.3.3.65		-	
>SNPN Cell Based MDT					YES	ignore
>>SNPN Cell ID List for MDT		1..<maxno ofCellIDfor MDT>			-	
>>>NR CGI	M		9.3.1.7		-	
>>>NID	M		9.3.3.42	Identifies an SNPN together with the PLMN Identity in the <i>NR CGI</i> IE.	-	
>SNPN TAI Based MDT					YES	ignore

>>SNPN TAI List for MDT		1..<maxno ofTAforMDT>			-	
>>>TAI	M		9.3.3.11		-	
>>>NID	M		9.3.3.42	Identifies an SNPN together with the PLMN Identity in the TAI IE.	-	
>SNPN Based MDT					YES	ignore
>>SNPN List for MDT		1..<maxno ofMDTSNPNs>			-	
>>>PLMN Identity	M		9.3.3.5		-	
>>>NID	M		9.3.3.42	Identifies an SNPN together with the PLMN Identity IE.	-	
>Geography Based				The geographical area can be used with NTN deployment.	YES	ignore
>>Geographical Area	M		9.3.3.68		-	
CHOICE MDT Mode	M				-	
>Immediate MDT						
>>Measurements to Activate	M		BITSTRING (SIZE(8))	Each position in the bitmap indicates a MDT measurement, as defined in TS 37.320 [41]. First Bit = M1, Second Bit = M2, Third Bit = M4, Fourth Bit = M5, Fifth Bit = M6, Sixth Bit = M7, Seventh Bit = logging of M1 from event triggered measurement reports according to existing RRM configuration, other bits reserved for future use. Value "1" indicates "activate" and value "0" indicates "do not activate".	-	
>>M1 Configuration	C-ifM1		9.3.1.171		-	
>>M4 Configuration	C-ifM4		9.3.1.172		-	
>>M5 Configuration	C-ifM5		9.3.1.173		-	
>>M6 Configuration	C-ifM6		9.3.1.174		-	
>>M7 Configuration	C-ifM7		9.3.1.175		-	
>>Bluetooth Measurement Configuration	O		9.3.1.177		-	
>>WLAN Measurement Configuration	O		9.3.1.178		-	
>>MDT Location Information	O		9.3.1.176		-	
>>Sensor Measurement Configuration	O		9.3.1.179		-	
>Logged MDT						

>>Logging Interval	M		ENUMERATED (320ms, 640ms, 1280ms, 2560ms, 5120ms, 10240ms, 20480ms, 30720ms, 40960ms, 61440ms, infinity, ...)	Corresponds to the <i>LoggingInterval</i> IE as defined in TS 38.331 [18].	-	
>>Logging Duration	M		ENUMERATED (10, 20, 40, 60, 90, 120, ...)	Corresponds to the <i>LoggingDuration</i> IE as defined in TS 38.331 [18]. Unit: [minute].	-	
>>CHOICE Report Type	M				-	
>>>Periodical			NULL			
>>>Event Triggered						
>>>>Event Trigger Logged MDT Configuration	M		9.3.1.180		-	
>>Bluetooth Measurement Configuration	O		9.3.1.177		-	
>>WLAN Measurement Configuration	O		9.3.1.178		-	
>>Sensor Measurement Configuration	O		9.3.1.179		-	
>>Area Scope of Neighbour Cells	O		9.3.1.182		-	
>>Early Measurement	O		ENUMERATED (true, ...)	This IE indicates whether the UE is allowed to log measurements on early measurement related frequencies in logged MDT as specified in TS 38.331 [18].	YES	ignore
Signalling Based MDT PLMN List	O		MDT PLMN List 9.3.1.168		-	
PNI-NPN Area Scope of MDT		0..1		This IE is ignored if the <i>Area Scope of MDT</i> IE is set to "PLMN Wide"	YES	ignore
>CAG List for MDT	M		9.3.3.65		-	
Network Slice Area Scope of MDT	O		9.3.3.67		YES	ignore

Range bound	Explanation
maxnoofCellIDforMDT	Maximum no. of Cell ID subject for MDT scope. Value is 32.
maxnoofTAforMDT	Maximum no. of TA subject for MDT scope. Value is 8.
maxnoofMDTSNPNS	Maximum no. of SNPNs in the MDT SNPN list. Value is 16.

Condition	Explanation
C-ifM1	This IE shall be present if the <i>Measurements to Activate</i> IE has the first bit set to "1".
C-ifM4	This IE shall be present if the <i>Measurements to Activate</i> IE has the third bit set to "1".
C-ifM5	This IE shall be present if the <i>Measurements to Activate</i> IE has the fourth bit set to "1".
C-ifM6	This IE shall be present if the <i>Measurements to Activate</i> IE has the fifth bit set to "1".
C-ifM7	This IE shall be present if the <i>Measurements to Activate</i> IE has the sixth bit set to "1".

9.3.1.170 MDT Configuration-EUTRA

This IE defines the MDT configuration parameters of EUTRA.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT Activation	M		ENUMERATED (Immediate MDT only, Logged MDT only, Immediate MDT and Trace, ...)	
CHOICE <i>Area Scope of MDT</i>	M			
> <i>Cell based</i>				
>>Cell ID List for MDT		1.. <i>maxnoofCellIDforMDT</i>		
>>>E-UTRA CGI	M		9.3.1.9	
> <i>TA based</i>				
>>TA List for MDT		1.. <i>maxnoofTAforMDT</i>		
>>>TAC	M		9.3.3.10	The TAI is derived using the current serving PLMN.
> <i>PLMN wide</i>			NULL	
> <i>TAI based</i>				
>>TAI List for MDT		1.. <i>maxnoofTAforMDT</i>		
>>>TAI	M		9.3.3.11	
MDT Mode	M		OCTET STRING	<i>MDTMode</i> IE defined in TS 36.413 [16].
Signalling Based MDT PLMN List	O		MDT PLMN List 9.3.1.168	

Range bound	Explanation
maxnoofCellIDforMDT	Maximum no. of Cell ID subject for MDT scope. Value is 32.
maxnoofTAforMDT	Maximum no. of TA subject for MDT scope. Value is 8.

9.3.1.171 M1 Configuration

This IE defines the parameters for M1 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M1 Reporting Trigger	M		ENUMERATED (periodic, A2event-triggered, A2event-triggered periodic, ...)		-	
M1 Threshold Event A2	C-ifM1A2trigger				-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>CHOICE <i>Threshold Type</i>	M				-	
>>RSRP						
>>>Threshold RSRP	M		INTEGER (0..127)	Corresponds to the <i>RSRP-Range</i> IE as defined in TS 38.331 [18].	-	
>>RSRQ						
>>>Threshold RSRQ	M		INTEGER (0..127)	Corresponds to the <i>RSRQ-Range</i> IE as defined in TS 38.331 [18].	-	
>>SINR						
>>>Threshold SINR	M		INTEGER (0..127)	Corresponds to the <i>SINR-Range</i> IE as defined in TS 38.331 [18].	-	
M1 Periodic Reporting	C- ifperiodic MDT				-	
>Report Interval	M		ENUMERATED (ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, min1, min6, min12, min30, min60)	Corresponds to the <i>ReportInterval</i> IE as defined in TS 38.331 [18]. The value min60 is not used in the specification.	-	
>Report Amount	M		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity)	Number of reports.	-	
>Extended Report Interval	O		ENUMERATED (ms20480, ms40960, ...)	This IE is the extension of the <i>Report Interval</i> IE. If this IE is present, the <i>Report Interval</i> IE is ignored.	YES	ignore
Include Beam Measurements Indication	O		ENUMERATED (true, ...)	To configure whether the UE should include beam level measurements.	YES	ignore
Beam Measurements Report Configuration	C- ifM1Bea mMeasIn d				YES	ignore
>Beam Measurements Report Quantity		0..1		This IE indicates the beam measurement quantity and corresponds to the <i>MeasReportQuantity</i> IE as defined in TS 38.331 [18].	-	
>>RSRP	M		ENUMERATED (true, ..., false)		-	
>>RSRQ	M		ENUMERATED (true, ..., false)		-	
>>SINR	M		ENUMERATED (true, ..., false)		-	
>MaxNrofRS-IndexesTo Report	O		INTEGER (1..64, ...)	Indicates the max number of beam measurements to be reported and corresponds to the	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
				<i>maxNrofRS-IndexesToReport</i> in the <i>ReportConfigNR</i> IE as defined in TS 38.331 [18].		

Condition	Explanation
C-ifM1A2trigger	This IE shall be present if the <i>M1 Reporting Trigger</i> IE is set to the value "A2event-triggered" or "A2event-triggered periodic".
C-ifperiodicMDT	This IE shall be present if the <i>M1 Reporting Trigger</i> IE is set to the value "periodic" or "A2event-triggered periodic".
C-ifM1BeamMeasInd	This IE shall be present if the <i>Include Beam Measurements Indication</i> IE is set to the value "true".

9.3.1.172 M4 Configuration

This IE defines the parameters for M4 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M4 Collection Period	M		ENUMERATED (ms1024, ms2048, ms5120, ms10240, min1, ...)		-	
M4 Links to Log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, ...)		-	
M4 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, ...)	Number of reports.	YES	ignore

9.3.1.173 M5 Configuration

This IE defines the parameters for M5 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M5 Collection Period	M		ENUMERATED (ms1024, ms2048, ms5120, ms10240, min1, ...)		-	
M5 Links to Log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, ...)		-	
M5 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, ...)	Number of reports	YES	ignore

9.3.1.174 M6 Configuration

This IE defines the parameters for M6 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M6 Report Interval	M		ENUMERATED (ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, ms20480, ms40960, min1, min6, min12, min30, ...)		-	
M6 Links to Log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, ...)		-	
M6 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, ...)	Number of reports	YES	ignore
Excess Packet Delay Threshold Configuration	O		9.3.1.244		YES	ignore

9.3.1.175 M7 Configuration

This IE defines the parameters for M7 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M7 Collection Period	M		INTEGER (1..60, ...)	Unit: minutes	-	
M7 Links to Log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, ...)		-	
M7 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, ...)	Number of reports	YES	ignore

9.3.1.176 MDT Location Information

This IE defines the MDT Location Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT Location Information	M		BITSTRING (SIZE(8))	Each position in the bitmap represents requested location information as defined in TS 37.320 [41]. First Bit = GNSS Other bits are reserved for future use and are ignored if received. Value "1" indicates "activate" and value "0" indicates "do not activate".

9.3.1.177 Bluetooth Measurement Configuration

This IE defines the parameters for Bluetooth measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Bluetooth Measurement Configuration	M		ENUMERATED (Setup, ...)	
Bluetooth Measurement Configuration Name List		0..1		This IE is present if the <i>Bluetooth Measurement Configuration</i> IE is set to "Setup".
>Bluetooth Measurement Configuration Name Item		1..<maxnoofBluetoothName>		
>>Bluetooth Measurement Configuration Name	M		OCTET STRING (SIZE (1..248))	
BT RSSI	O		ENUMERATED (true, ...)	In case of Immediate MDT, it corresponds to M8 measurement as defined in TS 37.320 [41].

Range bound	Explanation
maxnoofBluetoothName	Maximum no. of Bluetooth local name used for Bluetooth measurement collection. Value is 4.

9.3.1.178 WLAN Measurement Configuration

This IE defines the parameters for WLAN measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
WLAN Measurement Configuration	M		ENUMERATED (Setup, ...)	
WLAN Measurement Configuration Name List		0..1		This IE is present if the <i>Bluetooth Measurement Configuration</i> IE is set to "Setup".
>WLAN Measurement Configuration Name Item		1..<maxnoofWLANName>		
>>WLAN Measurement Configuration Name	M		OCTET STRING (SIZE (1..32))	
WLAN RSSI	O		ENUMERATED (true, ...)	In case of Immediate MDT, it corresponds to M8 as defined in TS 37.320 [41].
WLAN RTT	O		ENUMERATED (true, ...)	In case of Immediate MDT, it corresponds to M9 as defined in TS 37.320 [41].

Range bound	Explanation
maxnoofWLANName	Maximum no. of WLAN SSID used for WLAN measurement collection. Value is 4.

9.3.1.179 Sensor Measurement Configuration

This IE defines the parameters for Sensor measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Sensor Measurement Configuration	M		ENUMERATED (Setup, ...)	
Sensor Measurement Configuration Name List		<i>0..1</i>		
>Sensor Measurement Configuration Name Item		<i>1..<maxnoofSensorName></i>		
>>CHOICE <i>Sensor Name</i>	M			
>>>Uncompensated Barometric				
>>>>Uncompensated Barometric Configuration	M		ENUMERATED (true, ...)	
>>>UE speed				
>>>>UE Speed Configuration	M		ENUMERATED (true, ...)	
>>>UE orientation				
>>>>UE orientation Configuration	M		ENUMERATED (true, ...)	

Range bound	Explanation
maxnoofSensorName	Maximum no. of Sensor local name used for Sensor measurement collection. Value is 3

9.3.1.180 Event Trigger Logged MDT Configuration

This IE defines the event trigger logged MDT configuration.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Event trigger type</i>	M			
>Out-of-coverage				
>>Out-of-Coverage Configuration	M		ENUMERATED (true, ...)	
>L1 Event				
>>CHOICE <i>L1 Event Threshold</i>	M			
>>>RSRP				
>>>>Threshold RSRP	M		INTEGER (0..127)	Corresponds to the <i>RSRP-Range</i> IE as defined in TS 38.331 [18].
>>>RSRQ				
>>>>Threshold RSRQ	M		INTEGER (0..127)	Corresponds to the <i>RSRQ-Range</i> IE as defined in TS 38.331 [18].
>>Hysteresis	M		INTEGER (0..30)	This parameter is used within the entry and leave condition of an event triggered reporting condition.
>>Time to Trigger	M		ENUMERATED (ms0, ms40, ms64, ms80, ms100, ms128, ms160, ms256, ms320, ms480, ms512, ms640, ms1024, ms1280, ms2560, ms5120)	Time during which specific criteria for the event needs to be met in order to trigger a measurement report.

9.3.1.181 NR Frequency Info

This defines the carrier frequency and bands used in a cell for a given direction (UL or DL) in FDD or for both UL and DL directions in TDD.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
NR ARFCN	M		INTEGER (0.. maxNRARFCN)	RF Reference Frequency as defined in TS 38.104 [39], section 5.4.2.1. The frequency provided in this IE identifies the absolute frequency position of the reference resource block (Common RB 0) of the carrier. Its lowest subcarrier is also known as Point A.
NR Frequency Band List		1		This IE is not used in this specification and is ignored.
>NR Frequency Band Item		1..<maxnoofNRCellBands>		
>>NR Frequency Band	M		INTEGER (1.. 1024, ...)	Primary NR Operating Band as defined in TS 38.104 [39], section 5.4.2.3. The value 1 corresponds to n1, value 2 corresponds to NR operating band n2, etc.

Range bound	Explanation
maxNRARFCN	Maximum value of NRARFCNs. Value is 3279165.
maxnoofNRCellBands	Maximum no. of frequency bands supported for a NR cell. Value is 32.

9.3.1.182 Area Scope of Neighbour Cells

This IE defines the area scope of neighbour cells for logged MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Area Scope of Neighbour Cells Item	M	1..<maxnoofFreqforMDT>		
>NR Frequency Info	M		9.3.1.181	
>PCI List for MDT		0..<maxnoofNeighPCIforMDT>		
>>NR PCI	M		INTEGER (0..1007, ...)	NR Physical Cell ID

Range bound	Explanation
maxnoofFreqforMDT	Maximum no. of Frequency Information subject for MDT scope. Value is 8.
maxnoofNeighPCIforMDT	Maximum no. of Neighbour cells subject for MDT scope. Value is 32.

9.3.1.183 NPN Paging Assistance Information

This IE contains NPN Paging Assistance Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>NPN Paging Assistance Information</i>	M			
> <i>PNI-NPN Paging Assistance</i>				
>> <i>PNI-NPN Paging Assistance</i>	M		Allowed PNI-NPN List 9.3.3.45	

9.3.1.184 NPN Mobility Information

This IE indicates the access restrictions related to an NPN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>NPN Mobility Information</i>	M				-	
> <i>SNPN Mobility Information</i>					-	
>>Serving NID	M		NID 9.3.3.42		-	
>>Equivalent SNPNs List		0..1			YES	reject
>>>Equivalent SNPNs Item		1..<maxno ofESNPNs>		Allowed SNPNs in addition to Serving SNPN. This list corresponds to the list of "equivalent SNPNs" as defined in TS 24.501 [26]. This list is part of the roaming restriction information. Roaming restrictions apply to SNPNs other than the Serving SNPN and Equivalent SNPNs.	-	
>>>>PLMN Identity	M		9.3.3.5		-	
>>>>NID	M		9.3.3.42		-	
> <i>PNI-NPN Mobility Information</i>					-	
>>Allowed PNI-NPN List	M		9.3.3.45		-	

Range bound	Explanation
maxnoofESNPNs	Maximum no. of equivalent SNPNs. Value is 15.

9.3.1.185 Cell CAG Information

This IE provides information about support of closed access groups for a designated cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NG-RAN CGI	M		9.3.1.73	
Cell CAG List	M		9.3.3.47	

9.3.1.186 Target to Source Failure Transparent Container

This IE is used to transparently pass radio related information from the handover target to the handover source through the core network in case of failure of the preparation at the target; it is produced by the target RAN node and is transmitted to the source RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target to Source Failure Transparent Container	M		OCTET STRING	This IE includes a transparent container from the target RAN node to the source RAN node. The octets of the OCTET STRING are encoded according to the specifications of the target system. Note: In the current version of the specification, this IE may carry the <i>Target NG-RAN Node to Source NG-RAN Node Failure Transparent Container</i> IE.

9.3.1.187 Target NG-RAN Node to Source NG-RAN Node Failure Transparent Container

This IE is produced by the target NG-RAN node and is transmitted to the source NG-RAN node in case of preparation failure.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Cell CAG Information	O		9.3.1.185		-	
NGAP IE Support Information Response List	O		9.3.1.242		YES	ignore

9.3.1.188 DAPS Request Information

The *DAPS Indicator* IE indicates that the source NG-RAN node requests a DAPS Handover for the concerned DRB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DAPS Indicator	M		ENUMERATED (DAPS HO required, ...)	Indicates that DAPS Handover is requested

9.3.1.189 DAPS Response Information

The *DAPS Response Indicator* IE indicates the per DRB response to a requested DAPS Handover.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DAPS Response Indicator	M		ENUMERATED (DAPS HO accepted, DAPS HO not accepted, ...)	Indicates that DAPS Handover is accepted or not

9.3.1.190 Early Status Transfer Transparent Container

The *Early Status Transfer Transparent Container* IE is an information element that is produced by the source NG-RAN node and is transmitted to the target NG-RAN node. This IE is used for the NG DAPS handover case, and for the NG-based handover with time-based trigger condition case.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>Procedure Stage</i>	M				-	
> <i>First DL COUNT</i>						
>>DRBs Subject To Early Status Transfer List		1			-	
>>>DRBs Subject To Early Status Transfer Item		1..<maxno ofDRBs>			-	
>>>>DRB ID	M		9.3.1.53		-	
>>>>CHOICE <i>First DL COUNT</i>	M				-	
>>>>>12 bits						
>>>>>FIRST DL COUNT Value	M		COUNT Value for PDCP SN Length 12 9.3.1.109	PDCP-SN and Hyper frame number of the first DL SDU that the source NG-RAN node forwards to the target NG-RAN node in case of 12 bit long PDCP-SN	-	
>>>>>18 bits						
>>>>>>FIRST DL COUNT Value	M		COUNT Value for PDCP SN Length 18 9.3.1.110	PDCP-SN and Hyper frame number of the first DL SDU that the source NG-RAN node forwards to the target NG-RAN node in case of 18 bit long PDCP-SN	-	
> <i>DL Discarding</i>					YES	ignore
>>DRBs Subject To DL Discarding List		1			-	
>>>DRBs Subject To DL Discarding Item		1 .. <maxno of DRBs>			-	
>>>>DRB ID	M		9.3.1.53		-	
>>>>CHOICE <i>DL Discarding</i>	M				-	
>>>>>12 bits						
>>>>>>DISCARD DL COUNT Value	M		COUNT Value for PDCP SN Length 12 9.3.1.109	PDCP-SN and Hyper frame number for which the receiving NG-RAN node should discard forwarded DL SDUs associated with lower values in case of 12 bit long PDCP-SN	-	
>>>>>>18 bits						
>>>>>>>DISCARD DL COUNT Value	M		COUNT Value for PDCP SN Length 18	PDCP-SN and Hyper frame number for which	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
			9.3.1.110	the receiving NG-RAN node should discard forwarded DL SDUs associated with lower values in case of 18 bit long PDCP-SN		

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.

9.3.1.191 Extended Slice Support List

This IE indicates a list of supported slices.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Support Item		<i>1..<maxnoofExtSlicelItems></i>		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofExtSlicelItems	Maximum no. of signalled slice support items. Value is 65535.

9.3.1.192 UE Capability Info Request

This IE indicates the request to provide to the AMF the UE radio capability related information when retrieved from the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Capability Info Request	M		ENUMERATED (requested, ...)	

9.3.1.193 Extended RAN Node Name

This IE provides extended human readable name of the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Node Name Visible	O		VisibleString (SIZE(1..150, ...))	
RAN Node Name UTF8	O		UTF8String (SIZE(1..150, ...))	

9.3.1.194 MICO All PLMN

This IE indicates that the UE is configured with MICO mode by the AMF for the “all PLMN” as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MICO All PLMN	M		ENUMERATED (true, ...)	

9.3.1.195 Source Node ID

This IE identifies the source SN for the handover.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Source Node ID</i>	M			
> <i>Source E-UTRAN Node ID</i>				
>>Source en-gNB ID	M		Global gNB ID 9.3.1.6	This IE is used for handover from EN-DC to SA. The source en-gNB ID is the identity of the source SN.

9.3.1.196 E-UTRAN Composite Available Capacity Group

This IE indicates the overall available resource level in the cell in downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Composite Available Capacity Downlink	M		E-UTRAN Composite Available Capacity 9.3.1.197	For the downlink
Composite Available Capacity Uplink	M		E-UTRAN Composite Available Capacity 9.3.1.197	For the uplink

9.3.1.197 E-UTRAN Composite Available Capacity

This IE indicates the overall available resource level in the cell in either downlink or uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-UTRAN Cell Capacity Class Value	O		9.3.1.198	
E-UTRAN Capacity Value	M		9.3.1.199	'0' indicates no resource is available, Measured on a linear scale.

9.3.1.198 E-UTRAN Cell Capacity Class Value

This IE indicates the value that classifies the cell capacity with regards to the other cells. This IE only indicates resources that are configured for traffic purposes.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell Capacity Class Value	M		INTEGER (1..100, ...)	Value 1 indicates the minimum cell capacity, and 100 indicates the maximum cell capacity. There should be a linear relation between cell capacity and Cell Capacity Class Value.

9.3.1.199 E-UTRAN Capacity Value

This IE indicates the amount of resources that are available relative to the total E-UTRAN resources. The capacity value should be measured and reported so that the minimum E-UTRAN resource usage of existing services is reserved according to implementation. This IE can be weighted according to the ratio of cell capacity class values, if available.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Capacity Value	M		INTEGER (0..100)	Value 0 indicates no available capacity, and 100 indicates maximum available capacity. Capacity Value should be measured on a linear scale.

9.3.1.200 E-UTRAN Radio Resource Status

This IE indicates the usage of the PRBs for all traffic in downlink and uplink (TS 36.314 [49], TS 23.203 [50]) and the usage of PDCCH CCEs for downlink and uplink scheduling.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL GBR PRB Usage	M		INTEGER (0..100)	
UL GBR PRB Usage	M		INTEGER (0..100)	
DL Non-GBR PRB Usage	M		INTEGER (0..100)	
UL Non-GBR PRB Usage	M		INTEGER (0..100)	
DL Total PRB Usage	M		INTEGER (0..100)	
UL Total PRB Usage	M		INTEGER (0..100)	
DL Scheduling PDCCH CCE Usage	O		INTEGER (0..100)	
UL Scheduling PDCCH CCE Usage	O		INTEGER (0..100)	

9.3.1.201 Void

9.3.1.202 Void

9.3.1.203 Void

9.3.1.204 Void

9.3.1.205 NR Radio Resource Status

This IE indicates the usage of the PRBs per cell for MIMO for all traffic in Downlink and Uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL GBR PRB Usage for MIMO	M		INTEGER (0..100)	Per cell DL GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
UL GBR PRB Usage for MIMO	M		INTEGER (0..100)	Per cell UL GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
DL Non-GBR PRB Usage for MIMO	M		INTEGER (0..100)	Per cell DL Non-GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
UL Non-GBR PRB Usage for MIMO	M		INTEGER (0..100)	Per cell UL Non-GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
DL Total PRB Usage for MIMO	M		INTEGER (0..100)	Per cell DL Total PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
UL Total PRB usage for MIMO	M		INTEGER (0..100)	Per cell UL Total PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].

9.3.1.206 MBS Session ID

This IE uniquely identifies an MBS session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TMGI	M		OCTET STRING (SIZE(6))	Encoded as defined in TS 23.003 [23].
NID	O		9.3.3.42	

9.3.1.207 MBS Area Session ID

This IE indicates the Area Session ID for MBS session with location dependent context.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Area Session ID	M		INTEGER (0..65535, ...)	

9.3.1.208 MBS Service Area

This IE contains the MBS service area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Session Type</i>	M			
> <i>location independent</i>				
>>MBS Service Area Information	M		9.3.1.209	
> <i>location dependent</i>				
>>MBS Service Area Information List		1		
>>>MBS Service Area Information Item		1..<maxnoofMBSServiceAreaInformation>		
>>>>MBS Area Session ID	M		9.3.1.207	
>>>>MBS Service Area Information	M		9.3.1.209	

Range bound	Explanation
maxnoofMBSServiceAreaInformation	Maximum no. of MBS Service Area Information elements in the <i>MBS Service Area Information Location Dependent List</i> IE. Value is 256.

9.3.1.209 MBS Service Area Information

This IE contains MBS service area information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MBS Service Area Cell List		0..<maxnoofCellsforMBS>			-	
>NR CGI	M		9.3.1.7		-	
MBS Service Area TAI List		0..<maxnoofTAIforMBS>			-	
>TAI	M		9.3.3.11		-	
MBS Intended Service Area List		0..<maxnoofIntendedAreasforMBS>			YES	ignore
>Intended Service Area Coordinates	M		9.3.1.273		-	

Range bound	Explanation
maxnoofCellsforMBS	Maximum no. of cells allowed within one MBS Service Area. Value is 8192.
maxnoofTAIforMBS	Maximum no. of TAs allowed within one MBS Service Area. Value is 1024.
maxnoofIntendedAreasforMBS	Maximum no. of intended areas allowed within one MBS Service Area. Value is 65536.

9.3.1.210 MBS Support Indicator

This IE indicates whether MBS is supported for the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Support Indicator	M		ENUMERATED (multicast supported, ..., multicast supported with reception in RRC inactive)	

9.3.1.211 MBS Session Setup Request List

This IE provides information related to MBS sessions joined by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MBS Session Setup Request List		<i>1</i>			-	
>MBS Session Setup Request Item		<i>1..<maxno ofMBSSessions></i>			-	
>>MBS Session ID	M		9.3.1.206		-	
>>MBS Area Session ID	O		9.3.1.207		-	
>>Associated MBS QoS Flow Setup Request List		<i>0..1</i>			-	
>>>Associated MBS QoS Flow Setup Request Item		<i>1..<maxno ofMBSQoSflows></i>			-	
>>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51		-	
>>>>Associated Unicast QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51		-	
>>MBS Assistance Information	O		9.3.1.263		YES	ignore

Range bound	Explanation
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.
maxnoofMBSQoSflows	Maximum no. of MBS QoS flows allowed within one MBS session. Value is 64.

9.3.1.212 MBS Session Setup or Modify Request List

This IE provides information related to MBS sessions joined by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MBS Session Setup or Modify Request List		<i>1</i>			-	
>MBS Session Setup or Modify Request Item		<i>1..<maxno ofMBSSessions></i>			-	
>>MBS Session ID	M		9.3.1.206		-	
>>MBS Area Session ID	O		9.3.1.207		-	
>>Associated MBS QoS Flow Setup or Modify Request List		<i>0..1</i>			-	
>>>Associated MBS QoS Flow Setup or Modify Request Item		<i>1..<maxno ofMBSQoSflows></i>			-	
>>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51		-	
>>>>Associated Unicast QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51		-	
>>MBS QoS Flow To Release List	O		QoS Flow List with Cause 9.3.1.13	This IE indicates the MBS QoS Flow Identifiers of the MBS QoS Flows to be released.	-	
>>MBS Assistance Information	O		9.3.1.263		YES	ignore

Range bound	Explanation
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.
maxnoofMBSQoSflows	Maximum no. of MBS QoS flows allowed within one MBS session. Value is 64.

9.3.1.213 MBS Session Setup Response List

This IE contains a list of information related to MBS sessions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session Setup Response List		<i>1</i>		
>MBS Session Setup Response Item		<i>1..<maxnoofMBSSessions></i>		
>>MBS Session ID	M		9.3.1.206	
>>MBS Area Session ID	O		9.3.1.207	

Range bound	Explanation
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.

9.3.1.214 MBS Session Failed to Setup List

This IE contains a list of information related to MBS sessions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session Failed to Setup List		1		
>MBS Session Failed to Setup Item		1.. <i>maxnoofMBSessions</i>		
>>MBS Session ID	M		9.3.1.206	
>>MBS Area Session ID	O		9.3.1.207	
>>Cause	M		9.3.1.2	

Range bound	Explanation
maxnoofMBSessions	Maximum no. of MBS Sessions allowed within one PDU session. Value is 32.

9.3.1.215 MBS Session To Release List

This IE indicates MBS sessions to be removed.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session To Release List		1		
>MBS Session To Release Item		1.. <i>maxnoofMBSessions</i>		
>>MBS Session ID	M		9.3.1.206	
>>Cause	M		9.3.1.2	

Range bound	Explanation
maxnoofMBSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.

9.3.1.216 Multicast Group Paging Area

This IE contains a list of TAIs corresponding to the multicast group paging area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Area TAI List		1.. <i>maxnoofTAIforPaging</i>		
>TAI	M		9.3.3.11	

Range bound	Explanation
maxnoofTAIforPaging	Maximum no. of TAIs for multicast group paging. Value is 16.

9.3.1.217 MBS Session Status

This IE indicates whether the MBS session is activated or deactivated.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session Status	M		ENUMERATED (activated, deactivated, ...)	

9.3.1.218 MRB ID

This IE contains the MRB ID as specified in TS 38.401 [2].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MRB ID	M		INTEGER (1..512, ...)	

9.3.1.219 MRB Progress Information

This IE contains the MRB progress information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>PDCP SN Status</i>	M			
>12bits				
>>PDCP SN Length 12	M		INTEGER (0..4095)	
>18bits				
>>PDCP SN Length 18	M		INTEGER (0..262143)	

9.3.1.220 Time Synchronisation Assistance Information

This IE indicates 5G access stratum time distribution parameters as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Time Distribution Indication	M		ENUMERATED (enabled, disabled, ...)		-	
Uu Time Synchronisation Error Budget	C-ifEnabled		INTEGER (1..1000000, ...)	Expressed in units of 1ns.	-	
Clock Quality Reporting Control Information	O		9.3.1.250		YES	ignore

Condition	Explanation
ifEnabled	This IE shall be present if the <i>Time Distribution Indication</i> IE is set to the value "enabled".

9.3.1.221 Survival Time

This IE indicates the Survival Time of the TSC QoS flow as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Survival Time	M		INTEGER (0..1920000, ...)	Survival time expressed in units of 1 us.

9.3.1.222 QMC Deactivation

This IE indicates the QMC configurations to be deactivated.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoE Reference List		<i>1..<maxnoofUEAppLayerMeas></i>		
>QoE Reference	M		OCTET STRING (SIZE(6))	<i>QoE Reference</i> , as defined in clause 5.2 of TS 28.405 [45]. It consists of MCC+MNC+QMC ID, where the MCC and MNC are coming with the QMC activation request from the management system to identify one PLMN containing the management system, and QMC ID is a 3-bytes Octet String.

Range bound	Explanation
maxnoofUEAppLayerMeas	Maximum no. of UE application layer measurement. Value is 16.

9.3.1.223 QMC Configuration Information

This IE contains the configuration information for the QMC functionality.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Application Layer Measurement Information List		<i>1</i>		
>UE Application Layer Measurement Information Item		<i>1..<maxnoofUEAppLayerMeas></i>		
>>UE Application Layer Measurement Configuration Information	M		9.3.1.224	

Range bound	Explanation
maxnoofUEAppLayerMeas	Maximum no. of UE application layer measurements. Value is 16.

9.3.1.224 UE Application Layer Measurement Configuration Information

This IE defines configuration information for the QMC functionality.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QoE Reference	M		OCTET STRING (SIZE(6))	<i>QoE Reference</i> , as defined in clause 5.2 of TS 28.405 [45]. It consists of MCC+MNC+QMC ID, where the MCC and MNC are received with the QMC activation request from the management system to identify one PLMN containing the management system, and QMC ID is a 3-byte Octet String.	-	
Service Type	M		ENUMERATED (QMC for DASH streaming, QMC for MTSI, QMC for VR, ...)	This IE indicates the service type of QoE measurements.	-	
CHOICE Area Scope of QMC	O				-	
>Cell based						
>>Cell ID List for QMC		1..<maxno of CellID for QMC>			-	
>>>NG-RAN CGI	M		9.3.1.73	This IE can indicate an NR CGI or an E-UTRA CGI.	-	
>TA based						
>>TA List for QMC		1..<maxno of TA for QMC>			-	
>>>TAC	M		9.3.3.10	The TAI is derived using the current serving PLMN.	-	
>TAI based						
>>TAI List for QMC		1..<maxno of TAI for QMC>			-	
>>>TAI	M		9.3.3.11		-	
>PLMN area based						
>>PLMN List for QMC		1..<maxno of PLMN for QMC>			-	
>>>PLMN Identity	M		9.3.3.5		-	
Measurement Collection Entity IP Address	M		Transport Layer Address 9.3.2.4	The IP address of the entity receiving the QoE measurement report.	-	
QoE Measurement Status	O		ENUMERATED (ongoing, ...)	Indicates whether the QoE measurement has been started. Present in case of NG-based handover.	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Container for Application Layer Measurement Configuration	O		OCTET STRING (SIZE(1.. 8000))	Contains application layer measurement configuration, see Annex L in 26.247 [46], clause 16.5 in TS 26.114 [51] and clause 9 in TS 26.118 [52]. Present in case of initial QoE configuration, and shall be included in <i>Source to Target Transparent Container</i> IE for signalling-based QMC during NG-based handover.	-	
Measurement Configuration Application Layer ID	O		INTEGER (0..15, ...)	This IE is present only when the message containing it is NG-based handover related. The IE indicates the identity of the application layer measurement configuration and corresponds to the <i>MeasConfigAppLayerId</i> IE as defined in TS 38.331 [18].	-	
Slice Support List for QMC		0..1			-	
>Slice Support QMC Item		1..<maxno ofSNSSAIforQMC>			-	
>>S-NSSAI	M		9.3.1.24		-	
CHOICE MDT Alignment Information	O			Indicates the MDT measurements with which alignment is required.	-	
>S-based MDT						
>>NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	This IE is composed of the following: Trace Reference defined in TS 32.422 [11] (leftmost 6 octets, with PLMN information encoded as in 9.3.3.5), and Trace Recording Session Reference defined in TS 32.422 [11] (last 2 octets).	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Available RAN Visible QoE Metrics	O		9.3.1.225	Present in case of initial QoE configuration and in case of NG-based handover for signalling-based QoE measurement.	-	
Assistance Information for QoE Measurement	O		INTEGER (1..16, ...)	This IE indicates the suggested priority of the application layer measurement configuration. Values are ordered in decreasing order of priority, i.e., with 1 as the highest priority and 16 as the lowest priority.	YES	ignore
Communication Service Type	O		ENUMERATED (MBS broadcast, MBS multicast, ..., unicast)	This IE indicates the type of communication service for which the QoE measurement collection should be performed.	YES	ignore
QoE and RVQoE Reporting Paths	O		9.3.1.272	This IE indicates the SRBs currently used for QoE and RVQoE reporting for the present QoE measurement configuration.	YES	ignore

Range bound	Explanation
maxnoofCellIDforQMC	Maximum no. of Cell ID subject for QMC scope. Value is 32.
maxnoofTAforQMC	Maximum no. of TA subject for QMC scope. Value is 8.
maxnoofPLMNforQMC	Maximum no. of PLMNs in the PLMN list for QMC scope. Value is 16.
maxnoofSNSSAIforQMC	Maximum no. of S-NSSAIs in the S-NSSAI list for QMC scope. Value is 16.

9.3.1.225 Available RAN Visible QoE Metrics

This IE defines which RAN visible QoE metrics can be configured by the NG-RAN in the RAN visible QoE measurement.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Application Layer Buffer Level List	O		ENUMERATED (true, ...)	The IE defines whether the Buffer Level can be collected as a RAN visible QoE metric by NG-RAN from UE, for DASH streaming and VR service types.
Playout Delay for Media Startup	O		ENUMERATED (true, ...)	The IE defines whether the Playout delay can be collected as a RAN visible QoE metric by NG-RAN from UE, for DASH streaming and VR service types.

9.3.1.226 Void

9.3.1.227 NR Paging eDRX Information

This IE indicates the NR Paging eDRX parameters as defined in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR Paging eDRX Cycle	M		ENUMERATED (hfquarter, hfhalf, hf1, hf2, hf4, hf8, hf16, hf32, hf64, hf128, hf256, hf512, hf1024, ...)	$T_{\text{eDRX CN}}$ defined in TS 38.304 [12]. Unit: [number of hyperframes].
NR Paging Time Window	O		ENUMERATED (s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, ..., s17, s18, s19, s20, s21, s22, s23, s24, s25, s26, s27, s28, s29, s30, s31, s32)	PTW defined in TS 38.304 [12]. Unit: [1.28 seconds]

9.3.1.228 RedCap Indication

This IE is provided by the NG-RAN node to inform that the UE indicates Reduced Capability.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RedCap Indication	M		ENUMERATED (RedCap, ...)	

9.3.1.229 Target NSSAI Information

This IE contains the Target NSSAI and Index to RAT/Frequency Selection Priority as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target NSSAI	M		9.3.1.230	
Index to RAT/Frequency Selection Priority	M		9.3.1.61	

9.3.1.230 Target NSSAI

This IE contains the Target NSSAI as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target S-NSSAI List		1		
>Target S-NSSAI Item		1..<maxnoofTargetS-NSSAIs>		
>>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofTargetS-NSSAIs	Maximum no. of Target S-NSSAIs. Value is 8.

9.3.1.231 UE Slice Maximum Bit Rate List

This IE contains the UE Slice Maximum Bit Rate List as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Slice Maximum Bit Rate Item		<i>1..<maxnoof AllowedS-NSSAIs></i>		Applicable across all GBR and Non-GBR QoS flows.
>S-NSSAI	M		9.3.1.24	
>UE Slice Maximum Bit Rate Downlink	M		Bit Rate 9.3.1.4	This IE indicates the downlink UE-Slice-MBR as specified in TS 23.501 [9].
>UE Slice Maximum Bit Rate Uplink	M		Bit Rate 9.3.1.4	This IE indicates the uplink UE-Slice-MBR as specified in TS 23.501 [9].

Range bound	Explanation
maxnoofAllowedS-NSSAIs	Maximum no. of allowed S-NSSAI. Value is 8.

9.3.1.232 PEIPS Assistance Information

This IE provides the information related to CN paging subgrouping for a particular UE, as specified in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CN Subgroup ID	M		INTEGER (0..7, ...)	

9.3.1.233 5G ProSe Authorized

This IE provides information on the authorization status of the UE to use the 5G ProSe services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
5G ProSe Direct Discovery	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Direct Discovery	-	
5G ProSe Direct Communication	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Direct Communication	-	
5G ProSe Layer-2 UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 UE-to-Network Relay	-	
5G ProSe Layer-3 UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-3 UE-to-Network Relay	-	
5G ProSe Layer-2 Remote UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Remote UE	-	
5G ProSe Layer-2 Multipath	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the 5G ProSe Layer-2 Remote UE is authorized for 5G ProSe multipath transmission	YES	ignore
5G ProSe Layer-2 UE-to-UE Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 UE-to-UE Relay UE	YES	ignore
5G ProSe Layer-2 UE-to-UE Remote	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 UE-to-UE Remote UE.	YES	ignore
5G ProSe Layer-2 Multi-Hop UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Multi-Hop UE-to-Network Relay.	YES	ignore
5G ProSe Layer-2 Multi-Hop Intermediate UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Multi-Hop Intermediate UE-to-Network Relay.	YES	ignore
5G ProSe Layer-2 Multi-Hop Remote	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Multi-Hop Remote.	YES	ignore
5G ProSe Layer-3 Multi-Hop UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-3 Multi-Hop UE-to-Network Relay.	YES	ignore

9.3.1.234 5G ProSe PC5 QoS Parameters

This IE provides information on the 5G ProSe PC5 QoS parameters of the UE's sidelink communication for 5G ProSe services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
5G ProSe PC5 QoS Flow List		1		
>5G ProSe PC5 QoS Flow Item		1.. <i>maxnoofPC5QoSFlows</i>		
>>PQI	M		INTEGER (0..255, ...)	PQI is a special 5QI as specified in TS 23.501 [9].
>>5G ProSe PC5 Flow Bit Rates		0..1		Only applies for GBR QoS flows.
>>>Guaranteed Flow Bit Rate	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>>Maximum Flow Bit Rate	M		Bit Rate 9.3.1.4	Maximum Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>Range	O		ENUMERATED (m50, m80, m180, m200, m350, m400, m500, m700, m1000, ...)	Only applies for groupcast.
5G ProSe PC5 Link Aggregate Bit Rates	O		Bit Rate 9.3.1.4	Only applies for Non-GBR QoS flows.

Range bound	Explanation
maxnoofPC5QoSFlows	Maximum no. of PC5 QoS flows allowed towards one UE. Value is 2048.

9.3.1.235 Last Visited PSCell Information

The Last Visited PSCell Information may contain cell specific information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
PSCell ID	O		NG-RAN CGI 9.3.1.73	This IE is present when the SCG resources are configured for the UE.	-	
Time Stay	M		INTEGER (0..40950)	The duration of the time the UE stayed in the cell in 1/10 seconds. If the UE stays in a cell more than 4095s, this IE is set to 40950. Or the duration of the time when no SCG resources are configured for the UE.	-	
SCG Activation Time	O		INTEGER (0..40950)	The total duration of the time the UE stayed in the cell while the SCG was activated, in 1/10 seconds. If the duration is more than 4095s, this IE is set to 40950.	YES	ignore

9.3.1.236 MBS QoS Flows To Be Setup List

This IE contains a list of QoS flows and their associated QoS parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS QoS Flows Setup Request Item		<i>1..<maxnoofMBSQoSFlows></i>		
>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51	
>MBS QoS Flow Level QoS Parameters	M		QoS Flow Level QoS Parameters 9.3.1.12	

Range bound	Explanation
maxnoofMBSQoSflows	Maximum no. of MBS QoS flows allowed within one MBS session. Value is 64.

9.3.1.237 Reporting System

This IE lists the cells on the reporting system.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Reporting System</i>				
> <i>E-UTRAN</i>				
>> <i>E-UTRAN Cell To Report List</i>		1		
>>> <i>E-UTRAN Cell To Report Item</i>		1.. <i>maxnoofReportedCells</i>		
>>>>Cell ID	M		E-UTRA CGI 9.3.1.9	
> <i>NG-RAN</i>				
>> <i>NG-RAN Cell To Report List</i>		1		
>>> <i>NG-RAN Cell To Report Item</i>		1.. <i>maxnoofReportedCells</i>		
>>>>Cell ID	M		NG-RAN CGI 9.3.1.73	
> <i>No Reporting</i>			NULL	Resource status not available. Applicable to the <i>Inter-system Resource Status Reply</i> IE.

Range bound	Explanation
maxnoofReportedCells	Maximum no. of cells that can be reported. Value is 256.

9.3.1.238 TAI NSAG Support List

This IE contains the slice group mapping for all groups configured at the NG-RAN node per TAI.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TAI NSAG Support Item		1.. <i>maxnoofNSAGs</i>		
>NSAG ID	M		INTEGER (0..255, ...)	
>NSAG Slice Support List	M		Extended Slice Support List 9.3.1.191	Indicates the list of slices which belong to the NSAG.

Range bound	Explanation
maxnoofNSAGs	Maximum no. of Slice Groups for the TAI. Value is 256.

9.3.1.239 NGAP Protocol IE-Id

This IE uniquely identifies an NGAP Protocol IE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NGAP Protocol IE-Id	M		INTEGER (0..65535)	

9.3.1.240 NGAP Protocol IE Support Information

This IE provides information about support of functions associated to an NGAP Protocol IE-Id.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NGAP Protocol IE Support Information	M		ENUMERATED (supported, not-supported, ...)	

9.3.1.241 NGAP Protocol IE Presence Information

This IE provides information on whether an NGAP Protocol IE-Id was received within the message requesting the information (i.e. HANDOVER REQUEST).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NGAP Protocol IE Presence Information	M		ENUMERATED (present, not present)	

9.3.1.242 NGAP IE Support Information Response List

This IE provides information about support of functions associated to a list of NGAP Protocol IE-Ids.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NGAP IE Support Information Response Item		<i>1..<maxnoofIE SupportInfo></i>		
>NGAP Protocol IE-Id	M		9.3.1.239	
>NGAP Protocol IE Support Information	M		9.3.1.240	
>NGAP Protocol IE Presence Information	M		9.3.1.241	

Range bound	Explanation
maxnoofIESupportInfo	Maximum no. of IE Support Information. Value is 32.

9.3.1.243 MDT PLMN Modification List

This IE provides the modified list of PLMN allowed for MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT PLMN Modification List		<i>0..<maxnoofMDTPLMNs></i>		An empty list indicates there is no PLMN allowed for MDT.
>PLMN Identity	M		9.3.3.5	

Range bound	Explanation
maxnoofMDTPLMNs	Maximum no. of PLMNs in the MDT PLMN list. Value is 16.

9.3.1.244 Excess Packet Delay Threshold Configuration

This IE defines the parameters for Excess Packet Delay Threshold configuration to support the calculation of the PDCP Excess Packet Delay in the UL per DRB as specified in TS 38.314 [48].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Excess Packet Delay Threshold Item		<i>1..<maxnoofThresholdsForExcessPacketDelay></i>		
>5QI	M		INTEGER (0..255, ...)	Indicates the standardized or pre-configured 5QI as specified in TS 23.501 [9].
>Excess Packet Delay Threshold Value	M		ENUMERATED (ms0.25, ms0.5, ms1, ms2, ms4, ms5, ms10, ms20, ms30, ms40, ms50, ms60, ms70, ms80, ms90, ms100, ms150, ms300, ms500, ...)	

Range bound	Explanation
maxnoofThresholdsForExcessPacketDelay	Maximum no. of thresholds for Excess Packet Delay configuration. Value is 255.

9.3.1.245 Network Controlled Repeater Authorized

This IE provides the authorization status of the UE to act as a Network Controlled Repeater.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Network Controlled Repeater Authorized	M		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Network Controlled Repeater

9.3.1.246 Aerial UE Subscription Information

This IE is used by the NG-RAN node to know if the UE is allowed to use aerial function, refer to TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Aerial UE Subscription Information	M		ENUMERATED (allowed, not allowed,...)	

9.3.1.247 NR A2X Services Authorized

This IE provides information on the authorization status of the UE to use the NR sidelink for A2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Aerial UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Aerial UE
Aerial Controller UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Aerial Controller UE

9.3.1.248 LTE A2X Services Authorized

This IE provides information on the authorization status of the UE to use the LTE sidelink for A2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Aerial UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Aerial UE
Aerial Controller UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Aerial Controller UE

9.3.1.249 A2X PC5 QoS Parameters

This IE provides information on the PC5 QoS parameters of the UE's sidelink communication for A2X.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A2X PC5 QoS Flow List		<i>1</i>		
>A2X PC5 QoS Flow Item		<i>1..<maxnoofPC5QoSFlows></i>		
>>A2X PQI	M		INTEGER (0..255, ...)	PQI is a special 5QI as specified in TS 23.501 [9].
>>A2X PC5 Flow Bit Rates		<i>0..1</i>		Only applies for GBR QoS Flows.
>>>A2X Guaranteed Flow Bit Rate	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>>A2X Maximum Flow Bit Rate	M		Bit Rate 9.3.1.4	Maximum Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>A2X Range	O		ENUMERATED (m50, m80, m180, m200, m350, m400, m500, m700, m1000, ...)	Only applies for groupcast.
A2X PC5 Link Aggregate Bit Rates	O		Bit Rate 9.3.1.4	Only applies for non-GBR QoS Flows.

Range bound	Explanation
maxnoofPC5QoSFlows	Maximum no. of PC5 QoS flows allowed towards one UE. Value is 2048.

9.3.1.250 Clock Quality Reporting Control Information

This IE indicates the clock quality reporting control information as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Clock Quality Detail Level</i>	M			
>clock quality metrics			NULL	
>clock quality acceptance indication				
>>Clock Quality Acceptance Criteria	M		9.3.1.251	

9.3.1.251 Clock Quality Acceptance Criteria

This IE indicates the clock quality acceptance criteria as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Synchronisation State	O		BIT STRING { locked (0), holdover (1), freeRun (2) } (SIZE (8, ...))	Each position in the bitmap represents a synchronisation state. If a bit is set to "1", the respective synchronisation state is acceptable. If a bit is set to "0", the respective synchronisation state is not acceptable. Bits 3-7 reserved for future use.
Traceable to UTC	O		ENUMERATED (true, ...)	
Traceable to GNSS	O		ENUMERATED (true, ...)	
Clock Frequency Stability	O		BIT STRING (SIZE (16))	Indicates the offsetScaledLogVariance as specified in TS 23.501 [9].
Clock Accuracy	O		INTEGER (1..40000000, ...)	Clock accuracy expressed in units of 25 ns.
Parent Time Source	O		BIT STRING { syncE (0), pTP (1), gNSS (2), atomicClock (3), terrestrialRadio (4), serialTimeCode (5), nTP (6), handset (7), other (8) } (SIZE (16, ...))	Each position in the bitmap represents a parent time source. If a bit is set to "1", the respective parent time source is acceptable. If a bit is set to "0", the respective parent time source is not acceptable. Bits 9-15 reserved for future use.

9.3.1.252 RAN Timing Synchronisation Status Information

This IE indicates the RAN timing synchronisation status information provided towards the AMF as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Synchronisation State	O		ENUMERATED (locked, holdover, freeRun, ...)	
Traceable to UTC	O		ENUMERATED (true, false, ...)	
Traceable to GNSS	O		ENUMERATED (true, false, ...)	
Clock Frequency Stability	O		BIT STRING (SIZE (16))	Indicates the offsetScaledLogVariance as specified in TS 23.501 [9].
Clock Accuracy	O		9.3.1.253	
Parent Time Source	O		ENUMERATED (syncE, pTP, gNSS, atomicClock, terrestrialRadio, serialTimeCode, nTP, handSet, other, ...)	

9.3.1.253 Clock Accuracy

This IE indicates the clock accuracy as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Clock Accuracy</i>	M			
> <i>Value</i>				
>>Clock Accuracy Value	M		INTEGER (1..40000000, ...)	Indicates the absolute clock accuracy value. Unit in 25ns.
> <i>Index</i>				
>>Clock Accuracy Index	M		INTEGER (32..47, ...)	Indicates the clockAccuracy enumeration value specified in Table 5 of clause 7.6.2.6 of IEEE Std 1588 [55].

9.3.1.254 RAN TSS Scope

This IE indicates the scope of the RAN timing synchronisation status as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>RAN TSS scope</i>	M			
> <i>RAN node level</i>				
>>Global gNB ID	M		9.3.1.6	
> <i>cell list level</i>				
>>RAN TSS Cell List		1		
>>>RAN TSS Cell Item		1..<maxnoofCellsTSS>		
>>>>NR CGI	M		9.3.1.7	

Range bound	Explanation
maxnoofCellsTSS	Maximum no. of Cell IDs in the scope of the RAN timing synchronisation status. Value is 16384.

9.3.1.255 Burst Arrival Time Window

This IE indicates the burst arrival time window of the TSC QoS flow as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Burst Arrival Time Window Start	M		INTEGER (0..640000, ...)	Start of the burst arrival time window calculated with reference to the <i>Burst Arrival Time</i> IE, expressed in units of 1 us. Integer values are negative.
Burst Arrival Time Window End	M		INTEGER (0..640000, ...)	End of the burst arrival time window calculated with reference to the <i>Burst Arrival Time</i> IE, expressed in units of 1 us. Integer values are positive.

9.3.1.256 Periodicity Range

This IE indicates the periodicity range for the TSC QoS flow as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Periodicity Range</i>	M			
> <i>Periodicity Bound</i>				
>>Periodicity Lower Bound	M		Periodicity 9.3.1.132	
>>Periodicity Upper Bound	M		Periodicity 9.3.1.132	
> <i>Periodicity List</i>				
>> Allowed Periodicity List		1.. <i>maxnoofPeriodicities</i>		
>>>Allowed Periodicity	M		Periodicity 9.3.1.132	

Range bound	Explanation
maxnoofPeriodicities	Maximum no. of allowed periodicities. Value is 8.

9.3.1.257 TSC Traffic Characteristics Feedback

This IE provides the TSC traffic characteristics feedback of a TSC QoS flow (see TS 23.501 [9]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TSC Feedback Information Downlink	O		TSC Feedback Information 9.3.1.258	
TSC Feedback Information Uplink	O		TSC Feedback Information 9.3.1.258	

9.3.1.258 TSC Feedback Information

This IE provides the TSC feedback information for a TSC QoS flow in the uplink or downlink (see TS 23.501 [9]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Burst Arrival Time Offset	M		INTEGER (-640000..640000, ...)	Burst arrival time offset expressed in units of 1 us.
Adjusted Periodicity	O		Periodicity 9.3.1.132	Not applicable to reactive RAN feedback.

9.3.1.259 Mobile IAB Authorized

This IE provides information about the authorization status of the mobile IAB-node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Mobile IAB Authorized	M		ENUMERATED (authorized, not authorized, ...)	Indicates the mobile IAB-node authorization status.

9.3.1.260 Mobile IAB-MT User Location Information

This IE contains the NR CGI and the TAI for the co-located mobile IAB-MT of the mobile IAB-DU which serves the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR CGI	M		9.3.1.7	The NR CGI of the cell, which is the serving cell of the mobile IAB-MT co-located with the mobile IAB-DU that serves the UE.
TAI	M		9.3.3.11	The TAI supported by the cell, which is the serving cell of the mobile IAB-MT co-located with the mobile IAB-DU which serves the UE.

9.3.1.261 Partially Allowed NSSAI

This IE contains a list of S-NSSAI(s) partially permitted by the network for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Partially Allowed S-NSSAI Item		<i>1..<maxnoofAllowedS-NSSAIs></i>		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofPartiallyAllowedS-NSSAIs	Maximum no. of partially allowed S-NSSAI. Value is 8.

9.3.1.262 Associated Session ID

This IE is used to associate MBS Session IDs providing identical user data.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Associated Session ID	M		OCTET STRING	Coded as the <i>AssociatedSessionId</i> IE defined in TS 29.571 [35]. The gNB does not interpret the content of the <i>Associated Session ID</i> IE.

9.3.1.263 MBS Assistance Information

This IE provides MBS Assistance Information as specified in TS 38.300 [8] and TS 23.247 [44].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Assistance Information	M		ENUMERATED (true, ...)	

9.3.1.264 PDU Set QoS Information

This IE defines the PDU Set QoS Information to be applied to a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Set Packet Delay Budget	O		Extended Packet Delay Budget 9.3.1.135	PDU Set Delay Budget as specified in TS 23.501 [9].
PDU Set Error Rate	O		Packet Error Rate 9.3.1.81	PDU Set Error Rate as specified in TS 23.501 [9].
PDU Set Integrated Handling Information	O		ENUMERATED (true, false, ...)	PDU Set Integrated Handling Information as specified in TS 23.501 [9].

9.3.1.265 N6 Jitter Information

This IE indicates the N6 jitter information associated with the Periodicity in downlink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
N6 Jitter Lower Bound	M		INTEGER (-127..127)	Indicates the lower bound of the N6 jitter. The unit is: 0.5ms.
N6 Jitter Upper Bound	M		INTEGER (-127..127)	Indicates the upper bound of the N6 jitter. The unit is: 0.5ms.

9.3.1.266 ECN Marking or Congestion Information Reporting Request

This IE indicates to the NG-RAN node to perform ECN marking or to report information for ECN marking or to report congestion information for a QoS Flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>ECN Marking or Congestion Information Request</i>	M			
> <i>ECN Marking at RAN</i>				
>>ECN Marking at NG-RAN Request	M		ENUMERATED (ul, dl, both, stop, ...)	
> <i>ECN Marking at UPF</i>				
>>ECN Marking at UPF Request	M		ENUMERATED (ul, dl, both, stop, ...)	
> <i>Congestion Information</i>				
>>Congestion Information Request	M		ENUMERATED (ul, dl, both, stop, ...)	

9.3.1.267 ECN Marking or Congestion Information Reporting Status

This IE contains a list of QoS flows with activation status information for NG-RAN node to perform ECN marking or to report information for ECN marking or to report congestion information reporting.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
ECN Marking or Congestion Information Reporting Status Item		<i>1..<maxnoofQoSFlows></i>		
>QoS Flow Identifier	M		9.3.1.51	
>Activation Status	M		ENUMERATED (active, not Active, ...)	Indicates whether ECN marking at NG-RAN or reporting information for ECN marking at UPF or congestion information reporting is active or not active.

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.1.268 PDU Set based Handling Indicator

This IE indicates whether PDU Set based Handling is supported for the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Set based Handling Indicator	M		ENUMERATED (supported, ...)	

9.3.1.269 Ranging and Sidelink Positioning Service Information

This IE provides information on the UE for Ranging and Sidelink Positioning service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Ranging and Sidelink Positioning Authorized	M		ENUMERATED (authorized, not authorized, ...)	This IE indicates whether the UE is authorized to use RSPP communication resources and SL-PRS resources.
RSPP Transport QoS Parameters	O		9.3.1.270	This IE applies only if the UE is authorized for Ranging and Sidelink Positioning service.

9.3.1.270 RSPP Transport QoS Parameters

This IE provides information on the RSPP transport QoS parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RSPP Transport QoS Flow List		1		
>RSPP Transport QoS Flow Item		1..<maxnoofRSPPQoSFlows>		
>>PQI	M		INTEGER (0..255, ...)	PQI is a special 5QI as specified in TS 23.501 [9].
>>>RSPP Transport Bit Rates		0..1		Only applies for GBR QoS flows.
>>>>Guaranteed Flow Bit Rate	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate for the RSPP QoS flow. Details in TS 23.501 [9].
>>>>Maximum Flow Bit Rate	M		Bit Rate 9.3.1.4	Maximum Bit Rate for the RSPP QoS flow. Details in TS 23.501 [9].
>>>Range	O		ENUMERATED (m50, m80, m180, m200, m350, m400, m500, m700, m1000, ...)	Only applies for groupcast.
RSPP Transport Link Aggregate Bit Rates	O		Bit Rate 9.3.1.4	Only applies for Non-GBR QoS flows.

Range bound	Explanation
maxnoofRSPPQoSFlows	Maximum no. of RSPP QoS flows allowed towards one UE. Value is 2048.

9.3.1.271 User Plane Failure Indication

This IE is used to notify the NG-RAN node that a user plane failure occurred over NG-U interface.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
User Plane Failure Type	M		ENUMERATED (GTP-U Error Indication Received, UP Path Failure, ...)	Indicates the type of NG-U failure.
UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the UPF for which the failure occurred.
DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the NG-RAN node for which the failure occurred.

9.3.1.272 QoE and RVQoE Reporting Paths

This IE indicates the SRB currently used for receiving the QoE reports and RAN visible QoE reports.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoE Reporting Path	O		ENUMERATED (srb4, srb5, ...)	This IE indicates the SRB currently used for receiving the QoE reports.
RVQoE Reporting Path	O		ENUMERATED (srb4, srb5, ...)	This IE indicates the SRB currently used for receiving the RAN Visible QoE reports.

9.3.1.273 Intended Service Area Coordinates

This IE provides information on the Intended Service Area Coordinates for MBS Broadcast Service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Intended Service Area Coordinates	M		OCTET STRING	Includes the <i>areaCoordinates</i> contained in the <i>IntendedServiceAreaInfo</i> IE as defined in TS 38.331 [18]

9.3.1.274 GW Context Release Indication

This information element is set by the gNB to provide an indication that the AMF may release any resources related to the signalled NG UE context (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
GW Context Release Indication	M		ENUMERATED (true, ...)	

9.3.1.275 Requested NSSAI

This IE indicates the S-NSSAIs requested by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Requested NSSAI Item		<i>1..<maxnoof RequestedS-NSSAIs></i>		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofRequestedS-NSSAIs	Maximum no. of S-NSSAIs requested by the UE. Value is 8.

9.3.1.276 Reader Index

This IE represents identifier of a Reader within a gNB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reader Index	M		INTEGER (1..65535, ...)	

9.3.1.277 A-IoT Area ID

This IE is used to uniquely identify an A-IoT Area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
NID	O		9.3.3.42	
A-IoT Area Code	M		OCTET STRING (SIZE(3))	

9.3.1.278 A-IoT Support

This IE indicates the support of A-IoT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Support	M		ENUMERATED (A-IoT only, A-IoT and NR Uu, ...)	

9.3.1.279 LP-WUSPS Assistance Information

This IE provides the information related to LP-WUS paging subgrouping for a particular UE, as specified in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LP-WUS CN Subgroup ID	M		INTEGER (0..30, ...)	

9.3.1.280 Additional ULI

This IE contains the NR CGI and the TAI that reflect the location information of the WAB-node which serves the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR CGI	M		9.3.1.7	
TAI	M		9.3.3.11	

9.3.1.281 Available Bitrate Report Threshold List

This IE contains a list of available bitrate report thresholds. It is used for available bitrate report for UL and DL as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Available Bitrate Report Threshold Item		<i>1..<maxnoofThresholds></i>		
>Reporting Threshold	M		INTEGER (0.. 4,000,000,000, ...)	This IE indicates the Reporting threshold as specified in TS 23.501 [9]. The unit is Kbps.

Range bound	Explanation
maxnoofThresholds	Maximum no. of thresholds allowed to be provided by the SMF. Value is 8.

9.3.1.282 Aerial UE Flight information Reporting

This IE contains the aerial UE altitude reporting information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Altitude	M		9.3.1.284	
Time Stamp	M		9.3.1.75	Indicates the UTC time when the aerial UE flight information is received.

9.3.1.283 Aerial UE Flight Information Reporting Control

This IE indicates Aerial UE flight information reporting information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Higher Altitude Threshold	M		Altitude 9.3.1.284	Indicates the higher altitude threshold information for the Aerial UE reporting.
Lower Altitude Threshold	M		Altitude 9.3.1.284	Indicates the lower altitude threshold information for the Aerial UE reporting.
Aerial UE Reporting Periodicity	O		ENUMERATED {ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, ms20480, ms40960, min1, min6, min12, min30, ...}	Indicates the periodicity of the Aerial UE reporting.
CHOICE Area ID	M			
>TAI				
>>TAI	M		9.3.3.11	
>RAN Node				
>>Global RAN Node ID	M		9.3.1.5	
>NG-RAN CGI				
>>NG-RAN CGI	M		9.3.1.73	

9.3.1.284 Altitude

This IE contains Altitude information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Altitude	M		INTEGER (-420..10000, ...)	Aerial UE altitude information as specified in TS 38.331[18]. The unit of this IE is meter

9.3.2 Transport Network Layer Related IEs

9.3.2.1 QoS Flow per TNL Information List

This IE is used to provide a list of additional UP transport layer information for a split PDU session, along with the associated QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow per TNL Information Item		<i>1..<maxnoofMultiConnectivityMinusOne></i>		
>QoS Flow per TNL Information	M		9.3.2.8	

Range bound	Explanation
maxnoofMultiConnectivityMinusOne	Maximum no. of connectivity allowed for a UE minus one. Value is 3. The current version of the specification supports 1.

9.3.2.2 UP Transport Layer Information

This IE is used to provide the NG UP transport layer information associated with a PDU session for an NG-RAN node – UPF pair. In this release it corresponds to an IP address and a GTP Tunnel Endpoint Identifier.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
<i>CHOICE UP Transport Layer Information</i>	M			
>GTP tunnel				
>>Endpoint IP Address	M		Transport Layer Address 9.3.2.4	
>>GTP-TEID	M		9.3.2.5	

9.3.2.3 E-RAB ID

This IE is the identifier of the LTE E-RAB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-RAB ID	M		INTEGER (0..15, ...)	

9.3.2.4 Transport Layer Address

This IE is an IP address.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transport Layer Address	M		BIT STRING (SIZE(1..160, ...))	The Radio Network Layer is not supposed to interpret the address information. It should pass it to the Transport Layer for interpretation. For details, see TS 38.414 [14].

9.3.2.5 GTP-TEID

This IE is the GTP Tunnel Endpoint Identifier to be used for the user plane transport between the NG-RAN node and the UPF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
GTP-TEID	M		OCTET STRING (SIZE(4))	For details and range, see TS 29.281 [15].

9.3.2.6 CP Transport Layer Information

This IE is used to provide the NG control plane transport layer information associated with an NG-RAN node – AMF pair.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE CP Transport Layer Information	M				-	
>Endpoint-IP-address						
>>Endpoint IP Address	M		Transport Layer Address 9.3.2.4		-	
>Endpoint-IP-address-and-port					YES	reject
>>Endpoint IP Address	M		Transport Layer Address 9.3.2.4		-	
>>Port Number	M		OCTET STRING (SIZE(2))		-	

9.3.2.7 TNL Association List

This IE contains a list of TNL associations. It is used for example to indicate failed TNL association(s).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TNL Association Item		1..<maxnoofTNLAssociations>		
>TNL Association Address	M		CP Transport Layer Information 9.3.2.6	
>Cause	M		9.3.1.2	

Range bound	Explanation
maxnoofTNLAssociations	Maximum no. of TNL Associations between the NG-RAN node and the AMF. Value is 32.

9.3.2.8 QoS Flow per TNL Information

This IE indicates the NG-U transport layer information and associated list of QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UP Transport Layer Information	M		9.3.2.2	
Associated QoS Flow List	M		9.3.1.99	

9.3.2.9 TNL Association Usage

This IE indicates the usage of the TNL association.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TNL Association Usage	M		ENUMERATED (ue, non-ue, both, ...)	Indicates whether the TNL association is only used for UE-associated signalling, or non-UE-associated signalling, or both.

9.3.2.10 TNL Address Weight Factor

This IE indicates the weight factor of the TNL address.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TNL Address Weight Factor	M		INTEGER (0..255)	Value 0 indicates the TNL address is not permitted for the initial NGAP message. If the value for each TNL address is the same, it indicates the deployments that rely solely on 5GC-based load balancing.

9.3.2.11 UP Transport Layer Information Pair List

This IE is used to provide a list of uplink UP transport layer information and associated downlink UP transport layer information for a split PDU session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UP Transport Layer Information Pair Item		<i>1..<maxnoofMultiConnectivityMinusOne></i>		
>UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	
>DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	

Range bound	Explanation
maxnoofMultiConnectivityMinusOne	Maximum no. of connectivity allowed for a UE minus one. Value is 3. The current version of the specification supports 1.

9.3.2.12 UP Transport Layer Information List

This IE is used to provide a list of additional UP transport layer information for a split PDU session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UP Transport Layer Information Item		1..<maxno ofMultiConnectivityMinusOne>			-	
>NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2		-	
>Common Network Instance	O		9.3.1.120		YES	ignore

Range bound	Explanation
maxnoofMultiConnectivityMinusOne	Maximum no. of connectivity allowed for a UE minus one. Value is 3. The current version of the specification supports 1.

9.3.2.13 QoS Flow List with Data Forwarding

This IE is used to provide a list of QoS flows with indication if forwarding is accepted.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QoS Flow Item with Data Forwarding		1..<maxno ofQoSFlows>			-	
>QoS Flow Identifier	M		9.3.1.51		-	
>Data Forwarding Accepted	O		9.3.1.62	This IE is included for the QoS flows in the PDU session to be forwarded over the PDU session forwarding tunnel. It may be included for the QoS flows in the PDU session to be forwarded over the DRB forwarding tunnel(s).	-	
>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.152	Index to the currently fulfilled alternative QoS parameters set	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.2.14 URI

This IE is an URI.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
URI	M		VisibleString	String representing URI (Uniform Resource Identifier)

9.3.2.15 MBS Session TNL Information 5GC

This IE provides 5GC TNL information for location dependent and location independent MBS sessions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Session Type</i>	M			
> <i>location independent</i>				
>>Shared NG-U Multicast TNL Information	M		9.3.2.16	
> <i>location dependent</i>				
>>MBS Session TNL Information 5GC List		1		
>>>MBS Session TNL Information 5GC Item		1.. <i>maxnoofMBSServiceAreaInformation</i>		
>>>>MBS Area Session ID	M		9.3.1.207	
>>>>Shared NG-U Multicast TNL Information	M		9.3.2.16	

Range bound	Explanation
maxnoofMBSServiceAreaInformation	Maximum no. of MBS Service Area Information. Value is 256.

9.3.2.16 Shared NG-U Multicast TNL Information

This IE provides the shared NG UP transport layer information associated with an MBS session at the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IP Multicast Address	M		Transport Layer Address 9.3.2.4	
IP Source Address	M		Transport Layer Address 9.3.2.4	
GTP-TEID at 5GC	M		GTP-TEID 9.3.2.5	

9.3.2.17 MBS Session TNL Information NG-RAN

This IE provides NG-RAN TNL information for location dependent and location independent broadcast MBS sessions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Session Type</i>	M			
> <i>location independent</i>				
>>Shared NG-U Unicast TNL Information	M		UP Transport Layer Information 9.3.2.2	
> <i>location dependent</i>				
>>MBS Session TNL Information NG-RAN List		1		
>>>MBS Session TNL Information NG-RAN Item		1.. <i>maxnoofM BSServiceArea Information</i> >		
>>>>MBS Area Session ID	M		9.3.1.207	
>>>>Shared NG-U Unicast TNL Information	O		UP Transport Layer Information 9.3.2.2	

Range bound	Explanation
maxnoofMBSServiceAreaInformation	Maximum no. of MBS Service Area Information. Value is 256.

9.3.2.18 MBS NG-U Failure Indication

This IE provides information that the 5GC has detected an NG-U failure.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Session Type</i>	M			
> <i>location independent</i>				
>>MBS User Plane Failure Indication	M		9.3.2.19	
> <i>location dependent</i>				
>>MBS UP Failure Indication List		1		
>>>MBS UP Failure Indication Item		1.. <i>maxnoofM BSServiceArea Information</i> >		
>>>>MBS Area Session ID	M		9.3.1.207	
>>>>MBS User Plane Failure Indication	M		9.3.2.19	

Range bound	Explanation
maxnoofMBSServiceAreaInformation	Maximum no. of MBS Service Area Information. Value is 256.

9.3.2.19 MBS User Plane Failure Indication

This IE provides information that the 5GC has detected an NG-U failure for MBS session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS User Plane Failure Indication	M		ENUMERATED (NG-U path failure detected, ...)	

9.3.3 NAS Related IEs

9.3.3.1 AMF UE NGAP ID

This IE uniquely identifies the UE association over the NG interface, as described in TS 38.401 [2].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF UE NGAP ID	M		INTEGER (0..2 ⁴⁰ -1)	

9.3.3.2 RAN UE NGAP ID

This IE uniquely identifies the UE association over the NG interface within the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN UE NGAP ID	M		INTEGER (0..2 ³² -1)	

9.3.3.3 GUAMI

This IE indicates the AMF identity.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
AMF Region ID	M		BIT STRING (SIZE(8))	
AMF Set ID	M		9.3.3.12	
AMF Pointer	M		9.3.3.19	

9.3.3.4 NAS-PDU

This IE contains a 5GC – UE or UE – 5GC message that is transferred without interpretation in the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NAS-PDU	M		OCTET STRING	The content is defined in TS 24.501 [26].

9.3.3.5 PLMN Identity

This IE indicates the PLMN Identity.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		OCTET STRING (SIZE(3))	Digits 0 to 9 encoded 0000 to 1001, 1111 used as filler digit. Two digits per octet: - bits 4 to 1 of octet n encoding digit 2n-1 - bits 8 to 5 of octet n encoding digit 2n PLMN Identity consists of 3 digits from MCC followed by either: - a filler digit plus 2 digits from MNC (in case of 2 digit MNC) or - 3 digits from MNC (in case of 3 digit MNC).

9.3.3.6 SON Configuration Transfer

This IE contains the configuration information, used by e.g., SON functionality, and additionally includes the NG-RAN node identifier of the destination of this configuration information and the NG-RAN node identifier of the source of this information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Target RAN Node ID SON		<i>1</i>			-	
>Global RAN Node ID	M		9.3.1.5		-	
>Selected TAI	M		TAI 9.3.3.11		-	
>NR CGI	O		9.3.1.7	This IE is ignored if the <i>SON Information</i> IE contains the <i>SON Information Reply</i> IE.	YES	ignore
Source RAN Node ID		<i>1</i>			-	
>Global RAN Node ID	M		9.3.1.5		-	
>Selected TAI	M		TAI 9.3.3.11		-	
SON Information	M		9.3.3.7		-	
Xn TNL Configuration Info	C- ifSONInfo rmationR equest		9.3.3.9	Source NG-RAN node Xn TNL Configuration Info.	-	
Requested TNL Information	O		PLMN Identity 9.3.3.5	Indicates to the candidate NG-RAN node the PLMN ID for which the <i>Xn TNL Configuration Info</i> IE is requested if the <i>SON Information</i> IE is set to "Xn TNL Configuration Info". If the <i>SON Information Request</i> IE within the <i>SON Information</i> IE is set to any value other than "Xn TNL Configuration Info", the <i>Requested TNL Information</i> IE is ignored.	YES	ignore

Condition	Explanation
ifSONInformationRequest	This IE shall be present if the <i>SON Information</i> IE contains the <i>SON Information Request</i> IE set to the value "Xn TNL Configuration Info".

9.3.3.7 SON Information

This IE identifies the nature of the configuration information transferred, i.e., a request, a reply or a report.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE SON Information	M				-	
>SON Information Request						
>>SON Information Request	M		ENUMERATED (Xn TNL Configuration Info, ...)		-	
>SON Information Reply						
>>SON Information Reply	M		9.3.3.8		-	
>SON Information Report						
>>SON Information Report	M		9.3.3.35		YES	ignore

9.3.3.8 SON Information Reply

This IE contains the configuration information to be replied to the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Xn TNL Configuration Info	O		9.3.3.9	

9.3.3.9 Xn TNL Configuration Info

This IE is used for signalling Xn TNL Configuration information for automatic Xn SCTP association establishment.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Xn Transport Layer Addresses		<i>1..<maxno ofXnTLAs></i>			-	
>Transport Layer Address	M		9.3.2.4	Transport Layer Addresses for Xn SCTP endpoint.	-	
Xn Extended Transport Layer Addresses		<i>0..<maxno ofXnExtTLAs></i>			-	
>IP-Sec Transport Layer Address	O		Transport Layer Address 9.3.2.4	Transport Layer Addresses for IP-Sec endpoint.	-	
>Xn GTP Transport Layer Addresses		<i>0..<maxno ofXnGTP-TLAs></i>			-	
>>GTP Transport Layer Address	M		Transport Layer Address 9.3.2.4	GTP Transport Layer Addresses for GTP end-points (used for data forwarding over Xn).	-	
>Xn SCTP Transport Layer Addresses		<i>0..<maxno ofXnTLAs></i>			YES	ignore
>>Transport Layer Address SCTP	M		Transport Layer Address 9.3.2.4	Transport Layer Addresses for Xn SCTP endpoint.	-	

Range bound	Explanation
maxnoofXnTLAs	Maximum no. of Xn Transport Layer Addresses for an SCTP end-point. Value is 2.
maxnoofXnExtTLAs	Maximum no. of Xn Extended Transport Layer Addresses in the message. Value is 16.
maxnoofXnGTP-TLAs	Maximum no. of Xn GTP Transport Layer Addresses for a GTP end-point in the message. Value is 16.

9.3.3.10 TAC

This IE is used to uniquely identify a Tracking Area Code.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TAC	M		OCTET STRING (SIZE(3))	

9.3.3.11 TAI

This IE is used to uniquely identify a Tracking Area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
TAC	M		9.3.3.10	

9.3.3.12 AMF Set ID

This IE is used to uniquely identify an AMF Set within the AMF Region.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Set ID	M		BIT STRING (SIZE(10))	

9.3.3.13 Routing ID

This IE is used to identify an LMF or a TSCTSF within the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Routing ID	M		OCTET STRING	The maximum length is 16 octets, referring to the length of a Universally Unique Identifier (UUID) version 4 as specified in section 4.4 in IETF RFC 4122 [53]

9.3.3.14 NRPPa-PDU

This IE contains an NG-RAN node – LMF or LMF – NG-RAN node message that is transferred without interpretation in the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NRPPa-PDU	M		OCTET STRING	

9.3.3.15 RAN Paging Priority

This IE contains the service priority as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Paging Priority	M		INTEGER (1..256)	Values ordered in decreasing order of priority, i.e. with 1 as the highest priority and 256 as the lowest priority

9.3.3.16 EPS TAC

This IE is used to uniquely identify an EPS Tracking Area Code.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
EPS TAC	M		OCTET STRING (SIZE(2))	

9.3.3.17 EPS TAI

This IE is used to uniquely identify an EPS Tracking Area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
EPS TAC	M		9.3.3.16	

9.3.3.18 UE Paging Identity

This IE represents the Identity with which the UE is paged.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>UE Paging Identity</i>	M			
>5G-S-TMSI				
>>5G-S-TMSI	M		9.3.3.20	

9.3.3.19 AMF Pointer

This IE is used to identify one or more AMF(s) within the AMF Set.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Pointer	M		BIT STRING (SIZE(6))	

9.3.3.20 5G-S-TMSI

This IE is used for security reasons, to hide the identity of a subscriber.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Set ID	M		9.3.3.12	
AMF Pointer	M		9.3.3.19	
5G-TMSI	M		OCTET STRING (SIZE(4))	5G-TMSI is unique within the AMF that allocated it.

9.3.3.21 AMF Name

This IE is used to uniquely identify the AMF (see TS 38.300 [8]). It may also be used as a human readable name of the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Name	M		PrintableString (SIZE(1..150, ...))	

9.3.3.22 Paging Origin

This IE indicates whether Paging is originated due to the PDU sessions from the non-3GPP access.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Origin	M		ENUMERATED (non-3GPP, ...)	

9.3.3.23 UE Identity Index Value

This IE is used by the NG-RAN node to calculate the Paging Frame as specified in TS 38.304 [12] and TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>UE Identity Index Value</i>	M			
> <i>Index Length 10</i>				
>> <i>Index Length 10</i>	M		BIT STRING (SIZE(10))	Coded as specified in TS 38.304 [12] and TS 36.304 [29].

9.3.3.24 Periodic Registration Update Timer

This IE is used to assist NG-RAN to generate corresponding timer for periodic RNA update for RRC_INACTIVE UEs.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodic Registration Update Timer	M		BIT STRING (SIZE(8))	Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit for the Periodic Registration Update Timer as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 10 minutes 0 0 1 value is incremented in multiples of 1 hour 0 1 0 value is incremented in multiples of 10 hours 0 1 1 value is incremented in multiples of 2 seconds 1 0 0 value is incremented in multiples of 30 seconds 1 0 1 value is incremented in multiples of 1 minute 1 1 1 value indicates that the timer is deactivated. 1 1 0 value is incremented in multiples of 1 hour in this version of the protocol.

9.3.3.25 UE-associated Logical NG-connection List

This IE contains a list of UE-associated logical NG-connections.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE-associated Logical NG-connection Item		<i>1..<maxnoofNGConnectionsToReset></i>		
>AMF UE NGAP ID	O		9.3.3.1	
>RAN UE NGAP ID	O		9.3.3.2	

Range bound	Explanation
maxnoofNGConnectionsToReset	Maximum no. of UE-associated logical NG-connections allowed to reset in one message. Value is 65536.

9.3.3.26 NAS Security Parameters from NG-RAN

This IE provides security related parameters for inter-system handover from NG-RAN to E-UTRAN or from NG-RAN to UTRAN via the eNB to the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NAS Security Parameters from NG-RAN	M		OCTET STRING	Refers to the <i>N1 mode to S1 mode NAS transparent container</i> IE, the details of the IE definition and the encoding are specified in TS 24.501 [26].

9.3.3.27 Source to Target AMF Information Reroute

This IE is used to transparently pass information provided by NSSF from the source AMF to the target AMF through the NG-RAN node; it is produced by the source core network node and is transmitted to the target core network node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Configured NSSAI	O		OCTET STRING (SIZE(128))	The maximum number of S-NSSAI in Configured NSSAI is 16. This IE contains optional mapping S-NSSAI. When present, this IE shall be transmitted transparent from the source Core network node to the target Core network node. The octets of the OCTET STRING are encoded according to description in TS 29.531 [30].
Rejected NSSAI in PLMN	O		OCTET STRING (SIZE(32))	This IE contain the rejected NSSAI(s) in the PLMN. When present, this IE shall be transmitted transparent from the source Core network node to the target Core network node. The octets of the OCTET STRING are encoded according to description in TS 29.531 [30].
Rejected NSSAI in TA	O		OCTET STRING (SIZE(32))	This IE contain the rejected NSSAI(s) in the TA. When present, this IE shall be transmitted transparent from the source Core network node to the target Core network node. The octets of the OCTET STRING are encoded according to description in TS 29.531 [30].

9.3.3.28 RIM Information Transfer

This IE contains information used by the RIM functionality, and additionally includes the NG-RAN node identifier of the destination of the RIM information and the NG-RAN node identifier of the source of this information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target RAN Node ID RIM		<i>1</i>		
>Global RAN Node ID	M		9.3.1.5	
>Selected TAI	M		TAI 9.3.3.11	
Source RAN Node ID		<i>1</i>		
>Global RAN Node ID	M		9.3.1.5	
>Selected TAI	M		TAI 9.3.3.11	
RIM Information	M		9.3.3.29	

9.3.3.29 RIM Information

This IE contains the RIM information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target gNB Set ID	M		gNB Set ID 9.3.1.122	The victim gNB Set ID.
RIM-RS Detection	M		ENUMERATED (RS detected, RS disappeared, ...)	

9.3.3.30 LAI

This IE is used to uniquely identify a Location Area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LAI				
>PLMN Identity	M		9.3.3.5	
>LAC	M		OCTET STRING (SIZE(2))	0000 and FFFE not allowed.

9.3.3.31 Extended Connected Time

This IE indicates the minimum time the RAN should keep the UE in RRC_CONNECTED state regardless of inactivity, as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended Connected Time	M		INTEGER (0..255)	Minimum time the RAN should keep the UE in RRC_CONNECTED state. Unit is second. Value of "0" indicates that the AMF is aware of pending data traffic, but no specific time value is requested.

9.3.3.32 End Indication

This IE indicates that there are no further NAS PDUs to be transmitted for this UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
End Indication	M		ENUMERATED (no further data, further data exists, ...)	

9.3.3.33 Inter-system SON Configuration Transfer

This IE contains the configuration information, used by e.g., SON functionality, transmitted between an NG-RAN node and an eNB and additionally includes the node identifier of the destination of this configuration information and the node identifier of the source of this information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Transfer Type</i>	M			
> <i>from E-UTRAN to NG-RAN</i>				
>>Source eNB-ID		1		
>>>Global eNB ID	M		9.3.1.165	
>>>Selected EPS TAI	M		EPS TAI 9.3.3.17	
>>Target NG-RAN node ID		1		
>>>Global RAN Node ID	M		9.3.1.5	
>>>Selected TAI	M		TAI 9.3.3.11	
> <i>from NG-RAN to E-UTRAN</i>				
>>Source NG-RAN Node ID		1		
>>>Global RAN Node ID	M		9.3.1.5	
>>>Selected TAI	M		TAI 9.3.3.11	
>>Target eNB-ID		1		
>>>Global eNB ID	M		9.3.1.165	
>>>Selected EPS TAI	M		EPS TAI 9.3.3.17	
Inter-system SON Information	M		9.3.3.34	

9.3.3.34 Inter-system SON Information

This IE identifies the nature of the configuration information transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>Inter-system SON Information</i>	M				-	
> <i>Inter-system SON Information Report</i>						
>>Inter-system SON Information Report	M		9.3.3.36		-	
> <i>Inter-system SON Information Request</i>					YES	ignore
>>Inter-system SON Information Request	M		9.3.3.54		-	
> <i>Inter-system SON Information Reply</i>					YES	ignore
>>Inter-system SON Information Reply	M		9.3.3.55		-	

9.3.3.35 SON Information Report

This IE contains the configuration information to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE SON <i>Information Report</i>	M				-	
> <i>Failure Indication Information</i>						
>> <i>Failure Indication</i>	M		9.3.3.37		-	
> <i>HO Report Information</i>						
>> <i>HO Report</i>	M		9.3.3.39		-	
> <i>Successful HO Report Information</i>					YES	ignore
>> Successful HO Report List		1			-	
>>> Successful HO Report Item		1..<maxno of Successful HO Reports>			-	
>>>>Successful HO Report Container	M		OCTET STRING	Includes the <i>SuccessHO-Report</i> IE as defined in TS 38.331 [18].	-	
> <i>Successful PSCell Change Report Information</i>					YES	ignore
>> Successful PSCell Change Report List		1			-	
>>> Successful PSCell Change Report Item		1..<maxno of Successful PSCell Change Reports>			-	
>>>>Successful PSCell Change Report Container	M		OCTET STRING	Includes the <i>SuccessPSCell-Report</i> IE as defined in TS 38.331 [18].	-	

Range bound	Explanation
maxnoofSuccessfulHOReports	Maximum no. of Successful HO Reports. Value is 64.
maxnoofSuccessfulPSCellChangeReports	Maximum no. of Successful PSCell Change Reports. Value is 64.

9.3.3.36 Inter-system SON Information Report

This IE contains the configuration information to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>Inter-system SON Information Report</i>	M				-	
> <i>HO Report Information</i>						
>>Inter-system HO Report	M		9.3.3.40		-	
> <i>Failure Indication Information</i>						
>>Inter-system Failure Indication	M		9.3.3.38		-	
> <i>Energy Savings Indication</i>					YES	ignore
>>Inter-system Cell State Indication	M		9.3.3.57		-	
> <i>Resource Status Report</i>					YES	ignore
>>Inter-system Resource Status Report	M		9.3.3.60		-	

9.3.3.37 Failure Indication

This IE contains the failure indication to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE RLF Report Container	M		9.3.3.41	

9.3.3.38 Inter-system Failure Indication

This IE contains the failure indication to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE RLF Report Container	O		9.3.3.41	Only contains the LTE RLF report in this version of the specification.

9.3.3.39 HO Report

This IE contains the HO report to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Handover Report Type	M		ENUMERATED (HO too early, HO to wrong cell, Inter-system ping-pong, ...)		-	
Handover Cause	M		Cause 9.3.1.2	Indicates handover cause employed for handover from source cell	-	
Source Cell CGI	M		NG-RAN CGI 9.3.1.73	NG-RAN CGI of the source cell for handover procedure	-	
Target Cell CGI	M		NG-RAN CGI 9.3.1.73	NG-RAN CGI of the target cell for handover procedure. If the Handover Report Type is set to "Inter-system ping-pong", it contains the target cell of the inter system handover from the other system to NG-RAN node cell	-	
Re-establishment Cell CGI	C- ifHandov erReport Type HoToWro ngCell		NG-RAN CGI 9.3.1.73	NG-RAN CGI of the cell where UE attempted re-establishment or where the UE successfully re-connected after the failure	-	
Source Cell C-RNTI	O		BIT STRING (SIZE (16))	C-RNTI allocated at the source NG-RAN node	-	
Target Cell in E-UTRAN	C- ifHandov erReport Type Intersyste mpingpon g		E-UTRA CGI 9.3.1.9	E-UTRA CGI of the E-UTRAN target cell for handover procedure	-	
Mobility Information	O		BIT STRING (SIZE (16))	This IE is not used in the specification. If received, the IE is ignored.	-	
UE RLF Report Container	O		9.3.3.41	The UE RLF Report Container IE received in the FAILURE INDICATION message.	-	
Extended Mobility Information	O		BIT STRING (SIZE (32))	Corresponds to the <i>Mobility Information</i> IE provided in the HANDOVER REQUEST message from the source NG-RAN node	YES	ignore

Target Cell C-RNTI	O		BIT STRING (SIZE (16))	C-RNTI allocated at the target NG-RAN node.	YES	ignore
Time Since Failure	O		INTEGER (0..172800, ...)	Corresponds to the <i>TimeSinceFailure</i> IE received from the UE in RLF Report as defined in TS 36.331 [21].	YES	ignore

Condition	Explanation
ifHandoverReportTypeHoToWrongCell	This IE shall be present if the <i>Handover Report Type</i> IE is set to the value "HO to wrong cell"
ifHandoverReportTypeIntersystempingpong	This IE shall be present if the <i>Handover Report Type</i> IE is set to the value "Inter-system ping-pong"

9.3.3.40 Inter-system HO Report

This IE contains the inter-system HO report to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE <i>Handover Report Type</i>	M				-	
>Too early Inter-system HO						
>>Source Cell ID	M		E-UTRA CGI 9.3.1.9	CGI of the source cell for the HO.	-	
>>Failure Cell ID	M		NG-RAN CGI 9.3.1.73	CGI of the target cell for the HO.	-	
>>UE RLF Report Container	O		9.3.3.41		-	
>Inter-system Unnecessary HO						
>>Source Cell CGI	M		NG-RAN CGI 9.3.1.73	Source NR cell in NG-RAN	-	
>>Target Cell CGI	M		E-UTRA CGI 9.3.1.9	Target cell in E-UTRAN	-	
>>Early IRAT HO	M		ENUMERATED (true, false, ...)	Is set to "true" if the measurement period expired due to an inter-RAT handover towards NR executed within the configured measurement duration and otherwise set to "false"	-	
>>Candidate Cell List		1			-	
>>>Candidate Cell Item		1..<maxno ofCandidateCells>			-	
>>>>CHOICE <i>Candidate Cell Type</i>	M				-	
>>>>>Candidate CGI						
>>>>>Candidate Cell ID	M		NR CGI 9.3.1.7	This IE contains an NR CGI.	-	
>>>>>Candidate PCI						
>>>>>Candidate PCI	M		INTEGER (0..1007, ...)	This IE includes the NR Physical Cell Identifier of detected cells not included in the <i>Candidate Cell List</i> IE and for which an NR CGI could not be derived.	-	
>>>>>Candidate NR ARFCN	M		INTEGER (0..maxNRARFCN)	RF Reference Frequency as defined in TS 38.104 [39], section 5.4.2.1. The frequency provided in this IE identifies the absolute frequency position of the reference resource block (Common RB 0) of the carrier. Its lowest subcarrier is also known as Point A.	-	

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
>Inter-system Mobility Failure for Voice Fallback					YES	ignore
>>Source Cell ID	M		NG-RAN CGI 9.3.1.73	CGI of the source cell for HO	-	
>>Failure Cell ID	M		E-UTRA CGI 9.3.1.9	CGI of the failure cell for HO	-	
>>Re-connect Cell ID	O		E-UTRA CGI 9.3.1.9	E-UTRA CGI of the cell where UE attempted re-establishment or where the UE successfully re-connected after the failure.	-	
>>UE RLF Report Container	O		9.3.3.41	Only contains the LTE RLF report in this version of the specification.	-	

Range bound	Explanation
maxnoofCandidateCells	Maximum no. of candidate cells. Value is 32
maxNRARFCN	Maximum value of NRARFCNs. Value is 3279165.

9.3.3.41 UE RLF Report Container

This IE contains the RLF Report to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>RLF type</i>	M			
>NR				
>>NR UE RLF Report Container	M		OCTET STRING	Includes the <i>nr-RLF-Report</i> contained in the <i>UEInformationResponse</i> message as defined in TS 38.331 [18].
>LTE				
>>LTE UE RLF Report Container	M		OCTET STRING	Includes the <i>RLF-Report-r9</i> contained in the <i>UEInformationResponse</i> message as defined in TS 36.331 [21]

9.3.3.42 NID

This IE is used to identify (together with a PLMN identifier) a Stand-alone Non-Public Network.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NID	M		BIT STRING (SIZE(44))	Defined in TS 23.003 [23].

9.3.3.43 CAG ID

This IE is used to identify (together with a PLMN identifier) a Public Network Integrated NPN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CAG ID	M		BIT STRING (SIZE(32))	Defined in TS 23.003 [23].

9.3.3.44 NPN Support

For SNPN, this IE identifies a supported SNPN together with the associated PLMN ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>NPN Support</i>	M			
> <i>SNPN</i>				
>>NID	M		9.3.3.42	

9.3.3.45 Allowed PNI-NPN List

This IE contains information on allowed UE mobility in PNI-NPN including allowed PNI-NPNs and whether the UE is allowed to access non-CAG cells for each PLMN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Allowed PNI-NPN Item		<i>1..<maxnoofEPLMNs+1></i>		
>PLMN Identity	M		9.3.3.5	
>PNI-NPN Restricted	M		ENUMERATED (restricted, not-restricted, ...)	If set to "restricted", indicates that the UE is not allowed to access non-CAG cells for this PLMN.
>Allowed CAG List per PLMN		<i>1..<maxnoofAllowedCAGsperPLMN></i>		
>>CAG ID	M		9.3.3.43	

Range bound	Explanation
maxnoofEPLMNs+1	Maximum no. of equivalent PLMNs plus one serving PLMN. Value is 16.
maxnoofAllowedCAGsperPLMN	Maximum number of CAGs per PLMN in UE's Allowed PNI-NPN list. Value is 256.

9.3.3.46 NPN Access Information

This IE contains information to perform access control for NPN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>NPN Access Information</i>	M			
> <i>PNI-NPN Access Information</i>				
>>Cell CAG List	M		9.3.3.47	

9.3.3.47 Cell CAG List

This IE indicates the list of CAG IDs supported by a cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell CAG List		<i>1..<maxnoofCAGsperCell></i>		
>CAG ID	M		9.3.3.43	

Range bound	Explanation
maxnoofCAGsperCell	Maximum no. of CAGs per cell. Value is 64. Max is 12 in this release.

9.3.3.48 UL CP Security Information

This IE contains NAS level security information to enable UE authentication by the AMF as described in TS 33.401 [27].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UL NAS MAC	M		BIT STRING (SIZE(16))	Defined in TS 33.401 [27].
UL NAS Count	M		BIT STRING (SIZE(5))	Defined in TS 33.401 [27].

9.3.3.49 DL CP Security Information

This IE contains NAS level security information to be forwarded to the UE as described in TS 33.401 [27].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL NAS MAC	M		BIT STRING (SIZE(16))	Defined in TS 33.401 [27].

9.3.3.50 Configured TAC Indication

This IE indicates that in all NR cells served by the gNB, the TAC with which this IE is associated, is only configured but not broadcast.

NOTE: This IE is defined in accordance to the possibility foreseen in TS 38.331 [18] to not broadcast the TAC if the NR cell only supports PSCell/SCell functionality.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Configured TAC Indication	M		ENUMERATED (true, ...)	

9.3.3.51 Extended AMF Name

This IE provides extended human readable name of the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Name Visible	O		VisibleString (SIZE(1..150, ...))	
AMF Name UTF8	O		UTF8String (SIZE(1..150, ...))	

9.3.3.52 Extended UE Identity Index Value

This IE is used by the NG-RAN node to calculate the Paging Frame and Paging Occasion as specified in TS 36.304 [29], the Paging Frame and Paging Occasion for eDRX and the UE_ID based subgroup ID for PEI as specified in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended UE Identity Index Value	M		BIT STRING (SIZE(16))	Encoded as 5G-S-TMSI mod 65536.

9.3.3.53 NR NTN TAI Information

This IE contains the TAI information for NR NTN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Serving PLMN	M		PLMN Identity 9.3.3.5	Indicates the UE's serving PLMN.
TAC List in NR NTN		1..<maxnoofTACsinNTN>		Includes all TAC(s) broadcast in the cell, for the UE's serving PLMN.
>TAC	M		9.3.3.10	
UE Location Derived TAC in NR NTN	O		TAC 9.3.3.10	This IE contains TAC information derived from the actual UE location, if available.

Range bound	Explanation
maxnoofTACsinNTN	Maximum no. of TACs broadcast per cell. Value is 12.

9.3.3.54 Inter-system SON Information Request

This IE contains the request information to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Inter-system SON Information Request</i>	M			
>NG-RAN Cell Activation				
>>Inter-system Cell Activation Request	M		9.3.3.56	
>Resource Status				
>>Inter-system Resource Status Request	M		9.3.3.59	

9.3.3.55 Inter-system SON Information Reply

This IE contains the reply information to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Inter-system SON Information Reply</i>	M			
>NG-RAN Cell Activation				
>>Inter-system Cell Activation Reply	M		9.3.3.58	
>Resource Status				
>>Inter-system Resource Status Reply	M		9.3.3.61	

9.3.3.56 Inter-system Cell Activation Request

This IE contains request information for inter-system Cell Activation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Activation ID	M		INTEGER (0..16384, ...)	Allocated by the eNB.
Cells to Activate List		<i>1..<maxnoofCellsinNGRANNode></i>		
>NG-RAN CGI	M		9.3.1.73	

Range bound	Explanation
maxnoofCellsinNGRANNode	Maximum no. of cells that can be served by a NG-RAN node. Value is 16384.

9.3.3.57 Inter-system Cell State Indication

This IE contains notification information for inter-system Cell Activation and Deactivation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Notification Cell List		<i>1</i>		
>Notification Cell Item		<i>1..<maxnoofCellsinNGRANNode></i>		
>>NG-RAN CGI	M		9.3.1.73	
>>Notify Flag	M		ENUMERATED (Activated, Deactivated, ...)	

Range bound	Explanation
maxnoofCellsinNGRANNode	Maximum no. of cells that can be served by a NG-RAN node. Value is 16384.

9.3.3.58 Inter-system Cell Activation Reply

This IE contains reply information for inter-system Cell Activation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Activated Cells List		<i>1..<maxnoofCellsinNGRANNode></i>		
>NG-RAN CGI	M		9.3.1.73	
Activation ID	M		INTEGER (0..16384, ...)	Allocated by the eNB.

Range bound	Explanation
maxnoofCellsinNGRANNode	Maximum no. of cells that can be served by a NG-RAN node. Value is 16384.

9.3.3.59 Inter-system Resource Status Request

This IE contains information on the requested Inter-system Load Reporting reporting.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reporting System	M		9.3.1.237	
Report Characteristics	M		BITSTRING (SIZE(32))	Each position in the bitmap indicates measurement object the receiving node is requested to report. First Bit = Number of Active UEs, Second Bit = RRC connections, Third Bit = PRB. Other bits reserved for future use.
CHOICE <i>report type</i>	M			
> <i>Event based reporting</i>				
>>Inter-system Resource Threshold Low	M		INTEGER (0..100)	Inter-system Resource Status reporting is enabled when Composite Available Capacity is above this threshold or is disabled when Composite Available Capacity is below this threshold. The reporting node sends a report when the cell Composite Available Capacity becomes greater than or equal to the threshold.
>>Inter-system Resource Threshold High	M		INTEGER (0..100)	Inter-system Resource Status reporting is enabled when Composite Available Capacity is below this threshold or is disabled when Composite Available Capacity is above this threshold. The reporting node sends a report when the cell Composite Available Capacity becomes smaller than or equal to the threshold.
>>Number of Measurement Reporting Levels	M		ENUMERATED (2, 3, 4, 5, 10, ..., 0)	The reporting node divides the cell load scale into the indicated number of reporting levels, evenly distributed on a linear scale between the Inter-system Resource Threshold Low and the Inter-system Resource Threshold High. The reporting node sends a report each time the cell load changes from one reporting level to another. If the value is "0", the IE is ignored and only the Inter-system Resource Threshold Low and the Inter-system Resource Threshold High are used as the reporting levels.
> <i>Periodic Reporting</i>				
>>Reporting Periodicity	M		ENUMERATED (stop, single, 1000ms, 2000ms, 5000ms, 10000ms, ...)	Periodicity that can be used for reporting. If the value is "stop", the reporting node is asked to stop the periodic reporting. If the value is "single" there is only one report.

Range bound	Explanation
maxnoofReportedCells	Maximum no. of cells that can be reported. Value is 256.

9.3.3.60 Inter-system Resource Status Report

This IE contains the Inter-system load report.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Reporting System</i>	M			
> <i>E-UTRAN</i>				
>> <i>E-UTRAN Cell Report List</i>		1		
>>> <i>E-UTRAN Cell Report Item</i>		1.. <i>maxnoofReportedCells</i>		
>>>>Cell ID	M		E-UTRA CGI 9.3.1.9	
>>>>E-UTRAN Composite Available Capacity Group	M		9.3.1.196	Corresponds to the <i>Composite Available Capacity Group</i> IE as defined in TS 36.423 [40].
>>>>Number of Active UEs	O		INTEGER (0..16777215, ...)	Corresponds to the <i>Number of Active UEs</i> IE as defined in TS 36.314 [49].
>>>>RRC Connections	O		INTEGER (1..65536, ...)	Corresponds to the <i>RRC Connections</i> IE as defined in TS 38.423 [24].
>>>>E-UTRAN Radio Resource Status	O		9.3.1.200	
> <i>NG-RAN</i>				
>> <i>NG-RAN Cell Report List</i>		1		
>>> <i>NG-RAN Cell Report Item</i>		1.. <i>maxnoofReportedCells</i>		
>>>>Cell ID	M		NG-RAN CGI 9.3.1.73	
>>>>NR Composite Available Capacity Group	M		E-UTRAN Composite Available Capacity Group 9.3.1.196	Corresponds to the <i>Composite Available Capacity Group</i> IE as defined in TS 36.423 [40].
>>>>Number of Active UEs	O		INTEGER (0..16777215, ...)	Corresponds to the <i>Number of Active UEs</i> IE as defined in TS 38.423 [24].
>>>>RRC Connections	O		INTEGER (1..65536, ...)	Corresponds to the <i>RRC Connections</i> IE as defined in TS 38.423 [24].
>>>>NR Radio Resource Status	O		9.3.1.205	

Range bound	Explanation
maxnoofReportedCells	Maximum no. of cells that can be reported. Value is 256.

9.3.3.61 Inter-system Resource Status Reply

This IE indicates for which cell(s) the inter-system load reporting was successfully initiated.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reporting System	M		9.3.1.237	

9.3.3.62 Hashed UE Identity Index Value

This IE is the 13 Most Significant Bits (MSBs) of the Hashed ID defined in TS 38.304 [12] or TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Hashed UE Identity Index Value	M		BIT STRING (SIZE(13, ...))	

9.3.3.63 Paging Policy Differentiation

This IE provides paging policy differentiation information for a UE in RRC_INACTIVE state.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session for Paging List		1		
>PDU Session for Paging Item		1.. <i>maxnoofPDUSessions</i>		
>>PDU Session ID	M		9.3.1.50	
>>Paging Policy Differentiation List		1		
>>>Paging Policy Differentiation Item		1.. <i>maxnoofQoSFlows</i>		
>>>>QoS Flow Identifier	O		9.3.1.51	
>>>>Paging Policy Indicator	O		INTEGER (0..7, ...)	Used for paging policy differentiation as specified in TS 23.501 [9].
>>>>Allocation and Retention Priority	O		9.3.1.19	
>>>>5QI	O		INTEGER (0..255, ...)	Indicates the 5QI associated with the PDU session
>>>>DL Data Size	O		INTEGER (0..96000, ...)	Unit: Byte.

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.3.64 NR Paging Long eDRX Information for RRC INACTIVE

This IE indicates the NR Paging long eDRX parameters as defined in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR Paging Long eDRX Cycle for RRC INACTIVE	M		ENUMERATED (hf2, hf4, hf8, hf16, hf32, hf64, hf128, hf256, hf512, hf1024, ...)	$T_{eDRX, RAN}$ defined in TS 38.304 [12]. Unit: [number of hyperframes].
NR Paging Time Window for RRC_INACTIVE	M		ENUMERATED (s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, s17, s18, s19, s20, s21, s22, s23, s24, s25, s26, s27, s28, s29, s30, s31, s32, ...)	Unit: [1.28 seconds]

9.3.3.65 CAG List for MDT

This IE is used to identify the list of Public Network Integrated NPNs for MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CAG List for MDT		<i>1</i>		For logged MDT, this list contains a maximum of 144 Public Network Integrated NPNs with a maximum of 12 different PLMN identities, configurable to the <i>CAG-ConfigList</i> IE defined in TS 38.331 [18] where a PLMN ID may be repeated more than once.
>CAG List for MDT Item		<i>1..<maxnoofCAGforMDT></i>		
>>PLMN Identity	M		9.3.3.5	
>>CAG ID	M		9.3.3.43	

Range bound	Explanation
maxnoofCAGforMDT	Maximum no. of CAG IDs for MDT area scope. Value is 256.

9.3.3.66 AUN3 Device Access Information

This IE contains AUN3 device access information for the AUN3 device registration as described in TS 23.316 [34].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AUN3 Device Access	M		ENUMERATED (true, ...)	
Existing NG Connection with Same Location	M		ENUMERATED (true, false, ...)	

9.3.3.67 Network Slice Area Scope of MDT

This IE is used to identify the list of network slices for MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Network Slice List for MDT		<i>1</i>		
>Network Slice Item for MDT		<i>1..<maxnoofMDTPLMNs></i>		
>>PLMN Identity	M		9.3.3.5	
>>Slice MDT List		<i>1</i>		
>>>Slice MDT Item		<i>1..<maxnoofSliceItemsforMDT></i>		
>>>>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofMDTPLMNs	Maximum no. of PLMNs in the MDT PLMN list. Value is 16.
maxnoofSliceItemsforMDT	Maximum no. of S-NSSAIs for MDT area scope. Value is 1024.

9.3.3.68 Geographical Area

This IE is used to indicate the geographical area scope for NTN MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NTN Geographical Area		<i>1..<maxnoofAreaNTN></i>		
>CHOICE Area Type	M			
>>Circle				
>>>Reference Location	M		OCTET STRING	ReferenceLocation-r17 as defined in TS 38.331[18]
>>>Distance Radius	M		INTEGER(1..65535)	Each step represents 50m distance.
>>Polygon				
>>>Polygon	M		OCTET STRING	Defined in TS 37.355 [59]. The first/leftmost bit of the first octet contains the most significant bit.
MDT PLMN List	O		9.3.1.168	

Range bound	Explanation
maxnoofAreaNTN	Maximum no. the geographical area configurations. Value is 8.

9.3.3.69 A-IoT Correlation Identifier

This IE uniquely identifies an A-IoT session within an AIOTF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Correlation Identifier	M		INTEGER (0..65535, ...)	

9.3.3.70 AIOTF Identifier

This IE uniquely identifies the AIOTF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AIOTF Identifier	M		OCTET STRING (SIZE(16..36, ...))	Coded as the <i>NfInstanceId</i> IE defined in TS 29.571 [35].

9.3.3.71 Requested Service Area Information

This IE includes the Requested Service Area Information for the A-IoT inventory.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Requested Reader List		<i>0..1</i>		
>Requested Reader Item		<i>1..<maxnoofReaders></i>		
>>Global gNB ID	M		9.3.1.6	
>>Reader Index	M		9.3.1.276	
Requested A-IoT Area List		<i>0..1</i>		
>Requested A-IoT Area Item		<i>1..<maxnoofAloTAreas></i>		
>>A-IoT Area ID	M		9.3.1.277	

Range bound	Explanation
maxnoofReaders	Maximum no. of Readers. Value is 65535.
maxnoofAloTAreas	Maximum no. of A-IoT Areas. Value is 256.

9.3.3.72 Inventory Assistance Information

This IE includes the assistance information for the A-IoT Inventory Service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Expected D2R Message Size	M		INTEGER (1..256,...)	Unit: Byte. This IE indicates the expected size of the A-IoT NAS PDU carried in Inventory Report.
Approximate Number of Target A-IoT Devices	O		INTEGER (1..65535, ...)	
Time Interval	O		INTEGER (1..256, ...)	Unit: 0.1 second. This IE indicates the time interval as defined in TS 23.369 [60].

9.3.3.73 A-IoT Device Identification Requested

This IE represents the Requested Identity of the A-IoT device(s).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>Device Identification for Paging</i>	M			
>Single Device				
>>Single Device Identity	M		BIT STRING (SIZE(1..1024, ...))	Includes the A-IoT Identification Information defined in TS 23.003 [23].
>Group Devices				
>>Group Device Identity	M		BIT STRING (SIZE(1..1024, ...))	Includes the A-IoT Identification Information defined in TS 23.003 [23].
>All Devices			NULL	

9.3.3.74 RAN A-IoT Device NGAP ID

This IE uniquely identifies the A-IoT Device over the NG interface within the NG-RAN node within a A-IoT Session, as described in TS 38.401 [2].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN A-IoT Device NGAP ID	M		INTEGER (0..2 ³² -1)	

9.3.3.75 Command Assistance Information

This IE includes the assistance information for the A-IoT Command Service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Estimate of Expected D2R Message Size	M		INTEGER (1.. 256, ...)	Unit: Byte. This IE indicates the size of the A-IoT NAS PDU carried in Command Response.

9.3.3.76 AIOTF Name

This IE is used to uniquely identify the AIOTF. It may also be used as a human readable name of the AIOTF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AIOTF Name Visible	O		VisibleString (SIZE(1..150, ...))	
AIOTF Name UTF8	O		UTF8String (SIZE(1..150, ...))	

9.3.3.77 Further Extended UE Identity Index Value

This IE is used by the NG-RAN node to calculate the Paging Frame and Paging Occasion for eDRX and the UE_ID based subgroup ID for PEI and LP-WUS as specified in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Further Extended UE Identity Index Value	M		BIT STRING (SIZE(20))	Encoded as 5G-S-TMSI mod 1048576.

9.3.3.78 A-IoT Security Information

This IE contains A-IoT security information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE A-IoT Security Information	M			
>A-IoT RAND_n				
>>RAND_AIOT_n	M		BIT STRING (SIZE (128, ...))	Contains the RAND _{AIOT_n} as defined in TS 33.369 [61].

9.3.4 SMF Related IEs

9.3.4.1 PDU Session Resource Setup Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
PDU Session Aggregate Maximum Bit Rate	O		9.3.1.102	This IE shall be present when at least one Non-GBR QoS flow is being setup and is ignored otherwise.	YES	reject
UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs.	YES	reject
Additional UL NG-U UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	UPF endpoint of the additional NG-U transport bearer(s), for delivery of UL PDUs for split PDU session.	YES	reject
Data Forwarding Not Possible	O		9.3.1.63	This IE may be present in case of HANDOVER REQUEST message and is ignored otherwise.	YES	reject
PDU Session Type	M		9.3.1.52		YES	reject
Security Indication	O		9.3.1.27		YES	reject
Network Instance	O		9.3.1.113	This IE is ignored if the <i>Common Network Instance</i> IE is included.	YES	reject
QoS Flow Setup Request List		1			YES	reject
>QoS Flow Setup Request Item		1..<maxno of QoS Flows>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>QoS Flow Level QoS Parameters	M		9.3.1.12		-	
>>E-RAB ID	O		9.3.2.3		-	
>>TSC Traffic Characteristics	O		9.3.1.130	Traffic pattern information associated with the QFI. Details in TS 23.501 [9].	YES	ignore
>>Redundant QoS Flow Indicator	O		9.3.1.134	This IE indicates whether this QoS flow is requested for the redundant transmission.	YES	ignore
>>ECN Marking or Congestion Information Reporting Request	O		9.3.1.266		YES	ignore
Common Network Instance	O		9.3.1.120		YES	ignore
Direct Forwarding Path Availability	O		9.3.1.64	This IE may be present in case of inter-system handover and intra-system handover.	YES	ignore
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs for the redundant transmission.	YES	ignore

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Additional Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	UPF endpoint of the additional NG-U transport bearer(s), for delivery of redundant UL PDUs for split PDU session.	YES	ignore
Redundant Common Network Instance	O		Common Network Instance 9.3.1.120		YES	ignore
Redundant PDU Session Information	O		9.3.1.136		YES	ignore
MBS Session Setup Request List	O		9.3.1.211		YES	ignore
Downlink TL Container	O		OCTET STRING	Containing the Get Request message specified in TS 29.585 [56].	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.2 PDU Session Resource Setup Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL QoS Flow per TNL Information	M		QoS Flow per TNL Information 9.3.2.8	NG-RAN node endpoint of the NG-U transport bearer for delivery of DL PDUs, together with associated QoS flows.	-	
Additional DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of DL PDUs for split PDU session, together with associated QoS flows and corresponding to the <i>Additional UL NG-U UP TNL Information</i> IE in the <i>PDU Session Resource Setup Request Transfer</i> IE.	-	
Security Result	O		9.3.1.59		-	
QoS Flow Failed to Setup List	O		QoS Flow List with Cause 9.3.1.13		-	
Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information 9.3.2.8	NG-RAN node endpoint of the NG-U transport bearer(s) for delivery of DL PDUs of the indicated Redundant QoS Flow(s) and corresponding to the <i>Redundant UL NG-U UP TNL Information</i> IE in the <i>PDU Session Resource Setup Request Transfer</i> IE.	YES	ignore
Additional Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of redundant DL PDUs for split PDU session, together with associated QoS flows and corresponding to the <i>Additional Redundant UL NG-U UP TNL Information</i> IE in the <i>PDU Session Resource Setup Request Transfer</i> IE.	YES	ignore

Used RSN Information	O		Redundant PDU Session Information 9.3.1.136		YES	ignore
Global RAN Node ID of Secondary NG-RAN Node	O		Global RAN Node ID 9.3.1.5		YES	ignore
MBS Support Indicator	O		9.3.1.210		YES	ignore
MBS Session Setup Response List	O		9.3.1.213		YES	ignore
MBS Session Failed to Setup List	O		9.3.1.214		YES	ignore
QoS Flow TSC List		<i>0..1</i>			YES	ignore
>QoS Flow TSC Item		<i>1..<maxno ofQoSFlows></i>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>TSC Traffic Characteristics Feedback	O		9.3.1.257		-	
>>AN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Indicates the dynamic value of 5G-AN PDB in UL direction for the QoS flow as specified in TS 23.501 [9]	-	
Uplink TL Container	O		OCTET STRING	Containing the Get Response message specified in TS 29.585 [56].	YES	ignore
ECN Marking or Congestion Information Reporting Status	O		9.3.1.267		YES	ignore
PDU Set based Handling Indicator	O		9.3.1.268		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.3 PDU Session Resource Modify Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
PDU Session Aggregate Maximum Bit Rate	O		9.3.1.102		YES	reject
UL NG-U UP TNL Modify List		0..1			YES	reject
>UL NG-U UP TNL Modify Item		1..<maxno ofMultiConnectivity>		This IE(s) are included only for modification of an existing tunnel.	-	
>>UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs.	-	
>>DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the NG-RAN node.	-	
>>Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs for the redundant transmission.	YES	ignore
>>Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the NG-RAN node for the redundant transmission.	YES	ignore
Network Instance	O		9.3.1.113	This IE is ignored if the <i>Common Network Instance</i> IE is included.	YES	reject
QoS Flow Add or Modify Request List		0..1			YES	reject
>QoS Flow Add or Modify Request Item		1..<maxno ofQoSFlows>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>QoS Flow Level QoS Parameters	O		9.3.1.12		-	
>>E-RAB ID	O		9.3.2.3		-	
>>TSC Traffic Characteristics	O		9.3.1.130	Traffic pattern information associated with the QFI. Details in TS 23.501 [9].	YES	ignore
>>Redundant QoS Flow Indicator	O		9.3.1.134	This IE indicates whether this QoS flow is requested for the redundant transmission.	YES	ignore
>>UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	This IE indicates the UPF endpoint of NG-U transport bearer for the QoS flow.	YES	ignore
>>Downlink TL Container	O		OCTET STRING	Containing the Set Request message specified in TS 29.585 [56].	YES	ignore
>>ECN Marking or Congestion Information Reporting Request	O		9.3.1.266		YES	ignore
QoS Flow to Release List	O		QoS Flow List with Cause 9.3.1.13		YES	reject

Additional UL NG-U UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	UPF endpoint of the additional NG-U transport bearer(s) proposed for delivery of UL PDUs for split PDU session.	YES	reject
Common Network Instance	O		9.3.1.120		YES	ignore
Additional Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	UPF endpoint of the additional NG-U transport bearer(s) proposed for delivery of redundant UL PDUs for split PDU session.	YES	ignore
Redundant Common Network Instance	O		Common Network Instance 9.3.1.120		YES	ignore
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs for the redundant transmission of the Redundant QoS Flow(s).	YES	ignore
Security Indication	O		9.3.1.27		YES	ignore
MBS Session Setup or Modify Request List	O		9.3.1.212		YES	ignore
MBS Session To Release List	O		9.3.1.215		YES	ignore
User Plane Failure Indication	O		9.3.1.271		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.
maxnoofMultiConnectivity	Maximum no. of connectivity allowed for a UE. Value is 4. The current version of the specification supports up to 2 connectivity.

9.3.4.4 PDU Session Resource Modify Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.	-	
UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the 5GC node.	-	
QoS Flow Add or Modify Response List		0..1			-	
>QoS Flow Add or Modify Response Item		1..<maxno of QoS Flows>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.152	Index to the currently fulfilled alternative QoS parameters set	YES	Ignore
>>TSC Traffic Characteristics Feedback	O		9.3.1.257		YES	ignore
>>Uplink TL Container	O		OCTET STRING	Containing the Set Response message specified in TS 29.585 [56].	YES	ignore
>>DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of NG-U transport bearer for the QoS flow.	YES	ignore
>>AN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Indicates the dynamic value of 5G-AN PDB in UL direction for the QoS flow as specified in TS 23.501 [9].	YES	ignore
Additional DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of DL PDUs for split PDU session, together with associated QoS flows.	-	
QoS Flow Failed to Add or Modify List	O		QoS Flow List with Cause 9.3.1.13		-	
Additional NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer corresponding to the modified UPF endpoint received in the <i>PDU Session Resource Modify Request Transfer</i> IE in case of PDU session split.	YES	ignore

Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs for the redundant transmission.	YES	ignore
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the 5GC node for the redundant transmission.	YES	ignore
Additional Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of redundant DL PDUs for split PDU session, together with associated QoS flows.	YES	ignore
Additional Redundant NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer for delivery of redundant DL PDUs corresponding to the modified UPF endpoint(s) received in the <i>UL NG-U UP TNL Modify List</i> IE of the <i>PDU Session Resource Modify Request Transfer</i> IE in case of PDU session split.	YES	ignore
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore
MBS Support Indicator	O		9.3.1.210		YES	ignore
MBS Session Setup or Modify Response List	O		MBS Session Setup Response List 9.3.1.213		YES	ignore
MBS Session Failed to Setup or Modify List	O		MBS Session Failed to Setup List 9.3.1.214		YES	ignore
ECN Marking or Congestion Information Reporting Status	O		9.3.1.267		YES	ignore
PDU Set based Handling Indicator	O		9.3.1.268		YES	ignore
User Plane Failure Indication Report	O		ENUMERATED (QoS flows and tunnel to be released, QoS flows to be moved, new transport address allocated, ...)		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.5 PDU Session Resource Notify Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QoS Flow Notify List		<i>0..1</i>			-	
>QoS Flow Notify Item		<i>1..<maxno of QoSFlows></i>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Notification Cause	M		ENUMERATED (fulfilled, not fulfilled, ..., not fulfilled DL, not fulfilled UL)	Details in TS 23.501 [9].	-	
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Notify Index 9.3.1.153	Index to the currently fulfilled alternative QoS parameters set. Value 0 indicates that NG-RAN cannot even fulfil the lowest alternative parameters set.	YES	Ignore
>>TSC Traffic Characteristics Feedback	O		9.3.1.257		YES	ignore
QoS Flow Released List	O		QoS Flow List with Cause 9.3.1.13		-	
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore
QoS Flow Feedback List		<i>0..1</i>			YES	ignore
>QoS Flow Feedback Item		<i>1..<maxno of QoSFlows></i>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Update Feedback	O		BIT STRING { CN PDB DL(0), CN PDB UL(1)} (SIZE(8, ...))	Each position in the bitmap represents a QoS parameter. If a bit is set to "1", the respective parameter was not updated. If a bit is set to "0", the respective parameter was successfully updated. Bits 2-7 reserved for future use.	-	
>>CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	Indicates when the packet delay budget downlink was not updated in path switch that NG-RAN can offer this value	-	
>>CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Indicates when the packet delay budget uplink was not updated in path switch that NG-RAN can offer this value	-	

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.6 PDU Session Resource Modify Indication Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL QoS Flow per TNL Information	M		QoS Flow per TNL Information 9.3.2.8	NG-RAN node endpoint of the NG-U transport bearer for delivery of DL PDUs, together with associated QoS flows.	-	
Additional DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of DL PDUs for split PDU session, together with associated QoS flows	-	
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore
Security Result	O		9.3.1.59	Current UP security status	YES	ignore
Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information 9.3.2.8	NG-RAN node endpoint of the NG-U transport bearer for delivery of DL PDUs for the redundant transmission, together with associated QoS flows.	YES	ignore
Additional Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of Redundant DL PDUs for split PDU session, together with associated QoS flows.	YES	ignore
Global RAN Node ID of Secondary NG-RAN Node	O		Global RAN Node ID 9.3.1.5		YES	ignore
ECN Marking or Congestion Information Reporting Status	O		9.3.1.267		YES	ignore

9.3.4.7 PDU Session Resource Modify Confirm Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QoS Flow Modify Confirm List		1			-	
>QoS Flow Modify Confirm Item		1..<maxno of QoSFlows>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer corresponding to the <i>DL QoS Flow per TNL Information</i> IE received in the <i>PDU Session Resource Modify Indication Transfer</i> IE.	-	
Additional NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer indicated in the <i>PDU Session Resource Modify Indication Transfer</i> IE and the corresponding UPF endpoint for split PDU session.	-	
QoS Flow Failed to Modify List	O		QoS Flow List with Cause 9.3.1.13		-	
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer identified by the above redundant DL NG-U UP TNL Information IE for the redundant transmission.	YES	ignore
Additional Redundant NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer for the redundant transmission indicated in the <i>PDU Session Resource Modify Indication Transfer</i> IE and the corresponding UPF endpoint for split PDU session.	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.8 Path Switch Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.	-	
DL NG-U TNL Information Reused	O		ENUMERATED (true, ...)	Indicates that DL NG-U TNL Information has been reused.	-	
User Plane Security Information	O		9.3.1.60		-	
QoS Flow Accepted List		1		QoS flows associated with the <i>DL NG-U UP TNL Information</i> IE.	-	
>QoS Flow Accepted Item		<i>1..<maxno of QoS Flows></i>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.152	Index to the currently fulfilled alternative QoS parameters set.	YES	ignore
Additional DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of DL PDUs for split PDU session, together with associated QoS flows.	YES	ignore
Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of redundant DL PDUs.	YES	ignore
Redundant DL NG-U TNL Information Reused	O		ENUMERATED (true, ...)	Indicates that Redundant DL NG-U TNL Information has been reused.	YES	ignore
Additional Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of Redundant DL PDUs for split PDU session, together with associated QoS flows.	YES	ignore
Used RSN Information	O		Redundant PDU Session Information 9.3.1.136		YES	ignore
Global RAN Node ID of Secondary NG-RAN Node	O		Global RAN Node ID 9.3.1.5		YES	ignore
MBS Support Indicator	O		9.3.1.210		YES	ignore
PDU Set based Handling Indicator	O		9.3.1.268		YES	ignore
ECN Marking or Congestion Information Reporting Status	O		9.3.1.267		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.9 Path Switch Request Acknowledge Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer corresponding to the DL NG-U UP TNL Information IE received in the Path Switch Request Transfer IE.	-	
Security Indication	O		9.3.1.27		-	
Additional NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer indicated in the Path Switch Request Transfer IE and the corresponding UPF endpoint for split PDU session.	YES	ignore
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs for the redundant transmission.	YES	ignore
Additional Redundant NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer for the redundant transmission indicated in the Path Switch Request Transfer IE and the corresponding UPF endpoint for split PDU session.	YES	ignore
QoS Flow Parameters List		<i>0..1</i>			YES	ignore
>QoS Flow Parameters Item		<i>1..<maxno of QoS Flows></i>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Alternative QoS Parameters Set List	O		9.3.1.151	Indicates alternative sets of QoS parameters for the QoS flow.	-	
>>CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore

>>CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
>>Burst Arrival Time Downlink	O		Burst Arrival Time 9.3.1.133	Indicates the downlink Burst Arrival Time of the TSC QoS flow	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.10 Handover Command Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	To deliver forwarded DL PDUs per PDU session tunnel.	-	
QoS Flow to be Forwarded List		0..1			-	
>QoS Flow to be Forwarded Item		1..<maxno ofQoSFlows>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
Data Forwarding Response DRB List	O		9.3.1.77		-	
Additional DL Forwarding UP TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint to deliver forwarded DL PDUs for split PDU session tunnel, together with associated QoS flows to be forwarded.	YES	ignore
UL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	To deliver forwarded UL PDUs	YES	reject
Additional UL Forwarding UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	NG-RAN node endpoint to deliver forwarded UL PDUs for split PDU session tunnel.	YES	reject
Data Forwarding Response E-RAB List	O		9.3.1.121		YES	ignore
QoS Flow Failed to Setup List	O		QoS Flow List with Cause 9.3.1.13		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.11 Handover Request Acknowledge Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.	-	
DL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	To deliver forwarded DL PDUs.	-	
Security Result	O		9.3.1.59		-	
QoS Flow Setup Response List	M		QoS Flow List with Data Forwarding 9.3.2.13	QoS flows associated with the <i>DL NG-U UP TNL Information</i> IE.	-	
QoS Flow Failed to Setup List	O		QoS Flow List with Cause 9.3.1.13		-	
Data Forwarding Response DRB List	O		9.3.1.77		-	
Additional DL UP TNL Information for HO List		0..1			YES	ignore
>Additional DL UP TNL Information for HO Item		1..<maxno ofMultiConnectivityMinusOne>		Additional DL UP TNL Information for split PDU session, in the same order as the UPF endpoint of the additional NG-U transport bearer(s) received in the <i>Handover Request Transfer</i> IE of the Handover Request message.	-	
>>Additional DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the additional NG-U transport bearer for delivery of DL PDUs.	-	
>>Additional QoS Flow Setup Response List	M		QoS Flow List with Data Forwarding 9.3.2.13	QoS flows associated with the <i>Additional DL NG-U UP TNL Information</i> IE.	-	
>>Additional DL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint to deliver forwarded DL PDUs.	-	
>>Additional Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the additional NG-U transport bearer for delivery of redundant DL PDUs.	YES	ignore
UL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	To deliver forwarded UL PDUs	YES	reject
Additional UL Forwarding UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	NG-RAN node endpoint to deliver forwarded UL PDUs for split PDU session.	YES	reject

Data Forwarding Response E-RAB List	O		9.3.1.121		YES	ignore
Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs for the redundant transmission.	YES	ignore
Used RSN Information	O		Redundant PDU Session Information 9.3.1.136		YES	ignore
Global RAN Node ID of Secondary NG-RAN Node	O		Global RAN Node ID 9.3.1.5		YES	ignore
MBS Support Indicator	O		9.3.1.210		YES	ignore
PDU Set based Handling Indicator	O		9.3.1.268		YES	ignore
ECN Marking or Congestion Information Reporting Status	O		9.3.1.267		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.
maxnoofMultiConnectivityMinusOne	Maximum no. of connectivity allowed for a UE minus one. Value is 3. The current version of the specification supports 1.

9.3.4.12 PDU Session Resource Release Command Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Cause	M		9.3.1.2		-	
QoS Flow Additional Information List		0..1			YES	ignore
>QoS Flow Additional Information Item		1..<maxno ofQoSFlows>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Downlink TL Container	O		OCTET STRING	Containing the Set Request message specified in TS 29.585 [56].	-	

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.13 PDU Session Resource Notify Released Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Cause	M		9.3.1.2		-	
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore
User Plane Error Indicator	O		ENUMERATED (GTP-U Error Indication Received, ...)		YES	ignore

9.3.4.14 Handover Required Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Direct Forwarding Path Availability	O		9.3.1.64	

9.3.4.15 Path Switch Request Setup Failed Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.16 PDU Session Resource Setup Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.4.17 PDU Session Resource Modify Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.4.18 Handover Preparation Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.19 Handover Resource Allocation Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.4.20 Path Switch Request Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.21 PDU Session Resource Release Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore
QoS Flow Additional Information List		0..1			YES	ignore
>QoS Flow Additional Information Item		1..<maxno of QoS Flows>			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Uplink TL Container	O		OCTET STRING	Containing the Set Response message specified in TS 29.585 [56].	-	

Range bound	Explanation
maxno of QoS Flows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.22 PDU Session Resource Modify Indication Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.23 Secondary RAT Data Usage Report Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Secondary RAT Usage Information	O		9.3.1.114	

9.3.4.24 UE Context Resume Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Failed to Resume List	O		QoS Flow List with Cause 9.3.1.13	

9.3.4.25 UE Context Resume Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Failed to Resume List	O		QoS Flow List with Cause 9.3.1.13	

9.3.4.26 UE Context Suspend Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Suspend Indicator	O		ENUMERATED (true, ...)	

9.3.5 MB-SMF Related IEs

9.3.5.1 Void

9.3.5.2 Void

9.3.5.3 MBS Session Setup or Modification Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MBS Session TNL Information 5GC	O		9.3.2.15		YES	reject
MBS QoS Flows To Be Setup or Modified List	M		MBS QoS Flows To Be Setup List 9.3.1.236		YES	reject
MBS Session FSA ID List		<i>0..<maxno of MBSFS As></i>			YES	ignore
>MBS Frequency Selection Area Identity	M		OCTET STRING (SIZE(3))		-	
Supported UE Type List		<i>0..<maxno of UE Types></i>			YES	ignore
>Supported UE Type	M		ENUMERATED (Non-(e)RedCap UE, (e)RedCap UE, ...)		-	
MBS NG-U Failure Indication	O		9.3.2.18		YES	reject

Range bound	Explanation
maxnoofMBSQoSFlows	Maximum no. of QoS Flows allowed within one MBS session. Value is 64.
maxnoofMBSFSAs	Maximum no. of FSA IDs for one MBS session. Value is 64.
maxnoofUETypes	Maximum no. of associated UE types. Value is 8.

9.3.5.4 Void

9.3.5.5 MBS Session Setup or Modification Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session TNL Information NG-RAN	O		9.3.2.17	

9.3.5.6 MBS Session Setup or Modification Failure Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.5.7 MBS Distribution Setup Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MBS Session ID	M		9.3.1.206		-	
MBS Area Session ID	O		9.3.1.207		-	
Shared NG-U Unicast TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.	-	
TAI MBS Support List		0..1		Supported TAs in the NG-RAN node.	YES	ignore
>TAI MBS Support Item		1..<maxnoofSupportedTAIforMBS>			-	
>>TAI	M		9.3.3.11		-	

Range bound	Explanation
maxnoofSupportedTAIforMBS	Maximum no. of TAs supported in the NG-RAN node, indicated for MBS purposes. Value is 256.

9.3.5.8 MBS Distribution Setup Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
MBS Area Session ID	O		9.3.1.207	
Shared NG-U Multicast TNL Information	O		9.3.2.16	
MBS QoS Flows To Be Setup List	M		9.3.1.236	
MBS Session Status	M		9.3.1.217	
MBS Service Area	O		9.3.1.208	

Range bound	Explanation
maxnoofMBSQoSFlows	Maximum no. of QoS Flows allowed within one MBS session. Value is 64.

9.3.5.9 MBS Distribution Setup Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
MBS Area Session ID	O		9.3.1.207	
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.5.10 MBS Distribution Release Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
MBS Area Session ID	O		9.3.1.207	
Shared NG-U Unicast TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.
Cause	M		9.3.1.2	

9.3.5.11 Multicast Session Activation Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	

9.3.5.12 Multicast Session Deactivation Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	

9.3.5.13 Multicast Session Update Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MBS Session ID	M		9.3.1.206		YES	reject
MBS Service Area	O		9.3.1.208		YES	reject
MBS QoS Flows To Be Setup or Modified List	O		MBS QoS Flows To Be Setup List 9.3.1.236		YES	reject
MBS QoS Flow To Release List	O		QoS Flow List with Cause 9.3.1.13	This IE indicates the MBS QoS flow Identifiers of the MBS QoS flows to be released.	YES	reject
MBS Session TNL Information 5GC	O		9.3.2.15		YES	reject
MBS User Plane Failure Indication	O		9.3.2.19		YES	reject

Range bound	Explanation
maxnoofMBSQoSFlows	Maximum no. of QoS Flows allowed within one MBS session. Value is 64.

9.3.5.14 MBS Session Release Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session TNL Information NG-RAN	O		9.3.2.17	

9.3.5.15 Broadcast Transport Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
MBS Session TNL Information NG-RAN	M		9.3.2.17	

9.3.5.16 Broadcast Transport Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	

9.3.5.17 Broadcast Transport Failure Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.6 AIOTF Related IEs

9.3.6.1 Inventory Request Transfer

This IE provides the inventory request related information from the AIOTF to the NG-RAN node.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
A-IoT Device Identification Requested	M		9.3.3.73		YES	reject
Requested Service Area Information	M		9.3.3.71		YES	reject
Inventory Assistance Information	M		9.3.3.72		YES	reject
Follow on Command Indication	O		ENUMERATED (true, ...)	This IE indicates that there is command(s) to be transmitted.	YES	reject
A-IoT Security Information	M		9.3.3.78		YES	reject

9.3.6.2 Inventory Response Transfer

This IE provides the inventory response related information from the NG-RAN node to the AIOTF.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Correlation Identifier	M		9.3.3.69	
Criticality Diagnostics	O		9.3.1.3	

9.3.6.3 Inventory Failure Transfer

This IE provides the inventory failure related information from the NG-RAN node to the AIOTF.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Correlation Identifier	M		9.3.3.69	
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.6.4 Inventory Report Transfer

This IE provides the inventory report related information from the NG-RAN node to the AIOTF.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Correlation Identifier	M		9.3.3.69	
Global gNB ID	M		9.3.1.6	
Reader Report List		<i>0..1</i>		
>Reader Report Item		<i>1..<maxnoofReaders></i>		
>>Reader Index	M		9.3.1.276	
>>Device Report List		<i>1</i>		
>>>Device Report Item		<i>1..<maxnoofDevices></i>		
>>>>A-IoT NAS PDU	M		OCTET STRING	
>>>>RAN A-IoT Device NGAP ID	O		9.3.3.74	
Inventory Complete Indication	O		ENUMERATED (true, ...)	

Range bound	Explanation
maxnoofReaders	Maximum no. of Readers served by a gNB. Value is 65535.
maxnoofDevices	Maximum no. of Devices within a reader. Value is 65535.

9.3.6.5 Command Request Transfer

This IE provides the command request related information from the AIOTF to the NG-RAN node.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
RAN A-IoT Device NGAP ID	M		9.3.3.74		YES	reject
A-IoT NAS PDU	M		OCTET STRING		YES	reject
Command Assistance Information	M		9.3.3.75		YES	reject

9.3.6.6 Command Response Transfer

This IE provides the command response related information from the NG-RAN node to the AIOTF.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Correlation Identifier	M		9.3.3.69	
RAN A-IoT Device NGAP ID	M		9.3.3.74	
A-IoT NAS PDU	M		OCTET STRING	
Criticality Diagnostics	O		9.3.1.3	

9.3.6.7 Command Failure Transfer

This IE provides the command failure related information from the NG-RAN node to the AIOTF.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Correlation Identifier	M		9.3.3.69	
RAN A-IoT Device NGAP ID	M		9.3.3.74	
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.6.8 AIOT Session Release Command Transfer

This IE provides the A-IoT session release command related information from the AIOTF to the NG-RAN node.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
A-IoT Correlation Identifier	M		9.3.3.69		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.3.6.9 AIOT Session Release Complete Transfer

This IE provides the A-IoT session release complete related information from the NG-RAN node to the AIOTF.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Correlation Identifier	M		9.3.3.69	
Criticality Diagnostics	O		9.3.1.3	

9.3.6.10 AIOT Session Release Request Transfer

This IE provides the A-IoT session release request related information from the NG-RAN node to the AIOTF.

In indirect communication, this IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
A-IoT Correlation Identifier	M		9.3.3.69	
Cause	M		9.3.1.2	

9.4 Message and Information Element Abstract Syntax (with ASN.1)

9.4.1 General

NGAP ASN.1 definition conforms to ITU-T Rec. X.691 [4], ITU-T Rec. X.680 [5] and ITU-T Rec. X.681 [6].

The ASN.1 definition specifies the structure and content of NGAP messages. NGAP messages can contain any IEs specified in the object set definitions for that message without the order or number of occurrence being restricted by ASN.1. However, for this version of the standard, a sending entity shall construct an NGAP message according to the PDU definitions module and with the following additional rules:

- IEs shall be ordered (in an IE container) in the order they appear in object set definitions.
- Object set definitions specify how many times IEs may appear. An IE shall appear exactly once if the presence field in an object has value "mandatory". An IE may appear at most once if the presence field in an object has value "optional" or "conditional". If in a tabular format there is multiplicity specified for an IE (i.e., an IE list) then in the corresponding ASN.1 definition the list definition is separated into two parts. The first part defines an IE container list where the list elements reside. The second part defines list elements. The IE container list appears as an IE of its own. For this version of the standard an IE container list may contain only one kind of list elements.

NOTE: In the above "IE" means an IE in the object set with an explicit ID. If one IE needs to appear more than once in one object set, then the different occurrences will have different IE IDs.

If an NGAP message that is not constructed as defined above is received, this shall be considered as Abstract Syntax Error, and the message shall be handled as defined for Abstract Syntax Error in subclause 10.3.6.

9.4.2 Usage of private message mechanism for non-standard use

The private message mechanism for non-standard use may be used:

- for special operator- (and/or vendor) specific features considered not to be part of the basic functionality, i.e., the functionality required for a complete and high-quality specification in order to guarantee multivendor interoperability;
- by vendors for research purposes, e.g., to implement and evaluate new algorithms/features before such features are proposed for standardisation.

The private message mechanism shall not be used for basic functionality. Such functionality shall be standardised.

9.4.3 Elementary Procedure Definitions

```
-- ASN1START
-- *****
--
-- Elementary Procedure definitions
--
-- *****

NGAP-PDU-Descriptions {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-Access (22) modules (3) ngap (1) version1 (1) ngap-PDU-Descriptions (0)}

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- *****
--
-- IE parameter types from other modules.
--
-- *****

IMPORTS

    Criticality,
    ProcedureCode
FROM NGAP-CommonDataTypes

    AMFConfigurationUpdate,
    AMFConfigurationUpdateAcknowledge,
    AMFConfigurationUpdateFailure,
    AMFCPRelocationIndication,
    AMFStatusIndication,
    BroadcastSessionModificationFailure,
    BroadcastSessionModificationRequest,
    BroadcastSessionModificationResponse,
    BroadcastSessionReleaseRequest,
    BroadcastSessionReleaseRequired,
    BroadcastSessionReleaseResponse,
    BroadcastSessionSetupFailure,
    BroadcastSessionSetupRequest,
    BroadcastSessionSetupResponse,
    BroadcastSessionTransportFailure,
    BroadcastSessionTransportRequest,
    BroadcastSessionTransportResponse,
    CellTrafficTrace,
    ConnectionEstablishmentIndication,
    DeactivateTrace,
    DistributionReleaseRequest,
    DistributionReleaseResponse,
    DistributionSetupFailure,
    DistributionSetupRequest,
```

DistributionSetupResponse,
DownlinkNASTransport,
DownlinkNonUEAssociatedNRPPaTransport,
DownlinkRANConfigurationTransfer,
DownlinkRANEarlyStatusTransfer,
DownlinkRANStatusTransfer,
DownlinkUEAssociatedNRPPaTransport,
ErrorIndication,
HandoverCancel,
HandoverCancelAcknowledge,
HandoverCommand,
HandoverFailure,
HandoverNotify,
HandoverPreparationFailure,
HandoverRequest,
HandoverRequestAcknowledge,
HandoverRequired,
HandoverSuccess,
InitialContextSetupFailure,
InitialContextSetupRequest,
InitialContextSetupResponse,
InitialUEMessage,
LocationReport,
LocationReportingControl,
LocationReportingFailureIndication,
MTCommunicationHandlingRequest,
MTCommunicationHandlingResponse,
MTCommunicationHandlingFailure,
MulticastSessionActivationFailure,
MulticastSessionActivationRequest,
MulticastSessionActivationResponse,
MulticastSessionDeactivationRequest,
MulticastSessionDeactivationResponse,
MulticastSessionUpdateFailure,
MulticastSessionUpdateRequest,
MulticastSessionUpdateResponse,
MulticastGroupPaging,
NASNonDeliveryIndication,
NGReset,
NGResetAcknowledge,
NGRemovalFailure,
NGRemovalRequest,
NGRemovalResponse,
NGSetupFailure,
NGSetupRequest,
NGSetupResponse,
OverloadStart,
OverloadStop,
Paging,
PathSwitchRequest,
PathSwitchRequestAcknowledge,
PathSwitchRequestFailure,
PDUSessionResourceModifyConfirm,
PDUSessionResourceModifyIndication,

PDUSessionResourceModifyRequest,
PDUSessionResourceModifyResponse,
PDUSessionResourceNotify,
PDUSessionResourceReleaseCommand,
PDUSessionResourceReleaseResponse,
PDUSessionResourceSetupRequest,
PDUSessionResourceSetupResponse,
PrivateMessage,
PWSCancelRequest,
PWSCancelResponse,
PWSFailureIndication,
PWSRestartIndication,
RANConfigurationUpdate,
RANConfigurationUpdateAcknowledge,
RANConfigurationUpdateFailure,
RANCPRelocationIndication,
RANPagingRequest,
RerouteNASRequest,
RetrieveUEInformation,
RRCInactiveTransitionReport,
SecondaryRATDataUsageReport,
TimingSynchronisationStatusRequest,
TimingSynchronisationStatusResponse,
TimingSynchronisationStatusFailure,
TimingSynchronisationStatusReport,
TraceFailureIndication,
TraceStart,
UEContextModificationFailure,
UEContextModificationRequest,
UEContextModificationResponse,
UEContextReleaseCommand,
UEContextReleaseComplete,
UEContextReleaseRequest,
UEContextResumeRequest,
UEContextResumeResponse,
UEContextResumeFailure,
UEContextSuspendRequest,
UEContextSuspendResponse,
UEContextSuspendFailure,
UEInformationTransfer,
UERadioCapabilityCheckRequest,
UERadioCapabilityCheckResponse,
UERadioCapabilityIDMappingRequest,
UERadioCapabilityIDMappingResponse,
UERadioCapabilityInfoIndication,
UETNLABindingReleaseRequest,
UplinkNASTransport,
UplinkNonUEAssociatedNRPPaTransport,
UplinkRANConfigurationTransfer,
UplinkRANEarlyStatusTransfer,
UplinkRANStatusTransfer,
UplinkUEAssociatedNRPPaTransport,
WriteReplaceWarningRequest,
WriteReplaceWarningResponse,

UplinkRIMInformationTransfer,
DownlinkRIMInformationTransfer,
InventoryRequest,
InventoryResponse,
InventoryFailure,
InventoryReport,
CommandRequest,
CommandResponse,
CommandFailure,
AIOTSessionReleaseCommand,
AIOTSessionReleaseComplete,
AIOTSessionReleaseRequest

FROM NGAP-PDU-Contents

id-AMFConfigurationUpdate,
id-AMFCPrelocationIndication,
id-AMFStatusIndication,
id-BroadcastSessionModification,
id-BroadcastSessionRelease,
id-BroadcastSessionReleaseRequired,
id-BroadcastSessionSetup,
id-BroadcastSessionTransport,
id-CellTrafficTrace,
id-ConnectionEstablishmentIndication,
id-DeactivateTrace,
id-DistributionRelease,
id-DistributionSetup,
id-DownlinkNASTransport,
id-DownlinkNonUEAssociatedNRPPaTransport,
id-DownlinkRANConfigurationTransfer,
id-DownlinkRANEarlyStatusTransfer,
id-DownlinkRANStatusTransfer,
id-DownlinkRIMInformationTransfer,
id-DownlinkUEAssociatedNRPPaTransport,
id-ErrorIndication,
id-HandoverCancel,
id-HandoverNotification,
id-HandoverPreparation,
id-HandoverResourceAllocation,
id-HandoverSuccess,
id-InitialContextSetup,
id-InitialUEMessage,
id-LocationReport,
id-LocationReportingControl,
id-LocationReportingFailureIndication,
id-MTCommunicationHandling,
id-MulticastGroupPaging,
id-MulticastSessionActivation,
id-MulticastSessionDeactivation,
id-MulticastSessionUpdate,
id-NASNonDeliveryIndication,
id-NGReset,

id-NGRemoval,
id-NGSetup,
id-OverloadStart,
id-OverloadStop,
id-Paging,
id-PathSwitchRequest,
id-PDUSessionResourceModify,
id-PDUSessionResourceModifyIndication,
id-PDUSessionResourceNotify,
id-PDUSessionResourceRelease,
id-PDUSessionResourceSetup,
id-PrivateMessage,
id-PWSCancel,
id-PWSFailureIndication,
id-PWSRestartIndication,
id-RANConfigurationUpdate,
id-RANCPRelocationIndication,
id-RANPagingRequest,
id-RerouteNASRequest,
id-RetrieveUEInformation,
id-RRCInactiveTransitionReport,
id-SecondaryRATDataUsageReport,
id-TimingSynchronisationStatus,
id-TimingSynchronisationStatusReport,
id-TraceFailureIndication,
id-TraceStart,
id-UEContextModification,
id-UEContextRelease,
id-UEContextReleaseRequest,
id-UEContextResume,
id-UEContextSuspend,
id-UEInformationTransfer,
id-UERadioCapabilityCheck,
id-UERadioCapabilityIDMapping,
id-UERadioCapabilityInfoIndication,
id-UETNLABindingRelease,
id-UplinkNASTransport,
id-UplinkNonUEAssociatedNRPPaTransport,
id-UplinkRANConfigurationTransfer,
id-UplinkRANEarlyStatusTransfer,
id-UplinkRANStatusTransfer,
id-UplinkRIMInformationTransfer,
id-UplinkUEAssociatedNRPPaTransport,
id-WriteReplaceWarning,
id-InventoryRequest,
id-InventoryReport,
id-CommandRequest,
id-AIOTSessionRelease,
id-AIOTSessionReleaseRequest

FROM NGAP-Constants;

-- *****


```

--
-- Interface Elementary Procedure Class
--
-- *****

NGAP-ELEMENTARY-PROCEDURE ::= CLASS {
    &InitiatingMessage
    &SuccessfulOutcome                OPTIONAL,
    &UnsuccessfulOutcome              OPTIONAL,
    &procedureCode                    ProcedureCode UNIQUE,
    &criticality                      Criticality DEFAULT ignore
}

WITH SYNTAX {
    INITIATING MESSAGE                &InitiatingMessage
    [SUCCESSFUL OUTCOME                &SuccessfulOutcome]
    [UNSUCCESSFUL OUTCOME              &UnsuccessfulOutcome]
    PROCEDURE CODE                    &procedureCode
    [CRITICALITY                      &criticality]
}

-- *****
--
-- Interface PDU Definition
--
-- *****

NGAP-PDU ::= CHOICE {
    initiatingMessage                InitiatingMessage,
    successfulOutcome                SuccessfulOutcome,
    unsuccessfulOutcome              UnsuccessfulOutcome,
    ...
}

InitiatingMessage ::= SEQUENCE {
    procedureCode                    NGAP-ELEMENTARY-PROCEDURE.&procedureCode    ({NGAP-ELEMENTARY-PROCEDURES}),
    criticality                      NGAP-ELEMENTARY-PROCEDURE.&criticality        ({NGAP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value                            NGAP-ELEMENTARY-PROCEDURE.&InitiatingMessage  ({NGAP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

SuccessfulOutcome ::= SEQUENCE {
    procedureCode                    NGAP-ELEMENTARY-PROCEDURE.&procedureCode    ({NGAP-ELEMENTARY-PROCEDURES}),
    criticality                      NGAP-ELEMENTARY-PROCEDURE.&criticality        ({NGAP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value                            NGAP-ELEMENTARY-PROCEDURE.&SuccessfulOutcome  ({NGAP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

UnsuccessfulOutcome ::= SEQUENCE {
    procedureCode                    NGAP-ELEMENTARY-PROCEDURE.&procedureCode    ({NGAP-ELEMENTARY-PROCEDURES}),
    criticality                      NGAP-ELEMENTARY-PROCEDURE.&criticality        ({NGAP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value                            NGAP-ELEMENTARY-PROCEDURE.&UnsuccessfulOutcome ({NGAP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

-- *****
--

```

```

-- Interface Elementary Procedure List
--
-- *****

NGAP-ELEMENTARY-PROCEDURES NGAP-ELEMENTARY-PROCEDURE ::= {
    NGAP-ELEMENTARY-PROCEDURES-CLASS-1 |
    NGAP-ELEMENTARY-PROCEDURES-CLASS-2,
    ...
}

NGAP-ELEMENTARY-PROCEDURES-CLASS-1 NGAP-ELEMENTARY-PROCEDURE ::= {
    aMFConfigurationUpdate |
    broadcastSessionModification |
    broadcastSessionRelease |
    broadcastSessionSetup |
    broadcastSessionTransport |
    distributionSetup |
    distributionRelease |
    handoverCancel |
    handoverPreparation |
    handoverResourceAllocation |
    initialContextSetup |
    mTCommunicationHandling |
    multicastSessionActivation |
    multicastSessionDeactivation |
    multicastSessionUpdate |
    nGReset |
    nGSetup |
    pathSwitchRequest |
    pDUSessionResourceModify |
    pDUSessionResourceModifyIndication |
    pDUSessionResourceRelease |
    pDUSessionResourceSetup |
    pWSCancel |
    rANConfigurationUpdate |
    timingSynchronisationStatus |
    uEContextModification |
    uEContextRelease |
    uEContextResume |
    uEContextSuspend |
    uERadioCapabilityCheck |
    uERadioCapabilityIDMapping |
    writeReplaceWarning |
    nGRemoval |
    inventoryRequest |
    commandRequest |
    aIOTSessionRelease |
    ...
}

NGAP-ELEMENTARY-PROCEDURES-CLASS-2 NGAP-ELEMENTARY-PROCEDURE ::= {
    aMFPCRelocationIndication |
    aMFStatusIndication |
    broadcastSessionReleaseRequired |

```

```
cellTrafficTrace
connectionEstablishmentIndication
deactivateTrace
downlinkNASTransport
downlinkNonUEAssociatedNRPPaTransport
downlinkRANConfigurationTransfer
downlinkRANEarlyStatusTransfer
downlinkRANStatusTransfer
downlinkRIMInformationTransfer
downlinkUEAssociatedNRPPaTransport
errorIndication
handoverNotification
handoverSuccess
initialUEMessage
locationReport
locationReportingControl
locationReportingFailureIndication
multicastGroupPaging
nASNonDeliveryIndication
overloadStart
overloadStop
paging
pDUSessionResourceNotify
privateMessage
pWSFailureIndication
pWSRestartIndication
rANCPRelocationIndication
rANPagingRequest
rerouteNASRequest
retrieveUEInformation
rRCInactiveTransitionReport
secondaryRATDataUsageReport
timingSynchronisationStatusReport
traceFailureIndication
traceStart
uEContextReleaseRequest
uEInformationTransfer
uERadioCapabilityInfoIndication
uETNLABindingRelease
uplinkNASTransport
uplinkNonUEAssociatedNRPPaTransport
uplinkRANConfigurationTransfer
uplinkRANEarlyStatusTransfer
uplinkRANStatusTransfer
uplinkRIMInformationTransfer
uplinkUEAssociatedNRPPaTransport
inventoryReport
aIOTSessionReleaseRequest
...
}

-- *****
--
```

```
-- Interface Elementary Procedures
--
-- *****

aMFConfigurationUpdate NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      AMFConfigurationUpdate
    SUCCESSFUL OUTCOME      AMFConfigurationUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    AMFConfigurationUpdateFailure
    PROCEDURE CODE          id-AMFConfigurationUpdate
    CRITICALITY              reject
}

aMFPCRelocationIndication NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      aMFPCRelocationIndication
    PROCEDURE CODE          id-aMFPCRelocationIndication
    CRITICALITY              reject
}

aMFStatusIndication NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      aMFStatusIndication
    PROCEDURE CODE          id-aMFStatusIndication
    CRITICALITY              ignore
}

broadcastSessionModification NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastSessionModificationRequest
    SUCCESSFUL OUTCOME      BroadcastSessionModificationResponse
    UNSUCCESSFUL OUTCOME    BroadcastSessionModificationFailure
    PROCEDURE CODE          id-BroadcastSessionModification
    CRITICALITY              reject
}

broadcastSessionRelease NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastSessionReleaseRequest
    SUCCESSFUL OUTCOME      BroadcastSessionReleaseResponse
    PROCEDURE CODE          id-BroadcastSessionRelease
    CRITICALITY              reject
}

broadcastSessionReleaseRequired NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastSessionReleaseRequired
    PROCEDURE CODE          id-BroadcastSessionReleaseRequired
    CRITICALITY              reject
}

broadcastSessionSetup NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      BroadcastSessionSetupRequest
    SUCCESSFUL OUTCOME      BroadcastSessionSetupResponse
    UNSUCCESSFUL OUTCOME    BroadcastSessionSetupFailure
    PROCEDURE CODE          id-BroadcastSessionSetup
    CRITICALITY              reject
}

broadcastSessionTransport NGAP-ELEMENTARY-PROCEDURE ::= {
```

```
INITIATING MESSAGE      BroadcastSessionTransportRequest
SUCCESSFUL OUTCOME      BroadcastSessionTransportResponse
UNSUCCESSFUL OUTCOME    BroadcastSessionTransportFailure
PROCEDURE CODE          id-BroadcastSessionTransport
CRITICALITY             reject
}

cellTrafficTrace NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      CellTrafficTrace
  PROCEDURE CODE          id-CellTrafficTrace
  CRITICALITY             ignore
}

connectionEstablishmentIndication NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      ConnectionEstablishmentIndication
  PROCEDURE CODE          id-ConnectionEstablishmentIndication
  CRITICALITY             reject
}

deactivateTrace NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      DeactivateTrace
  PROCEDURE CODE          id-DeactivateTrace
  CRITICALITY             ignore
}

distributionSetup NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      DistributionSetupRequest
  SUCCESSFUL OUTCOME      DistributionSetupResponse
  UNSUCCESSFUL OUTCOME    DistributionSetupFailure
  PROCEDURE CODE          id-DistributionSetup
  CRITICALITY             reject
}

distributionRelease NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      DistributionReleaseRequest
  SUCCESSFUL OUTCOME      DistributionReleaseResponse
  PROCEDURE CODE          id-DistributionRelease
  CRITICALITY             reject
}

downlinkNASTransport NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      DownlinkNASTransport
  PROCEDURE CODE          id-DownlinkNASTransport
  CRITICALITY             ignore
}

downlinkNonUEAssociatedNRPPaTransport NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      DownlinkNonUEAssociatedNRPPaTransport
  PROCEDURE CODE          id-DownlinkNonUEAssociatedNRPPaTransport
  CRITICALITY             ignore
}

downlinkRANConfigurationTransfer NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      DownlinkRANConfigurationTransfer
```

```
    PROCEDURE CODE      id-DownlinkRANConfigurationTransfer
    CRITICALITY          ignore
}

downlinkRANEarlyStatusTransfer NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE   DownlinkRANEarlyStatusTransfer
    PROCEDURE CODE       id-DownlinkRANEarlyStatusTransfer
    CRITICALITY          ignore
}

downlinkRANStatusTransfer NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE   DownlinkRANStatusTransfer
    PROCEDURE CODE       id-DownlinkRANStatusTransfer
    CRITICALITY          ignore
}

downlinkUEAssociatedNRPPaTransport NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE   DownlinkUEAssociatedNRPPaTransport
    PROCEDURE CODE       id-DownlinkUEAssociatedNRPPaTransport
    CRITICALITY          ignore
}

errorIndication NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE   ErrorIndication
    PROCEDURE CODE       id-ErrorIndication
    CRITICALITY          ignore
}

handoverCancel NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE   HandoverCancel
    SUCCESSFUL OUTCOME   HandoverCancelAcknowledge
    PROCEDURE CODE       id-HandoverCancel
    CRITICALITY          reject
}

handoverNotification NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE   HandoverNotify
    PROCEDURE CODE       id-HandoverNotification
    CRITICALITY          ignore
}

handoverPreparation NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE   HandoverRequired
    SUCCESSFUL OUTCOME   HandoverCommand
    UNSUCCESSFUL OUTCOME HandoverPreparationFailure
    PROCEDURE CODE       id-HandoverPreparation
    CRITICALITY          reject
}

handoverResourceAllocation NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE   HandoverRequest
    SUCCESSFUL OUTCOME   HandoverRequestAcknowledge
    UNSUCCESSFUL OUTCOME HandoverFailure
    PROCEDURE CODE       id-HandoverResourceAllocation
}
```

```
    CRITICALITY          reject
  }

handoverSuccess NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE    HandoverSuccess
  PROCEDURE CODE        id-HandoverSuccess
  CRITICALITY           ignore
}

initialContextSetup NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE    InitialContextSetupRequest
  SUCCESSFUL OUTCOME    InitialContextSetupResponse
  UNSUCCESSFUL OUTCOME InitialContextSetupFailure
  PROCEDURE CODE        id-InitialContextSetup
  CRITICALITY           reject
}

initialUEMessage NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE    InitialUEMessage
  PROCEDURE CODE        id-InitialUEMessage
  CRITICALITY           ignore
}

locationReport NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE    LocationReport
  PROCEDURE CODE        id-LocationReport
  CRITICALITY           ignore
}

locationReportingControl NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE    LocationReportingControl
  PROCEDURE CODE        id-LocationReportingControl
  CRITICALITY           ignore
}

locationReportingFailureIndication NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE    LocationReportingFailureIndication
  PROCEDURE CODE        id-LocationReportingFailureIndication
  CRITICALITY           ignore
}

mTCommunicationHandling NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE    MTCommunicationHandlingRequest
  SUCCESSFUL OUTCOME    MTCommunicationHandlingResponse
  UNSUCCESSFUL OUTCOME MTCommunicationHandlingFailure
  PROCEDURE CODE        id-MTCommunicationHandling
  CRITICALITY           reject
}

multicastSessionActivation NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE    MulticastSessionActivationRequest
  SUCCESSFUL OUTCOME    MulticastSessionActivationResponse
  UNSUCCESSFUL OUTCOME MulticastSessionActivationFailure
  PROCEDURE CODE        id-MulticastSessionActivation
}
```

```
    CRITICALITY          reject
  }

multicastSessionDeactivation NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      MulticastSessionDeactivationRequest
  SUCCESSFUL OUTCOME       MulticastSessionDeactivationResponse
  PROCEDURE CODE          id-MulticastSessionDeactivation
  CRITICALITY             reject
}

multicastSessionUpdate NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      MulticastSessionUpdateRequest
  SUCCESSFUL OUTCOME       MulticastSessionUpdateResponse
  UNSUCCESSFUL OUTCOME    MulticastSessionUpdateFailure
  PROCEDURE CODE          id-MulticastSessionUpdate
  CRITICALITY             reject
}

multicastGroupPaging      NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      MulticastGroupPaging
  PROCEDURE CODE          id-MulticastGroupPaging
  CRITICALITY             ignore
}

NASNonDeliveryIndication NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      NASNonDeliveryIndication
  PROCEDURE CODE          id-NASNonDeliveryIndication
  CRITICALITY             ignore
}

NGReset NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      NGReset
  SUCCESSFUL OUTCOME       NGResetAcknowledge
  PROCEDURE CODE          id-NGReset
  CRITICALITY             reject
}

NGSetup NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      NGSetupRequest
  SUCCESSFUL OUTCOME       NGSetupResponse
  UNSUCCESSFUL OUTCOME    NGSetupFailure
  PROCEDURE CODE          id-NGSetup
  CRITICALITY             reject
}

NGRemoval NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      NGRemovalRequest
  SUCCESSFUL OUTCOME       NGRemovalResponse
  UNSUCCESSFUL OUTCOME    NGRemovalFailure
  PROCEDURE CODE          id-NGRemoval
  CRITICALITY             reject
}

overloadStart NGAP-ELEMENTARY-PROCEDURE ::= {
```



```
INITIATING MESSAGE      OverloadStart
PROCEDURE CODE          id-OverloadStart
CRITICALITY             ignore
}

overloadStop NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      OverloadStop
  PROCEDURE CODE          id-OverloadStop
  CRITICALITY             reject
}

paging NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      Paging
  PROCEDURE CODE          id-Paging
  CRITICALITY             ignore
}

pathSwitchRequest NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      PathSwitchRequest
  SUCCESSFUL OUTCOME      PathSwitchRequestAcknowledge
  UNSUCCESSFUL OUTCOME   PathSwitchRequestFailure
  PROCEDURE CODE          id-PathSwitchRequest
  CRITICALITY             reject
}

pDUSessionResourceModify NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      PDUSessionResourceModifyRequest
  SUCCESSFUL OUTCOME      PDUSessionResourceModifyResponse
  PROCEDURE CODE          id-PDUSessionResourceModify
  CRITICALITY             reject
}

pDUSessionResourceModifyIndication NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      PDUSessionResourceModifyIndication
  SUCCESSFUL OUTCOME      PDUSessionResourceModifyConfirm
  PROCEDURE CODE          id-PDUSessionResourceModifyIndication
  CRITICALITY             reject
}

pDUSessionResourceNotify NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      PDUSessionResourceNotify
  PROCEDURE CODE          id-PDUSessionResourceNotify
  CRITICALITY             ignore
}

pDUSessionResourceRelease NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      PDUSessionResourceReleaseCommand
  SUCCESSFUL OUTCOME      PDUSessionResourceReleaseResponse
  PROCEDURE CODE          id-PDUSessionResourceRelease
  CRITICALITY             reject
}

pDUSessionResourceSetup NGAP-ELEMENTARY-PROCEDURE ::= {
  INITIATING MESSAGE      PDUSessionResourceSetupRequest
```

```
    SUCCESSFUL OUTCOME      PDUSessionResourceSetupResponse
    PROCEDURE CODE          id-PDUSessionResourceSetup
    CRITICALITY              reject
}

privateMessage NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PrivateMessage
    PROCEDURE CODE          id-PrivateMessage
    CRITICALITY              ignore
}

pWSCancel NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PWSCancelRequest
    SUCCESSFUL OUTCOME      PWSCancelResponse
    PROCEDURE CODE          id-PWSCancel
    CRITICALITY              reject
}

pWSFailureIndication NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PWSFailureIndication
    PROCEDURE CODE          id-PWSFailureIndication
    CRITICALITY              ignore
}

pWSRestartIndication NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PWSRestartIndication
    PROCEDURE CODE          id-PWSRestartIndication
    CRITICALITY              ignore
}

rANConfigurationUpdate NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RANConfigurationUpdate
    SUCCESSFUL OUTCOME      RANConfigurationUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    RANConfigurationUpdateFailure
    PROCEDURE CODE          id-RANConfigurationUpdate
    CRITICALITY              reject
}

rANCPRelocationIndication NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RANCPRelocationIndication
    PROCEDURE CODE          id-RANCPRelocationIndication
    CRITICALITY              reject
}

rANPagingRequest NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RANPagingRequest
    PROCEDURE CODE          id-RANPagingRequest
    CRITICALITY              ignore
}

rerouteNASRequest NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RerouteNASRequest
    PROCEDURE CODE          id-RerouteNASRequest
    CRITICALITY              reject
}
```

```
}

retrieveUEInformation NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE    RetrieveUEInformation
    PROCEDURE CODE        id-RetrieveUEInformation
    CRITICALITY            reject
}

RRCInactiveTransitionReport NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE    RRCInactiveTransitionReport
    PROCEDURE CODE        id-RRCInactiveTransitionReport
    CRITICALITY            ignore
}

secondaryRATDataUsageReport NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE    SecondaryRATDataUsageReport
    PROCEDURE CODE        id-SecondaryRATDataUsageReport
    CRITICALITY            ignore
}

timingSynchronisationStatus NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE    TimingSynchronisationStatusRequest
    SUCCESSFUL OUTCOME     TimingSynchronisationStatusResponse
    UNSUCCESSFUL OUTCOME  TimingSynchronisationStatusFailure
    PROCEDURE CODE        id-TimingSynchronisationStatus
    CRITICALITY            reject
}

timingSynchronisationStatusReport NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE    TimingSynchronisationStatusReport
    PROCEDURE CODE        id-TimingSynchronisationStatusReport
    CRITICALITY            ignore
}

traceFailureIndication NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE    TraceFailureIndication
    PROCEDURE CODE        id-TraceFailureIndication
    CRITICALITY            ignore
}

traceStart NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE    TraceStart
    PROCEDURE CODE        id-TraceStart
    CRITICALITY            ignore
}

UEContextModification NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE    UEContextModificationRequest
    SUCCESSFUL OUTCOME     UEContextModificationResponse
    UNSUCCESSFUL OUTCOME  UEContextModificationFailure
    PROCEDURE CODE        id-UEContextModification
    CRITICALITY            reject
}
```

```
UEContextRelease NGAP-ELEMENTARY-PROCEDURE ::= {  
    INITIATING MESSAGE      UEContextReleaseCommand  
    SUCCESSFUL OUTCOME      UEContextReleaseComplete  
    PROCEDURE CODE          id-UEContextRelease  
    CRITICALITY              reject  
}  
  
UEContextReleaseRequest NGAP-ELEMENTARY-PROCEDURE ::= {  
    INITIATING MESSAGE      UEContextReleaseRequest  
    PROCEDURE CODE          id-UEContextReleaseRequest  
    CRITICALITY              ignore  
}  
  
UEContextResume NGAP-ELEMENTARY-PROCEDURE ::= {  
    INITIATING MESSAGE      UEContextResumeRequest  
    SUCCESSFUL OUTCOME      UEContextResumeResponse  
    UNSUCCESSFUL OUTCOME    UEContextResumeFailure  
    PROCEDURE CODE          id-UEContextResume  
    CRITICALITY              reject  
}  
  
UEContextSuspend NGAP-ELEMENTARY-PROCEDURE ::= {  
    INITIATING MESSAGE      UEContextSuspendRequest  
    SUCCESSFUL OUTCOME      UEContextSuspendResponse  
    UNSUCCESSFUL OUTCOME    UEContextSuspendFailure  
    PROCEDURE CODE          id-UEContextSuspend  
    CRITICALITY              reject  
}  
  
UEInformationTransfer NGAP-ELEMENTARY-PROCEDURE ::= {  
    INITIATING MESSAGE      UEInformationTransfer  
    PROCEDURE CODE          id-UEInformationTransfer  
    CRITICALITY              reject  
}  
  
UERadioCapabilityCheck NGAP-ELEMENTARY-PROCEDURE ::= {  
    INITIATING MESSAGE      UERadioCapabilityCheckRequest  
    SUCCESSFUL OUTCOME      UERadioCapabilityCheckResponse  
    PROCEDURE CODE          id-UERadioCapabilityCheck  
    CRITICALITY              reject  
}  
  
UERadioCapabilityIDMapping NGAP-ELEMENTARY-PROCEDURE ::= {  
    INITIATING MESSAGE      UERadioCapabilityIDMappingRequest  
    SUCCESSFUL OUTCOME      UERadioCapabilityIDMappingResponse  
    PROCEDURE CODE          id-UERadioCapabilityIDMapping  
    CRITICALITY              reject  
}  
  
UERadioCapabilityInfoIndication NGAP-ELEMENTARY-PROCEDURE ::= {  
    INITIATING MESSAGE      UERadioCapabilityInfoIndication  
    PROCEDURE CODE          id-UERadioCapabilityInfoIndication  
    CRITICALITY              ignore  
}
```

```
uETNLABindingRelease NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UETNLABindingReleaseRequest
    PROCEDURE CODE          id-UETNLABindingRelease
    CRITICALITY              ignore
}

uplinkNASTransport NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UplinkNASTransport
    PROCEDURE CODE          id-UplinkNASTransport
    CRITICALITY              ignore
}

uplinkNonUEAssociatedNRPPaTransport NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UplinkNonUEAssociatedNRPPaTransport
    PROCEDURE CODE          id-UplinkNonUEAssociatedNRPPaTransport
    CRITICALITY              ignore
}

uplinkRANConfigurationTransfer NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UplinkRANConfigurationTransfer
    PROCEDURE CODE          id-UplinkRANConfigurationTransfer
    CRITICALITY              ignore
}

uplinkRANEarlyStatusTransfer NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UplinkRANEarlyStatusTransfer
    PROCEDURE CODE          id-UplinkRANEarlyStatusTransfer
    CRITICALITY              reject
}

uplinkRANStatusTransfer NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UplinkRANStatusTransfer
    PROCEDURE CODE          id-UplinkRANStatusTransfer
    CRITICALITY              ignore
}

uplinkUEAssociatedNRPPaTransport NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UplinkUEAssociatedNRPPaTransport
    PROCEDURE CODE          id-UplinkUEAssociatedNRPPaTransport
    CRITICALITY              ignore
}

writeReplaceWarning NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      WriteReplaceWarningRequest
    SUCCESSFUL OUTCOME      WriteReplaceWarningResponse
    PROCEDURE CODE          id-WriteReplaceWarning
    CRITICALITY              reject
}

uplinkRIMInformationTransfer NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UplinkRIMInformationTransfer
    PROCEDURE CODE          id-UplinkRIMInformationTransfer
    CRITICALITY              ignore
}
```

```

}

downlinkRIMInformationTransfer NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DownlinkRIMInformationTransfer
    PROCEDURE CODE          id-DownlinkRIMInformationTransfer
    CRITICALITY              ignore
}

inventoryRequest NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      InventoryRequest
    SUCCESSFUL OUTCOME      InventoryResponse
    UNSUCCESSFUL OUTCOME   InventoryFailure
    PROCEDURE CODE          id-InventoryRequest
    CRITICALITY              reject
}

inventoryReport NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      InventoryReport
    PROCEDURE CODE          id-InventoryReport
    CRITICALITY              reject
}

commandRequest NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CommandRequest
    SUCCESSFUL OUTCOME      CommandResponse
    UNSUCCESSFUL OUTCOME   CommandFailure
    PROCEDURE CODE          id-CommandRequest
    CRITICALITY              reject
}

aiOTSessionRelease NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      AIOTSessionReleaseCommand
    SUCCESSFUL OUTCOME      AIOTSessionReleaseComplete
    PROCEDURE CODE          id-AIOTSessionRelease
    CRITICALITY              reject
}

aiOTSessionReleaseRequest NGAP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      AIOTSessionReleaseRequest
    PROCEDURE CODE          id-AIOTSessionReleaseRequest
    CRITICALITY              reject
}

END
-- ASN1STOP

```

9.4.4 PDU Definitions

```

-- ASN1START
-- *****
--
-- PDU definitions for NGAP.
--

```

```
-- *****

NGAP-PDU-Contents {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-Access (22) modules (3) ngap (1) version1 (1) ngap-PDU-Contents (1) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- *****
--
-- IE parameter types from other modules.
--
-- *****

IMPORTS

    A2X-PC5-QoS-Parameters,
    AerialUESubscriptionInformation,
    AdditionalULI,
    AIoT-CorrelationIdentifier,
    AIOTFIdentifier,
    AIOTFName,
    AIOTSessionReleaseCommandTransfer,
    AIOTSessionReleaseCompleteTransfer,
    AIOTSessionReleaseRequestTransfer,
    AIoT-Support,
    AllowedNSSAI,
    AMFName,
    AMFSetID,
    AMF-TNLAssociationSetupList,
    AMF-TNLAssociationToAddList,
    AMF-TNLAssociationToRemoveList,
    AMF-TNLAssociationToUpdateList,
    AMF-UE-NGAP-ID,
    AssistanceDataForPaging,
    AssociatedSessionID,
    AUN3DeviceAccessInfo,
    AuthenticatedIndication,
    BroadcastCancelledAreaList,
    BroadcastCompletedAreaList,
    BroadcastTransportFailureTransfer,
    BroadcastTransportRequestTransfer,
    BroadcastTransportResponseTransfer,
    CancelAllWarningMessages,
    Cause,
    CellIDListForRestart,
    CEmodeBrestricted,
    CEmodeBSupport-Indicator,
    CNAssistedRANTuning,
    CommandRequestTransfer,
    CommandResponseTransfer,
    CommandFailureTransfer, ConcurrentWarningMessageInd,
```

CoreNetworkAssistanceInformationForInactive,
CPTransportLayerInformation,
CriticalityDiagnostics,
DataCodingScheme,
DirectForwardingPathAvailability,
DL-CP-SecurityInformation,
DL-Signalling,
EarlyStatusTransfer-TransparentContainer,
EDT-Session,
EmergencyAreaIDListForRestart,
EmergencyFallbackIndicator,
EN-DCSONConfigurationTransfer,
EndIndication,
Enhanced-CoverageRestriction,
ERedCapIndication,
EUTRA-CGI,
EUTRA-PagingeDRXInformation,
Extended-AMFName,
Extended-ConnectedTime,
Extended-RANNodeName,
FiveGCAction,
FiveG-ProSeAuthorized,
FiveG-ProSePC5QoSParameters,
FiveG-S-TMSI,
GlobalRANNodeID,
GUAMI,
GUAMIList,
GUAMIType,
GW-Context-Release-Indication,
HandoverFlag,
HandoverType,
IAB-Authorized,
IABNodeIndication,
IAB-Supported,
IMSVoiceSupportIndicator,
IndexToRFSP,
InfoOnRecommendedCellsAndRANNodesForPaging,
IntersystemSONConfigurationTransfer,
InventoryRequestTransfer,
InventoryResponseTransfer,
InventoryFailureTransfer,
InventoryReportTransfer,
LAI,
LocationReportingRequestType,
LPWUSPSAssistanceInformation,
LPWUSDisableIndication,
LTE-A2X-ServicesAuthorized,
LTEM-Indication,
LTEUESidelinkAggregateMaximumBitrate,
LTEV2XServicesAuthorized,
MaskedIMEISV,
MBS-AreaSessionID,
MBS-DistributionReleaseRequestTransfer,
MBS-DistributionSetupRequestTransfer,

MBS-DistributionSetupResponseTransfer,
MBS-DistributionSetupUnsuccessfulTransfer,
MBS-ServiceArea,
MBS-SessionID,
MBSSessionReleaseResponseTransfer,
MBSSessionSetupOrModFailureTransfer,
MBSSessionSetupOrModRequestTransfer,
MBSSessionSetupOrModResponseTransfer,
MDTPLMNList,
MDTPLMNModificationList,
MessageIdentifier,
MobileIAB-Authorized,
MobileIABNodeIndication,
MobileIAB-Supported,
MobilityRestrictionList,
MulticastGroupPagingAreaList,
MulticastSessionActivationRequestTransfer,
MulticastSessionDeactivationRequestTransfer,
MulticastSessionUpdateRequestTransfer,
NAS-PDU,
NASSecurityParametersFromNGRAN,
NB-IoT-DefaultPagingDRX,
NB-IoT-PagingDRX,
NB-IoT-Paging-eDRXInfo,
NB-IoT-UEPriority,
NetworkControlledRepeaterAuthorized,
NewSecurityContextInd,
NGRAN-CGI,
NGRAN-TNLAssociationToRemoveList,
NGRANTraceID,
NID,
NoPDUSessionIndication,
NotifySourceNGRANNode,
NPN-AccessInformation,
NR-A2X-ServicesAuthorized,
NR-CGI,
NR-PagingeDRXInformation,
NRPPa-PDU,
NRUESidelinkAggregateMaximumBitrate,
NRV2XServicesAuthorized,
NumberOfBroadcastsRequested,
OverloadResponse,
OverloadStartNSSAIList,
PagingAssisDataforCEcapabUE,
PagingCause,
PagingDRX,
PagingOrigin,
PagingPolicyDifferentiation,
PagingPriority,
Partially-Allowed-NSSAI,
PC5QoSParameters,
PDUSessionAggregateMaximumBitRate,
PDUSessionListMTCommHReq,
PDUSessionResourceAdmittedList,

PDUSessionResourceFailedToModifyListModCfm,
PDUSessionResourceFailedToModifyListModRes,
PDUSessionResourceFailedToResumeListRESReq,
PDUSessionResourceFailedToResumeListRESRes,
PDUSessionResourceFailedToSetupListCxtFail,
PDUSessionResourceFailedToSetupListCxtRes,
PDUSessionResourceFailedToSetupListH0Ack,
PDUSessionResourceFailedToSetupListPSReq,
PDUSessionResourceFailedToSetupListSUSRes,
PDUSessionResourceHandoverList,
PDUSessionResourceListCxtRelCpl,
PDUSessionResourceListCxtRelReq,
PDUSessionResourceListH0Req,
PDUSessionResourceModifyListModCfm,
PDUSessionResourceModifyListModInd,
PDUSessionResourceModifyListModReq,
PDUSessionResourceModifyListModRes,
PDUSessionResourceNotifyList,
PDUSessionResourceReleasedListNot,
PDUSessionResourceReleasedListPSAck,
PDUSessionResourceReleasedListPSFail,
PDUSessionResourceReleasedListRelRes,
PDUSessionResourceResumeListRESReq,
PDUSessionResourceResumeListRESRes,
PDUSessionResourceSecondaryRATUsageList,
PDUSessionResourceSetupListCxtReq,
PDUSessionResourceSetupListCxtRes,
PDUSessionResourceSetupListH0Req,
PDUSessionResourceSetupListSUSReq,
PDUSessionResourceSetupListSUSRes,
PDUSessionResourceSuspendListSUSReq,
PDUSessionResourceSwitchedList,
PDUSessionResourceToBeSwitchedDLLList,
PDUSessionResourceToReleaseListH0Cmd,
PDUSessionResourceToReleaseListRelCmd,
PEIPAssistanceInformation,
PLMNIdentity,
PLMNSupportList,
PrivacyIndicator,
PWSFailedCellIDList,
QMCConfigInfo,
QMCDeactivation,
RAN-AIOT-Device-NGAP-ID,
RANNodeName,
RANPagingPriority,
RANStatusTransfer-TransparentContainer,
RANTimingSynchronisationStatusInfo,
RAN-TSSRequestType,
RAN-TSSScope,
RAN-UE-NGAP-ID,
RedCapIndication,
RedirectionVoiceFallback,
RelativeAMFCapacity,
RepetitionPeriod,

RequestedNSSAI,
ResetType,
RGLLevelWirelineAccessCharacteristics,
RIMInformationTransfer,
RoutingID,
RRCEstablishmentCause,
RRCInactiveTransitionReportRequest,
RRCState,
SecurityContext,
SecurityKey,
SerialNumber,
ServedGUAMICollection,
SliceSupportList,
SLPositioningRangingServiceInfo,
S-NSSAI,
SONConfigurationTransfer,
SourceToTarget-AMFInformationReroute,
SourceToTarget-TransparentContainer,
SRVCCOperationPossible,
SupportedTAList,
Suspend-Request-Indication,
Suspend-Response-Indication,
TAI,
TAICollectionForPaging,
TAICollectionForRestart,
TargetID,
TargetNSSAIInformation,
TargetToSource-Failure-TransparentContainer,
TargetToSource-TransparentContainer,
TimeSyncAssistanceInfo,
TimeToWait,
TNLAssociationList,
TraceActivation,
TrafficLoadReductionIndication,
TransportLayerAddress,
UEAggregateMaximumBitRate,
UE-associatedLogicalNG-connectionList,
UECapabilityInfoRequest,
UEContextRequest,
UE-DifferentiationInfo,
UE-NGAP-IDs,
UEPagingIdentity,
UEPresenceInAreaOfInterestList,
UERadioCapability,
UERadioCapabilityForPaging,
UERadioCapabilityID,
UERetentionInformation,
UESecurityCapabilities,
UESliceMaximumBitRateList,
UE-UP-CIoT-Support,
UL-CP-SecurityInformation,
UnavailableGUAMICollection,
URI-address,
UserLocationInformation,

WarningAreaCoordinates,
WarningAreaList,
WarningMessageContents,
WarningSecurityInfo,
WarningType,
WUS-Assistance-Information,
XrDeviceWith2Rx

FROM NGAP-IEs

PrivateIE-Container{},
ProtocolExtensionContainer{},
ProtocolIE-Container{},
ProtocolIE-ContainerList{},
ProtocolIE-ContainerPair{},
ProtocolIE-SingleContainer{},
NGAP-PRIVATE-IES,
NGAP-PROTOCOL-EXTENSION,
NGAP-PROTOCOL-IES,
NGAP-PROTOCOL-IES-PAIR

FROM NGAP-Containers

id-A2X-PC5-QoS-Parameters,
id-AerialUESubscriptionInformation,
id-AdditionalULI,
id-AIoT-CorrelationIdentifier,
id-AIOTFIdentifier,
id-AIOTFName,
id-AIOTSessionReleaseCommandTransfer,
id-AIOTSessionReleaseCompleteTransfer,
id-AIOTSessionReleaseRequestTransfer,
id-AIoT-Support,
id-AllowedNSSAI,
id-AMFName,
id-AMFOverloadResponse,
id-AMFSetID,
id-AMF-TNLAssociationFailedToSetupList,
id-AMF-TNLAssociationSetupList,
id-AMF-TNLAssociationToAddList,
id-AMF-TNLAssociationToRemoveList,
id-AMF-TNLAssociationToUpdateList,
id-AMFTrafficLoadReductionIndication,
id-AMF-UE-NGAP-ID,
id-AMF-UE-NGAP-ID2,
id-AssistanceDataForPaging,
id-AssociatedSessionID,
id-AUN3DeviceAccessInfo,
id-AuthenticatedIndication,
id-BroadcastCancelledAreaList,
id-BroadcastCompletedAreaList,
id-BroadcastTransportFailureTransfer,
id-BroadcastTransportRequestTransfer,
id-BroadcastTransportResponseTransfer,

id-CancelAllWarningMessages,
id-Cause,
id-CellIDListForRestart,
id-CEmodeBrestricted,
id-CEmodeBSupport-Indicator,
id-CNAssistedRANTuning,
id-CommandRequestTransfer,
id-CommandResponseTransfer,
id-CommandFailureTransfer,
id-ConcurrentWarningMessageInd,
id-CoreNetworkAssistanceInformationForInactive,
id-CriticalityDiagnostics,
id-DataCodingScheme,
id-DefaultPagingDRX,
id-DirectForwardingPathAvailability,
id-DL-CP-SecurityInformation,
id-DL-Signalling,
id-EarlyStatusTransfer-TransparentContainer,
id-EDT-Session,
id-EmergencyAreaIDListForRestart,
id-EmergencyFallbackIndicator,
id-ENDC-SONConfigurationTransferDL,
id-ENDC-SONConfigurationTransferUL,
id-EndIndication,
id-Enhanced-CoverageRestriction,
id-ERedCapIndication,
id-EUTRA-CGI,
id-EUTRA-PagingDRXInformation,
id-Extended-AMFName,
id-Extended-ConnectedTime,
id-Extended-RANNodeName,
id-FiveGCAction,
id-FiveG-ProSeAuthorized,
id-FiveG-ProSePC5QoSParameters,
id-FiveG-ProSeUEPC5AggregateMaximumBitRate,
id-FiveG-S-TMSI,
id-GlobalRANNodeID,
id-GUAMI,
id-GUAMIList,
id-GUAMIType,
id-GW-Context-Release-Indication,
id-HandoverFlag,
id-HandoverType,
id-IAB-Authorized,
id-IABNodeIndication,
id-IAB-Supported,
id-IMSVoiceSupportIndicator,
id-IndexToRFSP,
id-InfoOnRecommendedCellsAndRANNodesForPaging,
id-IntersystemSONConfigurationTransferDL,
id-IntersystemSONConfigurationTransferUL,
id-InventoryRequestTransfer,
id-InventoryResponseTransfer,
id-InventoryFailureTransfer,

id-InventoryReportTransfer,
id-LocationReportingRequestType,
id-LPWUSPSAssistanceInformation,
id-LPWUSDisableIndication,
id-LTE-A2X-ServicesAuthorized,
id-LTE-A2X-UE-PC5-AggregateMaximumBitRate,
id-LTEM-Indication,
id-LTEUESidelinkAggregateMaximumBitrate,
id-LTEV2XServicesAuthorized,
id-ManagementBasedMDTPLMNList,
id-ManagementBasedMDTPLMNModificationList,
id-MaskedIMEISV,
id-MBS-AreaSessionID,
id-MBS-DistributionReleaseRequestTransfer,
id-MBS-DistributionSetupRequestTransfer,
id-MBS-DistributionSetupResponseTransfer,
id-MBS-DistributionSetupUnsuccessfulTransfer,
id-MBS-ServiceArea,
id-MBS-SessionID,
id-MBSSessionModificationFailureTransfer,
id-MBSSessionModificationRequestTransfer,
id-MBSSessionModificationResponseTransfer,
id-MBSSessionReleaseResponseTransfer,
id-MBSSessionSetupFailureTransfer,
id-MBSSessionSetupRequestTransfer,
id-MBSSessionSetupResponseTransfer,
id-MessageIdentifier,
id-MobileIAB-Authorized,
id-MobileIABNodeIndication,
id-MobileIAB-Supported,
id-MobilityRestrictionList,
id-MulticastGroupPagingAreaList,
id-MulticastSessionActivationRequestTransfer,
id-MulticastSessionDeactivationRequestTransfer,
id-MulticastSessionUpdateRequestTransfer,
id-NASC,
id-NAS-PDU,
id-NASSecurityParametersFromNGRAN,
id-NB-IoT-DefaultPagingDRX,
id-NB-IoT-PagingDRX,
id-NB-IoT-Paging-eDRXInfo,
id-NB-IoT-UEPriority,
id-NetworkControlledRepeaterAuthorized,
id-NewAMF-UE-NGAP-ID,
id-NewGUAMI,
id-NewSecurityContextInd,
id-NGAP-Message,
id-NGRAN-CGI,
id-NGRAN-TNLAssociationToRemoveList,
id-NGRANTraceID,
id-NoPDUSessionIndication,
id-NotifySourceNGRANNode,
id-NPN-AccessInformation,
id-NR-A2X-ServicesAuthorized,

id-NR-A2X-UE-PC5-AggregateMaximumBitRate,
id-NR-PagingDRXInformation,
id-NRPPa-PDU,
id-NRUESidelinkAggregateMaximumBitrate,
id-NRV2XServicesAuthorized,
id-NumberOfBroadcastsRequested,
id-OldAMF,
id-OverloadStartNSSAList,
id-PagingAssisDataforCEcapabUE,
id-PagingCause,
id-PagingDRX,
id-PagingOrigin,
id-PagingPolicyDifferentiation,
id-PagingPriority,
id-Partially-Allowed-NSSAI,
id-PC5QoSParameters,
id-PDUSessionListMTCommHReq,
id-PDUSessionResourceAdmittedList,
id-PDUSessionResourceFailedToModifyListModCfm,
id-PDUSessionResourceFailedToModifyListModRes,
id-PDUSessionResourceFailedToResumeListRESReq,
id-PDUSessionResourceFailedToResumeListRESRes,
id-PDUSessionResourceFailedToSetupListCxtFail,
id-PDUSessionResourceFailedToSetupListCxtRes,
id-PDUSessionResourceFailedToSetupListHOfAck,
id-PDUSessionResourceFailedToSetupListPSReq,
id-PDUSessionResourceFailedToSetupListSUSRes,
id-PDUSessionResourceHandoverList,
id-PDUSessionResourceListCxtRelCpl,
id-PDUSessionResourceListCxtRelReq,
id-PDUSessionResourceListHOfRqd,
id-PDUSessionResourceModifyListModCfm,
id-PDUSessionResourceModifyListModInd,
id-PDUSessionResourceModifyListModReq,
id-PDUSessionResourceModifyListModRes,
id-PDUSessionResourceNotifyList,
id-PDUSessionResourceReleasedListNot,
id-PDUSessionResourceReleasedListPSAck,
id-PDUSessionResourceReleasedListPSFail,
id-PDUSessionResourceReleasedListRelRes,
id-PDUSessionResourceResumeListRESReq,
id-PDUSessionResourceResumeListRESRes,
id-PDUSessionResourceSecondaryRATUsageList,
id-PDUSessionResourceSetupListCxtReq,
id-PDUSessionResourceSetupListCxtRes,
id-PDUSessionResourceSetupListHOfReq,
id-PDUSessionResourceSetupListSUSReq,
id-PDUSessionResourceSetupListSUSRes,
id-PDUSessionResourceSuspendListSUSReq,
id-PDUSessionResourceSwitchedList,
id-PDUSessionResourceToBeSwitchedDLList,
id-PDUSessionResourceToReleaseListHOfCmd,
id-PDUSessionResourceToReleaseListRelCmd,
id-PEIPAssistanceInformation,

id-PLMNSupportList,
id-PrivacyIndicator,
id-PWSFailedCellIDList,
id-QMCConfigInfo,
id-QMCDeactivation,
id-RAN-AIOT-Device-NGAP-ID,
id-RANNodeName,
id-RANPagingPriority,
id-RANStatusTransfer-TransparentContainer,
id-RANTimingSynchronisationStatusInfo,
id-RAN-TSSRequestType,
id-RAN-TSSScope,
id-RAN-UE-NGAP-ID,
id-RedCapIndication,
id-RedirectionVoiceFallback,
id-RelativeAMFCapacity,
id-RepetitionPeriod,
id-RequestedNSSAI,
id-ResetType,
id-RGLevelWirelineAccessCharacteristics,
id-RIMInformationTransfer,
id-RoutingID,
id-RRCEstablishmentCause,
id-RRCInactiveTransitionReportRequest,
id-RRC-Resume-Cause,
id-RRCState,
id-SecurityContext,
id-SecurityKey,
id-SelectedNID,
id-SelectedPLMNIdentity,
id-SerialNumber,
id-ServedGUAMIList,
id-SliceSupportList,
id-S-NSSAI,
id-SONConfigurationTransferDL,
id-SONConfigurationTransferUL,
id-SourceAMF-GUAMI,
id-SourceAMF-UE-NGAP-ID,
id-SourceToTarget-AMFInformationReroute,
id-SourceToTarget-TransparentContainer,
id-SRVCCOperationPossible,
id-SupportedTAList,
id-Suspend-Request-Indication,
id-Suspend-Response-Indication,
id-Target-RAN-Node-ID,
id-TAI,
id-TAIListForPaging,
id-TAIListForRestart,
id-TargetID,
id-TargetNSSAIInformation,
id-TargettoSource-Failure-TransparentContainer,
id-TargetToSource-TransparentContainer,
id-TimeSyncAssistanceInfo,
id-TimeToWait,


```
id-TNGFIdentityInformation,
id-TraceActivation,
id-TraceCollectionEntityIPAddress,
id-TraceCollectionEntityURI,
id-TWIFIdentityInformation,
id-UEAggregateMaximumBitRate,
id-UE-associatedLogicalNG-connectionList,
id-UECapabilityInfoRequest,
id-UEContextRequest,
id-UE-DifferentiationInfo,
id-UE-NGAP-IDs,
id-UEPagingIdentity,
id-UEPresenceInAreaOfInterestList,
id-UERadioCapability,
id-UERadioCapability-EUTRA-Format,
id-UERadioCapabilityForPaging,
id-UERadioCapabilityID,
id-UERetentionInformation,
id-UESecurityCapabilities,
id-UESliceMaximumBitRateList,
id-UE-UP-CIoT-Support,
id-UL-CP-SecurityInformation,
id-UnavailableGUAMList,
id-UserLocationInformation,
id-W-AGFIdentityInformation,
id-WarningAreaCoordinates,
id-WarningAreaList,
id-WarningMessageContents,
id-WarningSecurityInfo,
id-WarningType,
id-WUS-Assistance-Information,
id-XrDeviceWith2Rx,
id-SLPositioningRangingServiceInfo,
id-ExtendedOldAMF,
id-Aerial-UE-FlightInformationOngoingReporting-failed
```

FROM NGAP-Constants;

```
-- *****
--
-- PDU SESSION MANAGEMENT ELEMENTARY PROCEDURES
--
-- *****
--
-- *****
--
-- PDU Session Resource Setup Elementary Procedure
--
-- *****
--
-- *****
--
-- PDU SESSION RESOURCE SETUP REQUEST
```

```

--
-- *****
PDU Session Resource Setup Request ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {PDU Session Resource Setup Request IEs} },
    ...
}

PDU Session Resource Setup Request IEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-PagingPriority       CRITICALITY ignore TYPE RAN-PagingPriority       PRESENCE optional  }|
    { ID id-NAS-PDU                 CRITICALITY reject TYPE NAS-PDU                 PRESENCE optional  }|
    { ID id-PDU Session Resource Setup List SReq CRITICALITY reject TYPE PDU Session Resource Setup List SReq PRESENCE mandatory }|
    { ID id-UE Aggregate Maximum Bit Rate CRITICALITY ignore TYPE UE Aggregate Maximum Bit Rate PRESENCE optional  }|
    { ID id-UE Slice Maximum Bit Rate List CRITICALITY ignore TYPE UE Slice Maximum Bit Rate List PRESENCE optional  },
    ...
}

-- *****
--
-- PDU SESSION RESOURCE SETUP RESPONSE
--
-- *****

PDU Session Resource Setup Response ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {PDU Session Resource Setup Response IEs} },
    ...
}

PDU Session Resource Setup Response IEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-PDU Session Resource Setup List SRes CRITICALITY ignore TYPE PDU Session Resource Setup List SRes PRESENCE optional  }|
    { ID id-PDU Session Resource Failed To Setup List SRes CRITICALITY ignore TYPE PDU Session Resource Failed To Setup List SRes PRESENCE optional  }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional  }|
    { ID id-User Location Information CRITICALITY ignore TYPE User Location Information PRESENCE optional  },
    ...
}

-- *****
--
-- PDU Session Resource Release Elementary Procedure
--
-- *****
-- *****
--
-- PDU SESSION RESOURCE RELEASE COMMAND

```

```

--
-- *****
PDU SessionResourceReleaseCommand ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {PDU SessionResourceReleaseCommandIEs} },
    ...
}

PDU SessionResourceReleaseCommandIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RANPagingPriority        CRITICALITY ignore TYPE RANPagingPriority        PRESENCE optional   }|
    { ID id-NAS-PDU                  CRITICALITY ignore TYPE NAS-PDU                  PRESENCE optional   }|
    { ID id-PDU SessionResourceToReleaseListRelCmd CRITICALITY reject TYPE PDU SessionResourceToReleaseListRelCmd PRESENCE mandatory },
    ...
}

-- *****
--
-- PDU SESSION RESOURCE RELEASE RESPONSE
--
-- *****

PDU SessionResourceReleaseResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {PDU SessionResourceReleaseResponseIEs} },
    ...
}

PDU SessionResourceReleaseResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-PDU SessionResourceReleasedListRelRes CRITICALITY ignore TYPE PDU SessionResourceReleasedListRelRes PRESENCE mandatory }|
    { ID id-UserLocationInformation    CRITICALITY ignore TYPE UserLocationInformation PRESENCE optional   }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional   },
    ...
}

-- *****
--
-- PDU Session Resource Modify Elementary Procedure
--
-- *****
-- *****
--
-- PDU SESSION RESOURCE MODIFY REQUEST
--
-- *****

PDU SessionResourceModifyRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {PDU SessionResourceModifyRequestIEs} },
    ...
}

```

```

PDUSessionResourceModifyRequestIES NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RANPagingPriority        CRITICALITY ignore TYPE RANPagingPriority        PRESENCE optional   }|
  { ID id-PDUSessionResourceModifyListModReq CRITICALITY reject TYPE PDUSessionResourceModifyListModReq PRESENCE mandatory },
  ...
}

-- *****
--
-- PDU SESSION RESOURCE MODIFY RESPONSE
--
-- *****

PDUSessionResourceModifyResponse ::= SEQUENCE {
  protocolIES      ProtocolIE-Container      { {PDUSessionResourceModifyResponseIES} },
  ...
}

PDUSessionResourceModifyResponseIES NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory
  }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory
  }|
  { ID id-PDUSessionResourceModifyListModRes CRITICALITY ignore TYPE PDUSessionResourceModifyListModRes PRESENCE optional
  }|
  { ID id-PDUSessionResourceFailedToModifyListModRes CRITICALITY ignore TYPE PDUSessionResourceFailedToModifyListModRes PRESENCE optional
  }|
  { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE optional
  }|
  { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional
  },
  ...
}

-- *****
--
-- PDU Session Resource Notify Elementary Procedure
--
-- *****

-- *****
--
-- PDU SESSION RESOURCE NOTIFY
--
-- *****

PDUSessionResourceNotify ::= SEQUENCE {
  protocolIES      ProtocolIE-Container      { {PDUSessionResourceNotifyIES} },
  ...
}

PDUSessionResourceNotifyIES NGAP-PROTOCOL-IES ::= {

```

```

    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-PDUSessionResourceNotifyList CRITICALITY reject TYPE PDUSessionResourceNotifyList PRESENCE optional }|
    { ID id-PDUSessionResourceReleasedListNot CRITICALITY ignore TYPE PDUSessionResourceReleasedListNot PRESENCE optional }|
    { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE optional },
    ...
}

-- *****
--
-- PDU Session Resource Modify Indication Elementary Procedure
--
-- *****
--
-- *****
--
-- PDU SESSION RESOURCE MODIFY INDICATION
--
-- *****

PDUSessionResourceModifyIndication ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {PDUSessionResourceModifyIndicationIES} },
    ...
}

PDUSessionResourceModifyIndicationIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-PDUSessionResourceModifyListModInd CRITICALITY reject TYPE PDUSessionResourceModifyListModInd PRESENCE mandatory }|
    { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE optional },
    ...
}

-- *****
--
-- PDU SESSION RESOURCE MODIFY CONFIRM
--
-- *****

PDUSessionResourceModifyConfirm ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {PDUSessionResourceModifyConfirmIES} },
    ...
}

PDUSessionResourceModifyConfirmIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-PDUSessionResourceModifyListModCfm CRITICALITY ignore TYPE PDUSessionResourceModifyListModCfm PRESENCE optional }|
    { ID id-PDUSessionResourceFailedToModifyListModCfm CRITICALITY ignore TYPE PDUSessionResourceFailedToModifyListModCfm PRESENCE optional }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

```

```
-- *****
--
-- UE CONTEXT MANAGEMENT ELEMENTARY PROCEDURES
--
-- *****
--
-- *****
--
-- Initial Context Setup Elementary Procedure
--
-- *****
--
-- *****
--
-- INITIAL CONTEXT SETUP REQUEST
--
-- *****
```

```
InitialContextSetupRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {InitialContextSetupRequestIEs} },
    ...
}
```

```
InitialContextSetupRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-OldAMF                  CRITICALITY reject TYPE AMFName                  PRESENCE optional   }|
    { ID id-UEAggregateMaximumBitRate CRITICALITY reject TYPE UEAggregateMaximumBitRate PRESENCE conditional }|
-- The above IE shall be present if the PDU Session Resource Setup List IE is present
    { ID id-CoreNetworkAssistanceInformationForInactive CRITICALITY ignore TYPE CoreNetworkAssistanceInformationForInactive PRESENCE optional }|
    { ID id-GUAMI                  CRITICALITY reject TYPE GUAMI                  PRESENCE mandatory }|
    { ID id-PDUSessionResourceSetupListCxtReq CRITICALITY reject TYPE PDUSessionResourceSetupListCxtReq PRESENCE optional }|
    { ID id-AllowedNSSAI            CRITICALITY reject TYPE AllowedNSSAI            PRESENCE mandatory }|
    { ID id-UESecurityCapabilities   CRITICALITY reject TYPE UESecurityCapabilities   PRESENCE mandatory }|
    { ID id-SecurityKey              CRITICALITY reject TYPE SecurityKey              PRESENCE mandatory }|
    { ID id-TraceActivation          CRITICALITY ignore TYPE TraceActivation          PRESENCE optional   }|
    { ID id-MobilityRestrictionList  CRITICALITY ignore TYPE MobilityRestrictionList  PRESENCE optional   }|
    { ID id-UERadioCapability        CRITICALITY ignore TYPE UERadioCapability        PRESENCE optional   }|
    { ID id-IndexToRFSP              CRITICALITY ignore TYPE IndexToRFSP              PRESENCE optional   }|
    { ID id-MaskedIMEISV             CRITICALITY ignore TYPE MaskedIMEISV             PRESENCE optional   }|
    { ID id-NAS-PDU                  CRITICALITY ignore TYPE NAS-PDU                  PRESENCE optional   }|
    { ID id-EmergencyFallbackIndicator CRITICALITY reject TYPE EmergencyFallbackIndicator PRESENCE optional   }|
    { ID id-RRCInactiveTransitionReportRequest CRITICALITY ignore TYPE RRCInactiveTransitionReportRequest PRESENCE optional   }|
    { ID id-UERadioCapabilityForPaging CRITICALITY ignore TYPE UERadioCapabilityForPaging PRESENCE optional   }|
    { ID id-RedirectionVoiceFallback CRITICALITY ignore TYPE RedirectionVoiceFallback PRESENCE optional   }|
    { ID id-LocationReportingRequestType CRITICALITY ignore TYPE LocationReportingRequestType PRESENCE optional   }|
    { ID id-CNAssistedRANTuning       CRITICALITY ignore TYPE CNAssistedRANTuning       PRESENCE optional   }|
    { ID id-SRVCCOperationPossible    CRITICALITY ignore TYPE SRVCCOperationPossible    PRESENCE optional   }|
    { ID id-IAB-Authorized            CRITICALITY ignore TYPE IAB-Authorized            PRESENCE optional   }|
    { ID id-Enhanced-CoverageRestriction CRITICALITY ignore TYPE Enhanced-CoverageRestriction PRESENCE optional   }|
    { ID id-Extended-ConnectedTime    CRITICALITY ignore TYPE Extended-ConnectedTime    PRESENCE optional   }|
    { ID id-UE-DifferentiationInfo    CRITICALITY ignore TYPE UE-DifferentiationInfo    PRESENCE optional   }|
    { ID id-NRV2XServicesAuthorized   CRITICALITY ignore TYPE NRV2XServicesAuthorized   PRESENCE optional   }|
}
```

```

{ ID id-LTEV2XServicesAuthorized          CRITICALITY ignore TYPE LTEV2XServicesAuthorized          PRESENCE optional }|
{ ID id-NRUESidelinkAggregateMaximumBitrate CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate PRESENCE optional }|
{ ID id-LTEUESidelinkAggregateMaximumBitrate CRITICALITY ignore TYPE LTEUESidelinkAggregateMaximumBitrate PRESENCE optional }|
{ ID id-PC5QoSParameters                   CRITICALITY ignore TYPE PC5QoSParameters                   PRESENCE optional }|
{ ID id-CEmodeBrestricted                   CRITICALITY ignore TYPE CEmodeBrestricted                   PRESENCE optional }|
{ ID id-UE-UP-CIoT-Support                  CRITICALITY ignore TYPE UE-UP-CIoT-Support                  PRESENCE optional }|
{ ID id-RGLeveWirelineAccessCharacteristics CRITICALITY ignore TYPE RGLeveWirelineAccessCharacteristics PRESENCE optional }|
{ ID id-ManagementBasedMDTPLMNList         CRITICALITY ignore TYPE MDTPLMNList                       PRESENCE optional }|
{ ID id-UERadioCapabilityID                 CRITICALITY reject  TYPE UERadioCapabilityID                 PRESENCE optional }|
{ ID id-TimeSyncAssistanceInfo              CRITICALITY ignore TYPE TimeSyncAssistanceInfo              PRESENCE optional }|
{ ID id-QMCConfigInfo                      CRITICALITY ignore TYPE QMCConfigInfo                      PRESENCE optional }|
{ ID id-TargetNSSAIInformation              CRITICALITY ignore TYPE TargetNSSAIInformation              PRESENCE optional }|
{ ID id-UESliceMaximumBitRateList          CRITICALITY ignore TYPE UESliceMaximumBitRateList          PRESENCE optional }|
{ ID id-FiveG-ProSeAuthorized              CRITICALITY ignore TYPE FiveG-ProSeAuthorized              PRESENCE optional }|
{ ID id-FiveG-ProSeUEPC5AggregateMaximumBitRate CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate PRESENCE optional }|
{ ID id-FiveG-ProSePC5QoSParameters        CRITICALITY ignore TYPE FiveG-ProSePC5QoSParameters        PRESENCE optional }|
{ ID id-NetworkControlledRepeaterAuthorized CRITICALITY ignore TYPE NetworkControlledRepeaterAuthorized PRESENCE optional }|
{ ID id-AerialUESubscriptionInformation     CRITICALITY ignore TYPE AerialUESubscriptionInformation     PRESENCE optional }|
{ ID id-NR-A2X-ServicesAuthorized          CRITICALITY ignore TYPE NR-A2X-ServicesAuthorized          PRESENCE optional }|
{ ID id-LTE-A2X-ServicesAuthorized          CRITICALITY ignore TYPE LTE-A2X-ServicesAuthorized          PRESENCE optional }|
{ ID id-NR-A2X-UE-PC5-AggregateMaximumBitRate CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate PRESENCE optional }|
{ ID id-LTE-A2X-UE-PC5-AggregateMaximumBitRate CRITICALITY ignore TYPE LTEUESidelinkAggregateMaximumBitrate PRESENCE optional }|
{ ID id-A2X-PC5-QoS-Parameters             CRITICALITY ignore TYPE A2X-PC5-QoS-Parameters             PRESENCE optional }|
{ ID id-MobileIAB-Authorized               CRITICALITY ignore TYPE MobileIAB-Authorized               PRESENCE optional }|
{ ID id-Partially-Allowed-NSSAI            CRITICALITY ignore TYPE Partially-Allowed-NSSAI            PRESENCE optional }|
{ ID id-SLPositioningRangingServiceInfo    CRITICALITY ignore TYPE SLPositioningRangingServiceInfo    PRESENCE optional }|
{ ID id-ExtendedOldAMF                    CRITICALITY ignore TYPE Extended-AMFName                    PRESENCE optional }|
{ ID id-AMF-UE-NGAP-ID2                   CRITICALITY ignore TYPE AMF-UE-NGAP-ID                    PRESENCE optional },
...
}

-- *****
--
-- INITIAL CONTEXT SETUP RESPONSE
-- *****

InitialContextSetupResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container    { {InitialContextSetupResponseIEs} },
    ...
}

InitialContextSetupResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory
    }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory
    }|
    { ID id-PDUSessionResourceSetupListCxtRes CRITICALITY ignore TYPE PDUSessionResourceSetupListCxtRes PRESENCE optional
    }|
    { ID id-PDUSessionResourceFailedToSetupListCxtRes CRITICALITY ignore TYPE PDUSessionResourceFailedToSetupListCxtRes PRESENCE optional
    }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional
    },
    ...
}

```

```

}

-- *****
--
-- INITIAL CONTEXT SETUP FAILURE
--
-- *****

InitialContextSetupFailure ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {InitialContextSetupFailureIEs} },
    ...
}

InitialContextSetupFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore  TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore  TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-PDUSessionResourceFailedToSetupListCxtFail CRITICALITY ignore  TYPE PDUSessionResourceFailedToSetupListCxtFail PRESENCE optional }|
    { ID id-Cause                   CRITICALITY ignore  TYPE Cause                           PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics   CRITICALITY ignore  TYPE CriticalityDiagnostics           PRESENCE optional }|
    ...
}

-- *****
--
-- UE Context Release Request Elementary Procedure
--
-- *****

-- *****
--
-- UE CONTEXT RELEASE REQUEST
--
-- *****

UEContextReleaseRequest ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {UEContextReleaseRequest-IEs} },
    ...
}

UEContextReleaseRequest-IEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject  TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject  TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-PDUSessionResourceListCxtRelReq CRITICALITY reject  TYPE PDUSessionResourceListCxtRelReq PRESENCE optional }|
    { ID id-Cause                   CRITICALITY ignore  TYPE Cause                           PRESENCE mandatory }|
    { ID id-GW-Context-Release-Indication CRITICALITY reject  TYPE GW-Context-Release-Indication PRESENCE optional }|
    ...
}

```



```

-- *****
--
-- UE Context Release Elementary Procedure
--
-- *****
--
-- *****
--
-- UE CONTEXT RELEASE COMMAND
--
-- *****

UEContextReleaseCommand ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container          { {UEContextReleaseCommand-IEs} },
    ...
}

UEContextReleaseCommand-IEs NGAP-PROTOCOL-IES ::= {
    { ID id-UE-NGAP-IDs          CRITICALITY reject TYPE UE-NGAP-IDs          PRESENCE mandatory }|
    { ID id-Cause                CRITICALITY ignore  TYPE Cause              PRESENCE mandatory },
    ...
}

-- *****
--
-- UE CONTEXT RELEASE COMPLETE
--
-- *****

UEContextReleaseComplete ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container          { {UEContextReleaseComplete-IEs} },
    ...
}

UEContextReleaseComplete-IEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE optional   }|
    { ID id-InfoOnRecommendedCellsAndRANNodesForPaging CRITICALITY ignore TYPE InfoOnRecommendedCellsAndRANNodesForPaging PRESENCE optional   }|
    { ID id-PDUSessionResourceListCxtRelCpl CRITICALITY reject TYPE PDUSessionResourceListCxtRelCpl PRESENCE optional   }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional   }|
    { ID id-PagingAssisDataforCEcapabUE CRITICALITY ignore TYPE PagingAssisDataforCEcapabUE PRESENCE optional   },
    ...
}

-- *****
--
-- UE Context Resume Elementary Procedure
--
-- *****
--
-- *****
--
-- UE CONTEXT RESUME REQUEST

```

```

--
-- *****
UEContextResumeRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {UEContextResumeRequestIEs} },
    ...
}

UEContextResumeRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RRC-Resume-Cause        CRITICALITY ignore TYPE RRCEstablishmentCause    PRESENCE mandatory }|
    { ID id-PDUSessionResourceResumeListRESReq CRITICALITY reject TYPE PDUSessionResourceResumeListRESReq PRESENCE optional }|
    { ID id-PDUSessionResourceFailedToResumeListRESReq CRITICALITY reject TYPE PDUSessionResourceFailedToResumeListRESReq PRESENCE optional }|
    { ID id-Suspend-Request-Indication CRITICALITY ignore TYPE Suspend-Request-Indication PRESENCE optional }|
    { ID id-InfoOnRecommendedCellsAndRANNodesForPaging CRITICALITY ignore TYPE InfoOnRecommendedCellsAndRANNodesForPaging PRESENCE optional }|
    { ID id-PagingAssisDataforCEcapabUE CRITICALITY ignore TYPE PagingAssisDataforCEcapabUE PRESENCE optional }|
    { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE optional },
    ...
}

-- *****
--
-- UE CONTEXT RESUME RESPONSE
--
-- *****

UEContextResumeResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {UEContextResumeResponseIEs} },
    ...
}

UEContextResumeResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-PDUSessionResourceResumeListRESRes CRITICALITY reject TYPE PDUSessionResourceResumeListRESRes PRESENCE optional }|
    { ID id-PDUSessionResourceFailedToResumeListRESRes CRITICALITY reject TYPE PDUSessionResourceFailedToResumeListRESRes PRESENCE optional }|
    { ID id-SecurityContext          CRITICALITY reject TYPE SecurityContext          PRESENCE optional }|
    { ID id-Suspend-Response-Indication CRITICALITY ignore TYPE Suspend-Response-Indication PRESENCE optional }|
    { ID id-Extended-ConnectedTime CRITICALITY ignore TYPE Extended-ConnectedTime PRESENCE optional }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- UE CONTEXT RESUME FAILURE
--
-- *****

UEContextResumeFailure ::= SEQUENCE {

```

```

    protocolIEs      ProtocolIE-Container      { { UEContextResumeFailureIEs} },
    ...
}

```

```

UEContextResumeFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-Cause                   CRITICALITY ignore TYPE Cause                   PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics   CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional   },
    ...
}

```

```

-- *****
--
-- UE Context Suspend Elementary Procedure
--
-- *****
--
-- *****
--
-- UE CONTEXT SUSPEND REQUEST
--
-- *****

```

```

UEContextSuspendRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {UEContextSuspendRequestIEs} },
    ...
}

```

```

UEContextSuspendRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory
    }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory
    }|
    { ID id-InfoOnRecommendedCellsAndRANNodesForPaging CRITICALITY ignore TYPE InfoOnRecommendedCellsAndRANNodesForPaging PRESENCE optional
    }|
    { ID id-PagingAssisDataforCEcapabUE CRITICALITY ignore TYPE PagingAssisDataforCEcapabUE PRESENCE optional
    }|
    { ID id-PDUSessionResourceSuspendListSUSReq       CRITICALITY reject TYPE PDUSessionResourceSuspendListSUSReq       PRESENCE optional
    }|
    { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE optional
    },
    ...
}

```

```

-- *****
--
-- UE CONTEXT SUSPEND RESPONSE
--
-- *****

```

```

UEContextSuspendResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {UEContextSuspendResponseIEs} },

```

```

    ...
}

UEContextSuspendResponseIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-SecurityContext          CRITICALITY reject TYPE SecurityContext        PRESENCE optional  }|
    { ID id-CriticalityDiagnostics    CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional  },
    ...
}

-- *****
--
-- UE CONTEXT SUSPEND FAILURE
--
-- *****

UEContextSuspendFailure ::= SEQUENCE {
    protocolIES      ProtocolIE-Container    { { UEContextSuspendFailureIES} },
    ...
}

UEContextSuspendFailureIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-Cause                   CRITICALITY ignore TYPE Cause                   PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics    CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional  },
    ...
}

-- *****
--
-- UE Context Modification Elementary Procedure
--
-- *****

-- *****
--
-- UE CONTEXT MODIFICATION REQUEST
--
-- *****

UEContextModificationRequest ::= SEQUENCE {
    protocolIES      ProtocolIE-Container    { {UEContextModificationRequestIES} },
    ...
}

UEContextModificationRequestIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RANPagingPriority        CRITICALITY ignore TYPE RANPagingPriority        PRESENCE optional  }|
    { ID id-SecurityKey              CRITICALITY reject TYPE SecurityKey              PRESENCE optional  }|
    { ID id-IndexToRFSP              CRITICALITY ignore TYPE IndexToRFSP              PRESENCE optional  }|
    { ID id-UEAggregateMaximumBitRate CRITICALITY ignore TYPE UEAggregateMaximumBitRate PRESENCE optional  }|

```

```

    { ID id-UESecurityCapabilities                CRITICALITY reject TYPE UESecurityCapabilities PRESENCE optional }|
    { ID id-CoreNetworkAssistanceInformationForInactive CRITICALITY ignore TYPE CoreNetworkAssistanceInformationForInactive PRESENCE optional }|
    { ID id-EmergencyFallbackIndicator            CRITICALITY reject TYPE EmergencyFallbackIndicator PRESENCE optional }|
    { ID id-NewAMF-UE-NGAP-ID                    CRITICALITY reject TYPE AMF-UE-NGAP-ID PRESENCE optional }|
    { ID id-RRCInactiveTransitionReportRequest    CRITICALITY ignore TYPE RRCInactiveTransitionReportRequest PRESENCE optional }|
    { ID id-NewGUAMI                             CRITICALITY reject TYPE GUAMI PRESENCE optional }|
    { ID id-CNAssistedRANTuning                  CRITICALITY ignore TYPE CNAssistedRANTuning PRESENCE optional }|
    { ID id-SRVCCOperationPossible               CRITICALITY ignore TYPE SRVCCOperationPossible PRESENCE optional }|
    { ID id-IAB-Authorized                      CRITICALITY ignore TYPE IAB-Authorized PRESENCE optional }|
    { ID id-NRV2XServicesAuthorized             CRITICALITY ignore TYPE NRV2XServicesAuthorized PRESENCE optional }|
    { ID id-LTEV2XServicesAuthorized            CRITICALITY ignore TYPE LTEV2XServicesAuthorized PRESENCE optional }|
    { ID id-NRUESidelinkAggregateMaximumBitrate CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate PRESENCE optional }|
    { ID id-LTEUESidelinkAggregateMaximumBitrate CRITICALITY ignore TYPE LTEUESidelinkAggregateMaximumBitrate PRESENCE optional }|
    { ID id-PC5QoSParameters                   CRITICALITY ignore TYPE PC5QoSParameters PRESENCE optional }|
    { ID id-UERadioCapabilityID                 CRITICALITY reject TYPE UERadioCapabilityID PRESENCE optional }|
    { ID id-RGLevelWirelineAccessCharacteristics CRITICALITY ignore TYPE RGLevelWirelineAccessCharacteristics PRESENCE optional }|
    { ID id-TimeSyncAssistanceInfo              CRITICALITY ignore TYPE TimeSyncAssistanceInfo PRESENCE optional }|
    { ID id-QMCConfigInfo                      CRITICALITY ignore TYPE QMCConfigInfo PRESENCE optional }|
    { ID id-QMCDeactivation                    CRITICALITY ignore TYPE QMCDeactivation PRESENCE optional }|
    { ID id-UESliceMaximumBitRateList           CRITICALITY ignore TYPE UESliceMaximumBitRateList PRESENCE optional }|
    { ID id-ManagementBasedMDTPLMNModificationList CRITICALITY ignore TYPE MDTPLMNModificationList PRESENCE optional }|
    { ID id-FiveG-ProSeAuthorized              CRITICALITY ignore TYPE FiveG-ProSeAuthorized PRESENCE optional }|
    { ID id-FiveG-ProSeUEPC5AggregateMaximumBitRate CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate PRESENCE optional }|
    { ID id-FiveG-ProSePC5QoSParameters         CRITICALITY ignore TYPE FiveG-ProSePC5QoSParameters PRESENCE optional }|
    { ID id-NetworkControlledRepeaterAuthorized CRITICALITY ignore TYPE NetworkControlledRepeaterAuthorized PRESENCE optional }|
    { ID id-AerialUESubscriptionInformation     CRITICALITY ignore TYPE AerialUESubscriptionInformation PRESENCE optional }|
    { ID id-NR-A2X-ServicesAuthorized           CRITICALITY ignore TYPE NR-A2X-ServicesAuthorized PRESENCE optional }|
    { ID id-LTE-A2X-ServicesAuthorized          CRITICALITY ignore TYPE LTE-A2X-ServicesAuthorized PRESENCE optional }|
    { ID id-NR-A2X-UE-PC5-AggregateMaximumBitRate CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate PRESENCE optional }|
    { ID id-LTE-A2X-UE-PC5-AggregateMaximumBitRate CRITICALITY ignore TYPE LTEUESidelinkAggregateMaximumBitrate PRESENCE optional }|
    { ID id-A2X-PC5-QoS-Parameters             CRITICALITY ignore TYPE A2X-PC5-QoS-Parameters PRESENCE optional }|
    { ID id-MobileIAB-Authorized                CRITICALITY ignore TYPE MobileIAB-Authorized PRESENCE optional }|
    { ID id-SLPositioningRangingServiceInfo     CRITICALITY ignore TYPE SLPositioningRangingServiceInfo PRESENCE optional },
    ...
}

-- *****
--
-- UE CONTEXT MODIFICATION RESPONSE
--
-- *****

UEContextModificationResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container    { {UEContextModificationResponseIEs} },
    ...
}

UEContextModificationResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RRCState                 CRITICALITY ignore TYPE RRCState                 PRESENCE optional }|
    { ID id-UserLocationInformation  CRITICALITY ignore TYPE UserLocationInformation  PRESENCE optional }|
    { ID id-CriticalityDiagnostics   CRITICALITY ignore TYPE CriticalityDiagnostics   PRESENCE optional },

```

```

    ...
}

-- *****
--
-- UE CONTEXT MODIFICATION FAILURE
--
-- *****

UEContextModificationFailure ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {UEContextModificationFailureIES} },
    ...
}

UEContextModificationFailureIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-Cause                   CRITICALITY ignore TYPE Cause                   PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics   CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional   },
    ...
}

-- *****
--
-- RRC INACTIVE TRANSITION REPORT
--
-- *****

RRCInactiveTransitionReport ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {RRCInactiveTransitionReportIES} },
    ...
}

RRCInactiveTransitionReportIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RRCState                CRITICALITY ignore TYPE RRCState                PRESENCE mandatory }|
    { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE mandatory },
    ...
}

-- *****
--
-- Retrieve UE Information
--
-- *****

RetrieveUEInformation ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { { RetrieveUEInformationIES } },
    ...
}

RetrieveUEInformationIES NGAP-PROTOCOL-IES ::= {
    { ID id-FiveG-S-TMSI            CRITICALITY reject TYPE FiveG-S-TMSI            PRESENCE mandatory },

```

```
...
}
-- *****
-- UE Information Transfer
--
-- *****

UEInformationTransfer ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    { { UEInformationTransferIEs} },
    ...
}

UEInformationTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-FiveG-S-TMSI          CRITICALITY reject TYPE FiveG-S-TMSI          PRESENCE mandatory }|
    { ID id-NB-IoT-UEPriority      CRITICALITY ignore TYPE NB-IoT-UEPriority      PRESENCE optional   }|
    { ID id-UERadioCapability      CRITICALITY ignore TYPE UERadioCapability      PRESENCE optional   }|
    { ID id-S-NSSAI               CRITICALITY ignore TYPE S-NSSAI               PRESENCE optional   }|
    { ID id-AllowedNSSAI          CRITICALITY ignore TYPE AllowedNSSAI          PRESENCE optional   }|
    { ID id-UE-DifferentiationInfo CRITICALITY ignore TYPE UE-DifferentiationInfo    PRESENCE optional   }|
    { ID id-MaskedIMEISV          CRITICALITY ignore TYPE MaskedIMEISV          PRESENCE optional   }|
    { ID id-Partially-Allowed-NSSAI CRITICALITY ignore TYPE Partially-Allowed-NSSAI    PRESENCE optional   },
    ...
}

-- *****
--
-- RAN CP Relocation Indication
--
-- *****

RANCPRelocationIndication ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container { { RANCPRelocationIndicationIEs} },
    ...
}

RANCPRelocationIndicationIEs NGAP-PROTOCOL-IES ::= {
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-FiveG-S-TMSI           CRITICALITY reject TYPE FiveG-S-TMSI           PRESENCE mandatory }|
    { ID id-EUTRA-CGI              CRITICALITY ignore TYPE EUTRA-CGI              PRESENCE mandatory }|
    { ID id-TAI                    CRITICALITY ignore TYPE TAI                    PRESENCE mandatory }|
    { ID id-UL-CP-SecurityInformation CRITICALITY reject TYPE UL-CP-SecurityInformation    PRESENCE mandatory },
    ...
}

-- *****
--
-- UE MOBILITY MANAGEMENT ELEMENTARY PROCEDURES
--
-- *****
-- *****
```

```

--
-- Handover Preparation Elementary Procedure
--
-- *****
--
-- *****
--
-- HANDOVER REQUIRED
--
-- *****

HandoverRequired ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {HandoverRequiredIEs} },
    ...
}

HandoverRequiredIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-HandoverType            CRITICALITY reject TYPE HandoverType            PRESENCE mandatory }|
    { ID id-Cause                   CRITICALITY ignore TYPE Cause                   PRESENCE mandatory }|
    { ID id-TargetID                CRITICALITY reject TYPE TargetID                PRESENCE mandatory }|
    { ID id-DirectForwardingPathAvailability CRITICALITY ignore TYPE DirectForwardingPathAvailability PRESENCE optional }|
    { ID id-PDUSessionResourceListHORqd CRITICALITY reject TYPE PDUSessionResourceListHORqd PRESENCE mandatory }|
    { ID id-SourceToTarget-TransparentContainer CRITICALITY reject TYPE SourceToTarget-TransparentContainer PRESENCE mandatory },
    ...
}

-- *****
--
-- HANDOVER COMMAND
--
-- *****

HandoverCommand ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {HandoverCommandIEs} },
    ...
}

HandoverCommandIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-HandoverType            CRITICALITY reject TYPE HandoverType            PRESENCE mandatory }|
    { ID id-NASSecurityParametersFromNGRAN CRITICALITY reject TYPE NASSecurityParametersFromNGRAN PRESENCE conditional }|
-- The above IE shall be present if the HandoverType IE is set to the value "5GStoEPPS" or "5GStoUTRAN"
    { ID id-PDUSessionResourceHandoverList CRITICALITY ignore TYPE PDUSessionResourceHandoverList PRESENCE optional }|
    { ID id-PDUSessionResourceToReleaseListHOCmd CRITICALITY ignore TYPE PDUSessionResourceToReleaseListHOCmd PRESENCE optional }|
    { ID id-TargetToSource-TransparentContainer CRITICALITY reject TYPE TargetToSource-TransparentContainer PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****

```



```

--
-- HANDOVER PREPARATION FAILURE
--
-- *****

HandoverPreparationFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {HandoverPreparationFailureIEs} },
    ...
}

HandoverPreparationFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-Cause                    CRITICALITY ignore TYPE Cause                    PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics    CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional }|
    { ID id-TargettoSource-Failure-TransparentContainer CRITICALITY ignore TYPE TargettoSource-Failure-TransparentContainer PRESENCE optional },
    ...
}

-- *****
--
-- Handover Resource Allocation Elementary Procedure
--
-- *****

-- *****
--
-- HANDOVER REQUEST
--
-- *****

HandoverRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {HandoverRequestIEs} },
    ...
}

HandoverRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-HandoverType            CRITICALITY reject TYPE HandoverType            PRESENCE mandatory }|
    { ID id-Cause                    CRITICALITY ignore TYPE Cause                    PRESENCE mandatory }|
    { ID id-UEAggregateMaximumBitRate CRITICALITY reject TYPE UEAggregateMaximumBitRate PRESENCE mandatory }|
    { ID id-CoreNetworkAssistanceInformationForInactive CRITICALITY ignore TYPE CoreNetworkAssistanceInformationForInactive PRESENCE optional }|
    { ID id-UESecurityCapabilities   CRITICALITY reject TYPE UESecurityCapabilities   PRESENCE mandatory }|
    { ID id-SecurityContext          CRITICALITY reject TYPE SecurityContext          PRESENCE mandatory }|
    { ID id-NewSecurityContextInd     CRITICALITY reject TYPE NewSecurityContextInd     PRESENCE optional }|
    { ID id-NASC                     CRITICALITY reject TYPE NAS-PDU                     PRESENCE optional }|
    { ID id-PDUSessionResourceSetupListH0Req CRITICALITY reject TYPE PDUSessionResourceSetupListH0Req PRESENCE mandatory }|
    { ID id-AllowedNSSAI             CRITICALITY reject TYPE AllowedNSSAI             PRESENCE mandatory }|
    { ID id-TraceActivation          CRITICALITY ignore TYPE TraceActivation          PRESENCE optional }|
    { ID id-MaskedIMEISV             CRITICALITY ignore TYPE MaskedIMEISV             PRESENCE optional }|
    { ID id-SourceToTarget-TransparentContainer CRITICALITY reject TYPE SourceToTarget-TransparentContainer PRESENCE mandatory }|
    { ID id-MobilityRestrictionList   CRITICALITY ignore TYPE MobilityRestrictionList   PRESENCE optional }|
}

```

{ ID id-LocationReportingRequestType	CRITICALITY ignore	TYPE LocationReportingRequestType	PRESENCE optional	}
{ ID id-RRCInactiveTransitionReportRequest	CRITICALITY ignore	TYPE RRCInactiveTransitionReportRequest	PRESENCE optional	}
{ ID id-GUAMI	CRITICALITY reject	TYPE GUAMI	PRESENCE mandatory	}
{ ID id-RedirectionVoiceFallback	CRITICALITY ignore	TYPE RedirectionVoiceFallback	PRESENCE optional	}
{ ID id-CNAssistedRANTuning	CRITICALITY ignore	TYPE CNAssistedRANTuning	PRESENCE optional	}
{ ID id-SRVCCOperationPossible	CRITICALITY ignore	TYPE SRVCCOperationPossible	PRESENCE optional	}
{ ID id-IAB-Authorized	CRITICALITY reject	TYPE IAB-Authorized	PRESENCE optional	}
{ ID id-Enhanced-CoverageRestriction	CRITICALITY ignore	TYPE Enhanced-CoverageRestriction	PRESENCE optional	}
{ ID id-UE-DifferentiationInfo	CRITICALITY ignore	TYPE UE-DifferentiationInfo	PRESENCE optional	}
{ ID id-NRV2XServicesAuthorized	CRITICALITY ignore	TYPE NRV2XServicesAuthorized	PRESENCE optional	}
{ ID id-LTEV2XServicesAuthorized	CRITICALITY ignore	TYPE LTEV2XServicesAuthorized	PRESENCE optional	}
{ ID id-NRUESidelinkAggregateMaximumBitrate	CRITICALITY ignore	TYPE NRUESidelinkAggregateMaximumBitrate	PRESENCE optional	}
{ ID id-LTEUESidelinkAggregateMaximumBitrate	CRITICALITY ignore	TYPE LTEUESidelinkAggregateMaximumBitrate	PRESENCE optional	}
{ ID id-PC5QoSParameters	CRITICALITY ignore	TYPE PC5QoSParameters	PRESENCE optional	}
{ ID id-CEmodeRestricted	CRITICALITY ignore	TYPE CEmodeRestricted	PRESENCE optional	}
{ ID id-UE-UP-CIoT-Support	CRITICALITY ignore	TYPE UE-UP-CIoT-Support	PRESENCE optional	}
{ ID id-ManagementBasedMDTPLMNList	CRITICALITY ignore	TYPE MDTPLMNList	PRESENCE optional	}
{ ID id-UERadioCapabilityID	CRITICALITY reject	TYPE UERadioCapabilityID	PRESENCE optional	}
{ ID id-Extended-ConnectedTime	CRITICALITY ignore	TYPE Extended-ConnectedTime	PRESENCE optional	}
{ ID id-TimeSyncAssistanceInfo	CRITICALITY ignore	TYPE TimeSyncAssistanceInfo	PRESENCE optional	}
{ ID id-UESliceMaximumBitRateList	CRITICALITY ignore	TYPE UESliceMaximumBitRateList	PRESENCE optional	}
{ ID id-FiveG-ProSeAuthorized	CRITICALITY ignore	TYPE FiveG-ProSeAuthorized	PRESENCE optional	}
{ ID id-FiveG-ProSeUEPC5AggregateMaximumBitRate	CRITICALITY ignore	TYPE NRUESidelinkAggregateMaximumBitrate	PRESENCE optional	}
{ ID id-FiveG-ProSePC5QoSParameters	CRITICALITY ignore	TYPE FiveG-ProSePC5QoSParameters	PRESENCE optional	}
{ ID id-AerialUESubscriptionInformation	CRITICALITY ignore	TYPE AerialUESubscriptionInformation	PRESENCE optional	}
{ ID id-NR-A2X-ServicesAuthorized	CRITICALITY ignore	TYPE NR-A2X-ServicesAuthorized	PRESENCE optional	}
{ ID id-LTE-A2X-ServicesAuthorized	CRITICALITY ignore	TYPE LTE-A2X-ServicesAuthorized	PRESENCE optional	}
{ ID id-NR-A2X-UE-PC5-AggregateMaximumBitRate	CRITICALITY ignore	TYPE NRUESidelinkAggregateMaximumBitrate	PRESENCE optional	}
{ ID id-LTE-A2X-UE-PC5-AggregateMaximumBitRate	CRITICALITY ignore	TYPE LTEUESidelinkAggregateMaximumBitrate	PRESENCE optional	}
{ ID id-A2X-PC5-QoS-Parameters	CRITICALITY ignore	TYPE A2X-PC5-QoS-Parameters	PRESENCE optional	}
{ ID id-MobileIAB-Authorized	CRITICALITY ignore	TYPE MobileIAB-Authorized	PRESENCE optional	}
{ ID id-NoPDUSessionIndication	CRITICALITY ignore	TYPE NoPDUSessionIndication	PRESENCE optional	}
{ ID id-Partially-Allowed-NSSAI	CRITICALITY ignore	TYPE Partially-Allowed-NSSAI	PRESENCE optional	}
{ ID id-SLPositioningRangingServiceInfo	CRITICALITY ignore	TYPE SLPositioningRangingServiceInfo	PRESENCE optional	}
{ ID id-AMF-UE-NGAP-ID2	CRITICALITY ignore	TYPE AMF-UE-NGAP-ID	PRESENCE optional	}
{ ID id-Target-RAN-Node-ID	CRITICALITY ignore	TYPE GlobalRANNodeID	PRESENCE optional	},
...				

}|

-- *****

--

-- HANDOVER REQUEST ACKNOWLEDGE

--

-- *****

HandoverRequestAcknowledge ::= SEQUENCE {

 protocolIEs ProtocolIE-Container { {HandoverRequestAcknowledgeIEs} },

 ...

}

HandoverRequestAcknowledgeIEs NGAP-PROTOCOL-IES ::= {

{ ID id-AMF-UE-NGAP-ID	CRITICALITY ignore	TYPE AMF-UE-NGAP-ID	PRESENCE mandatory	}
{ ID id-RAN-UE-NGAP-ID	CRITICALITY ignore	TYPE RAN-UE-NGAP-ID	PRESENCE mandatory	}
{ ID id-PDUSessionResourceAdmittedList	CRITICALITY ignore	TYPE PDUSessionResourceAdmittedList	PRESENCE mandatory	}

```

    { ID id-PDUSessionResourceFailedToSetupListH0Ack    CRITICALITY ignore TYPE PDUSessionResourceFailedToSetupListH0Ack    PRESENCE optional }|
    { ID id-TargetToSource-TransparentContainer          CRITICALITY reject  TYPE TargetToSource-TransparentContainer    PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore TYPE CriticalityDiagnostics        PRESENCE optional  }|
    { ID id-NPN-AccessInformation                      CRITICALITY reject  TYPE NPN-AccessInformation          PRESENCE optional  }|
    { ID id-RedCapIndication                          CRITICALITY ignore TYPE RedCapIndication              PRESENCE optional  }|
    { ID id-ERedCapIndication                          CRITICALITY ignore TYPE ERedCapIndication             PRESENCE optional  },
    ...
}

-- *****
--
-- HANDOVER FAILURE
--
-- *****

HandoverFailure ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { { HandoverFailureIEs} },
    ...
}

HandoverFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-Cause                  CRITICALITY ignore TYPE Cause                  PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics  CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional  }|
    { ID id-TargettoSource-Failure-TransparentContainer CRITICALITY ignore TYPE TargettoSource-Failure-TransparentContainer PRESENCE
optional },
    ...
}

-- *****
--
-- Handover Notification Elementary Procedure
--
-- *****

-- *****
--
-- HANDOVER NOTIFY
--
-- *****

HandoverNotify ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { { HandoverNotifyIEs} },
    ...
}

HandoverNotifyIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-UserLocationInformation  CRITICALITY ignore TYPE UserLocationInformation PRESENCE mandatory }|
    { ID id-NotifySourceNGRANNode    CRITICALITY ignore TYPE NotifySourceNGRANNode  PRESENCE optional  },
    ...
}

```

```
-- *****
--
-- Path Switch Request Elementary Procedure
--
-- *****
--
-- *****
--
-- PATH SWITCH REQUEST
--
-- *****
```

```
PathSwitchRequest ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { { PathSwitchRequestIES} },
    ...
}
```

```
PathSwitchRequestIES NGAP-PROTOCOL-IES ::= {
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-SourceAMF-UE-NGAP-ID    CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-UserLocationInformation  CRITICALITY ignore TYPE UserLocationInformation PRESENCE mandatory }|
    { ID id-UESecurityCapabilities   CRITICALITY ignore TYPE UESecurityCapabilities   PRESENCE mandatory }|
    { ID id-PDUSessionResourceToBeSwitchedDLList CRITICALITY reject TYPE PDUSessionResourceToBeSwitchedDLList PRESENCE mandatory }|
    { ID id-PDUSessionResourceFailedToSetupListPSReq CRITICALITY ignore TYPE PDUSessionResourceFailedToSetupListPSReq PRESENCE optional }|
    { ID id-RRC-Resume-Cause         CRITICALITY ignore TYPE RRCestablishmentCause   PRESENCE optional }|
    { ID id-RedCapIndication         CRITICALITY ignore TYPE RedCapIndication         PRESENCE optional }|
    { ID id-ERedCapIndication        CRITICALITY ignore TYPE ERedCapIndication        PRESENCE optional }|
    { ID id-SourceAMF-GUAMI          CRITICALITY ignore TYPE GUAMI                PRESENCE optional },
    ...
}
```

```
-- *****
--
-- PATH SWITCH REQUEST ACKNOWLEDGE
--
-- *****
```

```
PathSwitchRequestAcknowledge ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { { PathSwitchRequestAcknowledgeIES} },
    ...
}
```

```
PathSwitchRequestAcknowledgeIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-UESecurityCapabilities   CRITICALITY reject TYPE UESecurityCapabilities   PRESENCE optional }|
    { ID id-SecurityContext          CRITICALITY reject TYPE SecurityContext         PRESENCE mandatory }|
    { ID id-NewSecurityContextInd     CRITICALITY reject TYPE NewSecurityContextInd     PRESENCE optional }|
    { ID id-PDUSessionResourceSwitchedList CRITICALITY ignore TYPE PDUSessionResourceSwitchedList PRESENCE mandatory }|
    { ID id-PDUSessionResourceReleasedListPSAck CRITICALITY ignore TYPE PDUSessionResourceReleasedListPSAck PRESENCE optional }|
    { ID id-AllowedNSSAI            CRITICALITY reject TYPE AllowedNSSAI            PRESENCE mandatory }|
}
```

```

    { ID id-CoreNetworkAssistanceInformationForInactive    CRITICALITY ignore TYPE CoreNetworkAssistanceInformationForInactive    PRESENCE
optional    }|
    { ID id-RRCInactiveTransitionReportRequest            CRITICALITY ignore TYPE RRCInactiveTransitionReportRequest            PRESENCE optional }|
    { ID id-CriticalityDiagnostics                        CRITICALITY ignore TYPE CriticalityDiagnostics                        PRESENCE optional }|
    { ID id-RedirectionVoiceFallback                     CRITICALITY ignore TYPE RedirectionVoiceFallback                     PRESENCE optional }|
    { ID id-CNAssistedRANTuning                          CRITICALITY ignore TYPE CNAssistedRANTuning                          PRESENCE optional }|
    { ID id-SRVCCOperationPossible                       CRITICALITY ignore TYPE SRVCCOperationPossible                       PRESENCE optional }|
    { ID id-Enhanced-CoverageRestriction                 CRITICALITY ignore TYPE Enhanced-CoverageRestriction                 PRESENCE optional }|
    { ID id-Extended-ConnectedTime                      CRITICALITY ignore TYPE Extended-ConnectedTime                      PRESENCE optional }|
    { ID id-UE-DifferentiationInfo                      CRITICALITY ignore TYPE UE-DifferentiationInfo                      PRESENCE optional }|
    { ID id-NRV2XServicesAuthorized                     CRITICALITY ignore TYPE NRV2XServicesAuthorized                     PRESENCE optional }|
    { ID id-LTEV2XServicesAuthorized                    CRITICALITY ignore TYPE LTEV2XServicesAuthorized                    PRESENCE optional }|
    { ID id-NRUESidelinkAggregateMaximumBitrate         CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate         PRESENCE optional }|
    { ID id-LTEUESidelinkAggregateMaximumBitrate        CRITICALITY ignore TYPE LTEUESidelinkAggregateMaximumBitrate        PRESENCE optional }|
    { ID id-PC5QoSParameters                            CRITICALITY ignore TYPE PC5QoSParameters                            PRESENCE optional }|
    { ID id-CEmodeBrestricted                           CRITICALITY ignore TYPE CEmodeBrestricted                           PRESENCE optional }|
    { ID id-UE-UP-CIoT-Support                          CRITICALITY ignore TYPE UE-UP-CIoT-Support                          PRESENCE optional }|
    { ID id-UERadioCapabilityID                         CRITICALITY reject TYPE UERadioCapabilityID                         PRESENCE optional }|
    { ID id-ManagementBasedMDTPLMNList                 CRITICALITY ignore TYPE MDTPLMNList                 PRESENCE optional }|
    { ID id-TimeSyncAssistanceInfo                     CRITICALITY ignore TYPE TimeSyncAssistanceInfo                     PRESENCE optional }|
    { ID id-FiveG-ProSeAuthorized                      CRITICALITY ignore TYPE FiveG-ProSeAuthorized                      PRESENCE optional }|
    { ID id-FiveG-ProSeUEPC5AggregateMaximumBitRate    CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate    PRESENCE optional }|
    { ID id-FiveG-ProSePC5QoSParameters                CRITICALITY ignore TYPE FiveG-ProSePC5QoSParameters                PRESENCE optional }|
    { ID id-ManagementBasedMDTPLMNModificationList     CRITICALITY ignore TYPE MDTPLMNModificationList     PRESENCE optional }|
    { ID id-IAB-Authorized                             CRITICALITY ignore TYPE IAB-Authorized                             PRESENCE optional }|
    { ID id-AerialUESubscriptionInformation             CRITICALITY ignore TYPE AerialUESubscriptionInformation             PRESENCE optional }|
    { ID id-NR-A2X-ServicesAuthorized                  CRITICALITY ignore TYPE NR-A2X-ServicesAuthorized                  PRESENCE optional }|
    { ID id-LTE-A2X-ServicesAuthorized                  CRITICALITY ignore TYPE LTE-A2X-ServicesAuthorized                  PRESENCE optional }|
    { ID id-NR-A2X-UE-PC5-AggregateMaximumBitRate      CRITICALITY ignore TYPE NRUESidelinkAggregateMaximumBitrate      PRESENCE optional }|
    { ID id-LTE-A2X-UE-PC5-AggregateMaximumBitRate     CRITICALITY ignore TYPE LTEUESidelinkAggregateMaximumBitrate     PRESENCE optional }|
    { ID id-A2X-PC5-QoS-Parameters                    CRITICALITY ignore TYPE A2X-PC5-QoS-Parameters                    PRESENCE optional }|
    { ID id-MobileIAB-Authorized                      CRITICALITY ignore TYPE MobileIAB-Authorized                      PRESENCE optional }|
    { ID id-Partially-Allowed-NSSAI                   CRITICALITY ignore TYPE Partially-Allowed-NSSAI                   PRESENCE optional }|
    { ID id-SLPositioningRangingServiceInfo             CRITICALITY ignore TYPE SLPositioningRangingServiceInfo             PRESENCE optional }|
    { ID id-AMF-UE-NGAP-ID                             CRITICALITY ignore TYPE AMF-UE-NGAP-ID                             PRESENCE optional },
    ...
}

-- *****
--
-- PATH SWITCH REQUEST FAILURE
--
-- *****

PathSwitchRequestFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { { PathSwitchRequestFailureIEs} },
    ...
}

PathSwitchRequestFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID            CRITICALITY ignore TYPE AMF-UE-NGAP-ID            PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID            CRITICALITY ignore TYPE RAN-UE-NGAP-ID            PRESENCE mandatory }|
    { ID id-PDUSessionResourceReleasedListPSFail CRITICALITY ignore TYPE PDUSessionResourceReleasedListPSFail PRESENCE mandatory }|

```

```

    { ID id-CriticalityDiagnostics          CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- Handover Cancellation Elementary Procedure
--
-- *****

-- *****
--
-- HANDOVER CANCEL
--
-- *****

HandoverCancel ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { { HandoverCancelIES} },
    ...
}

HandoverCancelIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID      CRITICALITY reject TYPE AMF-UE-NGAP-ID PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID      CRITICALITY reject TYPE RAN-UE-NGAP-ID PRESENCE mandatory }|
    { ID id-Cause                CRITICALITY ignore TYPE Cause          PRESENCE mandatory },
    ...
}

-- *****
--
-- HANDOVER CANCEL ACKNOWLEDGE
--
-- *****

HandoverCancelAcknowledge ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { { HandoverCancelAcknowledgeIES} },
    ...
}

HandoverCancelAcknowledgeIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID      CRITICALITY ignore TYPE AMF-UE-NGAP-ID PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID      CRITICALITY ignore TYPE RAN-UE-NGAP-ID PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- HANDOVER SUCCESS ELEMENTARY PROCEDURE
--
-- *****

```

```

-- HANDOVER SUCCESS
--
-- *****

HandoverSuccess ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { { HandoverSuccessIEs } },
    ...
}

HandoverSuccessIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID      CRITICALITY reject TYPE AMF-UE-NGAP-ID      PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID      CRITICALITY reject TYPE RAN-UE-NGAP-ID      PRESENCE mandatory },
    ...
}

-- *****
--
-- UPLINK RAN EARLY STATUS TRANSFER ELEMENTARY PROCEDURE
--
-- *****

-- *****
--
-- Uplink RAN Early Status Transfer
--
-- *****

UplinkRANEarlyStatusTransfer ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {UplinkRANEarlyStatusTransferIEs} },
    ...
}

UplinkRANEarlyStatusTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID      CRITICALITY reject TYPE AMF-UE-NGAP-ID      PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID      CRITICALITY reject TYPE RAN-UE-NGAP-ID      PRESENCE mandatory }|
    { ID id-EarlyStatusTransfer-TransparentContainer CRITICALITY reject TYPE EarlyStatusTransfer-TransparentContainer PRESENCE mandatory },
    ...
}

-- *****
--
-- DOWNLINK RAN EARLY STATUS TRANSFER ELEMENTARY PROCEDURE
--
-- *****

-- *****
--
-- Downlink RAN Early Status Transfer
--
-- *****

DownlinkRANEarlyStatusTransfer ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {DownlinkRANEarlyStatusTransferIEs} },
    ...
}

```

```

}

DownlinkRANEarlyStatusTransferIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-EarlyStatusTransfer-TransparentContainer CRITICALITY reject TYPE EarlyStatusTransfer-TransparentContainer PRESENCE mandatory },
  ...
}

-- *****
--
-- Uplink RAN Status Transfer Elementary Procedure
--
-- *****

-- *****
--
-- UPLINK RAN STATUS TRANSFER
--
-- *****

UplinkRANStatusTransfer ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {UplinkRANStatusTransferIEs} },
  ...
}

UplinkRANStatusTransferIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RANStatusTransfer-TransparentContainer CRITICALITY reject TYPE RANStatusTransfer-TransparentContainer PRESENCE mandatory },
  ...
}

-- *****
--
-- Downlink RAN Status Transfer Elementary Procedure
--
-- *****

-- *****
--
-- DOWNLINK RAN STATUS TRANSFER
--
-- *****

DownlinkRANStatusTransfer ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {DownlinkRANStatusTransferIEs} },
  ...
}

DownlinkRANStatusTransferIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|

```



```
{ ID id-RANStatusTransfer-TransparentContainer    CRITICALITY reject  TYPE RANStatusTransfer-TransparentContainer    PRESENCE mandatory  },
...
}

-- *****
--
-- PAGING ELEMENTARY PROCEDURE
--
-- *****

-- *****
--
-- PAGING
--
-- *****

Paging ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container          { {PagingIEs} },
    ...
}

PagingIEs NGAP-PROTOCOL-IES ::= {
    { ID id-UEPagingIdentity          CRITICALITY ignore  TYPE UEPagingIdentity          PRESENCE mandatory  }|
    { ID id-PagingDRX                 CRITICALITY ignore  TYPE PagingDRX           PRESENCE optional   }|
    { ID id-TAIListForPaging           CRITICALITY ignore  TYPE TAIListForPaging     PRESENCE mandatory  }|
    { ID id-PagingPriority              CRITICALITY ignore  TYPE PagingPriority       PRESENCE optional   }|
    { ID id-UERadioCapabilityForPaging CRITICALITY ignore  TYPE UERadioCapabilityForPaging PRESENCE optional   }|
    { ID id-PagingOrigin               CRITICALITY ignore  TYPE PagingOrigin         PRESENCE optional   }|
    { ID id-AssistanceDataForPaging     CRITICALITY ignore  TYPE AssistanceDataForPaging PRESENCE optional   }|
    { ID id-NB-IoT-Paging-eDRXInfo      CRITICALITY ignore  TYPE NB-IoT-Paging-eDRXInfo PRESENCE optional   }|
    { ID id-NB-IoT-PagingDRX            CRITICALITY ignore  TYPE NB-IoT-PagingDRX     PRESENCE optional   }|
    { ID id-Enhanced-CoverageRestriction CRITICALITY ignore  TYPE Enhanced-CoverageRestriction PRESENCE optional   }|
    { ID id-WUS-Assistance-Information  CRITICALITY ignore  TYPE WUS-Assistance-Information PRESENCE optional   }|
    { ID id-EUTRA-PagingeDRXInformation CRITICALITY ignore  TYPE EUTRA-PagingeDRXInformation PRESENCE optional   }|
    { ID id-CEmodeBrestricted           CRITICALITY ignore  TYPE CEmodeBrestricted    PRESENCE optional   }|
    { ID id-NR-PagingeDRXInformation     CRITICALITY ignore  TYPE NR-PagingeDRXInformation PRESENCE optional   }|
    { ID id-PagingCause                 CRITICALITY ignore  TYPE PagingCause          PRESENCE optional   }|
    { ID id-PEIPsAssistanceInformation  CRITICALITY ignore  TYPE PEIPsAssistanceInformation PRESENCE optional   }|
    { ID id-LPWUSPSAssistanceInformation CRITICALITY ignore  TYPE LPWUSPSAssistanceInformation PRESENCE optional   }|
    { ID id-LPWUSDisableIndication      CRITICALITY ignore  TYPE LPWUSDisableIndication PRESENCE optional   },
    ...
}

-- *****
--
-- NAS TRANSPORT ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- INITIAL UE MESSAGE
--
-- *****
```

```

InitialUEMessage ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {InitialUEMessage-IEs} },
    ...
}

InitialUEMessage-IEs NGAP-PROTOCOL-IES ::= {
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-NAS-PDU                 CRITICALITY reject TYPE NAS-PDU                 PRESENCE mandatory }|
    { ID id-UserLocationInformation CRITICALITY reject TYPE UserLocationInformation PRESENCE mandatory }|
    { ID id-RRCEstablishmentCause   CRITICALITY ignore TYPE RRCEstablishmentCause   PRESENCE mandatory }|
    { ID id-FiveG-S-TMSI            CRITICALITY reject TYPE FiveG-S-TMSI            PRESENCE optional  }|
    { ID id-AMFSetID                CRITICALITY ignore TYPE AMFSetID                PRESENCE optional  }|
    { ID id-UEContextRequest        CRITICALITY ignore TYPE UEContextRequest        PRESENCE optional  }|
    { ID id-AllowedNSSAI            CRITICALITY reject TYPE AllowedNSSAI            PRESENCE optional  }|
    { ID id-SourceToTarget-AMFInformationReroute CRITICALITY ignore TYPE SourceToTarget-AMFInformationReroute PRESENCE optional }|
    { ID id-SelectedPLMNIdentity    CRITICALITY ignore TYPE PLMNIdentity            PRESENCE optional  }|
    { ID id-IABNodeIndication       CRITICALITY reject TYPE IABNodeIndication       PRESENCE optional  }|
    { ID id-CEmodeBSupport-Indicator CRITICALITY reject TYPE CEmodeBSupport-Indicator PRESENCE optional  }|
    { ID id-LTEM-Indication         CRITICALITY ignore TYPE LTEM-Indication         PRESENCE optional  }|
    { ID id-EDT-Session             CRITICALITY ignore TYPE EDT-Session             PRESENCE optional  }|
    { ID id-AuthenticatedIndication CRITICALITY ignore TYPE AuthenticatedIndication PRESENCE optional  }|
    { ID id-NPN-AccessInformation   CRITICALITY reject TYPE NPN-AccessInformation   PRESENCE optional  }|
    { ID id-RedCapIndication        CRITICALITY ignore TYPE RedCapIndication        PRESENCE optional  }|
    { ID id-SelectedNID             CRITICALITY ignore TYPE NID                     PRESENCE optional  }|
    { ID id-MobileIABNodeIndication CRITICALITY reject TYPE MobileIABNodeIndication PRESENCE optional  }|
    { ID id-Partially-Allowed-NSSAI CRITICALITY ignore TYPE Partially-Allowed-NSSAI PRESENCE optional  }|
    { ID id-ERedCapIndication       CRITICALITY ignore TYPE ERedCapIndication       PRESENCE optional  }|
    { ID id-AUN3DeviceAccessInfo    CRITICALITY ignore TYPE AUN3DeviceAccessInfo    PRESENCE optional  }|
    { ID id-GUAMI                  CRITICALITY ignore TYPE GUAMI                  PRESENCE optional  }|
    { ID id-GUAMIType              CRITICALITY ignore TYPE GUAMIType              PRESENCE optional  }|
    { ID id-RequestedNSSAI         CRITICALITY ignore TYPE RequestedNSSAI         PRESENCE optional },
    ...
}

-- *****
--
-- DOWNLINK NAS TRANSPORT
--
-- *****

DownlinkNASTransport ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {DownlinkNASTransport-IEs} },
    ...
}

DownlinkNASTransport-IEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-OldAMF                  CRITICALITY reject TYPE AMFName                PRESENCE optional  }|
    { ID id-RANPagingPriority        CRITICALITY ignore TYPE RANPagingPriority        PRESENCE optional  }|
    { ID id-NAS-PDU                 CRITICALITY reject TYPE NAS-PDU                 PRESENCE mandatory }|
    { ID id-MobilityRestrictionList CRITICALITY ignore TYPE MobilityRestrictionList PRESENCE optional  }|
    { ID id-IndexToRFSP             CRITICALITY ignore TYPE IndexToRFSP             PRESENCE optional  }|
}

```

```

    { ID id-UEAggregateMaximumBitRate      CRITICALITY ignore TYPE UEAggregateMaximumBitRate PRESENCE optional }|
    { ID id-AllowedNSSAI                    CRITICALITY reject TYPE AllowedNSSAI          PRESENCE optional }|
    { ID id-SRVCCOperationPossible          CRITICALITY ignore TYPE SRVCCOperationPossible PRESENCE optional }|
    { ID id-Enhanced-CoverageRestriction    CRITICALITY ignore TYPE Enhanced-CoverageRestriction PRESENCE optional }|
    { ID id-Extended-ConnectedTime         CRITICALITY ignore TYPE Extended-ConnectedTime PRESENCE optional }|
    { ID id-UE-DifferentiationInfo         CRITICALITY ignore TYPE UE-DifferentiationInfo PRESENCE optional }|
    { ID id-CEmodeRestricted                CRITICALITY ignore TYPE CEmodeRestricted PRESENCE optional }|
    { ID id-UERadioCapability               CRITICALITY ignore TYPE UERadioCapability PRESENCE optional }|
    { ID id-UECapabilityInfoRequest         CRITICALITY ignore TYPE UECapabilityInfoRequest PRESENCE optional }|
    { ID id-EndIndication                   CRITICALITY ignore TYPE EndIndication PRESENCE optional }|
    { ID id-UERadioCapabilityID             CRITICALITY reject TYPE UERadioCapabilityID PRESENCE optional }|
    { ID id-TargetNSSAIInformation          CRITICALITY ignore TYPE TargetNSSAIInformation PRESENCE optional }|
    { ID id-MaskedIMEISV                    CRITICALITY ignore TYPE MaskedIMEISV PRESENCE optional }|
    { ID id-Partially-Allowed-NSSAI        CRITICALITY ignore TYPE Partially-Allowed-NSSAI PRESENCE optional }|
    { ID id-MobileIAB-Authorized            CRITICALITY ignore TYPE MobileIAB-Authorized PRESENCE optional }|
    { ID id-ExtendedOldAMF                  CRITICALITY ignore TYPE Extended-AMFName PRESENCE optional },
    ...
}

-- *****
--
-- UPLINK NAS TRANSPORT
--
-- *****

UplinkNASTransport ::= SEQUENCE {
    protocolIES      ProtocolIE-Container    { {UplinkNASTransport-IES} },
    ...
}

UplinkNASTransport-IES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID      CRITICALITY reject TYPE AMF-UE-NGAP-ID      PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID      CRITICALITY reject TYPE RAN-UE-NGAP-ID      PRESENCE mandatory }|
    { ID id-NAS-PDU              CRITICALITY reject TYPE NAS-PDU              PRESENCE mandatory }|
    { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE mandatory }|
    { ID id-W-AGFIdentityInformation CRITICALITY reject TYPE OCTET STRING PRESENCE optional }|
    { ID id-TNGFIdentityInformation CRITICALITY reject TYPE OCTET STRING PRESENCE optional }|
    { ID id-TWIFIdentityInformation CRITICALITY reject TYPE OCTET STRING PRESENCE optional },
    ...
}

-- *****
--
-- NAS NON DELIVERY INDICATION
--
-- *****

NASNonDeliveryIndication ::= SEQUENCE {
    protocolIES      ProtocolIE-Container    { {NASNonDeliveryIndication-IES} },
    ...
}

NASNonDeliveryIndication-IES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID      CRITICALITY reject TYPE AMF-UE-NGAP-ID      PRESENCE mandatory }|

```

```

    { ID id-RAN-UE-NGAP-ID      CRITICALITY reject TYPE RAN-UE-NGAP-ID PRESENCE mandatory }|
    { ID id-NAS-PDU             CRITICALITY ignore TYPE NAS-PDU         PRESENCE mandatory }|
    { ID id-Cause               CRITICALITY ignore TYPE Cause           PRESENCE mandatory },
    ...
}

-- *****
--
-- REROUTE NAS REQUEST
--
-- *****

RerouteNASRequest ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {RerouteNASRequest-IEs} },
    ...
}

RerouteNASRequest-IEs NGAP-PROTOCOL-IES ::= {
    { ID id-RAN-UE-NGAP-ID      CRITICALITY reject TYPE RAN-UE-NGAP-ID PRESENCE mandatory }|
    { ID id-AMF-UE-NGAP-ID      CRITICALITY ignore TYPE AMF-UE-NGAP-ID PRESENCE optional }|
    { ID id-NGAP-Message        CRITICALITY reject TYPE OCTET STRING PRESENCE mandatory }|
    { ID id-AMFSetID            CRITICALITY reject TYPE AMFSetID PRESENCE mandatory }|
    { ID id-AllowedNSSAI        CRITICALITY reject TYPE AllowedNSSAI PRESENCE optional }|
    { ID id-SourceToTarget-AMFInformationReroute CRITICALITY ignore TYPE SourceToTarget-AMFInformationReroute PRESENCE optional }|
    { ID id-Partially-Allowed-NSSAI CRITICALITY ignore TYPE Partially-Allowed-NSSAI PRESENCE optional },
    ...
}

-- *****
--
-- INTERFACE MANAGEMENT ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- NG Removal Elementary Procedure
--
-- *****

-- *****
--
-- NG REMOVAL REQUEST
--
-- *****

NGRemovalRequest ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {NGRemovalRequestIEs} },
    ...
}

NGRemovalRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-GlobalRANNodeID     CRITICALITY reject TYPE GlobalRANNodeID PRESENCE mandatory },
    ...
}

```

```

}

-- *****
--
-- NG REMOVAL RESPONSE
--
-- *****

NGRemovalResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {NGRemovalResponseIEs} },
    ...
}

NGRemovalResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMFName          CRITICALITY reject  TYPE AMFName          PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional }|
    { ID id-Extended-AMFName   CRITICALITY ignore TYPE Extended-AMFName   PRESENCE optional },
    ...
}

-- *****
--
-- NG REMOVAL FAILURE
--
-- *****

NGRemovalFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {NGRemovalFailureIEs} },
    ...
}

NGRemovalFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-Cause          CRITICALITY ignore  TYPE Cause          PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- NG Setup Elementary Procedure
--
-- *****

-- *****
--
-- NG SETUP REQUEST
--
-- *****

NGSetupRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {NGSetupRequestIEs} },
    ...
}

```

```

NGSetupRequestIEs NGAP-PROTOCOL-IES ::= {
  { ID id-GlobalRANNodeID          CRITICALITY reject TYPE GlobalRANNodeID          PRESENCE mandatory }|
  { ID id-RANNodeName              CRITICALITY ignore TYPE RANNodeName              PRESENCE optional  }|
  { ID id-SupportedTAList          CRITICALITY reject TYPE SupportedTAList          PRESENCE mandatory }|
  { ID id-DefaultPagingDRX        CRITICALITY ignore TYPE PagingDRX              PRESENCE mandatory }|
  { ID id-UERetentionInformation    CRITICALITY ignore TYPE UERetentionInformation    PRESENCE optional  }|
  { ID id-NB-IoT-DefaultPagingDRX CRITICALITY ignore TYPE NB-IoT-DefaultPagingDRX    PRESENCE optional  }|
  { ID id-Extended-RANNodeName     CRITICALITY ignore TYPE Extended-RANNodeName     PRESENCE optional  }|
  { ID id-AIoT-Support             CRITICALITY reject TYPE AIoT-Support             PRESENCE optional  }|
  { ID id-AdditionalULI            CRITICALITY ignore TYPE AdditionalULI            PRESENCE optional  },
  ...
}

-- *****
--
-- NG SETUP RESPONSE
--
-- *****

NGSetupResponse ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {NGSetupResponseIEs} },
  ...
}

NGSetupResponseIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMFName          CRITICALITY reject TYPE AMFName          PRESENCE mandatory }|
  { ID id-ServedGUAMIList  CRITICALITY reject TYPE ServedGUAMIList  PRESENCE mandatory }|
  { ID id-RelativeAMFCapacity CRITICALITY ignore TYPE RelativeAMFCapacity PRESENCE mandatory }|
  { ID id-PLMNSupportList  CRITICALITY reject TYPE PLMNSupportList  PRESENCE mandatory }|
  { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional  }|
  { ID id-UERetentionInformation CRITICALITY ignore TYPE UERetentionInformation PRESENCE optional  }|
  { ID id-IAB-Supported     CRITICALITY ignore TYPE IAB-Supported     PRESENCE optional  }|
  { ID id-Extended-AMFName  CRITICALITY ignore TYPE Extended-AMFName  PRESENCE optional  }|
  { ID id-MobileIAB-Supported CRITICALITY ignore TYPE MobileIAB-Supported PRESENCE optional  }|
  { ID id-AIoTFIdentifier    CRITICALITY reject TYPE AIoTFIdentifier    PRESENCE optional  }|
  { ID id-AIoTFFName        CRITICALITY reject TYPE AIoTFName        PRESENCE optional  },
  ...
}

-- *****
--
-- NG SETUP FAILURE
--
-- *****

NGSetupFailure ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {NGSetupFailureIEs} },
  ...
}

NGSetupFailureIEs NGAP-PROTOCOL-IES ::= {
  { ID id-Cause          CRITICALITY ignore TYPE Cause          PRESENCE mandatory }|
  { ID id-TimeToWait     CRITICALITY ignore TYPE TimeToWait     PRESENCE optional  }|
  { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional  },

```

```

    ...
}

-- *****
--
-- RAN Configuration Update Elementary Procedure
--
-- *****

-- *****
--
-- RAN CONFIGURATION UPDATE
--
-- *****

RANConfigurationUpdate ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {RANConfigurationUpdateIEs} },
    ...
}

RANConfigurationUpdateIEs NGAP-PROTOCOL-IES ::= {
    { ID id-RANNodeName          CRITICALITY ignore TYPE RANNodeName          PRESENCE optional }|
    { ID id-SupportedTAList      CRITICALITY reject TYPE SupportedTAList      PRESENCE optional }|
    { ID id-DefaultPagingDRX     CRITICALITY ignore TYPE PagingDRX           PRESENCE optional }|
    { ID id-GlobalRANNodeID      CRITICALITY ignore TYPE GlobalRANNodeID      PRESENCE optional }|
    { ID id-NGRAN-TNLAssociationToRemoveList CRITICALITY reject TYPE NGRAN-TNLAssociationToRemoveList PRESENCE optional }|
    { ID id-NB-IoT-DefaultPagingDRX CRITICALITY ignore TYPE NB-IoT-DefaultPagingDRX PRESENCE optional }|
    { ID id-Extended-RANNodeName CRITICALITY ignore TYPE Extended-RANNodeName PRESENCE optional }|
    { ID id-AdditionalULI        CRITICALITY ignore TYPE AdditionalULI        PRESENCE optional }|
    { ID id-AIoT-Support         CRITICALITY reject TYPE AIoT-Support         PRESENCE optional },
    ...
}

-- *****
--
-- RAN CONFIGURATION UPDATE ACKNOWLEDGE
--
-- *****

RANConfigurationUpdateAcknowledge ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {RANConfigurationUpdateAcknowledgeIEs} },
    ...
}

RANConfigurationUpdateAcknowledgeIEs NGAP-PROTOCOL-IES ::= {
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- RAN CONFIGURATION UPDATE FAILURE
--
-- *****

```

```

RANConfigurationUpdateFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container    { {RANConfigurationUpdateFailureIEs} },
    ...
}

RANConfigurationUpdateFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-Cause          CRITICALITY ignore TYPE Cause          PRESENCE mandatory }|
    { ID id-TimeToWait     CRITICALITY ignore TYPE TimeToWait     PRESENCE optional  }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional  },
    ...
}

-- *****
--
-- AMF Configuration Update Elementary Procedure
--
-- *****

-- *****
--
-- AMF CONFIGURATION UPDATE
--
-- *****

AMFConfigurationUpdate ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container    { {AMFConfigurationUpdateIEs} },
    ...
}

AMFConfigurationUpdateIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMFName          CRITICALITY reject TYPE AMFName          PRESENCE optional  }|
    { ID id-ServedGUAMList   CRITICALITY reject TYPE ServedGUAMList   PRESENCE optional  }|
    { ID id-RelativeAMFCapacity CRITICALITY ignore TYPE RelativeAMFCapacity PRESENCE optional  }|
    { ID id-PLMNSupportList  CRITICALITY reject TYPE PLMNSupportList  PRESENCE optional  }|
    { ID id-AMF-TNLAssociationToAddList CRITICALITY ignore TYPE AMF-TNLAssociationToAddList PRESENCE optional  }|
    { ID id-AMF-TNLAssociationToRemoveList CRITICALITY ignore TYPE AMF-TNLAssociationToRemoveList PRESENCE optional  }|
    { ID id-AMF-TNLAssociationToUpdateList CRITICALITY ignore TYPE AMF-TNLAssociationToUpdateList PRESENCE optional  }|
    { ID id-Extended-AMFName CRITICALITY ignore TYPE Extended-AMFName PRESENCE optional  },
    ...
}

-- *****
--
-- AMF CONFIGURATION UPDATE ACKNOWLEDGE
--
-- *****

AMFConfigurationUpdateAcknowledge ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container    { {AMFConfigurationUpdateAcknowledgeIEs} },
    ...
}

AMFConfigurationUpdateAcknowledgeIEs NGAP-PROTOCOL-IES ::= {

```



```

    { ID id-AMF-TNLAssociationSetupList          CRITICALITY ignore TYPE AMF-TNLAssociationSetupList PRESENCE optional }|
    { ID id-AMF-TNLAssociationFailedToSetupList  CRITICALITY ignore TYPE TNLAssociationList      PRESENCE optional }|
    { ID id-CriticalityDiagnostics                CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- AMF CONFIGURATION UPDATE FAILURE
--
-- *****

AMFConfigurationUpdateFailure ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {AMFConfigurationUpdateFailureIES} },
    ...
}

AMFConfigurationUpdateFailureIES NGAP-PROTOCOL-IES ::= {
    { ID id-Cause                CRITICALITY ignore TYPE Cause                PRESENCE mandatory }|
    { ID id-TimeToWait           CRITICALITY ignore TYPE TimeToWait           PRESENCE optional  }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional  },
    ...
}

-- *****
--
-- AMF Status Indication Elementary Procedure
--
-- *****

-- *****
--
-- AMF STATUS INDICATION
--
-- *****

AMFStatusIndication ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {AMFStatusIndicationIES} },
    ...
}

AMFStatusIndicationIES NGAP-PROTOCOL-IES ::= {
    { ID id-UnavailableGUAMList  CRITICALITY reject TYPE UnavailableGUAMList  PRESENCE mandatory },
    ...
}

-- *****
--
-- NG Reset Elementary Procedure
--
-- *****
--
-- *****

```

```

-- NG RESET
--
-- *****

NGReset ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {NGResetIES} },
    ...
}

NGResetIES NGAP-PROTOCOL-IES ::= {
    { ID id-Cause          CRITICALITY ignore TYPE Cause          PRESENCE mandatory }|
    { ID id-ResetType      CRITICALITY reject TYPE ResetType      PRESENCE mandatory },
    ...
}

-- *****
--
-- NG RESET ACKNOWLEDGE
--
-- *****

NGResetAcknowledge ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {NGResetAcknowledgeIES} },
    ...
}

NGResetAcknowledgeIES NGAP-PROTOCOL-IES ::= {
    { ID id-UE-associatedLogicalNG-connectionList          CRITICALITY ignore TYPE UE-associatedLogicalNG-connectionList          PRESENCE optional
    }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore TYPE CriticalityDiagnostics          PRESENCE optional
    },
    ...
}

-- *****
--
-- Error Indication Elementary Procedure
--
-- *****

-- *****
--
-- ERROR INDICATION
--
-- *****

ErrorIndication ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {ErrorIndicationIES} },
    ...
}

ErrorIndicationIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE optional }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE optional },
    ...
}

```

```

    { ID id-Cause                                CRITICALITY ignore TYPE Cause                PRESENCE optional }|
    { ID id-CriticalityDiagnostics                CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional }|
    { ID id-FiveG-S-TMSI                         CRITICALITY ignore TYPE FiveG-S-TMSI          PRESENCE optional },
    ...
}

-- *****
--
-- OVERLOAD START
--
-- *****

OverloadStart ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {OverloadStartIEs} },
    ...
}

OverloadStartIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMFOverloadResponse                CRITICALITY reject TYPE OverloadResponse                PRESENCE optional }|
    { ID id-AMFTrafficLoadReductionIndication  CRITICALITY ignore TYPE TrafficLoadReductionIndication  PRESENCE optional }|
    { ID id-OverloadStartNSSAIList              CRITICALITY ignore TYPE OverloadStartNSSAIList          PRESENCE optional }|
    { ID id-GUAMIList                          CRITICALITY ignore TYPE GUAMIList                        PRESENCE optional },
    ...
}

-- *****
--
-- OVERLOAD STOP
--
-- *****

OverloadStop ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {OverloadStopIEs} },
    ...
}

OverloadStopIEs NGAP-PROTOCOL-IES ::= {
    { ID id-GUAMIList                          CRITICALITY ignore TYPE GUAMIList                        PRESENCE optional },
    ...
}

-- *****
--
-- CONFIGURATION TRANSFER ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- UPLINK RAN CONFIGURATION TRANSFER
--
-- *****

UplinkRANConfigurationTransfer ::= SEQUENCE {

```

```

    protocolIEs      ProtocolIE-Container      { {UplinkRANConfigurationTransferIEs} },
    ...
}

UplinkRANConfigurationTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-SONConfigurationTransferUL          CRITICALITY ignore TYPE SONConfigurationTransfer          PRESENCE optional    }|
    { ID id-ENDC-SONConfigurationTransferUL      CRITICALITY ignore TYPE EN-DCSONConfigurationTransfer      PRESENCE optional    }|
    { ID id-IntersystemSONConfigurationTransferUL CRITICALITY ignore TYPE IntersystemSONConfigurationTransfer PRESENCE optional    },
    ...
}

-- *****
--
-- DOWNLINK RAN CONFIGURATION TRANSFER
--
-- *****

DownlinkRANConfigurationTransfer ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {DownlinkRANConfigurationTransferIEs} },
    ...
}

DownlinkRANConfigurationTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-SONConfigurationTransferDL          CRITICALITY ignore TYPE SONConfigurationTransfer          PRESENCE optional    }|
    { ID id-ENDC-SONConfigurationTransferDL      CRITICALITY ignore TYPE EN-DCSONConfigurationTransfer      PRESENCE optional    }|
    { ID id-IntersystemSONConfigurationTransferDL CRITICALITY ignore TYPE IntersystemSONConfigurationTransfer PRESENCE optional    },
    ...
}

-- *****
--
-- WARNING MESSAGE TRANSMISSION ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- Write-Replace Warning Elementary Procedure
--
-- *****

-- *****
--
-- WRITE-REPLACE WARNING REQUEST
--
-- *****

WriteReplaceWarningRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {WriteReplaceWarningRequestIEs} },
    ...
}

WriteReplaceWarningRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MessageIdentifier                    CRITICALITY reject TYPE MessageIdentifier                    PRESENCE mandatory    }|

```

```

    { ID id-SerialNumber          CRITICALITY reject TYPE SerialNumber          PRESENCE mandatory }|
    { ID id-WarningAreaList        CRITICALITY ignore TYPE WarningAreaList        PRESENCE optional  }|
    { ID id-RepetitionPeriod        CRITICALITY reject TYPE RepetitionPeriod        PRESENCE mandatory }|
    { ID id-NumberOfBroadcastsRequested CRITICALITY reject TYPE NumberOfBroadcastsRequested PRESENCE mandatory }|
    { ID id-WarningType            CRITICALITY ignore TYPE WarningType            PRESENCE optional  }|
    { ID id-WarningSecurityInfo     CRITICALITY ignore TYPE WarningSecurityInfo     PRESENCE optional  }|
    { ID id-DataCodingScheme        CRITICALITY ignore TYPE DataCodingScheme        PRESENCE optional  }|
    { ID id-WarningMessageContents  CRITICALITY ignore TYPE WarningMessageContents  PRESENCE optional  }|
    { ID id-ConcurrentWarningMessageInd CRITICALITY reject TYPE ConcurrentWarningMessageInd PRESENCE optional  }|
    { ID id-WarningAreaCoordinates  CRITICALITY ignore TYPE WarningAreaCoordinates  PRESENCE optional  },
    ...
}

-- *****
--
-- WRITE-REPLACE WARNING RESPONSE
--
-- *****

WriteReplaceWarningResponse ::= SEQUENCE {
    protocolIES          ProtocolIE-Container      { {WriteReplaceWarningResponseIES} },
    ...
}

WriteReplaceWarningResponseIES NGAP-PROTOCOL-IES ::= {
    { ID id-MessageIdentifier    CRITICALITY reject TYPE MessageIdentifier    PRESENCE mandatory }|
    { ID id-SerialNumber         CRITICALITY reject TYPE SerialNumber         PRESENCE mandatory }|
    { ID id-BroadcastCompletedAreaList CRITICALITY ignore TYPE BroadcastCompletedAreaList PRESENCE optional  }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics  PRESENCE optional  },
    ...
}

-- *****
--
-- PWS Cancel Elementary Procedure
--
-- *****

-- *****
--
-- PWS CANCEL REQUEST
--
-- *****

PWSCancelRequest ::= SEQUENCE {
    protocolIES          ProtocolIE-Container      { {PWSCancelRequestIES} },
    ...
}

PWSCancelRequestIES NGAP-PROTOCOL-IES ::= {
    { ID id-MessageIdentifier    CRITICALITY reject TYPE MessageIdentifier    PRESENCE mandatory }|
    { ID id-SerialNumber         CRITICALITY reject TYPE SerialNumber         PRESENCE mandatory }|
    { ID id-WarningAreaList      CRITICALITY ignore TYPE WarningAreaList      PRESENCE optional  }|
    { ID id-CancelAllWarningMessages CRITICALITY reject TYPE CancelAllWarningMessages PRESENCE optional  },

```

```

    ...
}

-- *****
--
-- PWS CANCEL RESPONSE
--
-- *****

PWSCancelResponse ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {PWSCancelResponseIEs} },
    ...
}

PWSCancelResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MessageIdentifier      CRITICALITY reject TYPE MessageIdentifier      PRESENCE mandatory }|
    { ID id-SerialNumber           CRITICALITY reject TYPE SerialNumber           PRESENCE mandatory }|
    { ID id-BroadcastCancelledAreaList CRITICALITY ignore TYPE BroadcastCancelledAreaList PRESENCE optional }|
    { ID id-CriticalityDiagnostics  CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- PWS Restart Indication Elementary Procedure
--
-- *****

-- *****
--
-- PWS RESTART INDICATION
--
-- *****

PWSRestartIndication ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {PWSRestartIndicationIEs} },
    ...
}

PWSRestartIndicationIEs NGAP-PROTOCOL-IES ::= {
    { ID id-CellIDListForRestart    CRITICALITY reject TYPE CellIDListForRestart    PRESENCE mandatory }|
    { ID id-GlobalRANNodeID         CRITICALITY reject TYPE GlobalRANNodeID         PRESENCE mandatory }|
    { ID id-TAIListForRestart       CRITICALITY reject TYPE TAIListForRestart       PRESENCE mandatory }|
    { ID id-EmergencyAreaIDListForRestart CRITICALITY reject TYPE EmergencyAreaIDListForRestart PRESENCE optional },
    ...
}

-- *****
--
-- PWS Failure Indication Elementary Procedure
--
-- *****

```

```

--
-- PWS FAILURE INDICATION
--
-- *****

PWSFailureIndication ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {PWSFailureIndicationIES} },
    ...
}

PWSFailureIndicationIES NGAP-PROTOCOL-IES ::= {
    { ID id-PWSFailedCellIDList      CRITICALITY reject  TYPE PWSFailedCellIDList      PRESENCE mandatory }|
    { ID id-GlobalRANNodeID          CRITICALITY reject  TYPE GlobalRANNodeID          PRESENCE mandatory },
    ...
}

-- *****
--
-- NRPPA TRANSPORT ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- DOWNLINK UE ASSOCIATED NRPPA TRANSPORT
--
-- *****

DownlinkUEAssociatedNRPPaTransport ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {DownlinkUEAssociatedNRPPaTransportIES} },
    ...
}

DownlinkUEAssociatedNRPPaTransportIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject  TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject  TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RoutingID               CRITICALITY reject  TYPE RoutingID               PRESENCE mandatory }|
    { ID id-NRPPa-PDU               CRITICALITY reject  TYPE NRPPa-PDU               PRESENCE mandatory },
    ...
}

-- *****
--
-- UPLINK UE ASSOCIATED NRPPA TRANSPORT
--
-- *****

UplinkUEAssociatedNRPPaTransport ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {UplinkUEAssociatedNRPPaTransportIES} },
    ...
}

UplinkUEAssociatedNRPPaTransportIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject  TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|

```

```

    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RoutingID               CRITICALITY reject TYPE RoutingID             PRESENCE mandatory }|
    { ID id-NRPPa-PDU               CRITICALITY reject TYPE NRPPa-PDU             PRESENCE mandatory },
    ...
}

-- *****
--
-- DOWNLINK NON UE ASSOCIATED NRPPA TRANSPORT
--
-- *****

DownlinkNonUEAssociatedNRPPaTransport ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {DownlinkNonUEAssociatedNRPPaTransportIEs} },
    ...
}

DownlinkNonUEAssociatedNRPPaTransportIEs NGAP-PROTOCOL-IES ::= {
    { ID id-RoutingID          CRITICALITY reject TYPE RoutingID          PRESENCE mandatory }|
    { ID id-NRPPa-PDU          CRITICALITY reject TYPE NRPPa-PDU          PRESENCE mandatory },
    ...
}

-- *****
--
-- UPLINK NON UE ASSOCIATED NRPPA TRANSPORT
--
-- *****

UplinkNonUEAssociatedNRPPaTransport ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {UplinkNonUEAssociatedNRPPaTransportIEs} },
    ...
}

UplinkNonUEAssociatedNRPPaTransportIEs NGAP-PROTOCOL-IES ::= {
    { ID id-RoutingID          CRITICALITY reject TYPE RoutingID          PRESENCE mandatory }|
    { ID id-NRPPa-PDU          CRITICALITY reject TYPE NRPPa-PDU          PRESENCE mandatory },
    ...
}

-- *****
--
-- TRACE ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- TRACE START
--
-- *****

TraceStart ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {TraceStartIEs} },

```



```

    ...
}

TraceStartIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-TraceActivation          CRITICALITY ignore TYPE TraceActivation        PRESENCE mandatory },
    ...
}

-- *****
--
-- TRACE FAILURE INDICATION
--
-- *****

TraceFailureIndication ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {TraceFailureIndicationIES} },
    ...
}

TraceFailureIndicationIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-NGRANTraceID           CRITICALITY ignore TYPE NGRANTraceID           PRESENCE mandatory }|
    { ID id-Cause                  CRITICALITY ignore TYPE Cause                  PRESENCE mandatory },
    ...
}

-- *****
--
-- DEACTIVATE TRACE
--
-- *****

DeactivateTrace ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {DeactivateTraceIES} },
    ...
}

DeactivateTraceIES NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-NGRANTraceID           CRITICALITY ignore TYPE NGRANTraceID           PRESENCE mandatory },
    ...
}

-- *****
--
-- CELL TRAFFIC TRACE
--
-- *****

CellTrafficTrace ::= SEQUENCE {

```

```

    protocolIEs      ProtocolIE-Container      { {CellTrafficTraceIEs} },
    ...
}

CellTrafficTraceIEs NGAP-PROTOCOL-IES ::= {
    {ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    {ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    {ID id-NGRANTraceID            CRITICALITY ignore TYPE NGRANTraceID            PRESENCE mandatory }|
    {ID id-NGRAN-CGI               CRITICALITY ignore TYPE NGRAN-CGI               PRESENCE mandatory }|
    {ID id-TraceCollectionEntityIPAddress CRITICALITY ignore TYPE TransportLayerAddress PRESENCE mandatory }|
    {ID id-PrivacyIndicator         CRITICALITY ignore TYPE PrivacyIndicator         PRESENCE optional   }|
    {ID id-TraceCollectionEntityURI CRITICALITY ignore TYPE URI-address             PRESENCE optional   },
    ...
}

-- *****
--
-- LOCATION REPORTING ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- LOCATION REPORTING CONTROL
--
-- *****

LocationReportingControl ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {LocationReportingControlIEs} },
    ...
}

LocationReportingControlIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-LocationReportingRequestType CRITICALITY ignore TYPE LocationReportingRequestType PRESENCE mandatory },
    ...
}

-- *****
--
-- LOCATION REPORTING FAILURE INDICATION
--
-- *****

LocationReportingFailureIndication ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {LocationReportingFailureIndicationIEs} },
    ...
}

LocationReportingFailureIndicationIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-Cause                   CRITICALITY ignore TYPE Cause                   PRESENCE mandatory }|

```

```

    { ID id-Aerial-UE-FlightInformationOngoingReporting-failed
      PRESENCE optional},
    ...
  }

Aerial-UE-FlightInformationOngoingReporting-failed ::= ENUMERATED {
  true,
  ...
}

-- *****
--
-- LOCATION REPORT
-- *****

LocationReport ::= SEQUENCE {
  protocolIEs          ProtocolIE-Container      { {LocationReportIEs} },
  ...
}

LocationReportIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-UserLocationInformation CRITICALITY ignore TYPE UserLocationInformation PRESENCE mandatory }|
  { ID id-UEPresenceInAreaOfInterestList CRITICALITY ignore TYPE UEPresenceInAreaOfInterestList PRESENCE optional }|
  { ID id-LocationReportingRequestType CRITICALITY ignore TYPE LocationReportingRequestType PRESENCE mandatory },
  ...
}

-- *****
--
-- UE TNLA BINDING ELEMENTARY PROCEDURES
-- *****

-- *****
--
-- UE TNLA BINDING RELEASE REQUEST
-- *****

UETNLABindingReleaseRequest ::= SEQUENCE {
  protocolIEs          ProtocolIE-Container      { {UETNLABindingReleaseRequestIEs} },
  ...
}

UETNLABindingReleaseRequestIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory },
  ...
}

```

```

-- *****
--
-- UE RADIO CAPABILITY MANAGEMENT ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- UE RADIO CAPABILITY INFO INDICATION
--
-- *****

UERadioCapabilityInfoIndication ::= SEQUENCE {
    protocolIES          ProtocolIE-Container          { {UERadioCapabilityInfoIndicationIEs} },
    ...
}

UERadioCapabilityInfoIndicationIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-UERadioCapability        CRITICALITY ignore TYPE UERadioCapability        PRESENCE mandatory }|
    { ID id-UERadioCapabilityForPaging CRITICALITY ignore TYPE UERadioCapabilityForPaging PRESENCE optional }|
    { ID id-UERadioCapability-EUTRA-Format CRITICALITY ignore TYPE UERadioCapability PRESENCE optional }|
    { ID id-XrDeviceWith2Rx          CRITICALITY ignore TYPE XrDeviceWith2Rx          PRESENCE optional },
    ...
}

-- *****
--
-- UE Radio Capability Check Elementary Procedure
--
-- *****

-- *****
--
-- UE RADIO CAPABILITY CHECK REQUEST
--
-- *****

UERadioCapabilityCheckRequest ::= SEQUENCE {
    protocolIES          ProtocolIE-Container          { {UERadioCapabilityCheckRequestIEs} },
    ...
}

UERadioCapabilityCheckRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-UERadioCapability        CRITICALITY ignore TYPE UERadioCapability        PRESENCE optional }|
    { ID id-UERadioCapabilityID      CRITICALITY reject TYPE UERadioCapabilityID      PRESENCE optional },
    ...
}

-- *****
--

```

```

-- UE RADIO CAPABILITY CHECK RESPONSE
--
-- *****

UERadioCapabilityCheckResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {UERadioCapabilityCheckResponseIEs} },
    ...
}

UERadioCapabilityCheckResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-IMSVoiceSupportIndicator CRITICALITY reject TYPE IMSVoiceSupportIndicator PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics   CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional }|
    ...
}

-- *****
--
-- PRIVATE MESSAGE ELEMENTARY PROCEDURE
--
-- *****

-- *****
--
-- PRIVATE MESSAGE
--
-- *****

PrivateMessage ::= SEQUENCE {
    privateIEs      PrivateIE-Container      { { PrivateMessageIEs } },
    ...
}

PrivateMessageIEs NGAP-PRIVATE-IES ::= {
    ...
}

-- *****
--
-- DATA USAGE REPORTING ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- SECONDARY RAT DATA USAGE REPORT
--
-- *****

SecondaryRATDataUsageReport ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {SecondaryRATDataUsageReportIEs} },
    ...
}

```

```

}

SecondaryRATDataUsageReportIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore  TYPE AMF-UE-NGAP-ID          PRESENCE mandatory
  }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore  TYPE RAN-UE-NGAP-ID          PRESENCE mandatory
  }|
  { ID id-PDUSessionResourceSecondaryRATUsageList  CRITICALITY ignore  TYPE PDUSessionResourceSecondaryRATUsageList  PRESENCE mandatory
  }|
  { ID id-HandoverFlag            CRITICALITY ignore  TYPE HandoverFlag            PRESENCE optional
  }|
  { ID id-UserLocationInformation  CRITICALITY ignore  TYPE UserLocationInformation  PRESENCE optional
  },
  ...
}

-- *****
--
-- RIM INFORMATION TRANSFER ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- UPLINK RIM INFORMATION TRANSFER
--
-- *****

UplinkRIMInformationTransfer ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {UplinkRIMInformationTransferIEs} },
  ...
}

UplinkRIMInformationTransferIEs NGAP-PROTOCOL-IES ::= {
  { ID id-RIMInformationTransfer  CRITICALITY ignore  TYPE RIMInformationTransfer PRESENCE optional },
  ...
}

-- *****
--
-- DOWNLINK RIM INFORMATION TRANSFER
--
-- *****

DownlinkRIMInformationTransfer ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {DownlinkRIMInformationTransferIEs} },
  ...
}

DownlinkRIMInformationTransferIEs NGAP-PROTOCOL-IES ::= {
  { ID id-RIMInformationTransfer  CRITICALITY ignore  TYPE RIMInformationTransfer PRESENCE optional },
  ...
}

```

```

-- *****
--
-- Connection Establishment Indication
--
-- *****

ConnectionEstablishmentIndication ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container { {ConnectionEstablishmentIndicationIEs} },
    ...
}

ConnectionEstablishmentIndicationIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-UERadioCapability        CRITICALITY ignore TYPE UERadioCapability        PRESENCE optional }|
    { ID id-EndIndication            CRITICALITY ignore TYPE EndIndication            PRESENCE optional }|
    { ID id-S-NSSAI                  CRITICALITY ignore TYPE S-NSSAI                  PRESENCE optional }|
    { ID id-AllowedNSSAI             CRITICALITY ignore TYPE AllowedNSSAI             PRESENCE optional }|
    { ID id-UE-DifferentiationInfo    CRITICALITY ignore TYPE UE-DifferentiationInfo    PRESENCE optional }|
    { ID id-DL-CP-SecurityInformation CRITICALITY ignore TYPE DL-CP-SecurityInformation CRITICALITY ignore }|
    { ID id-NB-IoT-UEPriority         CRITICALITY ignore TYPE NB-IoT-UEPriority         PRESENCE optional }|
    { ID id-Enhanced-CoverageRestriction CRITICALITY ignore TYPE Enhanced-CoverageRestriction PRESENCE optional }|
    { ID id-CEmodeBrestricted         CRITICALITY ignore TYPE CEmodeBrestricted         PRESENCE optional }|
    { ID id-UERadioCapabilityID       CRITICALITY reject TYPE UERadioCapabilityID       PRESENCE optional }|
    { ID id-MaskedIMEISV              CRITICALITY ignore TYPE MaskedIMEISV              PRESENCE optional }|
    { ID id-OldAMF                   CRITICALITY reject TYPE AMFName                   PRESENCE optional }|
    { ID id-Partially-Allowed-NSSAI   CRITICALITY ignore TYPE Partially-Allowed-NSSAI   PRESENCE optional }|
    { ID id-ExtendedOldAMF            CRITICALITY ignore TYPE Extended-AMFName            PRESENCE optional }|
    ...
}

-- *****
--
-- UE RADIO CAPABILITY ID MAPPING ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- UE RADIO CAPABILITY ID MAPPING REQUEST
--
-- *****

UERadioCapabilityIDMappingRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container { {UERadioCapabilityIDMappingRequestIEs} },
    ...
}

UERadioCapabilityIDMappingRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-UERadioCapabilityID      CRITICALITY reject TYPE UERadioCapabilityID      PRESENCE mandatory },
    ...
}

```

```
-- *****
--
-- UE RADIO CAPABILITY ID MAPPING RESPONSE
--
-- *****

UERadioCapabilityIDMappingResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {UERadioCapabilityIDMappingResponseIEs} },
    ...
}

UERadioCapabilityIDMappingResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-UERadioCapabilityID          CRITICALITY reject TYPE UERadioCapabilityID          PRESENCE mandatory }|
    { ID id-UERadioCapability            CRITICALITY ignore TYPE UERadioCapability            PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics        CRITICALITY ignore TYPE CriticalityDiagnostics        PRESENCE optional   },
    ...
}

-- *****
--
-- AMF CP Relocation Indication
--
-- *****

AMFCPrelocationIndication ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container { { AMFCPrelocationIndicationIEs } },
    ...
}

AMFCPrelocationIndicationIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-S-NSSAI                 CRITICALITY ignore TYPE S-NSSAI                 PRESENCE optional   }| --this IE is not used and
ignored
    { ID id-AllowedNSSAI            CRITICALITY ignore TYPE AllowedNSSAI            PRESENCE optional   }| --this IE is not used and
ignored
    { ID id-Partially-Allowed-NSSAI CRITICALITY ignore TYPE Partially-Allowed-NSSAI PRESENCE optional   }, --this IE is not used and
ignored
    ...
}

-- *****
--
-- MBS SESSION MANAGEMENT ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- Broadcast Session Setup Elementary Procedure
--
-- *****
-- *****
```



```

--
-- BROADCAST SESSION SETUP REQUEST
--
-- *****

BroadcastSessionSetupRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionSetupRequestIEs} },
    ...
}

BroadcastSessionSetupRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID          CRITICALITY reject TYPE MBS-SessionID          PRESENCE mandatory }|
    { ID id-S-NSSAI                CRITICALITY reject TYPE S-NSSAI                PRESENCE mandatory }|
    { ID id-MBS-ServiceArea        CRITICALITY reject TYPE MBS-ServiceArea        PRESENCE mandatory }|
    { ID id-MBSSessionSetupRequestTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING MBSSessionSetupOrModRequestTransfer) PRESENCE mandatory }|
    { ID id-AssociatedSessionID    CRITICALITY ignore TYPE AssociatedSessionID    PRESENCE optional },
    ...
}

-- *****
--
-- BROADCAST SESSION SETUP RESPONSE
--
-- *****

BroadcastSessionSetupResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionSetupResponseIEs} },
    ...
}

BroadcastSessionSetupResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID          CRITICALITY reject TYPE MBS-SessionID          PRESENCE mandatory }|
    { ID id-MBSSessionSetupResponseTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING MBSSessionSetupOrModResponseTransfer) PRESENCE optional }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- BROADCAST SESSION SETUP FAILURE
--
-- *****

BroadcastSessionSetupFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionSetupFailureIEs} },
    ...
}

BroadcastSessionSetupFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID          CRITICALITY reject TYPE MBS-SessionID          PRESENCE mandatory }|
    { ID id-MBSSessionSetupFailureTransfer CRITICALITY ignore TYPE OCTET STRING (CONTAINING MBSSessionSetupOrModFailureTransfer) PRESENCE optional }|
}

```

```

    { ID id-Cause                                CRITICALITY ignore TYPE Cause                PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics                CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional   },
    ...
}

-- *****
--
-- Broadcast Session Modification Elementary Procedure
--
-- *****

-- *****
--
-- BROADCAST SESSION MODIFICATION REQUEST
--
-- *****

BroadcastSessionModificationRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionModificationRequestIEs} },
    ...
}

BroadcastSessionModificationRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                        CRITICALITY reject TYPE MBS-SessionID                PRESENCE mandatory   }|
    { ID id-MBS-ServiceArea                      CRITICALITY reject TYPE MBS-ServiceArea            PRESENCE optional   }|
    { ID id-MBSSessionModificationRequestTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING MBSSessionSetupOrModRequestTransfer) PRESENCE optional   },
    ...
}

-- *****
--
-- BROADCAST SESSION MODIFICATION RESPONSE
--
-- *****

BroadcastSessionModificationResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionModificationResponseIEs} },
    ...
}

BroadcastSessionModificationResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                        CRITICALITY reject TYPE MBS-SessionID                PRESENCE mandatory   }|
    { ID id-MBSSessionModificationResponseTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING MBSSessionSetupOrModResponseTransfer) PRESENCE optional   }|
    { ID id-CriticalityDiagnostics                CRITICALITY ignore TYPE CriticalityDiagnostics            PRESENCE optional   },
    ...
}

-- *****
--
-- BROADCAST SESSION MODIFICATION FAILURE
--
-- *****

```

```

BroadcastSessionModificationFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionModificationFailureIEs} },
    ...
}

BroadcastSessionModificationFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-MBSSessionModificationFailureTransfer  CRITICALITY ignore  TYPE OCTET STRING (CONTAINING MBSSessionSetupOrModFailureTransfer) PRESENCE
optional }|
    { ID id-Cause                        CRITICALITY ignore  TYPE Cause                        PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics        CRITICALITY ignore  TYPE CriticalityDiagnostics        PRESENCE optional }|
    ...
}

-- *****
--
-- Broadcast Session Release Elementary Procedure
--
-- *****

-- *****
--
-- BROADCAST SESSION RELEASE REQUEST
--
-- *****

BroadcastSessionReleaseRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionReleaseRequestIEs} },
    ...
}

BroadcastSessionReleaseRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-Cause                        CRITICALITY ignore  TYPE Cause                        PRESENCE mandatory }|
    ...
}

-- *****
--
-- Broadcast Session Release Required Elementary Procedure
--
-- *****

-- *****
--
-- BROADCAST SESSION RELEASE REQUIRED
--
-- *****

BroadcastSessionReleaseRequired ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionReleaseRequiredIEs} },
    ...
}

```

```

BroadcastSessionReleaseRequiredIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-Cause                        CRITICALITY ignore   TYPE Cause                        PRESENCE mandatory },
    ...
}

-- *****
--
-- BROADCAST SESSION RELEASE RESPONSE
-- *****

BroadcastSessionReleaseResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionReleaseResponseIEs} },
    ...
}

BroadcastSessionReleaseResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-MBSSessionReleaseResponseTransfer  CRITICALITY ignore   TYPE OCTET STRING (CONTAINING MBSSessionReleaseResponseTransfer)  PRESENCE optional }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore   TYPE CriticalityDiagnostics          PRESENCE optional },
    ...
}

-- *****
--
-- Broadcast Session Transport Elementary Procedure
-- *****

-- *****
--
-- BROADCAST SESSION TRANSPORT REQUEST
-- *****

BroadcastSessionTransportRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionTransportRequestIEs} },
    ...
}

BroadcastSessionTransportRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-BroadcastTransportRequestTransfer  CRITICALITY reject  TYPE OCTET STRING (CONTAINING BroadcastTransportRequestTransfer)  PRESENCE mandatory },
    ...
}

-- *****
--
-- BROADCAST SESSION TRANSPORT RESPONSE

```

```

--
-- *****

BroadcastSessionTransportResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionTransportResponseIEs} },
    ...
}

BroadcastSessionTransportResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID          CRITICALITY reject  TYPE MBS-SessionID          PRESENCE mandatory   }|
    { ID id-BroadcastTransportResponseTransfer  CRITICALITY reject  TYPE OCTET STRING (CONTAINING BroadcastTransportResponseTransfer) PRESENCE
mandatory   }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics  PRESENCE optional   },
    ...
}

-- *****
--
-- BROADCAST SESSION TRANSPORT FAILURE
--
-- *****

BroadcastSessionTransportFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {BroadcastSessionTransportFailureIEs} },
    ...
}

BroadcastSessionTransportFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID          CRITICALITY reject  TYPE MBS-SessionID          PRESENCE mandatory   }|
    { ID id-BroadcastTransportFailureTransfer  CRITICALITY ignore  TYPE OCTET STRING (CONTAINING BroadcastTransportFailureTransfer) PRESENCE
mandatory   }|
    { ID id-Cause                  CRITICALITY ignore  TYPE Cause                  PRESENCE mandatory   }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics  PRESENCE optional   },
    ...
}

-- *****
--
-- Distribution Setup Elementary Procedure
--
-- *****

-- *****
--
-- DISTRIBUTION SETUP REQUEST
--
-- *****

DistributionSetupRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {DistributionSetupRequestIEs} },
    ...
}

```

```

DistributionSetupRequestIES NGAP-PROTOCOL-IES ::= {
  { ID id-MBS-SessionID                CRITICALITY reject TYPE MBS-SessionID                PRESENCE mandatory }|
  { ID id-MBS-AreaSessionID            CRITICALITY reject TYPE MBS-AreaSessionID            PRESENCE optional   }|
  { ID id-MBS-DistributionSetupRequestTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING MBS-DistributionSetupRequestTransfer) PRESENCE mandatory },
  ...
}

-- *****
--
-- DISTRIBUTION SETUP RESPONSE
--
-- *****

DistributionSetupResponse ::= SEQUENCE {
  protocolIES      ProtocolIE-Container      { {DistributionSetupResponseIES} },
  ...
}

DistributionSetupResponseIES NGAP-PROTOCOL-IES ::= {
  { ID id-MBS-SessionID                CRITICALITY reject TYPE MBS-SessionID                PRESENCE mandatory }|
  { ID id-MBS-AreaSessionID            CRITICALITY reject TYPE MBS-AreaSessionID            PRESENCE optional   }|
  { ID id-MBS-DistributionSetupResponseTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING MBS-DistributionSetupResponseTransfer) PRESENCE mandatory },
  { ID id-CriticalityDiagnostics        CRITICALITY ignore TYPE CriticalityDiagnostics        PRESENCE optional   },
  ...
}

-- *****
--
-- DISTRIBUTION SETUP FAILURE
--
-- *****

DistributionSetupFailure ::= SEQUENCE {
  protocolIES      ProtocolIE-Container      { {DistributionSetupFailureIES} },
  ...
}

DistributionSetupFailureIES NGAP-PROTOCOL-IES ::= {
  { ID id-MBS-SessionID                CRITICALITY reject TYPE MBS-SessionID                PRESENCE mandatory }|
  { ID id-MBS-AreaSessionID            CRITICALITY reject TYPE MBS-AreaSessionID            PRESENCE optional   }|
  { ID id-MBS-DistributionSetupUnsuccessfulTransfer CRITICALITY ignore TYPE OCTET STRING (CONTAINING MBS-
DistributionSetupUnsuccessfulTransfer) PRESENCE mandatory }|
  { ID id-Cause                        CRITICALITY ignore TYPE Cause                        PRESENCE mandatory }|
  { ID id-CriticalityDiagnostics        CRITICALITY ignore TYPE CriticalityDiagnostics        PRESENCE optional   },
  ...
}

-- *****
--
-- Distribution Release Elementary Procedure
--

```

```

-- *****
-- *****
--
-- DISTRIBUTION RELEASE REQUEST
--
-- *****

DistributionReleaseRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {DistributionReleaseRequestIEs} },
    ...
}

DistributionReleaseRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-MBS-AreaSessionID            CRITICALITY reject  TYPE MBS-AreaSessionID            PRESENCE optional   }|
    { ID id-MBS-DistributionReleaseRequestTransfer  CRITICALITY reject  TYPE OCTET STRING (CONTAINING MBS-DistributionReleaseRequestTransfer) PRESENCE mandatory }|
    { ID id-Cause                        CRITICALITY ignore   TYPE Cause                        PRESENCE mandatory },
    ...
}

-- *****
--
-- DISTRIBUTION RELEASE RESPONSE
--
-- *****

DistributionReleaseResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {DistributionReleaseResponseIEs} },
    ...
}

DistributionReleaseResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-MBS-AreaSessionID            CRITICALITY reject  TYPE MBS-AreaSessionID            PRESENCE optional   }|
    { ID id-CriticalityDiagnostics        CRITICALITY ignore   TYPE CriticalityDiagnostics        PRESENCE optional   },
    ...
}

-- *****
--
-- Multicast Session Activation Elementary Procedure
--
-- *****

-- *****
--
-- MULTICAST SESSION ACTIVATION REQUEST
--
-- *****

MulticastSessionActivationRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {MulticastSessionActivationRequestIEs} },

```

```

    ...
}

MulticastSessionActivationRequestIES NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-MulticastSessionActivationRequestTransfer  CRITICALITY reject  TYPE OCTET STRING (CONTAINING
MulticastSessionActivationRequestTransfer) PRESENCE mandatory },
    ...
}

-- *****
--
-- MULTICAST SESSION ACTIVATION RESPONSE
--
-- *****

MulticastSessionActivationResponse ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {MulticastSessionActivationResponseIES} },
    ...
}

MulticastSessionActivationResponseIES NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics        CRITICALITY ignore  TYPE CriticalityDiagnostics        PRESENCE optional   },
    ...
}

-- *****
--
-- MULTICAST SESSION ACTIVATION FAILURE
--
-- *****

MulticastSessionActivationFailure ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {MulticastSessionActivationFailureIES} },
    ...
}

MulticastSessionActivationFailureIES NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-Cause                        CRITICALITY ignore  TYPE Cause                        PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics        CRITICALITY ignore  TYPE CriticalityDiagnostics        PRESENCE optional   },
    ...
}

-- *****
--
-- Multicast Session Deactivation Elementary Procedure
--
-- *****

```



```

-- *****
--
-- MULTICAST SESSION DEACTIVATION REQUEST
--
-- *****

MulticastSessionDeactivationRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {MulticastSessionDeactivationRequestIEs} },
    ...
}

MulticastSessionDeactivationRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory }|
    { ID id-MulticastSessionDeactivationRequestTransfer CRITICALITY reject  TYPE OCTET STRING (CONTAINING
MulticastSessionDeactivationRequestTransfer) PRESENCE mandatory },
    ...
}

-- *****
--
-- MULTICAST SESSION DEACTIVATION RESPONSE
--
-- *****

MulticastSessionDeactivationResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {MulticastSessionDeactivationResponseIEs} },
    ...
}

MulticastSessionDeactivationResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID                CRITICALITY reject  TYPE MBS-SessionID                PRESENCE mandatory
    }|
    { ID id-CriticalityDiagnostics        CRITICALITY ignore  TYPE CriticalityDiagnostics        PRESENCE optional
    },
    ...
}

-- *****
--
-- Multicast Session Update Elementary Procedure
--
-- *****

-- *****
--
-- MULTICAST SESSION UPDATE REQUEST
--
-- *****

MulticastSessionUpdateRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {MulticastSessionUpdateRequestIEs} },
    ...
}

```

```

MulticastSessionUpdateRequestIEs NGAP-PROTOCOL-IES ::= {
  { ID id-MBS-SessionID                               CRITICALITY reject TYPE MBS-SessionID           PRESENCE mandatory }|
  { ID id-MBS-AreaSessionID                             CRITICALITY reject TYPE MBS-AreaSessionID       PRESENCE optional }|
  { ID id-MulticastSessionUpdateRequestTransfer         CRITICALITY reject TYPE OCTET STRING (CONTAINING MulticastSessionUpdateRequestTransfer) PRESENCE mandatory },
  ...
}

-- *****
--
-- MULTICAST SESSION UPDATE RESPONSE
--
-- *****

MulticastSessionUpdateResponse ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {MulticastSessionUpdateResponseIEs} },
  ...
}

MulticastSessionUpdateResponseIEs NGAP-PROTOCOL-IES ::= {
  { ID id-MBS-SessionID                               CRITICALITY reject TYPE MBS-SessionID           PRESENCE mandatory }|
  { ID id-MBS-AreaSessionID                             CRITICALITY reject TYPE MBS-AreaSessionID       PRESENCE optional }|
  { ID id-CriticalityDiagnostics                         CRITICALITY ignore TYPE CriticalityDiagnostics   PRESENCE optional },
  ...
}

-- *****
--
-- MULTICAST SESSION UPDATE FAILURE
--
-- *****

MulticastSessionUpdateFailure ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {MulticastSessionUpdateFailureIEs} },
  ...
}

MulticastSessionUpdateFailureIEs NGAP-PROTOCOL-IES ::= {
  { ID id-MBS-SessionID                               CRITICALITY reject TYPE MBS-SessionID           PRESENCE mandatory }|
  { ID id-MBS-AreaSessionID                             CRITICALITY reject TYPE MBS-AreaSessionID       PRESENCE optional }|
  { ID id-Cause                                           CRITICALITY ignore TYPE Cause                     PRESENCE mandatory }|
  { ID id-CriticalityDiagnostics                         CRITICALITY ignore TYPE CriticalityDiagnostics   PRESENCE optional },
  ...
}

-- *****
--
-- MULTICAST GROUP PAGING ELEMENTARY PROCEDURE
--
-- *****

```

```

-- *****
--
-- MULTICAST GROUP PAGING
--
-- *****

MulticastGroupPaging ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {MulticastGroupPagingIES} },
    ...
}

MulticastGroupPagingIES NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID          CRITICALITY ignore TYPE MBS-SessionID          PRESENCE mandatory }|
    { ID id-MBS-ServiceArea        CRITICALITY ignore TYPE MBS-ServiceArea        PRESENCE optional   }|
    { ID id-MulticastGroupPagingAreaList CRITICALITY ignore TYPE MulticastGroupPagingAreaList PRESENCE mandatory },
    ...
}

-- *****
--
-- TIMING SYNCHRONISATION STATUS REPORTING ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- Timing Synchronisation Status Elementary Procedure
--
-- *****

-- *****
--
-- TIMING SYNCHRONISATION STATUS REQUEST
--
-- *****

TimingSynchronisationStatusRequest ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {TimingSynchronisationStatusRequestIES} },
    ...
}

TimingSynchronisationStatusRequestIES NGAP-PROTOCOL-IES ::= {
    { ID id-RoutingID              CRITICALITY reject TYPE RoutingID              PRESENCE mandatory }|
    { ID id-RAN-TSSRequestType     CRITICALITY reject TYPE RAN-TSSRequestType     PRESENCE mandatory },
    ...
}

-- *****
--
-- TIMING SYNCHRONISATION STATUS RESPONSE
--
-- *****

```

```

TimingSynchronisationStatusResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {TimingSynchronisationStatusResponseIEs} },
    ...
}

TimingSynchronisationStatusResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-RoutingID          CRITICALITY ignore TYPE RoutingID          PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- TIMING SYNCHRONISATION STATUS FAILURE
--
-- *****

TimingSynchronisationStatusFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { { TimingSynchronisationStatusFailureIEs} },
    ...
}

TimingSynchronisationStatusFailureIEs NGAP-PROTOCOL-IES ::= {
    { ID id-RoutingID          CRITICALITY ignore TYPE RoutingID          PRESENCE mandatory }|
    { ID id-Cause              CRITICALITY ignore TYPE Cause              PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
    ...
}

-- *****
--
-- Timing Synchronisation Status Reporting Elementary Procedure
--
-- *****

-- *****
--
-- TIMING SYNCHRONISATION STATUS REPORT
--
-- *****

TimingSynchronisationStatusReport ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {TimingSynchronisationStatusReportIEs} },
    ...
}

TimingSynchronisationStatusReportIEs NGAP-PROTOCOL-IES ::= {
    { ID id-RoutingID          CRITICALITY ignore TYPE RoutingID          PRESENCE mandatory }|
    { ID id-RANTimingSynchronisationStatusInfo CRITICALITY ignore TYPE RANTimingSynchronisationStatusInfo PRESENCE mandatory }|
    { ID id-RAN-TSSScope       CRITICALITY ignore TYPE RAN-TSSScope       PRESENCE mandatory },
    ...
}

```

```

-- *****
--
-- MT Communication Handling Elementary Procedure
--
-- *****

-- *****
--
-- MT COMMUNICATION HANDLING REQUEST
--
-- *****

MTCommunicationHandlingRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { { MTCommunicationHandlingRequestIEs} },
    ...
}

MTCommunicationHandlingRequestIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-FiveGCAction            CRITICALITY reject TYPE FiveGCAction            PRESENCE mandatory }|
    { ID id-PDUSessionListMTCommHReq CRITICALITY reject TYPE PDUSessionListMTCommHReq PRESENCE mandatory },
    ...
}

-- *****
--
-- MT COMMUNICATION HANDLING RESPONSE
--
-- *****

MTCommunicationHandlingResponse ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {MTCommunicationHandlingResponseIEs} },
    ...
}

MTCommunicationHandlingResponseIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics   CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional   },
    ...
}

-- *****
--
-- MT COMMUNICATION HANDLING FAILURE
--
-- *****

MTCommunicationHandlingFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { { MTCommunicationHandlingFailureIEs} },
    ...
}

```

```

MTCommunicationHandlingFailureIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY reject TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY reject TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-Cause                   CRITICALITY ignore TYPE Cause                   PRESENCE mandatory }|
  { ID id-CriticalityDiagnostics   CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE optional },
  ...
}

-- *****

-- RAN Paging Request procedure
--
-- *****

-- *****
--
-- RAN PAGING REQUEST
--
-- *****

RANPagingRequest ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { { RANPagingRequestIEs } },
  ...
}

RANPagingRequestIEs NGAP-PROTOCOL-IES ::= {
  { ID id-AMF-UE-NGAP-ID          CRITICALITY ignore TYPE AMF-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-RAN-UE-NGAP-ID          CRITICALITY ignore TYPE RAN-UE-NGAP-ID          PRESENCE mandatory }|
  { ID id-PagingPolicyDifferentiation CRITICALITY ignore TYPE PagingPolicyDifferentiation PRESENCE optional }|
  { ID id-DL-Signalling           CRITICALITY ignore TYPE DL-Signalling           PRESENCE optional },
  ...
}

-- *****

--
-- A-IoT ELEMENTARY PROCEDURES
--
-- *****

-- *****
--
-- Inventory Request Elementary Procedure
--
-- *****

-- *****
--
-- INVENTORY REQUEST
--
-- *****

InventoryRequest ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {InventoryRequestIEs} },

```

```

    ...
}

InventoryRequestIES NGAP-PROTOCOL-IES ::= {
    { ID id-AIOTFIdentifier          CRITICALITY reject TYPE AIOTFIdentifier          PRESENCE mandatory }|
    { ID id-AIoT-CorrelationIdentifier CRITICALITY reject TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
    { ID id-InventoryRequestTransfer  CRITICALITY reject TYPE OCTET STRING (CONTAINING InventoryRequestTransfer) PRESENCE mandatory },
    ...
}

-- *****
--
-- INVENTORY RESPONSE
--
-- *****

InventoryResponse ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {InventoryResponseIES} },
    ...
}

InventoryResponseIES NGAP-PROTOCOL-IES ::= {
    { ID id-AIOTFIdentifier          CRITICALITY reject TYPE AIOTFIdentifier          PRESENCE mandatory }|
    { ID id-AIoT-CorrelationIdentifier CRITICALITY reject TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
    { ID id-InventoryResponseTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING InventoryResponseTransfer) PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics    CRITICALITY ignore TYPE CriticalityDiagnostics    PRESENCE optional },
    ...
}

-- *****
--
-- INVENTORY FAILURE
--
-- *****

InventoryFailure ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {InventoryFailureIES} },
    ...
}

InventoryFailureIES NGAP-PROTOCOL-IES ::= {
    { ID id-AIOTFIdentifier          CRITICALITY ignore TYPE AIOTFIdentifier          PRESENCE mandatory }|
    { ID id-AIoT-CorrelationIdentifier CRITICALITY ignore TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
    { ID id-InventoryFailureTransfer  CRITICALITY ignore TYPE OCTET STRING (CONTAINING InventoryFailureTransfer ) PRESENCE mandatory }|
    { ID id-Cause                     CRITICALITY ignore TYPE Cause                     PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics     CRITICALITY ignore TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}

-- *****

-- Inventory Report procedure
--
-- *****

```

```

-- *****
--
-- INVENTORY REPORT
--
-- *****

InventoryReport ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { { InventoryReportIES} },
    ...
}

InventoryReportIES NGAP-PROTOCOL-IES ::= {
    { ID id-AIoTIdentifier          CRITICALITY reject TYPE AIOTIdentifier          PRESENCE mandatory }|
    { ID id-AIoT-CorrelationIdentifier CRITICALITY reject TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
    { ID id-InventoryReportTransfer  CRITICALITY reject TYPE OCTET STRING (CONTAINING InventoryReportTransfer ) PRESENCE mandatory },
    ...
}

-- *****
--
-- Command Request Elementary Procedure
--
-- *****

-- *****
--
-- COMMAND REQUEST
--
-- *****

CommandRequest ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {CommandRequestIES} },
    ...
}

CommandRequestIES NGAP-PROTOCOL-IES ::= {
    { ID id-AIoTIdentifier          CRITICALITY reject TYPE AIOTIdentifier          PRESENCE mandatory }|
    { ID id-AIoT-CorrelationIdentifier CRITICALITY reject TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
    { ID id-RAN-AIoT-Device-NGAP-ID  CRITICALITY reject TYPE RAN-AIoT-Device-NGAP-ID PRESENCE mandatory }|
    { ID id-CommandRequestTransfer  CRITICALITY reject TYPE OCTET STRING (CONTAINING CommandRequestTransfer) PRESENCE mandatory },
    ...
}

-- *****
--
-- COMMAND RESPONSE
--
-- *****

CommandResponse ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {CommandResponseIES} },
    ...
}

```



```

CommandResponseIES NGAP-PROTOCOL-IES ::= {
  { ID id-AIOTFIdentifier          CRITICALITY reject TYPE AIOTFIdentifier          PRESENCE mandatory }|
  { ID id-AIoT-CorrelationIdentifier CRITICALITY reject TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
  { ID id-RAN-AIoT-Device-NGAP-ID   CRITICALITY reject TYPE RAN-AIoT-Device-NGAP-ID   PRESENCE mandatory }|
  { ID id-CommandResponseTransfer  CRITICALITY reject TYPE OCTET STRING (CONTAINING CommandResponseTransfer) PRESENCE mandatory }|
  { ID id-CriticalityDiagnostics    CRITICALITY ignore TYPE CriticalityDiagnostics    PRESENCE optional   },
  ...
}

-- *****
--
-- COMMAND FAILURE
--
-- *****

CommandFailure ::= SEQUENCE {
  protocolIES      ProtocolIE-Container      { {CommandFailureIES} },
  ...
}

CommandFailureIES NGAP-PROTOCOL-IES ::= {
  { ID id-AIOTFIdentifier          CRITICALITY ignore TYPE AIOTFIdentifier          PRESENCE mandatory }|
  { ID id-AIoT-CorrelationIdentifier CRITICALITY ignore TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
  { ID id-RAN-AIoT-Device-NGAP-ID   CRITICALITY ignore TYPE RAN-AIoT-Device-NGAP-ID   PRESENCE mandatory }|
  { ID id-CommandFailureTransfer    CRITICALITY ignore TYPE OCTET STRING (CONTAINING CommandFailureTransfer ) PRESENCE mandatory }|
  { ID id-Cause                     CRITICALITY ignore TYPE Cause                     PRESENCE mandatory }|
  { ID id-CriticalityDiagnostics     CRITICALITY ignore TYPE CriticalityDiagnostics     PRESENCE optional   },
  ...
}

-- *****
--
-- A-IoT Session Release Elementary Procedure
--
-- *****

-- *****
--
-- AIOT SESSION RELEASE COMMAND
--
-- *****

AIOTSessionReleaseCommand ::= SEQUENCE {
  protocolIES      ProtocolIE-Container      { {AIOTSessionReleaseCommandIES} },
  ...
}

AIOTSessionReleaseCommandIES NGAP-PROTOCOL-IES ::= {
  { ID id-AIOTFIdentifier          CRITICALITY reject TYPE AIOTFIdentifier          PRESENCE mandatory }|
  { ID id-AIoT-CorrelationIdentifier CRITICALITY reject TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
  { ID id-AIoTSessionReleaseCommandTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING AIOTSessionReleaseCommandTransfer) PRESENCE
mandatory },
  ...
}

```

```

}

-- *****
--
-- AIOT SESSION RELEASE COMPLETE
--
-- *****

AIOTSessionReleaseComplete ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {AIOTSessionReleaseCompleteIES} },
    ...
}

AIOTSessionReleaseCompleteIES NGAP-PROTOCOL-IES ::= {
    { ID id-AIOTFIdentifier          CRITICALITY ignore TYPE AIOTFIdentifier          PRESENCE mandatory }|
    { ID id-AIoT-CorrelationIdentifier CRITICALITY ignore TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
    { ID id-AIOTSessionReleaseCompleteTransfer CRITICALITY ignore TYPE OCTET STRING (CONTAINING AIOTSessionReleaseCompleteTransfer) PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore TYPE CriticalityDiagnostics      PRESENCE optional },
    ...
}

-- *****
--
-- A-IoT Session Release Request Elementary Procedure
--
-- *****

-- *****
--
-- AIOT SESSION RELEASE REQUEST
--
-- *****

AIOTSessionReleaseRequest ::= SEQUENCE {
    protocolIES      ProtocolIE-Container      { {AIOTSessionReleaseRequestIES} },
    ...
}

AIOTSessionReleaseRequestIES NGAP-PROTOCOL-IES ::= {
    { ID id-AIOTFIdentifier          CRITICALITY reject TYPE AIOTFIdentifier          PRESENCE mandatory }|
    { ID id-AIoT-CorrelationIdentifier CRITICALITY reject TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
    { ID id-AIOTSessionReleaseRequestTransfer CRITICALITY reject TYPE OCTET STRING (CONTAINING AIOTSessionReleaseRequestTransfer) PRESENCE mandatory },
    ...
}

END
-- ASN1STOP

```

9.4.5 Information Element Definitions

```
-- ASN1START
```

```
-- *****
--
-- Information Element Definitions
--
-- *****

NGAP-IEs {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-Access (22) modules (3) ngap (1) version1 (1) ngap-IEs (2) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

IMPORTS

    id-AdditionalDLForwardingUPTNLInformation,
    id-AdditionalULForwardingUPTNLInformation,
    id-AdditionalDLQoSFlowPerTNLInformation,
    id-AdditionalDLUPTNLInformationForHOList,
    id-AdditionalNGU-UP-TNLInformation,
    id-AdditionalRedundantDL-NGU-UP-TNLInformation,
    id-AdditionalRedundantDLQoSFlowPerTNLInformation,
    id-AdditionalRedundantNGU-UP-TNLInformation,
    id-AdditionalRedundantUL-NGU-UP-TNLInformation,
    id-AdditionalUL-NGU-UP-TNLInformation,
    id-AIOTFIdentifier,
    id-AIoT-CorrelationIdentifier,
    id-AIoT-DeviceIdentificationRequested,
    id-AIoT-SecurityInfo,
    id-AIoT-RequestedServiceAreaInformation,
    id-AIoT-InventoryAssistanceInformation,
    id-AIoT-FollowonCommandIndication,
    id-RAN-AIoT-Device-NGAP-ID,
    id-AIoT-CommandAssistanceInformation,
    id-AIoT-NASPDU,
    id-AlternativeQoSParaSetList,
    id-AssistanceInformationQoE-Meas,
    id-AdditionalCancelledlocationReportingReferenceIDList,
    id-BurstArrivalTimeDownlink,
    id-Cause,
    id-CNPacketDelayBudgetDL,
    id-CNPacketDelayBudgetUL,
    id-CNTypeRestrictionsForEquivalent,
    id-CNTypeRestrictionsForServing,
    id-CommonNetworkInstance,
    id-ConfiguredTACIndication,
    id-CN-MT-CommunicationHandling,
    id-CurrentQoSParaSetIndex,
    id-DAPSRequestInfo,
    id-DAPSResponseInfoList,
    id-DataForwardingNotPossible,
    id-DataForwardingResponseERABList,
    id-DirectForwardingPathAvailability,
```

id-DL-NGU-UP-TNLInformation,
id-DownlinkTLCContainer,
id-EndpointIPAddressAndPort,
id-EnergySavingIndication,
id-ExtendedMobilityInformation,
id-ExtendedPacketDelayBudget,
id-ExtendedRATRestrictionInformation,
id-ExtendedReportIntervalMDT,
id-ExtendedSliceSupportList,
id-ExtendedTAISliceSupportList,
id-ExtendedUEIdentityIndexValue,
id-EUTRA-PagingDRXInformation,
id-EquivalentSNPNsList,
id-FurtherExtendedUEIdentityIndexValue,
id-GeographyBasedMDT,
id-GlobalCable-ID,
id-GlobalRANNodeID,
id-GlobalTNGF-ID,
id-GlobalTWIF-ID,
id-GlobalW-AGF-ID,
id-GUAMIType,
id-HashedUEIdentityIndexValue,
id-IncludeBeamMeasurementsIndication,
id-IntersystemSONInformationRequest,
id-IntersystemSONInformationReply,
id-IntersystemResourceStatusUpdate,
id-IntersystemMobilityFailureforVoiceFallback,
id-LastEUTRAN-PLMNIdentity,
id-LastVisitedPSCellList,
id-LocationReportingAdditionalInfo,
id-LPWUSPSAssistanceInformation,
id-LPWUSDisableIndication,
id-M4ReportAmount,
id-M5ReportAmount,
id-M6ReportAmount,
id-ExcessPacketDelayThresholdConfiguration,
id-M7ReportAmount,
id-MaximumIntegrityProtectedDataRate-DL,
id-MBS-AreaSessionID,
id-MBS-QoSFlowsToBeSetupList,
id-MBS-QoSFlowsToBeSetupModList,
id-MBS-QoSFlowToReleaseList,
id-MBS-ServiceArea,
id-MBS-SessionFSAIDList,
id-MBS-SessionID,
id-MBS-ActiveSessionInformation-SourceToTargetList,
id-MBS-ActiveSessionInformation-TargetToSourceList,
id-MBS-AssistanceInformation,
id-MBS-IntendedServiceAreaList,
id-MBS-SessionTNLInfo5GC,
id-MBS-SupportIndicator,
id-MBSSessionFailedtoSetupList,
id-MBSSessionFailedtoSetuporModifyList,
id-MBSSessionSetupResponseList,

id-MBSSessionSetuporModifyResponseList,
id-MBSSessionToReleaseList,
id-MBSSessionSetupRequestList,
id-MBSSessionSetuporModifyRequestList,
id-MDTConfiguration,
id-MicoAllPLMN,
id-NetworkInstance,
id-NetworkSliceAreaScopeofMDT,
id-NGAPIESupportInformationRequestList,
id-NGAPIESupportInformationResponseList,
id-NID,
id-NR-CGI,
id-NRNTNTAIInformation,
id-NPN-MobilityInformation,
id-NPN-PagingAssistanceInformation,
id-NPN-Support,
id-NR-PagingeDRXInformation,
id-OldAssociatedQosFlowList-ULendmarkerexpected,
id-OnboardingSupport,
id-PagingAssisDataforCEcapabUE,
id-PagingCauseIndicationForVoiceService,
id-PDUSessionAggregateMaximumBitRate,
id-PduSessionExpectedUEActivityBehaviour,
id-PDUSessionPairID,
id-PDUSessionResourceFailedToSetupListCxtFail,
id-PDUSessionResourceReleaseResponseTransfer,
id-PDUSessionType,
id-PEIPSassistanceInformation,
id-PSCellInformation,
id-PNI-NPNBasedMDT,
id-PNI-NPN-AreaScopeofMDT,
id-QMCConfigInfo,
id-QosFlowAdditionalInfoList,
id-QosFlowAddOrModifyRequestList,
id-QosFlowFailedToSetupList,
id-QosFlowFeedbackList,
id-QosFlowParametersList,
id-QosFlowSetupRequestList,
id-QosFlowToReleaseList,
id-QosMonitoringRequest,
id-QosMonitoringReportingFrequency,
id-SNPN-CellBasedMDT,
id-SNPN-TAIBasedMDT,
id-SNPN-BasedMDT,
id-SuccessfulHandoverReportList,
id-SupportedUETypeList,
id-UEContextReferenceAtSource,
id-RAT-Information,
id-RedundantCommonNetworkInstance,
id-RedundantDL-NGU-TNLInformationReused,
id-RedundantDL-NGU-UP-TNLInformation,
id-RedundantDLQosFlowPerTNLInformation,
id-RedundantPDUSessionInformation,
id-RedundantQosFlowIndicator,

id-RedundantUL-NGU-UP-TNLInformation,
id-SCTP-TLAs,
id-SecondaryRATUsageInformation,
id-SecurityIndication,
id-SecurityResult,
id-SgNB-UE-X2AP-ID,
id-S-NSSAI,
id-SONInformationReport,
id-SourceNodeID,
id-SourceNodeTNLAddrInfo,
id-SourceSN-to-TargetSN-QMCInfo,
id-SourceTNLAddrInfo,
id-SurvivalTime,
id-Selected-Target-SNP-Identity,
id-TNLAssociationTransportLayerAddressNGRAN,
id-TAIBSSupportList,
id-TAINSAGSupportList,
id-TargetHomeENB-ID,
id-TargetRNC-ID,
id-TimeBasedHandoverInformation,
id-TraceCollectionEntityURI,
id-TSCTrafficCharacteristics,
id-UEHistoryInformationFromTheUE,
id-UERadioCapabilityForPaging,
id-UERadioCapabilityForPagingOfNB-IoT,
id-UL-NGU-UP-TNLInformation,
id-UL-NGU-UP-TNLModifyList,
id-ULForwarding,
id-ULForwardingUP-TNLInformation,
id-UplinkTLContainer,
id-UsedRSNInformation,
id-UserLocationInformationTNGF,
id-UserLocationInformationTWIF,
id-UserLocationInformationW-AGF,
id-UserPlaneErrorIndicator,
id-EarlyMeasurement,
id-BeamMeasurementsReportConfiguration,
id-DLDiscarding,
id-TAI,
id-HFCNode-ID-new,
id-GlobalCable-ID-new,
id-FiveGProSeLayer2Multipath,
id-CandidateRelayUEInformationList,
id-FiveGProSeLayer2UEtoUERelay,
id-FiveGProSeLayer2UEtoUERemote,
id-SuccessfulPSCellChangeReportList,
id-TargetCellCRNTI,
id-TimeSinceFailure,
id-ClockQualityReportingControlInfo,
id-RANfeedbacktype,
id-QoSFlowTSCList,
id-TSCTrafficCharacteristicsFeedback,
id-ANPacketDelayBudgetUL,
id-CommServiceType,

id-MobileIAB-MTUserLocationInformation,
id-PDUsSetQoSParameters,
id-PDUsSetbasedHandlingIndicator,
id-N6JitterInformation,
id-ECNMarkingorCongestionInformationReportingRequest,
id-ECNMarkingorCongestionInformationReportingStatus,
id-MN-only-MDT-collection,
id-XrDeviceWith2Rx,
id-MaximumDataBurstVolume,
id-MBS-NGUFailureIndication,
id-UserPlaneFailureIndication,
id-UserPlaneFailureIndicationReport,
id-QoERVQoEReportingPaths,
id-UserLocationInformationN3IWF-without-PortNumber,
id-ExtendedBackupAMFName,
id-MBS-UP-FailureIndication,
id-RequestedTNLInfo,
id-FiveGProSeLayer2MHUEtoNetworkRelay,
id-FiveGProSeLayer2MHIntermediateUEtoNetworkRelay,
id-FiveGProSeLayer2MHRemote,
id-FiveGProSeLayer3MHUEtoNetworkRelay,
id-AdditionalULI,
id-PduSetDelayBudgetDownlink,
id-PduSetDelayBudgetUplink,
id-PduSetErrorRateDownlink,
id-PduSetErrorRateUplink,
id-DLDPDUsSetInformationMarkingSupportIndication,
id-MonitoringRequestonAvailableBitrate,
id-MMSID,
id-Indication-of-Bitrate-Adaptation,
id-SCGActivationTime,
id-Aerial-UE-FlightInformationReporting,
id-Aerial-UE-FlightInformationReportingControllist,
maxnoofAIoTAreas,
maxnoofAllowedAreas,
maxnoofAllowedCAGsperPLMN,
maxnoofAllowedS-NSSAIs,
maxnoofAreaNTN,
maxnoofAoIMinusOne,
maxnoofBluetoothName,
maxnoofBPLMNs,
maxnoofCAGforMDT,
maxnoofCAGsperCell,
maxnoofCandidateCells,
maxnoofCellIDforMDT,
maxnoofCellIDforQMC,
maxnoofCellIDforWarning,
maxnoofCellinAoI,
maxnoofCellinEAI,
maxnoofCellsforMBS,
maxnoofCellsingNB,
maxnoofCellsinngeNB,
maxnoofCellsinNGRANNode,
maxnoofCellinTAI,

maxnoofCellsInUEHistoryInfo,
maxnoofCellsUEMovingTrajectory,
maxnoofDevices,
maxnoofDRBs,
maxnoofEmergencyAreaID,
maxnoofEAIforRestart,
maxnoofEPLMNs,
maxnoofEPLMNsPlusOne,
maxnoofE-RABs,
maxnoofErrors,
maxnoofExtSliceItems,
maxnoofESNPns,
maxnoofForbTACs,
maxnoofFreqforMDT,
maxnoofGUAMIs,
maxnoofMBSFSAs,
maxnoofMBSQoSFlows,
maxnoofMBSServiceAreaInformation,
maxnoofMBSAreaSessionIDs,
maxnoofMBSSessions,
maxnoofMBSSessionsofUE,
maxnoofMDTPLMNs,
maxnoofMRBs,
maxnoofMultiConnectivity,
maxnoofMultiConnectivityMinusOne,
maxnoofNeighPCIforMDT,
maxnoofNGAPIESupportInfo,
maxnoofNGConnectionsToReset,
maxNRARFCN,
maxnoofNRCellBands,
maxnoofNSAGs,
maxnoofPagingAreas,
maxnoofPC5QoSFlows,
maxnoofPDUSessions,
maxnoofPLMNs,
maxnoofPLMNforQMC,
maxnoofQoSFlows,
maxnoofQoSParaSets,
maxnoofRANNodeInAoI,
maxnoofReaders,
maxnoofRecommendedCells,
maxnoofRecommendedRANNodes,
maxnoofAoI,
maxnoofFlightInfoReportControl,
maxnoofPSCellsPerPrimaryCellInUEHistoryInfo,
maxnoofReportedCells,
maxnoofSensorName,
maxnoofServedGUAMIs,
maxnoofSliceItems,
maxnoofMDTSNPns,
maxnoofSNSSAIforQMC,
maxnoofSuccessfulHOREports,
maxnoofTACs,
maxnoofTACsinNTN,


```

maxnoofTAforMDT,
maxnoofTAforQMC,
maxnoofTAIforInactive,
maxnoofSupportedTAIforMBS,
maxnoofTAIforMBS,
maxnoofTAIforPaging,
maxnoofTAIforRestart,
maxnoofTAIforWarning,
maxnoofTAIinAoI,
maxnoofTargetS-NSSAIs,
maxnoofTimePeriods,
maxnoofTNLAssociations,
maxnoofUEAppLayerMeas,
maxnoofUEsforPaging,
maxnoofUETypes,
maxnoofWLANName,
maxnoofXnExtTLAs,
maxnoofXnGTP-TLAs,
maxnoofXnTLAs,
maxnoofThresholdsForExcessPacketDelay,
maxnoofCandidateRelayUEs,
maxnoofSuccessfulPSCellChangeReports,
maxnoofCellsTSS,
maxnoofPeriodicities,
maxnoofPartiallyAllowedS-NSSAIs,
maxnoofRSP PQoSFlows,
maxnoofSliceItemsforMDT,
maxnoofIntendedAreasforMBS,
maxnoofRequestedSNSSAIs,
maxnoofThresholds

```

FROM NGAP-Constants

```

Criticality,
ProcedureCode,
ProtocolIE-ID,
TriggeringMessage

```

FROM NGAP-CommonDataTypes

```

ProtocolExtensionContainer{},
ProtocolIE-Container{},
NGAP-PROTOCOL-EXTENSION,
ProtocolIE-SingleContainer{},
NGAP-PROTOCOL-IES

```

FROM NGAP-Containers;

-- A

AdditionalDLUPTNLInformationForHOList ::= SEQUENCE (SIZE(1..maxnoofMultiConnectivityMinusOne)) OF AdditionalDLUPTNLInformationForHOItem

AdditionalDLUPTNLInformationForHOItem ::= SEQUENCE {
 additionalDL-NGU-UP-TNLInformation UPTransportLayerInformation,

```

        additionalQosFlowSetupResponseList      QosFlowListWithDataForwarding,
        additionalDLForwardingUPTNLInformation  UPTransportLayerInformation      OPTIONAL,
        iE-Extensions      ProtocolExtensionContainer { { AdditionalDLUPTNLInformationForHOItem-ExtIEs} } OPTIONAL,
        ...
    }

AdditionalDLUPTNLInformationForHOItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalRedundantDL-NGU-UP-TNLInformation CRITICALITY ignore EXTENSION UPTransportLayerInformation PRESENCE optional },
    ...
}

AdditionalQosFlowInformation ::= ENUMERATED {
    more-likely,
    ...
}

AerialUESubscriptionInformation ::= ENUMERATED {
    allowed,
    not-allowed,
    ...
}

AdditionalULI ::= SEQUENCE {
    nRCGI                      NR-CGI,
    tAI                        TAI,
    iE-Extensions      ProtocolExtensionContainer { { AdditionalULI-ExtIEs} } OPTIONAL,
    ...
}

AdditionalULI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

A2X-PC5-FlowBitRates ::= SEQUENCE {
    a2X-GuaranteedFlowBitRate      BitRate,
    a2X-MaximumFlowBitRate      BitRate,
    iE-Extensions      ProtocolExtensionContainer { {A2X-PC5-FlowBitRates-ExtIEs} } OPTIONAL,
    ...
}

A2X-PC5-FlowBitRates-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

A2X-PC5-QoS-Flow-List ::= SEQUENCE (SIZE(1..maxnoofPC5QoSFlows)) OF A2X-PC5-QoS-Flow-Item

A2X-PC5-QoS-Flow-Item ::= SEQUENCE {
    a2X-PQI                      INTEGER (0..255, ...),
    a2X-PC5-FlowBitRates      A2X-PC5-FlowBitRates      OPTIONAL,
    a2X-Range      ENUMERATED {m50, m80, m180, m200,
                                m350, m400, m500, m700,
                                m1000, ...}      OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {A2X-PC5-QoS-Flow-Item-ExtIEs} } OPTIONAL,
    ...
}

```

```

}

A2X-PC5-QoS-Flow-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

A2X-PC5-QoS-Parameters ::= SEQUENCE {
    a2X-PC5-QoS-Flow-List          A2X-PC5-QoS-Flow-List,
    a2X-PC5-LinkAggregateBitRates  BitRate                      OPTIONAL,
    iE-Extensions                  ProtocolExtensionContainer { {A2X-PC5-QoS-Parameters-ExtIEs} } OPTIONAL,
    ...
}

A2X-PC5-QoS-Parameters-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionalCancelledlocationReportingReferenceIDList ::= SEQUENCE (SIZE(1.. maxnoofAoIMinusOne)) OF
AdditionalCancelledlocationReportingReferenceIDItem

AdditionalCancelledlocationReportingReferenceIDItem ::= SEQUENCE {
    locationReportingReferenceIDToBeCancelled      LocationReportingReferenceID,
    iE-Extensions                  ProtocolExtensionContainer { { AdditionalCancelledlocationReportingReferenceIDItem-ExtIEs} } OPTIONAL,
    ...
}

AdditionalCancelledlocationReportingReferenceIDItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AIoTAreaID ::= SEQUENCE {
    pLMNidentity      PLMNIdentity,
    nID                NID                      OPTIONAL,
    aIoTAreaCode       OCTET STRING (SIZE (3)),
    iE-Extensions      ProtocolExtensionContainer { { AIoTAreaID-ExtIEs} } OPTIONAL,
    ...
}

AIoTAreaID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AIoT-CorrelationIdentifier ::= INTEGER (0..65535, ...)

AIoT-FollowonCommandIndication ::= ENUMERATED {true, ...}

RAN-AIoT-Device-NGAP-ID ::= INTEGER (0..4294967295)

AIoT-CommandAssistanceInformation ::= SEQUENCE {
    estimateofExpectedD2RMsgSize      INTEGER (1..256, ...),
    iE-Extensions                    ProtocolExtensionContainer { { AIoT-CommandAssistanceInformation-ExtIEs} } OPTIONAL,
    ...
}

```

```

AIoT-CommandAssistanceInformation-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AIoT-DeviceIdentificationRequested ::= CHOICE {
    singleDevice          BIT STRING (SIZE(1..1024, ...)),
    groupDevices          BIT STRING (SIZE(1..1024, ...)),
    allDevices            NULL,
    choice-Extensions     ProtocolIE-SingleContainer { { AIoT-DeviceIdentificationRequested-ExtIEs} }
}

AIoT-DeviceIdentificationRequested-ExtIEs  NGAP-PROTOCOL-IES ::= {
    ...
}

AIoT-DeviceReportList ::= SEQUENCE (SIZE(1..maxnoofDevices)) OF AIoT-DeviceReportItem

AIoT-DeviceReportItem ::= SEQUENCE {
    aIoT-NASPDU          AIoT-NASPDU,
    rAN-AIoT-Device-NGAP-ID  RAN-AIoT-Device-NGAP-ID          OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { { AIoT-DeviceReportItem-ExtIEs} }  OPTIONAL,
    ...
}

AIoT-DeviceReportItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AIOTFIdentifier ::= OCTET STRING (SIZE(16..36, ...))

AIOTFName ::= SEQUENCE {
    aIOTFNameVisibleString  AIOTFNameVisibleString          OPTIONAL,
    aIOTFNameUTF8String     AIOTFNameUTF8String              OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { AIOTFName-ExtIEs } }  OPTIONAL,
    ...
}

AIOTFName-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AIOTFNameVisibleString ::= VisibleString (SIZE(1..150, ...))

AIOTFNameUTF8String ::= UTF8String (SIZE(1..150, ...))

AIoT-InventoryAssistanceInformation ::= SEQUENCE {
    expectedD2RMsgSize      INTEGER (1..256, ...),
    approximateNoofTargetDevices  INTEGER (1..65535, ...)  OPTIONAL,
    timeInterval            INTEGER (1..256, ...)          OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { AIoT-InventoryAssistanceInformation-ExtIEs} }  OPTIONAL,
    ...
}

AIoT-InventoryAssistanceInformation-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {

```

```

    ...
}

AloT-NASPDU ::= OCTET STRING

AloT-SecurityInfo ::= CHOICE {
    a-IoT-RAND-n          BIT STRING (SIZE(128, ...)),
    choice-Extensions     ProtocolIE-SingleContainer { {AloT-SecurityInfo-ExtIEs} }
}

AloT-SecurityInfo-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

AloT-ReaderReportList ::= SEQUENCE (SIZE(1..maxnoofReaders)) OF AloT-ReaderReportItem

AloT-ReaderReportItem ::= SEQUENCE {
    readerIndex            AloT-ReaderIndex,
    deviceReportList       AloT-DeviceReportList,
    iE-Extensions          ProtocolExtensionContainer { { AloT-ReaderReportItem-ExtIEs} } OPTIONAL,
    ...
}

AloT-ReaderReportItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AloT-ReaderIndex ::= INTEGER (1..65535, ...)

AloT-RequestedReaderList ::= SEQUENCE (SIZE(1..maxnoofReaders)) OF AloT-RequestedReaderItem

AloT-RequestedReaderItem ::= SEQUENCE {
    globalgNB-ID           GlobalGNB-ID,
    readerIndex            AloT-ReaderIndex,
    iE-Extensions          ProtocolExtensionContainer { { AloT-RequestedReaderItem-ExtIEs} } OPTIONAL,
    ...
}

AloT-RequestedReaderItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AloT-RequestedAloTAreaList ::= SEQUENCE (SIZE(1..maxnoofAloTAreas)) OF AloTAreaID

AloT-RequestedServiceAreaInformation ::= SEQUENCE {
    requestedReaderList     AloT-RequestedReaderList OPTIONAL,
    requestedAloTAreaList   AloT-RequestedAloTAreaList OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { AloT-RequestedServiceAreaInformation-ExtIEs} } OPTIONAL,
    ...
}

AloT-RequestedServiceAreaInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

}

AIOTSessionReleaseCommandTransfer ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {AIOTSessionReleaseCommandTransferIEs} },
    ...
}

AIOTSessionReleaseCommandTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AIoT-CorrelationIdentifier      CRITICALITY reject TYPE AIoT-CorrelationIdentifier      PRESENCE mandatory }|
    { ID id-Cause                          CRITICALITY ignore TYPE Cause                      PRESENCE mandatory },
    ...
}

AIOTSessionReleaseCompleteTransfer ::= SEQUENCE {
    correlationIdentifier      AIoT-CorrelationIdentifier,
    criticalityDiagnostics     CriticalityDiagnostics OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { {AIOTSessionReleaseCompleteTransfer-ExtIEs} } OPTIONAL,
    ...
}

AIOTSessionReleaseCompleteTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AIOTSessionReleaseRequestTransfer ::= SEQUENCE {
    correlationIdentifier      AIoT-CorrelationIdentifier,
    cause                     Cause,
    iE-Extensions              ProtocolExtensionContainer { {AIOTSessionReleaseRequestTransfer-ExtIEs} } OPTIONAL,
    ...
}

AIOTSessionReleaseRequestTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AIoT-Support ::= ENUMERATED {aIoT-only, aIoT-and-NRUu, ...}

AllocationAndRetentionPriority ::= SEQUENCE {
    priorityLevelARP           PriorityLevelARP,
    pre-emptionCapability       Pre-emptionCapability,
    pre-emptionVulnerability    Pre-emptionVulnerability,
    iE-Extensions              ProtocolExtensionContainer { {AllocationAndRetentionPriority-ExtIEs} } OPTIONAL,
    ...
}

AllocationAndRetentionPriority-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

Allowed-CAG-List-per-PLMN ::= SEQUENCE (SIZE(1..maxnoofAllowedCAGsperPLMN)) OF CAG-ID

AllowedNSSAI ::= SEQUENCE (SIZE(1..maxnoofAllowedS-NSSAIs)) OF AllowedNSSAI-Item

AllowedNSSAI-Item ::= SEQUENCE {

```

```

    s-NSSAI          S-NSSAI,
    iE-Extensions    ProtocolExtensionContainer { {AllowedNSSAI-Item-ExtIEs} } OPTIONAL,
    ...
}

AllowedNSSAI-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

Allowed-PNI-NPN-List ::= SEQUENCE (SIZE(1..maxnoofEPLMNsPlusOne)) OF Allowed-PNI-NPN-Item

Allowed-PNI-NPN-Item ::= SEQUENCE {
    pLMNIdentity          PLMNIdentity,
    pNI-NPN-restricted    ENUMERATED {restricted, not-restricted, ...},
    allowed-CAG-List-per-PLMN Allowed-CAG-List-per-PLMN,
    iE-Extensions          ProtocolExtensionContainer { {Allowed-PNI-NPN-Item-ExtIEs} } OPTIONAL,
    ...
}

Allowed-PNI-NPN-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AllowedTACs ::= SEQUENCE (SIZE(1..maxnoofAllowedAreas)) OF TAC

AlternativeQoSParaSetIndex ::= INTEGER (1..8, ...)

AlternativeQoSParaSetNotifyIndex ::= INTEGER (0..8, ...)

AlternativeQoSParaSetList ::= SEQUENCE (SIZE(1..maxnoofQosParaSets)) OF AlternativeQoSParaSetItem

AlternativeQoSParaSetItem ::= SEQUENCE {
    alternativeQoSParaSetIndex AlternativeQoSParaSetIndex,
    guaranteedFlowBitRateDL    BitRate OPTIONAL,
    guaranteedFlowBitRateUL    BitRate OPTIONAL,
    packetDelayBudget           PacketDelayBudget OPTIONAL,
    packetErrorRate             PacketErrorRate OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { {AlternativeQoSParaSetItem-ExtIEs} } OPTIONAL,
    ...
}

AlternativeQoSParaSetItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-MaximumDataBurstVolume CRITICALITY ignore EXTENSION MaximumDataBurstVolume PRESENCE optional }|
    { ID id-PduSetDelayBudgetDownlink CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget PRESENCE optional }|
    { ID id-PduSetDelayBudgetUplink CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget PRESENCE optional }|
    { ID id-PduSetErrorRateDownlink CRITICALITY ignore EXTENSION PacketErrorRate PRESENCE optional }|
    { ID id-PduSetErrorRateUplink CRITICALITY ignore EXTENSION PacketErrorRate PRESENCE optional },
    ...
}

AssistanceInformationQoE-Meas ::= INTEGER (1..16, ...)

AMFName ::= PrintableString (SIZE(1..150, ...))

```

```

AMFNameVisibleString ::= VisibleString (SIZE(1..150, ...))

AMFNameUTF8String ::= UTF8String (SIZE(1..150, ...))

AMFPagingTarget ::= CHOICE {
    globalRANNodeID      GlobalRANNodeID,
    tAI                  TAI,
    choice-Extensions    ProtocolIE-SingleContainer { {AMFPagingTarget-ExtIEs} }
}

AMFPagingTarget-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

AMFPointer ::= BIT STRING (SIZE(6))

AMFRegionID ::= BIT STRING (SIZE(8))

AMFSetID ::= BIT STRING (SIZE(10))

AMF-TNLAssociationSetupList ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF AMF-TNLAssociationSetupItem

AMF-TNLAssociationSetupItem ::= SEQUENCE {
    aMF-TNLAssociationAddress      CPTransportLayerInformation,
    iE-Extensions                  ProtocolExtensionContainer { {AMF-TNLAssociationSetupItem-ExtIEs} } OPTIONAL,
    ...
}

AMF-TNLAssociationSetupItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AMF-TNLAssociationToAddList ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF AMF-TNLAssociationToAddItem

AMF-TNLAssociationToAddItem ::= SEQUENCE {
    aMF-TNLAssociationAddress      CPTransportLayerInformation,
    tNLAssociationUsage            TNLAssociationUsage OPTIONAL,
    tNLAddressWeightFactor        TNLAddressWeightFactor,
    iE-Extensions                  ProtocolExtensionContainer { {AMF-TNLAssociationToAddItem-ExtIEs} } OPTIONAL,
    ...
}

AMF-TNLAssociationToAddItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AMF-TNLAssociationToRemoveList ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF AMF-TNLAssociationToRemoveItem

AMF-TNLAssociationToRemoveItem ::= SEQUENCE {
    aMF-TNLAssociationAddress      CPTransportLayerInformation,
    iE-Extensions                  ProtocolExtensionContainer { {AMF-TNLAssociationToRemoveItem-ExtIEs} } OPTIONAL,
    ...
}

```



```

AMF-TNLAssociationToRemoveItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-TNLAssociationTransportLayerAddressNGRAN CRITICALITY reject  EXTENSION CPTransportLayerInformation PRESENCE optional},
    ...
}

AMF-TNLAssociationToUpdateList ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF AMF-TNLAssociationToUpdateItem

AMF-TNLAssociationToUpdateItem ::= SEQUENCE {
    aMF-TNLAssociationAddress      CPTransportLayerInformation,
    tNLAssociationUsage            TNLAssociationUsage            OPTIONAL,
    tNLAddressWeightFactor         TNLAddressWeightFactor        OPTIONAL,
    iE-Extensions                  ProtocolExtensionContainer { {AMF-TNLAssociationToUpdateItem-ExtIEs} } OPTIONAL,
    ...
}

AMF-TNLAssociationToUpdateItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AMF-UE-NGAP-ID ::= INTEGER (0..1099511627775)

AreaOfInterest ::= SEQUENCE {
    areaOfInterestTAIList          AreaOfInterestTAIList          OPTIONAL,
    areaOfInterestCellList         AreaOfInterestCellList         OPTIONAL,
    areaOfInterestRANNodeList      AreaOfInterestRANNodeList      OPTIONAL,
    iE-Extensions                  ProtocolExtensionContainer { {AreaOfInterest-ExtIEs} } OPTIONAL,
    ...
}

AreaOfInterest-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AreaOfInterestCellList ::= SEQUENCE (SIZE(1..maxnoofCellinAoI)) OF AreaOfInterestCellItem

AreaOfInterestCellItem ::= SEQUENCE {
    nGRAN-CGI                     NGRAN-CGI,
    iE-Extensions                  ProtocolExtensionContainer { {AreaOfInterestCellItem-ExtIEs} } OPTIONAL,
    ...
}

AreaOfInterestCellItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AreaOfInterestList ::= SEQUENCE (SIZE(1..maxnoofAoI)) OF AreaOfInterestItem

AreaOfInterestItem ::= SEQUENCE {
    areaOfInterest                 AreaOfInterest,
    locationReportingReferenceID    LocationReportingReferenceID,
    iE-Extensions                  ProtocolExtensionContainer { {AreaOfInterestItem-ExtIEs} } OPTIONAL,
    ...
}

```

```

AreaOfInterestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AreaOfInterestRANNodeList ::= SEQUENCE (SIZE(1..maxnoofRANNodeinAoI)) OF AreaOfInterestRANNodeItem

AreaOfInterestRANNodeItem ::= SEQUENCE {
    globalRANNodeID      GlobalRANNodeID,
    iE-Extensions        ProtocolExtensionContainer { {AreaOfInterestRANNodeItem-ExtIEs} } OPTIONAL,
    ...
}

AreaOfInterestRANNodeItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AreaOfInterestTAIList ::= SEQUENCE (SIZE(1..maxnoofTAIinAoI)) OF AreaOfInterestTAIItem

AreaOfInterestTAIItem ::= SEQUENCE {
    tAI                  TAI,
    iE-Extensions        ProtocolExtensionContainer { {AreaOfInterestTAIItem-ExtIEs} } OPTIONAL,
    ...
}

AreaOfInterestTAIItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AssistanceDataForPaging ::= SEQUENCE {
    assistanceDataForRecommendedCells      AssistanceDataForRecommendedCells OPTIONAL,
    pagingAttemptInformation                PagingAttemptInformation OPTIONAL,
    iE-Extensions                          ProtocolExtensionContainer { {AssistanceDataForPaging-ExtIEs} } OPTIONAL,
    ...
}

AssistanceDataForPaging-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-NPN-PagingAssistanceInformation CRITICALITY ignore EXTENSION NPN-PagingAssistanceInformation PRESENCE optional }|
    { ID id-PagingAssisDataforCEcapabUE CRITICALITY ignore EXTENSION PagingAssisDataforCEcapabUE PRESENCE optional },
    ...
}

AssistanceDataForRecommendedCells ::= SEQUENCE {
    recommendedCellsForPaging      RecommendedCellsForPaging,
    iE-Extensions                  ProtocolExtensionContainer { {AssistanceDataForRecommendedCells-ExtIEs} } OPTIONAL,
    ...
}

AssistanceDataForRecommendedCells-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AssociatedMBSQosFlowSetupRequestList ::= SEQUENCE (SIZE(1..maxnoofMBSQoSFlows)) OF AssociatedMBSQosFlowSetupRequestItem

AssociatedMBSQosFlowSetupRequestItem ::= SEQUENCE {

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```

    mBS-QosFlowIdentifier      QosFlowIdentifier,
    associatedUnicastQosFlowIdentifier QosFlowIdentifier,
    iE-Extensions              ProtocolExtensionContainer { { AssociatedMBSQosFlowSetupRequestItem-ExtIEs} } OPTIONAL,
    ...
}

AssociatedMBSQosFlowSetupRequestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AssociatedMBSQosFlowSetuporModifyRequestList ::= SEQUENCE (SIZE(1..maxnoofMBSQoSFlows)) OF AssociatedMBSQosFlowSetuporModifyRequestItem

AssociatedMBSQosFlowSetuporModifyRequestItem ::= SEQUENCE {
    mBS-QosFlowIdentifier      QosFlowIdentifier,
    associatedUnicastQosFlowIdentifier QosFlowIdentifier,
    iE-Extensions              ProtocolExtensionContainer { { AssociatedMBSQosFlowSetuporModifyRequestItem-ExtIEs} } OPTIONAL,
    ...
}

AssociatedMBSQosFlowSetuporModifyRequestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AssociatedQosFlowList ::= SEQUENCE (SIZE(1..maxnoofQoSFlows)) OF AssociatedQosFlowItem

AssociatedQosFlowItem ::= SEQUENCE {
    qosFlowIdentifier      QosFlowIdentifier,
    qosFlowMappingIndication ENUMERATED {ul, dl, ...} OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {AssociatedQosFlowItem-ExtIEs} } OPTIONAL,
    ...
}

AssociatedQosFlowItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-CurrentQoSParaSetIndex CRITICALITY ignore EXTENSION AlternativeQoSParaSetIndex PRESENCE optional },
    ...
}

AssociatedSessionID ::= OCTET STRING

AUN3DeviceAccessInfo ::= SEQUENCE {
    aUN3DeviceAccess          ENUMERATED {true, ...},
    existingNGConnectionwithSameLocation ENUMERATED {true, false, ...},
    iE-Extensions              ProtocolExtensionContainer { { AUN3DeviceAccessInfo-ExtIEs} } OPTIONAL,
    ...
}

AUN3DeviceAccessInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AuthenticatedIndication ::= ENUMERATED {true, ...}

AveragingWindow ::= INTEGER (0..4095, ...)

```

```

AreaScopeOfMDT-NR ::= CHOICE {
    cellBased          CellBasedMDT-NR,
    tABased            TABasedMDT,
    pLMNWide          NULL,
    tAIBased           TAIBasedMDT,
    choice-Extensions  ProtocolIE-SingleContainer { {AreaScopeOfMDT-NR-ExtIEs} }
}

AreaScopeOfMDT-NR-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-PNI-NPNBasedMDT    CRITICALITY ignore TYPE PNI-NPNBasedMDT    PRESENCE mandatory }|
    { ID id-SNP-CellBasedMDT   CRITICALITY ignore TYPE SNPN-CellBasedMDT   PRESENCE mandatory }|
    { ID id-SNP-TAIBasedMDT    CRITICALITY ignore TYPE SNPN-TAIBasedMDT    PRESENCE mandatory }|
    { ID id-SNP-BasedMDT       CRITICALITY ignore TYPE SNPN-BasedMDT       PRESENCE mandatory }|
    { ID id-GeographyBasedMDT  CRITICALITY ignore TYPE GeographyBasedMDT    PRESENCE optional },
    ...
}

AreaScopeOfMDT-EUTRA ::= CHOICE {
    cellBased          CellBasedMDT-EUTRA,
    tABased            TABasedMDT,
    pLMNWide          NULL,
    tAIBased           TAIBasedMDT,
    choice-Extensions  ProtocolIE-SingleContainer { {AreaScopeOfMDT-EUTRA-ExtIEs} }
}

AreaScopeOfMDT-EUTRA-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

AreaScopeOfNeighCellsList ::= SEQUENCE (SIZE(1..maxnoofFreqforMDT)) OF AreaScopeOfNeighCellsItem
AreaScopeOfNeighCellsItem ::= SEQUENCE {
    nrFrequencyInfo      NRFrequencyInfo,
    pciListForMDT        PCIListForMDT                                OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { { AreaScopeOfNeighCellsItem-ExtIEs} } OPTIONAL,
    ...
}

AreaScopeOfNeighCellsItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AreaScopeOfQMC ::= CHOICE {
    cellBased          CellBasedQMC,
    tABased            TABasedQMC,
    tAIBased           TAIBasedQMC,
    pLMNAreaBased      PLMNAreaBasedQMC,
    choice-Extensions  ProtocolIE-SingleContainer { { AreaScopeOfQMC-ExtIEs} }
}

AreaScopeOfQMC-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

```

```

AvailableBitrateReportThresholdList ::= SEQUENCE (SIZE(1..maxnoofThresholds)) OF AvailableBitrateReportThresholdItem

AvailableBitrateReportThresholdItem ::= SEQUENCE {
    reportingThreshold          ReportingThreshold,
    iE-Extensions              ProtocolExtensionContainer { { AvailableBitrateReportThresholdItem-ExtIEs} } OPTIONAL,
    ...
}

AvailableBitrateReportThresholdItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AvailableRANVisibleQoEMetrics ::= SEQUENCE {
    applicationLayerBufferLevelList    ENUMERATED {true, ...}    OPTIONAL,
    playoutDelayForMediaStartup         ENUMERATED {true, ...}    OPTIONAL,
    iE-Extensions                      ProtocolExtensionContainer { { AvailableRANVisibleQoEMetrics-ExtIEs} } OPTIONAL,
    ...
}

AvailableRANVisibleQoEMetrics-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

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BeamMeasurementsReportConfiguration ::= SEQUENCE {
    beamMeasurementsReportQuantity    BeamMeasurementsReportQuantity    OPTIONAL,
    maxNrofRS-IndexesToReport          MaxNrofRS-IndexesToReport          OPTIONAL,
    iE-Extensions                      ProtocolExtensionContainer { { BeamMeasurementsReportConfiguration-ExtIEs} } OPTIONAL,
    ...
}

BeamMeasurementsReportConfiguration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

BeamMeasurementsReportQuantity ::= SEQUENCE {
    rSRP                             ENUMERATED {true, ..., false},
    rSRQ                             ENUMERATED {true, ..., false},
    sINR                             ENUMERATED {true, ..., false},
    iE-Extensions                    ProtocolExtensionContainer { { BeamMeasurementsReportQuantity-ExtIEs} } OPTIONAL,
    ...
}

BeamMeasurementsReportQuantity-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

BitRate ::= INTEGER (0..4000000000000, ...)

BroadcastCancelledAreaList ::= CHOICE {
    cellIDCancelledEUTRA              CellIDCancelledEUTRA,
    tAICancelledEUTRA                 TAICancelledEUTRA,

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    emergencyAreaIDCancelledEUTRA      EmergencyAreaIDCancelledEUTRA,
    cellIDCancelledNR                  CellIDCancelledNR,
    tAICancelledNR                     TAICancelledNR,
    emergencyAreaIDCancelledNR         EmergencyAreaIDCancelledNR,
    choice-Extensions                  ProtocolIE-SingleContainer { {BroadcastCancelledAreaList-ExtIEs} }
}

BroadcastCancelledAreaList-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

BroadcastCompletedAreaList ::= CHOICE {
    cellIDBroadcastEUTRA              CellIDBroadcastEUTRA,
    tAIBroadcastEUTRA                 TAIBroadcastEUTRA,
    emergencyAreaIDBroadcastEUTRA     EmergencyAreaIDBroadcastEUTRA,
    cellIDBroadcastNR                 CellIDBroadcastNR,
    tAIBroadcastNR                     TAIBroadcastNR,
    emergencyAreaIDBroadcastNR         EmergencyAreaIDBroadcastNR,
    choice-Extensions                  ProtocolIE-SingleContainer { {BroadcastCompletedAreaList-ExtIEs} }
}

BroadcastCompletedAreaList-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

BroadcastPLMNList ::= SEQUENCE (SIZE(1..maxnoofBPLMNs)) OF BroadcastPLMNItem

BroadcastPLMNItem ::= SEQUENCE {
    plMNIdentity                      PLMNIdentity,
    tAISliceSupportList               SliceSupportList,
    iE-Extensions                     ProtocolExtensionContainer { {BroadcastPLMNItem-ExtIEs} } OPTIONAL,
    ...
}

BroadcastPLMNItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-NPN-Support                CRITICALITY reject EXTENSION NPN-Support PRESENCE optional}|
    {ID id-ExtendedTAISliceSupportList CRITICALITY reject EXTENSION ExtendedSliceSupportList PRESENCE optional}|
    {ID id-TAINSAGSupportList          CRITICALITY ignore EXTENSION TAINSAGSupportList PRESENCE optional},
    ...
}

BroadcastTransportFailureTransfer ::= SEQUENCE {
    mBS-SessionID                     MBS-SessionID,
    cause                             Cause,
    criticalityDiagnostics              CriticalityDiagnostics OPTIONAL,
    iE-Extensions                      ProtocolExtensionContainer { { BroadcastTransportFailureTransfer-ExtIEs} } OPTIONAL,
    ...
}

BroadcastTransportFailureTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastTransportRequestTransfer ::= SEQUENCE {

```

```

    mBS-SessionID          MBS-SessionID,
    mBS-SessionTNLInfoNGRAN MBS-SessionTNLInfoNGRAN,
    iE-Extensions          ProtocolExtensionContainer { {BroadcastTransportRequestTransfer-ExtIEs} } OPTIONAL,
    ...
}

BroadcastTransportRequestTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastTransportResponseTransfer ::= SEQUENCE {
    mBS-SessionID          MBS-SessionID,
    iE-Extensions          ProtocolExtensionContainer { {BroadcastTransportResponseTransfer-ExtIEs} } OPTIONAL,
    ...
}

BroadcastTransportResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

BluetoothMeasurementConfiguration ::= SEQUENCE {
    bluetoothMeasConfig      BluetoothMeasConfig,
    bluetoothMeasConfigNameList BluetoothMeasConfigNameList OPTIONAL,
    bt-rssi                  ENUMERATED {true, ...} OPTIONAL,
    iE-Extensions            ProtocolExtensionContainer { { BluetoothMeasurementConfiguration-ExtIEs } } OPTIONAL,
    ...
}

BluetoothMeasurementConfiguration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

BluetoothMeasConfigNameList ::= SEQUENCE (SIZE(1..maxnoofBluetoothName)) OF BluetoothMeasConfigNameItem

BluetoothMeasConfigNameItem ::= SEQUENCE {
    bluetoothName            BluetoothName,
    iE-Extensions            ProtocolExtensionContainer { { BluetoothMeasConfigNameItem-ExtIEs } } OPTIONAL,
    ...
}

BluetoothMeasConfigNameItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

BluetoothMeasConfig ::= ENUMERATED {setup, ...}

BluetoothName ::= OCTET STRING (SIZE (1..248))

BurstArrivalTime ::= OCTET STRING

BurstArrivalTimeWindow ::= SEQUENCE {
    burstArrivalTimeWindowStart INTEGER (0..640000, ...),
    burstArrivalTimeWindowEnd   INTEGER (0..640000, ...),
    iE-Extension                 ProtocolExtensionContainer { {BurstArrivalTimeWindow-ExtIEs} } OPTIONAL,

```

```

    ...
}

BurstArrivalTimeWindow-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- C

CAG-ID ::= BIT STRING (SIZE(32))

CandidateRelayUEInformationList ::= SEQUENCE (SIZE(1.. maxnoofCandidateRelayUEs)) OF CandidateRelayUEInformationItem

CandidateRelayUEInformationItem ::= SEQUENCE {
    candidateRelayUE-Id      CandidateRelayUE-ID,
    iE-Extensions           ProtocolExtensionContainer { { CandidateRelayUEInformationItem-ExtIEs} } OPTIONAL,
    ...
}

CandidateRelayUEInformationItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CandidateRelayUE-ID ::= BIT STRING (SIZE(24))

CancelAllWarningMessages ::= ENUMERATED {
    true,
    ...
}

CancelledCellsInEAI-EUTRA ::= SEQUENCE (SIZE(1..maxnoofCellInEAI)) OF CancelledCellsInEAI-EUTRA-Item

CancelledCellsInEAI-EUTRA-Item ::= SEQUENCE {
    eUTRA-CGI      EUTRA-CGI,
    numberOfBroadcasts    NumberOfBroadcasts,
    iE-Extensions     ProtocolExtensionContainer { {CancelledCellsInEAI-EUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

CancelledCellsInEAI-EUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CancelledCellsInEAI-NR ::= SEQUENCE (SIZE(1..maxnoofCellInEAI)) OF CancelledCellsInEAI-NR-Item

CancelledCellsInEAI-NR-Item ::= SEQUENCE {
    nR-CGI      NR-CGI,
    numberOfBroadcasts    NumberOfBroadcasts,
    iE-Extensions     ProtocolExtensionContainer { {CancelledCellsInEAI-NR-Item-ExtIEs} } OPTIONAL,
    ...
}

CancelledCellsInEAI-NR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```



```

}

CancelledCellsInTAI-EUTRA ::= SEQUENCE (SIZE(1..maxnoofCellInTAI)) OF CancelledCellsInTAI-EUTRA-Item

CancelledCellsInTAI-EUTRA-Item ::= SEQUENCE {
    eUTRA-CGI          EUTRA-CGI,
    numberOfBroadcasts  NumberOfBroadcasts,
    iE-Extensions      ProtocolExtensionContainer { {CancelledCellsInTAI-EUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

CancelledCellsInTAI-EUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CancelledCellsInTAI-NR ::= SEQUENCE (SIZE(1..maxnoofCellInTAI)) OF CancelledCellsInTAI-NR-Item

CancelledCellsInTAI-NR-Item ::= SEQUENCE{
    nR-CGI          NR-CGI,
    numberOfBroadcasts  NumberOfBroadcasts,
    iE-Extensions      ProtocolExtensionContainer { {CancelledCellsInTAI-NR-Item-ExtIEs} } OPTIONAL,
    ...
}

CancelledCellsInTAI-NR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CandidateCellList ::= SEQUENCE (SIZE(1.. maxnoofCandidateCells)) OF CandidateCellItem

CandidateCellItem ::= SEQUENCE{
    candidateCell      CandidateCell,
    iE-Extensions      ProtocolExtensionContainer { {CandidateCellItem-ExtIEs} } OPTIONAL,
    ...
}

CandidateCellItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CandidateCell ::= CHOICE {
    candidateCGI          CandidateCellID,
    candidatePCI          CandidatePCI,
    choice-Extensions      ProtocolIE-SingleContainer { { CandidateCell-ExtIEs} }
}

CandidateCell-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

CandidateCellID ::= SEQUENCE {
    candidateCellID      NR-CGI,
    iE-Extensions        ProtocolExtensionContainer { { CandidateCellID-ExtIEs} }
}

```

OPTIONAL,

```
    ...
  }

CandidateCellID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

CandidatePCI ::= SEQUENCE {
  candidatePCI          INTEGER (0..1007, ...),
  candidateNRARFCN      INTEGER (0..maxNRARFCN),
  iE-Extensions         ProtocolExtensionContainer { { CandidatePCI-ExtIEs } } OPTIONAL,
  ...
}

CandidatePCI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

Cause ::= CHOICE {
  radioNetwork          CauseRadioNetwork,
  transport             CauseTransport,
  nas                  CauseNas,
  protocol              CauseProtocol,
  misc                 CauseMisc,
  choice-Extensions    ProtocolIE-SingleContainer { {Cause-ExtIEs} }
}

Cause-ExtIEs NGAP-PROTOCOL-IES ::= {
  ...
}

CauseMisc ::= ENUMERATED {
  control-processing-overload,
  not-enough-user-plane-processing-resources,
  hardware-failure,
  om-intervention,
  unknown-PLMN-or-SNPN,
  unspecified,
  ...
}

CauseNas ::= ENUMERATED {
  normal-release,
  authentication-failure,
  deregister,
  unspecified,
  ...,
  uE-not-in-PLMN-serving-area,
  mobile-IAB-not-authorized,
  iAB-not-authorized,
  no-aiot-NAS-response
}

CauseProtocol ::= ENUMERATED {
```

```
transfer-syntax-error,  
abstract-syntax-error-reject,  
abstract-syntax-error-ignore-and-notify,  
message-not-compatible-with-receiver-state,  
semantic-error,  
abstract-syntax-error-falsely-constructed-message,  
unspecified,  
...  
}  
  
CauseRadioNetwork ::= ENUMERATED {  
    unspecified,  
    txnrelocoverall-expiry,  
    successful-handover,  
    release-due-to-ngran-generated-reason,  
    release-due-to-5gc-generated-reason,  
    handover-cancelled,  
    partial-handover,  
    ho-failure-in-target-5GC-ngran-node-or-target-system,  
    ho-target-not-allowed,  
    tngrelocoverall-expiry,  
    tngrelocprep-expiry,  
    cell-not-available,  
    unknown-targetID,  
    no-radio-resources-available-in-target-cell,  
    unknown-local-UE-NGAP-ID,  
    inconsistent-remote-UE-NGAP-ID,  
    handover-desirable-for-radio-reason,  
    time-critical-handover,  
    resource-optimisation-handover,  
    reduce-load-in-serving-cell,  
    user-inactivity,  
    radio-connection-with-ue-lost,  
    radio-resources-not-available,  
    invalid-qos-combination,  
    failure-in-radio-interface-procedure,  
    interaction-with-other-procedure,  
    unknown-PDU-session-ID,  
    unknown-qos-flow-ID,  
    multiple-PDU-session-ID-instances,  
    multiple-qos-flow-ID-instances,  
    encryption-and-or-integrity-protection-algorithms-not-supported,  
    ng-intra-system-handover-triggered,  
    ng-inter-system-handover-triggered,  
    xn-handover-triggered,  
    not-supported-5QI-value,  
    ue-context-transfer,  
    ims-voice-eps-fallback-or-rat-fallback-triggered,  
    up-integrity-protection-not-possible,  
    up-confidentiality-protection-not-possible,  
    slice-not-supported,  
    ue-in-rrc-inactive-state-not-reachable,  
    redirection,  
    resources-not-available-for-the-slice,
```

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    ue-max-integrity-protected-data-rate-reason,
    release-due-to-cn-detected-mobility,
    ...,
    n26-interface-not-available,
    release-due-to-pre-emption,
    multiple-location-reporting-reference-ID-instances,
    rsn-not-available-for-the-up,
    npn-access-denied,
    cag-only-access-denied,
    insufficient-ue-capabilities,
    redcap-ue-not-supported,
    unknown-MBS-Session-ID,
    indicated-MBS-session-area-information-not-served-by-the-gNB,
    inconsistent-slice-info-for-the-session,
    misaligned-association-for-multicast-unicast,
    eredcap-ue-not-supported,
    two-rx-xr-ue-not-supported,
    aerial-ue-flight-information-reporting-initiation-failure,
    unknown-RAN-AIoT-Device-NGAP-ID,
    requested-AIoT-service-area-information-not-served-by-the-gNB,
    unknown-AIoT-session,
    aiot-device-not-reachable,
    multiple-AIoT-session-ID-instances
}

CauseTransport ::= ENUMERATED {
    transport-resource-unavailable,
    unspecified,
    ...
}

Cell-CAGInformation ::= SEQUENCE {
    nGRAN-CGI          NGRAN-CGI,
    cellCAGList        CellCAGList,
    iE-Extensions      ProtocolExtensionContainer { {Cell-CAGInformation-ExtIEs} } OPTIONAL,
    ...
}

Cell-CAGInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CellCAGList ::= SEQUENCE (SIZE(1..maxnoofCAGSperCell)) OF CAG-ID

CellIDBroadcastEUTRA ::= SEQUENCE (SIZE(1..maxnoofCellIDforWarning)) OF CellIDBroadcastEUTRA-Item

CellIDBroadcastEUTRA-Item ::= SEQUENCE {
    eUTRA-CGI          EUTRA-CGI,
    iE-Extensions      ProtocolExtensionContainer { {CellIDBroadcastEUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

CellIDBroadcastEUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {

```

```

    ...
}

CellIDBroadcastNR ::= SEQUENCE (SIZE(1..maxnoofCellIDforWarning)) OF CellIDBroadcastNR-Item

CellIDBroadcastNR-Item ::= SEQUENCE {
    nR-CGI          NR-CGI,
    iE-Extensions   ProtocolExtensionContainer { {CellIDBroadcastNR-Item-ExtIEs} } OPTIONAL,
    ...
}

CellIDBroadcastNR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CellIDCancelledEUTRA ::= SEQUENCE (SIZE(1..maxnoofCellIDforWarning)) OF CellIDCancelledEUTRA-Item

CellIDCancelledEUTRA-Item ::= SEQUENCE {
    eUTRA-CGI          EUTRA-CGI,
    numberOfBroadcasts  NumberOfBroadcasts,
    iE-Extensions      ProtocolExtensionContainer { {CellIDCancelledEUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

CellIDCancelledEUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CellIDCancelledNR ::= SEQUENCE (SIZE(1..maxnoofCellIDforWarning)) OF CellIDCancelledNR-Item

CellIDCancelledNR-Item ::= SEQUENCE {
    nR-CGI          NR-CGI,
    numberOfBroadcasts  NumberOfBroadcasts,
    iE-Extensions      ProtocolExtensionContainer { {CellIDCancelledNR-Item-ExtIEs} } OPTIONAL,
    ...
}

CellIDCancelledNR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CellIDListForRestart ::= CHOICE {
    eUTRA-CGIListforRestart  EUTRA-CGIList,
    nR-CGIListforRestart     NR-CGIList,
    choice-Extensions        ProtocolIE-SingleContainer { {CellIDListForRestart-ExtIEs} }
}

CellIDListForRestart-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

CellSize ::= ENUMERATED {verysmall, small, medium, large, ...}

```

```

CellType ::= SEQUENCE {
    cellSize          CellSize,
    iE-Extensions      ProtocolExtensionContainer { {CellType-ExtIEs} }    OPTIONAL,
    ...
}

CellType-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CModeBSupport-Indicator ::= ENUMERATED {supported,...}

CModeBrestricted ::= ENUMERATED {
    restricted,
    not-restricted,
    ...
}

ClockAccuracy ::= CHOICE {
    clockAccuracyValue      INTEGER (1..400000000, ...),
    clockAccuracyIndex      INTEGER (32..47, ...),
    choice-Extensions      ProtocolIE-SingleContainer { { ClockAccuracy-ExtIEs} }
}

ClockAccuracy-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

ClockQualityAcceptanceIndication ::= SEQUENCE {
    clockQualityAcceptanceCriteria      ClockQualityAcceptanceCriteria,
    iE-Extensions      ProtocolExtensionContainer { {ClockQualityAcceptanceIndication-ExtIEs} } OPTIONAL,
    ...
}

ClockQualityAcceptanceIndication-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ClockQualityAcceptanceCriteria ::= SEQUENCE {
    synchronisationState      BIT STRING (SIZE(8, ...))          OPTIONAL,
    traceabletoUTC            ENUMERATED {true, ...}             OPTIONAL,
    traceabletoGNSS           ENUMERATED {true, ...}             OPTIONAL,
    clockFrequencyStability   BIT STRING (SIZE(16))              OPTIONAL,
    clockAccuracy             INTEGER (1..400000000, ...)         OPTIONAL,
    parentTimeSource          BIT STRING (SIZE(16, ...))         OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { { ClockQualityAcceptanceCriteria-ExtIEs} } OPTIONAL,
    ...
}

ClockQualityAcceptanceCriteria-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

ClockQualityReportingControlInfo ::= SEQUENCE {
    clockQualityDetailLevel    ClockQualityDetailLevel,
    iE-Extensions              ProtocolExtensionContainer { {ClockQualityReportingControlInfo-ExtIEs} } OPTIONAL,
    ...
}

ClockQualityReportingControlInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ClockQualityDetailLevel ::= CHOICE {
    clockQualityMetrics          NULL,
    clockQualityAcceptanceIndication ClockQualityAcceptanceIndication,
    choice-extensions            ProtocolIE-SingleContainer { {ClockQualityDetailLevel-ExtIEs} }
}

ClockQualityDetailLevel-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

CNAssistedRANTuning ::= SEQUENCE {
    expectedUEBehaviour          ExpectedUEBehaviour OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {CNAssistedRANTuning-ExtIEs} } OPTIONAL,
    ...
}

CNAssistedRANTuning-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CNsubgroupID ::= INTEGER (0..7, ...)

CNTypeRestrictionsForEquivalent ::= SEQUENCE (SIZE(1..maxnoofEPLMNs)) OF CNTypeRestrictionsForEquivalentItem

CNTypeRestrictionsForEquivalentItem ::= SEQUENCE {
    plmnIdentity                PLMNIdentity,
    cn-Type                     ENUMERATED {epc-forbidden, fiveGC-forbidden, ...},
    iE-Extensions                ProtocolExtensionContainer { {CNTypeRestrictionsForEquivalentItem-ExtIEs} } OPTIONAL,
    ...
}

CNTypeRestrictionsForEquivalentItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CNTypeRestrictionsForServing ::= ENUMERATED {
    epc-forbidden,
    ...
}

CommandRequestTransfer ::= SEQUENCE {
    protocolIEs                 ProtocolIE-Container { { CommandRequestTransferIEs } },
    ...
}

```

```

CommandRequestTransferIES NGAP-PROTOCOL-IES ::= {
  { ID id-AIoT-CorrelationIdentifier          CRITICALITY reject TYPE AIoT-CorrelationIdentifier PRESENCE mandatory }|
  { ID id-RAN-AIoT-Device-NGAP-ID             CRITICALITY reject TYPE RAN-AIoT-Device-NGAP-ID PRESENCE mandatory }|
  { ID id-AIoT-NASPDU                         CRITICALITY reject TYPE AIoT-NASPDU PRESENCE mandatory }|
  { ID id-AIoT-CommandAssistanceInformation    CRITICALITY reject TYPE AIoT-CommandAssistanceInformation PRESENCE mandatory },
  ...
}

CommandResponseTransfer ::= SEQUENCE {
  correlationIdentifier      AIoT-CorrelationIdentifier,
  rAN-AIoT-Device-NGAP-ID   RAN-AIoT-Device-NGAP-ID,
  aIoT-NASPDU              AIoT-NASPDU,
  criticalityDiagnostics    CriticalityDiagnostics OPTIONAL,
  iE-Extensions            ProtocolExtensionContainer { { CommandResponseTransfer-ExtIES} } OPTIONAL,
  ...
}

CommandResponseTransfer-ExtIES NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

CommandFailureTransfer ::= SEQUENCE {
  correlationIdentifier      AIoT-CorrelationIdentifier,
  rAN-AIoT-Device-NGAP-ID   RAN-AIoT-Device-NGAP-ID,
  cause                    Cause,
  criticalityDiagnostics    CriticalityDiagnostics OPTIONAL,
  iE-Extensions            ProtocolExtensionContainer { { CommandFailureTransfer-ExtIES} } OPTIONAL,
  ...
}

CommandFailureTransfer-ExtIES NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

CommonNetworkInstance ::= OCTET STRING

CompletedCellsInEAI-EUTRA ::= SEQUENCE (SIZE(1..maxnoofCellInEAI)) OF CompletedCellsInEAI-EUTRA-Item

CompletedCellsInEAI-EUTRA-Item ::= SEQUENCE {
  eUTRA-CGI      EUTRA-CGI,
  iE-Extensions  ProtocolExtensionContainer { {CompletedCellsInEAI-EUTRA-Item-ExtIES} } OPTIONAL,
  ...
}

CompletedCellsInEAI-EUTRA-Item-ExtIES NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

CompletedCellsInEAI-NR ::= SEQUENCE (SIZE(1..maxnoofCellInEAI)) OF CompletedCellsInEAI-NR-Item

CompletedCellsInEAI-NR-Item ::= SEQUENCE {
  nR-CGI      NR-CGI,
  iE-Extensions ProtocolExtensionContainer { {CompletedCellsInEAI-NR-Item-ExtIES} } OPTIONAL,

```



```
}
...
CompletedCellsInEAI-NR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CompletedCellsInTAI-EUTRA ::= SEQUENCE (SIZE(1..maxnoofCellInTAI)) OF CompletedCellsInTAI-EUTRA-Item

CompletedCellsInTAI-EUTRA-Item ::= SEQUENCE{
    eUTRA-CGI          EUTRA-CGI,
    iE-Extensions      ProtocolExtensionContainer { {CompletedCellsInTAI-EUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

CompletedCellsInTAI-EUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CompletedCellsInTAI-NR ::= SEQUENCE (SIZE(1..maxnoofCellInTAI)) OF CompletedCellsInTAI-NR-Item

CompletedCellsInTAI-NR-Item ::= SEQUENCE{
    nR-CGI          NR-CGI,
    iE-Extensions      ProtocolExtensionContainer { {CompletedCellsInTAI-NR-Item-ExtIEs} } OPTIONAL,
    ...
}

CompletedCellsInTAI-NR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ConcurrentWarningMessageInd ::= ENUMERATED {
    true,
    ...
}

ConfidentialityProtectionIndication ::= ENUMERATED {
    required,
    preferred,
    not-needed,
    ...
}

ConfidentialityProtectionResult ::= ENUMERATED {
    performed,
    not-performed,
    ...
}

ConfiguredTACIndication ::= ENUMERATED {
    true,
    ...
}
```

```

CoreNetworkAssistanceInformationForInactive ::= SEQUENCE {
    ueIdentityIndexValue          UEIdentityIndexValue,
    ueSpecificDRX                 PagingDRX,
    periodicRegistrationUpdateTimer PeriodicRegistrationUpdateTimer,
    mICOModeIndication            MICOmodeIndication,
    tAIListForInactive            TAIListForInactive,
    expectedUEBehaviour           ExpectedUEBehaviour,
    iE-Extensions                 ProtocolExtensionContainer { {CoreNetworkAssistanceInformationForInactive-ExtIEs} }
    ...
}

CoreNetworkAssistanceInformationForInactive-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-EUTRA-PagingDRXInformation          CRITICALITY ignore EXTENSION EUTRA-PagingDRXInformation          PRESENCE optional }|
    { ID id-ExtendedUEIdentityIndexValue        CRITICALITY ignore EXTENSION ExtendedUEIdentityIndexValue        PRESENCE optional }|
    { ID id-UERadioCapabilityForPaging          CRITICALITY ignore EXTENSION UERadioCapabilityForPaging          PRESENCE optional }|
    { ID id-MicoAllPLMN                        CRITICALITY ignore EXTENSION MicoAllPLMN                        PRESENCE optional }|
    { ID id-NR-PagingDRXInformation             CRITICALITY ignore EXTENSION NR-PagingDRXInformation             PRESENCE optional }|
    { ID id-PagingCauseIndicationForVoiceService CRITICALITY ignore EXTENSION PagingCauseIndicationForVoiceService PRESENCE optional }|
    { ID id-PEIPSassistanceInformation          CRITICALITY ignore EXTENSION PEIPSassistanceInformation          PRESENCE optional }|
    { ID id-HashedUEIdentityIndexValue         CRITICALITY ignore EXTENSION HashedUEIdentityIndexValue         PRESENCE optional }|
    { ID id-CN-MT-CommunicationHandling         CRITICALITY ignore EXTENSION CN-MT-CommunicationHandling         PRESENCE optional }|
    { ID id-LPWUSPSAssistanceInformation        CRITICALITY ignore EXTENSION LPWUSPSAssistanceInformation        PRESENCE optional }|
    { ID id-FurtherExtendedUEIdentityIndexValue CRITICALITY ignore EXTENSION FurtherExtendedUEIdentityIndexValue PRESENCE optional }|
    { ID id-LPWUSDisableIndication             CRITICALITY ignore EXTENSION LPWUSDisableIndication             PRESENCE optional },
    ...
}

CN-MT-CommunicationHandling ::= ENUMERATED {supported, ...}

COUNTValueForPDCP-SN12 ::= SEQUENCE {
    pDCP-SN12          INTEGER (0..4095),
    hFN-PDCP-SN12      INTEGER (0..1048575),
    iE-Extensions      ProtocolExtensionContainer { {COUNTValueForPDCP-SN12-ExtIEs} } OPTIONAL,
    ...
}

COUNTValueForPDCP-SN12-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

COUNTValueForPDCP-SN18 ::= SEQUENCE {
    pDCP-SN18          INTEGER (0..262143),
    hFN-PDCP-SN18      INTEGER (0..16383),
    iE-Extensions      ProtocolExtensionContainer { {COUNTValueForPDCP-SN18-ExtIEs} } OPTIONAL,
    ...
}

COUNTValueForPDCP-SN18-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CoverageEnhancementLevel ::= OCTET STRING

CPTransportLayerInformation ::= CHOICE {

```

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    endpointIPAddress      TransportLayerAddress,
    choice-Extensions      ProtocolIE-SingleContainer { {CPTransportLayerInformation-ExtIEs} }
}

CPTransportLayerInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-EndpointIPAddressAndPort      CRITICALITY reject  TYPE EndpointIPAddressAndPort      PRESENCE mandatory  },
    ...
}

CriticalityDiagnostics ::= SEQUENCE {
    procedureCode           ProcedureCode                               OPTIONAL,
    triggeringMessage        TriggeringMessage                         OPTIONAL,
    procedureCriticality     Criticality                                OPTIONAL,
    iEsCriticalityDiagnostics CriticalityDiagnostics-IE-List           OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer {{CriticalityDiagnostics-ExtIEs}} OPTIONAL,
    ...
}

CriticalityDiagnostics-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CriticalityDiagnostics-IE-List ::= SEQUENCE (SIZE(1..maxnoofErrors)) OF CriticalityDiagnostics-IE-Item

CriticalityDiagnostics-IE-Item ::= SEQUENCE {
    iECriticality           Criticality,
    iE-ID                   ProtocolIE-ID,
    typeOfError             TypeOfError,
    iE-Extensions           ProtocolExtensionContainer {{CriticalityDiagnostics-IE-Item-ExtIEs}} OPTIONAL,
    ...
}

CriticalityDiagnostics-IE-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

C-RNTI ::= BIT STRING (SIZE(16))

CellBasedMDT-NR ::= SEQUENCE {
    cellIdListforMDT        CellIdListforMDT-NR,
    iE-Extensions           ProtocolExtensionContainer { {CellBasedMDT-NR-ExtIEs} } OPTIONAL,
    ...
}

CellBasedMDT-NR-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CellIdListforMDT-NR ::= SEQUENCE (SIZE(1..maxnoofCellIDforMDT)) OF NR-CGI

CellBasedMDT-EUTRA ::= SEQUENCE {
    cellIdListforMDT        CellIdListforMDT-EUTRA,
    iE-Extensions           ProtocolExtensionContainer { {CellBasedMDT-EUTRA-ExtIEs} } OPTIONAL,

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```

    ...
}

CellBasedMDT-EUTRA-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CellBasedQMC ::= SEQUENCE {
    cellIdListforQMC      CellIdListforQMC,
    iE-Extensions         ProtocolExtensionContainer { {CellBasedQMC-ExtIEs} } OPTIONAL,
    ...
}

CellBasedQMC-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CellIdListforQMC ::= SEQUENCE (SIZE(1..maxnoofCellIDforQMC)) OF NGRAN-CGI

CellIdListforMDT-EUTRA ::= SEQUENCE (SIZE(1..maxnoofCellIDforMDT)) OF EUTRA-CGI

CAGListforMDT ::= SEQUENCE (SIZE(1.. maxnoofCAGforMDT)) OF CAGListforMDTItem

CAGListforMDTItem ::= SEQUENCE {
    plmnID                PLMNIdentity,
    cAGID                 CAG-ID,
    iE-Extensions         ProtocolExtensionContainer { {CAGListforMDTItem-ExtIEs} }   OPTIONAL,
    ...
}

CAGListforMDTItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- D

DataCodingScheme ::= BIT STRING (SIZE(8))

DataForwardingAccepted ::= ENUMERATED {
    data-forwarding-accepted,
    ...
}

DataForwardingNotPossible ::= ENUMERATED {
    data-forwarding-not-possible,
    ...
}

DataForwardingResponseDRBList ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF DataForwardingResponseDRBItem

DataForwardingResponseDRBItem ::= SEQUENCE {
    dRB-ID                DRB-ID,
    dLForwardingUP-TNLIInformation  UPTransportLayerInformation                                OPTIONAL,

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        uLForwardingUP-TNLInformation      UPTransportLayerInformation      OPTIONAL,
        iE-Extensions      ProtocolExtensionContainer {{DataForwardingResponseDRBItem-ExtIEs}}  OPTIONAL,
        ...
    }

DataForwardingResponseDRBItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DAPSRequestInfo ::= SEQUENCE {
    dAPSIndicator      ENUMERATED {daps-ho-required, ...},
    iE-Extensions      ProtocolExtensionContainer { {DAPSRequestInfo-ExtIEs} } OPTIONAL,
    ...
}

DAPSRequestInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DAPSResponseInfoList ::= SEQUENCE (SIZE(1.. maxnoofDRBs)) OF DAPSResponseInfoItem

DAPSResponseInfoItem ::= SEQUENCE {
    dRB-ID      DRB-ID,
    dAPSResponseInfo      DAPSResponseInfo,
    iE-Extension      ProtocolExtensionContainer { {DAPSResponseInfoItem-ExtIEs} }      OPTIONAL,
    ...
}

DAPSResponseInfoItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DAPSResponseInfo ::= SEQUENCE {
    dapsresponseindicator      ENUMERATED {daps-ho-accepted, daps-ho-not-accepted, ...},
    iE-Extensions      ProtocolExtensionContainer { { DAPSResponseInfo-ExtIEs} }      OPTIONAL,
    ...
}

DAPSResponseInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DataForwardingResponseERABList ::= SEQUENCE (SIZE(1..maxnoofE-RABs)) OF DataForwardingResponseERABListItem

DataForwardingResponseERABListItem ::= SEQUENCE {
    e-RAB-ID      E-RAB-ID,
    dLForwardingUP-TNLInformation      UPTransportLayerInformation,
    iE-Extensions      ProtocolExtensionContainer { {DataForwardingResponseERABListItem-ExtIEs} }      OPTIONAL,
    ...
}

DataForwardingResponseERABListItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

}

DelayCritical ::= ENUMERATED {
    delay-critical,
    non-delay-critical,
    ...
}

DL-CP-SecurityInformation ::= SEQUENCE {
    dl-NAS-MAC          DL-NAS-MAC,
    iE-Extensions       ProtocolExtensionContainer { { DL-CP-SecurityInformation-ExtIEs} } OPTIONAL,
    ...
}

DL-CP-SecurityInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DL-Signalling ::= ENUMERATED {true, ...}

DL-NAS-MAC ::= BIT STRING (SIZE (16))

DLForwarding ::= ENUMERATED {
    dl-forwarding-proposed,
    ...
}

DL-NGU-TNLInformationReused ::= ENUMERATED {
    true,
    ...
}

DLPDUSetInformationMarkingSupportIndication ::= ENUMERATED {true, ...}

DirectForwardingPathAvailability ::= ENUMERATED {
    direct-path-available,
    ...
}

DRB-ID ::= INTEGER (1..32, ...)

DRBsSubjectToStatusTransferList ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF DRBsSubjectToStatusTransferItem

DRBsSubjectToStatusTransferItem ::= SEQUENCE {
    dRB-ID          DRB-ID,
    dRBStatusUL     DRBStatusUL,
    dRBStatusDL     DRBStatusDL,
    iE-Extension     ProtocolExtensionContainer { {DRBsSubjectToStatusTransferItem-ExtIEs} } OPTIONAL,
    ...
}

DRBsSubjectToStatusTransferItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-OldAssociatedQosFlowList-ULendmarkerexpected    CRITICALITY ignore EXTENSION AssociatedQosFlowList    PRESENCE optional },

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```

    ...
}

DRBStatusDL ::= CHOICE {
    drbStatusDL12          DRBStatusDL12,
    drbStatusDL18          DRBStatusDL18,
    choice-Extensions      ProtocolIE-SingleContainer { {DRBStatusDL-ExtIEs} }
}

DRBStatusDL-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

DRBStatusDL12 ::= SEQUENCE {
    dL-COUNTValue          COUNTValueForPDCP-SN12,
    iE-Extension           ProtocolExtensionContainer { {DRBStatusDL12-ExtIEs} }    OPTIONAL,
    ...
}

DRBStatusDL12-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DRBStatusDL18 ::= SEQUENCE {
    dL-COUNTValue          COUNTValueForPDCP-SN18,
    iE-Extension           ProtocolExtensionContainer { {DRBStatusDL18-ExtIEs} }    OPTIONAL,
    ...
}

DRBStatusDL18-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DRBStatusUL ::= CHOICE {
    drbStatusUL12          DRBStatusUL12,
    drbStatusUL18          DRBStatusUL18,
    choice-Extensions      ProtocolIE-SingleContainer { {DRBStatusUL-ExtIEs} }
}

DRBStatusUL-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

DRBStatusUL12 ::= SEQUENCE {
    uL-COUNTValue          COUNTValueForPDCP-SN12,
    receiveStatusOfUL-PDCP-SDUs BIT STRING (SIZE(1..2048))    OPTIONAL,
    iE-Extension           ProtocolExtensionContainer { {DRBStatusUL12-ExtIEs} }    OPTIONAL,
    ...
}

DRBStatusUL12-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

DRBStatusUL18 ::= SEQUENCE {
    uL-COUNTValue          COUNTValueForPDCP-SN18,
    receiveStatusOfUL-PDCP-SDUs BIT STRING (SIZE(1..131072)) OPTIONAL,
    iE-Extension          ProtocolExtensionContainer { {DRBStatusUL18-ExtIEs} } OPTIONAL,
    ...
}

DRBStatusUL18-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DRBstoQosFlowsMappingList ::= SEQUENCE (SIZE(1..maxnoofDRBs)) OF DRBstoQosFlowsMappingItem

DRBstoQosFlowsMappingItem ::= SEQUENCE {
    dRB-ID                DRB-ID,
    associatedQosFlowList  AssociatedQosFlowList,
    iE-Extensions          ProtocolExtensionContainer { {DRBstoQosFlowsMappingItem-ExtIEs} } OPTIONAL,
    ...
}

DRBstoQosFlowsMappingItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-DAPSRequestInfo CRITICALITY ignore EXTENSION DAPSRequestInfo PRESENCE optional },
    ...
}

Dynamic5QIDescriptor ::= SEQUENCE {
    priorityLevelQos      PriorityLevelQos,
    packetDelayBudget      PacketDelayBudget,
    packetErrorRate        PacketErrorRate,
    fiveQI                FiveQI OPTIONAL,
    delayCritical          DelayCritical OPTIONAL,
    -- The above IE shall be present in case of GBR QoS flow
    averagingWindow        AveragingWindow OPTIONAL,
    -- The above IE shall be present in case of GBR QoS flow
    maximumDataBurstVolume MaximumDataBurstVolume OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {Dynamic5QIDescriptor-ExtIEs} } OPTIONAL,
    ...
}

Dynamic5QIDescriptor-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-ExtendedPacketDelayBudget CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget PRESENCE optional }|
    { ID id-CNPacketDelayBudgetDL CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget PRESENCE optional }|
    { ID id-CNPacketDelayBudgetUL CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget PRESENCE optional },
    ...
}

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EarlyMeasurement ::= ENUMERATED {true, ...}

EarlyStatusTransfer-TransparentContainer ::= SEQUENCE {
    procedureStage          ProcedureStageChoice,
    iE-Extensions          ProtocolExtensionContainer { {EarlyStatusTransfer-TransparentContainer-ExtIEs} } OPTIONAL,
    ...
}

```



```

}

EarlyStatusTransfer-TransparentContainer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ProcedureStageChoice ::= CHOICE {
    first-dl-count          FirstDLCount,
    choice-Extensions       ProtocolIE-SingleContainer { {ProcedureStageChoice-ExtIEs} }
}

ProcedureStageChoice-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-DLDiscarding          CRITICALITY ignore  TYPE DLDiscarding          PRESENCE mandatory },
    ...
}

DLDiscarding ::= SEQUENCE {
    drBsSubjectToDLDiscarding          DRBsSubjectToDLDiscarding-List,
    iE-Extension                       ProtocolExtensionContainer { {DLDiscarding-ExtIEs} } OPTIONAL,
    ...
}

DLDiscarding-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DRBsSubjectToDLDiscarding-List ::= SEQUENCE (SIZE (1..maxnoofDRBs)) OF DRBsSubjectToDLDiscarding-Item

DRBsSubjectToDLDiscarding-Item ::= SEQUENCE {
    drbID          DRB-ID,
    dlCount        DLCountChoice,
    iE-Extension   ProtocolExtensionContainer { { DRBsSubjectToDLDiscarding-Item-ExtIEs} } OPTIONAL,
    ...
}

DRBsSubjectToDLDiscarding-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DLCountChoice ::= CHOICE {
    count12bits          COUNTValueForPDCP-SN12,
    count18bits          COUNTValueForPDCP-SN18,
    choice-extension     ProtocolIE-SingleContainer { {DLCountChoice-ExtIEs} }
}

DLCountChoice-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

FirstDLCount ::= SEQUENCE {
    drBsSubjectToEarlyStatusTransfer          DRBsSubjectToEarlyStatusTransfer-List,
    iE-Extension          ProtocolExtensionContainer { {FirstDLCount-ExtIEs} }          OPTIONAL,
    ...
}

```

```

FirstDLCount-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

DRBsSubjectToEarlyStatusTransfer-List ::= SEQUENCE (SIZE (1.. maxnoofDRBs)) OF DRBsSubjectToEarlyStatusTransfer-Item

DRBsSubjectToEarlyStatusTransfer-Item ::= SEQUENCE {
    dRB-ID                DRB-ID,
    firstDLCOUNT          DRBStatusDL,
    iE-Extension          ProtocolExtensionContainer { { DRBsSubjectToEarlyStatusTransfer-Item-ExtIEs} } OPTIONAL,
    ...
}

DRBsSubjectToEarlyStatusTransfer-Item-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ECNMarkingorCongestionInformationReportingRequest ::= CHOICE {
    eCNMarkingAtRANRequest          ECNMarkingAtRANRequest,
    eCNMarkingAtUPFRequest          ECNMarkingAtUPFRequest,
    congestionInformationRequest     CongestionInformationRequest,
    choice-Extensions              ProtocolIE-SingleContainer { {ECNMarkingorCongestionInformationReportingRequest-ExtIEs} }
}

ECNMarkingorCongestionInformationReportingRequest-ExtIEs  NGAP-PROTOCOL-IES ::= {
    ...
}

ECNMarkingAtRANRequest ::= ENUMERATED {ul, dl, both, stop, ...}

ECNMarkingAtUPFRequest ::= ENUMERATED {ul, dl, both, stop, ...}

CongestionInformationRequest ::= ENUMERATED {ul, dl, both, stop, ...}

ECNMarkingorCongestionInformationReportingStatus ::= SEQUENCE (SIZE (1..maxnoofQosFlows)) OF ECNMarkingorCongestionInformationReportingStatus-Item

ECNMarkingorCongestionInformationReportingStatus-Item ::= SEQUENCE {
    qosFlowIdentifier            QosFlowIdentifier,
    activationStatus             ActivationStatus,
    iE-Extension                 ProtocolExtensionContainer { { ECNMarkingorCongestionInformationReportingStatus-Item-ExtIEs} } OPTIONAL,
    ...
}

ECNMarkingorCongestionInformationReportingStatus-Item-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ActivationStatus ::= ENUMERATED {
    active,
    not-active,
    ...
}

```

```

EDT-Session ::= ENUMERATED {
    true,
    ...
}

EmergencyAreaID ::= OCTET STRING (SIZE(3))

EmergencyAreaIDBroadcastEUTRA ::= SEQUENCE (SIZE(1..maxnoofEmergencyAreaID)) OF EmergencyAreaIDBroadcastEUTRA-Item

EmergencyAreaIDBroadcastEUTRA-Item ::= SEQUENCE {
    emergencyAreaID          EmergencyAreaID,
    completedCellsInEAI-EUTRA CompletedCellsInEAI-EUTRA,
    iE-Extensions            ProtocolExtensionContainer { {EmergencyAreaIDBroadcastEUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

EmergencyAreaIDBroadcastEUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EmergencyAreaIDBroadcastNR ::= SEQUENCE (SIZE(1..maxnoofEmergencyAreaID)) OF EmergencyAreaIDBroadcastNR-Item

EmergencyAreaIDBroadcastNR-Item ::= SEQUENCE {
    emergencyAreaID          EmergencyAreaID,
    completedCellsInEAI-NR    CompletedCellsInEAI-NR,
    iE-Extensions            ProtocolExtensionContainer { {EmergencyAreaIDBroadcastNR-Item-ExtIEs} } OPTIONAL,
    ...
}

EmergencyAreaIDBroadcastNR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EmergencyAreaIDCancelledEUTRA ::= SEQUENCE (SIZE(1..maxnoofEmergencyAreaID)) OF EmergencyAreaIDCancelledEUTRA-Item

EmergencyAreaIDCancelledEUTRA-Item ::= SEQUENCE {
    emergencyAreaID          EmergencyAreaID,
    cancelledCellsInEAI-EUTRA CancelledCellsInEAI-EUTRA,
    iE-Extensions            ProtocolExtensionContainer { {EmergencyAreaIDCancelledEUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

EmergencyAreaIDCancelledEUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EmergencyAreaIDCancelledNR ::= SEQUENCE (SIZE(1..maxnoofEmergencyAreaID)) OF EmergencyAreaIDCancelledNR-Item

EmergencyAreaIDCancelledNR-Item ::= SEQUENCE {
    emergencyAreaID          EmergencyAreaID,
    cancelledCellsInEAI-NR    CancelledCellsInEAI-NR,
    iE-Extensions            ProtocolExtensionContainer { {EmergencyAreaIDCancelledNR-Item-ExtIEs} } OPTIONAL,
    ...
}

```

```

EmergencyAreaIDCancelledNR-Item-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EmergencyAreaIDList ::= SEQUENCE (SIZE(1..maxnoofEmergencyAreaID)) OF EmergencyAreaID

EmergencyAreaIDListForRestart ::= SEQUENCE (SIZE(1..maxnoofEAIforRestart)) OF EmergencyAreaID

EmergencyFallbackIndicator ::= SEQUENCE {
    emergencyFallbackRequestIndicator  EmergencyFallbackRequestIndicator,
    emergencyServiceTargetCN           EmergencyServiceTargetCN           OPTIONAL,
    iE-Extensions                      ProtocolExtensionContainer { {EmergencyFallbackIndicator-ExtIEs} }  OPTIONAL,
    ...
}

EmergencyFallbackIndicator-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EmergencyFallbackRequestIndicator ::= ENUMERATED {
    emergency-fallback-requested,
    ...
}

EmergencyServiceTargetCN ::= ENUMERATED {
    fiveGC,
    epc,
    ...
}

ENB-ID ::= CHOICE {
    macroENB-ID          BIT STRING (SIZE(20)),
    homeENB-ID           BIT STRING (SIZE(28)),
    short-macroENB-ID    BIT STRING (SIZE(18)),
    long-macroENB-ID     BIT STRING (SIZE(21)),
    choice-Extensions    ProtocolIE-SingleContainer { { ENB-ID-ExtIEs} }
}

ENB-ID-ExtIEs  NGAP-PROTOCOL-IES ::= {
    ...
}

Enhanced-CoverageRestriction ::= ENUMERATED {restricted, ... }

Extended-ConnectedTime ::= INTEGER (0..255)

EN-DCSONConfigurationTransfer ::= OCTET STRING

EndpointIPAddressAndPort ::=SEQUENCE {
    endpointIPAddress TransportLayerAddress,
    portNumber         PortNumber,

```

```

    iE-Extensions      ProtocolExtensionContainer { { EndpointIPAddressAndPort-ExtIEs } } OPTIONAL
  }

EndIndication ::= ENUMERATED {
    no-further-data,
    further-data-exists,
    ...
}

EndpointIPAddressAndPort-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EquivalentPLMNs ::= SEQUENCE (SIZE(1..maxnoofEPLMNs)) OF PLMNIdentity

EPS-TAC ::= OCTET STRING (SIZE(2))

EPS-TAI ::= SEQUENCE {
    pLMNIdentity      PLMNIdentity,
    ePS-TAC            EPS-TAC,
    iE-Extensions      ProtocolExtensionContainer { {EPS-TAI-ExtIEs} } OPTIONAL,
    ...
}

EPS-TAI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RAB-ID ::= INTEGER (0..15, ...)

E-RABInformationList ::= SEQUENCE (SIZE(1..maxnoofE-RABs)) OF E-RABInformationItem

E-RABInformationItem ::= SEQUENCE {
    e-RAB-ID          E-RAB-ID,
    dLForwarding      DLForwarding OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {E-RABInformationItem-ExtIEs} } OPTIONAL,
    ...
}

E-RABInformationItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-SourceTNLAddrInfo      CRITICALITY ignore EXTENSION TransportLayerAddress PRESENCE optional}|
    {ID id-SourceNodeTNLAddrInfo  CRITICALITY ignore EXTENSION TransportLayerAddress PRESENCE optional},
    ...
}

ERedCapIndication ::= ENUMERATED {
    true,
    ...
}

EUTRACellIdentity ::= BIT STRING (SIZE(28))

EUTRA-CGI ::= SEQUENCE {
    pLMNIdentity      PLMNIdentity,

```

```

    eUTRACellIdentity      EUTRACellIdentity,
    iE-Extensions          ProtocolExtensionContainer { {EUTRA-CGI-ExtIEs} } OPTIONAL,
    ...
}

EUTRA-CGI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EUTRA-CGIList ::= SEQUENCE (SIZE(1..maxnoofCellsinngeNB)) OF EUTRA-CGI

EUTRA-CGIListForWarning ::= SEQUENCE (SIZE(1..maxnoofCellIDforWarning)) OF EUTRA-CGI

EUTRA-PagingeDRXInformation ::= SEQUENCE {
    eUTRA-paging-eDRX-Cycle      EUTRA-Paging-eDRX-Cycle,
    eUTRA-paging-Time-Window     EUTRA-Paging-Time-Window OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {EUTRA-PagingeDRXInformation-ExtIEs} } OPTIONAL,
    ...
}

EUTRA-PagingeDRXInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EUTRA-Paging-eDRX-Cycle ::= ENUMERATED {
    hfhalf, hf1, hf2, hf4, hf6,
    hf8, hf10, hf12, hf14, hf16,
    hf32, hf64, hf128, hf256,
    ...
}

EUTRA-Paging-Time-Window ::= ENUMERATED {
    s1, s2, s3, s4, s5,
    s6, s7, s8, s9, s10,
    s11, s12, s13, s14, s15, s16,
    ...
}

EUTRAEncryptionAlgorithms ::= BIT STRING (SIZE(16, ...))

EUTRAIntegrityProtectionAlgorithms ::= BIT STRING (SIZE(16, ...))

EventType ::= ENUMERATED {
    direct,
    change-of-serve-cell,
    ue-presence-in-area-of-interest,
    stop-change-of-serve-cell,
    stop-ue-presence-in-area-of-interest,
    cancel-location-reporting-for-the-ue,
    ...,
    change-of-serving-cell-and-UE-presence-in-the-Area-of-Interest,
    report-the-Aerial-UE-flight-information,
    cancel-the-Aerial-UE-flight-information-reporting
}

```

ExcessPacketDelayThresholdConfiguration ::= SEQUENCE (SIZE(1..maxnoofThresholdsForExcessPacketDelay)) OF ExcessPacketDelayThresholdItem

```
ExcessPacketDelayThresholdItem ::= SEQUENCE {
    fiveQi          FiveQI,
    excessPacketDelayThresholdValue    ExcessPacketDelayThresholdValue,
    iE-Extensions    ProtocolExtensionContainer { { ExcessPacketDelayThresholdItem-ExtIEs} } OPTIONAL,
    ...
}
```

```
ExcessPacketDelayThresholdItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
ExcessPacketDelayThresholdValue ::= ENUMERATED {
    ms0dot25, ms0dot5, ms1, ms2, ms4, ms5, ms10, ms20, ms30, ms40, ms50, ms60, ms70, ms80, ms90, ms100, ms150, ms300, ms500,
    ...
}
```

ExpectedActivityPeriod ::= INTEGER (1..30|40|50|60|80|100|120|150|180|181, ...)

```
ExpectedHOInterval ::= ENUMERATED {
    sec15, sec30, sec60, sec90, sec120, sec180, long-time,
    ...
}
```

ExpectedIdlePeriod ::= INTEGER (1..30|40|50|60|80|100|120|150|180|181, ...)

```
ExpectedUEActivityBehaviour ::= SEQUENCE {
    expectedActivityPeriod          ExpectedActivityPeriod          OPTIONAL,
    expectedIdlePeriod              ExpectedIdlePeriod              OPTIONAL,
    sourceOfUEActivityBehaviourInformation    SourceOfUEActivityBehaviourInformation    OPTIONAL,
    iE-Extensions    ProtocolExtensionContainer { {ExpectedUEActivityBehaviour-ExtIEs} } OPTIONAL,
    ...
}
```

```
ExpectedUEActivityBehaviour-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
ExpectedUEBehaviour ::= SEQUENCE {
    expectedUEActivityBehaviour    ExpectedUEActivityBehaviour    OPTIONAL,
    expectedHOInterval              ExpectedHOInterval              OPTIONAL,
    expectedUEMobility              ExpectedUEMobility              OPTIONAL,
    expectedUEMovingTrajectory      ExpectedUEMovingTrajectory      OPTIONAL,
    iE-Extensions    ProtocolExtensionContainer { {ExpectedUEBehaviour-ExtIEs} } OPTIONAL,
    ...
}
```

```
ExpectedUEBehaviour-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}
```

ExpectedUEMobility ::= ENUMERATED {

```

    stationary,
    mobile,
    ...
}

ExpectedUEMovingTrajectory ::= SEQUENCE (SIZE(1..maxnoofCellsUEMovingTrajectory)) OF ExpectedUEMovingTrajectoryItem

ExpectedUEMovingTrajectoryItem ::= SEQUENCE {
    nGRAN-CGI          NGRAN-CGI,
    timeStayedInCell   INTEGER (0..4095) OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {ExpectedUEMovingTrajectoryItem-ExtIEs} } OPTIONAL,
    ...
}

ExpectedUEMovingTrajectoryItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

Extended-AMFName ::= SEQUENCE {
    amFNameVisibleString AMFNameVisibleString OPTIONAL,
    amFNameUTF8String    AMFNameUTF8String    OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { { Extended-AMFName-ExtIEs } } OPTIONAL,
    ...
}

Extended-AMFName-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ExtendedPacketDelayBudget ::= INTEGER (1..65535, ..., 65536..109999)

Extended-RANNodeName ::= SEQUENCE {
    rANNodeNameVisibleString RANNodeNameVisibleString OPTIONAL,
    rANNodeNameUTF8String    RANNodeNameUTF8String    OPTIONAL,
    iE-Extensions            ProtocolExtensionContainer { { Extended-RANNodeName-ExtIEs } } OPTIONAL,
    ...
}

Extended-RANNodeName-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ExtendedRATRestrictionInformation ::= SEQUENCE {
    primaryRATRestriction    BIT STRING (SIZE(8, ..., 16)),
    secondaryRATRestriction  BIT STRING (SIZE(8, ...)),
    iE-Extensions            ProtocolExtensionContainer { {ExtendedRATRestrictionInformation-ExtIEs} } OPTIONAL,
    ...
}

ExtendedRATRestrictionInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```



```

ExtendedRNC-ID ::= INTEGER (4096..65535)

ExtendedSliceSupportList ::= SEQUENCE (SIZE(1..maxnoofExtSliceItems)) OF SliceSupportItem

ExtendedUEIdentityIndexValue ::= BIT STRING (SIZE(16))

EventTrigger ::= CHOICE {
    outOfCoverage          ENUMERATED {true, ...},
    eventL1LoggedMDTConfig EventL1LoggedMDTConfig,
    choice-Extensions      ProtocolIE-SingleContainer { { EventTrigger-ExtIEs} }
}

EventTrigger-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

EventL1LoggedMDTConfig ::= SEQUENCE {
    l1Threshold          MeasurementThresholdL1LoggedMDT,
    hysteresis           Hysteresis,
    timeToTrigger        TimeToTrigger,
    iE-Extensions        ProtocolExtensionContainer { { EventL1LoggedMDTConfig-ExtIEs} } OPTIONAL,
    ...
}

EventL1LoggedMDTConfig-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MeasurementThresholdL1LoggedMDT ::= CHOICE {
    threshold-RSRP        Threshold-RSRP,
    threshold-RSRQ        Threshold-RSRQ,
    choice-Extensions      ProtocolIE-SingleContainer { { MeasurementThresholdL1LoggedMDT-ExtIEs} }
}

MeasurementThresholdL1LoggedMDT-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

EquivalentSNPNsList ::= SEQUENCE (SIZE(1..maxnoofESNPNS)) OF EquivalentSNPNsItem

EquivalentSNPNsItem ::= SEQUENCE {
    plmnIdentity          PLMNIdentity,
    nID                   NID,
    iE-Extensions        ProtocolExtensionContainer { {EquivalentSNPNsItem-ExtIEs} } OPTIONAL,
    ...
}

EquivalentSNPNsItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- F

FailureIndication ::= SEQUENCE {

```

```

    uERLFReportContainer      UERLFReportContainer,
    iE-Extensions             ProtocolExtensionContainer { { FailureIndication-ExtIEs} } OPTIONAL,
    ...
}

FailureIndication-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

FiveGCAction ::= CHOICE {
    hLComActivate             HLComActivate,
    hLComDeactivate           HLComDeactivate,
    choice-Extensions         ProtocolIE-SingleContainer { {FiveGCAction-ExtIEs} }
}

FiveGCAction-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

FiveG-ProSeAuthorized ::= SEQUENCE {
    fiveGProSeDirectDiscovery      FiveGProSeDirectDiscovery                OPTIONAL,
    fiveGProSeDirectCommunication  FiveGProSeDirectCommunication            OPTIONAL,
    fiveGProSeLayer2UEtoNetworkRelay FiveGProSeLayer2UEtoNetworkRelay        OPTIONAL,
    fiveGProSeLayer3UEtoNetworkRelay FiveGProSeLayer3UEtoNetworkRelay        OPTIONAL,
    fiveGProSeLayer2RemoteUE       FiveGProSeLayer2RemoteUE                OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { {FiveG-ProSeAuthorized-ExtIEs} } OPTIONAL,
    ...
}

FiveG-ProSeAuthorized-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-FiveGProSeLayer2Multipath      CRITICALITY ignore EXTENSION FiveGProSeLayer2Multipath      PRESENCE optional }|
    { ID id-FiveGProSeLayer2UEtoUERelay    CRITICALITY ignore EXTENSION FiveGProSeLayer2UEtoUERelay    PRESENCE optional }|
    { ID id-FiveGProSeLayer2UEtoUERemote    CRITICALITY ignore EXTENSION FiveGProSeLayer2UEtoUERemote    PRESENCE optional }|
    { ID id-FiveGProSeLayer2MHUEtoNetworkRelay CRITICALITY ignore EXTENSION FiveGProSeLayer2MHUEtoNetworkRelay PRESENCE optional}|
    { ID id-FiveGProSeLayer2MHIntermediateUEtoNetworkRelay CRITICALITY ignore EXTENSION FiveGProSeLayer2MHIntermediateUEtoNetworkRelay PRESENCE optional}|
    { ID id-FiveGProSeLayer2MHRemote       CRITICALITY ignore EXTENSION FiveGProSeLayer2MHRemote       PRESENCE optional}|
    { ID id-FiveGProSeLayer3MHUEtoNetworkRelay CRITICALITY ignore EXTENSION FiveGProSeLayer3MHUEtoNetworkRelay PRESENCE optional},
    ...
}

FiveGProSeDirectDiscovery ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveGProSeDirectCommunication ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

FiveGProSeLayer2UEtoNetworkRelay ::= ENUMERATED {

```

```
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer3UEtoNetworkRelay ::= ENUMERATED {  
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer2RemoteUE ::= ENUMERATED {  
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer2Multipath ::= ENUMERATED {  
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer2UEtoUERelay ::= ENUMERATED {  
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer2UEtoUERemote ::= ENUMERATED {  
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer2MHUEtoNetworkRelay ::= ENUMERATED {  
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer2MHIntermediateUEtoNetworkRelay ::= ENUMERATED {  
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer2MHRemote ::= ENUMERATED {  
    authorized,  
    not-authorized,  
    ...  
}  
  
FiveGProSeLayer3MHUEtoNetworkRelay ::= ENUMERATED {
```

```

    authorized,
    not-authorized,
    ...
}

FiveG-ProSePC5QoSParameters ::= SEQUENCE {
    fiveGProSePC5QoSFlowList          FiveGProSePC5QoSFlowList,
    fiveGProSePC5LinkAggregateBitRates BitRate OPTIONAL,
    iE-Extensions                     ProtocolExtensionContainer { { FiveG-ProSePC5QoSParameters-ExtIEs} } OPTIONAL,
    ...
}

FiveG-ProSePC5QoSParameters-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

FiveGProSePC5QoSFlowList ::= SEQUENCE (SIZE(1..maxnoofPC5QoSFlows)) OF FiveGProSePC5QoSFlowItem

FiveGProSePC5QoSFlowItem ::= SEQUENCE {
    fiveGProSePQI          FiveQI,
    fiveGProSePC5FlowBitRates FiveGProSePC5FlowBitRates OPTIONAL,
    fiveGProSeRange        Range OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { FiveGProSePC5QoSFlowItem-ExtIEs} } OPTIONAL,
    ...
}

FiveGProSePC5QoSFlowItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

FiveGProSePC5FlowBitRates ::= SEQUENCE {
    fiveGProSeGuaranteedFlowBitRate BitRate,
    fiveGProSeMaximumFlowBitRate    BitRate,
    iE-Extensions                   ProtocolExtensionContainer { { FiveGProSePC5FlowBitRates-ExtIEs} } OPTIONAL,
    ...
}

FiveGProSePC5FlowBitRates-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

FiveG-S-TMSI ::= SEQUENCE {
    aMFSetID          AMFSetID,
    aMFPointer        AMFPointer,
    fiveG-S-TMSI      FiveG-S-TMSI,
    iE-Extensions     ProtocolExtensionContainer { {FiveG-S-TMSI-ExtIEs} } OPTIONAL,
    ...
}

FiveG-S-TMSI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

FiveG-TMSI ::= OCTET STRING (SIZE(4))

FiveQI ::= INTEGER (0..255, ...)

ForbiddenAreaInformation ::= SEQUENCE (SIZE(1.. maxnoofEPLMNsPlusOne)) OF ForbiddenAreaInformation-Item

ForbiddenAreaInformation-Item ::= SEQUENCE {
    pLMNIdentity          PLMNIdentity,
    forbiddenTACs         ForbiddenTACs,
    iE-Extensions         ProtocolExtensionContainer { {ForbiddenAreaInformation-Item-ExtIEs} } OPTIONAL,
    ...
}

ForbiddenAreaInformation-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ForbiddenTACs ::= SEQUENCE (SIZE(1..maxnoofForbTACs)) OF TAC

FromEUTRANToNGRAN ::= SEQUENCE {
    sourceeNBID           IntersystemSONeNBID,
    targetNGRANnodeID     IntersystemSONNGRANnodeID,
    iE-Extensions         ProtocolExtensionContainer { { FromEUTRANToNGRAN-ExtIEs} } OPTIONAL
}

FromEUTRANToNGRAN-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

FromNGRANToEUTRAN ::= SEQUENCE {
    sourceNGRANnodeID     IntersystemSONNGRANnodeID,
    targeteNBID           IntersystemSONeNBID,
    iE-Extensions         ProtocolExtensionContainer { { FromNGRANToEUTRAN-ExtIEs} } OPTIONAL
}

FromNGRANToEUTRAN-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

FurtherExtendedUEIdentityIndexValue ::= BIT STRING (SIZE(20))

-- G

GBR-QosInformation ::= SEQUENCE {
    maximumFlowBitRateDL   BitRate,
    maximumFlowBitRateUL   BitRate,
    guaranteedFlowBitRateDL BitRate,
    guaranteedFlowBitRateUL BitRate,
    notificationControl     NotificationControl OPTIONAL,
    maximumPacketLossRateDL PacketLossRate OPTIONAL,
    maximumPacketLossRateUL PacketLossRate OPTIONAL,
    iE-Extensions         ProtocolExtensionContainer { {GBR-QosInformation-ExtIEs} } OPTIONAL,
    ...
}

```

```

}

GBR-QosInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  { ID id-AlternativeQoSParaSetList CRITICALITY ignore EXTENSION AlternativeQoSParaSetList PRESENCE optional }|
  { ID id-MonitoringRequestonAvailableBitrate CRITICALITY ignore EXTENSION MonitoringRequestonAvailableBitrate PRESENCE optional},
  ...
}

GeographyBasedMDT ::= SEQUENCE {
  nTN-Geographical-Area NTN-Geographical-Area-List,
  nTN-PLMN-List MDTPLMNList OPTIONAL,
  iE-Extensions ProtocolExtensionContainer { {GeographyBasedMDT-ExtIEs} } OPTIONAL,
  ...
}

GeographyBasedMDT-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

GlobalCable-ID ::= OCTET STRING

GlobalCable-ID-new ::= SEQUENCE {
  globalCable-ID GlobalCable-ID,
  tAI TAI,
  iE-Extensions ProtocolExtensionContainer { { GlobalCable-ID-new-ExtIEs} } OPTIONAL,
  ...
}

GlobalCable-ID-new-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

GlobalENB-ID ::= SEQUENCE {
  pLMNIdentity PLMNIdentity,
  eNB-ID ENB-ID,
  iE-Extensions ProtocolExtensionContainer { {GlobalENB-ID-ExtIEs} } OPTIONAL,
  ...
}

GlobalENB-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

GlobalGNB-ID ::= SEQUENCE {
  pLMNIdentity PLMNIdentity,
  gNB-ID GNB-ID,
  iE-Extensions ProtocolExtensionContainer { {GlobalGNB-ID-ExtIEs} } OPTIONAL,
  ...
}

GlobalGNB-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

```

```

}

GlobalN3IWF-ID ::= SEQUENCE {
    pLMNIdentity      PLMNIdentity,
    n3IWF-ID          N3IWF-ID,
    iE-Extensions     ProtocolExtensionContainer { {GlobalN3IWF-ID-ExtIEs} } OPTIONAL,
    ...
}

GlobalN3IWF-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

GlobalLine-ID ::= SEQUENCE {
    globalLineIdentity GlobalLineIdentity,
    lineType           LineType                                OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {GlobalLine-ID-ExtIEs} }    OPTIONAL,
    ...
}

GlobalLine-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-TAI          CRITICALITY ignore EXTENSION TAI          PRESENCE optional },
    ...
}

GlobalLineIdentity ::= OCTET STRING

GlobalNgENB-ID ::= SEQUENCE {
    pLMNIdentity      PLMNIdentity,
    ngENB-ID          NgENB-ID,
    iE-Extensions     ProtocolExtensionContainer { {GlobalNgENB-ID-ExtIEs} } OPTIONAL,
    ...
}

GlobalNgENB-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

GlobalRANNodeID ::= CHOICE {
    globalGNB-ID      GlobalGNB-ID,
    globalNgENB-ID    GlobalNgENB-ID,
    globalN3IWF-ID    GlobalN3IWF-ID,
    choice-Extensions ProtocolIE-SingleContainer { {GlobalRANNodeID-ExtIEs} }
}

GlobalRANNodeID-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-GlobalTNGF-ID      CRITICALITY reject TYPE GlobalTNGF-ID      PRESENCE mandatory }|
    { ID id-GlobalTWIF-ID     CRITICALITY reject TYPE GlobalTWIF-ID     PRESENCE mandatory }|
    { ID id-GlobalW-AGF-ID    CRITICALITY reject TYPE GlobalW-AGF-ID    PRESENCE mandatory },
    ...
}

GlobalTNGF-ID ::= SEQUENCE {
    pLMNIdentity      PLMNIdentity,

```

```

    tNGF-ID          TNGF-ID,
    iE-Extensions    ProtocolExtensionContainer { { GlobalTNGF-ID-ExtIEs } } OPTIONAL,
    ...
}

GlobalTNGF-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

GlobalTWIF-ID ::= SEQUENCE {
    plMNIdentity      PLMNIdentity,
    twIF-ID           TWIF-ID,
    iE-Extensions     ProtocolExtensionContainer { { GlobalTWIF-ID-ExtIEs } } OPTIONAL,
    ...
}

GlobalTWIF-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

GlobalW-AGF-ID ::= SEQUENCE {
    plMNIdentity      PLMNIdentity,
    w-AGF-ID          W-AGF-ID,
    iE-Extensions     ProtocolExtensionContainer { { GlobalW-AGF-ID-ExtIEs } } OPTIONAL,
    ...
}

GlobalW-AGF-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-ID ::= CHOICE {
    gNB-ID            BIT STRING (SIZE(22..32)),
    choice-Extensions ProtocolIE-SingleContainer { { GNB-ID-ExtIEs } }
}

GNB-ID-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

GTP-TEID ::= OCTET STRING (SIZE(4))

GTPtunnel ::= SEQUENCE {
    transportLayerAddress TransportLayerAddress,
    gTP-TEID              GTP-TEID,
    iE-Extensions         ProtocolExtensionContainer { { GTPtunnel-ExtIEs } } OPTIONAL,
    ...
}

GTPtunnel-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```



```

GUAMI ::= SEQUENCE {
    pLMNIdentity      PLMNIdentity,
    aMFRegionID       AMFRegionID,
    aMFSetID           AMFSetID,
    aMFPointer         AMFPointer,
    iE-Extensions     ProtocolExtensionContainer { {GUAMI-ExtIEs} } OPTIONAL,
    ...
}

GUAMI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

GUAMIType ::= ENUMERATED {native, mapped, ...}

GUAMIList ::= SEQUENCE (SIZE(1..maxnoofGUAMIs)) OF GUAMIList-Item

GUAMIList-Item ::= SEQUENCE {
    gUAMI              GUAMI,
    iE-Extensions     ProtocolExtensionContainer { {GUAMIList-Item-ExtIEs} } OPTIONAL,
    ...
}

GUAMIList-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

GW-Context-Release-Indication ::= ENUMERATED {true, ...}

-- H

HandoverCommandTransfer ::= SEQUENCE {
    dLForwardingUP-TNLInformation      UPTransportLayerInformation      OPTIONAL,
    qosFlowToBeForwardedList           QosFlowToBeForwardedList         OPTIONAL,
    dataForwardingResponseDRBList       DataForwardingResponseDRBList    OPTIONAL,
    iE-Extensions                      ProtocolExtensionContainer { {HandoverCommandTransfer-ExtIEs} } OPTIONAL,
    ...
}

HandoverCommandTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalDLForwardingUP-TNLInformation      CRITICALITY ignore EXTENSION QosFlowPerTNLInformationList      PRESENCE optional }|
    { ID id-ULForwardingUP-TNLInformation                CRITICALITY reject  EXTENSION UPTransportLayerInformation      PRESENCE optional }|
    { ID id-AdditionalULForwardingUP-TNLInformation      CRITICALITY reject  EXTENSION UPTransportLayerInformationList    PRESENCE optional }|
    { ID id-DataForwardingResponseERABList              CRITICALITY ignore  EXTENSION DataForwardingResponseERABList    PRESENCE optional }|
    { ID id-QosFlowFailedToSetupList                    CRITICALITY ignore  EXTENSION QosFlowListWithCause              PRESENCE optional },
    ...
}

HandoverFlag ::= ENUMERATED {
    handover-preparation,
    ...
}

```

```

HandoverPreparationUnsuccessfulTransfer ::= SEQUENCE {
    cause                Cause,
    iE-Extensions        ProtocolExtensionContainer { {HandoverPreparationUnsuccessfulTransfer-ExtIEs} } OPTIONAL,
    ...
}

HandoverPreparationUnsuccessfulTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

HandoverRequestAcknowledgeTransfer ::= SEQUENCE {
    dL-NGU-UP-TNLInformation      UPTransportLayerInformation,
    dLForwardingUP-TNLInformation UPTransportLayerInformation OPTIONAL,
    securityResult                SecurityResult OPTIONAL,
    qosFlowSetupResponseList      QosFlowListWithDataForwarding,
    qosFlowFailedToSetupList      QosFlowListWithCause OPTIONAL,
    dataForwardingResponseDRBList DataForwardingResponseDRBList OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {HandoverRequestAcknowledgeTransfer-ExtIEs} } OPTIONAL,
    ...
}

HandoverRequestAcknowledgeTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalDLUPTNLInformationForHOList CRITICALITY ignore EXTENSION AdditionalDLUPTNLInformationForHOList PRESENCE optional }|
    { ID id-ULForwardingUP-TNLInformation CRITICALITY reject EXTENSION UPTransportLayerInformation PRESENCE optional }|
    { ID id-AdditionalULForwardingUP-TNLInformation CRITICALITY reject EXTENSION UPTransportLayerInformationList PRESENCE optional }|
    { ID id-DataForwardingResponseERABList CRITICALITY ignore EXTENSION DataForwardingResponseERABList PRESENCE optional }|
    { ID id-RedundantDL-NGU-UP-TNLInformation CRITICALITY ignore EXTENSION UPTransportLayerInformation PRESENCE optional }|
    { ID id-UsedRSNInformation CRITICALITY ignore EXTENSION RedundantPDUSessionInformation PRESENCE optional }|
    { ID id-GlobalRANNodeID CRITICALITY ignore EXTENSION GlobalRANNodeID PRESENCE optional }|
    { ID id-MBS-SupportIndicator CRITICALITY ignore EXTENSION MBS-SupportIndicator PRESENCE optional }|
    { ID id-PDUSetbasedHandlingIndicator CRITICALITY ignore EXTENSION PDUSetbasedHandlingIndicator PRESENCE optional }|
    { ID id-ECNMarkingorCongestionInformationReportingStatus CRITICALITY ignore EXTENSION ECNMarkingorCongestionInformationReportingStatus PRESENCE optional },
    PRESENCE optional },
    ...
}

HandoverRequiredTransfer ::= SEQUENCE {
    directForwardingPathAvailability DirectForwardingPathAvailability OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {HandoverRequiredTransfer-ExtIEs} } OPTIONAL,
    ...
}

HandoverRequiredTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

HandoverResourceAllocationUnsuccessfulTransfer ::= SEQUENCE {
    cause                Cause,
    criticalityDiagnostics CriticalityDiagnostics OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { {HandoverResourceAllocationUnsuccessfulTransfer-ExtIEs} } OPTIONAL,
    ...
}

HandoverResourceAllocationUnsuccessfulTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {

```

```

    ...
}

HandoverType ::= ENUMERATED {
    intra5gs,
    fivegs-to-eps,
    eps-to-5gs,
    ...,
    fivegs-to-utran
}

HashedUEIdentityIndexValue ::= BIT STRING (SIZE(13, ...))

HFCNode-ID ::= OCTET STRING

HFCNode-ID-new ::= SEQUENCE {
    hFCNode-ID          HFCNode-ID,
    tAI                 TAI,
    iE-Extensions       ProtocolExtensionContainer { { HFCNode-ID-new-ExtIEs} } OPTIONAL,
    ...
}

HFCNode-ID-new-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-GlobalCable-ID          CRITICALITY ignore EXTENSION GlobalCable-ID          PRESENCE optional },
    ...
}

HLComActivate ::= SEQUENCE {
    nR-Paging-Long-eDRX-Information-for-RRC-INACTIVE NR-Paging-Long-eDRX-Information-for-RRC-INACTIVE,
    iE-Extensions                                     ProtocolExtensionContainer { { HLComActivate-ExtIEs} } OPTIONAL,
    ...
}

HLComActivate-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

HLComDeactivate ::= SEQUENCE {
    uEReachabilityIndication      ENUMERATED {true, ...},
    iE-Extensions                 ProtocolExtensionContainer { { HLComDeactivate-ExtIEs} } OPTIONAL,
    ...
}

HLComDeactivate-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

HOReport ::= SEQUENCE {
    handoverReportType            ENUMERATED {ho-too-early, ho-to-wrong-cell, intersystem-ping-pong, ...},
    handoverCause                 Cause,
    sourcecellCGI                 NGRAN-CGI,
    targetcellCGI                 NGRAN-CGI,
    reestablishmentcellCGI        NGRAN-CGI OPTIONAL,

```

```

-- The above IE shall be present if the Handover Report Type IE is set to the value "HO to wrong cell"
  sourcecellC-RNTI          BIT STRING (SIZE(16))          OPTIONAL,
  targetcellE-UTRAN         EUTRA-CGI                     OPTIONAL,
-- The above IE shall be present if the Handover Report Type IE is set to the value "Inter System ping-pong"
  mobilityInformation        MobilityInformation           OPTIONAL,
  uERLFReportContainer       UERLFReportContainer          OPTIONAL,
  iE-Extensions              ProtocolExtensionContainer { { HOReport-ExtIEs} } OPTIONAL,
  ...
}

HOReport-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  { ID id-ExtendedMobilityInformation        CRITICALITY ignore EXTENSION ExtendedMobilityInformation PRESENCE optional }|
  { ID id-TargetCellCRNTI                   CRITICALITY ignore EXTENSION C-RNTI PRESENCE optional }|
  { ID id-TimeSinceFailure                   CRITICALITY ignore EXTENSION TimeSinceFailure PRESENCE optional },
  ...
}

Hysteresis ::=
  INTEGER (0..30)

-- I

IAB-Authorized ::= ENUMERATED {
  authorized,
  not-authorized,
  ...
}

MobileIAB-MTUserLocationInformation ::= SEQUENCE {
  nRCGI          NR-CGI,
  tAI            TAI,
  iE-Extensions  ProtocolExtensionContainer { { MobileIAB-MTUserLocationInformation-ExtIEs} } OPTIONAL,
  ...
}

MobileIAB-MTUserLocationInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

IAB-Supported ::= ENUMERATED {
  true,
  ...
}

IABNodeIndication ::= ENUMERATED {
  true,
  ...
}

IMSVoiceSupportIndicator ::= ENUMERATED {
  supported,
  not-supported,
  ...
}

```

```

IndexToRFSP ::= INTEGER (1..256, ...)

Indication-of-Bitrate-Adaptation ::= ENUMERATED {uplink, ...}

InfoOnRecommendedCellsAndRANNodesForPaging ::= SEQUENCE {
    recommendedCellsForPaging      RecommendedCellsForPaging,
    recommendRANNodesForPaging      RecommendedRANNodesForPaging,
    iE-Extensions                  ProtocolExtensionContainer { {InfoOnRecommendedCellsAndRANNodesForPaging-ExtIEs} } OPTIONAL,
    ...
}

InfoOnRecommendedCellsAndRANNodesForPaging-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

IntegrityProtectionIndication ::= ENUMERATED {
    required,
    preferred,
    not-needed,
    ...
}

IntegrityProtectionResult ::= ENUMERATED {
    performed,
    not-performed,
    ...
}

IntendedNumberOfPagingAttempts ::= INTEGER (1..16, ...)

InterfacesToTrace ::= BIT STRING (SIZE(8))

ImmediateMDTNr ::= SEQUENCE {
    measurementsToActivate      MeasurementsToActivate,
    m1Configuration             M1Configuration             OPTIONAL,
    -- The above IE shall be present if the Measurements to Activate IE has the first bit set to "1"
    m4Configuration             M4Configuration             OPTIONAL,
    -- The above IE shall be present if the Measurements to Activate IE has the third bit set to "1"
    m5Configuration             M5Configuration             OPTIONAL,
    -- The above IE shall be present if the Measurements to Activate IE has the fourth bit set to "1"
    m6Configuration             M6Configuration             OPTIONAL,
    -- The above IE shall be present if the Measurements to Activate IE has the fifth bit set to "1"
    m7Configuration             M7Configuration             OPTIONAL,
    -- The above IE shall be present if the Measurements to Activate IE has the sixth bit set to "1"
    bluetoothMeasurementConfiguration BluetoothMeasurementConfiguration OPTIONAL,
    wlanMeasurementConfiguration WLANMeasurementConfiguration OPTIONAL,
    mDT-Location-Info           MDT-Location-Info           OPTIONAL,
    sensorMeasurementConfiguration SensorMeasurementConfiguration OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { { ImmediateMDTNr-ExtIEs} } OPTIONAL,
    ...
}

```

```

ImmediateMDTnr-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

InterSystemFailureIndication ::= SEQUENCE {
    uERLFReportContainer      UERLFReportContainer OPTIONAL,
    iE-Extensions             ProtocolExtensionContainer { { InterSystemFailureIndication-ExtIEs} } OPTIONAL,
    ...
}

InterSystemFailureIndication-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

IntersystemSONConfigurationTransfer ::= SEQUENCE {
    transferType              IntersystemSONTransferType,
    intersystemSONInformation IntersystemSONInformation,
    iE-Extensions             ProtocolExtensionContainer { { IntersystemSONConfigurationTransfer-ExtIEs} } OPTIONAL,
    ...
}

IntersystemSONConfigurationTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

IntersystemSONTransferType ::= CHOICE {
    fromEUTRANToNGRAN          FromEUTRANToNGRAN,
    fromNGRANToEUTRAN          FromNGRANToEUTRAN,
    choice-Extensions          ProtocolIE-SingleContainer { { IntersystemSONTransferType-ExtIEs} }
}

IntersystemSONTransferType-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

IntersystemSONeNBID ::= SEQUENCE {
    globaleNBID                GlobalENB-ID,
    selectedEPSTAI             EPS-TAI,
    iE-Extensions              ProtocolExtensionContainer { { IntersystemSONeNBID-ExtIEs} } OPTIONAL,
    ...
}

IntersystemSONeNBID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

IntersystemSONNGRANnodeID ::= SEQUENCE {
    globalRANNodeID            GlobalRANNodeID,
    selectedTAI                TAI,
    iE-Extensions              ProtocolExtensionContainer { { IntersystemSONNGRANnodeID-ExtIEs} } OPTIONAL,
    ...
}

IntersystemSONNGRANnodeID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {

```

```

    ...
}

IntersystemSONInformation ::= CHOICE {
    intersystemSONInformationReport    IntersystemSONInformationReport,
    choice-Extensions                 ProtocolIE-SingleContainer { { IntersystemSONInformation-ExtIEs} }
}

IntersystemSONInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-IntersystemSONInformationRequest    CRITICALITY ignore    TYPE IntersystemSONInformationRequest    PRESENCE mandatory }|
    { ID id-IntersystemSONInformationReply       CRITICALITY ignore    TYPE IntersystemSONInformationReply    PRESENCE mandatory },
    ...
}

-- -----
-- INTER SYSTEM SON INFORMATION REQUEST
-- -----

IntersystemSONInformationRequest ::= CHOICE {
    nGRAN-CellActivation                IntersystemCellActivationRequest,
    resourceStatus                      IntersystemResourceStatusRequest,
    choice-Extensions                   ProtocolIE-SingleContainer { { IntersystemSONInformationRequest-ExtIEs} }
}

IntersystemSONInformationRequest-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

IntersystemCellActivationRequest ::= SEQUENCE {
    activationID                        INTEGER (0..16384, ...),
    cellsToActivateList                CellsToActivateList,
    iE-Extensions                      ProtocolExtensionContainer { { IntersystemCellActivationRequest-ExtIEs} } OPTIONAL,
    ...
}

IntersystemCellActivationRequest-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CellsToActivateList ::= SEQUENCE (SIZE(1..maxnoofCellsinNGRANNode)) OF NGRAN-CGI

-- -----
-- Inter System Resource Status Request
-- -----

IntersystemResourceStatusRequest ::= SEQUENCE {
    reportingSystem                    ReportingSystem,
    reportCharacteristics               ReportCharacteristics,
    reportType                         ReportType,
    iE-Extensions                      ProtocolExtensionContainer { { IntersystemResourceStatusRequest-ExtIEs} } OPTIONAL,
    ...
}

```

```

IntersystemResourceStatusRequest-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ReportingSystem ::= CHOICE {
    eUTRAN          EUTRAN-ReportingSystemIEs,
    nGRAN          NGRAN-ReportingSystemIEs,
    noReporting     NULL,
    choice-Extensions ProtocolIE-SingleContainer { { ReportingSystem-ExtIEs} }
}

ReportingSystem-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

EUTRAN-ReportingSystemIEs ::= SEQUENCE {
    eUTRAN-CellToReportList EUTRAN-CellToReportList,
    iE-Extensions          ProtocolExtensionContainer { {EUTRAN-ReportingSystemIEs-ExtIEs} } OPTIONAL,
    ...
}

EUTRAN-ReportingSystemIEs-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NGRAN-ReportingSystemIEs ::= SEQUENCE {
    nGRAN-CellToReportList NGRAN-CellToReportList,
    iE-Extensions          ProtocolExtensionContainer { {NGRAN-ReportingSystemIEs-ExtIEs} } OPTIONAL,
    ...
}

NGRAN-ReportingSystemIEs-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EUTRAN-CellToReportList ::= SEQUENCE (SIZE(1..maxnoofReportedCells)) OF EUTRAN-CellToReportItem

EUTRAN-CellToReportItem ::= SEQUENCE {
    eCGI          EUTRA-CGI,
    iE-Extensions ProtocolExtensionContainer { {EUTRAN-CellToReportItem-ExtIEs} } OPTIONAL,
    ...
}

EUTRAN-CellToReportItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NGRAN-CellToReportList ::= SEQUENCE (SIZE(1.. maxnoofReportedCells)) OF NGRAN-CellToReportItem

NGRAN-CellToReportItem ::= SEQUENCE {
    nGRAN-CGI          NGRAN-CGI,
    iE-Extensions      ProtocolExtensionContainer { {NGRAN-CellToReportItem-ExtIEs} } OPTIONAL,
    ...
}

```



```
}  
  
NGRAN-CellToReportItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {  
    ...  
}  
  
ReportCharacteristics ::= BIT STRING(SIZE(32))  
  
ReportType ::= CHOICE {  
    eventBasedReporting      EventBasedReportingIEs,  
    periodicReporting        PeriodicReportingIEs,  
    choice-Extensions        ProtocolIE-SingleContainer { { ReportType-ExtIEs}}  
}  
  
ReportType-ExtIEs NGAP-PROTOCOL-IES ::= {  
    ...  
}  
  
EventBasedReportingIEs ::= SEQUENCE {  
    intersystemResourceThresholdLow      IntersystemResourceThreshold,  
    intersystemResourceThresholdHigh     IntersystemResourceThreshold,  
    numberOfMeasurementReportingLevels    NumberOfMeasurementReportingLevels,  
    iE-Extensions                        ProtocolExtensionContainer { {EventBasedReportingIEs-ExtIEs} } OPTIONAL,  
    ...  
}  
  
EventBasedReportingIEs-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {  
    ...  
}  
  
IntersystemResourceThreshold ::= INTEGER(0..100)  
  
NumberOfMeasurementReportingLevels ::= ENUMERATED {n2, n3, n4, n5, n10, ..., n0}  
  
PeriodicReportingIEs ::= SEQUENCE {  
    reportingPeriodicity      ReportingPeriodicity,  
    iE-Extensions             ProtocolExtensionContainer { {PeriodicReportingIEs-ExtIEs} } OPTIONAL,  
    ...  
}  
  
PeriodicReportingIEs-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {  
    ...  
}  
  
ReportingPeriodicity ::= ENUMERATED {  
    stop,  
    single,  
    ms1000,  
    ms2000,  
    ms5000,  
    ms10000,  
    ...  
}
```

```

-----
-- INTER SYSTEM SON INFORMATION REPLY
-----

IntersystemSONInformationReply ::= CHOICE {
    nGRAN-CellActivation      IntersystemCellActivationReply,
    resourceStatus            IntersystemResourceStatusReply,
    choice-Extensions         ProtocolIE-SingleContainer { { IntersystemSONInformationReply-ExtIEs} }
}

IntersystemSONInformationReply-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

IntersystemCellActivationReply ::= SEQUENCE {
    activatedCellList         ActivatedCellList,
    activation-ID             INTEGER(0..16384, ...),
    iE-Extensions             ProtocolExtensionContainer { { IntersystemCellActivationReply-ExtIEs} } OPTIONAL,
    ...
}

IntersystemCellActivationReply-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ActivatedCellList ::= SEQUENCE (SIZE(1..maxnoofCellsinNGRANNode)) OF NGRAN-CGI

-----
-- Inter System Resource Status Reply
-----

IntersystemResourceStatusReply ::= SEQUENCE {
    reportingsystem           ReportingSystem,
    iE-Extensions             ProtocolExtensionContainer { { IntersystemResourceStatusReply-ExtIEs} } OPTIONAL,
    ...
}

IntersystemResourceStatusReply-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-----
-- INTER SYSTEM SON INFORMATION REPORT
-----

IntersystemSONInformationReport ::= CHOICE {
    hOReportInformation        InterSystemHOReport,
    failureIndicationInformation InterSystemFailureIndication,
    choice-Extensions          ProtocolIE-SingleContainer { { IntersystemSONInformationReport-ExtIEs} }
}

```

```

IntersystemSONInformationReport-ExtIEs NGAP-PROTOCOL-IES ::= {
  { ID id-EnergySavingIndication          CRITICALITY ignore TYPE IntersystemCellStateIndication PRESENCE mandatory }|
  { ID id-IntersystemResourceStatusUpdate  CRITICALITY ignore TYPE IntersystemResourceStatusReport PRESENCE mandatory },
  ...
}

IntersystemCellStateIndication ::= SEQUENCE {
  notificationCellList NotificationCellList,
  iE-Extensions          ProtocolExtensionContainer { { IntersystemCellStateIndication-ExtIEs} } OPTIONAL,
  ...
}

IntersystemCellStateIndication-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

NotificationCellList ::= SEQUENCE (SIZE(1.. maxnoofCellsinNGRANNode)) OF NotificationCell-Item

NotificationCell-Item ::= SEQUENCE {
  nGRAN-CGI          NGRAN-CGI,
  notifyFlag          ENUMERATED {activated, deactivated, ...},
  iE-Extensions       ProtocolExtensionContainer { { NotificationCell-Item-ExtIEs} } OPTIONAL,
  ...
}

NotificationCell-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

-- -----
-- Inter System Resource Status Report
-- -----

IntersystemResourceStatusReport ::= SEQUENCE {
  reportingSystem      ResourceStatusReportingSystem,
  iE-Extensions        ProtocolExtensionContainer { { IntersystemResourceStatusReport-ExtIEs} } OPTIONAL,
  ...
}

IntersystemResourceStatusReport-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

ResourceStatusReportingSystem ::= CHOICE {
  eUTRAN-ReportingStatus      EUTRAN-ReportingStatusIEs,
  nGRAN-ReportingStatus       NGRAN-ReportingStatusIEs,
  choice-Extensions           ProtocolIE-SingleContainer { { ResourceStatusReportingSystem-ExtIEs} }
}

ResourceStatusReportingSystem-ExtIEs NGAP-PROTOCOL-IES ::= {
  ...
}

EUTRAN-ReportingStatusIEs ::= SEQUENCE {

```

```

    eUTRAN-CellReportList          EUTRAN-CellReportList,
    iE-Extensions          ProtocolExtensionContainer { {EUTRAN-ReportingStatusIEs-ExtIEs} }  OPTIONAL,
    ...
}

EUTRAN-ReportingStatusIEs-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EUTRAN-CellReportList ::= SEQUENCE (SIZE(1..maxnoofReportedCells)) OF EUTRAN-CellReportItem

EUTRAN-CellReportItem ::= SEQUENCE {
    eCGI                      EUTRA-CGI,
    eUTRAN-CompositeAvailableCapacityGroup  EUTRAN-CompositeAvailableCapacityGroup,
    eUTRAN-NumberOfActiveUEs      EUTRAN-NumberOfActiveUEs                      OPTIONAL,
    eUTRAN-NoofRRCConnections      NGRAN-NoofRRCConnections                    OPTIONAL,
    eUTRAN-RadioResourceStatus      EUTRAN-RadioResourceStatus                  OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {EUTRAN-CellReportItem-ExtIEs} }  OPTIONAL,
    ...
}

EUTRAN-CellReportItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EUTRAN-CompositeAvailableCapacityGroup ::= SEQUENCE {
    dL-CompositeAvailableCapacity      CompositeAvailableCapacity,
    uL-CompositeAvailableCapacity      CompositeAvailableCapacity,
    iE-Extensions          ProtocolExtensionContainer { { EUTRAN-CompositeAvailableCapacityGroup-ExtIEs} }  OPTIONAL,
    ...
}

EUTRAN-CompositeAvailableCapacityGroup-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

CompositeAvailableCapacity ::= SEQUENCE {
    cellCapacityClassValue      INTEGER (1..100, ...)                      OPTIONAL,
    capacityValue              INTEGER (0..100),
    iE-Extensions          ProtocolExtensionContainer { {CompositeAvailableCapacity-ExtIEs} }  OPTIONAL,
    ...
}

CompositeAvailableCapacity-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

EUTRAN-NumberOfActiveUEs ::= INTEGER (0..16777215, ...)

EUTRAN-RadioResourceStatus ::= SEQUENCE {
    dL-GBR-PRB-usage      INTEGER (0..100),
    uL-GBR-PRB-usage      INTEGER (0..100),
    dL-non-GBR-PRB-usage  INTEGER (0..100),

```

```

    uL-non-GBR-PRB-usage          INTEGER (0..100),
    dL-Total-PRB-usage            INTEGER (0..100),
    uL-Total-PRB-usage            INTEGER (0..100),
    dL-scheduling-PDCCH-CCE-usage INTEGER (0..100)   OPTIONAL,
    uL-scheduling-PDCCH-CCE-usage INTEGER (0..100)   OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { {EUTRAN-RadioResourceStatus-ExtIEs} } OPTIONAL,
    ...
}

EUTRAN-RadioResourceStatus-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NGRAN-ReportingStatusIEs ::= SEQUENCE {
    nGRAN-CellReportList          NGRAN-CellReportList,
    iE-Extensions                 ProtocolExtensionContainer { {NGRAN-ReportingStatusIEs-ExtIEs} } OPTIONAL,
    ...
}

NGRAN-ReportingStatusIEs-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NGRAN-CellReportList ::= SEQUENCE (SIZE(1..maxnoofReportedCells)) OF NGRAN-CellReportItem

NGRAN-CellReportItem ::= SEQUENCE {
    nGRAN-CGI                     NGRAN-CGI,
    nGRAN-CompositeAvailableCapacityGroup EUTRAN-CompositeAvailableCapacityGroup,
    nGRAN-NumberOfActiveUEs        NGRAN-NumberOfActiveUEs OPTIONAL,
    nGRAN-NoofRRCConnections       NGRAN-NoofRRCConnections OPTIONAL,
    nGRAN-RadioResourceStatus      NGRAN-RadioResourceStatus OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { {NGRAN-CellReportItem-ExtIEs} } OPTIONAL,
    ...
}

NGRAN-CellReportItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NGRAN-NumberOfActiveUEs ::= INTEGER (0..16777215, ...)

NGRAN-NoofRRCConnections ::= INTEGER (1..65536, ...)

NGRAN-RadioResourceStatus ::= SEQUENCE {
    dL-GBR-PRB-usage-for-MIMO      INTEGER (0..100),
    uL-GBR-PRB-usage-for-MIMO      INTEGER (0..100),
    dL-non-GBR-PRB-usage-for-MIMO  INTEGER (0..100),
    uL-non-GBR-PRB-usage-for-MIMO  INTEGER (0..100),
    dL-Total-PRB-usage-for-MIMO    INTEGER (0..100),
    uL-Total-PRB-usage-for-MIMO    INTEGER (0..100),
    iE-Extensions                 ProtocolExtensionContainer { { NGRAN-RadioResourceStatus-ExtIEs} } OPTIONAL,
    ...
}

```

```

NGRAN-RadioResourceStatus-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

InterSystemHOReport ::= SEQUENCE {
    handoverReportType      InterSystemHandoverReportType,
    iE-Extensions           ProtocolExtensionContainer { { InterSystemHOReport-ExtIEs} }      OPTIONAL,
    ...
}

InterSystemHOReport-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

InterSystemHandoverReportType ::= CHOICE {
    tooearlyIntersystemHO      TooeearlyIntersystemHO,
    intersystemUnnecessaryHO    IntersystemUnnecessaryHO,
    choice-Extensions          ProtocolIE-SingleContainer { { InterSystemHandoverReportType-ExtIEs} }
}

InterSystemHandoverReportType-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-IntersystemMobilityFailureforVoiceFallback CRITICALITY ignore TYPE IntersystemMobilityFailureforVoiceFallback PRESENCE mandatory },
    ...
}

IntersystemMobilityFailureforVoiceFallback ::= SEQUENCE {
    sourcecellID              NGRAN-CGI,
    targetcellID              EUTRA-CGI,
    reconnectCellID           EUTRA-CGI                      OPTIONAL,
    uERLFReportContainer       UERLFReportContainer          OPTIONAL,
    iE-Extensions             ProtocolExtensionContainer { { IntersystemMobilityFailureforVoiceFallback-ExtIEs} }      OPTIONAL,
    ...
}

IntersystemMobilityFailureforVoiceFallback-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

IntersystemUnnecessaryHO ::= SEQUENCE {
    sourcecellID              NGRAN-CGI,
    targetcellID              EUTRA-CGI,
    earlyIRATHO               ENUMERATED {true, false, ...},
    candidateCellList          CandidateCellList,
    iE-Extensions             ProtocolExtensionContainer { { IntersystemUnnecessaryHO-ExtIEs} }      OPTIONAL,
    ...
}

IntersystemUnnecessaryHO-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

InventoryRequestTransfer ::= SEQUENCE {

```

```

    protocolIEs      ProtocolIE-Container      { { InventoryRequestTransferIEs} },
    ...
}

InventoryRequestTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-AIoT-CorrelationIdentifier          CRITICALITY reject TYPE AIoT-CorrelationIdentifier          PRESENCE mandatory }|
    { ID id-AIoT-DeviceIdentificationRequested   CRITICALITY reject TYPE AIoT-DeviceIdentificationRequested PRESENCE mandatory }|
    { ID id-AIoT-RequestedServiceAreaInformation CRITICALITY reject TYPE AIoT-RequestedServiceAreaInformation PRESENCE mandatory }|
    { ID id-AIoT-InventoryAssistanceInformation  CRITICALITY reject TYPE AIoT-InventoryAssistanceInformation PRESENCE mandatory }|
    { ID id-AIoT-FollowonCommandIndication       CRITICALITY reject TYPE AIoT-FollowonCommandIndication PRESENCE optional }|
    { ID id-AIoT-SecurityInfo                   CRITICALITY reject TYPE AIoT-SecurityInfo           PRESENCE mandatory },
    ...
}

InventoryResponseTransfer ::= SEQUENCE {
    correlationIdentifier      AIoT-CorrelationIdentifier,
    criticalityDiagnostics     CriticalityDiagnostics
    iE-Extensions              ProtocolExtensionContainer { { InventoryResponseTransfer-ExtIEs} }
    ...
}

InventoryResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

InventoryFailureTransfer ::= SEQUENCE {
    correlationIdentifier      AIoT-CorrelationIdentifier,
    cause                     Cause,
    criticalityDiagnostics     CriticalityDiagnostics
    iE-Extensions              ProtocolExtensionContainer { { InventoryFailureTransfer-ExtIEs} }
    ...
}

InventoryFailureTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

InventoryReportTransfer ::= SEQUENCE {
    correlationIdentifier      AIoT-CorrelationIdentifier,
    globalGNB-ID              GlobalGNB-ID,
    readerReportList           AIoT-ReaderReportList
    inventoryCompleteIndication ENUMERATED {true, ...}
    iE-Extensions              ProtocolExtensionContainer { { InventoryReportTransfer-ExtIEs} }
    ...
}

InventoryReportTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

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```

LAC ::= OCTET STRING (SIZE (2))

LAI ::= SEQUENCE {
    plMNIdentity      PLMNIdentity,
    lAC               LAC,
    iE-Extensions     ProtocolExtensionContainer { {LAI-ExtIEs} } OPTIONAL,
    ...
}

LAI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

LastVisitedCellInformation ::= CHOICE {
    nGRANCell        LastVisitedNGRANCellInformation,
    eUTRANCell       LastVisitedEUTRANCellInformation,
    uTRANCell        LastVisitedUTRANCellInformation,
    gERANCell        LastVisitedGERANCellInformation,
    choice-Extensions ProtocolIE-SingleContainer { {LastVisitedCellInformation-ExtIEs} }
}

LastVisitedCellInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

LastVisitedCellItem ::= SEQUENCE {
    lastVisitedCellInformation LastVisitedCellInformation,
    iE-Extensions             ProtocolExtensionContainer { {LastVisitedCellItem-ExtIEs} } OPTIONAL,
    ...
}

LastVisitedCellItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

LastVisitedEUTRANCellInformation ::= OCTET STRING

LastVisitedGERANCellInformation ::= OCTET STRING

LastVisitedNGRANCellInformation ::= SEQUENCE {
    globalCellID      NGRAN-CGI,
    cellType          CellType,
    timeUEStayedInCell TimeUEStayedInCell,
    timeUEStayedInCellEnhancedGranularity TimeUEStayedInCellEnhancedGranularity OPTIONAL,
    hOCauseValue       Cause OPTIONAL,
    iE-Extensions     ProtocolExtensionContainer { {LastVisitedNGRANCellInformation-ExtIEs} } OPTIONAL,
    ...
}

LastVisitedNGRANCellInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-LastVisitedPSCellList CRITICALITY ignore EXTENSION LastVisitedPSCellList PRESENCE optional },
    ...
}

```


LastVisitedPSCellList ::= SEQUENCE (SIZE(1..maxnoofPSCellsPerPrimaryCellinUEHistoryInfo)) OF LastVisitedPSCellInformation

```
LastVisitedPSCellInformation ::= SEQUENCE {
    pSCellID          NGRAN-CGI          OPTIONAL,
    timeStay          INTEGER (0..40950),
    iE-Extensions     ProtocolExtensionContainer { {LastVisitedPSCellInformation-ExtIEs} }  OPTIONAL,
    ...
}
```

```
LastVisitedPSCellInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-SCGActivationTime  CRITICALITY ignore  EXTENSION SCGActivationTime PRESENCE optional },
    ...
}
```

LastVisitedUTRANCellInformation ::= OCTET STRING

```
LineType ::= ENUMERATED {
    dsl,
    pon,
    ...
}
```

```
LocationReportingAdditionalInfo ::= ENUMERATED {
    includePSCell,
    ...
}
```

LocationReportingReferenceID ::= INTEGER (1..64, ...)

```
LocationReportingRequestType ::= SEQUENCE {
    eventType          EventType,
    reportArea         ReportArea,
    areaOfInterestList AreaOfInterestList          OPTIONAL,
    locationReportingReferenceIDToBeCancelled LocationReportingReferenceID  OPTIONAL,
    -- The above IE shall be present if the Event Type IE is set to the value "stop UE presence in the area of interest"
    iE-Extensions     ProtocolExtensionContainer { {LocationReportingRequestType-ExtIEs} }  OPTIONAL,
    ...
}
```

```
LocationReportingRequestType-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-LocationReportingAdditionalInfo CRITICALITY ignore  EXTENSION LocationReportingAdditionalInfo  PRESENCE optional }|
    { ID id-AdditionalCancelledlocationReportingReferenceIDList CRITICALITY reject  EXTENSION AdditionalCancelledlocationReportingReferenceIDList
    PRESENCE optional }|
    { ID id-Aerial-UE-FlightInformationReportingControllist CRITICALITY ignore  EXTENSION Aerial-UE-FlightInformationReportingControllist PRESENCE
    optional},
    ...
}
```

```
Aerial-UE-FlightInformationReportingControl ::= SEQUENCE {
    higher-Altitude-Threshold  Altitude,
    lower-Altitude-Threshold  Altitude,
    aerial-UE-report-periodicity AerialUEReportingPeriodicity          OPTIONAL,
```

```

    areaID                Area-ID,
    iE-Extensions         ProtocolExtensionContainer { {Aerial-UE-FlightInformationReportingControl-ExtIEs}} OPTIONAL,
    ...
}

Aerial-UE-FlightInformationReportingControl-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

Aerial-UE-FlightInformationReportingControlList ::= SEQUENCE (SIZE(1.. maxnoofFlightInfoReportControl)) OF Aerial-UE-
FlightInformationReportingControlItem

Aerial-UE-FlightInformationReportingControlItem ::= SEQUENCE {
    aerial-UE-FlightInformationReportingControl    Aerial-UE-FlightInformationReportingControl,
    iE-Extensions         ProtocolExtensionContainer { {Aerial-UE-FlightInformationReportingControlItem-ExtIEs}} OPTIONAL,
    ...
}

Aerial-UE-FlightInformationReportingControlItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

Area-ID ::= CHOICE {
    tAI                TAI,
    globalRANNodeID    GlobalRANNodeID,
    nR-CGI             NR-CGI,
    choice-Extensions  ProtocolIE-SingleContainer { {Area-ID-ExtIEs} }
}

Area-ID-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

LoggedMDTNr ::= SEQUENCE {
    loggingInterval      LoggingInterval,
    loggingDuration      LoggingDuration,
    loggedMDTTTrigger     LoggedMDTTTrigger,
    bluetoothMeasurementConfiguration    BluetoothMeasurementConfiguration    OPTIONAL,
    wlanMeasurementConfiguration    WLANMeasurementConfiguration    OPTIONAL,
    sensorMeasurementConfiguration    SensorMeasurementConfiguration    OPTIONAL,
    areaScopeOfNeighCellsList        AreaScopeOfNeighCellsList        OPTIONAL,
    iE-Extensions         ProtocolExtensionContainer { {LoggedMDTNr-ExtIEs} } OPTIONAL,
    ...
}

LoggedMDTNr-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-EarlyMeasurement          CRITICALITY ignore  EXTENSION EarlyMeasurement          PRESENCE optional },
    ...
}

LoggingInterval ::= ENUMERATED {

```

```

    ms320, ms640, ms1280, ms2560, ms5120, ms10240, ms20480, ms30720, ms40960, ms61440,
    infinity,
    ...
}

LoggingDuration ::= ENUMERATED {m10, m20, m40, m60, m90, m120, ...}

Links-to-log ::= ENUMERATED {
    uplink,
    downlink,
    both-uplink-and-downlink,
    ...
}

LoggedMDTTrigger ::= CHOICE{
    periodical          NULL,
    eventTrigger        EventTrigger,
    choice-Extensions   ProtocolIE-SingleContainer { {LoggedMDTTrigger-ExtIEs} }
}

LoggedMDTTrigger-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

LPWUSPSAssistanceInformation ::= SEQUENCE {
    lPWUSCnsubgroupID    LPWUSCnsubgroupID,
    iE-Extensions        ProtocolExtensionContainer { {LPWUSPSAssistanceInformation-ExtIEs} } OPTIONAL,
    ...
}

LPWUSPSAssistanceInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

LPWUSCnsubgroupID ::= INTEGER (0..30, ...)

LPWUSDisableIndication ::= ENUMERATED {true, ...}

LTEM-Indication ::= ENUMERATED {lte-m, ...}

LTEUERLFReportContainer ::= OCTET STRING

LTEV2XServicesAuthorized ::= SEQUENCE {
    vehicleUE            VehicleUE                                OPTIONAL,
    pedestrianUE         PedestrianUE                            OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { {LTEV2XServicesAuthorized-ExtIEs} } OPTIONAL,
    ...
}

LTEV2XServicesAuthorized-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

LTEUESidelinkAggregateMaximumBitrate ::= SEQUENCE {

```

```

    uESidelinkAggregateMaximumBitRate      BitRate,
    iE-Extensions      ProtocolExtensionContainer { {LTEUE-Sidelink-Aggregate-MaximumBitrates-ExtIEs} } OPTIONAL,
    ...
}

LTEUE-Sidelink-Aggregate-MaximumBitrates-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

LTE-A2X-ServicesAuthorized ::= SEQUENCE {
    aerialUE      ENUMERATED {authorized, not-authorized, ...}      OPTIONAL,
    aerialControllerUE      ENUMERATED {authorized, not-authorized, ...}      OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {LTE-A2X-ServicesAuthorized-ExtIEs} }      OPTIONAL,
    ...
}

LTE-A2X-ServicesAuthorized-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- M

MaskedIMEISV ::= BIT STRING (SIZE(64))

MaximumDataBurstVolume ::= INTEGER (0..4095, ..., 4096.. 2000000)

MessageIdentifier ::= BIT STRING (SIZE(16))

MaximumIntegrityProtectedDataRate ::= ENUMERATED {
    bitrate64kbs,
    maximum-UE-rate,
    ...
}

MBS-AreaSessionID ::= INTEGER (0..65535, ...)

CommServiceType ::= ENUMERATED {mbs-broadcast, mbs-multicast, ..., unicast}

MBS-DataForwardingResponseMRBList ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF MBS-DataForwardingResponseMRBItem

MBS-DataForwardingResponseMRBItem ::= SEQUENCE {
    mRB-ID      MRB-ID,
    dL-Forwarding-UPTNLInformation      UPTransportLayerInformation,
    mRB-ProgressInformation      MRB-ProgressInformation      OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { { MBS-DataForwardingResponseMRBItem-ExtIEs} }      OPTIONAL,
    ...
}

MBS-DataForwardingResponseMRBItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-MappingandDataForwardingRequestList ::= SEQUENCE (SIZE(1..maxnoofMRBs)) OF MBS-MappingandDataForwardingRequestItem

```

```

MBS-MappingandDataForwardingRequestItem ::= SEQUENCE {
    mRB-ID                MRB-ID,
    mBS-QoSFlowList        MBS-QoSFlowList,
    mRB-ProgressInformation MRB-ProgressInformation OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { MBS-MappingandDataForwardingRequestItem-ExtIEs} } OPTIONAL,
    ...
}

MBS-MappingandDataForwardingRequestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-QoSFlowList ::= SEQUENCE (SIZE(1..maxnoofMBSQoSFlows)) OF QoSFlowIdentifier

MRB-ProgressInformation ::= CHOICE {
    pDCP-SN-Length12    INTEGER (0..4095),
    pDCP-SN-Length18    INTEGER (0..262143),
    choice-Extensions   ProtocolIE-SingleContainer { { MRB-ProgressInformation-ExtIEs} }
}

MRB-ProgressInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

MBS-QoSFlowsToBeSetupList ::= SEQUENCE (SIZE(1.. maxnoofMBSQoSFlows)) OF MBS-QoSFlowsToBeSetupItem

MBS-QoSFlowsToBeSetupItem ::= SEQUENCE {
    mBSqosFlowIdentifier    QoSFlowIdentifier,
    mBSqosFlowLevelQosParameters QoSFlowLevelQosParameters,
    iE-Extensions          ProtocolExtensionContainer { {MBS-QoSFlowsToBeSetupItem-ExtIEs} } OPTIONAL,
    ...
}

MBS-QoSFlowsToBeSetupItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-ServiceArea ::= CHOICE {
    locationindependent    MBS-ServiceAreaInformation,
    locationdependent      MBS-ServiceAreaInformationList,
    choice-Extensions      ProtocolIE-SingleContainer { {MBS-ServiceArea-ExtIEs} }
}

MBS-ServiceArea-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

MBS-ServiceAreaInformationList ::= SEQUENCE (SIZE(1..maxnoofMBSServiceAreaInformation)) OF MBS-ServiceAreaInformationItem

MBS-ServiceAreaInformationItem ::= SEQUENCE {
    mBS-AreaSessionID      MBS-AreaSessionID,

```

```

    mBS-ServiceAreaInformation    MBS-ServiceAreaInformation,
    iE-Extensions                 ProtocolExtensionContainer { {MBS-ServiceAreaInformationItem-ExtIes} } OPTIONAL,
    ...
}

MBS-ServiceAreaInformationItem-ExtIes NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-ServiceAreaInformation ::= SEQUENCE {
    mBS-ServiceAreaCellList    MBS-ServiceAreaCellList                OPTIONAL,
    mBS-ServiceAreaTAIList     MBS-ServiceAreaTAIList                OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { {MBS-ServiceAreaInformation-ExtIes} } OPTIONAL,
    ...
}

MBS-ServiceAreaInformation-ExtIes NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-MBS-IntendedServiceAreaList    CRITICALITY ignore    EXTENSION MBS-IntendedServiceAreaList    PRESENCE optional },
    ...
}

MBS-ServiceAreaCellList ::= SEQUENCE (SIZE(1.. maxnoofCellsforMBS)) OF NR-CGI

MBS-ServiceAreaTAIList ::= SEQUENCE (SIZE(1.. maxnoofTAIforMBS)) OF TAI

MBS-IntendedServiceAreaList ::= SEQUENCE (SIZE(1..maxnoofIntendedAreasforMBS)) OF IntendedServiceAreaCoordinates

IntendedServiceAreaCoordinates ::= OCTET STRING

MBS-SessionID ::= SEQUENCE {
    tMGI                TMGI,
    nID                 NID
    iE-Extensions       ProtocolExtensionContainer { {MBS-SessionID-ExtIes} }    OPTIONAL,
    ...
}

MBS-SessionID-ExtIes NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-SessionFailedtoSetupList ::= SEQUENCE (SIZE(1.. maxnoofMBSsessions)) OF MBS-SessionFailedtoSetupItem

MBS-SessionFailedtoSetupItem ::= SEQUENCE {
    mBS-SessionID        MBS-SessionID,
    mBS-AreaSessionID    MBS-AreaSessionID                OPTIONAL,
    cause                Cause,
    iE-Extensions        ProtocolExtensionContainer { { MBS-SessionFailedtoSetupItem-ExtIes} }    OPTIONAL,
    ...
}

MBS-SessionFailedtoSetupItem-ExtIes NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

MBS-ActiveSessionInformation-SourcetoTargetList ::= SEQUENCE (SIZE(1..maxnoofMBSSessionsofUE)) OF MBS-ActiveSessionInformation-SourcetoTargetItem

```

MBS-ActiveSessionInformation-SourcetoTargetItem ::= SEQUENCE {
    mBS-SessionID           MBS-SessionID,
    mBS-AreaSessionID       MBS-AreaSessionID           OPTIONAL,
    mBS-ServiceArea         MBS-ServiceArea             OPTIONAL,
    mBS-QoSFlowsToBeSetupList MBS-QoSFlowsToBeSetupList,
    mBS-MappingandDataForwardingRequestList MBS-MappingandDataForwardingRequestList OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { MBS-ActiveSessionInformation-SourcetoTargetItem-ExtIEs} } OPTIONAL,
    ...
}

```

```

MBS-ActiveSessionInformation-SourcetoTargetItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

MBS-ActiveSessionInformation-TargettoSourceList ::= SEQUENCE (SIZE(1..maxnoofMBSSessionsofUE)) OF MBS-ActiveSessionInformation-TargettoSourceItem

```

MBS-ActiveSessionInformation-TargettoSourceItem ::= SEQUENCE {
    mBS-SessionID           MBS-SessionID,
    mBS-DataForwardingResponseMRBList MBS-DataForwardingResponseMRBList           OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { MBS-ActiveSessionInformation-TargettoSourceItem-ExtIEs} } OPTIONAL,
    ...
}

```

```

MBS-ActiveSessionInformation-TargettoSourceItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

MBS-AssistanceInformation ::= ENUMERATED {true, ...}

```

MBSSessionSetupOrModFailureTransfer ::= SEQUENCE {
    cause           Cause,
    criticalityDiagnostics CriticalityDiagnostics OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { MBSSessionSetupOrModFailureTransfer-ExtIEs} } OPTIONAL,
    ...
}

```

```

MBSSessionSetupOrModFailureTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

MBSSessionSetupResponseList ::= SEQUENCE (SIZE(1.. maxnoofMBSSessions)) OF MBSSessionSetupResponseItem

```

MBSSessionSetupResponseItem ::= SEQUENCE {
    mBS-SessionID           MBS-SessionID,
    mBS-AreaSessionID       MBS-AreaSessionID           OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { MBSSessionSetupResponseItem-ExtIEs} } OPTIONAL,
    ...
}

```

```

MBSSessionSetupResponseItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

MBSSessionSetupOrModRequestTransfer ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {MBSSessionSetupOrModRequestTransferIEs} },
    ...
}

MBSSessionSetupOrModRequestTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionTNLInfo5GC          CRITICALITY reject TYPE MBS-SessionTNLInfo5GC          PRESENCE optional    }|
    { ID id-MBS-QoSFlowsToBeSetupModList    CRITICALITY reject TYPE MBS-QoSFlowsToBeSetupList    PRESENCE mandatory   }|
    { ID id-MBS-SessionFSAIDList            CRITICALITY ignore TYPE MBS-SessionFSAIDList            PRESENCE optional    }|
    { ID id-SupportedUETypeList             CRITICALITY ignore TYPE SupportedUETypeList             PRESENCE optional    }|
    { ID id-MBS-NGUFailureIndication        CRITICALITY reject TYPE MBS-NGUFailureIndication        PRESENCE optional    },
    ...
}

SupportedUETypeList ::= SEQUENCE (SIZE(1.. maxnoofUETypes)) OF SupportedUEType

SupportedUEType ::= ENUMERATED {non-RedCap-eRedCap-UE, redCap-eRedCap-UE, ...}

MBS-SessionFSAIDList ::= SEQUENCE (SIZE(1.. maxnoofMBSFSAs)) OF MBS-SessionFSAID

MBS-SessionFSAID ::= OCTET STRING (SIZE(3))

MBSSessionReleaseResponseTransfer ::= SEQUENCE {
    mBS-SessionTNLInfoNGRAN      MBS-SessionTNLInfoNGRAN      OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {MBSSessionReleaseResponseTransfer-ExtIEs} } OPTIONAL,
    ...
}

MBSSessionReleaseResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBSSessionSetupOrModResponseTransfer ::= SEQUENCE {
    mBS-SessionTNLInfoNGRAN      MBS-SessionTNLInfoNGRAN      OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {MBSSessionSetupOrModResponseTransfer-ExtIEs} } OPTIONAL,
    ...
}

MBSSessionSetupOrModResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-SupportIndicator ::= ENUMERATED {
    multicast-supported,
    ...,
    multicast-supported-with-reception-in-RRC-inactive
}

MBS-SessionTNLInfo5GC ::= CHOICE {
    locationindependent          SharedNGU-MulticastTNLInformation,
    locationindependent          MBS-SessionTNLInfo5GCList,
    choice-Extensions            ProtocolIE-SingleContainer { {MBS-SessionTNLInfo5GC-ExtIEs} }
}

```



```

MBS-SessionTNLInfo5GC-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

MBS-SessionTNLInfo5GCList ::= SEQUENCE (SIZE(1..maxnoofMBSServiceAreaInformation)) OF MBS-SessionTNLInfo5GCItem

MBS-SessionTNLInfo5GCItem ::= SEQUENCE {
    mBS-AreaSessionID                MBS-AreaSessionID,
    sharedNGU-MulticastTNLInformation SharedNGU-MulticastTNLInformation,
    iE-Extensions                    ProtocolExtensionContainer { {MBS-SessionTNLInfo5GCItem-ExtIEs} } OPTIONAL,
    ...
}

MBS-SessionTNLInfo5GCItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-SessionTNLInfoNGRAN ::= CHOICE {
    locationindependent    UPTransportLayerInformation,
    locationdependent      MBS-SessionTNLInfoNGRANList,
    choice-Extensions      ProtocolIE-SingleContainer { {MBS-SessionTNLInfoNGRAN-ExtIEs} }
}

MBS-SessionTNLInfoNGRAN-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

MBS-SessionTNLInfoNGRANList ::= SEQUENCE (SIZE(1..maxnoofMBSServiceAreaInformation)) OF MBS-SessionTNLInfoNGRANItem

MBS-SessionTNLInfoNGRANItem ::= SEQUENCE {
    mBS-AreaSessionID                MBS-AreaSessionID,
    sharedNGU-UnicastTNLInformation  UPTransportLayerInformation OPTIONAL,
    iE-Extensions                    ProtocolExtensionContainer { {MBS-SessionTNLInfoNGRANItem-ExtIEs} } OPTIONAL,
    ...
}

MBS-SessionTNLInfoNGRANItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-DistributionReleaseRequestTransfer ::= SEQUENCE {
    mBS-SessionID                MBS-SessionID,
    mBS-AreaSessionID            MBS-AreaSessionID OPTIONAL,
    sharedNGU-UnicastTNLInformation UPTransportLayerInformation OPTIONAL,
    cause                        Cause,
    iE-Extensions                ProtocolExtensionContainer { { MBS-DistributionReleaseRequesTransfer-ExtIEs} } OPTIONAL,
    ...
}

MBS-DistributionReleaseRequesTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-DistributionSetupRequestTransfer ::= SEQUENCE {

```

```

    mBS-SessionID                MBS-SessionID,
    mBS-AreaSessionID            MBS-AreaSessionID                OPTIONAL,
    sharedNGU-UnicastTNLInformation  UPTransportLayerInformation  OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { { MBS-DistributionSetupRequestTransfer-ExtIEs} }  OPTIONAL,
    ...
}

MBS-DistributionSetupRequestTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
{ ID id-TAIMBSSupportList        CRITICALITY ignore  EXTENSION TAIMBSSupportList        PRESENCE optional  },
    ...
}

MBS-DistributionSetupResponseTransfer ::= SEQUENCE {
    mBS-SessionID                MBS-SessionID,
    mBS-AreaSessionID            MBS-AreaSessionID                OPTIONAL,
    sharedNGU-MulticastTNLInformation  SharedNGU-MulticastTNLInformation  OPTIONAL,
    mBS-QoSFlowsToBeSetupList        MBS-QoSFlowsToBeSetupList,
    mBSSessionStatus              MBSSESSIONSTATUS,
    mBS-ServiceArea              MBS-ServiceArea                OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {MBS-DistributionSetupResponseTransfer-ExtIEs} }  OPTIONAL,
    ...
}

MBS-DistributionSetupResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-DistributionSetupUnsuccessfulTransfer ::= SEQUENCE {
    mBS-SessionID                MBS-SessionID,
    mBS-AreaSessionID            MBS-AreaSessionID                OPTIONAL,
    cause                        Cause,
    criticalityDiagnostics        CriticalityDiagnostics          OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { { MBS-DistributionSetupUnsuccessfulTransfer-ExtIEs} }  OPTIONAL,
    ...
}

MBS-DistributionSetupUnsuccessfulTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-NGUFailureIndication ::= CHOICE {
    locationindependent          MBS-UP-FailureIndication,
    locationdependent            MBS-UP-FailureIndicationList,
    choice-Extensions            ProtocolIE-SingleContainer { {MBS-NGUFailureIndication-ExtIEs} }
}

MBS-NGUFailureIndication-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

MBS-UP-FailureIndicationList ::= SEQUENCE (SIZE(1..maxnoofMBSServiceAreaInformation)) OF MBS-UP-FailureIndicationItem

MBS-UP-FailureIndicationItem ::= SEQUENCE {

```

```

    mBS-AreaSessionID          MBS-AreaSessionID,
    mBS-UP-FailureIndication    MBS-UP-FailureIndication,
    iE-Extensions              ProtocolExtensionContainer { {MBS-UP-FailureIndicationItem-ExtIEs} }    OPTIONAL,
    ...
}

MBS-UP-FailureIndicationItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-UP-FailureIndication ::= ENUMERATED {ng-u-path-failure-detected, ...}

MBS-SessionSetupRequestList ::= SEQUENCE (SIZE(1..maxnoofMBSSessions)) OF MBS-SessionSetupRequestItem

MBS-SessionSetupRequestItem ::= SEQUENCE {
    mBS-SessionID                MBS-SessionID,
    mBS-AreaSessionID            MBS-AreaSessionID                                OPTIONAL,
    associatedMBSQosFlowSetupRequestList AssociatedMBSQosFlowSetupRequestList        OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { { MBS-SessionSetupRequestItem-ExtIEs} }    OPTIONAL,
    ...
}

MBS-SessionSetupRequestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-MBS-AssistanceInformation          CRITICALITY ignore  EXTENSION MBS-AssistanceInformation  PRESENCE optional },
    ...
}

MBS-SessionSetuporModifyRequestList ::= SEQUENCE (SIZE(1..maxnoofMBSSessions)) OF MBS-SessionSetuporModifyRequestItem

MBS-SessionSetuporModifyRequestItem ::= SEQUENCE {
    mBS-SessionID                MBS-SessionID,
    mBS-AreaSessionID            MBS-AreaSessionID                                OPTIONAL,
    associatedMBSQosFlowSetuporModifyRequestList AssociatedMBSQosFlowSetuporModifyRequestList    OPTIONAL,
    mBS-QosFlowToReleaseList      QosFlowListWithCause                          OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer {{MBS-SessionSetuporModifyRequestItem-ExtIEs}}    OPTIONAL,
    ...
}

MBS-SessionSetuporModifyRequestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-MBS-AssistanceInformation          CRITICALITY ignore  EXTENSION MBS-AssistanceInformation  PRESENCE optional },
    ...
}

MBS-SessionToReleaseList ::= SEQUENCE (SIZE(1..maxnoofMBSSessions)) OF MBS-SessionToReleaseItem

MBS-SessionToReleaseItem ::= SEQUENCE {
    mBS-SessionID                MBS-SessionID,
    cause                        Cause,
    iE-Extensions                ProtocolExtensionContainer { { MBS-SessionToReleaseItem-ExtIEs} }    OPTIONAL,
    ...
}

MBS-SessionToReleaseItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {

```

```

    ...
}

MBSSessionStatus ::= ENUMERATED {
    activated,
    deactivated,
    ...
}

MicoAllPLMN ::= ENUMERATED {
    true,
    ...
}

MICOModeIndication ::= ENUMERATED {
    true,
    ...
}

MobilityInformation ::= BIT STRING (SIZE(16))

ExtendedMobilityInformation ::= BIT STRING (SIZE(32))

MobilityRestrictionList ::= SEQUENCE {
    servingPLMN                PLMNIdentity,
    equivalentPLMNs            EquivalentPLMNs                OPTIONAL,
    rATRestrictions            RATRestrictions                OPTIONAL,
    forbiddenAreaInformation    ForbiddenAreaInformation        OPTIONAL,
    serviceAreaInformation      ServiceAreaInformation          OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { {MobilityRestrictionList-ExtIEs} } OPTIONAL,
    ...
}

MobilityRestrictionList-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-LastEUTRAN-PLMNIdentity          CRITICALITY ignore EXTENSION PLMNIdentity          PRESENCE optional }|
    { ID id-CNTTypeRestrictionsForServing      CRITICALITY ignore EXTENSION CNTTypeRestrictionsForServing PRESENCE optional }|
    { ID id-CNTTypeRestrictionsForEquivalent    CRITICALITY ignore EXTENSION CNTTypeRestrictionsForEquivalent PRESENCE optional }|
    { ID id-NPN-MobilityInformation            CRITICALITY reject  EXTENSION NPN-MobilityInformation PRESENCE optional },
    ...
}

MDT-AlignmentInfo ::= CHOICE {
    s-basedMDT                NGRANTraceID,
    choice-Extensions          ProtocolIE-SingleContainer { { MDT-AlignmentInfo-ExtIEs} }
}

MDT-AlignmentInfo-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

MDTPLMNList ::= SEQUENCE (SIZE(1..maxnoofMDTPLMNs)) OF PLMNIdentity

MDTPLMNModificationList ::= SEQUENCE (SIZE(0..maxnoofMDTPLMNs)) OF PLMNIdentity

```

```

MDT-Configuration ::= SEQUENCE {
    mdt-Config-NR          MDT-Configuration-NR          OPTIONAL,
    mdt-Config-EUTRA       MDT-Configuration-EUTRA       OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { MDT-Configuration-ExtIEs} } OPTIONAL,
    ...
}

MDT-Configuration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-MN-only-MDT-collection          CRITICALITY ignore  EXTENSION MN-only-MDT-collection          PRESENCE optional  },
    ...
}

MN-only-MDT-collection ::= ENUMERATED {mn-only, ...}

MDT-Configuration-NR ::= SEQUENCE {
    mdt-Activation          MDT-Activation,
    areaScopeOfMDT          AreaScopeOfMDT-NR,
    mDTModeNr              MDTModeNr,
    signallingBasedMDTPLMNList MDTPLMNList                OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { MDT-Configuration-NR-ExtIEs} }  OPTIONAL,
    ...
}

MDT-Configuration-NR-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-PNI-NPN-AreaScopeofMDT          CRITICALITY ignore  EXTENSION PNI-NPN-AreaScopeofMDT          PRESENCE optional  }|
    { ID id-NetworkSliceAreaScopeofMDT      CRITICALITY ignore  EXTENSION NetworkSliceAreaScopeofMDT      PRESENCE optional },
    ...
}

MDT-Configuration-EUTRA ::= SEQUENCE {
    mdt-Activation          MDT-Activation,
    areaScopeOfMDT          AreaScopeOfMDT-EUTRA,
    mDTMode                 MDTModeEutra,
    signallingBasedMDTPLMNList MDTPLMNList                OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { MDT-Configuration-EUTRA-ExtIEs} }  OPTIONAL,
    ...
}

MDT-Configuration-EUTRA-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MDT-Activation ::= ENUMERATED {
    immediate-MDT-only,
    logged-MDT-only,
    immediate-MDT-and-Trace,
    ...
}

MDTModeNr ::= CHOICE {
    immediateMDTNr          ImmediateMDTNr,
    loggedMDTNr             LoggedMDTNr,
    choice-Extensions       ProtocolIE-SingleContainer { {MDTModeNr-ExtIEs} }
}

```

```

}

MDTModeNr-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

MDTModeEutra ::= OCTET STRING

MeasurementsToActivate ::= BIT STRING(SIZE(8))

MobileIAB-Authorized ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

MobileIABNodeIndication ::= ENUMERATED {
    true,
    ...
}

MobileIAB-Supported ::= ENUMERATED {
    true,
    ...
}

MonitoringRequestonAvailableBitrate ::= SEQUENCE {
    monitoringRequest           MonitoringRequest,
    dlAvailableBitrateReportThresholds AvailableBitrateReportThresholdList OPTIONAL,
    -- The above IE shall be present if the Monitoring Request IE is set to the value "dl" or "both"
    ulAvailableBitrateReportThresholds AvailableBitrateReportThresholdList OPTIONAL,
    -- The above IE shall be present if the Monitoring Request IE is set to the value "ul" or "both"
    iE-Extensions               ProtocolExtensionContainer { { MonitoringRequestonAvailableBitrate-ExtIEs} } OPTIONAL,
    ...
}

MonitoringRequestonAvailableBitrate-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MonitoringRequest ::= ENUMERATED {ul, dl, both, stop, ...}

MRB-ID ::= INTEGER (1..512, ...)

MulticastSessionActivationRequestTransfer ::= SEQUENCE {
    mBS-SessionID              MBS-SessionID,
    iE-Extensions               ProtocolExtensionContainer { { MulticastSessionActivationRequestTransfer-ExtIEs} } OPTIONAL,
    ...
}

MulticastSessionActivationRequestTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

MulticastSessionDeactivationRequestTransfer ::= SEQUENCE {
    mBS-SessionID          MBS-SessionID,
    iE-Extensions          ProtocolExtensionContainer { { MulticastSessionDeactivationRequestTransfer-ExtIEs} } OPTIONAL,
    ...
}

MulticastSessionDeactivationRequestTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MulticastSessionUpdateRequestTransfer ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container          { {MulticastSessionUpdateRequestTransferIEs} },
    ...
}

MulticastSessionUpdateRequestTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-MBS-SessionID          CRITICALITY reject TYPE MBS-SessionID          PRESENCE mandatory }|
    { ID id-MBS-ServiceArea        CRITICALITY reject TYPE MBS-ServiceArea        PRESENCE optional }|
    { ID id-MBS-QoSFlowsToBeSetupModList CRITICALITY reject TYPE MBS-QoSFlowsToBeSetupList PRESENCE optional }|
    { ID id-MBS-QoSFlowToReleaseList CRITICALITY reject TYPE QoSFlowListWithCause PRESENCE optional }|
    { ID id-MBS-SessionTNLInfo5GC CRITICALITY reject TYPE MBS-SessionTNLInfo5GC PRESENCE optional }|
    { ID id-MBS-UP-FailureIndication CRITICALITY reject TYPE MBS-UP-FailureIndication PRESENCE optional },
    ...
}

MulticastGroupPagingAreaList ::= SEQUENCE (SIZE(1..maxnoofPagingAreas)) OF MulticastGroupPagingAreaItem

MulticastGroupPagingAreaItem ::= SEQUENCE {
    multicastGroupPagingArea          MulticastGroupPagingArea,
    uE-PagingList                    UE-PagingList                    OPTIONAL,
    iE-Extensions                    ProtocolExtensionContainer { { MulticastGroupPagingAreaItem-ExtIEs} } OPTIONAL,
    ...
}

MulticastGroupPagingAreaItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MBS-AreaTAIList ::= SEQUENCE (SIZE(1..maxnoofTAIforPaging)) OF TAI

MulticastGroupPagingArea ::= SEQUENCE {
    mBS-AreaTAIList          MBS-AreaTAIList,
    iE-Extensions          ProtocolExtensionContainer { { MulticastGroupPagingArea-ExtIEs} } OPTIONAL,
    ...
}

MulticastGroupPagingArea-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-PagingList ::= SEQUENCE (SIZE(1..maxnoofUEsforPaging)) OF UE-PagingItem

UE-PagingItem ::= SEQUENCE {

```

```

    uEIdentityIndexValue      UEIdentityIndexValue,
    pagingDRX                  PagingDRX
    iE-Extensions              ProtocolExtensionContainer { { UE-PagingItem-ExtIEs} } OPTIONAL,
    ...
}

UE-PagingItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

M1Configuration ::= SEQUENCE {
    m1reportingTrigger          M1ReportingTrigger,
    m1thresholdEventA2          M1ThresholdEventA2
    -- The above IE shall be present if the M1 Reporting Trigger IE is set to the value "A2event-triggered" or "A2event-triggered periodic"
    m1periodicReporting          M1PeriodicReporting
    -- The above IE shall be present if the M1 Reporting Trigger IE is set to the value "periodic" or "A2event-triggered periodic"
    iE-Extensions              ProtocolExtensionContainer { { M1Configuration-ExtIEs} } OPTIONAL,
    ...
}

M1Configuration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-IncludeBeamMeasurementsIndication CRITICALITY ignore EXTENSION IncludeBeamMeasurementsIndication PRESENCE optional }|
    { ID id-BeamMeasurementsReportConfiguration CRITICALITY ignore EXTENSION BeamMeasurementsReportConfiguration PRESENCE conditional },
    -- The above IE shall be present if the Include Beam Measurements Indication IE is set to the value "true"
    ...
}

IncludeBeamMeasurementsIndication ::= ENUMERATED {
    true,
    ...
}

MaxNrofRS-IndexesToReport ::= INTEGER (1..64, ...)

M1ReportingTrigger ::= ENUMERATED {
    periodic,
    a2eventtriggered,
    a2eventtriggered-periodic,
    ...
}

M1ThresholdEventA2 ::= SEQUENCE {
    m1ThresholdType            M1ThresholdType,
    iE-Extensions              ProtocolExtensionContainer { { M1ThresholdEventA2-ExtIEs} } OPTIONAL,
    ...
}

M1ThresholdEventA2-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

M1ThresholdType ::= CHOICE {
    threshold-RSRP              Threshold-RSRP,
    threshold-RSRQ              Threshold-RSRQ,

```



```

    threshold-SINR      Threshold-SINR,
    choice-Extensions   ProtocolIE-SingleContainer { {M1ThresholdType-ExtIEs} }
}

M1ThresholdType-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

M1PeriodicReporting ::= SEQUENCE {
    reportInterval      ReportIntervalMDT,
    reportAmount        ReportAmountMDT,
    iE-Extensions       ProtocolExtensionContainer { { M1PeriodicReporting-ExtIEs} } OPTIONAL,
    ...
}

M1PeriodicReporting-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-ExtendedReportIntervalMDT      CRITICALITY ignore  EXTENSION ExtendedReportIntervalMDT      PRESENCE optional},
    ...
}

M4Configuration ::= SEQUENCE {
    m4period            M4period,
    m4-links-to-log     Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { { M4Configuration-ExtIEs} } OPTIONAL,
    ...
}

M4Configuration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-M4ReportAmount      CRITICALITY ignore  EXTENSION M4ReportAmountMDT      PRESENCE optional      },
    ...
}

M4ReportAmountMDT ::= ENUMERATED {r1, r2, r4, r8, r16, r32, r64, infinity, ...}

M4period ::= ENUMERATED {ms1024, ms2048, ms5120, ms10240, min1, ... }

M5Configuration ::= SEQUENCE {
    m5period            M5period,
    m5-links-to-log     Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { { M5Configuration-ExtIEs} } OPTIONAL,
    ...
}

M5Configuration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-M5ReportAmount      CRITICALITY ignore  EXTENSION M5ReportAmountMDT      PRESENCE optional      },
    ...
}

M5ReportAmountMDT ::= ENUMERATED {r1, r2, r4, r8, r16, r32, r64, infinity, ...}

M5period ::= ENUMERATED {ms1024, ms2048, ms5120, ms10240, min1, ... }

M6Configuration ::= SEQUENCE {

```

```

    m6report-Interval    M6report-Interval,
    m6-links-to-log      Links-to-log,
    iE-Extensions        ProtocolExtensionContainer { { M6Configuration-ExtIEs} } OPTIONAL,
    ...
}

M6Configuration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-M6ReportAmount      CRITICALITY ignore  EXTENSION M6ReportAmountMDT      PRESENCE optional      }|
    { ID id-ExcessPacketDelayThresholdConfiguration CRITICALITY ignore  EXTENSION ExcessPacketDelayThresholdConfiguration PRESENCE optional },
    ...
}

M6ReportAmountMDT ::= ENUMERATED {r1, r2, r4, r8, r16, r32, r64, infinity, ...}

M6report-Interval ::= ENUMERATED {
    ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, ms20480, ms40960, min1, min6, min12, min30,
    ...
}

M7Configuration ::= SEQUENCE {
    m7period            M7period,
    m7-links-to-log      Links-to-log,
    iE-Extensions        ProtocolExtensionContainer { { M7Configuration-ExtIEs} } OPTIONAL,
    ...
}

M7Configuration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-M7ReportAmount      CRITICALITY ignore  EXTENSION M7ReportAmountMDT      PRESENCE optional      },
    ...
}

M7ReportAmountMDT ::= ENUMERATED {r1, r2, r4, r8, r16, r32, r64, infinity, ...}

M7period ::= INTEGER(1..60, ...)

MDT-circle ::= SEQUENCE {
    reference-location    OCTET STRING,
    distance-radius       INTEGER(1..65535),
    iE-Extensions         ProtocolExtensionContainer { {MDT-circle-ExtIEs} } OPTIONAL,
    ...
}

MDT-circle-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MDT-Location-Info ::= SEQUENCE {
    mDT-Location-Information MDT-Location-Information,
    iE-Extensions           ProtocolExtensionContainer { { MDT-Location-Info-ExtIEs} } OPTIONAL,
    ...
}

```

```
MDT-Location-Info-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

MDT-Location-Information ::= BIT STRING (SIZE (8))

MDT-polygon ::= OCTET STRING

MMSID ::= OCTET STRING (SIZE (1))

-- N

N3IWF-ID ::= CHOICE {
    n3IWF-ID          BIT STRING (SIZE(16)),
    choice-Extensions ProtocolIE-SingleContainer { {N3IWF-ID-ExtIEs} }
}

N3IWF-ID-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

N6JitterInformation ::= SEQUENCE {
    n6JitterLowerBound    INTEGER (-127..127),
    n6JitterUpperBound    INTEGER (-127..127),
    iE-Extensions         ProtocolExtensionContainer { { N6JitterInformation-ExtIEs} }    OPTIONAL,
    ...
}

N6JitterInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NAS-PDU ::= OCTET STRING

NASSecurityParametersFromNGRAN ::= OCTET STRING

NB-IoT-DefaultPagingDRX ::= ENUMERATED {
    rf128, rf256, rf512, rf1024,
    ...
}

NB-IoT-PagingDRX ::= ENUMERATED {
    rf32, rf64, rf128, rf256, rf512, rf1024,
    ...
}

NB-IoT-Paging-eDRXCycle ::= ENUMERATED {
    hf2, hf4, hf6, hf8, hf10, hf12, hf14, hf16, hf32, hf64, hf128, hf256, hf512, hf1024,
    ...
}

NB-IoT-Paging-TimeWindow ::= ENUMERATED {
    s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16,
    ...
}
```

```

}

NB-IoT-Paging-eDRXInfo ::= SEQUENCE {
    nB-IoT-Paging-eDRXCycle      NB-IoT-Paging-eDRXCycle,
    nB-IoT-Paging-TimeWindow      NB-IoT-Paging-TimeWindow      OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { { NB-IoT-Paging-eDRXInfo-ExtIEs} } OPTIONAL,
    ...
}

NB-IoT-Paging-eDRXInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NB-IoT-UEPriority ::= INTEGER (0..255, ...)

NetworkControlledRepeaterAuthorized ::= ENUMERATED {authorized, not-authorized, ...}

NetworkInstance ::= INTEGER (1..256, ...)

NetworkSliceListforMDT ::= SEQUENCE (SIZE(1.. maxnoofMDTPLMNs)) OF NetworkSliceItemforMDT

NetworkSliceItemforMDT ::= SEQUENCE {
    plmnID      PLMNIdentity,
    sliceMDTList      SliceMDTList,
    iE-Extensions      ProtocolExtensionContainer { {NetworkSliceItemforMDT-ExtIEs} } OPTIONAL,
    ...
}

NetworkSliceItemforMDT-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NetworkSliceAreaScopeofMDT ::= SEQUENCE {
    networkSliceListforMDT      NetworkSliceListforMDT,
    iE-Extensions      ProtocolExtensionContainer { {NetworkSliceAreaScopeofMDT-ExtIEs} } OPTIONAL,
    ...
}

NetworkSliceAreaScopeofMDT-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SliceMDTList ::= SEQUENCE (SIZE(1.. maxnoofSliceItemsforMDT)) OF SliceMDTItem

SliceMDTItem ::= SEQUENCE {
    sNSSAI      S-NSSAI,
    iE-Extensions      ProtocolExtensionContainer { { SliceMDTItem-ExtIEs} } OPTIONAL,
    ...
}

SliceMDTItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

NewSecurityContextInd ::= ENUMERATED {
    true,
    ...
}

NextHopChainingCount ::= INTEGER (0..7)

NextPagingAreaScope ::= ENUMERATED {
    same,
    changed,
    ...
}

NGAPISupportInformationRequestList ::= SEQUENCE (SIZE(1.. maxnoofNGAPISupportInfo)) OF NGAPISupportInformationRequestItem

NGAPISupportInformationRequestItem ::= SEQUENCE {
    ngap-ProtocolIE-Id      ProtocolIE-ID,
    iE-Extensions          ProtocolExtensionContainer { { NGAPISupportInformationRequestItem-ExtIEs } } OPTIONAL,
    ...
}

NGAPISupportInformationRequestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NGAPISupportInformationResponseList ::= SEQUENCE (SIZE(1.. maxnoofNGAPISupportInfo)) OF NGAPISupportInformationResponseItem

NGAPISupportInformationResponseItem ::= SEQUENCE {
    ngap-ProtocolIE-Id      ProtocolIE-ID,
    ngap-ProtocolIESupportInfo  ENUMERATED {supported, not-supported, ...},
    ngap-ProtocolIEPresenceInfo  ENUMERATED {present, not-present, ...},
    iE-Extensions          ProtocolExtensionContainer { { NGAPISupportInformationResponseItem-ExtIEs } } OPTIONAL,
    ...
}

NGAPISupportInformationResponseItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NgENB-ID ::= CHOICE {
    macroNgENB-ID          BIT STRING (SIZE(20)),
    shortMacroNgENB-ID     BIT STRING (SIZE(18)),
    longMacroNgENB-ID      BIT STRING (SIZE(21)),
    choice-Extensions      ProtocolIE-SingleContainer { {NgENB-ID-ExtIEs} }
}

NgENB-ID-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

NotifySourceNGRANNode ::= ENUMERATED {
    notifySource,

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```

    ...
}

NGRAN-CGI ::= CHOICE {
    nR-CGI          NR-CGI,
    eUTRA-CGI       EUTRA-CGI,
    choice-Extensions ProtocolIE-SingleContainer { {NGRAN-CGI-ExtIEs} }
}

NGRAN-CGI-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

NGRAN-TNLAssociationToRemoveList ::= SEQUENCE (SIZE(1..maxnoofTNLAAssociations)) OF NGRAN-TNLAssociationToRemoveItem

NGRAN-TNLAssociationToRemoveItem ::= SEQUENCE {
    tNLAssociationTransportLayerAddress      CPTransportLayerInformation,
    tNLAssociationTransportLayerAddressAMF    CPTransportLayerInformation      OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { NGRAN-TNLAssociationToRemoveItem-ExtIEs} } OPTIONAL
}

NGRAN-TNLAssociationToRemoveItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NGRANTraceID ::= OCTET STRING (SIZE(8))

NID ::= BIT STRING (SIZE(44))

NonDynamic5QIDescriptor ::= SEQUENCE {
    fiveQI          FiveQI,
    priorityLevelQos      PriorityLevelQos      OPTIONAL,
    averagingWindow      AveragingWindow      OPTIONAL,
    maximumDataBurstVolume      MaximumDataBurstVolume      OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {NonDynamic5QIDescriptor-ExtIEs} } OPTIONAL,
    ...
}

NonDynamic5QIDescriptor-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-CNPacketDelayBudgetDL    CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget    PRESENCE optional    }|
    { ID id-CNPacketDelayBudgetUL    CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget    PRESENCE optional    },
    ...
}

NoPDUSessionIndication ::= ENUMERATED {true, ...}

NotAllowedTACs ::= SEQUENCE (SIZE(1..maxnoofAllowedAreas)) OF TAC

NotificationCause ::= ENUMERATED {
    fulfilled,
    not-fulfilled,
    ...,
    not-fulfilled-DL,
    not-fulfilled-UL
}

```

```

}

NotificationControl ::= ENUMERATED {
    notification-requested,
    ...
}

NPN-AccessInformation ::= CHOICE {
    pNI-NPN-Access-Information    CellCAGList,
    choice-Extensions            ProtocolIE-SingleContainer { {NPN-AccessInformation-ExtIEs} }
}

NPN-AccessInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

NPN-MobilityInformation ::= CHOICE {
    sNPN-MobilityInformation      SNPN-MobilityInformation,
    pNI-NPN-MobilityInformation    PNI-NPN-MobilityInformation,
    choice-Extensions            ProtocolIE-SingleContainer { {NPN-MobilityInformation-ExtIEs} }
}

NPN-MobilityInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

NPN-PagingAssistanceInformation ::= CHOICE {
    pNI-NPN-PagingAssistance      Allowed-PNI-NPN-List,
    choice-Extensions            ProtocolIE-SingleContainer { {NPN-PagingAssistanceInformation-ExtIEs} }
}

NPN-PagingAssistanceInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

NPN-Support ::= CHOICE {
    sNPN                          NID,
    choice-Extensions            ProtocolIE-SingleContainer { {NPN-Support-ExtIEs} }
}

NPN-Support-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

NRCellIdentity ::= BIT STRING (SIZE(36))

NR-CGI ::= SEQUENCE {
    pLMNIdentity                PLMNIdentity,
    nRCellIdentity              NRCellIdentity,
    iE-Extensions               ProtocolExtensionContainer { {NR-CGI-ExtIEs} } OPTIONAL,
    ...
}

```

```

NR-CGI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NR-CGIList ::= SEQUENCE (SIZE(1..maxnoofCellsingNB)) OF NR-CGI

NR-CGIListForWarning ::= SEQUENCE (SIZE(1..maxnoofCellIDforWarning)) OF NR-CGI

NR-PagingeDRXInformation ::= SEQUENCE {
    nR-paging-eDRX-Cycle          NR-Paging-eDRX-Cycle,
    nR-paging-Time-Window        NR-Paging-Time-Window          OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {NR-PagingeDRXInformation-ExtIEs} }    OPTIONAL,
    ...
}

NR-PagingeDRXInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NR-Paging-eDRX-Cycle ::= ENUMERATED {
    hfquarter, hfhalf, hf1, hf2, hf4, hf8, hf16,
    hf32, hf64, hf128, hf256, hf512, hf1024,
    ...
}

NR-Paging-Long-eDRX-Information-for-RRC-INACTIVE ::= SEQUENCE {
    nR-paging-Long-eDRX-Cycle-for-RRC-INACTIVE          NR-Paging-Long-eDRX-Cycle-for-RRC-INACTIVE,
    nR-paging-Time-Window-for-RRC-INACTIVE              NR-Paging-Time-Window-for-RRC-INACTIVE,
    iE-Extensions                ProtocolExtensionContainer { {NR-Paging-Long-eDRX-Information-for-RRC-INACTIVE-ExtIEs} } OPTIONAL,
    ...
}

NR-Paging-Long-eDRX-Information-for-RRC-INACTIVE-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NR-Paging-Long-eDRX-Cycle-for-RRC-INACTIVE ::= ENUMERATED {
    hf2, hf4, hf8, hf16,
    hf32, hf64, hf128, hf256, hf512, hf1024,
    ...
}

NR-Paging-Time-Window ::= ENUMERATED {
    s1, s2, s3, s4, s5,
    s6, s7, s8, s9, s10,
    s11, s12, s13, s14, s15, s16,
    ...,
    s17, s18, s19, s20, s21, s22, s23, s24,
    s25, s26, s27, s28, s29, s30, s31, s32
}

NR-Paging-Time-Window-for-RRC-INACTIVE ::= ENUMERATED {
    s1, s2, s3, s4, s5,
    s6, s7, s8, s9, s10,

```



```

    s11, s12, s13, s14, s15, s16,
    s17, s18, s19, s20, s21, s22, s23, s24,
    s25, s26, s27, s28, s29, s30, s31, s32,
    ...
}

NREncryptionAlgorithms ::= BIT STRING (SIZE(16, ...))

NRIntegrityProtectionAlgorithms ::= BIT STRING (SIZE(16, ...))

NRMobilityHistoryReport ::= OCTET STRING

NRPPa-PDU ::= OCTET STRING

NRUERLFReportContainer ::= OCTET STRING

NRNTNTAIInformation ::= SEQUENCE {
    servingPLMN                PLMNIdentity,
    tACListInNRNTN             TACListInNRNTN,
    uELocationDerivedTACInNRNTN TAC                                OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { { NRNTNTAIInformation-ExtIEs} } OPTIONAL,
    ...
}

NRNTNTAIInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NumberOfBroadcasts ::= INTEGER (0..65535)

NumberOfBroadcastsRequested ::= INTEGER (0..65535)

NRARFCN ::= INTEGER (0.. maxNRARFCN)

NRFrequencyBand ::= INTEGER (1..1024, ...)

NRFrequencyBand-List ::= SEQUENCE (SIZE(1..maxnoofNRCellBands)) OF NRFrequencyBandItem

NRFrequencyBandItem ::= SEQUENCE {
    nr-frequency-band          NRFrequencyBand,
    iE-Extension               ProtocolExtensionContainer { {NRFrequencyBandItem-ExtIEs} }    OPTIONAL,
    ...
}

NRFrequencyBandItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NRFrequencyInfo ::= SEQUENCE {
    nrARFCN                    NRARFCN,
    frequencyBand-List         NRFrequencyBand-List,
    iE-Extension               ProtocolExtensionContainer { {NRFrequencyInfo-ExtIEs} }    OPTIONAL,
    ...
}

```

```

}

NRFrequencyInfo-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NR-PCI ::= INTEGER (0..1007, ...)

NRV2XServicesAuthorized ::= SEQUENCE {
    vehicleUE           VehicleUE                               OPTIONAL,
    pedestrianUE        PedestrianUE                             OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {NRV2XServicesAuthorized-ExtIEs} } OPTIONAL,
    ...
}

NRV2XServicesAuthorized-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

VehicleUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

PedestrianUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

NRUESidelinkAggregateMaximumBitrate ::= SEQUENCE {
    uESidelinkAggregateMaximumBitrate BitRate,
    iE-Extensions       ProtocolExtensionContainer { {NRUESidelinkAggregateMaximumBitrate-ExtIEs} } OPTIONAL,
    ...
}

NRUESidelinkAggregateMaximumBitrate-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

NSAG-ID ::= INTEGER (0..255, ...)

NR-A2X-ServiceAuthorized ::= SEQUENCE {
    aerialUE           ENUMERATED {authorized, not-authorized, ...}    OPTIONAL,
    aerialControllerUE ENUMERATED {authorized, not-authorized, ...}    OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {NR-A2X-ServiceAuthorized-ExtIEs} } OPTIONAL,
    ...
}

NR-A2X-ServiceAuthorized-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

NTN-Geographical-Area-List ::= SEQUENCE (SIZE(1..maxnoofAreaNTN)) OF NTN-Geographical-Area-Item

NTN-Geographical-Area-Item ::= CHOICE {
    circle                MDT-circle,
    polygon               MDT-polygon,
    choice-Extensions     ProtocolIE-SingleContainer { {NTN-Geographicla-Area-Item-ExtIEs} }
}

NTN-Geographicla-Area-Item-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

-- 0

OnboardingSupport ::= ENUMERATED {
    true,
    ...
}

OverloadAction ::= ENUMERATED {
    reject-non-emergency-mo-dt,
    reject-rrc-cr-signalling,
    permit-emergency-sessions-and-mobile-terminated-services-only,
    permit-high-priority-sessions-and-mobile-terminated-services-only,
    ...
}

OverloadResponse ::= CHOICE {
    overloadAction        OverloadAction,
    choice-Extensions     ProtocolIE-SingleContainer { {OverloadResponse-ExtIEs} }
}

OverloadResponse-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

OverloadStartNSSAIList ::= SEQUENCE (SIZE (1..maxnoofSliceItems)) OF OverloadStartNSSAIItem

OverloadStartNSSAIItem ::= SEQUENCE {
    sliceOverloadList      SliceOverloadList,
    sliceOverloadResponse  OverloadResponse OPTIONAL,
    sliceTrafficLoadReductionIndication TrafficLoadReductionIndication OPTIONAL,
    iE-Extensions         ProtocolExtensionContainer { {OverloadStartNSSAIItem-ExtIEs} } OPTIONAL,
    ...
}

OverloadStartNSSAIItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- P

PacketDelayBudget ::= INTEGER (0..1023, ...)

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```
PacketErrorRate ::= SEQUENCE {
    pERScalar      INTEGER (0..9, ...),
    pERExponent    INTEGER (0..9, ...),
    iE-Extensions  ProtocolExtensionContainer { {PacketErrorRate-ExtIEs} } OPTIONAL,
    ...
}

PacketErrorRate-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PacketLossRate ::= INTEGER (0..1000, ...)

PagingAssisDataforCEcapabUE ::= SEQUENCE {
    eUTRA-CGI      EUTRA-CGI,
    coverageEnhancementLevel CoverageEnhancementLevel,
    iE-Extensions  ProtocolExtensionContainer { { PagingAssisDataforCEcapabUE-ExtIEs} } OPTIONAL,
    ...
}

PagingAssisDataforCEcapabUE-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PagingAttemptInformation ::= SEQUENCE {
    pagingAttemptCount      PagingAttemptCount,
    intendedNumberOfPagingAttempts IntendedNumberOfPagingAttempts,
    nextPagingAreaScope      NextPagingAreaScope
    iE-Extensions            ProtocolExtensionContainer { {PagingAttemptInformation-ExtIEs} } OPTIONAL,
    ...
    OPTIONAL,
}

PagingAttemptInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PagingAttemptCount ::= INTEGER (1..16, ...)

PagingCause ::= ENUMERATED {
    voice,
    ...
}

PagingCauseIndicationForVoiceService ::= ENUMERATED {
    supported,
    ...
}

PagingDRX ::= ENUMERATED {
    v32,
    v64,
    v128,
    v256,
    ...
}
```

```

}

PagingOrigin ::= ENUMERATED {
    non-3gpp,
    ...
}

PagingPolicyDifferentiation ::= SEQUENCE {
    pduSessionForPagingList      PDU SessionForPagingList,
    iE-Extensions                ProtocolExtensionContainer { { PagingPolicyDifferentiation-ExtIEs} } OPTIONAL,
    ...
}

PagingPolicyDifferentiation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDU SessionForPagingList ::= SEQUENCE (SIZE(1..maxnoofPDU Sessions)) OF PDU SessionForPagingItem

PDU SessionForPagingItem ::= SEQUENCE {
    pduSessionID                PDU SessionID,
    pagingPolicyDifferentiationList PagingPolicyDifferentiationList,
    iE-Extensions                ProtocolExtensionContainer { { PDU SessionForPagingItem-ExtIEs} } OPTIONAL,
    ...
}

PDU SessionForPagingItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PagingPolicyDifferentiationList ::= SEQUENCE (SIZE(1..maxnoofQoS Flows)) OF PagingPolicyDifferentiationItem

PagingPolicyDifferentiationItem ::= SEQUENCE {
    qosFlowIdentifier            QoS FlowIdentifier OPTIONAL,
    pagingPolicyIndicator        PagingPolicyIndicator OPTIONAL,
    allocationAndRetentionPriority AllocationAndRetentionPriority OPTIONAL,
    fiveQI                      FiveQI OPTIONAL,
    dl-DataSize                  INTEGER (0..96000, ...) OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { { PagingPolicyDifferentiationItem-ExtIEs} } OPTIONAL,
    ...
}

PagingPolicyDifferentiationItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PagingPolicyIndicator ::= INTEGER (0..7, ...)

PagingPriority ::= ENUMERATED {
    priolevel1,
    priolevel2,
    priolevel3,
    priolevel4,

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    priolevel5,
    priolevel6,
    priolevel7,
    priolevel8,
    ...
}

PagingProbabilityInformation ::= ENUMERATED {
    p00, p05, p10, p15, p20, p25, p30, p35, p40, p45, p50, p55, p60, p65, p70, p75, p80, p85, p90, p95, p100,
    ...
}

ParentTimeSource ::= ENUMERATED {synce, ptp, gnss, atomicclock, terrestrialradio, serialtimecode, ntp, handset, other, ...}

Partially-Allowed-NSSAI ::= SEQUENCE (SIZE(1.. maxnoofPartiallyAllowedS-NSSAIs)) OF PartiallyAllowedNSSAI-Item

PartiallyAllowedNSSAI-Item ::= SEQUENCE {
    s-NSSAI          S-NSSAI,
    iE-Extensions    ProtocolExtensionContainer { {PartiallyAllowedNSSAI-Item-ExtIEs} } OPTIONAL,
    ...
}

PartiallyAllowedNSSAI-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PathSwitchRequestAcknowledgeTransfer ::= SEQUENCE {
    uL-NGU-UP-TNLInformation    UPTransportLayerInformation                OPTIONAL,
    securityIndication          SecurityIndication                        OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { {PathSwitchRequestAcknowledgeTransfer-ExtIEs} } OPTIONAL,
    ...
}

PathSwitchRequestAcknowledgeTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalNGU-UP-TNLInformation    CRITICALITY ignore EXTENSION UPTransportLayerInformationPairList PRESENCE optional }|
    { ID id-RedundantUL-NGU-UP-TNLInformation  CRITICALITY ignore EXTENSION UPTransportLayerInformation    PRESENCE optional }|
    { ID id-AdditionalRedundantNGU-UP-TNLInformation  CRITICALITY ignore EXTENSION UPTransportLayerInformationPairList PRESENCE optional }|
    { ID id-QosFlowParametersList             CRITICALITY ignore EXTENSION QosFlowParametersList      PRESENCE optional },
    ...
}

PathSwitchRequestSetupFailedTransfer ::= SEQUENCE {
    cause          Cause,
    iE-Extensions  ProtocolExtensionContainer { {PathSwitchRequestSetupFailedTransfer-ExtIEs} } OPTIONAL,
    ...
}

PathSwitchRequestSetupFailedTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PathSwitchRequestTransfer ::= SEQUENCE {
    dL-NGU-UP-TNLInformation    UPTransportLayerInformation,
    dL-NGU-TNLInformationReused DL-NGU-TNLInformationReused        OPTIONAL,

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    userPlaneSecurityInformation      UserPlaneSecurityInformation      OPTIONAL,
    qosFlowAcceptedList              QosFlowAcceptedList,
    iE-Extensions                    ProtocolExtensionContainer { {PathSwitchRequestTransfer-ExtIEs} } OPTIONAL,
    ...
}

PathSwitchRequestTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalDLQosFlowPerTNLInformation      CRITICALITY ignore EXTENSION QosFlowPerTNLInformationList      PRESENCE optional }|
    { ID id-RedundantDL-NGU-UP-TNLInformation        CRITICALITY ignore EXTENSION UPTransportLayerInformation      PRESENCE optional }|
    { ID id-RedundantDL-NGU-TNLInformationReused      CRITICALITY ignore EXTENSION DL-NGU-TNLInformationReused      PRESENCE optional }|
    { ID id-AdditionalRedundantDLQosFlowPerTNLInformation CRITICALITY ignore EXTENSION QosFlowPerTNLInformationList      PRESENCE optional }|
    { ID id-UsedRSNInformation                      CRITICALITY ignore EXTENSION RedundantPDUSessionInformation      PRESENCE optional }|
    { ID id-GlobalRANNodeID                        CRITICALITY ignore EXTENSION GlobalRANNodeID                      PRESENCE optional }|
    { ID id-MBS-SupportIndicator                    CRITICALITY ignore EXTENSION MBS-SupportIndicator                PRESENCE optional }|
    { ID id-PDUSetbasedHandlingIndicator            CRITICALITY ignore EXTENSION PDUSetbasedHandlingIndicator          PRESENCE optional }|
    { ID id-ECNMarkingorCongestionInformationReportingStatus CRITICALITY ignore EXTENSION ECNMarkingorCongestionInformationReportingStatus PRESENCE optional },
    ...
}

PathSwitchRequestUnsuccessfulTransfer ::= SEQUENCE {
    cause          Cause,
    iE-Extensions  ProtocolExtensionContainer { {PathSwitchRequestUnsuccessfulTransfer-ExtIEs} } OPTIONAL,
    ...
}

PathSwitchRequestUnsuccessfulTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PC5QoSParameters ::= SEQUENCE {
    pc5QoSFlowList      PC5QoSFlowList,
    pc5LinkAggregateBitRates BitRate
    iE-Extensions        ProtocolExtensionContainer { { PC5QoSParameters-ExtIEs} } OPTIONAL,
    ...
}

PC5QoSParameters-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PC5QoSFlowList ::= SEQUENCE (SIZE(1..maxnoofPC5QoSFlows)) OF PC5QoSFlowItem

PC5QoSFlowItem ::= SEQUENCE {
    pQI          FiveQI,
    pc5FlowBitRates PC5FlowBitRates
    range        Range
    iE-Extensions ProtocolExtensionContainer { { PC5QoSFlowItem-ExtIEs} } OPTIONAL,
    ...
}

PC5QoSFlowItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

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PC5FlowBitRates ::= SEQUENCE {
    guaranteedFlowBitRate      BitRate,
    maximumFlowBitRate         BitRate,
    iE-Extensions              ProtocolExtensionContainer { { PC5FlowBitRates-ExtIEs} }    OPTIONAL,
    ...
}

PC5FlowBitRates-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PCIListForMDT ::= SEQUENCE (SIZE(1.. maxnoofNeighPCIforMDT)) OF NR-PCI

PrivacyIndicator ::= ENUMERATED {
    immediate-MDT,
    logged-MDT,
    ...
}

PDUSessionAggregateMaximumBitRate ::= SEQUENCE {
    pDUSessionAggregateMaximumBitRateDL      BitRate,
    pDUSessionAggregateMaximumBitRateUL      BitRate,
    iE-Extensions              ProtocolExtensionContainer { {PDUSessionAggregateMaximumBitRate-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionAggregateMaximumBitRate-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionID ::= INTEGER (0..255)

PDUSessionPairID ::= INTEGER (0..255, ...)

PDUSessionResourceAdmittedList ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceAdmittedItem

PDUSessionResourceAdmittedItem ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    handoverRequestAcknowledgeTransfer OCTET STRING (CONTAINING HandoverRequestAcknowledgeTransfer),
    iE-Extensions              ProtocolExtensionContainer { {PDUSessionResourceAdmittedItem-ExtIEs} }    OPTIONAL,
    ...
}

PDUSessionResourceAdmittedItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToModifyListModCfm ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToModifyItemModCfm

PDUSessionResourceFailedToModifyItemModCfm ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    pDUSessionResourceModifyIndicationUnsuccessfulTransfer OCTET STRING (CONTAINING PDUSessionResourceModifyIndicationUnsuccessfulTransfer),

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    iE-Extensions      ProtocolExtensionContainer { {PDUSessionResourceFailedToModifyItemModCfm-ExtIes} } OPTIONAL,
    ...
}

PDUSessionResourceFailedToModifyItemModCfm-ExtIes NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToModifyListModRes ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToModifyItemModRes

PDUSessionResourceFailedToModifyItemModRes ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    pDUSessionResourceModifyUnsuccessfulTransfer OCTET STRING (CONTAINING PDUSessionResourceModifyUnsuccessfulTransfer),
    iE-Extensions      ProtocolExtensionContainer { {PDUSessionResourceFailedToModifyItemModRes-ExtIes} } OPTIONAL,
    ...
}

PDUSessionResourceFailedToModifyItemModRes-ExtIes NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToResumeListRESReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToResumeItemRESReq

PDUSessionResourceFailedToResumeItemRESReq ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    cause                       Cause,
    iE-Extensions      ProtocolExtensionContainer { {PDUSessionResourceFailedToResumeItemRESReq-ExtIes} } OPTIONAL,
    ...
}

PDUSessionResourceFailedToResumeItemRESReq-ExtIes NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToResumeListRESRes ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToResumeItemRESRes

PDUSessionResourceFailedToResumeItemRESRes ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    cause                       Cause,
    iE-Extensions      ProtocolExtensionContainer { {PDUSessionResourceFailedToResumeItemRESRes-ExtIes} } OPTIONAL,
    ...
}

PDUSessionResourceFailedToResumeItemRESRes-ExtIes NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToSetupListCxtFail ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToSetupItemCxtFail

PDUSessionResourceFailedToSetupItemCxtFail ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    pDUSessionResourceSetupUnsuccessfulTransfer OCTET STRING (CONTAINING PDUSessionResourceSetupUnsuccessfulTransfer),
    iE-Extensions      ProtocolExtensionContainer { {PDUSessionResourceFailedToSetupItemCxtFail-ExtIes} } OPTIONAL,

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```

    ...
}

PDUSessionResourceFailedToSetupItemCxtFail-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToSetupListCxtRes ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToSetupItemCxtRes

PDUSessionResourceFailedToSetupItemCxtRes ::= SEQUENCE {
    pDUSessionID PDUSessionID,
    pDUSessionResourceSetupUnsuccessfulTransfer OCTET STRING (CONTAINING PDUSessionResourceSetupUnsuccessfulTransfer),
    iE-Extensions ProtocolExtensionContainer { {PDUSessionResourceFailedToSetupItemCxtRes-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceFailedToSetupItemCxtRes-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToSetupListH0Ack ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToSetupItemH0Ack

PDUSessionResourceFailedToSetupItemH0Ack ::= SEQUENCE {
    pDUSessionID PDUSessionID,
    handoverResourceAllocationUnsuccessfulTransfer OCTET STRING (CONTAINING HandoverResourceAllocationUnsuccessfulTransfer),
    iE-Extensions ProtocolExtensionContainer { {PDUSessionResourceFailedToSetupItemH0Ack-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceFailedToSetupItemH0Ack-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToSetupListPSReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToSetupItemPSReq

PDUSessionResourceFailedToSetupItemPSReq ::= SEQUENCE {
    pDUSessionID PDUSessionID,
    pathSwitchRequestSetupFailedTransfer OCTET STRING (CONTAINING PathSwitchRequestSetupFailedTransfer),
    iE-Extensions ProtocolExtensionContainer { {PDUSessionResourceFailedToSetupItemPSReq-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceFailedToSetupItemPSReq-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceFailedToSetupListSUSRes ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceFailedToSetupItemSUSRes

PDUSessionResourceFailedToSetupItemSUSRes ::= SEQUENCE {
    pDUSessionID PDUSessionID,
    pDUSessionResourceSetupUnsuccessfulTransfer OCTET STRING (CONTAINING PDUSessionResourceSetupUnsuccessfulTransfer),
    iE-Extensions ProtocolExtensionContainer { {PDUSessionResourceFailedToSetupItemSUSRes-ExtIEs} } OPTIONAL,
    ...
}

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```

PDUSessionResourceFailedToSetupItemSRes-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceHandoverList ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceHandoverItem

PDUSessionResourceHandoverItem ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    handoverCommandTransfer      OCTET STRING (CONTAINING HandoverCommandTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceHandoverItem-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceHandoverItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceInformationList ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceInformationItem

PDUSessionResourceInformationItem ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    qosFlowInformationList       QosFlowInformationList,
    dRBsToQosFlowsMappingList   DRBsToQosFlowsMappingList OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceInformationItem-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceInformationItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceListCxtRelCpl ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceItemCxtRelCpl

PDUSessionResourceItemCxtRelCpl ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceItemCxtRelCpl-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceItemCxtRelCpl-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-PDUSessionResourceReleaseResponseTransfer  CRITICALITY ignore  EXTENSION OCTET STRING (CONTAINING
PDUSessionResourceReleaseResponseTransfer)  PRESENCE optional  },
    ...
}

PDUSessionResourceListCxtRelReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceItemCxtRelReq

PDUSessionResourceItemCxtRelReq ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceItemCxtRelReq-ExtIEs} } OPTIONAL,
    ...
}

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```

PDUSessionResourceItemCxtRelReq-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceListHORqd ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceItemHORqd

PDUSessionResourceItemHORqd ::= SEQUENCE {
    pduSessionID          PDUSessionID,
    handoverRequiredTransfer OCTET STRING (CONTAINING HandoverRequiredTransfer),
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceItemHORqd-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceItemHORqd-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionListMTCommHReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionListMTCommHReq-Item

PDUSessionListMTCommHReq-Item ::= SEQUENCE {
    pduSessionID          PDUSessionID,
    iE-Extensions          ProtocolExtensionContainer { { PDUSessionListMTCommHReq-Item-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionListMTCommHReq-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceModifyConfirmTransfer ::= SEQUENCE {
    qosFlowModifyConfirmList      QosFlowModifyConfirmList,
    uLNGU-UP-TNLInformation        UPTransportLayerInformation,
    additionalNG-UUPTNLInformation UPTransportLayerInformationPairList OPTIONAL,
    qosFlowFailedToModifyList      QosFlowListWithCause OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceModifyConfirmTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceModifyConfirmTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-RedundantUL-NGU-UP-TNLInformation      CRITICALITY ignore EXTENSION UPTransportLayerInformation      PRESENCE optional }|
    { ID id-AdditionalRedundantNGU-UP-TNLInformation CRITICALITY ignore EXTENSION UPTransportLayerInformationPairList PRESENCE optional },
    ...
}

PDUSessionResourceModifyIndicationUnsuccessfulTransfer ::= SEQUENCE {
    cause          Cause,
    iE-Extensions  ProtocolExtensionContainer { {PDUSessionResourceModifyIndicationUnsuccessfulTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceModifyIndicationUnsuccessfulTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

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PDUSessionResourceModifyRequestTransfer ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container      { {PDUSessionResourceModifyRequestTransferIEs} },
    ...
}

PDUSessionResourceModifyRequestTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-PDUSessionAggregateMaximumBitRate      CRITICALITY reject TYPE PDUSessionAggregateMaximumBitRate      PRESENCE optional }|
    { ID id-UL-NGU-UP-TNLModifyList                CRITICALITY reject TYPE UL-NGU-UP-TNLModifyList                PRESENCE optional }|
    { ID id-NetworkInstance                        CRITICALITY reject TYPE NetworkInstance                        PRESENCE optional }|
    { ID id-QosFlowAddOrModifyRequestList           CRITICALITY reject TYPE QosFlowAddOrModifyRequestList           PRESENCE optional }|
    { ID id-QosFlowToReleaseList                   CRITICALITY reject TYPE QosFlowListWithCause                   PRESENCE optional }|
    { ID id-AdditionalUL-NGU-UP-TNLInformation      CRITICALITY reject TYPE UPTransportLayerInformationList        PRESENCE optional }|
    { ID id-CommonNetworkInstance                  CRITICALITY ignore TYPE CommonNetworkInstance                  PRESENCE optional }|
    { ID id-AdditionalRedundantUL-NGU-UP-TNLInformation CRITICALITY ignore TYPE UPTransportLayerInformationList        PRESENCE optional }|
    { ID id-RedundantCommonNetworkInstance         CRITICALITY ignore TYPE CommonNetworkInstance                  PRESENCE optional }|
    { ID id-RedundantUL-NGU-UP-TNLInformation      CRITICALITY ignore TYPE UPTransportLayerInformation        PRESENCE optional }|
    { ID id-SecurityIndication                     CRITICALITY ignore TYPE SecurityIndication                     PRESENCE optional }|
    { ID id-MBSSessionSetuporModifyRequestList     CRITICALITY ignore TYPE MBSSessionSetuporModifyRequestList     PRESENCE optional }|
    { ID id-MBSSessionToReleaseList                CRITICALITY ignore TYPE MBSSessionToReleaseList                PRESENCE optional }|
    { ID id-UserPlaneFailureIndication             CRITICALITY ignore TYPE UserPlaneFailureIndication             PRESENCE optional },
    ...
}

PDUSessionResourceModifyResponseTransfer ::= SEQUENCE {
    dL-NGU-UP-TNLInformation      UPTransportLayerInformation      OPTIONAL,
    uL-NGU-UP-TNLInformation      UPTransportLayerInformation      OPTIONAL,
    qosFlowAddOrModifyResponseList QosFlowAddOrModifyResponseList  OPTIONAL,
    additionalDLQosFlowPerTNLInformation QosFlowPerTNLInformationList  OPTIONAL,
    qosFlowFailedToAddOrModifyList QosFlowListWithCause          OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {PDUSessionResourceModifyResponseTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceModifyResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalNGU-UP-TNLInformation      CRITICALITY ignore EXTENSION UPTransportLayerInformationPairList PRESENCE optional }|
    { ID id-RedundantDL-NGU-UP-TNLInformation   CRITICALITY ignore EXTENSION UPTransportLayerInformation      PRESENCE optional }|
    { ID id-RedundantUL-NGU-UP-TNLInformation   CRITICALITY ignore EXTENSION UPTransportLayerInformation      PRESENCE optional }|
    { ID id-AdditionalRedundantDLQosFlowPerTNLInformation CRITICALITY ignore EXTENSION QosFlowPerTNLInformationList PRESENCE optional }|
    { ID id-AdditionalRedundantNGU-UP-TNLInformation CRITICALITY ignore EXTENSION UPTransportLayerInformationPairList PRESENCE optional }|
    { ID id-SecondaryRATUsageInformation        CRITICALITY ignore EXTENSION SecondaryRATUsageInformation      PRESENCE optional }|
    { ID id-MBS-SupportIndicator                CRITICALITY ignore EXTENSION MBS-SupportIndicator            PRESENCE optional }|
    { ID id-MBSSessionSetuporModifyResponseList CRITICALITY ignore EXTENSION MBSSessionSetupResponseList      PRESENCE optional }|
    { ID id-MBSSessionFailedtoSetuporModifyList CRITICALITY ignore EXTENSION MBSSessionFailedtoSetupList      PRESENCE optional }|
    { ID id-ECNMarkingorCongestionInformationReportingStatus CRITICALITY ignore EXTENSION ECNMarkingorCongestionInformationReportingStatus PRESENCE optional }|
    { ID id-PDUSetbasedHandlingIndicator        CRITICALITY ignore EXTENSION PDUSetbasedHandlingIndicator          PRESENCE optional }|
    { ID id-UserPlaneFailureIndicationReport    CRITICALITY ignore EXTENSION UserPlaneFailureIndicationReport    PRESENCE optional },
    ...
}

PDUSessionResourceModifyIndicationTransfer ::= SEQUENCE {
    dLQosFlowPerTNLInformation      QosFlowPerTNLInformation,
    additionalDLQosFlowPerTNLInformation QosFlowPerTNLInformationList  OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {PDUSessionResourceModifyIndicationTransfer-ExtIEs} } OPTIONAL,

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}
...
}

PDUSessionResourceModifyIndicationTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  { ID id-SecondaryRATUsageInformation          CRITICALITY ignore EXTENSION SecondaryRATUsageInformation PRESENCE optional }|
  { ID id-SecurityResult                        CRITICALITY ignore EXTENSION SecurityResult          PRESENCE optional }|
  { ID id-RedundantDLQosFlowPerTNLInformation    CRITICALITY ignore EXTENSION QosFlowPerTNLInformation PRESENCE optional }|
  { ID id-AdditionalRedundantDLQosFlowPerTNLInformation CRITICALITY ignore EXTENSION QosFlowPerTNLInformationList PRESENCE optional }|
  { ID id-GlobalRANNodeID                      CRITICALITY ignore EXTENSION GlobalRANNodeID          PRESENCE optional }|
  { ID id-ECNMarkingorCongestionInformationReportingStatus CRITICALITY ignore EXTENSION ECNMarkingorCongestionInformationReportingStatus PRESENCE optional },
  ...
}

PDUSessionResourceModifyListModCfm ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceModifyItemModCfm

PDUSessionResourceModifyItemModCfm ::= SEQUENCE {
  pDUSessionID                PDUSessionID,
  pDUSessionResourceModifyConfirmTransfer OCTET STRING (CONTAINING PDUSessionResourceModifyConfirmTransfer),
  iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceModifyItemModCfm-ExtIEs} } OPTIONAL,
  ...
}

PDUSessionResourceModifyItemModCfm-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

PDUSessionResourceModifyListModInd ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceModifyItemModInd

PDUSessionResourceModifyItemModInd ::= SEQUENCE {
  pDUSessionID                PDUSessionID,
  pDUSessionResourceModifyIndicationTransfer OCTET STRING (CONTAINING PDUSessionResourceModifyIndicationTransfer),
  iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceModifyItemModInd-ExtIEs} } OPTIONAL,
  ...
}

PDUSessionResourceModifyItemModInd-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

PDUSessionResourceModifyListModReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceModifyItemModReq

PDUSessionResourceModifyItemModReq ::= SEQUENCE {
  pDUSessionID                PDUSessionID,
  nAS-PDU                    NAS-PDU OPTIONAL,
  pDUSessionResourceModifyRequestTransfer OCTET STRING (CONTAINING PDUSessionResourceModifyRequestTransfer),
  iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceModifyItemModReq-ExtIEs} } OPTIONAL,
  ...
}

PDUSessionResourceModifyItemModReq-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  { ID id-S-NSSAI                CRITICALITY reject EXTENSION S-NSSAI PRESENCE optional }|
  { ID id-PduSessionExpectedUEActivityBehaviour CRITICALITY ignore EXTENSION ExpectedUEActivityBehaviour PRESENCE optional },
  ...
}

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```

}

PDUSessionResourceModifyListModRes ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceModifyItemModRes

PDUSessionResourceModifyItemModRes ::= SEQUENCE {
    pduSessionID                PDUSessionID,
    pduSessionResourceModifyResponseTransfer OCTET STRING (CONTAINING PDUSessionResourceModifyResponseTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceModifyItemModRes-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceModifyItemModRes-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceModifyUnsuccessfulTransfer ::= SEQUENCE {
    cause                        Cause,
    criticalityDiagnostics       CriticalityDiagnostics OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceModifyUnsuccessfulTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceModifyUnsuccessfulTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceNotifyList ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceNotifyItem

PDUSessionResourceNotifyItem ::= SEQUENCE {
    pduSessionID                PDUSessionID,
    pduSessionResourceNotifyTransfer OCTET STRING (CONTAINING PDUSessionResourceNotifyTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceNotifyItem-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceNotifyItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceNotifyReleasedTransfer ::= SEQUENCE {
    cause                        Cause,
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceNotifyReleasedTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceNotifyReleasedTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-SecondaryRATUsageInformation    CRITICALITY ignore EXTENSION SecondaryRATUsageInformation PRESENCE optional }|
    { ID id-UserPlaneErrorIndicator         CRITICALITY ignore EXTENSION UserPlaneErrorIndicator PRESENCE optional },
    ...
}

UserPlaneErrorIndicator ::= ENUMERATED {
    gTP-U-error-indication-received,
    ...
}

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}

PDUSessionResourceNotifyTransfer ::= SEQUENCE {
    qosFlowNotifyList      QosFlowNotifyList                OPTIONAL,
    qosFlowReleasedList    QosFlowListWithCause             OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceNotifyTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceNotifyTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-SecondaryRATUsageInformation      CRITICALITY ignore EXTENSION SecondaryRATUsageInformation PRESENCE optional }|
    { ID id-QosFlowFeedbackList              CRITICALITY ignore EXTENSION QosFlowFeedbackList          PRESENCE optional },
    ...
}

PDUSessionResourceReleaseCommandTransfer ::= SEQUENCE {
    cause                  Cause,
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceReleaseCommandTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceReleaseCommandTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-QosFlowAdditionalInfoList          CRITICALITY ignore EXTENSION QosFlowAdditionalInfoListRelCom PRESENCE optional },
    ...
}

PDUSessionResourceReleasedListNot ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceReleasedItemNot

PDUSessionResourceReleasedItemNot ::= SEQUENCE {
    pduSessionID          PDUSessionID,
    pduSessionResourceNotifyReleasedTransfer OCTET STRING (CONTAINING PDUSessionResourceNotifyReleasedTransfer),
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceReleasedItemNot-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceReleasedItemNot-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceReleasedListPSAck ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceReleasedItemPSAck

PDUSessionResourceReleasedItemPSAck ::= SEQUENCE {
    pduSessionID          PDUSessionID,
    pathSwitchRequestUnsuccessfulTransfer OCTET STRING (CONTAINING PathSwitchRequestUnsuccessfulTransfer),
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceReleasedItemPSAck-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceReleasedItemPSAck-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceReleasedListPSFail ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceReleasedItemPSFail

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PDUSessionResourceReleasedItemPSFail ::= SEQUENCE {
    pduSessionID          PDUSessionID,
    pathSwitchRequestUnsuccessfulTransfer OCTET STRING (CONTAINING PathSwitchRequestUnsuccessfulTransfer),
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceReleasedItemPSFail-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceReleasedItemPSFail-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceReleasedListRelRes ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceReleasedItemRelRes

PDUSessionResourceReleasedItemRelRes ::= SEQUENCE {
    pduSessionID          PDUSessionID,
    pduSessionResourceReleaseResponseTransfer OCTET STRING (CONTAINING PDUSessionResourceReleaseResponseTransfer),
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceReleasedItemRelRes-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceReleasedItemRelRes-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceReleaseResponseTransfer ::= SEQUENCE {
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceReleaseResponseTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceReleaseResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-SecondaryRATUsageInformation          CRITICALITY ignore EXTENSION SecondaryRATUsageInformation          PRESENCE optional }|
    { ID id-QosFlowAdditionalInfoList              CRITICALITY ignore EXTENSION QosFlowAdditionalInfoListRelRes          PRESENCE optional },
    ...
}

PDUSessionResourceResumeListRESReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceResumeItemRESReq

PDUSessionResourceResumeItemRESReq ::= SEQUENCE {
    pduSessionID          PDUSessionID,
    ueContextResumeRequestTransfer OCTET STRING (CONTAINING UEContextResumeRequestTransfer),
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceResumeItemRESReq-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceResumeItemRESReq-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceResumeListRESRes ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceResumeItemRESRes

PDUSessionResourceResumeItemRESRes ::= SEQUENCE {
    pduSessionID          PDUSessionID,
    ueContextResumeResponseTransfer OCTET STRING (CONTAINING UEContextResumeResponseTransfer),
    iE-Extensions          ProtocolExtensionContainer { {PDUSessionResourceResumeItemRESRes-ExtIEs} } OPTIONAL,

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    ...
}

PDUSessionResourceResumeItemRESRes-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceSecondaryRATUsageList ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceSecondaryRATUsageItem

PDUSessionResourceSecondaryRATUsageItem ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    secondaryRATDataUsageReportTransfer OCTET STRING (CONTAINING SecondaryRATDataUsageReportTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceSecondaryRATUsageItem-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSecondaryRATUsageItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceSetupListCxtReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceSetupItemCxtReq

PDUSessionResourceSetupItemCxtReq ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    nAS-PDU                     NAS-PDU                                OPTIONAL,
    s-NSSAI                     S-NSSAI,
    pDUSessionResourceSetupRequestTransfer OCTET STRING (CONTAINING PDUSessionResourceSetupRequestTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceSetupItemCxtReq-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSetupItemCxtReq-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-PduSessionExpectedUEActivityBehaviour CRITICALITY ignore EXTENSION ExpectedUEActivityBehaviour PRESENCE optional },
    ...
}

PDUSessionResourceSetupListCxtRes ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceSetupItemCxtRes

PDUSessionResourceSetupItemCxtRes ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    pDUSessionResourceSetupResponseTransfer OCTET STRING (CONTAINING PDUSessionResourceSetupResponseTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceSetupItemCxtRes-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSetupItemCxtRes-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceSetupListH0Req ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceSetupItemH0Req

PDUSessionResourceSetupItemH0Req ::= SEQUENCE {
    pDUSessionID                PDUSessionID,
    s-NSSAI                     S-NSSAI,

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    handoverRequestTransfer      OCTET STRING (CONTAINING PDUSessionResourceSetupRequestTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceSetupItemHOREq-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSetupItemHOREq-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-PduSessionExpectedUEActivityBehaviour          CRITICALITY ignore  EXTENSION ExpectedUEActivityBehaviour PRESENCE optional },
    ...
}

PDUSessionResourceSetupListSUReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceSetupItemSUReq

PDUSessionResourceSetupItemSUReq ::= SEQUENCE {
    pduSessionID                PDUSessionID,
    pduSessionNAS-PDU            NAS-PDU                                OPTIONAL,
    s-NSSAI                      S-NSSAI,
    pduSessionResourceSetupRequestTransfer      OCTET STRING (CONTAINING PDUSessionResourceSetupRequestTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceSetupItemSUReq-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSetupItemSUReq-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-PduSessionExpectedUEActivityBehaviour          CRITICALITY ignore  EXTENSION ExpectedUEActivityBehaviour PRESENCE optional },
    ...
}

PDUSessionResourceSetupListSURES ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceSetupItemSURES

PDUSessionResourceSetupItemSURES ::= SEQUENCE {
    pduSessionID                PDUSessionID,
    pduSessionResourceSetupResponseTransfer      OCTET STRING (CONTAINING PDUSessionResourceSetupResponseTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceSetupItemSURES-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSetupItemSURES-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceSetupRequestTransfer ::= SEQUENCE {
    protocolIEs                ProtocolIE-Container { {PDUSessionResourceSetupRequestTransferIEs} },
    ...
}

PDUSessionResourceSetupRequestTransferIEs NGAP-PROTOCOL-IES ::= {
    { ID id-PDUSessionAggregateMaximumBitRate          CRITICALITY reject  TYPE PDUSessionAggregateMaximumBitRate          PRESENCE optional  }|
    { ID id-UL-NGU-UP-TNLInformation                  CRITICALITY reject  TYPE UPTransportLayerInformation          PRESENCE mandatory  }|
    { ID id-AdditionalUL-NGU-UP-TNLInformationList      CRITICALITY reject  TYPE UPTransportLayerInformationList      PRESENCE optional   }|
    { ID id-DataForwardingNotPossible                  CRITICALITY reject  TYPE DataForwardingNotPossible           PRESENCE optional   }|
    { ID id-PDUSessionType                             CRITICALITY reject  TYPE PDUSessionType                     PRESENCE mandatory  }|
    { ID id-SecurityIndication                         CRITICALITY reject  TYPE SecurityIndication                  PRESENCE optional   }|
    { ID id-NetworkInstance                           CRITICALITY reject  TYPE NetworkInstance                    PRESENCE optional   }|
    { ID id-QoSFlowSetupRequestList                    CRITICALITY reject  TYPE QoSFlowSetupRequestList            PRESENCE mandatory  }|
    { ID id-CommonNetworkInstance                     CRITICALITY ignore  TYPE CommonNetworkInstance              PRESENCE optional   }|

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    { ID id-DirectForwardingPathAvailability          CRITICALITY ignore TYPE DirectForwardingPathAvailability PRESENCE optional }|
    { ID id-RedundantUL-NGU-UP-TNLInformation          CRITICALITY ignore TYPE UPTransportLayerInformation PRESENCE optional }|
    { ID id-AdditionalRedundantUL-NGU-UP-TNLInformation CRITICALITY ignore TYPE UPTransportLayerInformationList PRESENCE optional }|
    { ID id-RedundantCommonNetworkInstance            CRITICALITY ignore TYPE CommonNetworkInstance PRESENCE optional }|
    { ID id-RedundantPDUSessionInformation            CRITICALITY ignore TYPE RedundantPDUSessionInformation PRESENCE optional }|
    { ID id-MBSSessionSetupRequestList                CRITICALITY ignore TYPE MBSSessionSetupRequestList PRESENCE optional }|
    { ID id-DownlinkTLCContainer                      CRITICALITY ignore TYPE TLCContainer PRESENCE optional },
    ...
}

PDUSessionResourceSetupResponseTransfer ::= SEQUENCE {
    dLQosFlowPerTNLInformation      QosFlowPerTNLInformation,
    additionalDLQosFlowPerTNLInformation QosFlowPerTNLInformationList OPTIONAL,
    securityResult                  SecurityResult OPTIONAL,
    qosFlowFailedToSetupList        QosFlowListWithCause OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {PDUSessionResourceSetupResponseTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSetupResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-RedundantDLQosFlowPerTNLInformation          CRITICALITY ignore EXTENSION QosFlowPerTNLInformation PRESENCE optional }|
    { ID id-AdditionalRedundantDLQosFlowPerTNLInformation CRITICALITY ignore EXTENSION QosFlowPerTNLInformationList PRESENCE optional }|
    { ID id-UsedRSNInformation                          CRITICALITY ignore EXTENSION RedundantPDUSessionInformation PRESENCE optional }|
    { ID id-GlobalRANNodeID                            CRITICALITY ignore EXTENSION GlobalRANNodeID PRESENCE optional }|
    { ID id-MBS-SupportIndicator                        CRITICALITY ignore EXTENSION MBS-SupportIndicator PRESENCE optional }|
    { ID id-MBSSessionSetupResponseList                 CRITICALITY ignore EXTENSION MBSSessionSetupResponseList PRESENCE optional }|
    { ID id-MBSSessionFailedToSetupList                 CRITICALITY ignore EXTENSION MBSSessionFailedToSetupList PRESENCE optional }|
    { ID id-QoSFlowTSCList                             CRITICALITY ignore EXTENSION QoSFlowTSCList PRESENCE optional }|
    { ID id-UplinkTLCContainer                          CRITICALITY ignore EXTENSION TLCContainer PRESENCE optional }|
    { ID id-ECNMarkingorCongestionInformationReportingStatus CRITICALITY ignore EXTENSION ECNMarkingorCongestionInformationReportingStatus PRESENCE optional }|
    PRESENCE optional }|
    { ID id-PDUSetbasedHandlingIndicator                CRITICALITY ignore EXTENSION PDUSetbasedHandlingIndicator PRESENCE optional },
    ...
}

PDUSessionResourceSetupUnsuccessfulTransfer ::= SEQUENCE {
    cause                Cause,
    criticalityDiagnostics CriticalityDiagnostics OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { {PDUSessionResourceSetupUnsuccessfulTransfer-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSetupUnsuccessfulTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceSuspendListSUSReq ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceSuspendItemSUSReq

PDUSessionResourceSuspendItemSUSReq ::= SEQUENCE {
    pDUSessionID          PDUSessionID,
    ueContextSuspendRequestTransfer OCTET STRING (CONTAINING UEContextSuspendRequestTransfer),
    iE-Extensions         ProtocolExtensionContainer { {PDUSessionResourceSuspendItemSUSReq-ExtIEs} } OPTIONAL,
    ...
}

```

```

PDUSessionResourceSuspendItemSUSReq-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceSwitchedList ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceSwitchedItem

PDUSessionResourceSwitchedItem ::= SEQUENCE {
    pduSessionID                PDUSessionID,
    pathSwitchRequestAcknowledgeTransfer  OCTET STRING (CONTAINING PathSwitchRequestAcknowledgeTransfer),
    iE-Extensions                ProtocolExtensionContainer { { PDUSessionResourceSwitchedItem-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceSwitchedItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-PduSessionExpectedUEActivityBehaviour          CRITICALITY ignore  EXTENSION ExpectedUEActivityBehaviour PRESENCE optional },
    ...
}

PDUSessionResourceToBeSwitchedDLList ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceToBeSwitchedDLItem

PDUSessionResourceToBeSwitchedDLItem ::= SEQUENCE {
    pduSessionID                PDUSessionID,
    pathSwitchRequestTransfer    OCTET STRING (CONTAINING PathSwitchRequestTransfer),
    iE-Extensions                ProtocolExtensionContainer { { PDUSessionResourceToBeSwitchedDLItem-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceToBeSwitchedDLItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceToReleaseListHOCmd ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceToReleaseItemHOCmd

PDUSessionResourceToReleaseItemHOCmd ::= SEQUENCE {
    pduSessionID                PDUSessionID,
    handoverPreparationUnsuccessfulTransfer  OCTET STRING (CONTAINING HandoverPreparationUnsuccessfulTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceToReleaseItemHOCmd-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionResourceToReleaseItemHOCmd-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionResourceToReleaseListRelCmd ::= SEQUENCE (SIZE(1..maxnoofPDUSessions)) OF PDUSessionResourceToReleaseItemRelCmd

PDUSessionResourceToReleaseItemRelCmd ::= SEQUENCE {
    pduSessionID                PDUSessionID,
    pduSessionResourceReleaseCommandTransfer  OCTET STRING (CONTAINING PDUSessionResourceReleaseCommandTransfer),
    iE-Extensions                ProtocolExtensionContainer { {PDUSessionResourceToReleaseItemRelCmd-ExtIEs} } OPTIONAL,
    ...
}

```

```

PDUSessionResourceToReleaseItemRelCmd-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUSessionType ::= ENUMERATED {
    ipv4,
    ipv6,
    ipv4v6,
    ethernet,
    unstructured,
    ...
}

PDUSessionUsageReport ::= SEQUENCE {
    rATType          ENUMERATED {nr, eutra, ..., nr-unlicensed, e-utra-unlicensed},
    pduSessionTimedReportList  VolumeTimedReportList,
    iE-Extensions    ProtocolExtensionContainer { {PDUSessionUsageReport-ExtIEs} } OPTIONAL,
    ...
}

PDUSessionUsageReport-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUsetQoSParameters ::= SEQUENCE {
    ulPDUsetQoSInformation  PDUsetQoSInformation  OPTIONAL,
    dlPDUsetQoSInformation  PDUsetQoSInformation  OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { PDUsetQoSParameters-ExtIEs } }  OPTIONAL,
    ...
}

PDUsetQoSParameters-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUsetQoSInformation ::= SEQUENCE {
    pduSetDelayBudget      ExtendedPacketDelayBudget  OPTIONAL,
    pduSetErrorRate        PacketErrorRate            OPTIONAL,
    pduSetIntegratedHandlingInformation  ENUMERATED {true, false, ...}  OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { PDUsetQoSInformation-ExtIEs } } OPTIONAL,
    ...
}

PDUsetQoSInformation-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PDUsetbasedHandlingIndicator ::= ENUMERATED {supported, ...}

PEIPsAssistanceInformation ::= SEQUENCE {
    cNsubgroupID          CNsubgroupID,
    iE-Extensions          ProtocolExtensionContainer { {PEIPsAssistanceInformation-ExtIEs} } OPTIONAL,
    ...
}

```

```

PEIPSSassistanceInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

Periodicity ::= INTEGER (0..640000, ...)

PeriodicityBound ::= SEQUENCE {
    periodicityLowerBound      Periodicity,
    periodicityUpperBound     Periodicity,
    iE-Extensions             ProtocolExtensionContainer { {PeriodicityBound-ExtIEs} } OPTIONAL,
    ...
}

PeriodicityBound-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PeriodicityList ::= SEQUENCE {
    allowedPeriodicityList    AllowedPeriodicityList,
    iE-Extensions             ProtocolExtensionContainer { {PeriodicityList-ExtIEs} } OPTIONAL,
    ...
}

PeriodicityList-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

AllowedPeriodicityList ::= SEQUENCE (SIZE(1..maxnoofPeriodicities)) OF Periodicity

PeriodicityRange ::= CHOICE {
    periodicityBound          PeriodicityBound,
    periodicityList           AllowedPeriodicityList,
    choice-extensions         ProtocolIE-SingleContainer { {PeriodicityRange-ExtIEs} }
}

PeriodicityRange-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

PeriodicRegistrationUpdateTimer ::= BIT STRING (SIZE(8))

PLMNIdentity ::= OCTET STRING (SIZE(3))

PLMNAreaBasedQMC ::= SEQUENCE {
    plmnListforQMC           PLMNListforQMC,
    iE-Extensions            ProtocolExtensionContainer { {PLMNAreaBasedQMC-ExtIEs} } OPTIONAL,
    ...
}

PLMNAreaBasedQMC-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

PLMNListforQMC ::= SEQUENCE (SIZE(1..maxnoofPLMNforQMC)) OF PLMNIdentity

PLMNSupportList ::= SEQUENCE (SIZE(1..maxnoofPLMNs)) OF PLMNSupportItem

PLMNSupportItem ::= SEQUENCE {
    pLMNIdentity          PLMNIdentity,
    sliceSupportList      SliceSupportList,
    iE-Extensions         ProtocolExtensionContainer { {PLMNSupportItem-ExtIEs} } OPTIONAL,
    ...
}

PLMNSupportItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-NPN-Support          CRITICALITY reject EXTENSION NPN-Support          PRESENCE optional }|
    { ID id-ExtendedSliceSupportList CRITICALITY reject EXTENSION ExtendedSliceSupportList PRESENCE optional }|
    { ID id-OnboardingSupport     CRITICALITY ignore EXTENSION OnboardingSupport     PRESENCE optional },
    ...
}

PNI-NPN-MobilityInformation ::= SEQUENCE {
    allowed-PNI-NPI-List      Allowed-PNI-NPN-List,
    iE-Extensions            ProtocolExtensionContainer { {PNI-NPN-MobilityInformation-ExtIEs} } OPTIONAL,
    ...
}

PNI-NPN-MobilityInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PortNumber ::= OCTET STRING (SIZE(2))

Pre-emptionCapability ::= ENUMERATED {
    shall-not-trigger-pre-emption,
    may-trigger-pre-emption,
    ...
}

Pre-emptionVulnerability ::= ENUMERATED {
    not-pre-emptable,
    pre-emptable,
    ...
}

PriorityLevelARP ::= INTEGER (1..15)

PriorityLevelQos ::= INTEGER (1..127, ...)

PWSFailedCellIDList ::= CHOICE {
    eUTRA-CGI-PWSFailedList      EUTRA-CGIList,
    nR-CGI-PWSFailedList         NR-CGIList,
    choice-Extensions            ProtocolIE-SingleContainer { {PWSFailedCellIDList-ExtIEs} }
}

PWSFailedCellIDList-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

```



```

}

PNI-NPN-AreaScopeofMDT ::= SEQUENCE {
    cAGListforMDT      CAGListforMDT,
    iE-Extensions      ProtocolExtensionContainer { {PNI-NPN-AreaScopeofMDT-ExtIEs} } OPTIONAL,
    ...
}

PNI-NPN-AreaScopeofMDT-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

PNI-NPNBasedMDT ::= SEQUENCE {
    cAGListforMDT      CAGListforMDT,
    iE-Extensions      ProtocolExtensionContainer { {PNI-NPNBasedMDT-ExtIEs} } OPTIONAL,
    ...
}

PNI-NPNBasedMDT-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- Q

QMCCfgInfo ::= SEQUENCE {
    ueAppLayerMeasInfoList      UEAppLayerMeasInfoList,
    iE-Extensions      ProtocolExtensionContainer { { QMCCfgInfo-ExtIEs} } OPTIONAL,
    ...
}

QMCCfgInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QMCDeactivation ::= SEQUENCE {
    qoEReferenceList      QoEReferenceList,
    iE-Extensions      ProtocolExtensionContainer { { QMCDeactivation-ExtIEs} } OPTIONAL,
    ...
}

QMCDeactivation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QoEReferenceList ::= SEQUENCE (SIZE(1..maxnoofUEAppLayerMeas)) OF QoEReference

QoEReference ::= OCTET STRING (SIZE(6))

QoERVQoEReportingPaths ::= SEQUENCE {
    qoEReportingPath      ENUMERATED{srb4, srb5, ...} OPTIONAL,
    rvQoEReportingPath      ENUMERATED{srb4, srb5, ...} OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {QoERVQoEReportingPaths-ExtIEs} } OPTIONAL,
    ...
}

```

```

QoERVQoEReportingPaths-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QosCharacteristics ::= CHOICE {
    nonDynamic5QI      NonDynamic5QIDescriptor,
    dynamic5QI         Dynamic5QIDescriptor,
    choice-Extensions  ProtocolIE-SingleContainer { {QosCharacteristics-ExtIEs} }
}

QosCharacteristics-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

QosFlowAcceptedList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowAcceptedItem

QosFlowAcceptedItem ::= SEQUENCE {
    qosFlowIdentifier      QosFlowIdentifier,
    iE-Extensions          ProtocolExtensionContainer { {QosFlowAcceptedItem-ExtIEs} } OPTIONAL,
    ...
}

QosFlowAcceptedItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-CurrentQoSParaSetIndex CRITICALITY ignore EXTENSION AlternativeQoSParaSetIndex PRESENCE optional },
    ...
}

QosFlowAdditionalInfoListRelCom ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowAdditionalInfoItemRelCom

QosFlowAdditionalInfoItemRelCom ::= SEQUENCE {
    qosFlowIdentifier      QosFlowIdentifier,
    downlinkTLContainer    TLContainer OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {QosFlowAdditionalInfoItemRelCom-ExtIEs} } OPTIONAL,
    ...
}

QosFlowAdditionalInfoItemRelCom-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QosFlowAdditionalInfoListRelRes ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowAdditionalInfoItemRelRes

QosFlowAdditionalInfoItemRelRes ::= SEQUENCE {
    qosFlowIdentifier      QosFlowIdentifier,
    uplinkTLContainer      TLContainer OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {QosFlowAdditionalInfoItemRelRes-ExtIEs} } OPTIONAL,
    ...
}

QosFlowAdditionalInfoItemRelRes-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

QosFlowAddOrModifyRequestList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowAddOrModifyRequestItem

QosFlowAddOrModifyRequestItem ::= SEQUENCE {
 qosFlowIdentifier QosFlowIdentifier,
 qosFlowLevelQosParameters QosFlowLevelQosParameters OPTIONAL,
 e-RAB-ID E-RAB-ID OPTIONAL,
 iE-Extensions ProtocolExtensionContainer { {QosFlowAddOrModifyRequestItem-ExtIEs} } OPTIONAL,
 ...
 }

QosFlowAddOrModifyRequestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
 {ID id-TSCTrafficCharacteristics CRITICALITY ignore EXTENSION TSCTrafficCharacteristics PRESENCE optional }|
 {ID id-RedundantQosFlowIndicator CRITICALITY ignore EXTENSION RedundantQosFlowIndicator PRESENCE optional }|
 {ID id-UL-NGU-UP-TNLInformation CRITICALITY ignore EXTENSION UPTransportLayerInformation PRESENCE optional }|
 {ID id-DownlinkTLContainer CRITICALITY ignore EXTENSION TLContainer PRESENCE optional }|
 {ID id-ECNMarkingorCongestionInformationReportingRequest CRITICALITY ignore EXTENSION ECNMarkingorCongestionInformationReportingRequest PRESENCE optional },
 ...
 }

QosFlowAddOrModifyResponseList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowAddOrModifyResponseItem

QosFlowAddOrModifyResponseItem ::= SEQUENCE {
 qosFlowIdentifier QosFlowIdentifier,
 iE-Extensions ProtocolExtensionContainer { {QosFlowAddOrModifyResponseItem-ExtIEs} } OPTIONAL,
 ...
 }

QosFlowAddOrModifyResponseItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
 { ID id-CurrentQoSParaSetIndex CRITICALITY ignore EXTENSION AlternativeQoSParaSetIndex PRESENCE optional }|
 { ID id-TSCTrafficCharacteristicsFeedback CRITICALITY ignore EXTENSION TSCTrafficCharacteristicsFeedback PRESENCE optional }|
 { ID id-UplinkTLContainer CRITICALITY ignore EXTENSION TLContainer PRESENCE optional }|
 { ID id-DL-NGU-UP-TNLInformation CRITICALITY ignore EXTENSION UPTransportLayerInformation PRESENCE optional }|
 { ID id-ANPacketDelayBudgetUL CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget PRESENCE optional },
 ...
 }

QosFlowFeedbackList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowFeedbackItem

QosFlowFeedbackItem ::= SEQUENCE {
 qosFlowIdentifier QosFlowIdentifier,
 updateFeedback UpdateFeedback OPTIONAL,
 cNpacketDelayBudgetDL ExtendedPacketDelayBudget OPTIONAL,
 cNpacketDelayBudgetUL ExtendedPacketDelayBudget OPTIONAL,
 iE-Extensions ProtocolExtensionContainer { {QosFlowFeedbackItem-ExtIEs} } OPTIONAL,
 ...
 }

QosFlowFeedbackItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
 ...
 }

```

QosFlowIdentifier ::= INTEGER (0..63, ...)

QosFlowInformationList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowInformationItem

QosFlowInformationItem ::= SEQUENCE {
    qosFlowIdentifier      QosFlowIdentifier,
    dlForwarding           DLForwarding                                OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {QosFlowInformationItem-ExtIEs} } OPTIONAL,
    ...
}

QosFlowInformationItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-ULForwarding      CRITICALITY ignore EXTENSION ULForwarding          PRESENCE optional}|
    {ID id-SourceTNLAddrInfo CRITICALITY ignore EXTENSION TransportLayerAddress PRESENCE optional}|
    {ID id-SourceNodeTNLAddrInfo CRITICALITY ignore EXTENSION TransportLayerAddress PRESENCE optional},
    ...
}

QosFlowLevelQosParameters ::= SEQUENCE {
    qosCharacteristics      QosCharacteristics,
    allocationAndRetentionPriority AllocationAndRetentionPriority,
    gBR-QosInformation      GBR-QosInformation                                OPTIONAL,
    reflectiveQosAttribute   ReflectiveQosAttribute                        OPTIONAL,
    additionalQosFlowInformation AdditionalQosFlowInformation              OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { {QosFlowLevelQosParameters-ExtIEs} } OPTIONAL,
    ...
}

QosFlowLevelQosParameters-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-QosMonitoringRequest      CRITICALITY ignore EXTENSION QosMonitoringRequest          PRESENCE optional}|
    {ID id-QosMonitoringReportingFrequency CRITICALITY ignore EXTENSION QosMonitoringReportingFrequency PRESENCE optional}|
    {ID id-PDUsSetQoSParameters      CRITICALITY ignore EXTENSION PDUsSetQoSParameters          PRESENCE optional}|
    {ID id-DLPDUsSetInformationMarkingSupportIndication CRITICALITY ignore EXTENSION DLPDUsSetInformationMarkingSupportIndication PRESENCE optional}|
    { ID id-MMSID                    CRITICALITY ignore EXTENSION MMSID                      PRESENCE optional}|
    { ID id-Indication-of-Bitrate-Adaptation CRITICALITY ignore EXTENSION Indication-of-Bitrate-Adaptation PRESENCE optional},
    ...
}

QosMonitoringRequest ::= ENUMERATED {ul, dl, both, ..., stop}

QosMonitoringReportingFrequency ::= INTEGER (1..1800, ...)

QosFlowList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowIdentifier

QosFlowListWithCause ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowWithCauseItem

QosFlowWithCauseItem ::= SEQUENCE {
    qosFlowIdentifier      QosFlowIdentifier,
    cause                  Cause,
    iE-Extensions          ProtocolExtensionContainer { {QosFlowWithCauseItem-ExtIEs} } OPTIONAL,
    ...
}

```

```

QosFlowWithCauseItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QosFlowModifyConfirmList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowModifyConfirmItem

QosFlowModifyConfirmItem ::= SEQUENCE {
    qosFlowIdentifier          QosFlowIdentifier,
    iE-Extensions              ProtocolExtensionContainer { {QosFlowModifyConfirmItem-ExtIEs} }    OPTIONAL,
    ...
}

QosFlowModifyConfirmItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QosFlowNotifyList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowNotifyItem

QosFlowNotifyItem ::= SEQUENCE {
    qosFlowIdentifier          QosFlowIdentifier,
    notificationCause          NotificationCause,
    iE-Extensions              ProtocolExtensionContainer { {QosFlowNotifyItem-ExtIEs} }    OPTIONAL,
    ...
}

QosFlowNotifyItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-CurrentQoSParaSetIndex          CRITICALITY ignore EXTENSION AlternativeQoSParaSetNotifyIndex PRESENCE optional }|
    { ID id-TSCTrafficCharacteristicsFeedback CRITICALITY ignore EXTENSION TSCTrafficCharacteristicsFeedback PRESENCE optional },
    ...
}

QosFlowParametersList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowParametersItem

QosFlowParametersItem ::= SEQUENCE {
    qosFlowIdentifier          QosFlowIdentifier,
    alternativeQoSParaSetList  AlternativeQoSParaSetList    OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { {QosFlowParametersItem-ExtIEs} }    OPTIONAL,
    ...
}

QosFlowParametersItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-CNPacketDelayBudgetDL          CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget PRESENCE optional }|
    { ID id-CNPacketDelayBudgetUL          CRITICALITY ignore EXTENSION ExtendedPacketDelayBudget PRESENCE optional }|
    { ID id-BurstArrivalTimeDownlink        CRITICALITY ignore EXTENSION BurstArrivalTime PRESENCE optional },
    ...
}

QosFlowPerTNLInformation ::= SEQUENCE {
    uPTransportLayerInformation  UPTransportLayerInformation,
    associatedQosFlowList        AssociatedQosFlowList,
    iE-Extensions              ProtocolExtensionContainer { { QosFlowPerTNLInformation-ExtIEs} }    OPTIONAL,
    ...
}

```

```

QosFlowPerTNLInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QosFlowPerTNLInformationList ::= SEQUENCE (SIZE(1..maxnoofMultiConnectivityMinusOne)) OF QosFlowPerTNLInformationItem

QosFlowPerTNLInformationItem ::= SEQUENCE {
    qosFlowPerTNLInformation          QosFlowPerTNLInformation,
    iE-Extensions                     ProtocolExtensionContainer { { QosFlowPerTNLInformationItem-ExtIEs} } OPTIONAL,
    ...
}

QosFlowPerTNLInformationItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QosFlowSetupRequestList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowSetupRequestItem

QosFlowSetupRequestItem ::= SEQUENCE {
    qosFlowIdentifier                 QosFlowIdentifier,
    qosFlowLevelQosParameters         QosFlowLevelQosParameters,
    e-RAB-ID                         E-RAB-ID OPTIONAL,
    iE-Extensions                     ProtocolExtensionContainer { {QosFlowSetupRequestItem-ExtIEs} } OPTIONAL,
    ...
}

QosFlowSetupRequestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-TSCTrafficCharacteristics CRITICALITY ignore EXTENSION TSCTrafficCharacteristics PRESENCE optional }|
    {ID id-RedundantQosFlowIndicator CRITICALITY ignore EXTENSION RedundantQosFlowIndicator PRESENCE optional }|
    {ID id-ECNMarkingorCongestionInformationReportingRequest CRITICALITY ignore EXTENSION ECNMarkingorCongestionInformationReportingRequest PRESENCE optional },
    ...
}

QosFlowListWithDataForwarding ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowItemWithDataForwarding

QosFlowItemWithDataForwarding ::= SEQUENCE {
    qosFlowIdentifier                 QosFlowIdentifier,
    dataForwardingAccepted             DataForwardingAccepted OPTIONAL,
    iE-Extensions                     ProtocolExtensionContainer { {QosFlowItemWithDataForwarding-ExtIEs} } OPTIONAL,
    ...
}

QosFlowItemWithDataForwarding-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-CurrentQosParaSetIndex CRITICALITY ignore EXTENSION AlternativeQosParaSetIndex PRESENCE optional },
    ...
}

QosFlowToBeForwardedList ::= SEQUENCE (SIZE(1..maxnoofQosFlows)) OF QosFlowToBeForwardedItem

QosFlowToBeForwardedItem ::= SEQUENCE {
    qosFlowIdentifier                 QosFlowIdentifier,
    iE-Extensions                     ProtocolExtensionContainer { {QosFlowToBeForwardedItem-ExtIEs} } OPTIONAL,
    ...
}

```

```

}

QoSFlowToBeForwardedItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QoSFlowTSCList ::= SEQUENCE (SIZE(1..maxnoofQoSFlows)) OF QoSFlowTSCItem

QoSFlowTSCItem ::= SEQUENCE {
    qosFlowIdentifier          QoSFlowIdentifier,
    tSCTrafficCharacteristicsFeedback  TSCTrafficCharacteristicsFeedback  OPTIONAL,
    aNPacketDelayBudgetUL      ExtendedPacketDelayBudget  OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { {QoSFlowTSCItem-ExtIEs} }  OPTIONAL,
    ...
}

QoSFlowTSCItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

QoSFlowsUsageReportList ::= SEQUENCE (SIZE(1..maxnoofQoSFlows)) OF QoSFlowsUsageReport-Item

QoSFlowsUsageReport-Item ::= SEQUENCE {
    qosFlowIdentifier          QoSFlowIdentifier,
    rATType                    ENUMERATED {nr, eutra, ..., nr-unlicensed, e-utra-unlicensed},
    qosFlowsTimedReportList    VolumeTimedReportList,
    iE-Extensions              ProtocolExtensionContainer { {QoSFlowsUsageReport-Item-ExtIEs} }  OPTIONAL,
    ...
}

QoSFlowsUsageReport-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- R

RANfeedbacktype ::= CHOICE {
    proactive                RANfeedbacktype-proactive,
    reactive                 RANfeedbacktype-reactive,
    choice-extensions        ProtocolIE-SingleContainer { {RANfeedbacktype-ExtIEs} }
}

RANfeedbacktype-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

RANfeedbacktype-proactive ::= SEQUENCE {
    burstArrivalTimeWindow    BurstArrivalTimeWindow,
    periodicityRange           PeriodicityRange  OPTIONAL,
    iE-Extension               ProtocolExtensionContainer { {RANfeedbacktype-proactive-ExtIEs} }  OPTIONAL,
    ...
}

RANfeedbacktype-proactive-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {

```

```

}
...
}
RANfeedbacktype-reactive ::= SEQUENCE {
    capabilityForBATAdaptation  ENUMERATED {true, ...},
    iE-Extension                ProtocolExtensionContainer { {RANfeedbacktype-reactive-ExtIEs} }  OPTIONAL,
    ...
}
RANfeedbacktype-reactive-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}
Range ::= ENUMERATED {m50, m80, m180, m200, m350, m400, m500, m700, m1000, ...}
RANnodeName ::= PrintableString (SIZE(1..150, ...))
RANnodeNameVisibleString ::= VisibleString (SIZE(1..150, ...))
RANnodeNameUTF8String ::= UTF8String (SIZE(1..150, ...))
RANpagingPriority ::= INTEGER (1..256)
RANstatusTransfer-TransparentContainer ::= SEQUENCE {
    dRBsSubjectToStatusTransferList  DRBsSubjectToStatusTransferList,
    iE-Extensions                    ProtocolExtensionContainer { {RANstatusTransfer-TransparentContainer-ExtIEs} }  OPTIONAL,
    ...
}
RANstatusTransfer-TransparentContainer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}
RANtimingSynchronisationStatusInfo ::= SEQUENCE {
    synchronisationState      ENUMERATED {locked, holdover, freerun, ...}  OPTIONAL,
    traceabletoUTC            ENUMERATED {true, false, ...}  OPTIONAL,
    traceabletoGNSS           ENUMERATED {true, false, ...}  OPTIONAL,
    clockFrequencyStability   BIT STRING (SIZE(16))  OPTIONAL,
    clockAccuracy             ClockAccuracy  OPTIONAL,
    parentTimeSource          ParentTimeSource  OPTIONAL,
    iE-Extensions            ProtocolExtensionContainer { {RANtimingSynchronisationStatusInfo-ExtIEs} }  OPTIONAL,
    ...
}
RANtimingSynchronisationStatusInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}
RAN-TSSRequestType ::= ENUMERATED {start, stop, ...}
RAN-TSSScope ::= CHOICE {
    rANnodeLevel      GlobalGNB-ID,
    cellListLevel     RANTSSCellList,
}

```



```

    choice-Extensions      ProtocolIE-SingleContainer { { RAN-TSSScope-ExtIEs } }
  }

RAN-TSSScope-ExtIEs NGAP-PROTOCOL-IES ::= {
  ...
}

RANTSSCellList ::= SEQUENCE (SIZE(1..maxnoofCellsTSS)) OF RANTSSCellItem

RANTSSCellItem ::= SEQUENCE {
  nRCGI          NR-CGI,
  iE-Extensions  ProtocolExtensionContainer { {RANTSSCellItem-ExtIEs} } OPTIONAL,
  ...
}

RANTSSCellItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

RAN-UE-NGAP-ID ::= INTEGER (0..4294967295)

RAT-Information ::= ENUMERATED {
  unlicensed,
  nb-IoT,
  ...,
  nR-LEO,
  nR-MEO,
  nR-GEO,
  nR-OTHERSAT
}

RATRestrictions ::= SEQUENCE (SIZE(1..maxnoofEPLMNsPlusOne)) OF RATRestrictions-Item

RATRestrictions-Item ::= SEQUENCE {
  pLMNIdentity          PLMNIdentity,
  rATRestrictionInformation RATRestrictionInformation,
  iE-Extensions          ProtocolExtensionContainer { {RATRestrictions-Item-ExtIEs} } OPTIONAL,
  ...
}

RATRestrictions-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  { ID id-ExtendedRATRestrictionInformation CRITICALITY ignore EXTENSION ExtendedRATRestrictionInformation PRESENCE optional },
  ...
}

RATRestrictionInformation ::= BIT STRING (SIZE(8, ...))

RecommendedCellsForPaging ::= SEQUENCE {
  recommendedCellList      RecommendedCellList,
  iE-Extensions            ProtocolExtensionContainer { {RecommendedCellsForPaging-ExtIEs} } OPTIONAL,
  ...
}

RecommendedCellsForPaging-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {

```

```

    ...
}

RecommendedCellList ::= SEQUENCE (SIZE(1..maxnoofRecommendedCells)) OF RecommendedCellItem

RecommendedCellItem ::= SEQUENCE {
    nGRAN-CGI          NGRAN-CGI,
    timeStayedInCell   INTEGER (0..4095) OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {RecommendedCellItem-ExtIEs} } OPTIONAL,
    ...
}

RecommendedCellItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

RecommendedRANNodesForPaging ::= SEQUENCE {
    recommendedRANNodeList RecommendedRANNodeList,
    iE-Extensions          ProtocolExtensionContainer { {RecommendedRANNodesForPaging-ExtIEs} } OPTIONAL,
    ...
}

RecommendedRANNodesForPaging-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

RecommendedRANNodeList ::= SEQUENCE (SIZE(1..maxnoofRecommendedRANNodes)) OF RecommendedRANNodeItem

RecommendedRANNodeItem ::= SEQUENCE {
    amFPagingTarget AMFPagingTarget,
    iE-Extensions  ProtocolExtensionContainer { {RecommendedRANNodeItem-ExtIEs} } OPTIONAL,
    ...
}

RecommendedRANNodeItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

RedCapIndication ::= ENUMERATED {
    redcap,
    ...
}

RedirectionVoiceFallback ::= ENUMERATED {
    possible,
    not-possible,
    ...
}

RedundantPDUSessionInformation ::= SEQUENCE {
    rSN          RSN,
    iE-Extensions ProtocolExtensionContainer { {RedundantPDUSessionInformation-ExtIEs} } OPTIONAL,
    ...
}

```

```

RedundantPDUSessionInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-PDUSessionPairID    CRITICALITY ignore  EXTENSION PDUSessionPairID  PRESENCE optional },
    ...
}

RedundantQosFlowIndicator ::= ENUMERATED {true, false}

ReflectiveQosAttribute ::= ENUMERATED {
    subject-to,
    ...
}

RelativeAMFCapacity ::= INTEGER (0..255)

ReportArea ::= ENUMERATED {
    cell,
    ...
}

ReportingThreshold ::= INTEGER (0.. 4000000000, ...)

RepetitionPeriod ::= INTEGER (0..131071)

ResetAll ::= ENUMERATED {
    reset-all,
    ...
}

ReportAmountMDT ::= ENUMERATED {
    r1, r2, r4, r8, r16, r32, r64, rinfinity
}

ReportIntervalMDT ::= ENUMERATED {
    ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, min1, min6, min12, min30, min60
}

ExtendedReportIntervalMDT ::= ENUMERATED {
    ms20480, ms40960, ...
}

RequestedNSSAI ::= SEQUENCE (SIZE(1..maxnoofRequestedSNSSAIs)) OF RequestedNSSAIItem

RequestedNSSAIItem ::= SEQUENCE {
    s-NSSAI          S-NSSAI,
    iE-Extensions    ProtocolExtensionContainer { {RequestedNSSAIItem-ExtIEs} }  OPTIONAL,
    ...
}

RequestedNSSAIItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ResetType ::= CHOICE {

```

```

    nG-Interface          ResetAll,
    partOfNG-Interface    UE-associatedLogicalNG-connectionList,
    choice-Extensions      ProtocolIE-SingleContainer { {ResetType-ExtIEs} }
}

ResetType-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

RGLevelWirelineAccessCharacteristics ::= OCTET STRING

RNC-ID ::= INTEGER (0..4095)

RoutingID ::= OCTET STRING

RRCContainer ::= OCTET STRING

RRCEstablishmentCause ::= ENUMERATED {
    emergency,
    highPriorityAccess,
    mt-Access,
    mo-Signalling,
    mo-Data,
    mo-VoiceCall,
    mo-VideoCall,
    mo-SMS,
    mps-PriorityAccess,
    mcs-PriorityAccess,
    ...,
    notAvailable,
    mo-ExceptionData
}

RRCInactiveTransitionReportRequest ::= ENUMERATED {
    subsequent-state-transition-report,
    single-rrc-connected-state-report,
    cancel-report,
    ...
}

RRCState ::= ENUMERATED {
    inactive,
    connected,
    ...
}

RSN ::= ENUMERATED {v1, v2, ...}

RIMInformationTransfer ::= SEQUENCE {
    targetRANNodeID-RIM      TargetRANNodeID-RIM,
    sourceRANNodeID          SourceRANNodeID,
    rIMInformation            RIMInformation,
    iE-Extensions            ProtocolExtensionContainer { {RIMInformationTransfer-ExtIEs} } OPTIONAL,
    ...
}

```

```

}

RIMInformationTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

RIMInformation ::= SEQUENCE {
    targetgNBSetID      GNBSetID,
    rIM-RSDetection     ENUMERATED {rs-detected, rs-disappeared, ...},
    iE-Extensions       ProtocolExtensionContainer { {RIMInformation-ExtIEs} } OPTIONAL,
    ...
}

RIMInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

GNBSetID ::= BIT STRING (SIZE(22))

RSPPQoSFlowList ::= SEQUENCE (SIZE(1..maxnoofRSPPQoSFlows)) OF RSPPQoSFlowItem

RSPPQoSFlowItem ::= SEQUENCE {
    pQI                 FiveQI,
    rSPPFlowBitRates    RSPPFlowBitRates OPTIONAL,
    range               Range OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { RSPPQoSFlowItem-ExtIEs} } OPTIONAL,
    ...
}

RSPPQoSFlowItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

RSPPFlowBitRates ::= SEQUENCE {
    guaranteedFlowBitRate BitRate,
    maximumFlowBitRate    BitRate,
    iE-Extensions         ProtocolExtensionContainer { { RSPPFlowBitRates-ExtIEs} } OPTIONAL,
    ...
}

RSPPFlowBitRates-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

RequestedTNLInfo ::= SEQUENCE {
    pLMNIdentity          PLMNIdentity,
    iE-Extensions         ProtocolExtensionContainer { { RequestedTNLInfo-ExtIEs} } OPTIONAL,
    ...
}

RequestedTNLInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

-- S

ScheduledCommunicationTime ::= SEQUENCE {
    dayOfWeek          BIT STRING (SIZE(7))                OPTIONAL,
    timeOfDayStart      INTEGER (0..86399, ...)             OPTIONAL,
    timeOfDayEnd        INTEGER (0..86399, ...)             OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { ScheduledCommunicationTime-ExtIEs} } OPTIONAL,
    ...
}

ScheduledCommunicationTime-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SCTP-TLAs ::= SEQUENCE (SIZE(1..maxnoofXnTLAs)) OF TransportLayerAddress

SD ::= OCTET STRING (SIZE(3))

SecondaryRATUsageInformation ::= SEQUENCE {
    pduSessionUsageReport      PDUUsageReport                OPTIONAL,
    qosFlowsUsageReportList     QoSFlowsUsageReportList       OPTIONAL,
    iE-Extension               ProtocolExtensionContainer { {SecondaryRATUsageInformation-ExtIEs} } OPTIONAL,
    ...
}

SecondaryRATUsageInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SecondaryRATDataUsageReportTransfer ::= SEQUENCE {
    secondaryRATUsageInformation      SecondaryRATUsageInformation                OPTIONAL,
    iE-Extensions                    ProtocolExtensionContainer { {SecondaryRATDataUsageReportTransfer-ExtIEs} } OPTIONAL,
    ...
}

SecondaryRATDataUsageReportTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SecurityContext ::= SEQUENCE {
    nextHopChainingCount      NextHopChainingCount,
    nextHopNH                 SecurityKey,
    iE-Extensions            ProtocolExtensionContainer { {SecurityContext-ExtIEs} } OPTIONAL,
    ...
}

SecurityContext-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SecurityIndication ::= SEQUENCE {

```

```

    integrityProtectionIndication      IntegrityProtectionIndication,
    confidentialityProtectionIndication ConfidentialityProtectionIndication,
    maximumIntegrityProtectedDataRate-UL MaximumIntegrityProtectedDataRate OPTIONAL,
-- The above IE shall be present if the Integrity Protection Indication IE is set to the value "required" or "preferred"
    iE-Extensions      ProtocolExtensionContainer { {SecurityIndication-ExtIEs} } OPTIONAL,
    ...
}

SecurityIndication-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-MaximumIntegrityProtectedDataRate-DL CRITICALITY ignore EXTENSION MaximumIntegrityProtectedDataRate PRESENCE optional },
    ...
}

SecurityKey ::= BIT STRING (SIZE(256))

SecurityResult ::= SEQUENCE {
    integrityProtectionResult      IntegrityProtectionResult,
    confidentialityProtectionResult ConfidentialityProtectionResult,
    iE-Extensions      ProtocolExtensionContainer { {SecurityResult-ExtIEs} } OPTIONAL,
    ...
}

SecurityResult-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SensorMeasurementConfiguration ::= SEQUENCE {
    sensorMeasConfig      SensorMeasConfig,
    sensorMeasConfigNameList      SensorMeasConfigNameList OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {SensorMeasurementConfiguration-ExtIEs} } OPTIONAL,
    ...
}

SensorMeasurementConfiguration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SensorMeasConfigNameList ::= SEQUENCE (SIZE(1..maxnoofSensorName)) OF SensorMeasConfigNameItem

SensorMeasConfigNameItem ::= SEQUENCE {
    sensorNameConfig      SensorNameConfig,
    iE-Extensions      ProtocolExtensionContainer { { SensorMeasConfigNameItem-ExtIEs } } OPTIONAL,
    ...
}

SensorMeasConfigNameItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SensorMeasConfig ::= ENUMERATED {setup,...}

SensorNameConfig ::= CHOICE {
    uncompensatedBarometricConfig      ENUMERATED {true, ...},
    ueSpeedConfig      ENUMERATED {true, ...},

```

```

    ueOrientationConfig      ENUMERATED {true, ...},
    choice-Extensions        ProtocolIE-SingleContainer { {SensorNameConfig-ExtIEs} }
}

SensorNameConfig-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

SerialNumber ::= BIT STRING (SIZE(16))

ServedGUAMIList ::= SEQUENCE (SIZE(1..maxnoofServedGUAMIs)) OF ServedGUAMIItem

ServedGUAMIItem ::= SEQUENCE {
    GUAMI                GUAMI,
    backupAMFName         AMFName
    iE-Extensions        ProtocolExtensionContainer { {ServedGUAMIItem-ExtIEs} } OPTIONAL,
    ...
}

ServedGUAMIItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-GUAMIType      CRITICALITY ignore EXTENSION GUAMITYPE      PRESENCE optional }|
    {ID id-ExtendedBackupAMFName CRITICALITY ignore EXTENSION Extended-AMFName PRESENCE optional},
    ...
}

ServiceAreaInformation ::= SEQUENCE (SIZE(1.. maxnoofEPLMNsPlusOne)) OF ServiceAreaInformation-Item

ServiceAreaInformation-Item ::= SEQUENCE {
    PLMNIdentity          PLMNIdentity,
    allowedTACs           AllowedTACs
    notAllowedTACs        NotAllowedTACs
    iE-Extensions         ProtocolExtensionContainer { {ServiceAreaInformation-Item-ExtIEs} } OPTIONAL,
    ...
}

ServiceAreaInformation-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

ServiceType ::= ENUMERATED {streaming, mTSI, vR, ...}

SgNB-UE-X2AP-ID ::= INTEGER (0..4294967295)

SharedNGU-MulticastTNLInformation ::= SEQUENCE {
    iP-MulticastAddress    TransportLayerAddress,
    iP-SourceAddress       TransportLayerAddress,
    gTP-TEID              GTP-TEID,
    iE-Extensions         ProtocolExtensionContainer { {SharedNGU-MulticastTNLInformation-ExtIEs} } OPTIONAL,
    ...
}

SharedNGU-MulticastTNLInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```



```

SliceOverloadList ::= SEQUENCE (SIZE(1..maxnoofSliceItems)) OF SliceOverloadItem

SliceOverloadItem ::= SEQUENCE {
    s-NSSAI          S-NSSAI,
    iE-Extensions    ProtocolExtensionContainer { {SliceOverloadItem-ExtIEs} }    OPTIONAL,
    ...
}

SliceOverloadItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SliceSupportList ::= SEQUENCE (SIZE(1..maxnoofSliceItems)) OF SliceSupportItem

SliceSupportItem ::= SEQUENCE {
    s-NSSAI          S-NSSAI,
    iE-Extensions    ProtocolExtensionContainer { {SliceSupportItem-ExtIEs} }    OPTIONAL,
    ...
}

SliceSupportItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SliceSupportListQMC ::= SEQUENCE (SIZE(1..maxnoofSNSSAIforQMC)) OF SliceSupportQMC-Item

SliceSupportQMC-Item ::= SEQUENCE {
    s-NSSAI          S-NSSAI,
    iE-Extensions    ProtocolExtensionContainer { {SliceSupportQMC-Item-ExtIEs} }    OPTIONAL,
    ...
}

SliceSupportQMC-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SNPN-MobilityInformation ::= SEQUENCE {
    serving-NID      NID,
    iE-Extensions    ProtocolExtensionContainer { {SNPN-MobilityInformation-ExtIEs} }    OPTIONAL,
    ...
}

SNPN-MobilityInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-EquivalentSNPNsList CRITICALITY reject EXTENSION EquivalentSNPNsList    PRESENCE optional},
    ...
}

S-NSSAI ::= SEQUENCE {
    sST              SST,
    sD              SD,
    iE-Extensions    ProtocolExtensionContainer { { S-NSSAI-ExtIEs} }    OPTIONAL,
    ...
}

```

```

S-NSSAI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SONConfigurationTransfer ::= SEQUENCE {
    targetRANNodeID-SON      TargetRANNodeID-SON,
    sourceRANNodeID          SourceRANNodeID,
    SONInformation            SONInformation,
    xnTNLConfigurationInfo   XnTNLConfigurationInfo OPTIONAL,
    -- The above IE shall be present if the SON Information Request IE set to the value "Xn TNL Configuration Info"
    iE-Extensions            ProtocolExtensionContainer { {SONConfigurationTransfer-ExtIEs} } OPTIONAL,
    ...
}

SONConfigurationTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-RequestedTNLInfo    CRITICALITY ignore EXTENSION RequestedTNLInfo PRESENCE optional},
    ...
}

SONInformation ::= CHOICE {
    SONInformationRequest      SONInformationRequest,
    SONInformationReply        SONInformationReply,
    choice-Extensions          ProtocolIE-SingleContainer { {SONInformation-ExtIEs} }
}

SONInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-SONInformationReport CRITICALITY ignore TYPE SONInformationReport PRESENCE mandatory },
    ...
}

SONInformationReply ::= SEQUENCE {
    xnTNLConfigurationInfo     XnTNLConfigurationInfo OPTIONAL,
    iE-Extensions              ProtocolExtensionContainer { {SONInformationReply-ExtIEs} } OPTIONAL,
    ...
}

SONInformationReply-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SONInformationReport ::= CHOICE {
    failureIndicationInformation FailureIndication,
    hOReportInformation          HOReport,
    choice-Extensions            ProtocolIE-SingleContainer { { SONInformationReport-ExtIEs} }
}

SONInformationReport-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-SuccessfulHandoverReportList CRITICALITY ignore TYPE SuccessfulHandoverReportList PRESENCE mandatory }|
    { ID id-SuccessfulPSCellChangeReportList CRITICALITY ignore TYPE SuccessfulPSCellChangeReportList PRESENCE mandatory },
    ...
}

SNPN-CellBasedMDT ::= SEQUENCE {

```

```

        SNPNCellIdListforMDT      SNPNCellIdListforMDT,
        iE-Extensions      ProtocolExtensionContainer { {SNPN-CellBasedMDT-ExtIEs} } OPTIONAL,
        ...
    }

SNPN-CellBasedMDT-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SNPNCellIdListforMDT ::= SEQUENCE (SIZE(1..maxnoofCellIDforMDT)) OF SNPNCellIdListforMDTItem

SNPNCellIdListforMDTItem ::= SEQUENCE {
    nR-CGI                      NR-CGI,
    nID                        NID,
    iE-Extensions      ProtocolExtensionContainer {{ SNPNCellIdListforMDTItem-ExtIEs}}      OPTIONAL,
    ...
}

SNPNCellIdListforMDTItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SNPN-TAIBasedMDT ::= SEQUENCE {
    SNPNTAIListforMDT      SNPNTAIListforMDT,
    iE-Extensions      ProtocolExtensionContainer { {SNPN-TAIBasedMDT-ExtIEs} } OPTIONAL,
    ...
}

SNPN-TAIBasedMDT-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SNPNTAIListforMDT ::= SEQUENCE (SIZE(1.. maxnoofTAforMDT)) OF SNPNTAIListforMDTItem

SNPNTAIListforMDTItem ::= SEQUENCE {
    tAI                      TAI,
    nID                      NID,
    iE-Extensions      ProtocolExtensionContainer {{ SNPNTAIListforMDTItem-ExtIEs}}      OPTIONAL,
    ...
}

SNPNTAIListforMDTItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SNPN-BasedMDT ::= SEQUENCE {
    SNPNListforMDT      SNPNListforMDT,
    iE-Extensions      ProtocolExtensionContainer { {SNPN-BasedMDT-ExtIEs} } OPTIONAL,
    ...
}

SNPN-BasedMDT-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

}

SNPNListforMDT ::= SEQUENCE (SIZE(1.. maxnoofMDTSNPNs)) OF SNPNListforMDTItem

SNPNListforMDTItem ::= SEQUENCE {
    pLMNIdentity          PLMNIdentity,
    nID                   NID,
    iE-Extensions         ProtocolExtensionContainer {{ SNPNListforMDTItem-ExtIEs}} OPTIONAL,
    ...
}

SNPNListforMDTItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-----
-- SON Information Report
-----

SuccessfulHandoverReportList ::= SEQUENCE (SIZE(1..maxnoofSuccessfulHOReports)) OF SuccessfulHandoverReport-Item

SuccessfulHandoverReport-Item ::= SEQUENCE {
    successfulHOReportContainer OCTET STRING,
    iE-Extensions              ProtocolExtensionContainer { { SuccessfulHandoverReport-Item-ExtIEs} } OPTIONAL,
    ...
}

SuccessfulHandoverReport-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SuccessfulPSCellChangeReportList ::= SEQUENCE (SIZE(1..maxnoofSuccessfulPSCellChangeReports)) OF SuccessfulPSCellChangeReport-Item

SuccessfulPSCellChangeReport-Item ::= SEQUENCE {
    successfulPSCellChangeReportContainer OCTET STRING,
    iE-Extensions              ProtocolExtensionContainer { { SuccessfulPSCellChangeReport-Item-ExtIEs} } OPTIONAL,
    ...
}

SuccessfulPSCellChangeReport-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SONInformationRequest ::= ENUMERATED {
    xn-TNL-configuration-info,
    ...
}

SourceNGRANNode-ToTargetNGRANNode-TransparentContainer ::= SEQUENCE {
    rRCContainer          RRCContainer,
    pduSessionResourceInformationList PDU Session Resource Information List OPTIONAL,
    e-RABInformationList   E-RAB Information List OPTIONAL,
    targetCell-ID          NGRAN-CGI,
    indexToRFSP            IndexToRFSP OPTIONAL,

```

```

    uEHistoryInformation          UEHistoryInformation,
    iE-Extensions      ProtocolExtensionContainer { {SourceNGRANNode-ToTargetNGRANNode-TransparentContainer-ExtIEs} } OPTIONAL,
    ...
}

SourceNGRANNode-ToTargetNGRANNode-TransparentContainer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-SgNB-UE-X2AP-ID          CRITICALITY ignore EXTENSION SgNB-UE-X2AP-ID          PRESENCE optional }|
    { ID id-UEHistoryInformationFromTheUE CRITICALITY ignore EXTENSION UEHistoryInformationFromTheUE PRESENCE optional }|
    { ID id-SourceNodeID              CRITICALITY ignore EXTENSION SourceNodeID              PRESENCE optional }|
    { ID id-UEContextReferenceAtSource CRITICALITY ignore EXTENSION RAN-UE-NGAP-ID          PRESENCE optional }|
    { ID id-MBS-ActiveSessionInformation-SourceToTargetList CRITICALITY ignore EXTENSION MBS-ActiveSessionInformation-SourceToTargetList
    PRESENCE optional }|
    { ID id-QMCCConfigInfo            CRITICALITY ignore EXTENSION QMCCConfigInfo            PRESENCE optional }|
    { ID id-NGAPIESupportInformationRequestList CRITICALITY ignore EXTENSION NGAPIESupportInformationRequestList PRESENCE optional }|
    { ID id-CandidateRelayUEInformationList CRITICALITY reject EXTENSION CandidateRelayUEInformationList PRESENCE optional }|
    { ID id-TimeBasedHandoverInformation CRITICALITY ignore EXTENSION TimeBasedHandoverInformation PRESENCE optional }|
    { ID id-SourceSN-to-TargetSN-QMCInfo CRITICALITY ignore EXTENSION QMCCConfigInfo            PRESENCE optional },
    ...
}

SourceNodeID ::= CHOICE {
    sourceengNB-ID      GlobalGNB-ID,
    choice-Extensions  ProtocolIE-SingleContainer { { SourceNodeID-ExtIEs} }
}

SourceNodeID-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

SourceOfUEActivityBehaviourInformation ::= ENUMERATED {
    subscription-information,
    statistics,
    ...
}

SourceRANNodeID ::= SEQUENCE {
    globalRANNodeID      GlobalRANNodeID,
    selectedTAI          TAI,
    iE-Extensions      ProtocolExtensionContainer { {SourceRANNodeID-ExtIEs} } OPTIONAL,
    ...
}

SourceRANNodeID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SourceToTarget-TransparentContainer ::= OCTET STRING
-- This IE includes a transparent container from the source RAN node to the target RAN node.
-- The octets of the OCTET STRING are encoded according to the specifications of the target system.

SourceToTarget-AMFInformationReroute ::= SEQUENCE {
    configuredNSSAI      ConfiguredNSSAI          OPTIONAL,
    rejectedNSSAIinPLMN  RejectedNSSAIinPLMN        OPTIONAL,
    rejectedNSSAIinTA    RejectedNSSAIinTA          OPTIONAL,

```

```

    iE-Extensions      ProtocolExtensionContainer { {SourceToTarget-AMFInformationReroute-ExtIEs} } OPTIONAL,
    ...
}

SourceToTarget-AMFInformationReroute-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- This IE includes information from the source Core node to the target Core node for reroute information provide by NSSF.
-- The octets of the OCTET STRING are encoded according to the specifications of the Core network.

SRVCCOperationPossible ::= ENUMERATED {
    possible,
    notPossible,
    ...
}

ConfiguredNSSAI ::= OCTET STRING (SIZE(128))

RejectedNSSAIinPLMN ::= OCTET STRING (SIZE(32))

RejectedNSSAIinTA ::= OCTET STRING (SIZE(32))

SST ::= OCTET STRING (SIZE(1))

SupportedTAList ::= SEQUENCE (SIZE(1..maxnoofTACs)) OF SupportedTAItem

SupportedTAItem ::= SEQUENCE {
    tAC                      TAC,
    broadcastPLMNList        BroadcastPLMNList,
    iE-Extensions            ProtocolExtensionContainer { {SupportedTAItem-ExtIEs} } OPTIONAL,
    ...
}

SupportedTAItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-ConfiguredTACIndication      CRITICALITY ignore  EXTENSION ConfiguredTACIndication  PRESENCE optional }|
    {ID id-RAT-Information              CRITICALITY reject  EXTENSION RAT-Information        PRESENCE optional },
    ...
}

SuspendIndicator ::= ENUMERATED {
    true,
    ...
}

Suspend-Request-Indication ::= ENUMERATED {
    suspend-requested,
    ...
}

Suspend-Response-Indication ::= ENUMERATED {
    suspend-indicated,
    ...
}

```

```

SurvivalTime ::= INTEGER (0..1920000, ...)

SLPositioningRangingServiceInfo ::= SEQUENCE{
    SLPositioningRangingAuthorized      SLPositioningRangingAuthorized,
    SLPositioningRangingQoSParameters  SLPositioningRangingQoSParameters  OPTIONAL,
    iE-Extensions                      ProtocolExtensionContainer { {SLPositioningRangingServiceInfo-ExtIEs} } OPTIONAL,
    ...
}

SLPositioningRangingServiceInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SLPositioningRangingAuthorized ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

SLPositioningRangingQoSParameters ::= SEQUENCE {
    rSPPQoSFlowList                RSPPQoSFlowList,
    rSPPLinkAggregateBitRates      BitRate                                OPTIONAL,
    iE-Extensions                  ProtocolExtensionContainer { { SLPositioningRangingQoSParameters-ExtIEs} }  OPTIONAL,
    ...
}

SLPositioningRangingQoSParameters-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

SCGActivationTime ::= INTEGER (0..40950)

-- T

TAC ::= OCTET STRING (SIZE(3))

TACListInNRNTN ::= SEQUENCE (SIZE(1..maxnoofTACsinNTN)) OF TAC

TAI ::= SEQUENCE {
    pLMNIdentity      PLMNIdentity,
    tAC                TAC,
    iE-Extensions      ProtocolExtensionContainer { {TAI-ExtIEs} } OPTIONAL,
    ...
}

TAI-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIBroadcastEUTRA ::= SEQUENCE (SIZE(1..maxnoofTAIforWarning)) OF TAIBroadcastEUTRA-Item

TAIBroadcastEUTRA-Item ::= SEQUENCE {

```

```

    tAI
    completedCellsInTAI-EUTRA    TAI,
    iE-Extensions    ProtocolExtensionContainer { {TAIBroadcastEUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

TAIBroadcastEUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIBroadcastNR ::= SEQUENCE (SIZE(1..maxnoofTAIforWarning)) OF TAIBroadcastNR-Item

TAIBroadcastNR-Item ::= SEQUENCE {
    tAI
    completedCellsInTAI-NR    TAI,
    iE-Extensions    ProtocolExtensionContainer { {TAIBroadcastNR-Item-ExtIEs} } OPTIONAL,
    ...
}

TAIBroadcastNR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAICancelledEUTRA ::= SEQUENCE (SIZE(1..maxnoofTAIforWarning)) OF TAICancelledEUTRA-Item

TAICancelledEUTRA-Item ::= SEQUENCE {
    tAI
    cancelledCellsInTAI-EUTRA    TAI,
    iE-Extensions    ProtocolExtensionContainer { {TAICancelledEUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

TAICancelledEUTRA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAICancelledNR ::= SEQUENCE (SIZE(1..maxnoofTAIforWarning)) OF TAICancelledNR-Item

TAICancelledNR-Item ::= SEQUENCE {
    tAI
    cancelledCellsInTAI-NR    TAI,
    iE-Extensions    ProtocolExtensionContainer { {TAICancelledNR-Item-ExtIEs} } OPTIONAL,
    ...
}

TAICancelledNR-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIListForInactive ::= SEQUENCE (SIZE(1..maxnoofTAIforInactive)) OF TAIListForInactiveItem

TAIListForInactiveItem ::= SEQUENCE {
    tAI
    iE-Extensions    TAI,
    ProtocolExtensionContainer { {TAIListForInactiveItem-ExtIEs} } OPTIONAL,

```



```

    ...
}

TAIListForInactiveItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIListForPaging ::= SEQUENCE (SIZE(1..maxnoofTAIforPaging)) OF TAIListForPagingItem

TAIListForPagingItem ::= SEQUENCE {
    tAI          TAI,
    iE-Extensions ProtocolExtensionContainer { {TAIListForPagingItem-ExtIEs} } OPTIONAL,
    ...
}

TAIListForPagingItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIListForRestart ::= SEQUENCE (SIZE(1..maxnoofTAIforRestart)) OF TAI

TAIListForWarning ::= SEQUENCE (SIZE(1..maxnoofTAIforWarning)) OF TAI

TAINSAGSupportList ::= SEQUENCE (SIZE(1..maxnoofNSAGs)) OF TAINSAGSupportItem

TAINSAGSupportItem ::= SEQUENCE {
    nSAG-ID          NSAG-ID,
    nSAGSliceSupportList ExtendedSliceSupportList,
    iE-Extensions    ProtocolExtensionContainer { {TAINSAGSupportItem-ExtIEs} } OPTIONAL,
    ...
}

TAINSAGSupportItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIMBSSupportItem ::= SEQUENCE {
    tAI          TAI,
    iE-Extensions ProtocolExtensionContainer { {TAIMBSSupportItem-ExtIEs} } OPTIONAL,
    ...
}

TAIMBSSupportItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIMBSSupportList ::= SEQUENCE (SIZE(1..maxnoofSupportedTAIforMBS)) OF TAIMBSSupportItem

TargeteNB-ID ::= SEQUENCE {
    globalENB-ID      GlobalNgENB-ID,
    selected-EPS-TAI  EPS-TAI,
    iE-Extensions    ProtocolExtensionContainer { {TargeteNB-ID-ExtIEs} } OPTIONAL,
    ...
}

```

```

}

TargeteNB-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TargetHomeENB-ID ::= SEQUENCE {
    plMNidentity          PLMNIdentity,
    homeENB-ID           BIT STRING (SIZE(28)),
    selected-EPS-TAI     EPS-TAI,
    iE-Extensions        ProtocolExtensionContainer { {TargetHomeENB-ID-ExtIEs} } OPTIONAL,
    ...
}

TargetHomeENB-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TargetID ::= CHOICE {
    targetRANNodeID       TargetRANNodeID,
    targeteNB-ID         TargeteNB-ID,
    choice-Extensions     ProtocolIE-SingleContainer { {TargetID-ExtIEs} }
}

TargetID-ExtIEs NGAP-PROTOCOL-IES ::= {
    {ID id-TargetRNC-ID    CRITICALITY reject TYPE TargetRNC-ID    PRESENCE mandatory }|
    {ID id-TargetHomeENB-ID CRITICALITY reject TYPE TargetHomeENB-ID PRESENCE mandatory },
    ...
}

TargetNGRANNode-ToSourceNGRANNode-TransparentContainer ::= SEQUENCE {
    rRCContainer          RRCContainer,
    iE-Extensions        ProtocolExtensionContainer { {TargetNGRANNode-ToSourceNGRANNode-TransparentContainer-ExtIEs} } OPTIONAL,
    ...
}

TargetNGRANNode-ToSourceNGRANNode-TransparentContainer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-DAPSResponseInfoList          CRITICALITY ignore EXTENSION DAPSResponseInfoList          PRESENCE optional
    }|
    { ID id-DirectForwardingPathAvailability CRITICALITY ignore EXTENSION DirectForwardingPathAvailability PRESENCE optional
    }|
    { ID id-MBS-ActiveSessionInformation-TargettoSourceList CRITICALITY ignore EXTENSION MBS-ActiveSessionInformation-TargettoSourceList
    PRESENCE optional }|
    { ID id-NGAPIESupportInformationResponseList          CRITICALITY ignore EXTENSION NGAPIESupportInformationResponseList          PRESENCE optional
    },
    ...
}

TargetNGRANNode-ToSourceNGRANNode-FailureTransparentContainer ::= SEQUENCE {
    cell-CAGInformation    Cell-CAGInformation          OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { {TargetNGRANNode-ToSourceNGRANNode-FailureTransparentContainer-ExtIEs} } OPTIONAL,
    ...
}

```

```

TargetNGRANNode-ToSourceNGRANNode-FailureTransparentContainer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-NGAPIESupportInformationResponseList CRITICALITY ignore EXTENSION NGAPIESupportInformationResponseList PRESENCE optional },
    ...
}

```

```

TargetNSSAI ::= SEQUENCE (SIZE(1..maxnoofTargetS-NSSAIs)) OF TargetNSSAI-Item

```

```

TargetNSSAI-Item ::= SEQUENCE {
    s-NSSAI S-NSSAI,
    iE-Extensions ProtocolExtensionContainer { {TargetNSSAI-Item-ExtIEs} } OPTIONAL,
    ...
}

```

```

TargetNSSAI-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

TargetNSSAIInformation ::= SEQUENCE {
    targetNSSAI TargetNSSAI,
    indexToRFSP IndexToRFSP,
    iE-Extensions ProtocolExtensionContainer { {TargetNSSAIInformation-Item-ExtIEs} } OPTIONAL,
    ...
}

```

```

TargetNSSAIInformation-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

TargetRANNodeID ::= SEQUENCE {
    globalRANNodeID GlobalRANNodeID,
    selectedTAI TAI,
    iE-Extensions ProtocolExtensionContainer { {TargetRANNodeID-ExtIEs} } OPTIONAL,
    ...
}

```

```

TargetRANNodeID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-Selected-Target-SNPN-Identity CRITICALITY reject EXTENSION NID PRESENCE optional },
    ...
}

```

```

TargetRANNodeID-RIM ::= SEQUENCE {
    globalRANNodeID GlobalRANNodeID,
    selectedTAI TAI,
    iE-Extensions ProtocolExtensionContainer { {TargetRANNodeID-RIM-ExtIEs} } OPTIONAL,
    ...
}

```

```

TargetRANNodeID-RIM-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

```

```

TargetRANNodeID-SON ::= SEQUENCE {
    globalRANNodeID GlobalRANNodeID,
    selectedTAI TAI,

```

```

    iE-Extensions      ProtocolExtensionContainer { {TargetRANNodeID-SON-ExtIEs} } OPTIONAL,
    ...
}

TargetRANNodeID-SON-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-NR-CGI      CRITICALITY ignore  EXTENSION NR-CGI PRESENCE optional },
    ...
}

TargetRNC-ID ::= SEQUENCE {
    LAI                LAI,
    rNC-ID              RNC-ID,
    extendedRNC-ID      ExtendedRNC-ID                                OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {TargetRNC-ID-ExtIEs} }  OPTIONAL,
    ...
}

TargetRNC-ID-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TargetToSource-TransparentContainer ::= OCTET STRING
-- This IE includes a transparent container from the target RAN node to the source RAN node.
-- The octets of the OCTET STRING are encoded according to the specifications of the target system.

TargettoSource-Failure-TransparentContainer ::= OCTET STRING
-- This IE includes a transparent container from the target RAN node to the source RAN node.
-- The octets of the OCTET STRING are encoded according to the specifications of the target system (if applicable).

TimeBasedHandoverInformation ::= SEQUENCE {
    hOWindowStart      HandoverWindowStart,
    hOWindowDuration    HandoverWindowDuration,
    iE-Extensions      ProtocolExtensionContainer { {TimeBasedHandoverInformation-ExtIEs} }  OPTIONAL,
    ...
}

TimeBasedHandoverInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

HandoverWindowStart ::= INTEGER (0..549755813887)

HandoverWindowDuration ::= INTEGER (1..6000)

TimerApproachForGUAMIRemoval ::= ENUMERATED {
    apply-timer,
    ...
}

TimeSinceFailure ::= INTEGER (0..172800, ...)

TimeStamp ::= OCTET STRING (SIZE(4))

TimeSyncAssistanceInfo ::= SEQUENCE {

```

```

    timeDistributionIndication      ENUMERATED {enabled, disabled, ...},
    uUTimeSyncErrorBudget          INTEGER (1..1000000, ...) OPTIONAL,
-- The above IE shall be present if the Time Distribution Indication IE is set to the value "enabled"
    iE-Extensions                  ProtocolExtensionContainer { {TimeSyncAssistanceInfo-ExtIEs} } OPTIONAL,
    ...
}

TimeSyncAssistanceInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-ClockQualityReportingControlInfo    CRITICALITY ignore    EXTENSION ClockQualityReportingControlInfo    PRESENCE optional},
    ...
}

TimeToWait ::= ENUMERATED {v1s, v2s, v5s, v10s, v20s, v60s, ...}

TimeUEStayedInCell ::= INTEGER (0..4095)

TimeUEStayedInCellEnhancedGranularity ::= INTEGER (0..40950)

TMGI ::= OCTET STRING (SIZE(6))

TNAP-ID ::= OCTET STRING

TNGF-ID ::= CHOICE {
    tNGF-ID                BIT STRING (SIZE(32, ...)),
    choice-Extensions      ProtocolIE-SingleContainer { {TNGF-ID-ExtIEs} }
}

TNGF-ID-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

TNLAddressWeightFactor ::= INTEGER (0..255)

TNLAssociationList ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF TNLAssociationItem

TNLAssociationItem ::= SEQUENCE {
    tNLAssociationAddress      CPTransportLayerInformation,
    cause                      Cause,
    iE-Extensions              ProtocolExtensionContainer { {TNLAssociationItem-ExtIEs} } OPTIONAL,
    ...
}

TNLAssociationItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TNLAssociationUsage ::= ENUMERATED {
    ue,
    non-ue,
    both,
    ...
}

TooearlyIntersystemHO ::= SEQUENCE {

```

```

    sourcecellID          EUTRA-CGI,
    failurecellID          NGRAN-CGI,
    uERLFReportContainer  UERLFReportContainer OPTIONAL,
    iE-Extensions          ProtocolExtensionContainer { { TooearyIntersystemHO-ExtIEs} } OPTIONAL,
    ...
}

TooearyIntersystemHO-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TraceActivation ::= SEQUENCE {
    nGRANTraceID          NGRANTraceID,
    interfacesToTrace      InterfacesToTrace,
    traceDepth             TraceDepth,
    traceCollectionEntityIPaddress TransportLayerAddress,
    iE-Extensions          ProtocolExtensionContainer { {TraceActivation-ExtIEs} } OPTIONAL,
    ...
}

TraceActivation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-MDTConfiguration      CRITICALITY ignore EXTENSION MDT-Configuration PRESENCE optional }|
    { ID id-TraceCollectionEntityURI CRITICALITY ignore EXTENSION URI-address PRESENCE optional },
    ...
}

TraceDepth ::= ENUMERATED {
    minimum,
    medium,
    maximum,
    minimumWithoutVendorSpecificExtension,
    mediumWithoutVendorSpecificExtension,
    maximumWithoutVendorSpecificExtension,
    ... ,
    minimumOnlyVendorSpecificTraceRecord,
    mediumOnlyVendorSpecificTraceRecord,
    maximumOnlyVendorSpecificTraceRecord
}

TrafficLoadReductionIndication ::= INTEGER (1..99)

TransportLayerAddress ::= BIT STRING (SIZE(1..160, ...))

TypeOfError ::= ENUMERATED {
    not-understood,
    missing,
    ...
}

TAIBasedMDT ::= SEQUENCE {
    tAIListforMDT          TAIListforMDT,
    iE-Extensions          ProtocolExtensionContainer { {TAIBasedMDT-ExtIEs} } OPTIONAL,
    ...
}

```

```

TAIBasedMDT-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIListforMDT ::= SEQUENCE (SIZE(1..maxnoofTAforMDT)) OF TAI

TAIBasedQMC ::= SEQUENCE {
    tAIListforQMC      TAIListforQMC,
    iE-Extensions      ProtocolExtensionContainer { {TAIBasedQMC-ExtIEs} } OPTIONAL,
    ...
}

TAIBasedQMC-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIListforQMC ::= SEQUENCE (SIZE(1..maxnoofTAforQMC)) OF TAI

TABasedQMC ::= SEQUENCE {
    tAIListforQMC      TAListforQMC,
    iE-Extensions      ProtocolExtensionContainer { {TABasedQMC-ExtIEs} } OPTIONAL,
    ...
}

TABasedQMC-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAListforQMC ::= SEQUENCE (SIZE(1..maxnoofTAforQMC)) OF TAC

TABasedMDT ::= SEQUENCE {
    tAListforMDT      TAListforMDT,
    iE-Extensions      ProtocolExtensionContainer { {TABasedMDT-ExtIEs} } OPTIONAL,
    ...
}

TABasedMDT-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TAListforMDT ::= SEQUENCE (SIZE(1..maxnoofTAforMDT)) OF TAC

Threshold-RSRP ::= INTEGER(0..127)

Threshold-RSRQ ::= INTEGER(0..127)

Threshold-SINR ::= INTEGER(0..127)

TimeToTrigger ::= ENUMERATED {ms0, ms40, ms64, ms80, ms100, ms128, ms160, ms256, ms320, ms480, ms512, ms640, ms1024, ms1280, ms2560, ms5120}

TLContainer ::= OCTET STRING

```

```

TWAP-ID ::= OCTET STRING

TWIF-ID ::= CHOICE {
    twif-ID          BIT STRING (SIZE(32, ...)),
    choice-Extensions ProtocolIE-SingleContainer { {TWIF-ID-ExtIEs} }
}

TWIF-ID-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

TSCAssistanceInformation ::= SEQUENCE {
    periodicity          Periodicity,
    burstArrivalTime     BurstArrivalTime
    iE-Extensions        ProtocolExtensionContainer { {TSCAssistanceInformation-ExtIEs} } OPTIONAL,
    ...
}

TSCAssistanceInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-SurvivalTime          CRITICALITY ignore EXTENSION SurvivalTime          PRESENCE optional}|
    { ID id-RANfeedbacktype       CRITICALITY ignore EXTENSION RANfeedbacktype       PRESENCE optional}|
    { ID id-N6JitterInformation    CRITICALITY ignore EXTENSION N6JitterInformation    PRESENCE optional},
    ...
}

TSTrafficCharacteristics ::= SEQUENCE {
    tSCAssistanceInformationDL      TSCAssistanceInformation          OPTIONAL,
    tSCAssistanceInformationUL      TSCAssistanceInformation          OPTIONAL,
    iE-Extensions                  ProtocolExtensionContainer { {TSTrafficCharacteristics-ExtIEs} } OPTIONAL,
    ...
}

TSTrafficCharacteristics-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TSTrafficCharacteristicsFeedback ::= SEQUENCE {
    tSCFeedbackInformationDL        TSCFeedbackInformation          OPTIONAL,
    tSCFeedbackInformationUL        TSCFeedbackInformation          OPTIONAL,
    iE-Extensions                  ProtocolExtensionContainer { { TSTrafficCharacteristicsFeedback-ExtIEs} } OPTIONAL,
    ...
}

TSTrafficCharacteristicsFeedback-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

TSCFeedbackInformation ::= SEQUENCE {
    burstArrivalTimeOffset          INTEGER (-640000..640000, ...),
    adjustedPeriodicity             Periodicity                      OPTIONAL,
    iE-Extensions                  ProtocolExtensionContainer { { TSCFeedbackInformation-ExtIEs} } OPTIONAL,
    ...
}

```



```

TSCFeedbackInformation-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- U

UEAggregateMaximumBitRate ::= SEQUENCE {
    ueAggregateMaximumBitRateDL    BitRate,
    ueAggregateMaximumBitRateUL    BitRate,
    iE-Extensions                 ProtocolExtensionContainer { {UEAggregateMaximumBitRate-ExtIEs} } OPTIONAL,
    ...
}

UEAggregateMaximumBitRate-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UEAppLayerMeasInfoList ::= SEQUENCE (SIZE(1..maxnoofUEAppLayerMeas)) OF UEAppLayerMeasInfoItem

UEAppLayerMeasInfoItem ::= SEQUENCE {
    ueAppLayerMeasConfigInfo        UEAppLayerMeasConfigInfo,
    iE-Extensions                 ProtocolExtensionContainer { { UEAppLayerMeasInfoItem-ExtIEs} } OPTIONAL,
    ...
}

UEAppLayerMeasInfoItem-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UEAppLayerMeasConfigInfo ::= SEQUENCE {
    qoEReference                    QoEReference,
    serviceType                    ServiceType,
    areaScopeOfQMC                  AreaScopeOfQMC                                OPTIONAL,
    measCollEntityIPAddress          TransportLayerAddress,
    qoEMeasurementStatus             ENUMERATED {ongoing,...}                OPTIONAL,
    containerForAppLayerMeasConfig   OCTET STRING (SIZE(1..8000))           OPTIONAL,
    measConfigAppLayerID             INTEGER (0..15, ...)                   OPTIONAL,
    sliceSupportListQMC              SliceSupportListQMC                    OPTIONAL,
    mDT-AlignmentInfo               MDT-AlignmentInfo                       OPTIONAL,
    availableRANVisibleQoEMetrics    AvailableRANVisibleQoEMetrics          OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { { UEAppLayerMeasConfigInfo-ExtIEs} } OPTIONAL,
    ...
}

UEAppLayerMeasConfigInfo-ExtIEs  NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-AssistanceInformationQoE-Meas    CRITICALITY ignore  EXTENSION AssistanceInformationQoE-Meas    PRESENCE optional }|
    { ID id-CommServiceType                 CRITICALITY ignore  EXTENSION CommServiceType        PRESENCE optional }|
    { ID id-QoERVQoEReportingPaths          CRITICALITY ignore  EXTENSION QoERVQoEReportingPaths  PRESENCE optional },
    ...
}

UE-associatedLogicalNG-connectionList ::= SEQUENCE (SIZE(1..maxnoofNGConnectionsToReset)) OF UE-associatedLogicalNG-connectionItem

```

```

UE-associatedLogicalNG-connectionItem ::= SEQUENCE {
    aMF-UE-NGAP-ID      AMF-UE-NGAP-ID                      OPTIONAL,
    rAN-UE-NGAP-ID      RAN-UE-NGAP-ID                      OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {UE-associatedLogicalNG-connectionItem-ExtIEs} } OPTIONAL,
    ...
}

UE-associatedLogicalNG-connectionItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UECapabilityInfoRequest ::= ENUMERATED {
    requested,
    ...
}

UEContextRequest ::= ENUMERATED {requested, ...}

UEContextResumeRequestTransfer ::= SEQUENCE {
    qosFlowFailedToResumeList      QosFlowListWithCause      OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { {UEContextResumeRequestTransfer-ExtIEs} } OPTIONAL,
    ...
}

UEContextResumeRequestTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UEContextResumeResponseTransfer ::= SEQUENCE {
    qosFlowFailedToResumeList      QosFlowListWithCause      OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { {UEContextResumeResponseTransfer-ExtIEs} } OPTIONAL,
    ...
}

UEContextResumeResponseTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UEContextSuspendRequestTransfer ::= SEQUENCE {
    suspendIndicator               SuspendIndicator          OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { {UEContextSuspendRequestTransfer-ExtIEs} } OPTIONAL,
    ...
}

UEContextSuspendRequestTransfer-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-DifferentiationInfo ::= SEQUENCE {
    periodicCommunicationIndicator ENUMERATED {periodically, ondemand, ... }      OPTIONAL,
    periodicTime                  INTEGER (1..3600, ...)                        OPTIONAL,
    scheduledCommunicationTime     ScheduledCommunicationTime                    OPTIONAL,
    stationaryIndication           ENUMERATED {stationary, mobile, ...}           OPTIONAL,

```

```

    trafficProfile                ENUMERATED {single-packet, dual-packets, multiple-packets, ...} OPTIONAL,
    batteryIndication             ENUMERATED {battery-powered, battery-powered-not-rechargeable-or-replaceable, not-battery-powered, ...}
    OPTIONAL,
    iE-Extensions                 ProtocolExtensionContainer { { UE-DifferentiationInfo-ExtIEs } } OPTIONAL,
    ...
}

UE-DifferentiationInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UEHistoryInformation ::= SEQUENCE (SIZE(1..maxnoofCellsInUEHistoryInfo)) OF LastVisitedCellItem

UEHistoryInformationFromTheUE ::= CHOICE {
    nR                            NRMobilityHistoryReport,
    choice-Extensions             ProtocolIE-SingleContainer { {UEHistoryInformationFromTheUE-ExtIEs} }
}

UEHistoryInformationFromTheUE-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

UEIdentityIndexValue ::= CHOICE {
    indexLength10                 BIT STRING (SIZE(10)),
    choice-Extensions             ProtocolIE-SingleContainer { {UEIdentityIndexValue-ExtIEs} }
}

UEIdentityIndexValue-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

UE-NGAP-IDs ::= CHOICE {
    uE-NGAP-ID-pair              UE-NGAP-ID-pair,
    aMF-UE-NGAP-ID              aMF-UE-NGAP-ID,
    choice-Extensions            ProtocolIE-SingleContainer { {UE-NGAP-IDs-ExtIEs} }
}

UE-NGAP-IDs-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

UE-NGAP-ID-pair ::= SEQUENCE{
    aMF-UE-NGAP-ID              aMF-UE-NGAP-ID,
    rAN-UE-NGAP-ID              rAN-UE-NGAP-ID,
    iE-Extensions               ProtocolExtensionContainer { {UE-NGAP-ID-pair-ExtIEs} } OPTIONAL,
    ...
}

UE-NGAP-ID-pair-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UEPagingIdentity ::= CHOICE {
    fiveG-S-TMSI                FiveG-S-TMSI,

```

```

    choice-Extensions      ProtocolIE-SingleContainer { {UEPagingIdentity-ExtIEs} }
  }

UEPagingIdentity-ExtIEs NGAP-PROTOCOL-IES ::= {
  ...
}

UEPresence ::= ENUMERATED {in, out, unknown, ...}

UEPresenceInAreaOfInterestList ::= SEQUENCE (SIZE(1..maxnoofAoI)) OF UEPresenceInAreaOfInterestItem

UEPresenceInAreaOfInterestItem ::= SEQUENCE {
  locationReportingReferenceID      LocationReportingReferenceID,
  uEPresence                        UEPresence,
  iE-Extensions                    ProtocolExtensionContainer { {UEPresenceInAreaOfInterestItem-ExtIEs} } OPTIONAL,
  ...
}

UEPresenceInAreaOfInterestItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

UERadioCapability ::= OCTET STRING

UERadioCapabilityForPaging ::= SEQUENCE {
  uERadioCapabilityForPagingOfNR      UERadioCapabilityForPagingOfNR      OPTIONAL,
  uERadioCapabilityForPagingOfEUTRA    UERadioCapabilityForPagingOfEUTRA    OPTIONAL,
  iE-Extensions                      ProtocolExtensionContainer { {UERadioCapabilityForPaging-ExtIEs} } OPTIONAL,
  ...
}

UERadioCapabilityForPaging-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  { ID id-UERadioCapabilityForPagingOfNB-IoT      CRITICALITY ignore EXTENSION UERadioCapabilityForPagingOfNB-IoT      PRESENCE optional },
  ...
}

UERadioCapabilityForPagingOfNB-IoT ::= OCTET STRING

UERadioCapabilityForPagingOfNR ::= OCTET STRING

UERadioCapabilityForPagingOfEUTRA ::= OCTET STRING

UERadioCapabilityID ::= OCTET STRING

UERetentionInformation ::= ENUMERATED {
  ues-retained,
  ...
}

UERLFReportContainer ::= CHOICE {
  nR      NRUERLFReportContainer,
  lTE      LTEUERLFReportContainer,
  choice-Extensions      ProtocolIE-SingleContainer { {UERLFReportContainer-ExtIEs} }
}

```

```

UERLFReportContainer-ExtIES NGAP-PROTOCOL-IES ::= {
    ...
}

UESecurityCapabilities ::= SEQUENCE {
    nRencryptionAlgorithms          NRencryptionAlgorithms,
    nRintegrityProtectionAlgorithms NRintegrityProtectionAlgorithms,
    eUTRAEncryptionAlgorithms      EUTRAEncryptionAlgorithms,
    eUTRAIntegrityProtectionAlgorithms EUTRAIntegrityProtectionAlgorithms,
    iE-Extensions                  ProtocolExtensionContainer { {UESecurityCapabilities-ExtIES} } OPTIONAL,
    ...
}

UESecurityCapabilities-ExtIES NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UESliceMaximumBitRateList ::= SEQUENCE (SIZE(1..maxnoofAllowedS-NSSAIs)) OF UESliceMaximumBitRateItem

UESliceMaximumBitRateItem ::= SEQUENCE {
    s-NSSAI          S-NSSAI,
    uESliceMaximumBitRateDL BitRate,
    uESliceMaximumBitRateUL BitRate,
    iE-Extensions    ProtocolExtensionContainer { { UESliceMaximumBitRateItem-ExtIES} } OPTIONAL,
    ...
}

UESliceMaximumBitRateItem-ExtIES NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-UP-CIoT-Support ::= ENUMERATED {supported, ...}

UL-CP-SecurityInformation ::= SEQUENCE {
    ul-NAS-MAC          UL-NAS-MAC,
    ul-NAS-Count        UL-NAS-Count,
    iE-Extensions       ProtocolExtensionContainer { { UL-CP-SecurityInformation-ExtIES} } OPTIONAL,
    ...
}

UL-CP-SecurityInformation-ExtIES NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-NAS-MAC ::= BIT STRING (SIZE (16))

UL-NAS-Count ::= BIT STRING (SIZE (5))

UL-NGU-UP-TNLModifyList ::= SEQUENCE (SIZE(1..maxnoofMultiConnectivity)) OF UL-NGU-UP-TNLModifyItem

UL-NGU-UP-TNLModifyItem ::= SEQUENCE {
    uL-NGU-UP-TNLInformation      UPTransportLayerInformation,

```

```

    dL-NGU-UP-TNLInformation      UPTransportLayerInformation,
    iE-Extensions      ProtocolExtensionContainer { {UL-NGU-UP-TNLModifyItem-ExtIEs} } OPTIONAL,
    ...
}

UL-NGU-UP-TNLModifyItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-RedundantUL-NGU-UP-TNLInformation      CRITICALITY ignore EXTENSION UPTransportLayerInformation      PRESENCE optional      }|
    { ID id-RedundantDL-NGU-UP-TNLInformation      CRITICALITY ignore EXTENSION UPTransportLayerInformation      PRESENCE optional      },
    ...
}

UnavailableGUAMIList ::= SEQUENCE (SIZE(1..maxnoofServedGUAMIs)) OF UnavailableGUAMIItem

UnavailableGUAMIItem ::= SEQUENCE {
    gUAMI                      GUAMI,
    timerApproachForGUAMIRemoval      TimerApproachForGUAMIRemoval      OPTIONAL,
    backupAMFName                  AMFName      OPTIONAL,
    iE-Extensions      ProtocolExtensionContainer { {UnavailableGUAMIItem-ExtIEs} }      OPTIONAL,
    ...
}

UnavailableGUAMIItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    {ID id-ExtendedBackupAMFName      CRITICALITY ignore EXTENSION Extended-AMFName      PRESENCE optional},
    ...
}

ULForwarding ::= ENUMERATED {
    ul-forwarding-proposed,
    ...
}

UpdateFeedback ::= BIT STRING (SIZE(8, ...))

UPTransportLayerInformation ::= CHOICE {
    gTPTunnel      GTPTunnel,
    choice-Extensions      ProtocolIE-SingleContainer { {UPTransportLayerInformation-ExtIEs} }
}

UPTransportLayerInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

UPTransportLayerInformationList ::= SEQUENCE (SIZE(1..maxnoofMultiConnectivityMinusOne)) OF UPTransportLayerInformationItem

UPTransportLayerInformationItem ::= SEQUENCE {
    nGU-UP-TNLInformation      UPTransportLayerInformation,
    iE-Extensions      ProtocolExtensionContainer { {UPTransportLayerInformationItem-ExtIEs} } OPTIONAL,
    ...
}

UPTransportLayerInformationItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-CommonNetworkInstance      CRITICALITY ignore EXTENSION CommonNetworkInstance      PRESENCE optional      },
    ...
}

```

```

UPTransportLayerInformationPairList ::= SEQUENCE (SIZE(1..maxnoofMultiConnectivityMinusOne)) OF UPTransportLayerInformationPairItem

UPTransportLayerInformationPairItem ::= SEQUENCE {
    uL-NGU-UP-TNLInformation      UPTransportLayerInformation,
    dL-NGU-UP-TNLInformation      UPTransportLayerInformation,
    iE-Extensions                 ProtocolExtensionContainer { {UPTransportLayerInformationPairItem-ExtIEs} } OPTIONAL,
    ...
}

UPTransportLayerInformationPairItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

URI-address ::= VisibleString

UserLocationInformation ::= CHOICE {
    userLocationInformationEUTRA      UserLocationInformationEUTRA,
    userLocationInformationNR         UserLocationInformationNR,
    userLocationInformationN3IWF-with-PortNumber UserLocationInformationN3IWF-with-PortNumber,
    choice-Extensions                 ProtocolIE-SingleContainer { {UserLocationInformation-ExtIEs} }
}

UserLocationInformation-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-UserLocationInformationTNGF      CRITICALITY ignore TYPE UserLocationInformationTNGF      PRESENCE mandatory }|
    { ID id-UserLocationInformationTWIF      CRITICALITY ignore TYPE UserLocationInformationTWIF      PRESENCE mandatory }|
    { ID id-UserLocationInformationW-AGF     CRITICALITY ignore TYPE UserLocationInformationW-AGF     PRESENCE mandatory }|
    { ID id-UserLocationInformationN3IWF-without-PortNumber CRITICALITY ignore TYPE UserLocationInformationN3IWF-without-PortNumber PRESENCE mandatory },
    ...
}

UserLocationInformationEUTRA ::= SEQUENCE {
    eUTRA-CGI      EUTRA-CGI,
    tAI            TAI,
    timeStamp      TimeStamp
    iE-Extensions  ProtocolExtensionContainer { {UserLocationInformationEUTRA-ExtIEs} } OPTIONAL,
    ...
}

UserLocationInformationEUTRA-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-PSCellInformation      CRITICALITY ignore EXTENSION NGRAN-CGI      PRESENCE optional},
    ...
}

UserLocationInformationN3IWF-with-PortNumber ::= SEQUENCE {
    iPAddress      TransportLayerAddress,
    portNumber     PortNumber,
    iE-Extensions  ProtocolExtensionContainer { {UserLocationInformationN3IWF-with-PortNumber-ExtIEs} } OPTIONAL,
    ...
}

UserLocationInformationN3IWF-with-PortNumber-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {

```

```

    { ID id-TAI          CRITICALITY ignore  EXTENSION TAI          PRESENCE optional },
    ...
}
UserLocationInformationN3IWF-without-PortNumber ::= SEQUENCE {
    ipAddress            TransportLayerAddress,
    tAI                  TAI          OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { { UserLocationInformationN3IWF-without-PortNumber-ExtIEs} } OPTIONAL,
    ...
}

UserLocationInformationN3IWF-without-PortNumber-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

UserLocationInformationTNGF ::= SEQUENCE {
    tNAP-ID              TNAP-ID,
    ipAddress            TransportLayerAddress,
    portNumber           PortNumber          OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { {UserLocationInformationTNGF-ExtIEs} } OPTIONAL,
    ...
}

UserLocationInformationTNGF-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-TAI          CRITICALITY ignore  EXTENSION TAI          PRESENCE optional },
    ...
}

UserLocationInformationTWIF ::= SEQUENCE {
    tWAP-ID              TWAP-ID,
    ipAddress            TransportLayerAddress,
    portNumber           PortNumber          OPTIONAL,
    iE-Extensions        ProtocolExtensionContainer { {UserLocationInformationTWIF-ExtIEs} } OPTIONAL,
    ...
}

UserLocationInformationTWIF-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-TAI          CRITICALITY ignore  EXTENSION TAI          PRESENCE optional },
    ...
}

UserLocationInformationW-AGF ::= CHOICE {
    globalLine-ID        GlobalLine-ID,
    hFCNode-ID           HFCNode-ID,
    choice-Extensions    ProtocolIE-SingleContainer { { UserLocationInformationW-AGF-ExtIEs} }
}

UserLocationInformationW-AGF-ExtIEs NGAP-PROTOCOL-IES ::= {
    { ID id-GlobalCable-ID CRITICALITY ignore TYPE GlobalCable-ID PRESENCE mandatory }|
    { ID id-HFCNode-ID-new CRITICALITY ignore TYPE HFCNode-ID-new PRESENCE mandatory }|
    { ID id-GlobalCable-ID-new CRITICALITY ignore TYPE GlobalCable-ID-new PRESENCE mandatory },
    ...
}

UserLocationInformationNR ::= SEQUENCE {

```



```

nR-CGI          NR-CGI,
tAI             TAI,
timeStamp      TimeStamp
iE-Extensions   ProtocolExtensionContainer { {UserLocationInformationNR-ExtIEs} } OPTIONAL,
...
}

UserLocationInformationNR-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  { ID id-PSCellInformation          CRITICALITY ignore EXTENSION NGRAN-CGI          PRESENCE optional }|
  { ID id-NID                      CRITICALITY reject  EXTENSION NID              PRESENCE optional }|
  { ID id-NRNTNTAIInformation       CRITICALITY ignore EXTENSION NRNTNTAIInformation PRESENCE optional }|
  { ID id-MobileIAB-MTUserLocationInformation CRITICALITY ignore EXTENSION MobileIAB-MTUserLocationInformation PRESENCE optional }|
  { ID id-AdditionalULI             CRITICALITY ignore EXTENSION AdditionalULI      PRESENCE optional }|
  { ID id-Aerial-UE-FlightInformationReporting CRITICALITY ignore EXTENSION Aerial-UE-FlightInformationReporting PRESENCE optional },
  ...
}

Aerial-UE-FlightInformationReporting ::= SEQUENCE {
  altitude      Altitude,
  timeStamp     TimeStamp,
  iE-Extensions ProtocolExtensionContainer { {Aerial-UE-FlightInformationReporting-ExtIEs} } OPTIONAL,
  ...
}

Aerial-UE-FlightInformationReporting-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

Altitude ::= INTEGER (-420..10000, ...)

AerialUEReportingPeriodicity ::= ENUMERATED {ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, ms20480, ms40960, min1, min6, min12, min30, ...}

UserPlaneFailureIndication ::= SEQUENCE {
  userPlaneFailureType      UserPlaneFailureType,
  uL-NGU-UP-TNLInformation  UPTransportLayerInformation,
  dL-NGU-UP-TNLInformation  UPTransportLayerInformation,
  iE-Extensions             ProtocolExtensionContainer { { UserPlaneFailureIndication-ExtIEs} } OPTIONAL,
  ...
}

UserPlaneFailureIndication-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
  ...
}

UserPlaneFailureIndicationReport ::= ENUMERATED {
  qos-flows-and-tunnel-to-be-released,
  qos-flows-to-be-moved,
  new-transport-address-allocated,
  ...
}

```

```

UserPlaneFailureType ::= ENUMERATED {
    gtp-u-error-indication-received,
    up-path-failure,
    ...
}

UserPlaneSecurityInformation ::= SEQUENCE {
    securityResult          SecurityResult,
    securityIndication      SecurityIndication,
    iE-Extensions          ProtocolExtensionContainer { {UserPlaneSecurityInformation-ExtIEs} } OPTIONAL,
    ...
}

UserPlaneSecurityInformation-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- V

VolumeTimedReportList ::= SEQUENCE (SIZE(1..maxnoofTimePeriods)) OF VolumeTimedReport-Item

VolumeTimedReport-Item ::= SEQUENCE {
    startTimeStamp          OCTET STRING (SIZE(4)),
    endTimeStamp            OCTET STRING (SIZE(4)),
    usageCountUL             INTEGER (0..18446744073709551615),
    usageCountDL             INTEGER (0..18446744073709551615),
    iE-Extensions          ProtocolExtensionContainer { {VolumeTimedReport-Item-ExtIEs} } OPTIONAL,
    ...
}

VolumeTimedReport-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- W

W-AGF-ID ::= CHOICE {
    w-AGF-ID                BIT STRING (SIZE(16, ...)),
    choice-Extensions       ProtocolIE-SingleContainer { {W-AGF-ID-ExtIEs} }
}

W-AGF-ID-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

WarningAreaCoordinates ::= OCTET STRING (SIZE(1..1024))

WarningAreaList ::= CHOICE {
    eUTRA-CGIListForWarning  EUTRA-CGIListForWarning,
    nR-CGIListForWarning     NR-CGIListForWarning,
    tAIListForWarning        TAIListForWarning,
    emergencyAreaIDList      EmergencyAreaIDList,
    choice-Extensions        ProtocolIE-SingleContainer { {WarningAreaList-ExtIEs} }
}

```

```

WarningAreaList-ExtIEs NGAP-PROTOCOL-IES ::= {
    ...
}

WarningMessageContents ::= OCTET STRING (SIZE(1..9600))

WarningSecurityInfo ::= OCTET STRING (SIZE(50))

WarningType ::= OCTET STRING (SIZE(2))

WLANMeasurementConfiguration ::= SEQUENCE {
    wlanMeasConfig          WLANMeasConfig,
    wlanMeasConfigNameList  WLANMeasConfigNameList
    wlan-rssi               ENUMERATED {true, ...} OPTIONAL,
    wlan-rtt                ENUMERATED {true, ...} OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { WLANMeasurementConfiguration-ExtIEs } } OPTIONAL,
    ...
}

WLANMeasurementConfiguration-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

WLANMeasConfigNameList ::= SEQUENCE (SIZE(1..maxnoofWLANName)) OF WLANMeasConfigNameItem

WLANMeasConfigNameItem ::= SEQUENCE {
    WLANName                WLANName,
    iE-Extensions           ProtocolExtensionContainer { { WLANMeasConfigNameItem-ExtIEs } } OPTIONAL,
    ...
}

WLANMeasConfigNameItem-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

WLANMeasConfig ::= ENUMERATED {setup, ...}

WLANName ::= OCTET STRING (SIZE (1..32))

WUS-Assistance-Information ::= SEQUENCE {
    pagingProbabilityInformation PagingProbabilityInformation,
    iE-Extensions           ProtocolExtensionContainer { { WUS-Assistance-Information-ExtIEs } } OPTIONAL,
    ...
}

WUS-Assistance-Information-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

-- X

XnExtTLAs ::= SEQUENCE (SIZE(1..maxnoofXnExtTLAs)) OF XnExtTLA-Item

```

```

XnExtTLA-Item ::= SEQUENCE {
    iPsecTLA                TransportLayerAddress    OPTIONAL,
    gTP-TLAs                XnGTP-TLAs              OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { {XnExtTLA-Item-ExtIEs} } OPTIONAL,
    ...
}

XnExtTLA-Item-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    { ID id-SCTP-TLAs        CRITICALITY ignore   EXTENSION SCTP-TLAs    PRESENCE optional },
    ...
}

XnGTP-TLAs ::= SEQUENCE (SIZE(1..maxnoofXnGTP-TLAs)) OF TransportLayerAddress

XnTLAs ::= SEQUENCE (SIZE(1..maxnoofXnTLAs)) OF TransportLayerAddress

XnTNLConfigurationInfo ::= SEQUENCE {
    xnTransportLayerAddresses    XnTLAs,
    xnExtendedTransportLayerAddresses    XnExtTLAs                OPTIONAL,
    iE-Extensions                ProtocolExtensionContainer { {XnTNLConfigurationInfo-ExtIEs} } OPTIONAL,
    ...
}

XnTNLConfigurationInfo-ExtIEs NGAP-PROTOCOL-EXTENSION ::= {
    ...
}

XrDeviceWith2Rx ::= ENUMERATED {true, ...}

-- Y
-- Z

END
-- ASN1STOP

```

9.4.6 Common Definitions

```

-- ASN1START
-- *****
--
-- Common definitions
--
-- *****

NGAP-CommonDataTypes {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-Access (22) modules (3) ngap (1) version1 (1) ngap-CommonDataTypes (3) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

```

```

Criticality      ::= ENUMERATED { reject, ignore, notify }

Presence        ::= ENUMERATED { optional, conditional, mandatory }

PrivateIE-ID    ::= CHOICE {
    local        INTEGER (0..65535),
    global       OBJECT IDENTIFIER
}

ProcedureCode   ::= INTEGER (0..255)

ProtocolExtensionID ::= INTEGER (0..65535)

ProtocolIE-ID   ::= INTEGER (0..65535)

TriggeringMessage ::= ENUMERATED { initiating-message, successful-outcome, unsuccessful-outcome }

END
-- ASN1STOP

```

9.4.7 Constant Definitions

```

-- ASN1START
-- *****
--
-- Constant definitions
--
-- *****

NGAP-Constants {
    itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
    ngran-Access (22) modules (3) ngap (1) version1 (1) ngap-Constants (4) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- *****
--
-- IE parameter types from other modules.
--
-- *****

IMPORTS

    ProcedureCode,
    ProtocolIE-ID
FROM NGAP-CommonDataTypes;

-- *****
--
-- Elementary Procedures

```

```
--
-- *****
id-AMFConfigurationUpdate      ProcedureCode ::= 0
id-AMFStatusIndication         ProcedureCode ::= 1
id-CellTrafficTrace            ProcedureCode ::= 2
id-DeactivateTrace             ProcedureCode ::= 3
id-DownlinkNASTransport        ProcedureCode ::= 4
id-DownlinkNonUEAssociatedNRPPaTransport ProcedureCode ::= 5
id-DownlinkRANConfigurationTransfer ProcedureCode ::= 6
id-DownlinkRANStatusTransfer   ProcedureCode ::= 7
id-DownlinkUEAssociatedNRPPaTransport ProcedureCode ::= 8
id-ErrorIndication             ProcedureCode ::= 9
id-HandoverCancel              ProcedureCode ::= 10
id-HandoverNotification        ProcedureCode ::= 11
id-HandoverPreparation         ProcedureCode ::= 12
id-HandoverResourceAllocation  ProcedureCode ::= 13
id-InitialContextSetup        ProcedureCode ::= 14
id-InitialUEMessage            ProcedureCode ::= 15
id-LocationReportingControl     ProcedureCode ::= 16
id-LocationReportingFailureIndication ProcedureCode ::= 17
id-LocationReport              ProcedureCode ::= 18
id-NASNonDeliveryIndication     ProcedureCode ::= 19
id-NGReset                     ProcedureCode ::= 20
id-NGSetup                     ProcedureCode ::= 21
id-OverloadStart               ProcedureCode ::= 22
id-OverloadStop                ProcedureCode ::= 23
id-Paging                      ProcedureCode ::= 24
id-PathSwitchRequest           ProcedureCode ::= 25
id-PDUSessionResourceModify    ProcedureCode ::= 26
id-PDUSessionResourceModifyIndication ProcedureCode ::= 27
id-PDUSessionResourceRelease   ProcedureCode ::= 28
id-PDUSessionResourceSetup     ProcedureCode ::= 29
id-PDUSessionResourceNotify    ProcedureCode ::= 30
id-PrivateMessage              ProcedureCode ::= 31
id-PWSCancel                   ProcedureCode ::= 32
id-PWSFailureIndication        ProcedureCode ::= 33
id-PWSRestartIndication        ProcedureCode ::= 34
id-RANConfigurationUpdate      ProcedureCode ::= 35
id-RerouteNASRequest           ProcedureCode ::= 36
id-RRCInactiveTransitionReport ProcedureCode ::= 37
id-TraceFailureIndication      ProcedureCode ::= 38
id-TraceStart                  ProcedureCode ::= 39
id-UEContextModification       ProcedureCode ::= 40
id-UEContextRelease            ProcedureCode ::= 41
id-UEContextReleaseRequest     ProcedureCode ::= 42
id-UERadioCapabilityCheck      ProcedureCode ::= 43
id-UERadioCapabilityInfoIndication ProcedureCode ::= 44
id-UETNLABindingRelease        ProcedureCode ::= 45
id-UplinkNASTransport          ProcedureCode ::= 46
id-UplinkNonUEAssociatedNRPPaTransport ProcedureCode ::= 47
id-UplinkRANConfigurationTransfer ProcedureCode ::= 48
id-UplinkRANStatusTransfer     ProcedureCode ::= 49
id-UplinkUEAssociatedNRPPaTransport ProcedureCode ::= 50
```

id-WriteReplaceWarning	ProcedureCode ::= 51
id-SecondaryRATDataUsageReport	ProcedureCode ::= 52
id-UplinkRIMInformationTransfer	ProcedureCode ::= 53
id-DownlinkRIMInformationTransfer	ProcedureCode ::= 54
id-RetrieveUEInformation	ProcedureCode ::= 55
id-UEInformationTransfer	ProcedureCode ::= 56
id-RANCPRelocationIndication	ProcedureCode ::= 57
id-UEContextResume	ProcedureCode ::= 58
id-UEContextSuspend	ProcedureCode ::= 59
id-UERadioCapabilityIDMapping	ProcedureCode ::= 60
id-HandoverSuccess	ProcedureCode ::= 61
id-UplinkRANEarlyStatusTransfer	ProcedureCode ::= 62
id-DownlinkRANEarlyStatusTransfer	ProcedureCode ::= 63
id-AMFCPrelocationIndication	ProcedureCode ::= 64
id-ConnectionEstablishmentIndication	ProcedureCode ::= 65
id-BroadcastSessionModification	ProcedureCode ::= 66
id-BroadcastSessionRelease	ProcedureCode ::= 67
id-BroadcastSessionSetup	ProcedureCode ::= 68
id-DistributionSetup	ProcedureCode ::= 69
id-DistributionRelease	ProcedureCode ::= 70
id-MulticastSessionActivation	ProcedureCode ::= 71
id-MulticastSessionDeactivation	ProcedureCode ::= 72
id-MulticastSessionUpdate	ProcedureCode ::= 73
id-MulticastGroupPaging	ProcedureCode ::= 74
id-BroadcastSessionReleaseRequired	ProcedureCode ::= 75
id-TimingSynchronisationStatus	ProcedureCode ::= 76
id-TimingSynchronisationStatusReport	ProcedureCode ::= 77
id-MTCommunicationHandling	ProcedureCode ::= 78
id-RANPagingRequest	ProcedureCode ::= 79
id-BroadcastSessionTransport	ProcedureCode ::= 80
id-NGRemoval	ProcedureCode ::= 81
id-InventoryRequest	ProcedureCode ::= 82
id-InventoryReport	ProcedureCode ::= 83
id-CommandRequest	ProcedureCode ::= 84
id-AIOTSessionRelease	ProcedureCode ::= 85
id-AIOTSessionReleaseRequest	ProcedureCode ::= 86

```
-- *****
--
-- Extension constants
--
-- *****
```

maxPrivateIES	INTEGER ::= 65535
maxProtocolExtensions	INTEGER ::= 65535
maxProtocolIES	INTEGER ::= 65535

```
-- *****
--
-- Lists
--
-- *****
```

maxnoofAllowedAreas	INTEGER ::= 16
maxnoofAllowedCAGsperPLMN	INTEGER ::= 256
maxnoofAllowedS-NSSAIs	INTEGER ::= 8
maxnoofBluetoothName	INTEGER ::= 4
maxnoofBPLMNs	INTEGER ::= 12
maxnoofCAGsperCell	INTEGER ::= 64
maxnoofCellIDforMDT	INTEGER ::= 32
maxnoofCellIDforWarning	INTEGER ::= 65535
maxnoofCellinAoI	INTEGER ::= 256
maxnoofCellinEAI	INTEGER ::= 65535
maxnoofCellinTAI	INTEGER ::= 65535
maxnoofCellsforMBS	INTEGER ::= 8192
maxnoofCellsingNB	INTEGER ::= 16384
maxnoofCellsinngeNB	INTEGER ::= 256
maxnoofCellsinNGRANNode	INTEGER ::= 16384
maxnoofCellsinUEHistoryInfo	INTEGER ::= 16
maxnoofCellsUEMovingTrajectory	INTEGER ::= 16
maxnoofDRBs	INTEGER ::= 32
maxnoofEmergencyAreaID	INTEGER ::= 65535
maxnoofEAIforRestart	INTEGER ::= 256
maxnoofEPLMNs	INTEGER ::= 15
maxnoofEPLMNsPlusOne	INTEGER ::= 16
maxnoofE-RABs	INTEGER ::= 256
maxnoofErrors	INTEGER ::= 256
maxnoofExtSliceItems	INTEGER ::= 65535
maxnoofForbTACs	INTEGER ::= 4096
maxnoofFreqforMDT	INTEGER ::= 8
maxnoofGUAMIs	INTEGER ::= 1024
maxnoofMBSAreaSessionIDs	INTEGER ::= 256
maxnoofMBSFSAs	INTEGER ::= 64
maxnoofMBSQoSFlows	INTEGER ::= 64
maxnoofMBSSessions	INTEGER ::= 32
maxnoofMBSSessionsofUE	INTEGER ::= 256
maxnoofMBSServiceAreaInformation	INTEGER ::= 256
maxnoofMDTPLMNs	INTEGER ::= 16
maxnoofMRBs	INTEGER ::= 32
maxnoofMultiConnectivity	INTEGER ::= 4
maxnoofMultiConnectivityMinusOne	INTEGER ::= 3
maxnoofNeighPCIforMDT	INTEGER ::= 32
maxnoofNGAPIESupportInfo	INTEGER ::= 32
maxnoofNGConnectionsToReset	INTEGER ::= 65536
maxnoofNRCellBands	INTEGER ::= 32
maxnoofNSAGs	INTEGER ::= 256
maxnoofPagingAreas	INTEGER ::= 64
maxnoofPC5QoSFlows	INTEGER ::= 2048
maxnoofPDUSessions	INTEGER ::= 256
maxnoofPLMNs	INTEGER ::= 12
maxnoofPSCellsPerPrimaryCellinUEHistoryInfo	INTEGER ::= 8
maxnoofQoSFlows	INTEGER ::= 64
maxnoofQoSParaSets	INTEGER ::= 8
maxnoofRANNodeinAoI	INTEGER ::= 64
maxnoofRecommendedCells	INTEGER ::= 16
maxnoofRecommendedRANNodes	INTEGER ::= 16

maxnoofAoI	INTEGER ::= 64
maxnoofAoIMinusOne	INTEGER ::= 63
maxnoofFlightInfoReportControl	INTEGER ::= 64
maxnoofReportedCells	INTEGER ::= 256
maxnoofSensorName	INTEGER ::= 3
maxnoofServedGUAMIs	INTEGER ::= 256
maxnoofSliceItems	INTEGER ::= 1024
maxnoofSuccessfulH0Reports	INTEGER ::= 64
maxnoofTACs	INTEGER ::= 256
maxnoofTACsinNTN	INTEGER ::= 12
maxnoofTAforMDT	INTEGER ::= 8
maxnoofTAIforInactive	INTEGER ::= 16
maxnoofSupportedTAIforMBS	INTEGER ::= 256
maxnoofTAIforMBS	INTEGER ::= 1024
maxnoofTAIforPaging	INTEGER ::= 16
maxnoofTAIforRestart	INTEGER ::= 2048
maxnoofTAIforWarning	INTEGER ::= 65535
maxnoofTAIinAoI	INTEGER ::= 16
maxnoofTimePeriods	INTEGER ::= 2
maxnoofTNLAssociations	INTEGER ::= 32
maxnoofUEsforPaging	INTEGER ::= 4096
maxnoofUETypes	INTEGER ::= 8
maxnoofWLANName	INTEGER ::= 4
maxnoofXnExtTLAs	INTEGER ::= 16
maxnoofXnGTP-TLAs	INTEGER ::= 16
maxnoofXnTLAs	INTEGER ::= 2
maxnoofCandidateCells	INTEGER ::= 32
maxnoofTargets-NSSAIs	INTEGER ::= 8
maxNRARFCN	INTEGER ::= 3279165
maxnoofCellIDforQMC	INTEGER ::= 32
maxnoofPLMNforQMC	INTEGER ::= 16
maxnoofUEAppLayerMeas	INTEGER ::= 16
maxnoofSNSSAIforQMC	INTEGER ::= 16
maxnoofTAforQMC	INTEGER ::= 8
maxnoofThresholdsForExcessPacketDelay	INTEGER ::= 255
maxnoofESNPNs	INTEGER ::= 15
maxnoofCandidateRelayUEs	INTEGER ::= 32
maxnoofSuccessfulPSCellChangeReports	INTEGER ::= 64
maxnoofCellsTSS	INTEGER ::= 16384
maxnoofPeriodicities	INTEGER ::= 8
maxnoofCAGforMDT	INTEGER ::= 256
maxnoofMDTSNPNs	INTEGER ::= 16
maxnoofPartiallyAllowedS-NSSAIs	INTEGER ::= 8
maxnoofRSPPQoSFlows	INTEGER ::= 2048
maxnoofSliceItemsforMDT	INTEGER ::= 1024
maxnoofAreaNTN	INTEGER ::= 8
maxnoofIntendedAreasforMBS	INTEGER ::= 65536
maxnoofRequestedSNSSAIs	INTEGER ::= 8
maxnoofReaders	INTEGER ::= 65535
maxnoofDevices	INTEGER ::= 65535
maxnoofAIoTAreas	INTEGER ::= 256
maxnoofThresholds	INTEGER ::= 8

```
-- *****  
--  
-- IEs  
--  
-- *****
```

id-AllowedNSSAI	ProtocolIE-ID ::= 0
id-AMFName	ProtocolIE-ID ::= 1
id-AMFOverloadResponse	ProtocolIE-ID ::= 2
id-AMFSetID	ProtocolIE-ID ::= 3
id-AMF-TNLAssociationFailedToSetupList	ProtocolIE-ID ::= 4
id-AMF-TNLAssociationSetupList	ProtocolIE-ID ::= 5
id-AMF-TNLAssociationToAddList	ProtocolIE-ID ::= 6
id-AMF-TNLAssociationToRemoveList	ProtocolIE-ID ::= 7
id-AMF-TNLAssociationToUpdateList	ProtocolIE-ID ::= 8
id-AMFTrafficLoadReductionIndication	ProtocolIE-ID ::= 9
id-AMF-UE-NGAP-ID	ProtocolIE-ID ::= 10
id-AssistanceDataForPaging	ProtocolIE-ID ::= 11
id-BroadcastCancelledAreaList	ProtocolIE-ID ::= 12
id-BroadcastCompletedAreaList	ProtocolIE-ID ::= 13
id-CancelAllWarningMessages	ProtocolIE-ID ::= 14
id-Cause	ProtocolIE-ID ::= 15
id-CellIDListForRestart	ProtocolIE-ID ::= 16
id-ConcurrentWarningMessageInd	ProtocolIE-ID ::= 17
id-CoreNetworkAssistanceInformationForInactive	ProtocolIE-ID ::= 18
id-CriticalityDiagnostics	ProtocolIE-ID ::= 19
id-DataCodingScheme	ProtocolIE-ID ::= 20
id-DefaultPagingDRX	ProtocolIE-ID ::= 21
id-DirectForwardingPathAvailability	ProtocolIE-ID ::= 22
id-EmergencyAreaIDListForRestart	ProtocolIE-ID ::= 23
id-EmergencyFallbackIndicator	ProtocolIE-ID ::= 24
id-EUTRA-CGI	ProtocolIE-ID ::= 25
id-FiveG-S-TMSI	ProtocolIE-ID ::= 26
id-GlobalRANNodeID	ProtocolIE-ID ::= 27
id-GUAMI	ProtocolIE-ID ::= 28
id-HandoverType	ProtocolIE-ID ::= 29
id-IMSVoiceSupportIndicator	ProtocolIE-ID ::= 30
id-IndexToRFSP	ProtocolIE-ID ::= 31
id-InfoOnRecommendedCellsAndRANNodesForPaging	ProtocolIE-ID ::= 32
id-LocationReportingRequestType	ProtocolIE-ID ::= 33
id-MaskedIMEISV	ProtocolIE-ID ::= 34
id-MessageIdentifier	ProtocolIE-ID ::= 35
id-MobilityRestrictionList	ProtocolIE-ID ::= 36
id-NASC	ProtocolIE-ID ::= 37
id-NAS-PDU	ProtocolIE-ID ::= 38
id-NASSecurityParametersFromNGRAN	ProtocolIE-ID ::= 39
id-NewAMF-UE-NGAP-ID	ProtocolIE-ID ::= 40
id-NewSecurityContextInd	ProtocolIE-ID ::= 41
id-NGAP-Message	ProtocolIE-ID ::= 42
id-NGRAN-CGI	ProtocolIE-ID ::= 43
id-NGRANTraceID	ProtocolIE-ID ::= 44
id-NR-CGI	ProtocolIE-ID ::= 45
id-NRPPa-PDU	ProtocolIE-ID ::= 46

id-NumberOfBroadcastsRequested	ProtocolIE-ID ::= 47
id-OldAMF	ProtocolIE-ID ::= 48
id-OverloadStartNSSAList	ProtocolIE-ID ::= 49
id-PagingDRX	ProtocolIE-ID ::= 50
id-PagingOrigin	ProtocolIE-ID ::= 51
id-PagingPriority	ProtocolIE-ID ::= 52
id-PDUSessionResourceAdmittedList	ProtocolIE-ID ::= 53
id-PDUSessionResourceFailedToModifyListModRes	ProtocolIE-ID ::= 54
id-PDUSessionResourceFailedToSetupListCxtRes	ProtocolIE-ID ::= 55
id-PDUSessionResourceFailedToSetupListHOAck	ProtocolIE-ID ::= 56
id-PDUSessionResourceFailedToSetupListPSReq	ProtocolIE-ID ::= 57
id-PDUSessionResourceFailedToSetupListSURES	ProtocolIE-ID ::= 58
id-PDUSessionResourceHandoverList	ProtocolIE-ID ::= 59
id-PDUSessionResourceListCxtRelCpl	ProtocolIE-ID ::= 60
id-PDUSessionResourceListHORqd	ProtocolIE-ID ::= 61
id-PDUSessionResourceModifyListModCfm	ProtocolIE-ID ::= 62
id-PDUSessionResourceModifyListModInd	ProtocolIE-ID ::= 63
id-PDUSessionResourceModifyListModReq	ProtocolIE-ID ::= 64
id-PDUSessionResourceModifyListModRes	ProtocolIE-ID ::= 65
id-PDUSessionResourceNotifyList	ProtocolIE-ID ::= 66
id-PDUSessionResourceReleasedListNot	ProtocolIE-ID ::= 67
id-PDUSessionResourceReleasedListPSAck	ProtocolIE-ID ::= 68
id-PDUSessionResourceReleasedListPSFail	ProtocolIE-ID ::= 69
id-PDUSessionResourceReleasedListRelRes	ProtocolIE-ID ::= 70
id-PDUSessionResourceSetupListCxtReq	ProtocolIE-ID ::= 71
id-PDUSessionResourceSetupListCxtRes	ProtocolIE-ID ::= 72
id-PDUSessionResourceSetupListHOReq	ProtocolIE-ID ::= 73
id-PDUSessionResourceSetupListSUReq	ProtocolIE-ID ::= 74
id-PDUSessionResourceSetupListSURES	ProtocolIE-ID ::= 75
id-PDUSessionResourceToBeSwitchedDLList	ProtocolIE-ID ::= 76
id-PDUSessionResourceSwitchedList	ProtocolIE-ID ::= 77
id-PDUSessionResourceToReleaseListHOCmd	ProtocolIE-ID ::= 78
id-PDUSessionResourceToReleaseListRelCmd	ProtocolIE-ID ::= 79
id-PLMNSupportList	ProtocolIE-ID ::= 80
id-PWSFailedCellIDList	ProtocolIE-ID ::= 81
id-RANNodeName	ProtocolIE-ID ::= 82
id-RANPagingPriority	ProtocolIE-ID ::= 83
id-RANStatusTransfer-TransparentContainer	ProtocolIE-ID ::= 84
id-RAN-UE-NGAP-ID	ProtocolIE-ID ::= 85
id-RelativeAMFCapacity	ProtocolIE-ID ::= 86
id-RepetitionPeriod	ProtocolIE-ID ::= 87
id-ResetType	ProtocolIE-ID ::= 88
id-RoutingID	ProtocolIE-ID ::= 89
id-RRCEstablishmentCause	ProtocolIE-ID ::= 90
id-RRCInactiveTransitionReportRequest	ProtocolIE-ID ::= 91
id-RRCState	ProtocolIE-ID ::= 92
id-SecurityContext	ProtocolIE-ID ::= 93
id-SecurityKey	ProtocolIE-ID ::= 94
id-SerialNumber	ProtocolIE-ID ::= 95
id-ServedGUAMList	ProtocolIE-ID ::= 96
id-SliceSupportList	ProtocolIE-ID ::= 97
id-SONConfigurationTransferDL	ProtocolIE-ID ::= 98
id-SONConfigurationTransferUL	ProtocolIE-ID ::= 99
id-SourceAMF-UE-NGAP-ID	ProtocolIE-ID ::= 100

id-SourceToTarget-TransparentContainer	ProtocolIE-ID ::= 101
id-SupportedTAList	ProtocolIE-ID ::= 102
id-TAIListForPaging	ProtocolIE-ID ::= 103
id-TAIListForRestart	ProtocolIE-ID ::= 104
id-TargetID	ProtocolIE-ID ::= 105
id-TargetToSource-TransparentContainer	ProtocolIE-ID ::= 106
id-TimeToWait	ProtocolIE-ID ::= 107
id-TraceActivation	ProtocolIE-ID ::= 108
id-TraceCollectionEntityIPAddress	ProtocolIE-ID ::= 109
id-UEAggregateMaximumBitRate	ProtocolIE-ID ::= 110
id-UE-associatedLogicalNG-connectionList	ProtocolIE-ID ::= 111
id-UEContextRequest	ProtocolIE-ID ::= 112
id-UE-NGAP-IDs	ProtocolIE-ID ::= 114
id-UEPagingIdentity	ProtocolIE-ID ::= 115
id-UEPresenceInAreaOfInterestList	ProtocolIE-ID ::= 116
id-UERadioCapability	ProtocolIE-ID ::= 117
id-UERadioCapabilityForPaging	ProtocolIE-ID ::= 118
id-UESecurityCapabilities	ProtocolIE-ID ::= 119
id-UnavailableGUAMList	ProtocolIE-ID ::= 120
id-UserLocationInformation	ProtocolIE-ID ::= 121
id-WarningAreaList	ProtocolIE-ID ::= 122
id-WarningMessageContents	ProtocolIE-ID ::= 123
id-WarningSecurityInfo	ProtocolIE-ID ::= 124
id-WarningType	ProtocolIE-ID ::= 125
id-AdditionalUL-NGU-UP-TNLInformation	ProtocolIE-ID ::= 126
id-DataForwardingNotPossible	ProtocolIE-ID ::= 127
id-DL-NGU-UP-TNLInformation	ProtocolIE-ID ::= 128
id-NetworkInstance	ProtocolIE-ID ::= 129
id-PDUSessionAggregateMaximumBitRate	ProtocolIE-ID ::= 130
id-PDUSessionResourceFailedToModifyListModCfm	ProtocolIE-ID ::= 131
id-PDUSessionResourceFailedToSetupListCxtFail	ProtocolIE-ID ::= 132
id-PDUSessionResourceListCxtRelReq	ProtocolIE-ID ::= 133
id-PDUSessionType	ProtocolIE-ID ::= 134
id-QosFlowAddOrModifyRequestList	ProtocolIE-ID ::= 135
id-QosFlowSetupRequestList	ProtocolIE-ID ::= 136
id-QosFlowToReleaseList	ProtocolIE-ID ::= 137
id-SecurityIndication	ProtocolIE-ID ::= 138
id-UL-NGU-UP-TNLInformation	ProtocolIE-ID ::= 139
id-UL-NGU-UP-TNLModifyList	ProtocolIE-ID ::= 140
id-WarningAreaCoordinates	ProtocolIE-ID ::= 141
id-PDUSessionResourceSecondaryRATUsageList	ProtocolIE-ID ::= 142
id-HandoverFlag	ProtocolIE-ID ::= 143
id-SecondaryRATUsageInformation	ProtocolIE-ID ::= 144
id-PDUSessionResourceReleaseResponseTransfer	ProtocolIE-ID ::= 145
id-RedirectionVoiceFallback	ProtocolIE-ID ::= 146
id-UERetentionInformation	ProtocolIE-ID ::= 147
id-S-NSSAI	ProtocolIE-ID ::= 148
id-PSCellInformation	ProtocolIE-ID ::= 149
id-LastEUTRAN-PLMNIdentity	ProtocolIE-ID ::= 150
id-MaximumIntegrityProtectedDataRate-DL	ProtocolIE-ID ::= 151
id-AdditionalDLForwardingUPTNLInformation	ProtocolIE-ID ::= 152
id-AdditionalDLUPTNLInformationForHOList	ProtocolIE-ID ::= 153
id-AdditionalNGU-UP-TNLInformation	ProtocolIE-ID ::= 154
id-AdditionalDLQosFlowPerTNLInformation	ProtocolIE-ID ::= 155

id-SecurityResult	ProtocolIE-ID ::= 156
id-ENDC-SONConfigurationTransferDL	ProtocolIE-ID ::= 157
id-ENDC-SONConfigurationTransferUL	ProtocolIE-ID ::= 158
id-OldAssociatedQosFlowList-ULendmarkerexpected	ProtocolIE-ID ::= 159
id-CNTypeRestrictionsForEquivalent	ProtocolIE-ID ::= 160
id-CNTypeRestrictionsForServing	ProtocolIE-ID ::= 161
id-NewGUAMI	ProtocolIE-ID ::= 162
id-ULForwarding	ProtocolIE-ID ::= 163
id-ULForwardingUP-TNLInformation	ProtocolIE-ID ::= 164
id-CNAssistedRANTuning	ProtocolIE-ID ::= 165
id-CommonNetworkInstance	ProtocolIE-ID ::= 166
id-NGRAN-TNLAssociationToRemoveList	ProtocolIE-ID ::= 167
id-TNLAssociationTransportLayerAddressNGRAN	ProtocolIE-ID ::= 168
id-EndpointIPAddressAndPort	ProtocolIE-ID ::= 169
id-LocationReportingAdditionalInfo	ProtocolIE-ID ::= 170
id-SourceToTarget-AMFInformationReroute	ProtocolIE-ID ::= 171
id-AdditionalULForwardingUPTNLInformation	ProtocolIE-ID ::= 172
id-SCTP-TLAs	ProtocolIE-ID ::= 173
id-SelectedPLMNIdentity	ProtocolIE-ID ::= 174
id-RIMInformationTransfer	ProtocolIE-ID ::= 175
id-GUAMIType	ProtocolIE-ID ::= 176
id-SRVCCOperationPossible	ProtocolIE-ID ::= 177
id-TargetRNC-ID	ProtocolIE-ID ::= 178
id-RAT-Information	ProtocolIE-ID ::= 179
id-ExtendedRATRestrictionInformation	ProtocolIE-ID ::= 180
id-QosMonitoringRequest	ProtocolIE-ID ::= 181
id-SgNB-UE-X2AP-ID	ProtocolIE-ID ::= 182
id-AdditionalRedundantDL-NGU-UP-TNLInformation	ProtocolIE-ID ::= 183
id-AdditionalRedundantDLQosFlowPerTNLInformation	ProtocolIE-ID ::= 184
id-AdditionalRedundantNGU-UP-TNLInformation	ProtocolIE-ID ::= 185
id-AdditionalRedundantUL-NGU-UP-TNLInformation	ProtocolIE-ID ::= 186
id-CNPacketDelayBudgetDL	ProtocolIE-ID ::= 187
id-CNPacketDelayBudgetUL	ProtocolIE-ID ::= 188
id-ExtendedPacketDelayBudget	ProtocolIE-ID ::= 189
id-RedundantCommonNetworkInstance	ProtocolIE-ID ::= 190
id-RedundantDL-NGU-TNLInformationReused	ProtocolIE-ID ::= 191
id-RedundantDL-NGU-UP-TNLInformation	ProtocolIE-ID ::= 192
id-RedundantDLQosFlowPerTNLInformation	ProtocolIE-ID ::= 193
id-RedundantQosFlowIndicator	ProtocolIE-ID ::= 194
id-RedundantUL-NGU-UP-TNLInformation	ProtocolIE-ID ::= 195
id-TSCTrafficCharacteristics	ProtocolIE-ID ::= 196
id-RedundantPDUSessionInformation	ProtocolIE-ID ::= 197
id-UsedRSNInformation	ProtocolIE-ID ::= 198
id-IAB-Authorized	ProtocolIE-ID ::= 199
id-IAB-Supported	ProtocolIE-ID ::= 200
id-IABNodeIndication	ProtocolIE-ID ::= 201
id-NB-IoT-PagingDRX	ProtocolIE-ID ::= 202
id-NB-IoT-Paging-eDRXInfo	ProtocolIE-ID ::= 203
id-NB-IoT-DefaultPagingDRX	ProtocolIE-ID ::= 204
id-Enhanced-CoverageRestriction	ProtocolIE-ID ::= 205
id-Extended-ConnectedTime	ProtocolIE-ID ::= 206
id-PagingAssisDataforCEcapabUE	ProtocolIE-ID ::= 207
id-WUS-Assistance-Information	ProtocolIE-ID ::= 208
id-UE-DifferentiationInfo	ProtocolIE-ID ::= 209

id-NB-IoT-UEPriority	ProtocolIE-ID ::= 210
id-UL-CP-SecurityInformation	ProtocolIE-ID ::= 211
id-DL-CP-SecurityInformation	ProtocolIE-ID ::= 212
id-TAI	ProtocolIE-ID ::= 213
id-UERadioCapabilityForPagingOfNB-IoT	ProtocolIE-ID ::= 214
id-LTEV2XServicesAuthorized	ProtocolIE-ID ::= 215
id-NRV2XServicesAuthorized	ProtocolIE-ID ::= 216
id-LTEUESidelinkAggregateMaximumBitrate	ProtocolIE-ID ::= 217
id-NRUESidelinkAggregateMaximumBitrate	ProtocolIE-ID ::= 218
id-PC5QoSParameters	ProtocolIE-ID ::= 219
id-AlternativeQoSParaSetList	ProtocolIE-ID ::= 220
id-CurrentQoSParaSetIndex	ProtocolIE-ID ::= 221
id-CEmodeBrestricted	ProtocolIE-ID ::= 222
id-EUTRA-PagingeDRXInformation	ProtocolIE-ID ::= 223
id-CEmodeBSupport-Indicator	ProtocolIE-ID ::= 224
id-LTEM-Indication	ProtocolIE-ID ::= 225
id-EndIndication	ProtocolIE-ID ::= 226
id-EDT-Session	ProtocolIE-ID ::= 227
id-UECapabilityInfoRequest	ProtocolIE-ID ::= 228
id-PDUSessionResourceFailedToResumeListRESReq	ProtocolIE-ID ::= 229
id-PDUSessionResourceFailedToResumeListRESRes	ProtocolIE-ID ::= 230
id-PDUSessionResourceSuspendListSUSReq	ProtocolIE-ID ::= 231
id-PDUSessionResourceResumeListRESReq	ProtocolIE-ID ::= 232
id-PDUSessionResourceResumeListRESRes	ProtocolIE-ID ::= 233
id-UE-UP-CIoT-Support	ProtocolIE-ID ::= 234
id-Suspend-Request-Indication	ProtocolIE-ID ::= 235
id-Suspend-Response-Indication	ProtocolIE-ID ::= 236
id-RRC-Resume-Cause	ProtocolIE-ID ::= 237
id-RGLevelWirelineAccessCharacteristics	ProtocolIE-ID ::= 238
id-W-AGFIdentityInformation	ProtocolIE-ID ::= 239
id-GlobalTNGF-ID	ProtocolIE-ID ::= 240
id-GlobalTWIF-ID	ProtocolIE-ID ::= 241
id-GlobalW-AGF-ID	ProtocolIE-ID ::= 242
id-UserLocationInformationW-AGF	ProtocolIE-ID ::= 243
id-UserLocationInformationTNGF	ProtocolIE-ID ::= 244
id-AuthenticatedIndication	ProtocolIE-ID ::= 245
id-TNGFIdentityInformation	ProtocolIE-ID ::= 246
id-TWIFIdentityInformation	ProtocolIE-ID ::= 247
id-UserLocationInformationTWIF	ProtocolIE-ID ::= 248
id-DataForwardingResponseERABList	ProtocolIE-ID ::= 249
id-IntersystemSONConfigurationTransferDL	ProtocolIE-ID ::= 250
id-IntersystemSONConfigurationTransferUL	ProtocolIE-ID ::= 251
id-SONInformationReport	ProtocolIE-ID ::= 252
id-UEHistoryInformationFromTheUE	ProtocolIE-ID ::= 253
id-ManagementBasedMDTPLMNList	ProtocolIE-ID ::= 254
id-MDTConfiguration	ProtocolIE-ID ::= 255
id-PrivacyIndicator	ProtocolIE-ID ::= 256
id-TraceCollectionEntityURI	ProtocolIE-ID ::= 257
id-NPN-Support	ProtocolIE-ID ::= 258
id-NPN-AccessInformation	ProtocolIE-ID ::= 259
id-NPN-PagingAssistanceInformation	ProtocolIE-ID ::= 260
id-NPN-MobilityInformation	ProtocolIE-ID ::= 261
id-TargettoSource-Failure-TransparentContainer	ProtocolIE-ID ::= 262
id-NID	ProtocolIE-ID ::= 263

id-UERadioCapabilityID	ProtocolIE-ID ::= 264
id-UERadioCapability-EUTRA-Format	ProtocolIE-ID ::= 265
id-DAPSRequestInfo	ProtocolIE-ID ::= 266
id-DAPSResponseInfoList	ProtocolIE-ID ::= 267
id-EarlyStatusTransfer-TransparentContainer	ProtocolIE-ID ::= 268
id-NotifySourceNGRANNode	ProtocolIE-ID ::= 269
id-ExtendedSliceSupportList	ProtocolIE-ID ::= 270
id-ExtendedTAISliceSupportList	ProtocolIE-ID ::= 271
id-ConfiguredTACIndication	ProtocolIE-ID ::= 272
id-Extended-RANNodeName	ProtocolIE-ID ::= 273
id-Extended-AMFName	ProtocolIE-ID ::= 274
id-GlobalCable-ID	ProtocolIE-ID ::= 275
id-QosMonitoringReportingFrequency	ProtocolIE-ID ::= 276
id-QosFlowParametersList	ProtocolIE-ID ::= 277
id-QosFlowFeedbackList	ProtocolIE-ID ::= 278
id-BurstArrivalTimeDownlink	ProtocolIE-ID ::= 279
id-ExtendedUEIdentityIndexValue	ProtocolIE-ID ::= 280
id-PduSessionExpectedUEActivityBehaviour	ProtocolIE-ID ::= 281
id-MicoAllPLMN	ProtocolIE-ID ::= 282
id-QosFlowFailedToSetupList	ProtocolIE-ID ::= 283
id-SourceTNLAddrInfo	ProtocolIE-ID ::= 284
id-ExtendedReportIntervalMDT	ProtocolIE-ID ::= 285
id-SourceNodeID	ProtocolIE-ID ::= 286
id-NRNTNTAIInformation	ProtocolIE-ID ::= 287
id-UEContextReferenceAtSource	ProtocolIE-ID ::= 288
id-LastVisitedPSCellList	ProtocolIE-ID ::= 289
id-IntersystemSONInformationRequest	ProtocolIE-ID ::= 290
id-IntersystemSONInformationReply	ProtocolIE-ID ::= 291
id-EnergySavingIndication	ProtocolIE-ID ::= 292
id-IntersystemResourceStatusUpdate	ProtocolIE-ID ::= 293
id-SuccessfulHandoverReportList	ProtocolIE-ID ::= 294
id-MBS-AreaSessionID	ProtocolIE-ID ::= 295
id-MBS-QoSFlowsToBeSetupList	ProtocolIE-ID ::= 296
id-MBS-QoSFlowsToBeSetupModList	ProtocolIE-ID ::= 297
id-MBS-ServiceArea	ProtocolIE-ID ::= 298
id-MBS-SessionID	ProtocolIE-ID ::= 299
id-MBS-DistributionReleaseRequestTransfer	ProtocolIE-ID ::= 300
id-MBS-DistributionSetupRequestTransfer	ProtocolIE-ID ::= 301
id-MBS-DistributionSetupResponseTransfer	ProtocolIE-ID ::= 302
id-MBS-DistributionSetupUnsuccessfulTransfer	ProtocolIE-ID ::= 303
id-MulticastSessionActivationRequestTransfer	ProtocolIE-ID ::= 304
id-MulticastSessionDeactivationRequestTransfer	ProtocolIE-ID ::= 305
id-MulticastSessionUpdateRequestTransfer	ProtocolIE-ID ::= 306
id-MulticastGroupPagingAreaList	ProtocolIE-ID ::= 307
id-MBS-SupportIndicator	ProtocolIE-ID ::= 309
id-MBSSessionFailedtoSetupList	ProtocolIE-ID ::= 310
id-MBSSessionFailedtoSetuporModifyList	ProtocolIE-ID ::= 311
id-MBSSessionSetupResponseList	ProtocolIE-ID ::= 312
id-MBSSessionSetuporModifyResponseList	ProtocolIE-ID ::= 313
id-MBSSessionSetupFailureTransfer	ProtocolIE-ID ::= 314
id-MBSSessionSetupRequestTransfer	ProtocolIE-ID ::= 315
id-MBSSessionSetupResponseTransfer	ProtocolIE-ID ::= 316
id-MBSSessionToReleaseList	ProtocolIE-ID ::= 317
id-MBSSessionSetupRequestList	ProtocolIE-ID ::= 318

id-MBSSessionSetuporModifyRequestList	ProtocolIE-ID ::= 319
id-MBS-ActiveSessionInformation-SourcetoTargetList	ProtocolIE-ID ::= 323
id-MBS-ActiveSessionInformation-TargettoSourceList	ProtocolIE-ID ::= 324
id-OnboardingSupport	ProtocolIE-ID ::= 325
id-TimeSyncAssistanceInfo	ProtocolIE-ID ::= 326
id-SurvivalTime	ProtocolIE-ID ::= 327
id-QMCConfigInfo	ProtocolIE-ID ::= 328
id-QMCDeactivation	ProtocolIE-ID ::= 329
id-PDUSessionPairID	ProtocolIE-ID ::= 331
id-NR-PagingeDRXInformation	ProtocolIE-ID ::= 332
id-RedCapIndication	ProtocolIE-ID ::= 333
id-TargetNSSAIInformation	ProtocolIE-ID ::= 334
id-UESliceMaximumBitRateList	ProtocolIE-ID ::= 335
id-M4ReportAmount	ProtocolIE-ID ::= 336
id-M5ReportAmount	ProtocolIE-ID ::= 337
id-M6ReportAmount	ProtocolIE-ID ::= 338
id-M7ReportAmount	ProtocolIE-ID ::= 339
id-IncludeBeamMeasurementsIndication	ProtocolIE-ID ::= 340
id-ExcessPacketDelayThresholdConfiguration	ProtocolIE-ID ::= 341
id-PagingCause	ProtocolIE-ID ::= 342
id-PagingCauseIndicationForVoiceService	ProtocolIE-ID ::= 343
id-PEIPsAssistanceInformation	ProtocolIE-ID ::= 344
id-FiveG-ProSeAuthorized	ProtocolIE-ID ::= 345
id-FiveG-ProSeUEPC5AggregateMaximumBitRate	ProtocolIE-ID ::= 346
id-FiveG-ProSePC5QoSParameters	ProtocolIE-ID ::= 347
id-MBSSessionModificationFailureTransfer	ProtocolIE-ID ::= 348
id-MBSSessionModificationRequestTransfer	ProtocolIE-ID ::= 349
id-MBSSessionModificationResponseTransfer	ProtocolIE-ID ::= 350
id-MBS-QoSFlowToReleaseList	ProtocolIE-ID ::= 351
id-MBS-SessionTNLInfo5GC	ProtocolIE-ID ::= 352
id-TAINSAGSupportList	ProtocolIE-ID ::= 353
id-SourceNodeTNLAddrInfo	ProtocolIE-ID ::= 354
id-NGAPIESupportInformationRequestList	ProtocolIE-ID ::= 355
id-NGAPIESupportInformationResponseList	ProtocolIE-ID ::= 356
id-MBS-SessionFSAIDList	ProtocolIE-ID ::= 357
id-MBSSessionReleaseResponseTransfer	ProtocolIE-ID ::= 358
id-ManagementBasedMDTPLMNModificationList	ProtocolIE-ID ::= 359
id-EarlyMeasurement	ProtocolIE-ID ::= 360
id-BeamMeasurementsReportConfiguration	ProtocolIE-ID ::= 361
id-HFCNode-ID-new	ProtocolIE-ID ::= 362
id-GlobalCable-ID-new	ProtocolIE-ID ::= 363
id-TargetHomeENB-ID	ProtocolIE-ID ::= 364
id-HashedUEIdentityIndexValue	ProtocolIE-ID ::= 365
id-ExtendedMobilityInformation	ProtocolIE-ID ::= 366
id-NetworkControlledRepeaterAuthorized	ProtocolIE-ID ::= 367
id-AdditionalCancelledlocationReportingReferenceIDList	ProtocolIE-ID ::= 368
id-Selected-Target-SNPN-Identity	ProtocolIE-ID ::= 369
id-EquivalentSNPNsList	ProtocolIE-ID ::= 370
id-SelectedNID	ProtocolIE-ID ::= 371
id-SupportedUETypesList	ProtocolIE-ID ::= 372
id-AerialUESubscriptionInformation	ProtocolIE-ID ::= 373
id-NR-A2X-ServicesAuthorized	ProtocolIE-ID ::= 374
id-LTE-A2X-ServicesAuthorized	ProtocolIE-ID ::= 375
id-NR-A2X-UE-PC5-AggregateMaximumBitRate	ProtocolIE-ID ::= 376

id-LTE-A2X-UE-PC5-AggregateMaximumBitRate	ProtocolIE-ID ::= 377
id-A2X-PC5-QoS-Parameters	ProtocolIE-ID ::= 378
id-FiveGProSeLayer2Multipath	ProtocolIE-ID ::= 379
id-FiveGProSeLayer2UEtoUERelay	ProtocolIE-ID ::= 380
id-FiveGProSeLayer2UEtoUERemote	ProtocolIE-ID ::= 381
id-CandidateRelayUEInformationList	ProtocolIE-ID ::= 382
id-SuccessfulPSCellChangeReportList	ProtocolIE-ID ::= 383
id-IntersystemMobilityFailureforVoiceFallback	ProtocolIE-ID ::= 384
id-TargetCellCRNTI	ProtocolIE-ID ::= 385
id-TimeSinceFailure	ProtocolIE-ID ::= 386
id-RANTimingSynchronisationStatusInfo	ProtocolIE-ID ::= 387
id-RAN-TSSRequestType	ProtocolIE-ID ::= 388
id-RAN-TSSScope	ProtocolIE-ID ::= 389
id-ClockQualityReportingControlInfo	ProtocolIE-ID ::= 390
id-RANfeedbacktype	ProtocolIE-ID ::= 391
id-QoSFlowTSCList	ProtocolIE-ID ::= 392
id-TSCTrafficCharacteristicsFeedback	ProtocolIE-ID ::= 393
id-DownlinkTLCContainer	ProtocolIE-ID ::= 394
id-UplinkTLCContainer	ProtocolIE-ID ::= 395
id-ANPacketDelayBudgetUL	ProtocolIE-ID ::= 396
id-QoSFlowAdditionalInfoList	ProtocolIE-ID ::= 397
id-AssistanceInformationQoE-Meas	ProtocolIE-ID ::= 398
id-CommServiceType	ProtocolIE-ID ::= 399
id-MobileIAB-Authorized	ProtocolIE-ID ::= 400
id-MobileIAB-MTUserLocationInformation	ProtocolIE-ID ::= 401
id-MobileIABNodeIndication	ProtocolIE-ID ::= 402
id-NoPDUSessionIndication	ProtocolIE-ID ::= 403
id-MobileIAB-Supported	ProtocolIE-ID ::= 404
id-CN-MT-CommunicationHandling	ProtocolIE-ID ::= 405
id-FiveGCAction	ProtocolIE-ID ::= 406
id-PagingPolicyDifferentiation	ProtocolIE-ID ::= 407
id-DL-Signalling	ProtocolIE-ID ::= 408
id-PNI-NPN-AreaScopeofMDT	ProtocolIE-ID ::= 409
id-PNI-NPNBasedMDT	ProtocolIE-ID ::= 410
id-SNPN-CellBasedMDT	ProtocolIE-ID ::= 411
id-SNPN-TAIBasedMDT	ProtocolIE-ID ::= 412
id-SNPN-BasedMDT	ProtocolIE-ID ::= 413
id-Partially-Allowed-NSSAI	ProtocolIE-ID ::= 414
id-AssociatedSessionID	ProtocolIE-ID ::= 415
id-MBS-AssistanceInformation	ProtocolIE-ID ::= 416
id-BroadcastTransportFailureTransfer	ProtocolIE-ID ::= 417
id-BroadcastTransportRequestTransfer	ProtocolIE-ID ::= 418
id-BroadcastTransportResponseTransfer	ProtocolIE-ID ::= 419
id-TimeBasedHandoverInformation	ProtocolIE-ID ::= 420
id-DLDiscarding	ProtocolIE-ID ::= 421
id-PDUsetQoSParameters	ProtocolIE-ID ::= 422
id-PDUsetbasedHandlingIndicator	ProtocolIE-ID ::= 423
id-N6JitterInformation	ProtocolIE-ID ::= 424
id-ECNMarkingorCongestionInformationReportingRequest	ProtocolIE-ID ::= 425
id-ECNMarkingorCongestionInformationReportingStatus	ProtocolIE-ID ::= 426
id-ERedCapIndication	ProtocolIE-ID ::= 427
id-XrDeviceWith2Rx	ProtocolIE-ID ::= 428
id-UserPlaneErrorIndicator	ProtocolIE-ID ::= 429
id-SLPositioningRangingServiceInfo	ProtocolIE-ID ::= 430

id-PDUSessionListMTCommHReq	ProtocolIE-ID ::= 431
id-MaximumDataBurstVolume	ProtocolIE-ID ::= 432
id-MN-only-MDT-collection	ProtocolIE-ID ::= 433
id-MBS-NGUFailureIndication	ProtocolIE-ID ::= 434
id-UserPlaneFailureIndication	ProtocolIE-ID ::= 435
id-UserPlaneFailureIndicationReport	ProtocolIE-ID ::= 436
id-SourceSN-to-TargetSN-QMCInfo	ProtocolIE-ID ::= 437
id-QoERVQoEReportingPaths	ProtocolIE-ID ::= 438
id-UserLocationInformationN3IWF-without-PortNumber	ProtocolIE-ID ::= 439
id-AUN3DeviceAccessInfo	ProtocolIE-ID ::= 440
id-TAIBSSSupportList	ProtocolIE-ID ::= 441
id-ExtendedBackupAMFName	ProtocolIE-ID ::= 442
id-ExtendedOldAMF	ProtocolIE-ID ::= 443
id-MBS-UP-FailureIndication	ProtocolIE-ID ::= 444
id-RequestedTNLInfo	ProtocolIE-ID ::= 445
id-Aerial-UE-FlightInformationOngoingReporting-failed	ProtocolIE-ID ::= 446
id-NetworkSliceAreaScopeofMDT	ProtocolIE-ID ::= 447
id-GeographyBasedMDT	ProtocolIE-ID ::= 448
id-MBS-IntendedServiceAreaList	ProtocolIE-ID ::= 449
id-AMF-UE-NGAP-ID2	ProtocolIE-ID ::= 450
id-GUAMIList	ProtocolIE-ID ::= 451
id-GW-Context-Release-Indication	ProtocolIE-ID ::= 452
id-SourceAMF-GUAMI	ProtocolIE-ID ::= 453
id-RequestedNSSAI	ProtocolIE-ID ::= 454
id-Target-RAN-Node-ID	ProtocolIE-ID ::= 455
id-InventoryRequestTransfer	ProtocolIE-ID ::= 456
id-InventoryResponseTransfer	ProtocolIE-ID ::= 457
id-InventoryFailureTransfer	ProtocolIE-ID ::= 458
id-InventoryReportTransfer	ProtocolIE-ID ::= 459
id-CommandRequestTransfer	ProtocolIE-ID ::= 460
id-CommandResponseTransfer	ProtocolIE-ID ::= 461
id-CommandFailureTransfer	ProtocolIE-ID ::= 462
id-AIOTSessionReleaseCommandTransfer	ProtocolIE-ID ::= 463
id-AIOTSessionReleaseCompleteTransfer	ProtocolIE-ID ::= 464
id-AIOTSessionReleaseRequestTransfer	ProtocolIE-ID ::= 465
id-AIoT-CorrelationIdentifier	ProtocolIE-ID ::= 466
id-AIOTFIdentifier	ProtocolIE-ID ::= 467
id-AIoT-DeviceIdentificationRequested	ProtocolIE-ID ::= 468
id-AIoT-RequestedServiceAreaInformation	ProtocolIE-ID ::= 469
id-AIoT-InventoryAssistanceInformation	ProtocolIE-ID ::= 470
id-AIoT-FollowonCommandIndication	ProtocolIE-ID ::= 471
id-RAN-AIoT-Device-NGAP-ID	ProtocolIE-ID ::= 472
id-AIoT-CommandAssistanceInformation	ProtocolIE-ID ::= 473
id-AIoT-NASPDU	ProtocolIE-ID ::= 474
id-AIoT-Support	ProtocolIE-ID ::= 475
id-AIOTFName	ProtocolIE-ID ::= 476
id-LPWUSPSAssistanceInformation	ProtocolIE-ID ::= 477
id-FurtherExtendedUEIdentityIndexValue	ProtocolIE-ID ::= 478
id-FiveGProSeLayer2MHUEtoNetworkRelay	ProtocolIE-ID ::= 479
id-FiveGProSeLayer2MHIntermediateUEtoNetworkRelay	ProtocolIE-ID ::= 480
id-FiveGProSeLayer2MHRemote	ProtocolIE-ID ::= 481
id-FiveGProSeLayer3MHUEtoNetworkRelay	ProtocolIE-ID ::= 482
id-AdditionalULI	ProtocolIE-ID ::= 483
id-PduSetDelayBudgetDownlink	ProtocolIE-ID ::= 484

id-PduSetDelayBudgetUplink	ProtocolIE-ID ::= 485
id-PduSetErrorRateDownlink	ProtocolIE-ID ::= 486
id-PduSetErrorRateUplink	ProtocolIE-ID ::= 487
id-DLPDUSetInformationMarkingSupportIndication	ProtocolIE-ID ::= 488
id-MonitoringRequestonAvailableBitrate	ProtocolIE-ID ::= 489
id-MMSID	ProtocolIE-ID ::= 490
id-Indication-of-Bitrate-Adaptation	ProtocolIE-ID ::= 491
id-SCGActivationTime	ProtocolIE-ID ::= 492
id-Aerial-UE-FlightInformationReporting	ProtocolIE-ID ::= 493
id-Aerial-UE-FlightInformationReportingControllist	ProtocolIE-ID ::= 494
id-LPWUSDisableIndication	ProtocolIE-ID ::= 495
id-AIoT-SecurityInfo	ProtocolIE-ID ::= 496

END
-- ASN1STOP

9.4.8 Container Definitions

```
-- ASN1START
-- *****
--
-- Container definitions
--
-- *****

NGAP-Containers {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
ngran-Access (22) modules (3) ngap (1) version1 (1) ngap-Containers (5) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- *****
--
-- IE parameter types from other modules.
--
-- *****

IMPORTS

    Criticality,
    Presence,
    PrivateIE-ID,
    ProtocolExtensionID,
    ProtocolIE-ID
FROM NGAP-CommonDataTypes

    maxPrivateIEs,
    maxProtocolExtensions,
```

```

    maxProtocolIEs
FROM NGAP-Constants;

-- *****
--
-- Class Definition for Protocol IEs
--
-- *****

NGAP-PROTOCOL-IES ::= CLASS {
    &id                ProtocolIE-ID                UNIQUE,
    &criticality        Criticality,
    &Value,
    &presence            Presence
}
WITH SYNTAX {
    ID                &id
    CRITICALITY        &criticality
    TYPE                &Value
    PRESENCE            &presence
}

-- *****
--
-- Class Definition for Protocol IEs
--
-- *****

NGAP-PROTOCOL-IES-PAIR ::= CLASS {
    &id                ProtocolIE-ID                UNIQUE,
    &firstCriticality    Criticality,
    &FirstValue,
    &secondCriticality    Criticality,
    &SecondValue,
    &presence            Presence
}
WITH SYNTAX {
    ID                &id
    FIRST CRITICALITY    &firstCriticality
    FIRST TYPE            &FirstValue
    SECOND CRITICALITY    &secondCriticality
    SECOND TYPE            &SecondValue
    PRESENCE            &presence
}

-- *****
--
-- Class Definition for Protocol Extensions
--
-- *****

NGAP-PROTOCOL-EXTENSION ::= CLASS {
    &id                ProtocolExtensionID            UNIQUE,
    &criticality        Criticality,

```

```

    &Extension,
    &presence      Presence
}
WITH SYNTAX {
    ID              &id
    CRITICALITY     &criticality
    EXTENSION       &Extension
    PRESENCE        &presence
}

-- *****
--
-- Class Definition for Private IEs
--
-- *****

NGAP-PRIVATE-IES ::= CLASS {
    &id              PrivateIE-ID,
    &criticality     Criticality,
    &Value,
    &presence        Presence
}
WITH SYNTAX {
    ID              &id
    CRITICALITY     &criticality
    TYPE            &Value
    PRESENCE        &presence
}

-- *****
--
-- Container for Protocol IEs
--
-- *****

ProtocolIE-Container {NGAP-PROTOCOL-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (0..maxProtocolIEs)) OF
    ProtocolIE-Field {{IEsSetParam}}

ProtocolIE-SingleContainer {NGAP-PROTOCOL-IES : IEsSetParam} ::=
    ProtocolIE-Field {{IEsSetParam}}

ProtocolIE-Field {NGAP-PROTOCOL-IES : IEsSetParam} ::= SEQUENCE {
    id              NGAP-PROTOCOL-IES.&id              ({IEsSetParam}),
    criticality     NGAP-PROTOCOL-IES.&criticality      ({IEsSetParam}{@id}),
    value           NGAP-PROTOCOL-IES.&Value            ({IEsSetParam}{@id})
}

-- *****
--
-- Container for Protocol IE Pairs
--
-- *****

```

```

ProtocolIE-ContainerPair {NGAP-PROTOCOL-IES-PAIR : IEsSetParam} ::=
    SEQUENCE (SIZE (0..maxProtocolIEs)) OF
        ProtocolIE-FieldPair {{IEsSetParam}}

ProtocolIE-FieldPair {NGAP-PROTOCOL-IES-PAIR : IEsSetParam} ::= SEQUENCE {
    id                NGAP-PROTOCOL-IES-PAIR.&id                ({IEsSetParam}),
    firstCriticality   NGAP-PROTOCOL-IES-PAIR.&firstCriticality   ({IEsSetParam}{@id}),
    firstValue         NGAP-PROTOCOL-IES-PAIR.&firstValue         ({IEsSetParam}{@id}),
    secondCriticality  NGAP-PROTOCOL-IES-PAIR.&secondCriticality  ({IEsSetParam}{@id}),
    secondValue        NGAP-PROTOCOL-IES-PAIR.&secondValue        ({IEsSetParam}{@id})
}

-- *****
--
-- Container Lists for Protocol IE Containers
--
-- *****

ProtocolIE-ContainerList {INTEGER : lowerBound, INTEGER : upperBound, NGAP-PROTOCOL-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (lowerBound..upperBound)) OF
        ProtocolIE-SingleContainer {{IEsSetParam}}

ProtocolIE-ContainerPairList {INTEGER : lowerBound, INTEGER : upperBound, NGAP-PROTOCOL-IES-PAIR : IEsSetParam} ::=
    SEQUENCE (SIZE (lowerBound..upperBound)) OF
        ProtocolIE-ContainerPair {{IEsSetParam}}

-- *****
--
-- Container for Protocol Extensions
--
-- *****

ProtocolExtensionContainer {NGAP-PROTOCOL-EXTENSION : ExtensionSetParam} ::=
    SEQUENCE (SIZE (1..maxProtocolExtensions)) OF
        ProtocolExtensionField {{ExtensionSetParam}}

ProtocolExtensionField {NGAP-PROTOCOL-EXTENSION : ExtensionSetParam} ::= SEQUENCE {
    id                NGAP-PROTOCOL-EXTENSION.&id                ({ExtensionSetParam}),
    criticality       NGAP-PROTOCOL-EXTENSION.&criticality       ({ExtensionSetParam}{@id}),
    extensionValue     NGAP-PROTOCOL-EXTENSION.&Extension        ({ExtensionSetParam}{@id})
}

-- *****
--
-- Container for Private IEs
--
-- *****

PrivateIE-Container {NGAP-PRIVATE-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (1..maxPrivateIEs)) OF
        PrivateIE-Field {{IEsSetParam}}

PrivateIE-Field {NGAP-PRIVATE-IES : IEsSetParam} ::= SEQUENCE {
    id                NGAP-PRIVATE-IES.&id                ({IEsSetParam}),

```

```
        criticality      NGAP-PRIVATE-IES.&criticality  ({IEsSetParam}{@id}),
        value            NGAP-PRIVATE-IES.&Value       ({IEsSetParam}{@id})
    }

END
-- ASN1STOP
```

9.5 Message Transfer Syntax

NGAP shall use the ASN.1 Basic Packed Encoding Rules (BASIC-PER) Aligned Variant as transfer syntax as specified in ITU-T Rec. X.691 [4].

9.6 Timers

$T_{NGRELOCprep}$

- Specifies the maximum time for the Handover Preparation procedure in the source NG-RAN node.

$T_{NGRELOCoverall}$

- Specifies the maximum time for the protection of the overall handover procedure in the source NG-RAN node.

$T_{XnRELOCoverall}$

- Specified in TS 38.423 [24].

10 Handling of Unknown, Unforeseen and Erroneous Protocol Data

10.1 General

Protocol Error cases can be divided into three classes:

- Transfer Syntax Error.
- Abstract Syntax Error.
- Logical Error.

Protocol errors can occur in the following functions within a receiving node:

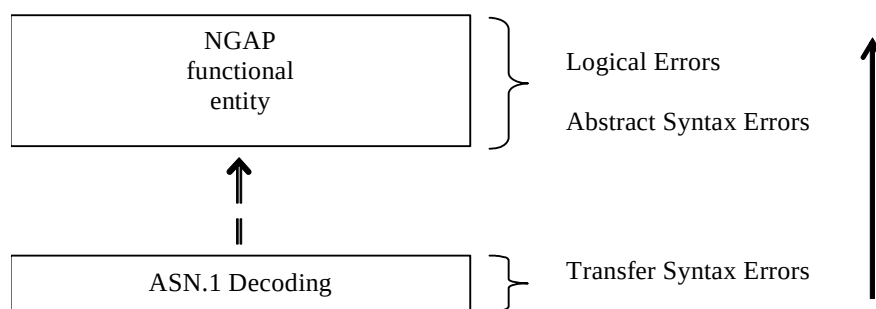


Figure 10.1-1: Protocol Errors in NGAP.

The information stated in subclauses 10.2, 10.3 and 10.4, to be included in the message used when reporting an error, is what at minimum shall be included. Other optional information elements within the message may also be included, if available. This is also valid for the case when the reporting is done with a response message. The latter is an exception to what is stated in subclause 4.1.

10.2 Transfer Syntax Error

A Transfer Syntax Error occurs when the receiver is not able to decode the received physical message. Transfer syntax errors are always detected in the process of ASN.1 decoding. If a Transfer Syntax Error occurs, the receiver should initiate Error Indication procedure with appropriate cause value for the Transfer Syntax protocol error.

Examples for Transfer Syntax Errors are:

- Violation of value ranges in ASN.1 definition of messages. E.g., if an IE has a defined value range of 0 to 10 (ASN.1: INTEGER (0..10)), and 12 will be received, then this will be treated as a transfer syntax error.
- Violation in list element constraints. E.g., if a list is defined as containing 1 to 10 elements, and 12 elements will be received, then this case will be handled as a transfer syntax error.
- Missing mandatory elements in ASN.1 SEQUENCE definitions (as sent by the originator of the message).
- Wrong order of elements in ASN.1 SEQUENCE definitions (as sent by the originator of the message).

10.3 Abstract Syntax Error

10.3.1 General

An Abstract Syntax Error occurs when the receiving functional NGAP entity:

1. receives IEs or IE groups that cannot be understood (unknown IE ID);

2. receives IEs for which the logical range is violated (e.g., ASN.1 definition: 0 to 15, the logical range is 0 to 10, while values 11 to 15 are undefined), and 12 will be received; this case will be handled as an abstract syntax error using criticality information sent by the originator of the message);
3. does not receive IEs or IE groups but according to the specified presence of the concerning object, the IEs or IE groups should have been present in the received message.
4. receives IEs or IE groups that are defined to be part of that message in wrong order or with too many occurrences of the same IE or IE group;
5. receives IEs or IE groups but according to the conditional presence of the concerning object and the specified condition, the IEs or IE groups should not have been present in the received message.
6. receives IEs or IE groups for a functionality that is not supported.

Cases 1, 2 and 6 (not comprehended IE/IE group) are handled based on received Criticality information. Case 3 (missing IE/IE group) is handled based on Criticality information and Presence information for the missing IE/IE group specified in the version of the specification used by the receiver. Case 4 (IEs or IE groups in wrong order or with too many occurrences) and Case 5 (erroneously present conditional IEs or IE groups) result in rejecting the procedure.

If an Abstract Syntax Error occurs, the receiver shall read the remaining message and shall then for each detected Abstract Syntax Error that belong to cases 1-3 and 6 act according to the Criticality Information and Presence Information for the IE/IE group due to which Abstract Syntax Error occurred in accordance with subclauses 10.3.4 and 10.3.5. The handling of cases 4 and 5 is specified in subclause 10.3.6.

10.3.2 Criticality Information

In the NGAP messages there is criticality information set for individual IEs and/or IE groups. This criticality information instructs the receiver how to act when receiving an IE or an IE group that is not comprehended, i.e., the entire item (IE or IE group) which is not (fully or partially) comprehended shall be treated in accordance with its own criticality information as specified in subclause 10.3.4.

In addition, the criticality information is used in case of the missing IE/IE group abstract syntax error (see subclause 10.3.5).

The receiving node shall take different actions depending on the value of the Criticality Information. The three possible values of the Criticality Information for an IE/IE group are:

- Reject IE.
- Ignore IE and Notify Sender.
- Ignore IE.

The comprehension of different IEs or IE groups within a standard version or between standard versions is not mandated. Any IE or IE group that is not supported shall be considered not comprehended, even if another IE or IE group for that EP from that standard version is comprehended, and action based on criticality shall be applied.

The comprehension of different EPs within a standard version or between different standard versions is not mandated. Any EP that is not supported shall be considered not comprehended, even if another EP from that standard version is comprehended, and action based on criticality shall be applied.

10.3.3 Presence Information

For many IEs/IE groups which are optional according to the ASN.1 transfer syntax, NGAP specifies separately if the presence of these IEs/IE groups is optional or mandatory with respect to RNS application by means of the presence field of the concerning object of class NGAP-PROTOCOL-IES, NGAP-PROTOCOL-IES-PAIR, NGAP-PROTOCOL-EXTENSION or NGAP-PRIVATE-IES.

The presence field of the indicated classes supports three values:

1. Optional;
2. Conditional;

3. Mandatory.

If an IE/IE group is not included in a received message and the presence of the IE/IE group is mandatory or the presence is conditional and the condition is true according to the version of the specification used by the receiver, an abstract syntax error occurs due to a missing IE/IE group.

If an IE/IE group is included in a received message and the presence of the IE/IE group is conditional and the condition is false according to the version of the specification used by the receiver, an abstract syntax error occurs due to this erroneously present conditional IE/IE group.

10.3.4 Not comprehended IE/IE group

10.3.4.1 Procedure Code

The receiving node shall treat the different types of received criticality information of the *Procedure Code* IE according to the following:

Reject IE:

- If a message is received with a *Procedure Code* IE marked with "*Reject IE*" which the receiving node does not comprehend, the receiving node shall reject the procedure using the Error Indication procedure.

Ignore IE and Notify Sender:

- If a message is received with a *Procedure Code* IE marked with "*Ignore IE and Notify Sender*" which the receiving node does not comprehend, the receiving node shall ignore the procedure and initiate the Error Indication procedure.

Ignore IE:

- If a message is received with a *Procedure Code* IE marked with "*Ignore IE*" which the receiving node does not comprehend, the receiving node shall ignore the procedure.

When using the Error Indication procedure to reject a procedure or to report an ignored procedure it shall include the *Procedure Code* IE, the *Triggering Message* IE, and the *Procedure Criticality* IE in the *Criticality Diagnostics* IE.

10.3.4.1A Type of Message

When the receiving node cannot decode the *Type of Message* IE, the Error Indication procedure shall be initiated with an appropriate cause value.

10.3.4.2 IEs other than the Procedure Code and Type of Message

The receiving node shall treat the different types of received criticality information of an IE/IE group other than the *Procedure Code* IE and *Type of Message* IE according to the following:

Reject IE:

- If a message *initiating* a procedure is received containing one or more IEs/IE group marked with "*Reject IE*" which the receiving node does not comprehend; none of the functional requests of the message shall be executed. The receiving node shall reject the procedure and report the rejection of one or more IEs/IE group using the message normally used to report unsuccessful outcome of the procedure. In case the information received in the initiating message was insufficient to determine a value for all IEs that are required to be present in the message used to report the unsuccessful outcome of the procedure, the receiving node shall instead terminate the procedure and initiate the Error Indication procedure.
- If a message *initiating* a procedure that does not have a message to report unsuccessful outcome is received containing one or more IEs/IE groups marked with "*Reject IE*" which the receiving node does not comprehend, the receiving node shall terminate the procedure and initiate the Error Indication procedure.
- If a *response* message is received containing one or more IEs marked with "*Reject IE*", that the receiving node does not comprehend, the receiving node shall consider the procedure as unsuccessfully terminated and initiate local error handling.

Ignore IE and Notify Sender:

- If a message *initiating* a procedure is received containing one or more IEs/IE groups marked with "*Ignore IE and Notify Sender*" which the receiving node does not comprehend, the receiving node shall ignore the content of the not comprehended IEs/IE groups, continue with the procedure as if the not comprehended IEs/IE groups were not received (except for the reporting) using the understood IEs/IE groups, and report in the response message of the procedure that one or more IEs/IE groups have been ignored. In case the information received in the initiating message was insufficient to determine a value for all IEs that are required to be present in the response message, the receiving node shall instead terminate the procedure and initiate the Error Indication procedure.
- if a message *initiating* a procedure that does not have a message to report the outcome of the procedure is received containing one or more IEs/IE groups marked with "*Ignore IE and Notify Sender*" which the receiving node does not comprehend, the receiving node shall ignore the content of the not comprehended IEs/IE groups, continue with the procedure as if the not comprehended IEs/IE groups were not received (except for the reporting) using the understood IEs/IE groups, and initiate the Error Indication procedure to report that one or more IEs/IE groups have been ignored.
- If a *response* message is received containing one or more IEs/IE groups marked with "*Ignore IE and Notify Sender*" which the receiving node does not comprehend, the receiving node shall ignore the content of the not comprehended IEs/IE groups, continue with the procedure as if the not comprehended IEs/IE groups were not received (except for the reporting) using the understood IEs/IE groups and initiate the Error Indication procedure.

Ignore IE:

- If a message *initiating* a procedure is received containing one or more IEs/IE groups marked with "*Ignore IE*" which the receiving node does not comprehend, the receiving node shall ignore the content of the not comprehended IEs/IE groups and continue with the procedure as if the not comprehended IEs/IE groups were not received using the understood IEs/IE groups.
- If a *response* message is received containing one or more IEs/IE groups marked with "*Ignore IE*" which the receiving node does not comprehend, the receiving node shall ignore the content of the not comprehended IEs/IE groups and continue with the procedure as if the not comprehended IEs/IE groups were not received using the understood IEs/IE groups.

When reporting not comprehended IEs/IE groups marked with "*Reject IE*" or "*Ignore IE and Notify Sender*" using a response message defined for the procedure, the *Information Element Criticality Diagnostics* IE shall be included in the *Criticality Diagnostics* IE for each reported IE/IE group.

When reporting not comprehended IEs/IE groups marked with "*Reject IE*" or "*Ignore IE and Notify Sender*" using the Error Indication procedure, the *Procedure Code* IE, the *Triggering Message* IE, *Procedure Criticality* IE, and the *Information Element Criticality Diagnostics* IE shall be included in the *Criticality Diagnostics* IE for each reported IE/IE group.

10.3.5 Missing IE or IE group

The receiving node shall treat the missing IE/IE group according to the criticality information for the missing IE/IE group in the received message specified in the version of this specification used by the receiver:

Reject IE:

- if a received message *initiating* a procedure is missing one or more IEs/IE groups with specified criticality "*Reject IE*"; none of the functional requests of the message shall be executed. The receiving node shall reject the procedure and report the missing IEs/IE groups using the message normally used to report unsuccessful outcome of the procedure. In case the information received in the initiating message was insufficient to determine a value for all IEs that are required to be present in the message used to report the unsuccessful outcome of the procedure, the receiving node shall instead terminate the procedure and initiate the Error Indication procedure.
- if a received message *initiating* a procedure that does not have a message to report unsuccessful outcome is missing one or more IEs/IE groups with specified criticality "*Reject IE*", the receiving node shall terminate the procedure and initiate the Error Indication procedure.
- if a received *response* message is missing one or more IEs/IE groups with specified criticality "*Reject IE*", the receiving node shall consider the procedure as unsuccessfully terminated and initiate local error handling.

Ignore IE and Notify Sender:

- if a received message *initiating* a procedure is missing one or more IEs/IE groups with specified criticality "*Ignore IE and Notify Sender*", the receiving node shall ignore that those IEs are missing and continue with the procedure based on the other IEs/IE groups present in the message and report in the response message of the procedure that one or more IEs/IE groups were missing. In case the information received in the initiating message was insufficient to determine a value for all IEs that are required to be present in the response message, the receiving node shall instead terminate the procedure and initiate the Error Indication procedure.
- if a received message *initiating* a procedure that does not have a message to report the outcome of the procedure is missing one or more IEs/IE groups with specified criticality "*Ignore IE and Notify Sender*", the receiving node shall ignore that those IEs are missing and continue with the procedure based on the other IEs/IE groups present in the message and initiate the Error Indication procedure to report that one or more IEs/IE groups were missing.
- if a received *response* message is missing one or more IEs/IE groups with specified criticality "*Ignore IE and Notify Sender*", the receiving node shall ignore that those IEs are missing and continue with the procedure based on the other IEs/IE groups present in the message and initiate the Error Indication procedure to report that one or more IEs/IE groups were missing.

Ignore IE:

- if a received message *initiating* a procedure is missing one or more IEs/IE groups with specified criticality "*Ignore IE*", the receiving node shall ignore that those IEs are missing and continue with the procedure based on the other IEs/IE groups present in the message.
- if a received *response* message is missing one or more IEs/IE groups with specified criticality "*Ignore IE*", the receiving node shall ignore that those IEs/IE groups are missing and continue with the procedure based on the other IEs/IE groups present in the message.

When reporting missing IEs/IE groups with specified criticality "*Reject IE*" or "*Ignore IE and Notify Sender*" using a response message defined for the procedure, the *Information Element Criticality Diagnostics* IE shall be included in the *Criticality Diagnostics* IE for each reported IE/IE group.

When reporting missing IEs/IE groups with specified criticality "*Reject IE*" or "*Ignore IE and Notify Sender*" using the Error Indication procedure, the *Procedure Code* IE, the *Triggering Message* IE, *Procedure Criticality* IE, and the *Information Element Criticality Diagnostics* IE shall be included in the *Criticality Diagnostics* IE for each reported IE/IE group.

10.3.6 IEs or IE groups received in wrong order or with too many occurrences or erroneously present

If a message with IEs or IE groups in wrong order or with too many occurrences is received or if IEs or IE groups with a conditional presence are present when the condition is not met (i.e., erroneously present), the receiving node shall behave according to the following:

- If a message *initiating* a procedure is received containing IEs or IE groups in wrong order or with too many occurrences or erroneously present, none of the functional requests of the message shall be executed. The receiving node shall reject the procedure and report the cause value "Abstract Syntax Error (Falsely Constructed Message)" using the message normally used to report unsuccessful outcome of the procedure. In case the information received in the initiating message was insufficient to determine a value for all IEs that are required to be present in the message used to report the unsuccessful outcome of the procedure, the receiving node shall instead terminate the procedure and initiate the Error Indication procedure.
- If a message *initiating* a procedure that does not have a message to report unsuccessful outcome is received containing IEs or IE groups in wrong order or with too many occurrences or erroneously present, the receiving node shall terminate the procedure and initiate the Error Indication procedure, and use cause value "Abstract Syntax Error (Falsely Constructed Message)".
- If a *response* message is received containing IEs or IE groups in wrong order or with too many occurrences or erroneously present, the receiving node shall consider the procedure as unsuccessfully terminated and initiate local error handling.

When determining the correct order only the IEs specified in the specification version used by the receiver shall be considered.

10.4 Logical Error

Logical error situations occur when a message is comprehended correctly, but the information contained within the message is not valid (i.e., semantic error), or describes a procedure which is not compatible with the state of the receiver. In these conditions, the following behaviour shall be performed (unless otherwise specified) as defined by the class of the elementary procedure, irrespective of the criticality information of the IEs/IE groups containing the erroneous values.

Class 1:

Where the logical error occurs in a request message of a class 1 procedure, and the procedure has a message to report this unsuccessful outcome, this message shall be sent with an appropriate cause value. Typical cause values are:

- Semantic Error.
- Message not compatible with receiver state.

Where the logical error is contained in a request message of a class 1 procedure, and the procedure does not have a message to report this unsuccessful outcome, the procedure shall be terminated and the Error Indication procedure shall be initiated with an appropriate cause value. The *Procedure Code* IE and the *Triggering Message* IE within the *Criticality Diagnostics* IE shall then be included in order to identify the message containing the logical error.

Where the logical error exists in a response message of a class 1 procedure, the procedure shall be considered as unsuccessfully terminated and local error handling shall be initiated.

Class 2:

Where the logical error occurs in a message of a class 2 procedure, the procedure shall be terminated and the Error Indication procedure shall be initiated with an appropriate cause value. The *Procedure Code* IE and the *Triggering Message* IE within the *Criticality Diagnostics* IE shall then be included in order to identify the message containing the logical error.

10.5 Exceptions

The error handling for all the cases described hereafter shall take precedence over any other error handling described in the other subclauses of clause 10.

- If any type of error (Transfer Syntax Error, Abstract Syntax Error or Logical Error) is detected in the ERROR INDICATION message, it shall not trigger the Error Indication procedure in the receiving Node but local error handling.
- In case a response message or Error Indication message needs to be returned, but the information necessary to determine the receiver of that message is missing, the procedure shall be considered as unsuccessfully terminated and local error handling shall be initiated.
- If an error that terminates a procedure occurs, the returned cause value shall reflect the error that caused the termination of the procedure even if one or more abstract syntax errors with criticality "ignore and notify" have earlier occurred within the same procedure.
- If an AP ID error is detected, the error handling as described in subclause 10.6 shall be applied.

10.6 Handling of AP ID

NOTE: The "first message", the "first returned message" and the "last message" as used below correspond to messages for a UE-associated logical connection. The "first message" has a new AP ID from the sending node and the "first returned message" is the first response message, which has a new AP ID from the node sending the "first returned message". Thereafter the two AP IDs are included in all messages over the UE-associated logical connection unless otherwise allowed by the specification. The "last message" is a message sent by a node in order to complete the termination of a given UE-associated logical connection, such that no other messages for the same connection are expected in either direction. The nodes should ensure as far as possible that previously allocated AP ID are not immediately reused.

If a node receives a first returned message that includes an unknown local AP ID, the receiving node shall initiate an Error Indication procedure with inclusion of the received AP IDs from the peer node and an appropriate cause value. Both nodes shall initiate a local release of any established UE-associated logical connection (for the same NG interface) having these AP IDs as local or remote identifier.

If a node receives a message (other than the first or first returned messages) including an erroneous AP ID that is either an unknown local AP ID, or an inconsistent remote AP ID (i.e. it is different to the remote AP ID stored previously for this UE-associated logical connection) for the same NG interface:

- if this message is not the last message for this UE-associated logical connection, the node shall initiate an Error Indication procedure with inclusion of the received AP ID(s) from the peer node and an appropriate cause value. Both nodes shall initiate a local release of any established UE-associated logical connection (for the same NG interface) having the erroneous AP ID as either the local or remote identifier.
- if this message is the last message for this UE-associated logical connection, the receiving node shall initiate a local release of any established UE-associated logical connection (for the same NG interface) having the erroneous AP ID as either the local or remote identifier.

Annex A (informative): Change history

Change history							
Date	Meeting	Tdoc	CR	Rev	Cat	Subject/Comment	New version
2017-04	R3#95b	R3-171209	-	-	-	TS skeleton	0.0.0
2017-04	R3#95b	R3-171311	-	-	-	Incorporated agreed TPs from R3#95b	0.0.1
2017-05	R3#96	R3-171480	-	-	-	Update of title page and change history	0.0.2
2017-05	R3#96	R3-171975	-	-	-	Incorporated agreed TPs from R3#96	0.1.0
2017-07	R3 NR#2	R3-172604	-	-	-	Incorporated agreed TPs from R3 NR#2 Adhoc	0.2.0
2017-08	R3#97	R3-173447	-	-	-	Incorporated agreed TPs from R3#97	0.3.0
2017-10	R3#97b	R3-174239	-	-	-	Incorporated agreed TPs from R3#97b	0.4.0
2017-12	R3#98	R3-175056	-	-	-	Incorporated agreed TPs from R3#98	0.5.0
2018-01	R3 NR#1	R3-180651	-	-	-	Incorporated agreed TPs from R3 NR Adhoc 1801	0.6.0
2018-03	R3#99	R3-181588	-	-	-	Incorporated agreed TPs from R3#99	0.7.0
2018-04	R3#99b	R3-182524	-	-	-	Incorporated agreed TPs from R3#99b	0.8.0
2018-05	R3#100	R3-183592	-	-	-	Incorporated agreed TPs from R3#100	0.9.0
2018-06	RAN#80	RP-180737	-	-	-	For approval	1.0.0
2018-06	RAN#80	-	-	-	-	Specification approved at TSG-RAN and placed under change control	15.0.0
2018-09	RAN#81	RP-181922	0001	2	F	NR Corrections (38.413 Baseline CR covering RAN3-101 agreements)	15.1.0
2018-12	RAN#82	RP-182448	0003	2	F	Baseline CR for TS 38.413	15.2.0
2019-03	RAN#83	RP-190556	0005	3	F	NGAP Corrections for UP Security Handling in DC during PDU Session Lifetime	15.3.0
2019-03	RAN#83	RP-190555	0008	2	F	Separate UL/DL limits for UE's maximum IP rate	15.3.0
2019-03	RAN#83	RP-190554	0009	1	F	Data volume reporting for MR-DC with 5GC	15.3.0
2019-03	RAN#83	RP-190554	0010	3	F	Correction of PDU Session split at handover	15.3.0
2019-03	RAN#83	RP-190556	0011	1	F	Correction of EPS Voice Fallback	15.3.0
2019-03	RAN#83	RP-190556	0012	-	F	Correction of slice support over NG	15.3.0
2019-03	RAN#83	RP-190556	0014	1	F	Rapporteur updates for TS 38.413	15.3.0
2019-03	RAN#83	RP-190556	0015	-	F	Correction of User Location Information IE presence in HANDOVER NOTIFY	15.3.0
2019-03	RAN#83	RP-190556	0019	1	F	Correction to RRC state report	15.3.0
2019-03	RAN#83	RP-190555	0021	-	F	Support of RAN initiated multiple SCTP associations	15.3.0
2019-03	RAN#83	RP-190556	0023	-	F	Corrections on RAN/AMF Configuration Update	15.3.0
2019-03	RAN#83	RP-190556	0024	2	F	Correction of EPC interworking	15.3.0
2019-03	RAN#83	RP-190556	0025	1	F	Correction of Emergency Fallback	15.3.0
2019-03	RAN#83	RP-190202	0027	3	F	Transfer of the PSCell information to Core Network	15.3.0
2019-03	RAN#83	RP-190558	0028	1	F	Release due to pre-emption	15.3.0
2019-03	RAN#83	RP-190558	0029	-	F	Handling of APID for the first returned message	15.3.0
2019-03	RAN#83	RP-190556	0037	-	F	Clarification on the usage of TNL information	15.3.0
2019-03	RAN#83	RP-190556	0044	1	F	NG Setup Correction and UE context retention	15.3.0
2019-03	RAN#83	RP-190556	0045	1	F	UE AMBR handling in PDU Session Resource Setup procedure	15.3.0
2019-03	RAN#83	RP-190556	0046	1	F	Remove the second tunnel in the PDU session split, 5GC initiated	15.3.0
2019-03	RAN#83	RP-190556	0048	1	F	When NG-RAN node fails to set up a QoS flow for IMS voice	15.3.0
2019-03	RAN#83	RP-190556	0052	-	F	Correction of ASN.1 for PDU Session Resource Modify Response	15.3.0
2019-03	RAN#83	RP-190556	0053	1	F	Cause value in RRC fallback case	15.3.0
2019-03	RAN#83	RP-190556	0058	2	F	S-NSSAI update during EPS to 5GS handover	15.3.0
2019-03	RAN#83	RP-190561	0064	1	F	Introduction of TNL Address discovery for EN-DC (using new container)	15.3.0
2019-03	RAN#83	RP-190200	0066	-	F	Correction of ASN.1 for SON Configuration Transfer and PDU Session Resource Modify Request Transfer	15.3.0
2019-07	RAN#84	RP-191394	0099	1	F	Rapporteur updates for TS 38.413	15.4.0
2019-07	RP-84	RP-191397	0041	2	F	Support of ongoing re-mapping on source side during SDAP mobility	15.4.0
2019-07	RP-84	RP-191397	0067	1	F	NGAP Further Clarification of S-NSSAI Update for EPS to 5GS HO	15.4.0
2019-07	RP-84	RP-191394	0071	-	F	CR38413 for Clarification on PDU Session resource modify	15.4.0
2019-07	RP-84	RP-191397	0075	1	F	Correction of Core Network Type Restrictions	15.4.0
2019-07	RP-84	RP-191394	0077	1	F	Correction of PDU Session Release	15.4.0
2019-07	RP-84	RP-191395	0084	2	F	Removal of multiple SCTP associations PS: This CR was not implemented as it was not based on the latest version of the spec.	15.4.0
2019-07	RP-84	RP-191394	0095	-	F	Correction on Error Indication procedure	15.4.0
2019-07	RP-84	RP-191394	0096	-	F	Location Report Request Type	15.4.0
2019-07	RP-84	RP-191394	0101	2	F	GUAMI update in case of AMF change	15.4.0
2019-07	RP-84	RP-191397	0102	2	F	Data forwarding and QoS flow remapping	15.4.0
2019-07	RP-84	RP-191397	0111	1	F	Correction of CN Assistance Information	15.4.0
2019-07	RP-84	RP-191397	0112	-	F	Correction of Network Instance	15.4.0
2019-07	RP-84	RP-191394	0117	1	F	Correction of AMF UE NGAP ID	15.4.0
2019-07	RP-84	RP-191394	0130	1	F	Adding PSCell to the User Location Information	15.4.0
2019-07	RP-84	RP-191394	0135	-	F	Correction on Handover Command message	15.4.0
2019-07	RP-84	RP-191394	0148	-	F	Correction of duplicated descriptions on additional UL tunnel information	15.4.0
2019-09	RP-85	RP-192167	0084	4	F	Removal of multiple SCTP associations	15.5.0

2019-09	RP-85	RP-192166	0161	2	F	Correction of secured signalling connection	15.5.0
2019-09	RP-85	RP-192167	0178	1	F	PDU Session fail in Path Switch Request procedure	15.5.0
2019-09	RP-85	RP-192167	0195	2	F	Reroute NSSF provided information	15.5.0
2019-09	RP-85	RP-192166	0199		F	Correction of Handover Command message	15.5.0
2019-09	RP-85	RP-192167	0220	1	F	NGAP correction of Initial Context Setup procedure text	15.5.0
2019-09	RP-85	RP-192167	0226	1	F	Rapporteur cleanup of IE semantics descriptions	15.5.0
2019-12	RP-86	RP-192915	0256	1	F	Correction of NAS transparent container	15.6.0
2019-12	RP-86	RP-192915	0258	1	F	Missing procedural texts for NG interface	15.6.0
2019-12	RP-86	RP-192915	0261		F	Correction of Handover Command	15.6.0
2019-12	RP-86	RP-192915	0262	1	F	Correction of S-NSSAI coding	15.6.0
2019-12	RP-86	RP-192916	0269	1	F	Correction of Port Number IE in tabular	15.6.0
2019-12	RP-86	RP-192915	0276	2	F	Enable inclusion of the Backup AMF Name IE	15.6.0
2019-12	RP-86	RP-192916	0281		F	Correction of NG Handover	15.6.0
2019-12	RP-86	RP-192896	0286	3	F	Addition of abnormal cases for location report procedure	15.6.0
2019-12	RP-86	RP-192916	0300	2	F	CR to 38.413 on clarifications to Xn TNL Configuration Info	15.6.0
2019-12	RP-86	RP-192916	0303		F	CR for Clarification on purpose of path switch request	15.6.0
2019-12	RP-86	RP-193055	0304	-	F	Correction of Xn TNL Configuration Info	15.6.0
2019-12	RP-86	RP-192912	0051	7	B	Support of Direct Data forwarding for handover between 4G and 5G	16.0.0
2019-12	RP-86	RP-192908	0137	6	B	CR to 38.413 for signalling design for RIM	16.0.0
2019-12	RP-86	RP-192916	0143	3	B	The GUAMI and GUMMEI usage for EPS/5GS interworking	16.0.0
2019-12	RP-86	RP-192913	0266	1	C	Extending the MDBV Range	16.0.0
2020-03	RP-87-e	RP-200424	0234	6	B	Support of SRVCC from 5G to 3G	16.1.0
2020-03	RP-87-e	RP-200422	0291	2	B	Introduction of NR-U	16.1.0
2020-03	RP-87-e	RP-200425	0314	1	F	Addition of the PSCell information in the path update procedure	16.1.0
2020-03	RP-87-e	RP-200428	0317		A	Correction of Warning Security Information in ETWS primary notification	16.1.0
2020-03	RP-87-e	RP-200429	0319		A	Correction of tabular for Xn TNL Configuration Info	16.1.0
2020-03	RP-87-e	RP-200425	0320	1	F	NGAP Rapporteur corrections	16.1.0
2020-03	RP-87-e	RP-200475	0329	4	B	E2E delay measurement for QoS monitoring for URLLC	16.1.0
2020-03	RP-87-e	RP-200419	0331	1	B	Inter-system direct forwarding with shared SgNB/gNB	16.1.0
2020-03	RP-87-e	RP-200428	0335		A	Correction of RAN paging priority	16.1.0
2020-03	RP-87-e	RP-200428	0337	1	A	PDU session resource in UE context release	16.1.0
2020-03	RP-87-e	RP-200423	0347	2	B	Introducing Radio Capability Optimisation (RACS) (The CR is not implemented. The CR was marked agreed by mistake while the WI is not yet complete)	16.1.0
2020-07	RP-88-e	RP-201077	0063	13	B	BL CR to 38.413: Support for IAB	16.2.0
2020-07	RP-88-e	RP-201079	0082	15	B	Introduction of NR IIOT support to TS 38.413	16.2.0
2020-07	RP-88-e	RP-201088	0120	10	B	Introduction of NB-IoT Paging and eDRX aspects	16.2.0
2020-07	RP-88-e	RP-201086	0153	11	B	Common CP/UP aspects of CIoT UEs when connected to 5GC	16.2.0
2020-07	RP-88-e	RP-201335	0156	11	B	Introduction of NB-IoT related NG-AP procedures	16.2.0
2020-07	RP-88-e	RP-201088	0157	9	B	Introduction of CP UP NB-IoT Others	16.2.0
2020-07	RP-88-e	RP-201074	0168	10	B	Support of NR V2X over NG	16.2.0
2020-07	RP-88-e	RP-201087	0172	10	B	Introduction of eMTC connected to 5GC	16.2.0
2020-07	RP-88-e	RP-201086	0173	8	B	Introduction of Control Plane CIoT 5GS Optimisation for NB-IOT and eMTC	16.2.0
2020-07	RP-88-e	RP-201086	0188	10	B	Introduction of Suspend-Resume	16.2.0
2020-07	RP-88-e	RP-201081	0192	11	B	CR for introducing WWC in RAN	16.2.0
2020-07	RP-88-e	RP-201082	0237	10	B	Addition of SON features	16.2.0
2020-07	RP-88-e	RP-201082	0280	7	B	Addition of MDT feature	16.2.0
2020-07	RP-88-e	RP-201080	0290	9	B	Introduction of Non-Public Networks	16.2.0
2020-07	RP-88-e	RP-201079	0313	4	B	Support of Ethernet Header Compression	16.2.0
2020-07	RP-88-e	RP-201078	0347	6	B	Introducing Radio Capability Optimisation (RACS)	16.2.0
2020-07	RP-88-e	RP-201091	0357	2	A	Clarification the usage of the New AMF UE NGAP ID included in the UE CONTEXT MODIFICATION REQUEST message	16.2.0
2020-07	RP-88-e	RP-201075	0362	5	B	Baseline CR for introducing Rel-16 NR mobility enhancement	16.2.0
2020-07	RP-88-e	RP-201083	0364		F	ASN.1 Correction of the Data Forwarding Response E-RAB List IE	16.2.0
2020-07	RP-88-e	RP-201085	0365		F	NGAP Rapporteur corrections	16.2.0
2020-07	RP-88-e	RP-201091	0371	1	A	Correction of Revoke E-RAB ID	16.2.0
2020-07	RP-88-e	RP-200795	0372	3	F	Voice fallback triggered by PDU session resource setup	16.2.0
2020-07	RP-88-e	RP-201091	0379	1	A	Correction on AS rekeying handling	16.2.0
2020-07	RP-88-e	RP-201090	0389	3	A	Correction to PDU SESSION RESOURCE MODIFY CONFIRM	16.2.0
2020-07	RP-88-e	RP-201092	0392	1	A	Selected PLMN ID for untrusted non-3GPP access	16.2.0
2020-07	RP-88-e	RP-201090	0395	2	A	Correstion on PDU Session Resrouce Modification Procedures	16.2.0
2020-07	RP-88-e	RP-201085	0401	1	F	QoS monitoring for URLLC	16.2.0
2020-07	RP-88-e	RP-201090	0408	4	F	Correction of S-NSSAI range	16.2.0
2020-09	RP-89-e	RP-201955	0383	3	F	Support of PSCell/SCell-only operation mode	16.3.0
2020-09	RP-89-e	RP-201945	0396	4	B	Update of the NRPPa Transport procedure to support NR positioning	16.3.0
2020-09	RP-89-e	RP-201948	0416	-	F	NGAP tabular corrections and asn.1 review	16.3.0
2020-09	RP-89-e	RP-201950	0417	1	F	Rapporteur cleanup of NGAP	16.3.0
2020-09	RP-89-e	RP-201955	0425	1	F	Correction of NAS PDU in PDU Session Modify	16.3.0
2020-09	RP-89-e	RP-201948	0427	1	F	Correction of NPN CAG Cells and non-CAG Cells	16.3.0

2020-09	RP-89-e	RP-201955	0443	1	A	Failure case of user location report	16.3.0
2020-09	RP-89-e	RP-201955	0445	1	A	Multiple location reporting requests and report	16.3.0
2020-09	RP-89-e	RP-201955	0462	-	F	Correction of asn.1 in NGAP Elementary Procedure List	16.3.0
2020-09	RP-89-e	RP-201955	0463	1	F	Corrections to 38.413 on node name type	16.3.0
2020-12	RP-90-e	RP-202314	0410	1	F	Correction on Coverage Enhancement Restrictions	16.4.0
2020-12	RP-90-e	RP-202314	0411	2	F	Correction on immediate suspension	16.4.0
2020-12	RP-90-e	RP-202310	0414	1	F	Add the support for updating RG Level Wireline Access Characteristics and Global Cable ID	16.4.0
2020-12	RP-90-e	RP-202314	0483	2	F	Correction of usage of the Extended Connected Time	16.4.0
2020-12	RP-90-e	RP-202312	0484	1	F	Support of release on CAG subscription change	16.4.0
2020-12	RP-90-e	RP-202313	0485	-	F	Removal of duplicate import	16.4.0
2020-12	RP-90-e	RP-202311	0486	1	F	Correction of Redundant Tunnel Setup	16.4.0
2020-12	RP-90-e	RP-202314	0499	1	F	CR38413 for clarification on UE-associated signalling for NBIOT procedures in Rel-16	16.4.0
2020-12	RP-90-e	RP-202315	0501	1	F	CR38413 for clarification on UE-associated signalling in Rel-16	16.4.0
2020-12	RP-90-e	RP-202315	0505	1	A	Clarification on an abnormal condition in PDU Session Resource Modify Procedure	16.4.0
2020-12	RP-90-e	RP-202313	0507	1	F	Introduction of reporting frequency for Qos monitoring for URLLC	16.4.0
2020-12	RP-90-e	RP-202310	0511	1	F	Introducing AQP in path switch request acknowledge message	16.4.0
2020-12	RP-90-e	RP-202312	0512	1	F	Introducing UE radio capability ID in Connection Establishment Indication	16.4.0
2020-12	RP-90-e	RP-202314	0514		F	Correction of RAN CP Relocation	16.4.0
2021-03	RP-91-e	RP-210239	0355	5	A	Clarification of AS re-keying in the UE Context Modification procedure	16.5.0
2021-03	RP-91-e	RP-210235	0508	4	F	Introducing QoS parameters update at Xn handover	16.5.0
2021-03	RP-91-e	RP-210239	0520	2	F	Including the Redundant UL NG-U UP TNL Information in the Modify Request	16.5.0
2021-03	RP-91-e	RP-210232	0533	1	F	Correction of SNPN failures	16.5.0
2021-03	RP-91-e	RP-210239	0534	1	F	Update on QoS monitoring control	16.5.0
2021-03	RP-91-e	RP-210237	0537	2	F	Correction on RAT Information Handling	16.5.0
2021-03	RP-91-e	RP-210230	0541	-	F	Correction to NRPPa Transport procedure description	16.5.0
2021-03	RP-91-e	RP-210237	0544	-	F	Correction on UE identity index for eMTC UE in RRC_INACTIVE	16.5.0
2021-03	RP-91-e	RP-210235	0557	2	F	Clarification of Secondary RAT in mobility restrictions	16.5.0
2021-06	RP-92-e	RP-211315	0477	3	F	Clarification on TAI Slice Support List	16.6.0
2021-06	RP-92-e	RP-211333	0522	3	F	Introducing Maximum Integrity Protected Data Rate after EPC to 5GC handover	16.6.0
2021-06	RP-92-e	RP-211333	0547	2	F	Supporting use of UE Radio Capability for Paging in RRC_INACTIVE	16.6.0
2021-06	RP-92-e	RP-211333	0556	2	A	Interactions with other procedures for the UE TNLA BINDING RELEASE	16.6.0
2021-06	RP-92-e	RP-211333	0574	1	F	Correction of PDU Session Resource Modification	16.6.0
2021-06	RP-92-e	RP-211334	0583	1	A	Correction on Abnormal Conditions in Handover Preparation Procedure for R16	16.6.0
2021-06	RP-92-e	RP-211334	0603	2	F	Cause value on NG for insufficient UE capabilities CR 38.413	16.6.0
2021-06	RP-92-e	RP-211333	0610	-	F	Correction on the use of the Core Network Assistance Information for RRC_INACTIVE IE	16.6.0
2021-06	RP-92-e	RP-211324	0614	2	F	Correction on Extended UE Identity Index Value	16.6.0
2021-09	RP-93-e	RP-211882	0376	7	F	NAS Non-Delivery	16.7.0
2021-09	RP-93-e	RP-211881	0431	4	F	Correction on Expected UE activity behaviour	16.7.0
2021-09	RP-93-e	RP-211881	0626		F	Correction of MICO mode	16.7.0
2021-09	RP-93-e	RP-211876	0631	-	F	Correcting Presence of the Cell CAG Information IE in ASN.1	16.7.0
2021-09	RP-93-e	RP-211882	0633	1	A	Deactivation of the MICO mode indication	16.7.0
2021-09	RP-93-e	RP-211881	0637	1	F	Correction of NAS PDU Non Delivery	16.7.0
2021-09	RP-93-e	RP-211882	0641	1	F	On IEs with reject criticality in the source and target transparent container	16.7.0
2021-09	RP-93-e	RP-211881	0645	1	F	Missing QoS Flows not Admitted List in HANDOVER COMMAND	16.7.0
2021-09	RP-93-e	RP-211883	0646	-	F	Clarification on the specified maximum length of the Routing ID Octet String	16.7.0
2021-09	RP-93-e	RP-211882	0660		F	Correction CR on Network instance	16.7.0
2021-12	RP-94-e	RP-212863	0663	1	F	Correction of Data Volume Report	16.8.0
2021-12	RP-94-e	RP-212863	0671	1	F	Adding reference for coding of Common Network Instance	16.8.0
2021-12	RP-94-e	RP-212869	0679	1	F	Clarification of UE with CAG information in UE subscription	16.8.0
2021-12	RP-94-e	RP-212868	0688	1	F	(NGAP CR) support the UE Radio Capability for Paging in RACS context	16.8.0
2021-12	RP-94-e	RP-212871	0693	1	F	Redundant network instance for split PDU session	16.8.0
2021-12	RP-94-e	RP-212864	0713	1	F	Allocation of TNL addresses for intra-system data forwarding	16.8.0
2022-03	RP-95-e	RP-220278	0619	3	F	Support of dynamic ACL during handover and dual connectivity	16.9.0
2022-03	RP-95-e	RP-220243	0691	2	F	Direct data forwarding for 4G to 5G handover	16.9.0
2022-03	RP-95-e	RP-220280	0727	1	F	Propagation of user consent related information during Xn inter-PLMN handover	16.9.0
2022-03	RP-95-e	RP-220278	0735		F	Correction on PDU Session Resource Setup procedure	16.9.0
2022-03	RP-95-e	RP-220279	0742	2	F	Correction of SNPN setup failure	16.9.0

2022-03	RP-95-e	RP-220280	0746	1	F	Value range misalignment for MDT M1, M8 and M9 configuration	16.9.0
2022-03	RP-95-e	RP-220279	0751	1	F	Clarification of the usage of an IE in case of DAPS HO	16.9.0
2022-03	RP-95-e	RP-220277	0752	1	F	Correction of intra-system Data Forwarding	16.9.0
2022-03	RP-95-e	RP-220243	0760	2	F	Direct data forwarding for mobility between DC and SA	16.9.0
2022-03	RP-95-e	RP-220225	0490	10	B	Introduction of NTN	17.0.0
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2025-12	RAN#110	RP-253347	1373	1	F	Correction on the presence of Reader Report List	19.1.0
2025-12	RAN#110	RP-253341	1380		D	Rapporteur corrections	19.1.0
2025-12	RAN#110	RP-253358	1387	1	A	Correction of URCP for RAN paging	19.1.0
2026-03	RAN#111	RP-260276	1377	3	A	Correction on prioritized alternative QoS profile	19.2.0
2026-03	RAN#111	RP-260273	1411	-	A	Beam measurement report quantity	19.2.0
2026-03	RAN#111	RP-260269	1412	2	F	Correction on command failure in A-IoT	19.2.0
2026-03	RAN#111	RP-260269	1413	-	F	Correction on A-IoT Device Identification Requested	19.2.0
2026-03	RAN#111	RP-260269	1415	2	F	Correction on the inclusion of security parameter for A-IoT	19.2.0
2026-03	RAN#111	RP-260269	1422	1	F	Correction on A-IoT related NGAP procedures	19.2.0
2026-03	RAN#111	RP-260268	1430	3	F	Correction on Aerial UE Flight Information Reporting Control IE	19.2.0
2026-03	RAN#111	RP-260269	1432	-	F	Correction on the description of NGAP Services	19.2.0
2026-06	RAN#112	RP-261201	1420	3	F	Correction on NR Femto awareness at CN	19.3.0
2026-06	RAN#112	RP-261204	1421	2	F	Correction on the LP-WUS disable indication	19.3.0
2026-06	RAN#112	RP-261201	1438	1	F	Correction on A-IoT aspects	19.3.0
2026-06	RAN#112	RP-261211	1451	2	A	Correction on QoE communication service type	19.3.0
2026-06	RAN#112	RP-261206	1454	1	F	Correction on Error Indication procedure	19.3.0
2026-06	RAN#112	RP-261201	1455	0	F	Miscellaneous corrections on A-IoT aspects	19.3.0
2026-06	RAN#112	RP-261206	1457	1	F	Clarification on Reporting Area	19.3.0

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